POWER SEMICONDUCTOR CONTROLLED DRIVES

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ING. ELECTRONICA

GOPAL K. DUBEY

LIST OF PRINCIPAL SYMBOLS

a	Per unit frequency i.e. ratio of operating to rated frequency
a_{T1}	Stator to rotor turns ratio
a_{T2}	AC side to converter side transformer turns ratio
a_T	a_{T1}/a_{T2}
B	Viscous friction coefficient
E	Voltage induced in the armature of a dc motor or in the stator of an induction motor, V
f	Frequency, Hz
f_b	Base frequency, Hz
f_c	Frequency of the carrier wave, Hz
I_a	Average value of the armature current of a dc motor, A
i_a	Instantaneous value of the armature current of a dc motor, A
Δi_a	Ripple in armature current, A
I_{ac}	Average value of the armature current at critical speed ω_{mc}, A
I_d	Average value of the dc link current of a converter, A
i_d	Instantaneous value of the dc link current of a converter, A
I_F	Field current of a synchronous motor, A
I_f	Per phase ac equivalent of I_F for a synchronous motor (also field current of a dc motor), A
I'_f	$I_f(X_m/X_s)$ for a synchronous motor, A
I_m	Magnetising current, A
I'_m	The ratio (V/jX_s) for a synchronous motor, A
I_M	Rated value of I'_m for a synchronous motor, A
I_r	Fundamental component of rotor current, A
I'_r	Fundamental component of rotor current referred to stator, A
I_{rms}	R.M.S. value of a non-sinusoidal current, A
I_s	Fundamental component of induction motor stator (or synchronous motor armature) current, A
i_s	Instantaneous value of ac source current, A
I_{sk}	kth harmonic in induction motor stator (or synchronous motor armature) current, A
J	Polar moment of inertia of motor-load system referred to the motor shaft, Kg-m^2
K	$K_e\Phi$
K_e	dc motor constant (also eddy current coefficient)
K_f	Φ/I_a
L_a	Armature circuit inductance of a dc motor, H
L_c	Commutating inductance of a synchronous motor, H
L_d	DC link inductance, H
L_s	Synchronous inductance of a synchronous motor, H
m	Modulation index
N	Speed, rpm
N_s	Synchronous speed, rpm
N_{sl}	Slip-speed, rpm
p	Number of poles
P_e	Rotor circuit electrical power, W
P_{cr}	Rotor circuit copper loss, W
P_g	Air-gap power, W
P_m	Develope mechanical power, W
P_r	Power recovered, W
R_a	Armature circuit resistance of a dc motor, Ω
R_B	Braking resistance, Ω
R_r	Rotor resistance, Ω
R'_r	Rotor resistance referred to stator, Ω
R_s	Stator resistance, Ω
R_t	Thevenin's equivalent resistance, Ω
s	Slip per unit (also Laplace operator)
s_k	kth harmonic slip per unit
s_m	Slip at the maximum torque
t	Time, S
T	Torque developed, N-m (also chopper period, S)

T_a	Torque developed, N-m
T_{ac}	Torque at the critical speed, ω_{mc}, N-m
T_F	Friction torque, N-m
T_L	Load torque, N-m
T_M	Torque required to do the useful mechanical work, N-m
T_{max}	Breakdown/Pull-out torque, N-m
u	Commutation overlap angle, rad.
V	Source voltage, V
v	Instantaneous value of dc source voltage, V
V_a	Average value of the dc motor armature voltage (also rectifier output voltage), V
v_a	Instantaneous value of the dc motor armature voltage, V
v_{AO}	Instantaneous inverter output voltage between phase A and the central point of the dc source, V
V_1	Fundamental component in a non-sinusoidal voltage, V
V_d	Average value of dc link voltage, V
v_d	Instantaneous value of dc link voltage, V
V_f	Excitation emf, V
V_k	kth harmonic component in a non-sinusoidal voltage, V
V_m	Peak value of an ac source voltage, V
V_r	Injected voltage, V
v_s	Instantaneous ac source voltage, V
X_m	Magnetising reactance, Ω
X_r	Rotor leakage reactance, Ω
X_r'	Rotor leakage reactance referred to stator, Ω
X_s	Stator leakage reactance of an induction motor or synchronous reactance of a synchronous motor, Ω
X_t	Thevenin's equivalent reactance, Ω
Z	$[R^2 + (\omega L_a)^2]^{1/2}$
α	Firing angle, rad
β	Commutation lead angle of a load commutated inverter or the angle at which the dc motor armature current drops to zero, rad
β_{min}	Minimum value of commutation lead angle, rad
γ	Margin angle for a load commutated inverter (also the angle at which the induced emf of a converter fed dc motor equals the ac source voltage), rad
γ_m	Minimum value of margin angle, rad
Θ	Phase of I_r' with air-gap emf E, rad
τ_a	Armature time constant, S
τ_m	Mechanical time constant, S
ϕ	Power factor angle of a synchronous motor or an induction motor, rad
ϕ_1	Fundamental power factor angle, rad
ϕ_r	Phase of injected voltage V_r with respect to source voltage V, rad
ϕ_s	Phase of stator current with respect to the source voltage V, rad
Φ	Flux per pole, Webers
ψ	$\tan^{-1}(\omega L_a/R_a)$
ω	Source frequency, rad/sec
ω_m	Speed, rad/sec
ω_{mb}	Base speed, rad/sec
ω_{mc}	Speed on the boundary between continuous and discontinuous conductions, rad/sec
ω_{mo}	Ideal no load speed, rad/sec
ω_{ms}	Synchronous speed, rad/sec
ω_{sl}	Slip speed, rad/sec
δ	Duty ratio of a chopper, torque angle of a synchronous motor fed from a voltage source
δ'	Torque angle of a synchronous motor fed from a current source
Variable*	Reference value of a variable
$\overline{\text{Variable}}$	Phasor

Power Semiconductor Controlled Drives

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Preface

The book describes fundamentals common to all electrical drives: dc drives fed by controlled rectifiers and choppers; squirrel-cage induction motor drives controlled by ac voltage controllers, inverters, and cycloconverters; slip-power controlled wound-rotor induction motor drives; and inverter and cycloconverter controlled synchronous motor drives, including brushless dc and ac motor drives. Both open-loop and closed-loop drives are discussed. Apart from thyristor converters, GTO and power transistor converters are also covered. Attention is also given to the interaction between the motor and converter, and to the relative performance features of various drives. Emphasis is placed on fundamentals, and conventional methods of motor analysis are used so a reader not familiar with complicated methods of motor analysis can also follow the text.

A number of solved examples are given to demonstrate drive performance features, as well as analysis and design methods. Unsolved problems are given at the end of each chapter. Answers to these problems are given at the end of the book.

A list of references is provided at the end of each chapter. A bibliography listing relevant and related books is also given at the end of the book.

It is hoped that the book will be useful to practicing engineers, university teachers, and students. It can be used as a textbook for a one-semester senior or graduate course. While a senior-level course may cover only sixty to seventy percent of the text, the entire book can be used for a graduate course. The reader is assumed to have a basic knowledge of power electronics, electrical machines, and control systems. The course may also run concurrently with a first course on power electronics.

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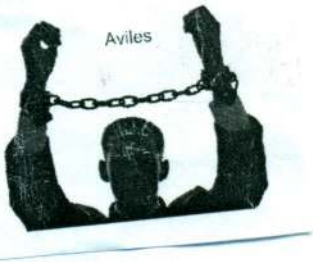

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Answers to Selected Problems

CHAPTER 1

1.1 A: unstable, B: unstable, C: stable, D: stable.

1.2 **(a)** $\omega_{mA} = \dfrac{a + \sqrt{a^2 - 4bc}}{2c}$, $\omega_{mB} = \dfrac{a - \sqrt{a^2 - 4bc}}{2c}$;
For positive and real equilibrium speeds $a^2 > 4bc$;
(b) ω_{mA} is stable and ω_{mB} is unstable.

1.3 A: unstable, B: stable, C: unstable, D: stable, E: unstable, F: stable, G: unstable, H: stable.

1.4 **(a)** $\omega_{mA} = \dfrac{a + \sqrt{a^2 - 4c(d-b)}}{2c}$, $\omega_{mB} = \dfrac{a - \sqrt{a^2 - 4c(d-b)}}{2c}$
For positive and real speeds $d > b$ and $a^2 > 4c(d - b)$;
(b) ω_{mA} is stable and ω_{mB} is unstable.

1.5 3 S.

CHAPTER 2

2.1 **(a)** decrease to 500 rpm; **(b)** decrease to 25 A.

2.2 **(a)** Increase to 2000 rpm; **(b)** Increase to 400 A.

2.3 11 Ω.

2.4 1000 rpm, 200 A.

2.5 **(a)** 139.4 V; **(b)** 80 percent.

2.6 562.5 rpm.

2.7 52.5 V.

2.8 1.44.
2.9 1118 rpm (clockwise), 223.6 A in reverse direction.
2.11 1022 rpm.
2.12 0.89 Ω.
2.13 8.64 Ω.
2.14 **(a)** 2.16 Ω, **(b)** 384 N-m, **(c)** 211 N-m.

CHAPTER 3

3.1 **(a)** 230/260; **(b)** **(i)** 53.2°, **(ii)** 130°, **(iii)** 117.4°
3.2 **(a)** 230/260; **(b)** **(i)** $\alpha = 86.3°$, $\alpha_n = 0°$, **(ii)** $\alpha = 180°$, $\alpha_n = 106.2°$, **(iii)** $\alpha = 180°$, $\alpha_n = 85.3°$.
3.3 **(a)** mode IV, 8.1 N-m; **(b)** mode I, 30 N-m; **(c)** mode V, 31.9 N-m; **(d)** mode VI, 16.3 N-m.
3.4 **(a)** mode II, 13.9 N-m; **(b)** mode I, 24 N-m; **(c)** mode VI, 12.3 N-m; **(d)** mode V, 23.9 N-m.
3.5 $\alpha = 30°$: 878 rpm, 558.3 rpm, 140 N-m; $\alpha = 120°$: 760 rpm, 462 rpm, 153 N-m.
3.6 **(a)** 2485 rpm, 1003 rpm, 15 N-m; **(b)** 0 rpm, −1408 rpm, 18 N-m.
3.7 **(a)** mode I, 71.8°; **(b)** mode I, 64.6°; **(c)** mode V, 111°.
3.8 **(a)** mode I, $\alpha = 112°$, $\alpha_n = 0°$; **(b)** mode I, $\alpha = 98°$, $\alpha_n = 0°$; **(c)** mode V, $\alpha = 180°$, $\alpha_n = 73.8°$.
3.9 **(a)** mode I, 155 rpm; **(b)** mode I, 184 rpm; **(c)** mode V, −359 rpm.
3.10 **(a)** −80 rpm; **(b)** 370 rpm.
3.11 **(a)** 460/180 V; **(b)** **(i)** 18.9°, **(ii)** 124.3°, **(iii)** 117.4°.
3.12 **(a)** 460/180 V; **(b)** **(i)** $\alpha = 18.9°$; **(ii)** $\alpha = 124.3°$; **(iii)** $\alpha_n = 57.4°$, $\alpha = 120°$.
3.13 **(a)** 416 rpm; **(b)** −824 rpm; **(c)** 495 rpm.
3.14 **(a)** 87 rpm; **(b)** 484 rpm.
3.15 **(a)** mode II, 841 rpm; **(b)** mode I, 443 rpm; **(c)** mode III, 1111 rpm.
3.16 **(a)** mode I, 61.7°; **(b)** mode I, 41°; **(c)** mode III, 110°; **(d)** mode III, 110°.
3.17 0.7, 88 percent, 98.66 percent.
3.18 6.66 N-m, 16.9 mH.
3.19 96 mH, 0 rpm, 9.36 N-m.
3.20 22.2 percent, 4.94 mH.
3.21 **(a)** 6.74 A; **(b)** 0.78, 80.6 percent; **(c)** **(i)** 98 percent, **(ii)** 96 percent.

CHAPTER 4

4.1 0.57, 1.69 A.
4.2 Few points on the characteristic are:

N(rpm)	1141	453	343.6	292.5
T_a(N-m)	22.5	162	338.8	528.5

4.3 **(a)** 643 rpm, 2.3 KW; **(b)** 0.96 A; **(c)** 971.4 rpm, 1214 rpm.
4.4 **(a)** 410 rpm; **(b)** 0.458.

4.5 **(a)** $\Delta i_a = \frac{E}{2R}\left[\frac{S+P}{R_a+R_B} - \frac{P+S}{R_a}\right]$,

$$T_a = \frac{K^2\omega_m}{R_a}\left[\frac{R_a+\delta R_B}{R_a+R_B} - \frac{R_B\tau_a' P}{(R_a+R_B)T} + \frac{R_B P\tau_a'}{RT}\left(\frac{Q}{R_a} + \frac{S}{R_a+R_B}\right)\right],$$

where $P = \exp\{-(1-\delta)T/\tau_a'\} - 1$, $Q = 1 - \{\exp - \delta T/\tau_a\}$,
$R = 1 - \exp\{-\delta T/\tau_a - (1-\delta)T/\tau_a'\}$, and
$S = \exp\{-\delta T/\tau_a\} - \exp\{-\delta T/\tau_a - (1-\delta)T/\tau_a'\}$;

(b) $T_a = (0.12N)$ N-m, N is speed in rpm.

4.6 **(a)** For $45.5 \le N \le 500$, $\delta = (0.42N - 19.12) \div 230$, and for $0 \le N \le 45.5$, $\delta = 0$, N is speed in rpm;

(b) 0.558,

$i_a = 217.6 - 218.2e^{-36.76t}$ A, $0 \le t \le 1.395 \times 10^{-3}$ S
$= 274e^{-36.76(t-0.001395)} - 263.6$ A,
1.395×10^{-3}S $\le t \le 2.5 \times 10^{-3}$S. Devices conduct as follows: D_2: 0 to 0.075 mS, S_1: 0.075 mS to 1.395 mS, D_1: 1.395 mS to 2.448 mS and S_2: 2.448 mS to 2.5 mS.

4.8 **(a)** 0.918; **(b)** 0.1515.

CHAPTER 6

6.1 **(a)** 59.17 A, 324 N-m, 0.9, 87.9 percent; **(b)** 0.266, 3.91; **(c)** 1.87; **(d)** 2.72 kW.

6.3 **(a)** 0 to −833 N-m, 1200 to 1275.4 rpm; **(b)** 79.6 kW; **(c)** 1216 rpm.

6.4 0.136, 3.94.

6.5

I_m(A)	8.16	4.9	2.86	1.71	0.9	0.368
N(rpm)	122	224.8	313	456	866	4549
T(N-m)	26.8	37.5	33.85	24.97	13.5	2.58

6.6 **(a)** 88.7 V, 7.4 A, no; **(b)** 62.75 V. 5.25 A, no.

6.7 **(a)** 1.88 Ω per phase; **(b)** 0.41 Ω per phase.

6.8 **(a)** Few points are tabulated below:

a	0.2	0.6	1.0	1.6	2.0
Motoring T_{max}(N-m)	62.8	143.5	182	82	53
Braking T_{max}(N-m)	−1170.5	−512	−403.6	−136	−83

(b) Few points are:

a	0.2	0.6	1.0
T_s(N-m)	53.4	90.4	80.56

6.9 117 percent, no, (V/f) ratio should be increased to 2.36.

6.10 Few points are

f(Hz)	6	12	18	24	36	48	60
V(V)	17	29.3	41.5	53.7	78	102.6	127

6.11 Few points are given here

f(Hz)	6	12	18	24	36	48	60
V(V)	30	43	54.5	65	85.9	106.4	127

For constant motoring breakdown torque: (V/f) = 2.9,
For constant (V/f) control: (V/f) = 2.1,
Braking breakdown torques: for (V/f) = 2.9, −1435 N-m
for (V/f) = 2.1, −761.3 N-m.

6.12 316 rpm.

6.13 (a) 0.4, 2.46; (b) 0.99, 0.445.

6.14 29.3 N-m, 1642 rpm.

6.15 (a) 42.67 Hz, (b) 820 rpm; (c) 51.3 Hz, 25.46 A; (d) 51.3 Hz, 26 A.

6.16 (a) 37.3 Hz; (b) 938.6 rpm, 25.56 A; (c) 939.6 rpm, 26.1 A.

6.17 20.25 V.

6.18 $8.94\angle -8.4°$ V or $45\angle -95.4°$ V; former will be preferred because of more efficient operation.

6.19 (a) (i) 1135.2 rpm, (ii) 1157.5 rpm; (b) (i) 1234.8 rpm, (ii) 1264.2 rpm.

6.21 (a) Few points are

I_2, p.u.	0.147	0.5	0.8	1.2	1.6	2.0	2.5	3.0
$\lvert\omega_{s\ell}\rvert$, rad/sec	0	3.97	6.59	10.2	14.0	18.3	24.6	32.8

The slip speed will be positive for motoring and negative for braking;
(b) motoring: 1120 rpm, 50.17 A; Braking: 1280 rpm, 50.17 A;
(c) motoring: 34.6 Hz, 32.3 A; Braking: 32 Hz, 32.3 A.

6.22 43 N-m, 10.48 A, 82.8 percent.

CHAPTER 7

7.2 1.169, 0.73

CHAPTER 8

8.1 (1) 41.33 Hz, 75.7 A, 48°; (2) 860 rpm, 75.7 A, 60.6°; (3) 10.2, 1.025.

8.2 (1) 41.4 Hz, 49°, 77.2 A, 91.1 percent, 0.81, 0.626; (2) 1169 rpm, 61 A, 93 percent, 0.81, 0.6; (3) 0.4 and 2.4 for motoring and braking respectively, 37.5 percent.

8.3 (1) slip speeds are 4.19 rad/sec and 1.99 rad/sec at rated and half rated torques respectively for all frequencies;
(2) Few points are tabulated below:

30 Hz	$\omega_{s\ell}$, rad/sec	6.28	10.47	15.7	20.94
	T, N-m	257	344	372.7	357.8
60 Hz	$\omega_{s\ell}$, rad/sec	5	10.33	15.1	
	T, N-m	147	181.2	170.6	

8.4 783 rpm, 40.5°, 15.33 A, 0.73, 15.63 A, 93 percent, 0.816.

8.5 (1) 182.2 N-m, 54.4 Hz, 49.33 A, 94.3 percent, 0.82;
(2) 1765 rpm, 165 N-m, 46.8 A, 94.8 percent, 0.86;
(3) 3143 rpm.

8.6 (1) 128.55 N-m, 67.9 Hz; (2) 2364 rpm, 92 N-m.

8.8 (1) 629.6 rpm; (2) 43 Hz; (3) 1477.5 rpm; (4) 70.8 Hz.

8.9 (1) 43.4 Hz, 185 A, 86.62 kW, 0.78, 92 percent;
(2) 631 rpm, 182.3 A; 58.68 kW, 0.77, 88.8 percent.

8.10 64 Hz, 78.7 kW, 109 A.

8.11 8.84 Ω.

8.12 (1) 102A, 125 A, 42.2°; (2) 40.4 Hz, 86.2 A, 0.66; (3) 736.6 rpm, 92.4 A, 113.2 A, 67.9°.

8.13 2243 rpm, 41.68 A, 51 A, 47.3°, 98.9 percent

8.14 0.75.

8.15 1208.8 rpm, 82.7 A, 132.7°.

8.16 32.86 Hz, 0.185.

CHAPTER 9

9.1 **(1)** 0.9 Ω; **(2)** 605.3 N-m.

9.3 **(1)** R = 4 or 0.003 Ω; **(2)** For R = 4 Ω, δ = 0.685 or 0.99; **(3)** For R = 4 Ω, N = −261 rpm or −16200 rpm.

9.4 **(1)** 0.85; **(2)** 888 or 144 rpm.

9.5 **(1)** 0.31; **(2)** 311.4 N-m, 0.29; **(3)** 0.126.

9.6 **(1)** 0.508; **(2)** 287 N-m, 0.57; **(3)** 394 N-m, 0.42.

9.8 **(1)** 0.305; **(2)** 126°; **(3)** 994.8 rpm.

CHAPTER 10

10.1 **(1)** 15.2 A, 61.4 N-m; **(2)** 59.7°; **(3)** 0.93 (lagging), 8.45 A; **(4)** Unity power factor operation not possible.

10.2 **(1)** 5.23, 62.75°; **(2)** 0.71 (leading), 616.8 A, 99.6 percent; **(3)** 140.5 A.

10.3 15.5 A, 0.84.

10.4 53373 N-m, 200 A.

10.5 *Below base speed:* I'_m, power factor and T are held constant at 19.24 A, 0.8 and 79.6 N-m respectively. V and P_m change linearly from their zero values at zero speed to 254 V and 15 kW respectively at base speed.
Above base speed:

N(rpm)	T(N-m)	P_m(kW)	cos ϕ
2160	67.7	15.3	0.817
2461	59.6	15.37	0.82

For 0.8 of full load power, theoretically, there is no restriction on the maximum speed.

10.6 24.47 N-m, yes at this torque the power factor will be unity for all speeds below base speed.

10.7 7800 rpm, 18.37 N-m, power factor will vary with speed.

10.8 *Below base speed:* I'_m, T and I_F are held constant at 47 A, 3979 N-m and 13.33 A respectively. P_m and V will increase linearly with speed from their zero values at zero speed to 500 kW and 3810.5 V, respectively, at 1200 rpm.
Above base speed: V and P_m will be constant at 3810.5 V and 500 kW respectively. For other parameters few points are:

N(rpm)	T(N-m)	I'_m(A)	I_F(A)
1440	3315.8	39.17	12.2
1680	2842	33.6	11.4
2040	2340.6	27.6	10.7
2400	1990	23.5	10.3

10.9 11.6 A.

CHAPTER 11

11.1 **(1)** 41.6°, 48029 N-m, 11 kV; **(2)** 2347 rpm; **(3)** 169°.

11.2 **(1)** 53.6°; **(2)** 0.66, 43.64°, 2704.7 N-m, 2934.6 V.

11.3 **(1)** 177.4°; **(2)** 0.16, 555.5 N-m, 717.4 V, 76.9°.

11.4 **(1)** 66.9°; **(2)** 0.6, 25327 N-m; **(3)** 0.43, 3630 N-m.

11.5 **(1)** $\delta'_{oc} = 191°$; **(2)** 0, 0 N-m; **(3)** 0.187, −1578.7 N-m.

11.8 Three points on the speed-torque curves are:

I_{rms}	Constant margin angle		Constant commutation lead angle		Constant no-load torque angle	
	N	T	N	T	N	T
1.5 × rated	1020	10254	1020	10254	1020	10254
rated	945.6	7345	1116	6059	2340	2863.5
0	820	0	1361	0	17807	0

11.9 For $I_s = 10$ A, $\delta' = 131°$, T = 28.75 N-m, V = 184 V.

11.10 **(1)** 155.3 N-m; **(2)** 573 V (line).

11.11 118 N-m, 0.99.

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LIST OF PRINCIPAL SYMBOLS

a	Per unit frequency i.e. ratio of operating to rated frequency
a_{T1}	Stator to rotor turns ratio
a_{T2}	AC side to converter side transformer turns ratio
a_T	a_{T1}/a_{T2}
B	Viscous friction coefficient
E	Voltage induced in the armature of a dc motor or in the stator of an induction motor, V
f	Frequency, Hz
f_b	Base frequency, Hz
f_c	Frequency of the carrier wave, Hz
I_a	Average value of the armature current of a dc motor, A
i_a	Instantaneous value of the armature current of a dc motor, A
Δi_a	Ripple in armature current, A
I_{ac}	Average value of the armature current at critical speed ω_{mc}, A
I_d	Average value of the dc link current of a converter, A
i_d	Instantaneous value of the dc link current of a converter, A
I_F	Field current of a synchronous motor, A
I_f	Per phase ac equivalent of I_F for a synchronous motor (also field current of a dc motor), A
I_f'	$I_f(X_m/X_s)$ for a synchronous motor, A
I_m	Magnetising current, A
I_m'	The ratio (V/jX_s) for a synchronous motor, A
I_M	Rated value of I_m' for a synchronous motor, A
I_r	Fundamental component of rotor current, A
I_r'	Fundamental component of rotor current referred to stator, A
I_{rms}	R.M.S. value of a non-sinusoidal current, A
I_s	Fundamental component of induction motor stator (or synchronous motor armature) current, A
i_s	Instantaneous value of ac source current, A
I_{sk}	kth harmonic in induction motor stator (or synchronous motor armature) current, A
J	Polar moment of inertia of motor-load system referred to the motor shaft, Kg-m^2
K	$K_e\Phi$
K_e	dc motor constant (also eddy current coefficient)
K_f	Φ/I_a
L_a	Armature circuit inductance of a dc motor, H
L_c	Commutating inductance of a synchronous motor, H
L_d	DC link inductance, H
L_s	Synchronous inductance of a synchronous motor, H
m	Modulation index
N	Speed, rpm
N_s	Synchronous speed, rpm
N_{sl}	Slip-speed, rpm
p	Number of poles
P_e	Rotor circuit electrical power, W
P_{cr}	Rotor circuit copper loss, W
P_g	Air-gap power, W
P_m	Develope mechanical power, W
P_r	Power recovered, W
R_a	Armature circuit resistance of a dc motor, Ω
R_B	Braking resistance, Ω
R_r	Rotor resistance, Ω
R_r'	Rotor resistance referred to stator, Ω
R_s	Stator resistance, Ω
R_t	Thevenin's equivalent resistance, Ω
s	Slip per unit (also Laplace operator)
s_k	kth harmonic slip per unit
s_m	Slip at the maximum torque
t	Time, S
T	Torque developed, N-m (also chopper period, S)

T_a	Torque developed, N-m
T_{ac}	Torque at the critical speed, ω_{mc}, N-m
T_F	Friction torque, N-m
T_L	Load torque, N-m
T_M	Torque required to do the useful mechanical work, N-m
T_{max}	Breakdown/Pull-out torque, N-m
u	Commutation overlap angle, rad.
V	Source voltage, V
v	Instantaneous value of dc source voltage, V
V_a	Average value of the dc motor armature voltage (also rectifier output voltage), V
v_a	Instantaneous value of the dc motor armature voltage, V
v_{AO}	Instantaneous inverter output voltage between phase A and the central point of the dc source, V
V_1	Fundamental component in a non-sinusoidal voltage, V
V_d	Average value of dc link voltage, V
v_d	Instantaneous value of dc link voltage, V
V_f	Excitation emf, V
V_k	kth harmonic component in a non-sinusoidal voltage, V
V_m	Peak value of an ac source voltage, V
V_r	Injected voltage, V
v_s	Instantaneous ac source voltage, V
X_m	Magnetising reactance, Ω
X_r	Rotor leakage reactance, Ω
X'_r	Rotor leakage reactance referred to stator, Ω
X_s	Stator leakage reactance of an induction motor or synchronous reactance of a synchronous motor, Ω
X_t	Thevenin's equivalent reactance, Ω $[R^2 + (\omega L_a)^2]^{1/2}$
α	Firing angle, rad
β	Commutation lead angle of a load commutated inverter or the angle at which the dc motor armature current drops to zero, rad
β_{min}	Minimum value of commutation lead angle, rad
γ	Margin angle for a load commutated inverter (also the angle at which the induced emf of a converter fed dc motor equals the ac source voltage), rad
γ_{min}	Minimum value of margin angle, rad
Θ_r	Phase of I'_r with air-gap emf E, rad
τ_a	Armature time constant, S
τ_m	Mechanical time constant, S
ϕ	Power factor angle of a synchronous motor or an induction motor, rad
ϕ_1	Fundamental power factor angle, rad
ϕ_r	Phase of injected voltage V_r with respect to source voltage V, rad
ϕ_s	Phase of stator current with respect to the source voltage V, rad
Φ	Flux per pole, Webers
ψ	$\tan^{-1}(\omega L_a/R_a)$
ω	Source frequency, rad/sec
ω_m	Speed, rad/sec
ω_{mb}	Base speed, rad/sec
ω_{mc}	Speed on the boundary between continuous and discontinuous conductions, rad/sec
ω_{mo}	Ideal no load speed, rad/sec
ω_{ms}	Synchronous speed, rad/sec
ω_{sl}	Slip speed, rad/sec
δ	Duty ratio of a chopper, torque angle of a synchronous motor fed from a voltage source
δ'	Torque angle of a synchronous motor fed from a current source
Variable*	Reference value of a variable
$\overline{\text{Variable}}$	Phasor

1

Power Semiconductor Controlled Drives: An Introduction

This chapter describes the common features of electrical drives controlled by power semiconductor converters.

1.1 THE POWER SEMICONDUCTOR DRIVE AND ITS ELEMENTS

The block diagram of an electrical drive controlled by a power semiconductor converter is shown in figure 1.1. The load is usually a machine designed to accomplish a certain task. For example, machines like trains, rolling mills, machine tools, paper mills, cranes, excavators, and so on constitute a load on the electrical drive. Usually the requirements of the load can be specified in terms of speed and torque demands. As examples, some requirements of loads for transient operations and normal running are listed in section 1.5. The requirements of some specific loads are described in section 1.2.3.

For a suitable drive to be selected for a specific load (or application), complete information about the load requirements should first be obtained. A motor having speed-torque and speed-current characteristics that suit the load requirements is chosen. A motor will have characteristics compatible to the load if it satisfies the speed and torque requirements of the load without exceeding the current limitation imposed either by the motor rating or the source capacity. The characteristic obtained when the supply is maintained at rated conditions is called the natural characteristic. For example, the speed-torque characteristic of an induction motor obtained at the rated terminal voltage and frequency will be its natural characteristic. Usually, the natural speed-torque characteristic is not compatible with the load requirements. Therefore, a power semiconductor converter is interposed between the source and the motor.

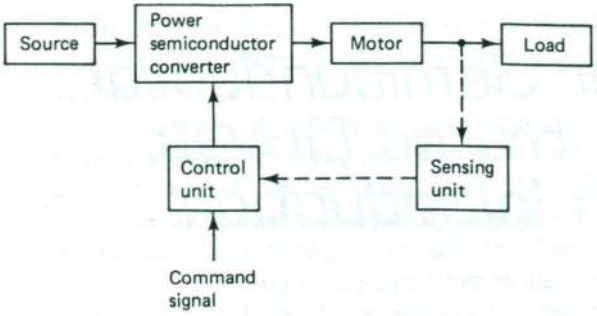

Figure 1.1 Block diagram of an electric drive.

The power semiconductor convertor, which will from now on simply be called a converter, controls the flow of power from the source to the motor in such a way that the motor speed-torque and speed-current characteristics become compatible with the load requirements. The controls for the converter are built in a control unit which operates at much lower voltage and power levels. The control unit consists of linear and digital integrated circuits and transistors. It may also consist of a microprocessor when sophisticated control is required. The command signal, which adjusts the operating point of the drive, forms an input to the control unit. The low voltage control unit is electrically isolated from the power circuit (or converter-motor circuit) for two reasons: In the absence of isolation, a malfunction may lead to the application of the power circuit voltage to the control unit. This may damage the control unit and may be detrimental to the safety of the operator who applies the command signal. Converters generate a considerable amount of harmonics. In the absence of isolation, the harmonics can enter the control unit and interfere with its operation. Sensing of certain parameters—such as converter current, motor speed, and so on—is usually required either for protection or for closed-loop operation. This function is performed by the sensing unit. When sensing the power circuit's electrical parameters—such as the converter current, voltage, and so on—isolation is provided between the power circuit and the control circuit for the reasons just stated.

The complete drive system—consisting of the load, motor, converter, source, control unit, and sensing circuit—must be treated as an integrated system. The operating point of the motor may be disturbed due to a change in any of these elements of the drive. Any such disturbance may produce changes which may affect the entire system. Choice of various elements is also interrelated. For example, the source may be determined by the necessity of using a particular motor, or the motor may have to be selected to suit the source available. Usually the converter is selected to make the motor characteristics suitable to the load; however, it has to be chosen taking into account the capacity of the source and rules governing its use. For example, it may be necessary to limit the current inrush during starting or other transient operations, to avoid voltage fluctuations on the supply lines, even when such a limitation is not required for the safe and satisfactory operation of the motor or the load. Further, a converter capable of imparting necessary characteristics to the motor most economically may have to be abandoned because harmonics are generated in the supply lines or due to the power factor deteriorating beyond permissible limits.

1.2 DYNAMICS OF THE MOTOR LOAD SYSTEM

This section is concerned with the presentation of the dynamic relations applicable to all types of motors and loads.

1.2.1 Fundamental Torque Equations

A motor generally drives a load (machine) through some transmission system. While the motor always rotates, the load may rotate or may undergo a translational motion. The load speed may be different from that of the motor; if the load has many parts, their speeds may be different, and while some may rotate, others may go through a translational motion. It is convenient, however, to represent the motor load system by an equivalent rotational system, as shown in figure 1.2. The following notation is adapted:

J = polar moment of inertia of the motor-load system referred to the motor shaft, Kg-m^2

ω_m = instantaneous angular velocity of the motor shaft, rad/sec

T = developed torque of the motor, N-m

T_L = the load (resisting) torque, referred to the motor shaft, N-m

Any motor-load system can be described by the following fundamental torque equation:

$$T = T_L + J\frac{d\omega_m}{dt} \tag{1.1}$$

Equation (1.1) shows that the torque developed by the motor is counterbalanced by a load torque T_L and the dynamic torque $Jd\omega_m/dt$. The torque component $Jd\omega_m/dt$ is called the dynamic torque because it is present only during the transient operations.

The drive accelerates or decelerates depending on whether T is greater or less than T_L. During the acceleration period, the motor should supply not only the load torque but an additional component $Jd\omega_m/dt$ to overcome the inertia of the drive. In applications having a load with large inertia, such as trains, the motor torque must exceed the load torque by a large amount to get an adequate amount of acceleration. Similarly, in applications requiring fast response, the motor torque should be maintained at the highest value and the motor-load system should be designed to have the lowest possible inertia. When the speed increases, the kinetic energy of the drive given by $\frac{1}{2}J\omega_m^2$ also increases; and, therefore, in addition to the energy supplied to the load, the motor should also supply the kinetic energy. During the deceleration period, the dynamic torque $Jd\omega_m/dt$ changes its sign, and thus assists the motor torque T in maintaining the motion of the drive by extracting energy from the stored kinetic energy. When the load has a high inertia, the motor should produce a large

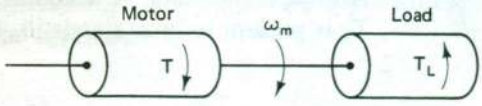

Figure 1.2 Equivalent motor-load system.

braking torque (negative T) to get adequate deceleration. When fast response is required, the braking torque should be maintained at the highest value and the motor-load system should be designed with the lowest possible inertia.

When, for a short time, the load torque T_L exceeds the maximum torque capability of the motor running at a given speed, deceleration occurs and the dynamic torque assists the motor torque in maintaining the motion. In some applications, involving a large torque of relavtively short duration followed by a no-load or light-load period of sufficient duration, the dynamic torque component is used so that a motor of smaller rating can be used. For example, in a pressing machine, a large torque of short duration is required during the pressing operation; otherwise the torque is nearly zero. A flywheel is mounted on the motor shaft to increase the equivalent inertia J. During the no-load period, the drive accelerates to store the kinetic energy. During the pressing operation, the load torque is much higher compared to the motor torque. The deceleration occurs, producing a dynamic torque. The dynamic torque and the motor torque together are able to produce the torque required for the pressing operation. In the absence of the flywheel, the motor will be required to supply the entire torque required for the pressing operation, and therefore the motor rating has to be much higher.

1.2.2 Components of the Load Torque

The load torque T_L can be further divided into the following components:

1. Friction torque T_F: The friction will be present at the motor shaft and also in the various parts of the load. The friction torque T_F is the equivalent value of various friction torques referred to the motor shaft.
2. Windage torque T_W: When a motor runs, the wind generates a torque opposing the motion. This is known as the windage torque.
3. Torque required to do the useful mechanical work, T_M: The nature of this torque depends on the type of load. It may be constant and independent of speed, it may be some function of speed, it may be time invariant or time variant, and its nature may also vary with the change in the load's mode of operation.

The variation of friction torque with speed is shown in figure 1.3a. Its value at standstill is much higher than its value at slightly above zero speed. Friction at zero speed is called stiction or static friction. For the drive to start, the motor torque should at least exceed stiction. The friction torque can be resolved into three components as shown in figure 1.3b. The component T_V which varies linearly with speed is called viscous friction and is given by the following equation:

$$T_V = B\omega_m \tag{1.2}$$

where B is the viscous friction coefficient.

The component T_C which is independent of speed is known as coulomb friction. A third component T_S accounts for the additional torque present at standstill. Since T_S is present only at standstill, it is not taken into account in the dynamic analysis.

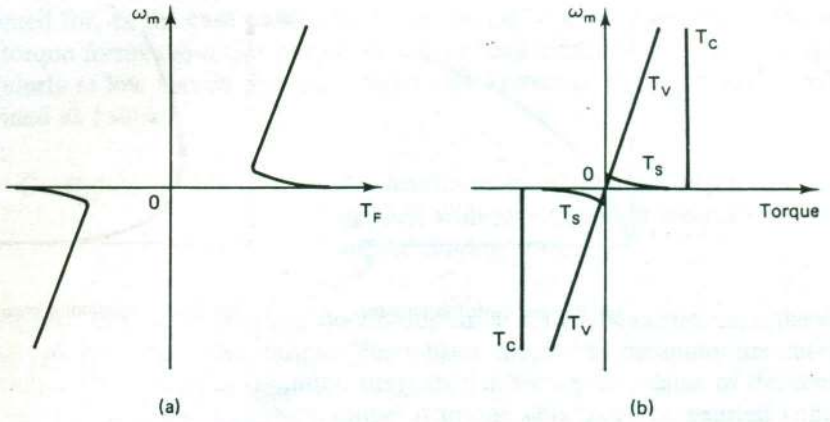

Figure 1.3 Friction torque and its components.

The windage torque T_W, which is proportional to speed squared, is given by the following equation:

$$T_W = C\omega_m^2 \tag{1.3}$$

where C is a constant.

From the preceding discussion, for finite speeds,

$$T_L = T_M + B\omega_m + T_C + C\omega_m^2 \tag{1.4}$$

In many applications $(T_C + C\omega_m^2)$ is very small compared to $B\omega_m$ and negligible compared to T_M. To simplify the analysis, the term $(T_C + C\omega_m^2)$ is approximately accounted for by updating the value of the viscous friction coefficient, B. With this approximation, from equation (1.1)

$$T = J\frac{d\omega_m}{dt} + T_M + B\omega_m \tag{1.5}$$

When T_M and T are constant or proportional to speed, equation (1.5) will be a first-order linear differential equation which can be solved analytically. Otherwise, equation (1.5) will be a nonlinear differential equation which should be solved numerically. Theoretically, the transients decay to zero in infinite time, which is not true in practice. To resolve this anomaly, the transient operation is considered over when a 95 percent change in speed has already taken place. For example, when the speed is changed from ω_{m1} to ω_{m2}, the equilibrium is assumed to have been reached when the speed becomes equal to $\omega_{m1} + 0.95(\omega_{m2} - \omega_{m1})$.

1.2.3 Some Common Load Torques

It is interesting to know the speed-torque requirements of few specific applications. Figure 1.4 shows speed-torque plots for some applications.

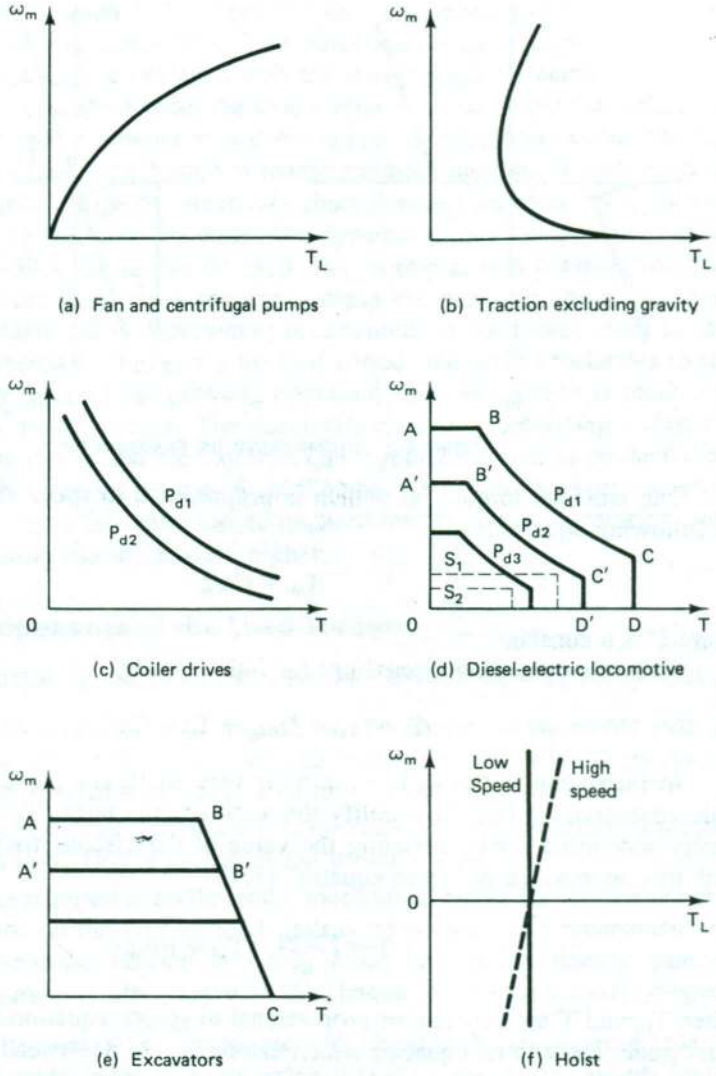

Figure 1.4 Some examples of load torque requirements.

In the case of centrifugal pumps, blowers, fans, and other loads involving the turbulent flow of fluid, the load-torque varies as the square of speed, as shown in figure 1.4a. This is nothing but the windage torque given by equation (1.3). The windage is also a predominant component at high speeds for trains, cars, and so on.

The variation of the traction load torque with speed, excluding the torque due to gravity, is shown in figure 1.4b. It is applicable to electric trains and road vehicles. It is comprised of the windage, viscous friction, coulomb friction, and stiction. When deciding about the torque requirements of the driving motor, the torque components that are needed to provide acceleration and to overcome gravity must also be

accounted for. In the case of electric trains, owing to very large inertia, the accelerating torque forms the major proportion of the total torque in the accelerating range, particularly at low speeds. An important factor known as the coefficient of adhesion is defined as follows:

$$\text{Coefficient of adhesion} = [(\text{the maximum tractive effort which can be applied without slipping at wheels/weight on the driving wheels})]$$

The tractive effort is nothing but the driving force at the wheel rim, and, therefore, it is proportional to the motor torque. The wheels coupled to the motor are called driving wheels. The foregoing equation suggests that for a given value of the coefficient of adhesion, there is a maximum value of torque which can be exerted without the driving wheels slipping. Slipping is always to be avoided as it damages the track and reduces its life. The value of the coefficient of adhesion depends on the conditions of the rail and wheel surface. In long distance trains, the distance between the consecutive stations is generally great. Acceleration and deceleration times form only a very small proportion of the total time of travel between the two stations. Therefore, the average speed mainly depends on the maximum speed and the acceleration is allowed to be low to suit the passengers' convenience. In this case, locomotives are used to drive the trains. Then the maximum weight on the driving wheels can be at the most equal to the weight of the locomotive. Therefore, the maximum torque that can be applied without the wheels slipping is small.

In the case of suburban trains, the distance between the consecutive stations is usually very small. The acceleration and deceleration times form a major proportion of the total traveling time. To get a high average speed, it is necessary to reduce the acceleration time. For a given value of the coefficient of adhesion, the acceleration can be increased only by increasing the weight on the driving wheels. Hence, instead of a locomotive, motorized train cars are used. Each motorized train car has its own driving motor. The usual pattern is to use the motorized train car and trailer cars in a ratio of 1:2. This allows a much higher weight to come on the driving wheels, compared to a locomotive, and hence, much faster acceleration can be obtained.

The coefficient of adhesion also depends on the speed-torque characteristic of the motor. When a wheel slips, the speed of the driving motor increases. If the torque drops by a large amount, for a given increase in speed, the wheel regains its grip immediately. Thus a motor speed-torque characteristic with a low regulation of speed is preferred to prevent wheel slippage. A smooth change of torque is also desirable to prevent wheel slippage. This is easily achieved when power semiconductor converters are used.

Figure 1.4c shows the speed-torque curves for applications where the driving motor is required to operate at a constant power. One such example is a coiler drive which is used in steel strip, paper, and plastic mills. In a reversible cold rolling steel strip mill, the strip to be rolled is received by the mill in the form of a reel wound on a mandrel. The mandrel is mounted on one side of the rolling stands. The purpose of the rolling stand is to reduce the cross section and to improve the surface finish. Another mandrel is mounted on the other side of the rolling stands. After rolling, the

strip is wound on this mandrel. As the rolling process progresses, the strip uncoils from one mandrel and coils on the other mandrel. To maintain even, good quality rolling and coiling, it is necessary to keep the strip's tension constant on both sides. Since the rolling stands receive and give out the strip at a fixed linear speed, the coiling and uncoiling is done at a constant power. Thus, the motor driving the coiling reel must develop a constant motoring power, and the motor coupled to the uncoiling reel must develop a constant braking power. The amount of power changes with the cross section and the material of the strip.

Figure 1.4d shows the required motor speed-torque characteristics of a diesel electric locomotive. A diesel electric locomotive employs a dc motor fed by a dc generator driven by a diesel engine. Instead of a dc generator, an alternator followed by a diode rectifier may also be used. Since the speed-torque curves of the diesel engine are not compatible with the traction requirements, it cannot be directly coupled to the driving wheels. By interposing the dc generator-motor set, speed-torque curves compatible with the traction requirements are obtained. The generator-motor system essentially acts as a torque converter, similar to a gear system in a car, but of course with a much superior performance. The diesel engine runs at full speed developing constant power. When the fuel injection is set at maximum, it develops full power. To make full use of the diesel engine power, the dc motor should develop constant power over the range of speed. Care is also taken to ensure that the diesel engine is not overloaded, otherwise it will simply stall. This is achieved by the constant power characteristic BC. The part CD is obtained by imposing a limit on the maximum motor torque. This is done for three reasons: (1) to limit the motor-generator current within a safe limit, (2) to prevent wheel slippage when the coefficient of adhesion is low, and (3) to prevent too high a motor torque which may lead to a draw-bar fracture when the coefficient of adhesion is high. The part AB is required to limit the motor speed within the permissible values and also to limit the terminal voltages of the motor and generator within rated values. For lower fuel injections, we get the inner characteristics, for example B′C′ shown in the figure. The different speed and torque limits, A′B′ and C′D′, are imposed to take care of varying conditions of the track related to the gradient and the adhesive coefficient. The characteristics S_1 and S_2 are provided for shunting purposes.

The characteristics shown in figure 1.4d are realized by interposing a converter between the source and the generator field.

The motor speed-torque requirements for excavators are shown in figure 1.4e. The purpose of the excavator is to dig earth. While digging, it may come across a rock. The motor will then simply stop. In such a situation, the motor torque must be limited to prevent mechanical damage to the excavator, and the motor current should also be restricted within the converter rating. This explains why the portion BC has the nature shown in the figure. The characteristics AB, A′B′, A″B″, and so on are provided to take care of the varying nature of the material to be excavated.

The crane-hoist characteristics are shown in figure 1.4f. In a low speed hoist, the torque is mainly due to gravity, which is constant and independent of speed. In high speed hoists, the viscous friction and windage also form an appreciable proportion of the load torque.

1.2.4 Classification of Load Torques

Various load torques can be classified into two broad categories:

1. Active load torques.
2. Passive load torques.

Load torques which have the potential to drive the motor under equilibrium conditions are called active load torques. Such load torques usually retain their sign when the direction of the drive rotation is changed. Torque due to the force of gravity and torques due to tension, compression, and torsion undergone by an elastic body come under this category. Let us consider the example of an electric train. When the train climbs up, the active torque due to gravity opposes the motion. Therefore, the driving motor has to generate extra torque to overcome the torque due to gravity. On the other hand, when the train goes down a steep grade, it is driven by the torque due to gravity. The motor produces braking torque to limit the speed within the safe values. This confirms the features of the active load torque just stated.

Load torques which always oppose the motion and change their sign on the reversal of motion are called passive load torques. Torques due to friction, cutting, and so on are in this category.

1.2.5 Electric Braking

In electric braking, the motor is made to work as a generator producing a negative torque. The electric braking may be required due to the following reasons:

1. If a motor running at some speed is disconnected from the supply, the only opposing torque will be the load torque T_L. The motor will stop only after the kinetic energy stored in its inertia is dissipated. When either the load torque is small or the inertia is large, the motor takes a long time to stop. In applications requiring frequent stops, the stopping time must be reduced by introducing additional opposing torque by the use of electric braking.
2. In some applications, such as traction, rapid emergency stops are essential to prevent accidents. The electric braking helps in achieving quick and smooth stops.
3. There are applications where accurate stops are required, such as in lifts, machine tools, ingot buggy control, and the screwdown mechanism in rolling mills. Electric braking allows accurate steps without subjecting mechanical parts to unduly large stress.
4. In certain applications involving active loads, the drive speed will reach dangerous values if the braking force is not provided by the motor. For example, in a hoist application when a loaded hoist is being lowered, the motor should provide a braking force to hold the speed within safe limits. Similarly in traction, when a train goes down a steep gradient, a braking force is required to hold the train speed within safe limits.

The braking force can also be obtained by using mechanical brakes. However, this leads to wear and tear of the mechanical parts. It is not as smooth as electric braking. Furthermore, by using regenerative braking (explained in later chapters), the energy is usefully employed instead of being wasted as with mechanical brakes.

1.2.6 Speed Torque Conventions and Multiquadrant Operation

In electrical machines, the torque is usually expressed as a function of speed (or slip), and therefore the speed is treated as an independent variable and the torque as a dependent variable. Accordingly, the speed is plotted on the 'X' axis and the torque on the 'Y' axis. In drives, the torque is considered the independent variable, and therefore is plotted on the 'X' axis. The speed, which is now the dependent variable, is plotted on the 'Y' axis. The reason for this convention is as follows. When selecting a drive, the first thing to be noted is the torque the motor should produce for the drive to be operable. The question of finding the drive speed arises only if the drive operates. Hence the torque is regarded as the independent variable.

For the proper use of equation (1.5) and the consideration of the multiquadrant operation of drives, it is useful to establish suitable conventions about the signs of torque and speed. The motor speed is considered positive when rotating in the forward direction. For drives which operate only in one direction, the forward speed will be their normal speed. For reversible drives the forward speed is chosen arbitrarily. Then the rotation in the opposite direction gives the reverse speed which is assigned the negative sign. The positive motor torque is defined as the torque which produces acceleration or a positive rate of change of speed in the forward direction. It tends to drive the motor in the forward direction. According to equation (1.5), the positive load torque is opposite in direction to the positive motor torque. Motor torque in the direction opposite to the positive motor torque will produce deceleration or a positive rate of change of speed in the reverse direction. It tends to drive the motor in the reverse direction.

A motor operates in two modes—motoring and braking. In motoring, it converts electrical energy to mechanical energy, which supports its motion. In braking, it works as a generator converting mechanical energy to electrical energy, which is consumed in some part of the circuit; and therefore it opposes the motion. The motor can provide motoring and braking operations in both forward and reverse directions.

Figure 1.5 shows the torque and speed coordinates for both positive and negative signs, chosen according to the convention just explained. In quadrant I, the power, which is the product of speed and torque is positive. Hence, the machine works as a motor supplying the mechanical energy. The operation in quadrant I is called forward motoring. In quadrant II, the power is negative. Hence, the machine works under braking opposing the motion. The operation in quadrant II is called forward braking. When coupled to a passive load, the deceleration will be caused both by the machine and the load. Consequently, the drive cannot have an equilibrium speed in this quadrant. Operation in this quadrant will take place only under transients. The same will happen when coupled to an active load torque with a positive sign. However, if the active load torque is negative, the drive can have an equi-

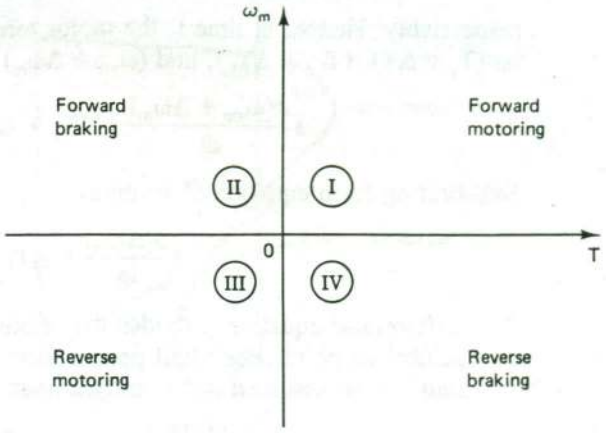

Figure 1.5 Multiquadrant operation of drives.

librium speed in this quadrant. Then the drive will also operate in steady-state in this quadrant. For example, when an electric train moves down a steep gradient, the torque due to gravity supports the motion and the braking opposes the motion. By the adjustment of the braking torque, the train is made to run at a desired speed.

Following the preceding arguments, it can be noted that quadrant III gives reverse motoring and quadrant IV provides reverse braking.

1.2.7 Steady-State Stability

According to equation (1.1), the equilibrium speed of a motor-load system is obtained when the motor torque equals the load torque. This is the speed at which the drive will normally operate in steady-state provided it is a speed of stable equilibrium. Equilibrium speed will be viewed as the stable speed provided that the operation will be restored to this speed after any small departure from it due to any disturbance in the motor-load system. The stability of an equilibrium point can be readily investigated by using the concept of steady-state stability. In this concept, the stability of an equilibrium point is evaluated from the steady-state speed-torque curves of the motor and the load. It is assumed that any departure from the equilibrium point, due to any disturbance, will be along these curves. This in effect means that the motor is assumed to be in electrical equilibrium for all operating points. The basis of this assumption is that the electrical time constant of a motor is usually negligible compared to its mechanical time constant.

At a given equilibrium operating point, let the motor torque, load torque, and drive speed be denoted by T_e, T_{Le}, and ω_{me} respectively. Then

$$T_e = T_{Le} \quad \text{and} \quad \frac{d\omega_{me}}{dt} = 0 \tag{1.6}$$

A disturbance in the supply, load, or any part of the drive will cause perturbations in the motor torque, load torque, and drive speed. At a time t, measured from the instant the disturbance is caused, let these perturbations be denoted by ΔT, ΔT_L, and $\Delta\omega_m$

respectively. Hence, at time t, the motor torque, load torque, and motor speed will be $(T_e + \Delta T)$, $(T_{Le} + \Delta T_L)$, and $(\omega_{me} + \Delta\omega_m)$ respectively. Now from equation (1.1)

$$J\frac{d(\omega_{me} + \Delta\omega_m)}{dt} + (T_{Le} + \Delta T_L) - (T_e + \Delta T) = 0 \tag{1.7}$$

Substituting from equation (1.6) gives

$$J\frac{d(\Delta\omega_m)}{dt} + \Delta T_L - \Delta T = 0 \tag{1.8}$$

This differential equation provides the relation between small perturbations around an equilibrium point. For small perturbations, the speed-torque curves of the motor and load can be assumed to be straight lines. Thus

$$\Delta T = \left(\frac{dT}{d\omega_m}\right)\Delta\omega_m \quad \text{and} \quad \Delta T_L = \left(\frac{dT_L}{d\omega_m}\right)\Delta\omega_m \tag{1.9}$$

Where $(dT/d\omega_m)$ and $(dT_L/d\omega_m)$ are the slopes of the steady-state speed-torque curves of the motor and the load at the operating point under consideration. Substituting from equations (1.9) into equation (1.8) and rearranging the terms gives

$$J\frac{d(\Delta\omega_m)}{dt} + \left[\frac{dT_L}{d\omega_m} - \frac{dT}{d\omega_m}\right]\Delta\omega_m = 0 \tag{1.10}$$

This is a first-order linear differential equation. If the initial deviation in speed at $t = 0$ is $(\Delta\omega_m)_0$, then the solution of equation (1.10) will be

$$\Delta\omega_m = (\Delta\omega_m)_o e^{-\frac{1}{J}\left(\frac{dT_L}{d\omega_m} - \frac{dT}{d\omega_m}\right)t} \tag{1.11}$$

An equilibrium point will be stable when $\Delta\omega_m$ approaches 0 as t approaches infinity. For this to happen, the exponent in equation (1.11) must be negative. This gives

$$\left(\frac{dT_L}{d\omega_m} - \frac{dT}{d\omega_m}\right) > 0 \tag{1.12}$$

Equation (1.12) suggests that for an increase in speed, the load torque must exceed the motor torque so that deceleration takes place and the operation returns to the equilibrium speed. Similarly, for a decrease in speed, the motor torque must exceed the load torque so that acceleration occurs and the operation returns to the equilibrium speed.

Let us examine the equilibrium points A and B, which are obtained when an induction motor drives the load T_{L1}, as shown in figure 1.6. Let us first examine point A for the steady-state stability. A small increase in speed makes the load torque greater than the motor torque. Deceleration occurs and the operation is restored to point A. Similarly, a small decrease in speed causes the motor torque to exceed the load torque. Acceleration occurs and the operation is restored to point A. Thus, A is a stable equilibrium point. Let us next examine the stability of the equilibrium point B. A small increase in speed causes the motor torque to exceed the load

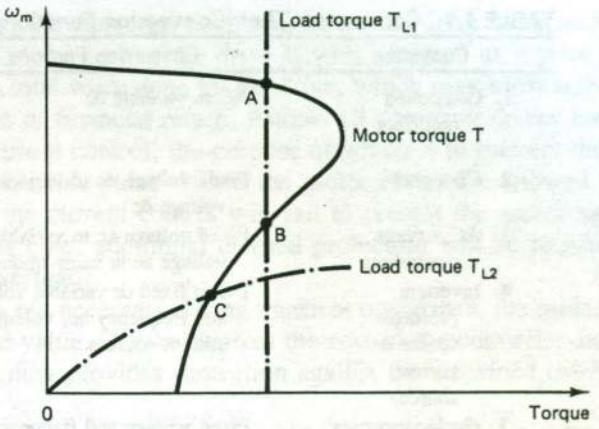

Figure 1.6 Steady-state stability of equilibrium points.

torque. Acceleration takes place and the operating point moves away from B. Similarly, a small decrease in speed makes the load torque greater than the motor torque, causing deceleration and the operating point to drift away from B. Thus B is an unstable equilibrium point.

Let us now consider the equilibrium point C which is obtained when the motor drives the load-torque T_{L2}. The load T_{L2} has a characteristic similar to the fan load (fig. 1.4a). Examination of point C shows that it is a stable equilibrium point. Note that points B and C lie on the same part of the motor speed-torque curve. However, C provides the stable operation but not B. This shows that the stability of an equilibrium point depends not on either the motor characteristic alone or the load characteristic alone but on the relative nature of the two.

Notice that the part of the motor characteristic on which points B and C are situated has a positive slope. Such a motor characteristic gives unstable operation with most loads, and therefore, it is sometimes called a statically unstable characteristic.

1.3 THE CONVERTER MOTOR SYSTEM

This section describes types of converters and features common to all variable speed drives controlled by power semiconductor converters.

1.3.1 Types of Converters

The motors commonly used in variable speed drives are induction motors, dc motors, and synchronous motors. For the control of induction motors, a fixed frequency ac supply with variable voltage or a variable frequency ac supply with variable voltage or current is required. Synchronous motors need a variable frequency supply with variable voltage or current. For the control of dc motors, a variable voltage dc supply is required. The variable voltage dc supply is also used for the control of induction and synchronous motors.

The supply that is generally available is the fixed voltage and fixed frequency ac supply. Sometimes a fixed voltage dc supply may be available. To satisfy the supply needs of various motors, a number of power semiconductor converters have been

TABLE 1.1 Converters, Their Conversion Functions and Applications

Converter	Conversion Function	Applications
1. Controlled rectifiers	AC to variable dc	Control of dc motors, induction motors and synchronous motors
2. Choppers	Fixed voltage dc to variable voltage dc	Control of dc motors and induction motors
3. AC voltage controllers	Fixed voltage ac to variable voltage ac at same frequency	Control of induction motors
4. Inverters (voltage source or current source)	DC to fixed or variable voltage and frequency ac, voltage or current sources	Control of induction motors and synchronous motors
5. Cycloconverters	Fixed voltage and frequency ac to variable voltage and frequency ac	Induction motors and synchronous motors

developed. These converters along with their conversion functions and applications are listed in table 1.1. A variable speed drive may use a single converter or more than one converter. Further, each converter listed in the table may be realized using different circuits. These circuits may not only differ in performance but also in terms of their capability for motor control in various quadrants. Usually, the lesser the number of quadrants covered, the lower the cost and the simpler the converter circuit.

While various features of individual converters and their circuits will be discussed in later chapters, some features common to all converters are presented in the following sections. Some of the features cannot be explained at this stage. Therefore, they would be simply stated. The detailed explanations will be given in later chapters.

1.3.2 Ratings of Converters and Motors

During transient operations such as starting, braking, speed reversal, speed changing, and so on, the motor current can be allowed to be higher than its rated current, due to its large thermal capacity. For example, depending on the design of the motor and the duration of the transient operation, a dc motor may be allowed to carry 2 to 3.5 times the rated current. The flow of current equal to the maximum permissible value allows full use of the motor torque capability both during motoring and braking. Consequently, the transients are completed in the shortest posssible period, and the motor does not stall due to short-time overloads. When fast response during transient operations is not necessary, then the motor current need not be allowed to exceed the rated current.

Converters use semiconductor elements which do not have any capacity for overload, due to their low thermal capacity. Therefore, their current rating is chosen as equal to the maximum current that may be required to flow through the motor. When the motor current is allowed to be K times ($K > 1$) the rated current, to get fast response during transient operations, the converter rating must be K times the

motor rating. Consequently, the converter cost, and hence the drive cost, increases substantially. The increase in the cost of the drive is well accepted as a price to be paid for the increase in the total work done by the drive, which may ultimately lead to an increase in production or financial return. Almost all converter drives are provided with some kind of current control, the purpose of which is to prevent the current from exceeding a permissible value. When the motor current is allowed to be K times its rated current, the current control will fail to protect the motor against sustained overloads. In this case, additional thermal protection will be required to protect against sustained overloads.

When fast response is not necessary during transient operations, the motor current is restricted to its rated value. This minimizes the cost of the converter and the drive. The current control now provides protection against the sustained overloads as well.

The continuous torque and power limitations of a drive in the four quadrants of operation are shown by the solid lines in figure 1.7 for speeds below and above base speed ω_{mb}. The base speed is the highest drive speed available at the rated flux (or without weakening of flux). From standstill to base speed, both for motoring and braking operations and for rotation in either direction, operation at the rated current imposes a limitation on the maximum available torque. The available power increases linearly with speed and generally reaches maximum value—equal to the continuous power rating of the motor—at base speed. Usually the motor is operated at a reduced voltage below base speed. Above base speed, the motor terminal voltage is maintained at the rated value. Motor operation at the rated current (and rated volt-

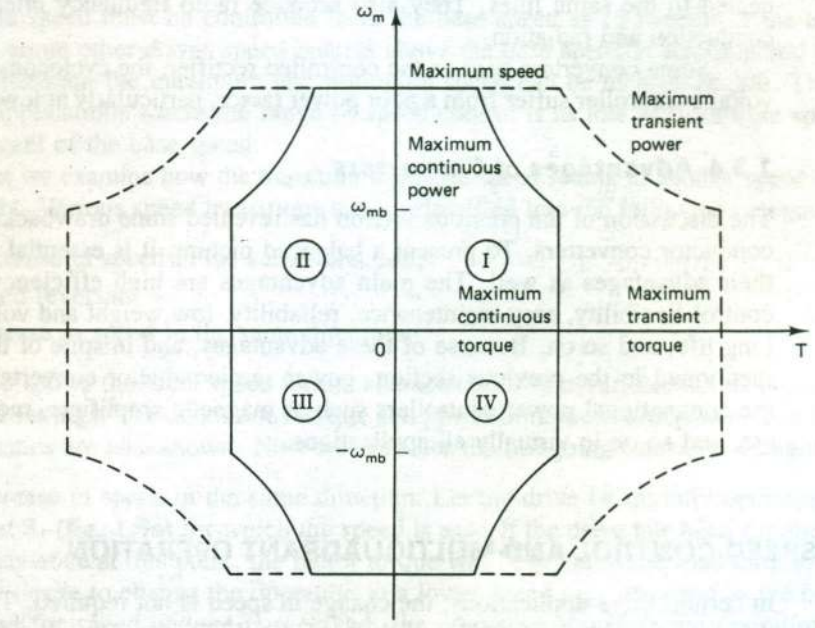

Figure 1.7 Maximum, continuous and transient, torque and power limitations of drives in four quadrants.

age) imposes a limitation on the maximum developed power, and the maximum available torque decreases with the increase in speed. One of the factors which imposes a limitation on the maximum speed is the mechanical strength of the motor-load system. Other factors, which are applicable to specific motors, are described in the relevant chapters.

As explained in the previous section, when fast response is desired, the motor current is allowed to exceed the rated current during transient operations. When the converter rating is chosen to match the motor transient current rating, the torque and power limitations shown by the dotted lines are obtained (fig. 1.7). During transient operation, the current may be forced to reach maximum permissible value, thus causing the drive to operate on the dotted curves. This is done to make full use of the motor torque and power capabilities.

1.3.3 Harmonics and Power Factor

All power semiconductor converters have harmonics in their output voltage and current. The contribution of these harmonics to the power developed by the motor is negligible. The harmonic currents increase the rms value of the motor current and distort the flux. Consequently, the copper and core losses are increased. To prevent the motor temperature from exceeding a safe value, the load on the motor must be less than rated. In other words, the motor has to be derated in the presence of harmonics.

All power semiconductor converters also have harmonics in their input currents. These harmonics cause line voltage fluctuations, adversely affecting other loads connected to the same lines. They also produce radio frequency interference through conduction and radiation.

Some converters such as the controlled rectifier, the cycloconverter, and the ac voltage controller suffer from a poor power factor, particularly at low output voltages.

1.3.4 Advantages of Converters

The discussion of the previous section has revealed some drawbacks of power semiconductor converters. To present a balanced picture, it is essential that we consider their advantages as well. The main advantages are high efficiency, fast response, control flexibility, easy maintenance, reliability, low weight and volume, less noise, long life, and so on. Because of these advantages, and in spite of the disadvantages mentioned in the previous section, power semiconductor converters have replaced the conventional power controllers such as magnetic amplifiers, mercury-arc rectifiers, and so on in virtually all applications.

1.4 SPEED CONTROL AND MULTIQUADRANT OPERATION

In certain drive applications, the change in speed is not required. The driving motor then operates at the rated conditions of supply on the natural speed-torque curve. Such drives are called single speed or constant speed drives. Here no converter is connected between the source and the motor during normal running. However, a converter may be employed during starting and also sometimes during braking.

Some drives operate at discrete speed settings. These drives are called multi-speed drives. In some other drives, stepless control of speed may be required. These drives and also the multi-speed drives are called variable speed drives. There are applications where the speeds of a number of motors must be controlled simultaneously. These motors are fed from a common converter. Such drives are called multi-motor drives. In all these drives, the converter which is used for speed control is also employed for starting control and sometimes also for braking control.

A variable speed drive is called a constant power drive if the maximum power capability of the drive does not change with the change in speed setting. The corresponding mode of operation is called the constant power mode or region. It must be noted that the term "constant power" refers to the maximum power capability of the drive and not to the actual power output which may vary from no load to full load. The constant torque drive and constant torque mode or region are defined in the same way.

Ideally, it is desired that for a given speed setting, the motor speed should remain constant as the load torque is changed from no load to full load. In practice, the speed drops with an increase in the load torque. The speed regulation is defined as

$$\text{speed regulation} = \frac{\text{no load speed} - \text{full load speed}}{\text{full load speed}} \tag{1.13}$$

If the open-loop control fails to provide the desired speed regulation, the drive is operated as a closed-loop speed control system.

The range of desired speed variation depends on the application. In some drives, the speed must be controlled from the base speed to 10 percent of the base speed. In some other drives speed control above the base speed is also required and the ratio between the maximum and minimum speed can be as high as 200. There are also applications where the range of speed control is as low as from base speed to 80 percent of the base speed.

Next we examine how the transition from one speed setting to another speed setting occurs. Various speed transitions can be classified into the following categories:

1. Decrease in speed in the same direction.
2. Speed reversal.
3. Increase in speed in the same direction.

Figure 1.8 shows the ideal speed torque characteristics of a variable speed drive for few speed settings. The continuous torque and power limitations are shown. The load characteristics are also shown. Now we consider the foregoing transition categories

1. Decrease in speed in the same direction: Let the drive be initially operating at point S_1 (fig. 1.9a) for which the speed is ω_{m1}. If the drive has been running in steady-state at this point, the motor torque will be equal to the load torque. We now desire to change the operation to a lower speed ω_{m2}. As soon as the command for speed change is received, the converter transfers the operation to point A in the braking quadrant. Since the transition from S_1 to A is done in a very brief time, the motor speed does not change appreciably due to its inertia. Now the motor is braked at the maximum current, which involves braking at

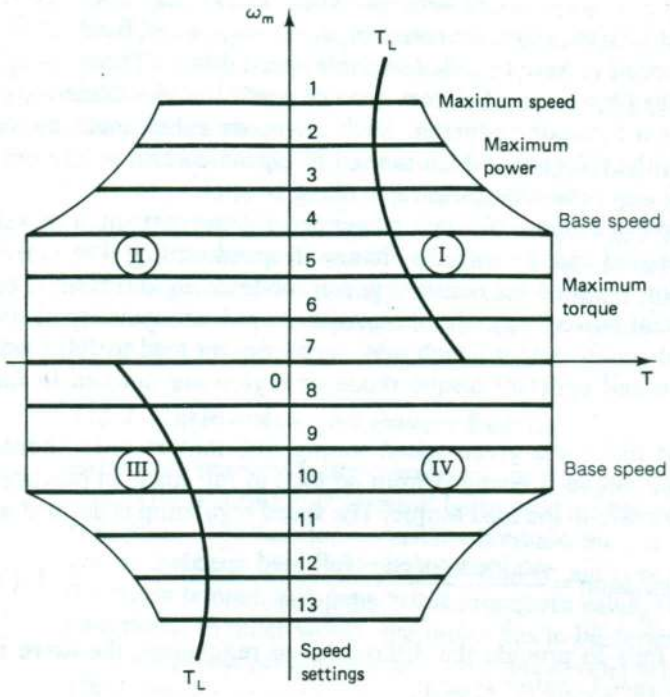

Figure 1.8 Motor speed-torque curves for different speed settings along with load characteristic.

the maximum power from A to B and braking at the maximum torque from B to C. At C the desired speed is reached. The converter therefore transfers the operation back to motoring, and the drive settles at point S_2—at which the motor torque equals the load torque. When the transition from S_1 to S_2 must be done in the least possible time, the braking is carried out at the maximum transient power and torque limits shown by the dotted lines in figure 1.7.

2. Speed reversal: Let the speed of the drive be changed from ω_{m1} to $-\omega_{m2}$, which means the transfer of the operation from S_1 to S_2^-, as shown in figure 1.9b. Soon after the command for the speed change is received, the operation is transferred to point A in the braking quadrant. The motor is braked to zero speed, initially at maximum power and then at maximum torque. At zero speed the converter transfers the operation to reverse motoring. The drive is accelerated at maximum torque to point D for which the motor speed is $-\omega_{m2}$. At D, the motor is permitted to change its torque. The drive finally settles at S_2^- for which the motor torque equals the load torque. Figure 1.9b also shows the transition from speed ω_{m1} to $-\omega_{m1}$, which occurs along the path S_1, A, B, C, D, E, F, and S_1^-. In the speed reversal also, when the transition from a given operating point to another operating point is to be achieved in a minimum time, braking and acceleration are done at the maximum transient torque and power limits, shown by the dotted lines in figure 1.7.

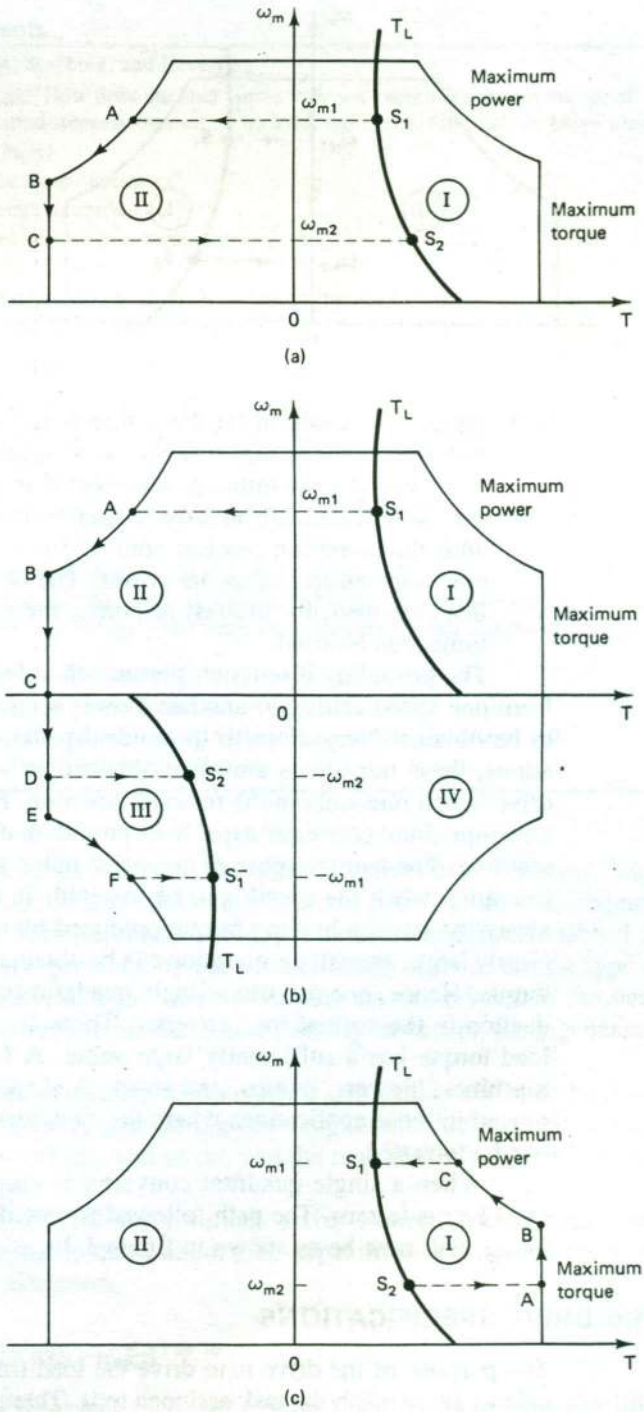

Figure 1.9 Various speed transitions: (a) Speed reduction, (b) Speed reversal and (c) Increase in speed.

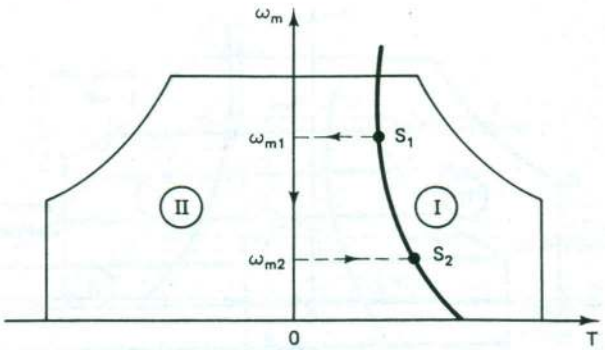

Figure 1.10 Speed transitions when single quadrant converter is used.

3. Increase in speed in the same direction: Figure 1.9c shows the transition from point S_2 at speed ω_{m2} to point S_1 at speed ω_{m1}. When the speed command is received, the operation is transferred to point A in the same quadrant. The drive is accelerated initially at maximum torque and then at maximum power until the operation reaches point C for which the speed is ω_{m1}. The motor is now allowed to change its power. The operation finally settles at point S_1. In this case also, for the fast response, the maximum transient torque and power limits can be used.

The preceding discussion presents the ideal paths to be followed for transition from one speed setting to another. Power semiconductor converters allow transition to be obtained very close to these ideal paths. Because of the economic considerations, these transitions are often obtained in a different manner. Let us consider a drive which runs only in the forward direction. For the ideal transition just described, a two-quadrant converter capable of providing operation in quadrants I and II will be required. The main purpose of operation in the second quadrant is to provide fast deceleration when the speed is to be lowered. In this operation, a passive load torque always assists the braking torque produced by the motor. If the load torque is sufficiently large, enough deceleration can be obtained in the absence of the motor-braking torque. Hence, one can use a single-quadrant converter. This allows considerable reduction in the cost of the converter. There are a number of applications where the load torque has a sufficiently large value. A few examples are paper and printing machines, blowers, pumps, and so on. A single-quadrant converter can also be employed in those applications where the slow response to decrease in speed command can be tolerated.

When a single-quadrant converter is employed, the motor torque can at the most be made zero. The path followed for the decrease in speed command from ω_{m1} to ω_{m2} will now be as shown in figure 1.10.

1.5 DRIVE SPECIFICATIONS

The purpose of the drive is to drive the load (machinery) in such a manner that it is able to accomplish the task assigned to it. Therefore, most of the drive specifications are governed by the load requirements. The load requirements can be divided into

TABLE 1.2 Load Requirements

Requirements Related to Starting, Stopping, and Reversing

1. Nature and range of load torque: How does the load torque vary with speed for its various speed settings? Is the machine started/stopped/reversed on no load, light load, full load, or heavy load?
2. What is the inertia—low or high?
3. How much time is allowed for these operations?
4. How frequently are these operations carried out?
5. Is the smooth acceleration and deceleration necessary?
6. Is an accurate stop required?
7. Should there be provision for quick stop or reversal during emergency?
8. Are these operations done manually or automatically?

Requirements Related to the Normal Running

1. Nature and range of the load torque and power: How does the load torque vary with speed for its various speed settings? What are the maximum and minimum values of torque and power? Is the load continuous or intermittent?
2. Will it be required to run just at one speed or at few discrete speeds?
3. Is an adjustable speed required? What is the desired range of speed? Should it be controlled steplessly or in steps?
4. What is the permissible regulation of speed in (2) and (3)?
5. Will the load be overhauling (driving the motor as a generator) at any part of the cycle? Will it be supplying a substantial amount of energy while overhauling or only a small amount of energy?

Requirements Related to Location and Environment

1. Does the location allow easy access for maintenance?
2. Is it going to be used in an environment consisting of dust, flammable gases, and so on?
3. Is it going to be submerged in water or any other liquid?

these categories: (1) requirements related to the transient operations of starting, stopping, and reversing, (2) requirements related to normal running, and (3) requirements related to location and environment. These requirements are listed in table 1.2.

Other specifications are governed by the source. Some of these are the type of available source, its capacity, and the rules governing its use related to the peak current, reactive power, power factor, harmonics, ability to accept the regenerated power, and so on.

Usually more than one converter-motor system can satisfy the specifications just listed. The one which is best, considering cost (capital and running), reliability, efficiency, life, maintenance needs, and so on, and the requirements for the load and source, is selected.

In the next chapters, we will study various drives controlled by power semiconductor converters and come to know about their capabilities and limitations in relation to the preceding requirements.

1.6 POWER SEMICONDUCTOR DEVICES[1 to 10]

The present section briefly describes the characteristics and the relative capabilities of the power semiconductor control devices commonly used in drives.

While discussing the relative capabilities, terms like low, medium, and high power drives will be used. They will refer to drive ratings from 0 to 100 kW, 100 kW to 750 kW, and 750 kW and above, respectively.

1.6.1 Thyristors

The circuit symbol and the $V_A - I_A$ characteristic of a thyristor are shown in figures 1.11a and b, respectively. In the absence of the gate current I_g, the current carried by the thyristor for the anode to cathode voltage V_A, less than the forward break over voltage V_{BO} and greater than the reverse breakdown voltage, is close to zero. Hence, it behaves as an open switch. However, if V_A is made greater than the forward breakover voltage, the thyristor voltage drops to a very small value (1.2 to 2.6 V, depending on the voltage rating), and the thyristor behaves as a closed switch. The forward breakover voltage decreases with an increase in the gate current. If the gate current is made sufficiently large, the breakover voltage becomes practically zero.

The thyristor is used essentially as a controlled switch. For a given application, a thyristor is chosen such that the peak forward and reverse voltages to which it may be subjected during normal operation are less than its forward breakover voltage and greater than its reverse breakdown voltage. This ensures that the thyristor remains off in the absence of the gate current. The gate current is applied at the instant turn-on is desired. The thyristor turns on provided $V_A > 0$. After turn-on, when I_A reaches a value known as the *latching current,* the thyristor continues to conduct even after the gate signal has been removed. Hence, only a pulse of current is required for turn-on.

A pulse transformer PT (or optical isolater) provides isolation between the gate and the firing circuit (fig. 1.12). A pulse train of frequency 5 to 10 kHz is used instead of a single pulse. The use of a pulse train allows a reduction in the size of the pulse transformer. The diodes in the gate circuit clamp the negative voltage and prevent the reverse current from flowing through the gate circuit when the thyristor is

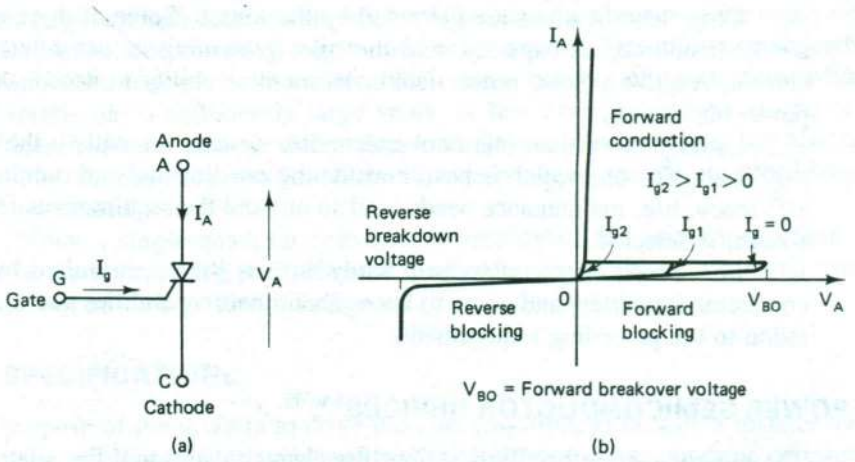

Figure 1.11 Thyristor: (a) Circuit symbol, (b) $V_A - I_A$ characteristics.

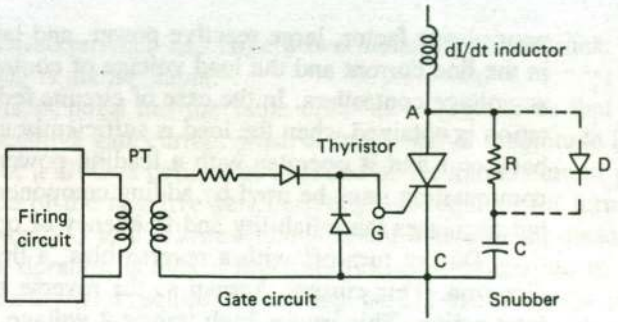

Figure 1.12 Thyristor protections.

conducting; otherwise, it may be damaged.[1] Because of the high current gain of the thyristor, the gate current pulse amplitude is very small compared to the thyristor current rating. For example, for a thyristor rated 200 A (rms), the gate current pulse amplitude may be of the order of 500 mA.

During the initial part of turn-on, only a small area near the gate conducts. The conduction spreads across the whole cross section with a finite velocity. If the anode current builds up faster than the spreading velocity, a large current will flow in a narrow region. Consequently, the thyristor will be damaged due to local heating. Hence, thyristors are assigned a dI/dt rating. It may typically be in the range of 20 to 500 A/μ sec. Fast-rising gate currents are employed to improve the dI/dt rating. An air-core inductor of few turns in series with the thyristor (fig. 1.12) is enough to restrict the rate of rise of the anode current within a safe limit.

The spurious turn-on of a thyristor can occur because the rate of rise of the anode to cathode voltage is greater than a critical value called the dV/dt rating. The dV/dt rating may typically be in the range of 20 to 1000 V per μsec. An RC snubber circuit (fig. 1.12) is connected in parallel with the thyristor to limit the dV/dt. It also protects the thyristor from voltage surges. Sometimes a diode, shown by the dotted lines, is added to make the snubber action more effective.

The major limitation of the thyristor is that it cannot be turned off by the gate signal. For turn-off — also known as commutation — the anode current, I_A, must be reduced below a value which is known as the *holding current*. Subsequently, it should be subjected to a reverse bias of sufficient duration for it to regain forward voltage blocking capability. The turn-off time ranges from 5 to 100 μsec. The slower thyristors with turn-off times 50 μsec and above are employed in ac to dc converters and ac voltage controllers. They are called converter grade thyristors. Thyristors with fast turn-off times are called inverter grade thyristors. They are used in inverters and choppers. Their turn-off times are generally in the range of 5 to 50 μsec depending on the voltage rating. The lower turn-off times are obtained at the expense of a larger conduction drop.

In the circuits powered by an ac source, the foregoing conditions for the thyristor turn-off are reached during the negative half cycle. The turn-off achieved with the help of the line voltage is called the line commutation. The line commutation has the advantages of simplicity and reliability. However, the restriction that the commutation can be done only after the line voltage reverses has been responsible for the

poor power factor, large reactive power, and large low frequency harmonic content in the line current and the load voltage of controlled rectifiers, cycloconverters, and ac voltage controllers. In the case of circuits fed by a dc source, the natural commutation is obtained when the load is sufficiently under-damped or when the load has a back emf and it operates with a leading power factor. For other loads, the forced commutation must be used by adding components. This not only increases the cost, but decreases the reliability and frequency of operation, and increases the losses.

During turn-off with a reverse bias, a thyristor carries current in the reverse direction. This current, known as the reverse recovery current, stops flowing by a snap action. This causes high transient voltage to be generated by the anode circuit inductance across other thyristors in the circuit. The snubber circuit of figure 1.12 is able to provide protection against the transient voltage by providing a path for the inductive current to flow.

Thyristors are assigned average and rms current ratings. The current should not be allowed to exceed these ratings even for a short duration, because of the low thermal capacity of thyristors. However, they have the ability to withstand short duration current surges. A thyristor may withstand a surge lasting three cycles (at 60 Hz) with its magnitude up to around ten times the rated rms current. Thyristors are protected against sustained overloads by the fast-acting closed-loop current control of converters. Fast fuse links are also provided. They are set to operate only when the current control cannot provide protection.

Thyristors are commercially available up to 4000 V, 1500 A (av), and 1200 V, 3000 A (av). High-power drives usually employ thyristors. They are also widely used in medium and low power drives, particularly in drives employing ac to dc converters and ac voltage controllers.

1.6.2 Asymmetrical Thyristors

In voltage-source inverters, and some choppers and thyristor commutation circuits, a diode is connected antiparallel with the thyristor. In these applications, the reverse voltage blocking capability of the thyristor is not of much significance and can be allowed to be small. The extra freedom made available can be used by the thyristor designer to reduce turn-off time, turn-on time, and conduction drop. Alternatively, it can be used to increase the forward blocking voltage. Asymmetrical thyristors are designed with a reverse blocking voltage of around 30 V. The turn-off times are typically in the range of 3 to 30 μsec, and they are substantially lower compared to a symmetrical thyristor of the same ratings.

A reverse conducting thyristor is obtained by monolithically integrating an antiparallel diode with an asymmetrical thyristor on the same chip. This eliminates the stray inductance between the thyristor and diode connections, and reduces the heatsink size. The disadvantage is that the ratio of thyristor and diode current ratings cannot be altered.

1.6.3 Gate-Turn-Off Thyristors

As the name suggests, a gate-turn-off thyristor (GTO) is a special thyristor where the turn-off can be achieved by a negative gate current pulse. GTOs have been in existence since the mid-1960s. It is only recently that they have become commercially

available with improved characteristics and large power handling capabilities. Ratings up to 2500 V and 2000 A are available.

The turn-on gate current pulse has the same order of magnitude as that of a thyristor. For turn-off, a negative gate current pulse of much higher magnitude is required. The turn-off gain of a GTO is defined as the ratio of the anode current prior to turn-off to the amplitude of the negative gate current pulse required for turn-off. The gain is very low: typically 3 to 5. However, this is not a significant disadvantage, noting that the pulse duration is only a few microseconds. Compared to this, the forced commutation of a thyristor requires a current pulse 1.2 to 2 times the anode current and of duration 2 to 3 times the thyristor turn-off time. Hence, the thyristor commutation circuit loss is much higher than the loss of the GTO turn-off circuit. Consequently, a GTO converter is more highly efficient than a forced-commutated thyristor converter. Due to the absence of bulky commutation circuits, a GTO converter has a lower volume than a forced-commutated thyristor converter.

Because of the faster switching speed, a GTO converter can operate at a higher frequency than a thyristor converter. This reduces filter size and gives more efficient and improved drive performance.

Due to the preceding advantages, the GTO is finding an increasing number of applications in medium and high (lower range) power drives employing inverters and choppers. They have opened up the possibility of operating ac to dc converters with improved power factors and reduced reactive power, harmonics, and current ripple.

The firing circuit is required to generate two sets of pulses. One for turn-on and another for turn-off. If the GTO with an unshorted emitter structure is used, a reverse gate bias of a few volts may also be provided to enhance the dV/dt capability which is otherwise low. The anode shorted emitter structures have dV/dt rating comparable to that for thyristors, but they cannot block the reverse voltage.

Polarized Snubber and Turn-Off Characteristics: The snubber for a GTO is required to perform the following two functions, in addition to limiting dV/dt during its off state: to limit dV/dt during and after turn-off to prevent retriggering, and to reduce device power dissipation during turn-off. Because of the more stringent requirements of a GTO snubber, a polarized snubber is used and the snubber capacitor size is several times higher than that of a thyristor. Figure 1.13 shows a GTO chopper with a polarized snubber. The stray inductances are also shown, as they have a considerable effect on the turn-off characteristics, which are shown in figure 1.14.

The turn-off is initiated by applying a negative gate current pulse I_g at $t = 0$. The finite rate of rise of the gate pulse is due to the lead inductance and the finite switching times of the semiconductor devices used in the gating circuit. The load current can be assumed constant during turn-off time because of its large time constant. During the saturation time t_s, the GTO remains in the on-state conducting full anode current with a small voltage drop equal to the conduction drop. After t_s, the GTO begins to block the anode current. At the same time, the gate-cathode junction begins to recover and the gate current begins to decrease. The rate of fall of the anode current is rather large (>300 A/μs). Because the load current is constant, the current through the snubber is forced to rise fast. Consequently, a voltage spike is produced across the GTO, the magnitude of which depends on the stray inductance L_s and the turn-on characteristic of the snubber diode D. It is important to minimize

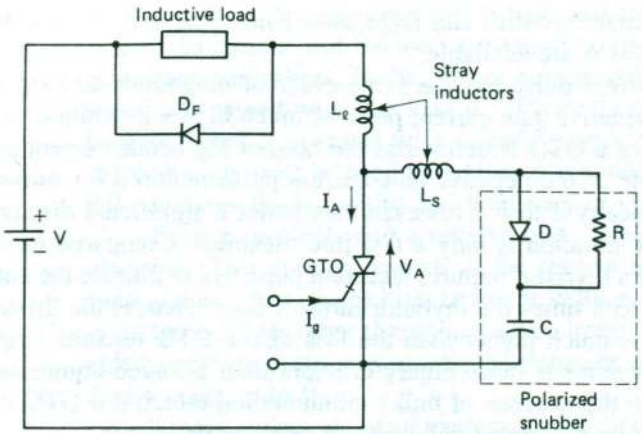

Figure 1.13 GTO chopper with polarized snubber.

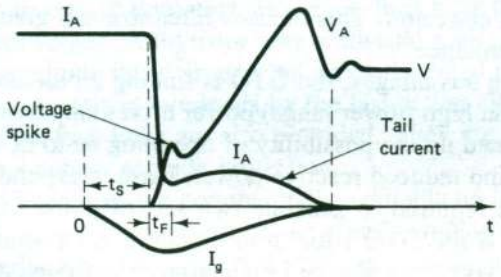

Figure 1.14 GTO turn-off characteristics.

the voltage spike, because it causes power dissipation in a localized region of the GTO. This is achieved by using a fast-turn-on diode for D and mounting the snubber as close as possible to the GTO to minimize stray inductance. It may further be noted that during the fall time t_F, the anode voltage is rising when the anode current is falling. A high power dissipation occurs during this period. The snubber capacitor limits this power by limiting the anode voltage. After the fall time, a tail of current is produced. The anode voltage rises at a rate determined by the snubber capacitor and overshoots before settling to the source voltage V. The tail current and overshoot voltage can be decreased by increasing the snubber capacitor and consequently decreasing the device power dissipation, but at the expense of the snubber loss.

1.6.4 Power Transistors

The circuit symbol of a NPN transistor is shown in figure 1.15a. The collector current I_c versus collector-emitter voltage V_{CE} characteristics, with base current I_B as a parameter, are shown in figure 1.15b. The PNP transistor has characteristics similar to the NPN transistor, with the current and voltage directions being reversed. Increasing the collector-emitter voltage beyond a certain level leads to avalanche breakdown. A reversal of the collector-emitter voltage will break down the base-emitter junction at a much lower voltage than the forward breakover voltage. When employed in applications requiring reverse voltage blocking capability, a diode is connected in series with the transistor. This, however, increases the on-state losses

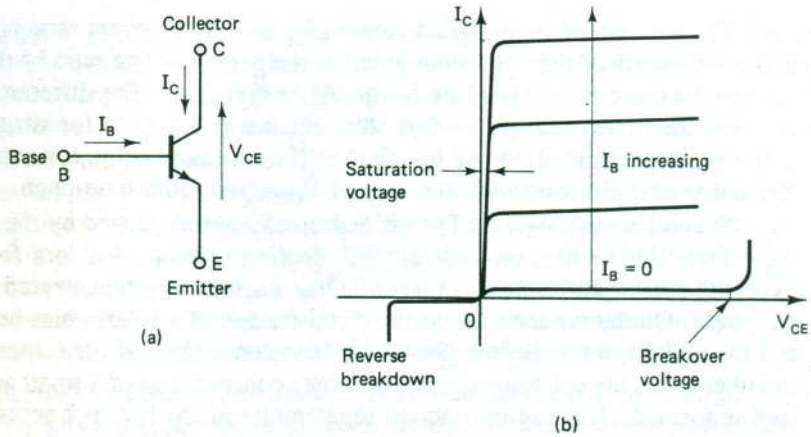

Figure 1.15 NPN transistor: (a) Circuit symbol, (b) $I_C - V_{CE}$ characteristics.

considerably. Within the working range, the collector current is a function of the base current. The ratio I_C/I_B is the transistor current gain. At high current levels, the power transistors typically operate at a current gain of 10 to 20.

In convertors, the power transistor is operated as a controlled switch. With zero base current, it blocks the voltage with a small leakage current (fig. 1.15b), and thus operates in the off-state. With a base current, which takes it into saturation, the power transistor conducts a large collector current with a small voltage drop (0.6 to 1.1 V); hence the device operates in the on-state. Unlike the thyristor, it requires a continuous base drive to remain in the on-state. Since the current gain is low, the base current magnitude is large. This creates some problems at high current levels. The transistor gain is not constant but changes substantially with the collector current. Ideally, the base current should be just enough to keep it under saturation. A lower current will desaturate it, thus increasing the power dissipation. A higher current will tend to reduce turn-on time but cause higher base dissipation and an increase in turn-off time, as explained later. Thus, ideally the base current should track the collector current to keep the transistor just under saturation for all operating points. Alternatively, the base current can be chosen for the highest collector current, leading to overdrive for lower collector currents and the consequent disadvantages just stated.

Although the transistor can be turned off by making the base current zero, a negative base drive is used to get fast turn-off. The turn-off time consists of two parts: the storage time and the fall time. During the storage time, the negative base drive sweeps out the minority carriers in the base. The collector junction remains forward biased and the transistor continues to conduct. During the fall time, the collector begins to block and turn-off is completed. A negative base drive reduces the storage time. Hence, the largest negative base drive that can be allowed without exceeding emitter-base avalanche breakdown voltage is used. It may be noted that the overdrive of the base during the on-state increases the storage time.

Transistors can switch considerably faster than thyristors. Switching times of less than 1 to 2 μsec are possible.

The transistors are assigned continuous and peak current ratings which are limited on the basis of the maximum junction temperature. The ratio of the maximum to continuous current rating is low compared to thyristors. The difference is due to the fact that the voltage drop in a thyristor remains small, even for very large transient currents, but not in a power transistor. The voltage rating consists of maximum (breakdown) collector-to-emitter voltage V_{CEO} (sus) with base open.

Second Breakdown: The second breakdown is caused by the localized heating effects during turn-on and turn-off. During turn-on, due to a forward bias between the base and emitter, the collector current is concentrated at the emitter periphery. On the other hand, during turn-off, due to a reverse bias between the base and the emitter, the collector current tends to concentrate at the center of the emitter. In either case, the collector current becomes concentrated in a small area where a hot spot is formed. The rise in junction temperature at the hot spot accentuates the current concentration due to the negative temperature coefficient of the transistor drop and leads to thermal runaway. It may be noted that the second breakdown is different from the avalanche breakdown, which is defined as the first breakdown.

The manufacturers supply safe operating area curves for turn-on and turn-off, taking into account various limitations of the transistor, including the second breakdown. The turn-on and turn-off load lines should be selected such that they lie within the respective areas and minimize turn-on and turn-off dissipations.

Polarized-Snubber: A transistor snubber performs three functions: to shift the device switching power loss to the snubber circuit, to avoid second breakdown, and to control dV/dt. The dV/dt and dI/dt ratings for a transistor are somewhat higher than for a GTO but less than those for a thyristor of the same rating. Because of the stringent snubber requirements, a polarized snubber with a large capacitor is used, as in the case of a GTO. The operation of the polarized snubber has been explained with the help of the chopper circuit shown in figure 1.16. Initially, the transistor is off and the load current is freewheeling through diode D_F. The snubber capacitor is charged to supply voltage V. The application of the base drive turns on the transistor. The supply voltage is absorbed by the inductor L and the load current is gradually transferred to the transistor. In the absence of L, the diode reverse recovery current, with an impressed voltage V, would flow through the transistor, increasing the dissipation and possibly leading to a second breakdown. The value of L

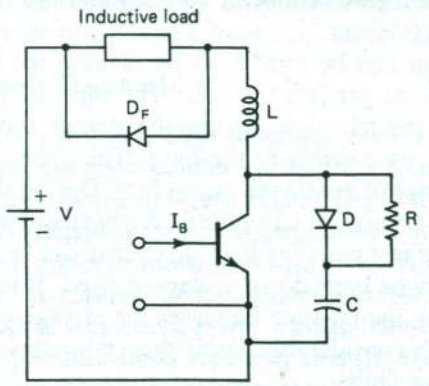

Figure 1.16 Transistor chopper with polarized snubber.

required is 1 to 2 μH. The source usually will have this amount of inductance. In that case, L can be omitted. During the on-state, C will be discharged to a transistor conduction drop through resistor R, dumping almost all its stored energy in R. A negative base drive is applied when the turn-off is desired. The transistor turns off after the storage and fall time. The load current is diverted to the snubber and the collector-emitter voltage rises slowly at a rate determined by the value of C. When the capacitor voltage becomes equal to the supply voltage, the load current is transferred to D_F. The capacitor voltage, after a small overshoot, settles to V. In the absence of C, the collector voltage will rise to the supply voltage soon after the collector current begins to fall. Because high V_{CE} and I_C exist simultaneously, the transistor dissipation will increase substantially, causing a second breakdown.

Darlington Power Transistor: A substantial improvement in gain (100–300), with a consequent reduction in the base drive, is obtained by connecting two transistors in cascade to form what is known as a Darlington connection as shown in figure 1.17. The two transistors can be incorporated in the same chip. A bypass diode D may also be added as shown by the dotted lines. Resistors R_1 and R_2 are low-ohm resistors for reducing the collector leakage current and to provide bias voltages across the base emitter junctions. A diode D_1 may be added to provide a low-impedance path for the reverse base current of transistor T_2 for fast turn-off. Alternatively, E_1 can be used for this purpose. The increase in gain is achieved at the expense of higher conduction drop (1.7 to 2.2 V) and slight decrease in switching frequency.

Comparison with Thyristor: The advantages of the transistor over the thyristor are (1) much higher switching frequency leading to improved and more efficient operation of converters; (2) can provide current limit protection by the base drive circuit; (3) does not require forced commutation circuit and saves the associated switching losses, cost, weight, and volume; and (4) low conduction drop.

The relative disadvantages are (1) requires large and continuous base drive; (2) cannot withstand reverse voltage; (3) the ratio of maximum to continuous current is low; and (4) the on resistance decreases with the increasing temperature, making the paralleling of the device difficult.

The power transistors, including the Darlington transistors, suitable for drive application are available up to the ratings of 120 V, 750 A and 1000 V, 60 A.

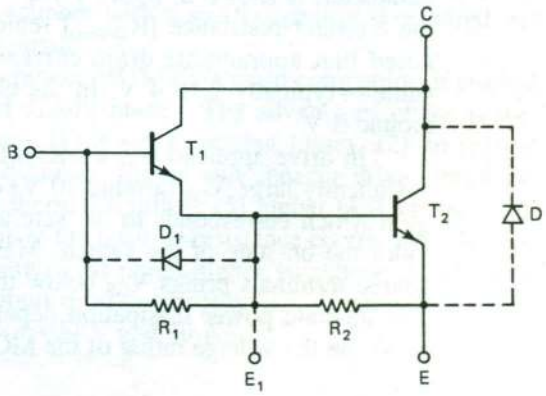

Figure 1.17 Darlington power transistor.

Because of the high switching frequency and more efficient operation, power transistors have succeeded in replacing thyristors in a number of low and medium power (up to around 200 kW) drives employing inverters and choppers.

Comparison with GTO: The advantages (1), (2), and (3), and disadvantages (1) through (4) just mentioned are applicable when the transistor is compared with the GTO. Both do not require a commutation circuit and therefore both have the associated advantages mentioned in the previous section. The drive power (sum of the power required for turn-on and turn-off) is lower for the GTO due to the use of a pulse both during turn-on and turn-off. It is more appropriate to compare them for this feature, considering the duty cycle. The transistor is efficient for a short duty cycle, while the GTO is for a long duty cycle. In the case of a short-circuit or fault leading to over currents, the transistor will come out of saturation and may fail by second breakdown. The chances of the GTO failure will be less, because the controllable current is much higher—typically two to six times the rated rms current. The controllable current is the maximum on-state current that can be turned off by the negative gate pulse.

Power transistors find wide application in low and medium (lower range) power inverter and chopper drives. They provide a superior and cheaper drive for low voltage applications (230 V and less). The rapid increase in the on-state resistance with the voltage rating has limited the device's rating below 1000 V for power applications. For high-current applications involving 450 V and above they cannot be used. Naturally, GTOs find application in this range.

1.6.5 Power MOSFETs

At present, the power handling capacity of the power metal-oxide semiconductor field-effect transistor (MOSFET) is low—typically 1000 V and 50 A. Its drive applications are also few. However, because of its superior characteristics, a possibility of its wide application at low power levels exists.

The circuit symbol is shown in figure 1.18a for an N-channel MOSFET. The device has an integral antiparallel diode which permits reverse current of the same magnitude as that of the main device. The nature of the drain current I_D versus drain-to-source voltage V_{DS} characteristics, with gate-to-source voltage V_{GS} as a parameter, is shown in figure 1.18b. These characteristics have two distinct regions: the constant resistance [$R_{DS(on)}$] region and the constant current region. It may be noted that appreciable drain current does not flow until V_{GS} crosses a threshold value—typically 2 to 4 V. In the characteristics shown, the threshold voltage is around 3 V.

In drive applications, a MOSFET is operated as a switch. The application of sufficiently large V_{GS} (around 10 V) ensures operation in the constant resistance region which corresponds to the saturation region for the power transistor. This provides the on-state of the switch. Making V_{GS} zero or simply shorting the gate and source terminals brings V_{GS} below the threshold value and turns off the MOSFET. The on-state power dissipation depends on the value of $R_{DS(on)}$, which in turn depends on the voltage rating of the MOSFET and increases substantially with temper-

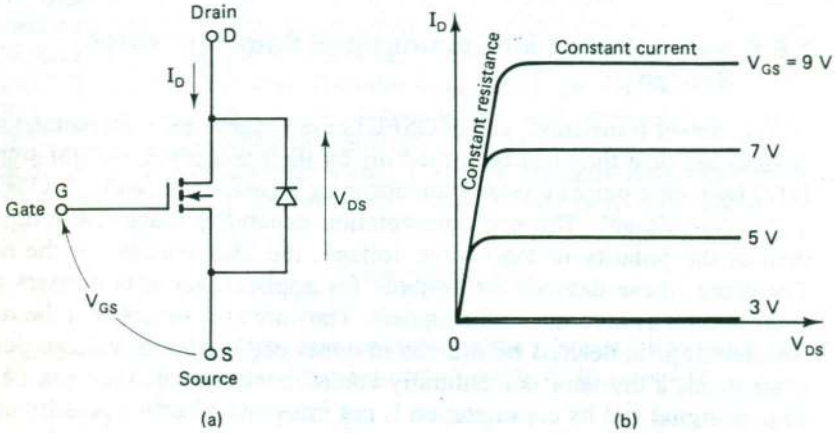

Figure 1.18 Power MOSFET: (a) Circuit symbol, (b) $I_D - V_{DS}$ characteristics.

ature. Because of the large value of $R_{DS(on)}$, the conduction loss of the MOSFET is high.

Unlike the power transistor which is a current controlled device, the MOSFET is a voltage controlled device. Since the gate terminal is electrically isolated from the source by a silicon oxide layer, the gate circuit impedance is very high and the device power gain is very large. Because of the very high input impedance, it can be driven directly from CMOS gates.

Since, unlike the power transistor, it is a majority-carrier device, there is no inherent storage time delay associated with it. The turn-on and turn-off times depend on the ability of the gate drive circuit to charge and discharge a tiny input capacitor C_{iss} which is the sum of the capacitances between gate and source, and gate and drain. Typical switching times are 150 to 200 nS for a 500 V 26 A device. Because of the low switching times and practically zero gate drive power, the switching loss of a MOSFET is almost nil. Power MOSFETs do not suffer from the problem of a second breakdown. Any high current concentration in a region increases the resistance at that point, due to the positive resistance coefficient. This forces current to be uniformly distributed over the area. The snubber is now required to provide only dV/dt protection, and hence, an RC snubber with a small capacitor is required and the snubber switching losses are very small.

Since the power transistor is the power MOSFET's main competitor, it is good to know their relative advantages and disadvantages. The advantages of the power MOSFET over the power transistor are (1) lower switching losses, (2) no second breakdown, (3) much larger gain and hence, simple and cheaper drive circuitary, (4) better reliability, ruggedness, and thermal stability, (5) higher peak current handling capability, (6) easy to parallel due to the positive resistance coefficient, (7) relatively more linear transfer characteristics, (8) much higher switching speeds. The only disadvantage is a higher conduction drop—typically 4.5 V compared to 1 V for the power transistor—for ratings of 400 V and 10 A.

1.6.6 Symbol of a Self-Commutated Semiconductor Switch

GTOs, power transistors, and MOSFETs are termed self-commutated semiconductor devices because they can be turned off by their respective control signals (that is, a GTO by a gate pulse, a power transistor by a base drive, and a MOSFET by a gate-to-source voltage). The self-commutation capability makes their turn-off independent of the polarity of the source voltage, the load voltage, or the nature of load. Therefore, these devices are suitable for applications in converters fed from a dc source, such as inverters and choppers. They are also suitable for the rectifiers where commutation is desired before the reversal of the source voltage polarity. On the other hand, a thyristor is a naturally commutated device. It cannot be turned off by its gate signal and its commutation is not independent of the polarity of the source or load voltage or the nature of the load. However, when a thyristor is employed with a forced commutation circuit, its turn-off can be made independent of the polarity of the source voltage, load voltage, and the nature of the load. Further, its commutation takes place by the application of a low-power gate signal to the thyristor of the commutation circuit. Hence, a thyristor with a forced commutation circuit behaves like a self-commutated semiconductor device. Since a thyristor with a forced commutation circuit, GTO, power transistor, and MOSFET essentially operates as a switch, it can be classified as a self-commutated semiconductor switch. A common symbol for the self-commutated semiconductor switch is shown in figure 1.19. The diode indicates the direction in which the switch can conduct current. The control signal is denoted by CS, which can be a voltage signal or current signal depending on the device.

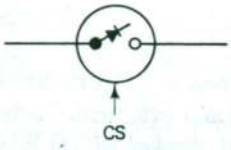

Figure 1.19 Symbol of a self-commutated semiconductor switch.

The foregoing classification and common symbol are useful because there are many applications, such as inverters, choppers, and so on, where a GTO, power transistor, MOSFET, or a thyristor with forced commutation circuit may be used.

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PROBLEMS

1.1 Figure P1.1 shows the speed-torque curves of the motor and load in the four quadrants. Comment on the stability of the equilibrium points A, B, C, and D.

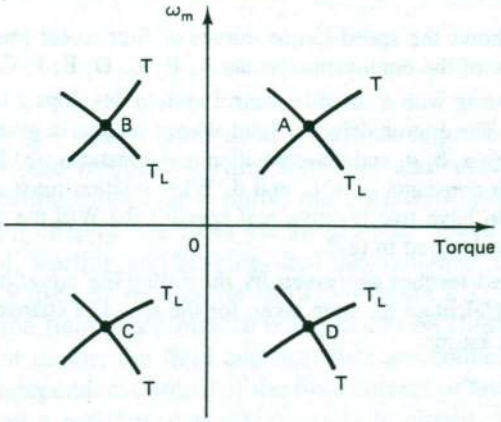
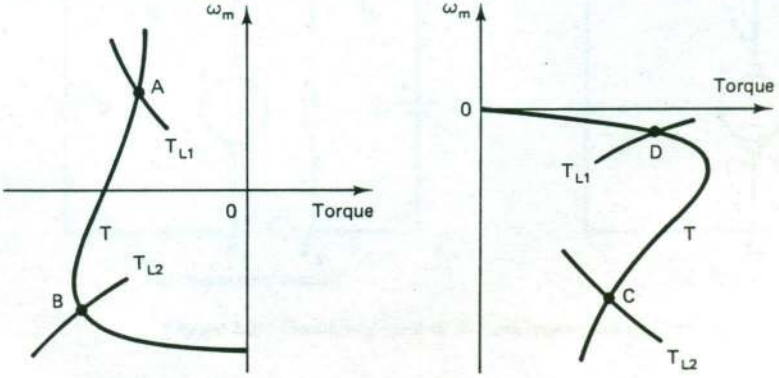


Figure P1.1

1.2 A motor operating with a suitable control system develops a torque given by the relation $T = a\omega_m - b$. The motor drives a load for which the torque is given by the expression $T_L = c\omega_m^2$, where a, b, and c are positive real constants. (a) Find the equilibrium speeds. What relation must exist between the constants for the drive to have two positive real speeds? (b) Determine the stability of the equilibrium points.



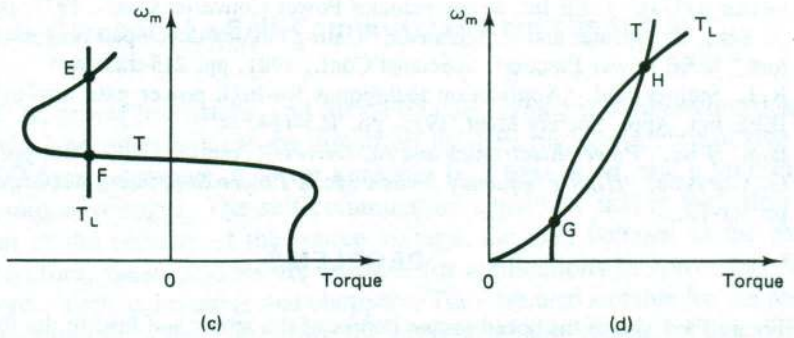


Figure P1.2

1.3 Figure P1.2 shows the speed-torque curves of four motor-load combinations. Comment on the stability of the equilibrium points A, B, C, D, E, F, G, and H.

1.4 A motor operating with a suitable control system develops a torque given by the relation $T = a\omega_m + b$. The motor drives a load whose torque is given by the expression $T_L = c\omega_m^2 + d$, where a, b, c, and d are positive real constants. (a) Find the equilibrium speeds in terms of the constants a, b, c, and d. What relation must exist between the constants for the drive to have two positive real speeds? (b) Will the drive be stable at the equilibrium points obtained in (a)?

1.5 Motor and load torques are given by the following equations: $T = 10$ N-m and $T_L = 0.1\ \omega_m$ N-m. Calculate the time taken for the speed to change from 0 to the equilibrium speed. $J = 0.1$ kg-m^2.

10

Synchronous Motors

There are many types of synchronous motors. Prominent among these are wound-field, permanent magnet, and reluctance motors. The stator, which has a 3-phase winding, is known as an armature.

The rotor of a wound-field motor has a dc field winding fed from a dc source through brushes and slip rings. When the brushes and slip rings are not acceptable because of frequent maintenance requirements, they are replaced by a brushless excitation system.

Generally, the rotor of a synchronous motor has salient poles. The high-speed and high-power machines, with two or four poles, employ cylindrical rotors. They are mainly used as alternators. In the present text, cylindrical rotor theory is primarily used because it is simple and also adequate for the analysis of salient pole motors, unless very accurate analysis is needed.

The feature which makes a wound-field motor different from other synchronous motors is that its power factor can be controlled by controlling its field excitation.

In low- and medium-power motors, field excitation can be provided using permanent magnets, thus dispensing with the field winding losses, dc source, brushes, and slip rings. But then the ability to control the power factor is lost. Permanent magnet motors are usually designed to operate at unity power factor at full load. At light loads they operate at a low (leading) power factor and efficiency.

The rotor of a synchronous reluctance motor has salient poles but without a field winding. The air-gap flux is produced solely by the magnetizing current drawn from the armature supply. Consequently, the machine has a low (lagging) power factor. It is suitable for applications requiring low torque and power.

Synchronous motors have a number of advantages over dc and induction motors. Due to the absence of a commutator, synchronous motors are not subjected to

the limitations of dc motors, such as a limit on maximum speed, voltage, and power ratings, frequent maintenance, and an inability to operate in explosive and contaminated environments. They also have lower weight, volume, and inertia compared to a dc motor for the same rating. The wound-field and permanent magnet synchronous motors have a higher full-load efficiency and power factor than an induction motor. Wound-field motors can be designed for a higher power rating than induction motors. Since the air-gap flux is not produced solely by the magnetizing current drawn from the armature, a larger air-gap suiting the mechanical design can be chosen. The speed can be precisely controlled by frequency control and can be made independent of variations in supply voltage, field current, and load. The ability to control the power factor gives them additional advantages described in the next chapter. However, compared to a squirrel-cage induction motor, they have a higher cost and size for the same rating and require more maintenance.

The present chapter describes various types of synchronous motors, their analysis and performance, the effect of harmonics, and variable frequency control from an independent oscillator. The self-control which is widely used in variable speed synchronous motor drives is described in the next chapter.

10.1 WOUND-FIELD CYLINDRICAL ROTOR MOTOR

The cylindrical rotor machine has its rotor in cylindrical form with a distributed dc field winding embedded in the rotor slots. It is characterized by a uniform air-gap. This type of rotor construction is employed in high-speed machines because of greater mechanical strength. The rotor has another winding known as the damper or amortisseur winding, which is like a squirrel-cage winding in an induction motor. The balanced three-phase winding on the armature (stator) has the same number of poles as the dc winding on the rotor.

When the armature is supplied by a balanced three-phase ac source of frequency ω radians per second (of f hertz), an air-gap flux wave rotating at a synchronous speed ω_{ms} radians per sec. is produced. Where

$$\omega_{ms} = \frac{2}{p}\omega = \frac{4\pi f}{p} \text{ rad./sec.} \tag{10.1}$$

and p = number of poles.

When excited, the field produced by the dc winding on the rotor interacts with the air-gap flux wave to produce torque. For the production of a steady unidirectional torque, the air-gap flux wave and the rotor field must travel at the same speed. Since the field produced by the dc winding revolves with the rotor, the rotor must run exactly at synchronous speed. Hence, a synchronous motor fed by a constant frequency source operates at a constant steady-state speed regardless of the load, supply voltage, or dc field winding current. When running in the steady state, the

relative speed between the air-gap flux wave and the rotor damper winding is zero. Hence, no voltages are induced in the damper winding. Thus, it does not have any influence on the steady-state operation of the motor.

10.1.1 Equivalent Circuits

The symbol of a wound-field motor is shown in figure 10.1a and its phasor diagram showing the relationship between various field components and induced voltages is shown in figure 10.1b. The rotor is running at synchronous speed. The distributed field winding on the rotor produces the field mmf $\bar{F}$, sinusoidally distributed in space and revolving at synchronous speed. The three-phase armature winding connected to a balanced three-phase supply carries a phase current I_s and produces the armature reaction mmf $\bar{A}$, which is also sinusoidally distributed in space and rotates at synchronous speed in the same direction as $\bar{F}$. The mmf $\bar{R}$, which is the resultant of mmfs $\bar{F}$ and $\bar{A}$ produces the air-gap flux wave $\bar{\Phi}_a$ which also rotates at synchronous speed and consequently induces back emf $\bar{E}$ in a phase of the machine. $\bar{E}$ is 90° out of phase with respect to $\bar{\Phi}_a$. $\bar{E}$ is known as the *air-gap emf*. The field emf $\bar{F}$ alone can be imagined to induce a voltage $\bar{V}_f$ in the armature. Its magnitude is equal to the open circuit voltage induced in the armature. It is known as the *excitation emf*. It is in quadrature with $\bar{F}$.

Based on the phasor diagram of figure 10.1b, the equivalent circuit shown inside the dotted lines in figure 10.2a is obtained. I_f is the fundamental frequency armature phase current which accounts for the field mmf $\bar{F}$ in the sense that when flowing through the three-phase armature winding it will produce a rotating mmf identical to $\bar{F}$ and consequently produce the same excitation emf $\bar{V}_f$. The magnetizing current $\bar{I}_m$ required to produce the air-gap flux Φ_a is the sum of phasors $\bar{I}_s$ and $\bar{I}_f$. The air-gap flux causes the emf $\bar{E}$ to be induced in the armature phase. X_m is the magnetizing reactance. The complete phasor diagram of the machine is obtained when R_s and $X_{s\ell}$, which are, respectively, the per-phase resistance and leakage reactance of the armature winding, are added. Figure 10.2b shows the phasor diagram based on this equivalent circuit. Here E_t is the total emf induced in the machine by the air-gap flux and the leakage flux. The equivalent current I_f is obtained as follows.

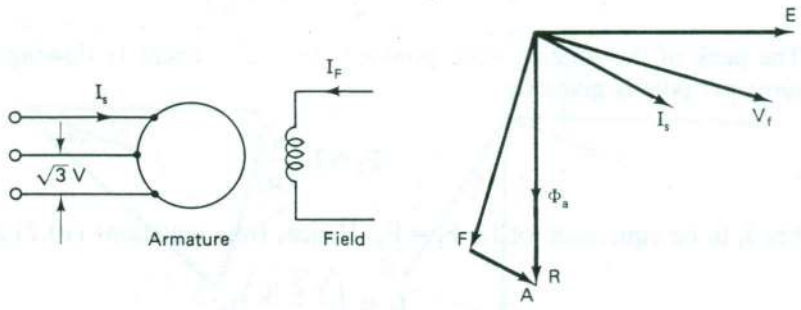


Figure 10.1 Synchronous motor.

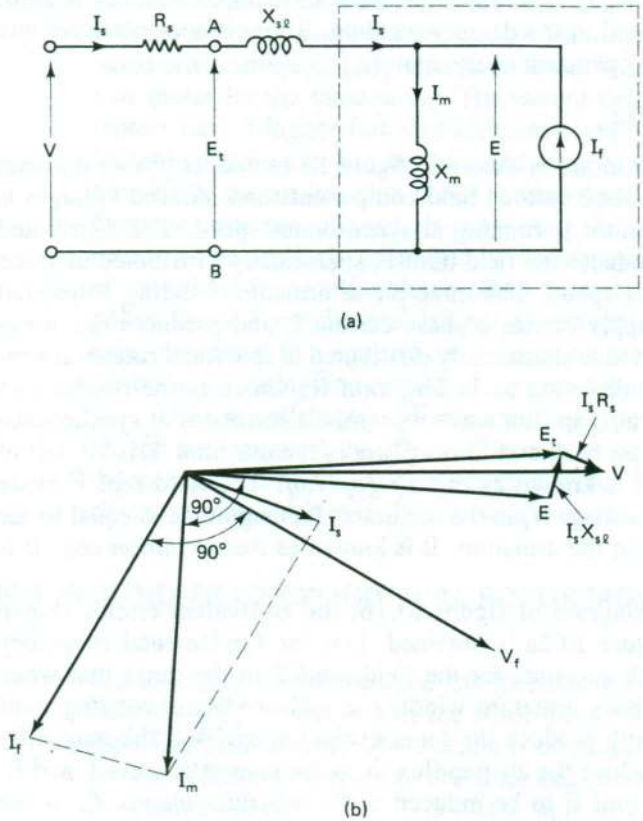


Figure 10.2 Per phase equivalent circuit and phasor diagram of a cylindrical rotor synchronous machine.

Let the number of turns in the armature phase winding and rotor field winding be n_a and n_f, respectively. The peak of rotating mmf produced by the rms current I_f flowing through (n_a/p) turns per pole is

$$F_1 = \frac{3}{2}(\sqrt{2}\,I_f)\left(\frac{n_a}{p}\right) \tag{10.2}$$

The peak of the rotating mmf produced by a dc current I_F flowing through (n_f/p) turns per pole is given by

$$F_2 = I_F\left(\frac{n_f}{p}\right) \tag{10.3}$$

For I_f to be equivalent of I_F, $F_1 = F_2$. Hence, from equations (10.2) and (10.3)

$$I_f = \left(\frac{\sqrt{2}}{3}\frac{n_f}{n_a}\right)I_F$$

or

$$I_f = nI_F \tag{10.4}$$

where n is constant for a given machine and is given by

$$n = \left(\frac{\sqrt{2}}{3}\frac{n_f}{n_a}\right) \tag{10.5}$$

Since the armature winding resistance R_s is very small, it can be ignored, except for the calculation of loss and efficiency. With this assumption, $E_t = V$ in figure 10.2b. Replacing the circuit on the right of terminals A and B by Thevenin's and Norton's equivalent circuits yields the equivalent circuits shown in figure 10.3a and b, respectively. Where

$$X_s = X_m + X_{s\ell} \tag{10.6}$$

$$\bar{V}_f = jX_m\bar{I}_f \tag{10.7}$$

$$\bar{I}'_f = \frac{X_m}{X_s}\bar{I}_f = \frac{\bar{V}_f}{jX_s} \tag{10.8}$$

and

$$\bar{I}'_m = \bar{I}_s + \bar{I}'_f = \frac{\bar{V}}{jX_s} \tag{10.9}$$

The reactance X_s is known as *synchronous reactance*. If $\bar{V}$ is taken as a reference phasor and the angle between $\bar{V}$ and $\bar{V}_f$ is denoted by δ, then

$$\bar{V}_f = V_f\angle -\delta \tag{10.10}$$

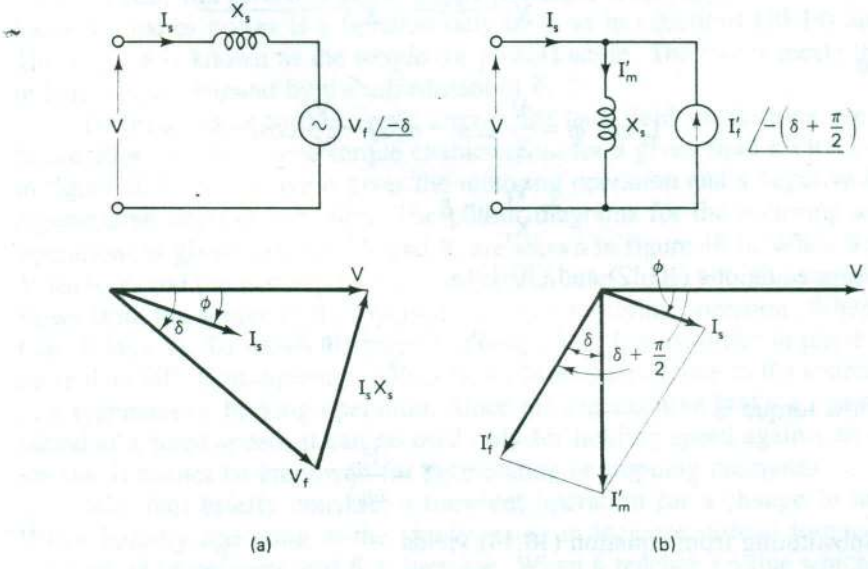


Figure 10.3 Simplified equivalent circuits and phasor diagrams of a cylindrical rotor synchronous machine.

and

$$\bar{I}_f' = \frac{\bar{V}_f}{jX_s} = I_f' \angle -(\delta + \pi/2) \tag{10.11}$$

The magnetizing reactance X_m is a nonlinear function of I_m, because of saturation. When the motor is operated in a narrow range of I_m, a suitable average value of the magnetizing reactance can be chosen for this range and used in the equivalent circuits. With this assumption, the analysis is only approximate. It is, however, widely used because of its simplicity. Here also, X_m and X_s are assumed constant.

10.1.2 Performance Equations for Operation from a Voltage Source

From figure 10.3a, the power input to the motor is expressed as

$$P_{in} = 3VI_s \cos \phi$$

Since the armature loss has been neglected, this is also the air-gap power. As no power is transferred to the rotor circuit, all of the air-gap power is converted into mechanical power; hence

$$P_m = 3VI_s \cos \phi \tag{10.12}$$

Again from figure 10.3a,

$$I_s = \frac{V\angle 0 - V_f \angle -\delta}{jX_s}$$

$$= \frac{V}{X_s}\angle -\pi/2 - \frac{V_f}{X_s}\angle -(\delta + \pi/2)$$

or

$$I_s \cos \phi = \frac{V}{X_s}\cos(-\pi/2) - \frac{V_f}{X_s}\cos(-\delta - \pi/2)$$

$$= \frac{V_f}{X_s}\sin \delta \tag{10.13}$$

From equations (10.12) and (10.13),

$$P_m = \frac{3VV_f}{X_s}\sin \delta \tag{10.14}$$

Now torque is

$$T = \frac{P_m}{\omega_{ms}}$$

Substituting from equation (10.14) yields

$$T = \frac{3}{\omega_{ms}} \cdot \frac{VV_f}{X_s} \cdot \sin \delta \tag{10.15}$$

Alternative expressions for P_m and T are obtained by substituting from equations (10.8) and (10.9) into equations (10.14) and (10.15). Thus,

$$P_m = 3VI'_f \sin \delta \tag{10.16}$$

$$= 3X_s I'_m I'_f \sin \delta \tag{10.17}$$

$$T = \frac{3}{\omega_{ms}} VI'_f \sin \delta \tag{10.18}$$

$$= \frac{3X_s}{\omega_{ms}} I'_m I'_f \sin \delta \tag{10.19}$$

Equations (10.14) to (10.19) have been derived for the case when $\overline{V}_f$ lags behind $\overline{V}$, by substituting the appropriate sign for the angle of lag. Hence, in these equations, δ should be taken positive when $\overline{V}_f$ lags behind $\overline{V}$.

10.2 OPERATION OF A WOUND-FIELD CYLINDRICAL ROTOR MOTOR FROM A CONSTANT VOLTAGE AND FREQUENCY SOURCE

This is the conventional operation of synchronous motors in the sense that in the past synchronous motors were employed only in constant speed drives where they were fed from a source of constant voltage and frequency. Some salient features of this operation are discussed here.

10.2.1 Motoring and Regenerative Braking Operations

For a given field excitation I_F, V_f is constant according to equations (10.4) and (10.7). Then, for given values of supply voltage, frequency, and field current, the motor torque or power is a function only of δ, as in equations (10.14) and (10.15). The angle δ is known as the torque (or power) angle. The motor meets the changes in load torque demand by the adjustment of δ.

Torque versus torque angle curves for two field excitations are shown in figure 10.4, and the speed torque characteristic for a given field excitation is shown in figure 10.5. A positive δ gives the motoring operation and a negative δ gives the regenerative braking operation. The phasor diagrams for the motoring and braking operations at given values of V and V_f are shown in figure 10.6. When δ is positive, V leads V_f and the motor power factor angle ϕ is less than 90°. Consequently, power flows from the source to the machine, giving a motoring operation. When δ is negative, V lags V_f. To satisfy Kirchhoff's voltage law, I_s must differ in phase from V by more than 90°. Consequently, power flows from the machine to the source, resulting in a regenerative braking operation. Since the regenerative braking operation is obtained at a fixed speed, it can be used only for holding speed against an active load torque. It cannot be employed for decelerating or stopping the motor.

We shall briefly consider a transient operation for a change in load torque. When initially operating in the steady state, an increase in load torque causes the rotor speed to decrease and δ to increase. When δ reaches a value which makes the motor torque equal the load torque, the rotor speed is lower than synchronous speed.

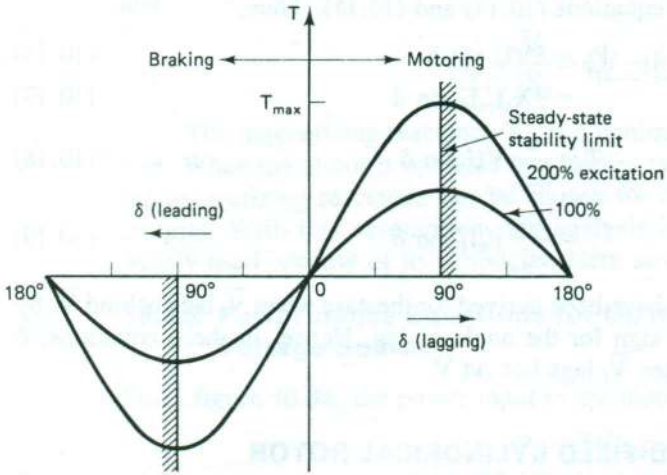


Figure 10.4 Torque-angle curves of a cylindrical rotor synchronous machine.

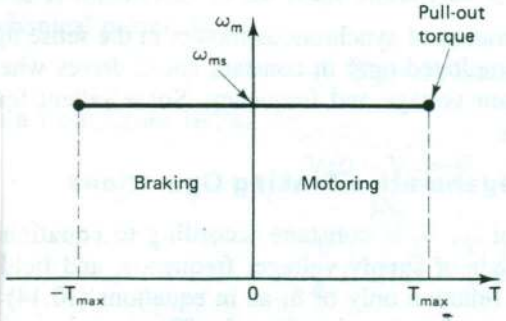
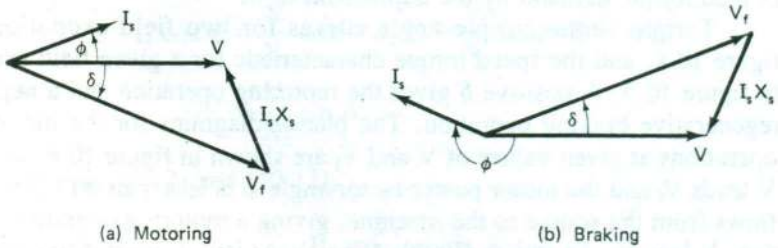


Figure 10.5 Speed-torque characteristic with a fixed frequency supply.



(a) Motoring (b) Braking

Figure 10.6 Phasor diagrams for a fixed I_F.

Consequently, δ continues to increase. Since the motor torque is now higher than the load torque, the rotor accelerates. When the rotor speed reaches synchronous speed, δ is larger than required; hence, the rotor continues to accelerate beyond synchronous speed. Thus, the rotor oscillates around synchronous speed and the required δ value before reaching steady state. Similar oscillations are produced for a disturbance in V or I_F or frequency. These oscillations are known as hunting. The damper winding is provided to damp these oscillations. When the rotor runs at synchronous speed, the relative speed between the air-gap rotating wave and the rotor is

zero; consequently, no voltage is induced in the damper winding and its presence has no influence on the motor operation. During hunting, the rotor speed deviates from the synchronous speed. Consequently, voltages are induced in the damper winding and torque is produced by induction motor action. The torque direction is such that it opposes any deviation from synchronous speed, consequently damping the oscillations.

From equations (10.14) and (10.15) and figure 10.4, for given values of supply voltage, frequency, and field excitation, the maximum torque or power, known as *pull-out torque* (T_{max}) or *pull-out power,* is reached at $\delta = \pm 90°$. Thus,

$$T_{max} = \frac{3}{\omega_{ms}} \left(\frac{V V_f}{X_s} \right) \tag{10.20}$$

Any increase of load torque beyond the motoring pull-out torque ($\delta = 90°$) will cause the rotor to slow down. Synchronous motor action will be lost since it will be impossible to restore the rotor speed to synchronous speed. This phenomenon is known as *pulling out of step* or *losing synchronism*. Automatic circuit breakers are provided to disconnect the motor from the supply when such a situation arises. In regenerative braking, the pull-out torque is reached at $\delta = -90°$. Care should be taken not to exceed the pull-out torque otherwise the machine speed may reach dangerously high values. Thus, the pull-out torque represents the maximum short time overloading capacity or the steady-state stability limit of the motor for motoring and braking operations. The pull-out torque can be increased by an increase in the field excitation or the supply voltage.

The stability limit for step changes in torque is lower than the pull-out torque. When the load torque is suddenly increased, the rotor slows down to allow δ to increase. When δ reaches a value δ_1, for which the motor torque is equal to the load torque, the rotor speed is lower than the synchronous speed. Consequently δ continues to increase, while the rotor accelerates. If the rotor speed fails to reach the synchronous speed before δ exceeds $(180° - \delta_1)$, the rotor will pull out of step, even through the load torque is less than pull-out torque.

10.2.2 Power Factor Control and V-Curves

From figure 10.3b,

$$\bar{I}_s = \bar{I}'_m - \bar{I}'_f \tag{10.21}$$

which gives

$$I_s \sin \phi = -I'_m + I'_f \cos \delta \tag{10.22}$$

$$I_s \cos \phi = I'_f \sin \delta \tag{10.23}$$

Now let us examine the motor operation at a constant torque and variable field excitation. From equation (10.18), for a constant torque, $I'_f \sin \delta$ is constant. Consequently the in-phase component of armature current $I_s \cos \phi$ is also constant from equation (10.23). At a given source voltage, I'_m is also constant. This suggests that the phasors $\bar{I}_s$ and $\bar{I}'_f$ lie on vertical loci AA′ and BB′, respectively, and their sum has a fixed value $\bar{I}'_m$ as shown in figure 10.7. The figure shows the phasor diagrams

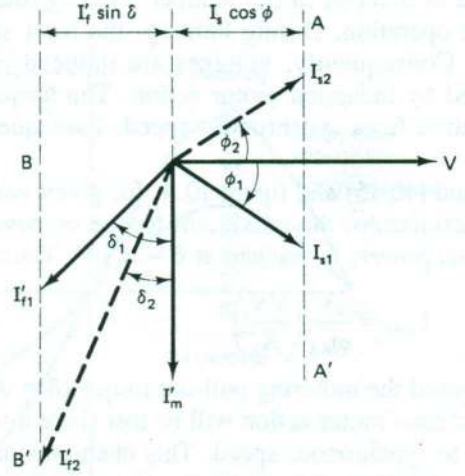


Figure 10.7 Synchronous motor power factor control by the control of field excitation.

for two values of I'_f, by continuous lines for I'_{f1} and by chain-dotted lines for I'_{f2}. For I'_{f1}, $(-I'_m + I'_{f1} \cos \delta) < 0$; consequently, according to equation (10.22), the machine draws a lagging reactive component of current from the line, resulting in a lagging power factor. For I'_{f2}, $(-I'_m + I'_{f2} \cos \delta) > 0$; hence, the machine draws a leading reactive component of current from the line, giving a leading power factor. The machine operates at unity power factor with a field current, which satisfies the following equation:

$$-I'_m + I'_f \cos \delta = 0 \tag{10.24}$$

Thus, when operating at a constant torque, the machine operates at a lagging power factor for a low value of field current I_F. An increase in I_F improves the power factor and reduces armature current. The minimum value of the armature current is reached when the power factor becomes unity. A further increase in the field current causes the machine to operate with a leading power factor and increased armature current. Figure 10.8 shows the plots between I_s and I_F for different values of torque. These curves are commonly known as V curves because of their shape. The machine is said to be *under-excited* when operating at a lagging power factor and *over-excited* when operating at a leading power factor.

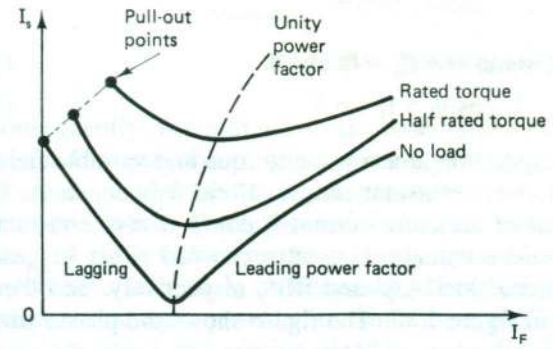


Figure 10.8 V-curves of a synchronous motor.

When operating at a given I_F and variable torque, the pull-out torque is reached at $\delta = 90°$. From figure 10.7, at pull-out (that is, at $\delta = 90°$), the entire magnetizing current I'_m is drawn from the source; consequently, the machine always operates at a lagging power factor. The locus of pull-out points is also shown in figure 10.8.

Example 10.1

A 3-phase 6600-V, 6 pole, 60 Hz, 1100 kW, Y-connected wound-field synchronous motor has the following parameters:

$X_m = 30\ \Omega, \quad X_{s\ell} = 6\ \Omega, \quad R_s = 1.2\ \Omega,$

field winding resistance $= 5\ \Omega, \quad n = 2$

When operating at the rated power and unity power factor, calculate

1. The field current and torque angle at full load.
2. The pull-out torque.
3. The power factor, armature current, and efficiency at half the rated torque and rated field current.
4. The field current to get unity power factor at half the rated torque.

Neglect friction, windage, and core loss.

Solution:

$$\text{Synchronous speed } \omega_{ms} = \frac{4\pi f}{p} = \frac{4\pi \times 60}{6} = 125.7 \text{ rad/sec.}$$

$$\text{Rated torque} = \frac{1100 \times 10^3}{125.7} = 8751 \text{ N-m}$$

$$V = 6600/\sqrt{3} = 3810.5 \text{ V}$$

$$\bar{I}'_m = \frac{V}{jX_s} = \frac{3810.5}{j36} = 105.8\underline{/-90°} \text{ A}$$

1. Since the full-load power factor is unity,

$$\sqrt{3} \times 6600 I_s = 1100 \times 10^3 + 3I_s^2 \times 1.2$$

which gives $\bar{I}_s = 99\underline{/0°}$ A

$$\bar{I}'_f = \bar{I}'_m - \bar{I}_s = 105.8\underline{/-90°} - 99\underline{/0°} = -99 - j105.8 = 144.9\underline{/-133°} \text{ A}$$

From equations (10.8) and (10.4),

$$I_F = \frac{X_s}{X_m}\frac{I'_f}{n} = \frac{36}{30} \times \frac{144.9}{2} = 86.94 \text{ A}$$

From equation (10.16),

$$\sin\delta = \frac{P_m}{3VI'_f} = \frac{1100 \times 10^3}{3 \times \frac{6600}{\sqrt{3}} \times 144.9} = 0.66$$

or $\delta = 41.6°$ (electrical) $= 13.9°$ (mechanical).

2. From equation (10.18), the pull-out torque is

$$T_{max} = \frac{3}{\omega_{ms}} V \times I_f' = \frac{3}{125.7} \times \frac{6600}{\sqrt{3}} \times 144.9 = 13178 \text{ N-m}$$

3. Half of rated torque = 4375.5 N-m

From equation (10.18)

$$4375.5 = \frac{3}{125.7} \cdot \frac{6600}{\sqrt{3}} \cdot 144.9 \sin \delta$$

or $\sin \delta = 0.33, \qquad \cos \delta = 0.94$

From equations (10.22) and (10.23),

$$I_s \sin \phi = -I_m' + I_f' \cos \delta = -105.8 + 144.9 \times 0.94 = 30.4$$

$$I_s \cos \phi = I_f' \sin \delta = 144.9 \times 0.33 = 47.8$$

Hence $I_s = \sqrt{(30.4)^2 + (47.8)^2} = 56.6$ A

$$\cos \phi = \frac{47.8}{56.6} = 0.84 \text{ (leading)}$$

Power output $P_m = 4375.5 \times 125.7$ W = 550 kW

Losses $= 3 \times (56.6)^2 \times 1.2 + 86.94^2 \times 5 = 49.32$ kW

$$\text{Efficiency} = \frac{550}{550 + 49.32} = 0.92$$

4. Power input at half rated torque and unity power factor

$$= \sqrt{3} \times 6600 I_s = 550 \times 10^3 + 3I_s^2 \times 1.2$$

which gives $I_s = 48.5$ A

Since the power factor is unity, from equations (10.22) and (10.23),

$$I_f' \cos \delta = I_m' = 105.8$$

$$I_f' \sin \delta = 48.5$$

Hence, $I_f' = \sqrt{105.8^2 + 48.5^2} = 116$ A

$$I_F = \frac{X_s}{X_m} \times \frac{I_f'}{n} = \frac{36}{30} \times \frac{116}{2} = 69.6 \text{ A}.$$

Example 10.2

The motor of example 10.1 is operating under regenerative braking. Calculate

1. The braking torque at unity power factor, and the rated armature current and terminal voltage. Also calculate the field current and torque angle.
2. What the power factor and the armature current will be when the driving load torque is 8000 N-m and the field current has the rated value.

Solution: From example 10.1: Rated $I_f' = 144.9$ A

$\omega_{ms} = 125.7$ rad/sec and $\bar{I}_m' = 105.8\underline{/-90°}$ A

1. The rated armature current = $99\underline{/180°}$ A

$$\text{Developed Power} = \sqrt{3} \times 6600 \times 99 + 3 \times 99^2 \times 1.2$$
$$= 1167 \times 10^3 \text{ W}$$

$$\text{Braking torque} = \frac{1167 \times 10^3}{125.7} = 9284 \text{ N-m}$$

From equations (10.22) and (10.23) at unity power factor,

$$I_f' \cos \delta = I_m' = 105.8$$
$$I_f' \sin \delta = I_s \cos \phi = 99 \cos 180° = -99$$

Hence, $I_f' = 145$ A

$$\delta = \tan^{-1}\left(\frac{-99}{105.8}\right) = -43° \text{ (electrical).}$$

2. From equation (10.18)

$$-8000 = \frac{3}{125.7} \times \frac{6600}{\sqrt{3}} \times 144.9 \sin \delta$$

or $\sin \delta = -0.6$ or $\cos \delta = 0.8$

From equations (10.22) and (10.23),

$$I_s \sin \phi = -105.8 + 144.9 \times 0.8 = 10.12$$
$$I_s \cos \phi = 144.9 \times (-0.6) = -86.94$$

Hence, $I_s = \sqrt{10.12^2 + 86.94^2} = 87.53$ A

$$\cos \phi = -86.94/87.53 = -0.99$$

10.3 CYLINDRICAL ROTOR MOTOR OPERATION FROM A CURRENT SOURCE

In figure 10.9, the equivalent circuit and the phasor diagram of figure 10.3b have been redrawn, taking the armature current as the reference vector. The angle δ' between the phasors I_f' and I_s is related to δ by the following equation:

$$\delta' = \frac{\pi}{2} + \delta - \phi \tag{10.25}$$

Now,

$$\overline{V} = jX_s\bar{I}_m' = jX_s(I_s\underline{/0} + I_f'\underline{/-\delta'})$$
$$= X_sI_f' \sin \delta' + jX_s(I_s + I_f' \cos \delta')$$

Power developed is

$$P_m = 3R_e(\bar{I}_s\overline{V}^*) = 3X_sI_sI_f' \sin \delta' \tag{10.26}$$

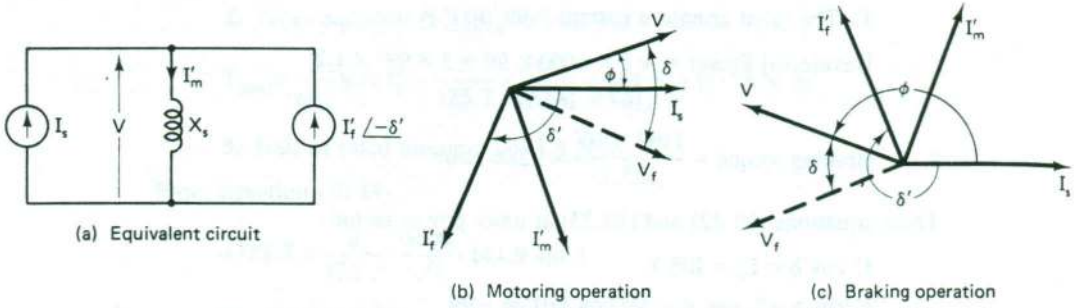


Figure 10.9 Synchronous motor fed by a current source.

and torque is

$$T = \frac{P_m}{\omega_{ms}} = 3\left(\frac{X_s}{\omega_{ms}}\right) I_s I'_f \sin \delta' \tag{10.27}$$

An alternative expression for torque is obtained by substituting from equation (10.25) into equation (10.27), giving

$$T = 3\left(\frac{X_s}{\omega_{ms}}\right) I_s I'_f \cos (\delta - \phi) \tag{10.28}$$

From equations (10.27) or (10.28), the pull-out torque is

$$T_{max} = 3\left(\frac{X_s}{\omega_{ms}}\right) I'_f I_s \tag{10.29}$$

The pull-out torque can be increased by increasing either I_s or I'_f.

For given values of I_s and I'_f, the torque is a function of the angle δ'. Hence δ' is defined as the torque angle. A torque versus torque angle curve is shown in figure 10.10. The motoring operation is obtained for $0 \le \delta' \le \pi$, and the braking operation is obtained for $\pi \le \delta' \le 2\pi$. A given value of motoring torque is obtained at

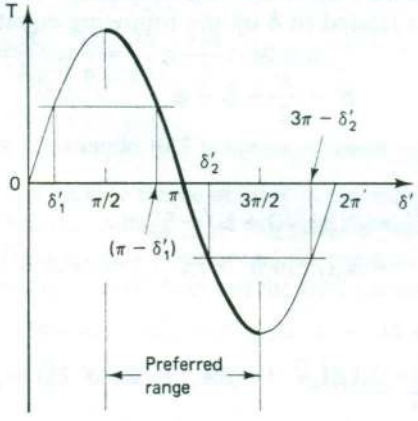


Figure 10.10 Torque—δ' curves of a synchronous machine fed by a current source.

two values of δ' — that is, δ'_1 and $(\pi - \delta'_1)$. Similarly, a given braking torque is available at two values of δ' — that is, $(\pi + \delta'_2)$ and $(3\pi - \delta'_2)$. Figure 10.11 shows the locus of I'_m as δ' is varied from 0 to π with I'_f and I_s constant. Phasors I'_m, I'_f, and V are shown for two values of δ' which yield the same value of torque. The following may be noted:

1. When δ' is less than $\pi/2$, the machine always operates at a lagging power factor. However, when δ' is greater than $\pi/2$, the machine may operate at either a lagging or a leading power factor.
2. For δ' greater than $\pi/2$, I'_m and V have lower values and the power factor is higher.
3. Because of the large value of I'_m for δ' less than $\pi/2$, the machine may operate under heavy saturation.

For these reasons, during the motoring operation, the preferred range of δ' is $\pi/2 < \delta' < \pi$. A similar analysis of the braking operation will show that the preferred range of δ' is $\pi < \delta' < 3\pi/2$. The preferred range of operation of δ' is also indicated in figure 10.10.

10.4 WOUND-FIELD SALIENT POLE MOTOR OPERATING FROM A VOLTAGE SOURCE OF CONSTANT FREQUENCY

The salient pole motor has field poles projecting out from the rotor core. It is characterized by different air-gap reluctances in the direct and quadrature axes. The tendency of the rotor poles to align with the rotating air-gap mmf wave at a position of minimum reluctance (or maximum flux) causes reluctance torque to be produced, in addition to the synchronous torque present in a cylindrical rotor motor. A damper winding is also provided on the rotor poles.

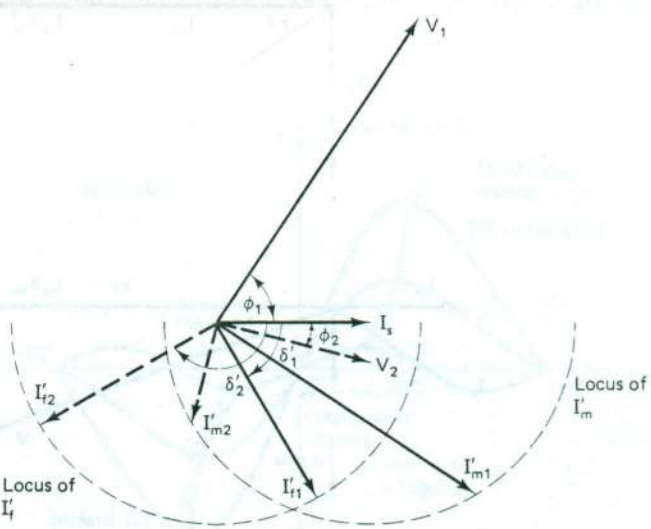


Figure 10.11 Phasor diagram showing variation of I'_m and V with δ'.

In the normal operating range of the motor, the reluctance torque is very small compared to the synchronous torque; therefore, the motor performance can be calculated with satisfactory accuracy by cylindrical rotor theory as presented in previous sections. When more accurate results are required, the analysis presented next can be used.

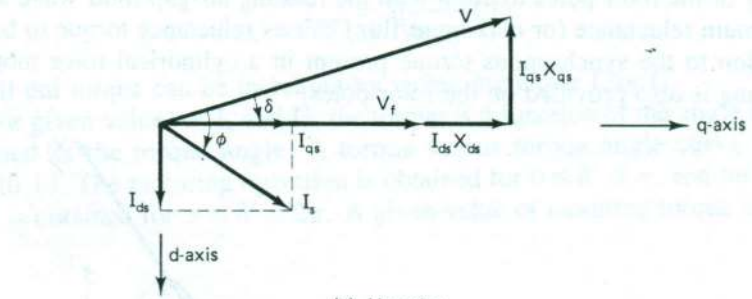
Because of the different air-gap reluctances in the direct and quadrature axes, the magnetizing reactances, and therefore, the synchronous reactances are also different for the direct and quadrature axes. Let the synchronous reactances in the direct and quadrature axes be denoted by X_{ds} and X_{qs}, respectively. Because of the difference in the reactance values, it is difficult to obtain a simple equivalent circuit. The machine is best described directly in terms of a phasor diagram. Figure 10.12 shows the phasor diagrams for motoring and braking operations, neglecting armature resistance. I_{ds} and I_{qs} are the direct and quadrature axis components of armature current I_s. The relationship between $\bar{I}_f'$ and $\bar{V}_f$ is obtained by replacing X_s by X_{ds} in equation (10.8). Thus,

$$\bar{I}_f' = \frac{\bar{V}_f}{jX_{ds}} \tag{10.30}$$

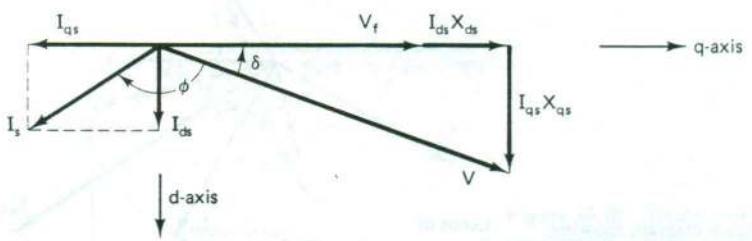
From the phasor diagram of figure 10.12a the following equations are obtained:

$$I_s \cos \phi = I_{qs} \cos \delta - I_{ds} \sin \delta \tag{10.31}$$

$$I_{ds} = \frac{V \cos \delta - V_f}{X_{ds}} \tag{10.32}$$



(a) Motoring



(b) Braking

Figure 10.12 Phasor diagrams of a salient pole motor.

$$I_{qs} = \frac{V \sin \delta}{X_{qs}} \tag{10.33}$$

Substituting from equations (10.32) and (10.33) into equation (10.31) yields

$$I_s \cos \phi = \frac{V_f \sin \delta}{X_{ds}} + \frac{V(X_{ds} - X_{qs})}{2X_{ds}X_{qs}} \sin 2\delta \tag{10.34}$$

The developed power is given by equation (10.12). Substituting from equation (10.34) into equation (10.12) gives

$$P_m = 3\left[\frac{VV_f}{X_{ds}} \sin \delta + \frac{V^2(X_{ds} - X_{qs})}{2X_{ds}X_{qs}} \sin 2\delta\right] \tag{10.35}$$

Now

$$T = \frac{P_m}{\omega_{ms}} = \frac{3}{\omega_{ms}}\left[\frac{VV_f}{X_{ds}} \sin \delta + \frac{V^2(X_{ds} - X_{qs})}{2X_{ds}X_{qs}} \sin 2\delta\right] \tag{10.36}$$

The torque expression has two components. The first is the synchronous torque. This is similar to the expression in equation (10.15), except that X_s is replaced by X_{ds}. This component increases with field excitation. The second component is due to the reluctance torque. It is independent of field excitation. The torque versus torque angle curves for three values of field excitation are shown in figure 10.13. The steady-state stability limits, which are reached at the peak values of torque, are also shown in the figure.

The expressions for operation from a current source can be derived following a similar approach.

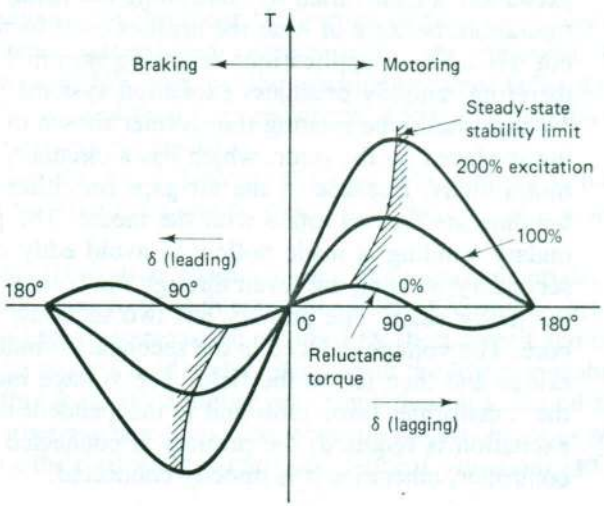


Figure 10.13 Torque-angle curves of a salient-pole synchronous machine.

10.5 STARTING AND BRAKING WHEN FED FROM A CONSTANT FREQUENCY SOURCE

A synchronous motor fed from a constant frequency source is not self-starting. It is normally started as an induction motor, with the damper winding serving as a squirrel-cage rotor. The field winding is shorted through a discharge resistance to keep interwinding voltages within safe values. When near synchronous speed, the field is excited. The synchronous torque assisted by the damper winding torque is able to pull the rotor into step with the rotating field.

The starting current is high and the starting torque is low. Any attempt to decrease the starting current and increase the starting torque by increasing the damper winding resistance makes the damper winding less effective during pull-in and hunting. In large size motors, the high starting current (5 to 10 times the full-load current) causes a large dip in the supply voltage, consequently further reducing the starting torque which is already low.

Since the frequency is constant, regenerative braking can be employed only for holding an active load at synchronous speed. For stopping, dynamic braking is used. Dynamic braking is obtained by disconnecting the armature from the source and connecting it to a three-phase resistor. The machine works as a synchronous generator and dissipates the generated energy in the braking resistor. Since the synchronous reactance is higher than the braking resistance for most of the speed range, the variation in armature current and torque is not large; therefore, a single section of resistance is enough. Plugging is not used because of the high current, which is far beyond the capability of the damper winding unless it is specially designed.

10.6 BRUSHLESS EXCITATION OF WOUND-FIELD MACHINES

Here we are concerned mainly with the excitation of synchronous motors for variable speed applications. One commonly used method is to convert ac to dc in a converter and apply it to the field winding through slip rings and brushes. The field excitation is then varied by controlling the firing angle of the converter. In this configuration, because of wear the brushes must be replaced at regular intervals. This is not desirable in applications requiring minimal maintenance. These applications, therefore, employ brushless excitation systems.[3] One such system makes use of a 1-phase shell type rotating transformer shown in figure 10.14. The secondary winding is placed on the rotor, which has a circular cross section and is mounted on the motor shaft. Because of the air-gaps on either side, the rotor and the secondary winding are free to rotate with the motor. The portion of the shaft inside the secondary winding is made hollow to avoid eddy currents. The connections from the secondary winding are taken through the hollow shaft to a diode bridge mounted on the motor shaft. The primary has two sections, which are placed on the stationary core. The voltage induced in the secondary winding is rectified by the rotating diode bridge and then fed to the field. The voltage induced in the secondary depends on the transformer turns ratio and is independent of rotor speed. When variable field excitation is required, the primary is connected to the ac mains through a voltage controller; otherwise it is directly connected.

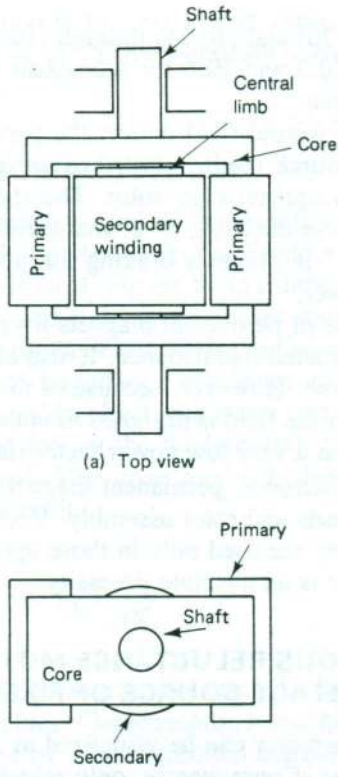


Figure 10.14 Rotary transformer for brushless excitation.

10.7 PERMANENT MAGNET SYNCHRONOUS MOTOR OPERATING FROM A FIXED FREQUENCY SOURCE

In permanent magnet synchronous motors, field excitation is obtained by suitably mounting the permanent magnets on the rotor.[4,5] Ferrites or rare earth (cobalt-samarium) magnets, with linear magnetization characteristics, are employed. Ferrites are commonly used because of low cost, but the machine becomes bulky due to low remnance. Rare earths, because of their high remnance, allow a large reduction in weight and size, but they are very expensive.

The armature (stator) has the usual 3-phase winding. The damper winding is also provided on the rotor to serve the same purposes as in the case of a wound-field motor.

Because of the linear magnetization characteristics of magnet materials, the equivalent circuit of figures 10.3 and 10.9, with constant values of V_f and I'_f, are applicable. The equivalent circuit parameters can be obtained by short-circuit and open circuit tests, as in the case of a wound-field motor. Because of the low permeability of the magnet material, the flux path has a higher reluctance than in a wound-field motor; therefore, the magnetizing reactance, and consequently the synchronous reactance, has a lower value. Since the equivalent circuits are identical, equations (10.12)

through (10.20) and (10.25) through (10.29) and all characteristics described in sections 10.1–10.3 and 10.5 for a constant field operation of a wound-field motor are also applicable.

Like a wound-field motor, the permanent magnet motor fed from a constant frequency source is also started as an induction motor, with the damper winding serving as a squirrel-cage rotor. The starting performance of a permanent magnet motor is worse than that of a wound-field machine. Since the excitation cannot be "turned off," it causes a braking torque to be produced. As a result, the starting torque is lower.

The use of permanent magnets for excitation eliminates brushes and slip rings, and the associated maintenance. It also eliminates the field copper loss and the need for a dc source. However, because of the constant field, the power factor cannot be controlled. If the field is designed to obtain a unity power factor at full load, the motor operates at a very low power factor (leading) at light loads, resulting in poor efficiency. Furthermore, permanent magnet motors are expensive because of the high cost of magnets and rotor assembly. Because of the high cost and poor starting performance they are used only in those applications where their synchronous machine characteristic is an absolute necessity.

10.8 SYNCHRONOUS RELUCTANCE MOTOR OPERATING FROM A VOLTAGE SOURCE OF FIXED FREQUENCY

A reluctance motor can be visualized as a salient pole synchronous motor without a field winding. Consequently, only reluctance torque is present. The torque expression is obtained by substituting $V_f = 0$ or ignoring the synchronous torque in equation (10.36). Thus,

$$T = \frac{3V^2}{\omega_{ms}} \left[\frac{X_{ds} - X_{qs}}{2X_{ds}X_{qs}} \right] \sin 2\delta \tag{10.37}$$

The machine is capable of providing both motoring and braking operations. The pull-out torque (T_{max}) or steady-state stability limit is reached at $\delta = 45°$, giving

$$T_{max} = \frac{3V^2}{\omega_{ms}} \left[\frac{X_{ds} - X_{qs}}{2X_{ds}X_{qs}} \right] \tag{10.38}$$

The rotor is constructed with flux barriers so that the flux in the q-axis of the machine is much less than that in the d-axis. This makes $X_{ds} \gg X_{qs}$; consequently, the pull-out torque is increased. The torque angle curve shown in figure 10.13 for zero excitation is applicable to this machine. The speed-torque curve shown in figure 10.5 is also applicable, except that the pull-out torque is much lower.

Because of the absence of the field excitation, the entire magnetizing current required to produce the air-gap flux has to come from the armature supply. Consequently, the machine has a low lagging power factor—between 0.65 to 0.75 at full load.

The reluctance motor fed from a constant frequency source is also started as an induction motor with the damper winding serving as a squirrel-cage rotor. Consequently, the starting current is high but the torque is low.

The reluctance motor is suitable for low-power applications where its synchronous machine characteristic is of absolute necessity, and the low torque and poor power factor are acceptable.

10.9 OPERATION WITH NONSINUSOIDAL SUPPLIES

Various points discussed in section 6.8 regarding the operation of an induction motor with nonsinusoidal supplies are also applicable to synchronous motors, provided that the appropriate equivalent circuit is used. Figure 10.15a shows the kth harmonic per phase equivalent circuit of a wound-field synchronous motor without a damper winding. It is obtained from the fundamental equivalent circuit of figure 10.3a by substituting appropriate values of impedance and ignoring the fundamental frequency voltage V_f. Because of the skin effect, the stator resistance has a somewhat higher value for the kth harmonic then at the fundamental. Since R_{sk} is very small compared to kX_s, it can be ignored. This gives the equivalent circuit shown in figure 10.15c. From the equivalent circuit,

$$I_{sk} = \frac{V_k}{kX_s} \tag{10.39}$$

When fed by a nonsinusoidal voltage source, because of a large value of kX_s, harmonic currents have very small values. Consequently, harmonics have little effect on the motor performance. When fed by a nonsinusoidal current source, a small value of I_{sk} can produce a large harmonic voltage, causing considerable distortion in the machine terminal voltage. The undersirable low-speed torque pulsations and derating caused by the harmonic losses will also be present.

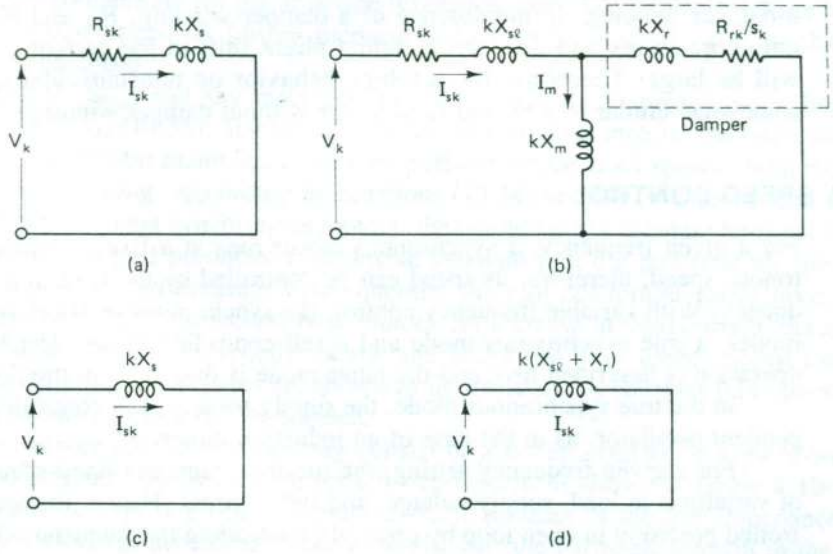


Figure 10.15 kth harmonic equivalent circuits of a synchronous motor.

The per-phase harmonic equivalent circuit of a wound-field motor with a damper winding is shown in figure 10.15b. It is obtained from the fundamental equivalent circuit of figure 10.2a by ignoring the fundamental frequency current source I_f and accounting for the effect of the damper winding. Because of the difference in speed between the rotating field produced by the kth harmonic and the rotor speed, the damper winding acts in the same way as the squirrel-cage rotor of an induction motor, and consequently has a similar equivalent circuit. As explained in section 6.8.2, slip s_k is close to 1 for all values of k. Since kX_m is very large and R_{sk} and R_{rk} are small compared to $kX_{s\ell}$ and kX_r, the equivalent circuit of figure 10.15b can be approximated by the equivalent circuit of figure 10.15d. Now,

$$I_{sk} = \frac{V_k}{k(X_{s\ell} + X_r)} \tag{10.40}$$

Note that $(X_{s\ell} + X_r) \ll X_s$.

The machine behavior is similar to an induction motor. When fed by a nonsinusoidal voltage source, the flux essentially remains sinusoidal, and the torque is produced mainly by the fundamental. Harmonics are able to produce sixth and twelfth harmonic frequency torque pulsations. The machine losses increase due to harmonic currents, leading to its derating. When fed by a nonsinusoidal current source, harmonics flow through the low impedance path offered by the damper winding (fig. 10.15b). Hence, I_m and consequently the flux and terminal voltage are somewhat sinusoidal. Torque is only produced by the fundamental component of the current. Torque pulsations and machine derating are also present.

The equivalent circuit of figure 10.15b and the approximate equivalent circuit of figure 10.15d are applicable to permanent magnet synchronous motors with and without damper windings. When a damper winding is present, R_{rk} and X_r will represent the combined effect of damper winding and core loss in magnets. The behavior of the machine on nonsinusoidal supplies will be similar to a wound-field motor with a damper winding. In the absence of a damper winding, R_{rk} and X_r will represent only core losses and the corresponding phase shift in the current, and their values will be large. Therefore, the machine behavior on nonsinusoidal supplies will be somewhat similar to a wound-field motor without damper windings.

10.10 SPEED CONTROL

For a given frequency, a synchronous motor runs at a fixed speed equal to synchronous speed; therefore, its speed can be controlled by the control of its supply frequency. With variable frequency control, the synchronous motor may operate in two modes: a true synchronous mode and a self-controlled mode. The former mode of operation is described here and the latter mode is discussed in the next chapter.

In the true synchronous mode, the supply frequency is controlled from an independent oscillator, as in the case of an induction motor.

For a given frequency setting, the machine runs at a fixed speed, independent of variations in load, supply voltage, and field current. Hence, the speed can be controlled precisely in open loop by precisely controlling the frequency. When operating in steady state, a gradual increase in frequency causes the armature field speed to be-

come somewhat greater than the rotor speed and the torque angle to increase. The motor accelerates to follow the changes in frequency. When the frequency reaches a new value, the machine settles at the new speed after hunting oscillations which are damped by the damper winding. On the other hand, a gradual decrease in frequency causes the armature field speed to become somewhat lower than the rotor speed. Consequently, the torque angle becomes negative. The motor decelerates under regenerative braking following the changes in frequency. When the frequency reaches a new setting, the machine settles at a new speed after oscillations which are damped by the damper winding.

The foregoing discussion suggests that the frequency must be changed gradually to allow the rotor to track the changes in the revolving field speed, otherwise the motor may pull out of step. For slowly changing torques, the motor will be steady-state stable up to the pull-out torque. For sudden changes in the load torque, it will be stable for the lower range of torques.

The motor can also be started by increasing the frequency slowly from its zero value. It draws a much lower current and produces a much higher torque compared to starting as an induction motor. The machine can also be braked down to zero speed by regenerative braking. A given phase sequence thus provides operation in quadrants I and II. Its reversal at standstill gives operation in quadrants III and IV.

In practice, the true synchronous mode of operation is employed only with voltage source inverters. Hence, it is described here only for operation with voltage source inverters.

As in the case of an induction motor, the common control strategy is to operate the motor at a constant air-gap flux up to the base speed and at a constant terminal voltage above the base speed. The air-gap flux depends on the value of the magnetizing current I'_m. Hence, constant air-gap flux operation below base speed is achieved by operating the motor with a constant (V/f) ratio. At low speeds, the (V/f) ratio is increased to compensate for the armature resistance drop. The terminal voltage saturates at the base speed. A further increase in frequency causes the flux to change inversely with speed.

Since V_f, ω_{ms}, and any reactance are proportional to frequency, according to equations (10.15), (10.36), and (10.38), for all synchronous motors, the operation at a constant (V/f) ratio maintains a constant pull-out torque at all speeds, both during motoring and braking. According to equations (10.14) and (10.35), for wound-field and permanent magnet synchronous motors, the operation at a constant terminal voltage above base speed produces a somewhat constant pull-out power at all speeds, both for motoring and braking. Consequently, the pull-out torque varies inversely with speed. In the case of wound-field motors, an increase in field current increases the pull-out torque for all speeds, below and above the base speed. Speed torque curves for different frequencies are shown in figure 10.16. A constant flux operation has been assumed below the base speed.

The voltage source inverter may consist of a 6-step inverter or a pulse-width modulated inverter. The waveform for a 6-step inverter is shown in figure 8.1b and the harmonic content is given by equation (8.2). Because of the low impedance of the damper winding, according to equation (10.40), the harmonic content in the armature current will be high. The harmonics will cause losses and torque pulsations.

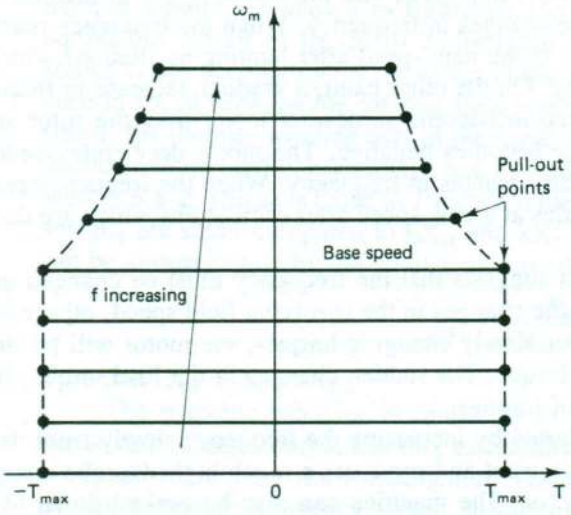


Figure 10.16 Speed-torque curves of a synchronous motor with variable frequency control.

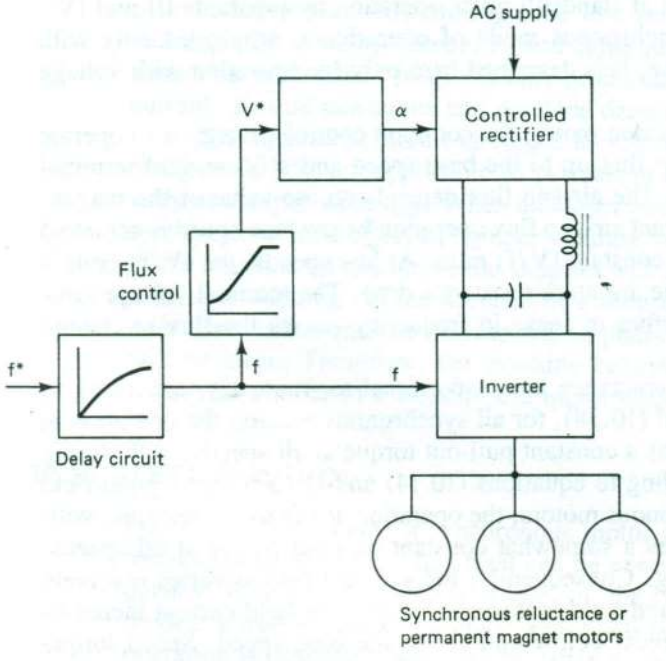


Figure 10.17 Variable frequency control of multiple synchronous motors.

The harmonic currents can be reduced by increasing the impedance of the damper windings but then they will be less effective during hunting. When the torque pulsations are unacceptable, a pulse-width modulated inverter may be used.

Because of the hunting and stability problems, the true synchronous mode is employed only in multiple synchronous reluctance and permanent magnet motor drives requiring accurate speed tracking between the motors — for example, in fiber spinning mills, paper mills, textile mills, and so on. A drive for such applications is shown in figure 10.17. The frequency command f* is applied to the inverter through

a delay circuit to ensure that the change in inverter frequency is slow enough for the motor speed to track the revolving field speed. A flux controller changes the machine terminal voltage with frequency to maintain a constant flux below base speed and a constant terminal voltage above base speed for all motors. Since all the motors are connected in parallel to a common inverter, their speeds are uniquely related to the command frequency.

The variable speed synchronous motor drives are generally operated in the self-controlled mode, which is described in the next chapter.

Example 10.3

The machine of example 10.1 is now controlled from a variable frequency source. The (V/f) ratio is maintained constant up to base speed. Calculate

1. The armature current, torque angle, and power factor at full-load torque, half the rated speed, and the rated field current.
2. The armature current and power factor at half the rated speed, half full-load torque, and the rated field current.
3. The torque and field current for the rated armature current, 1500 rpm, and the unity power factor.

Solution: From example 10.1, $I'_m = 105.8$ A, Rated power factor = 1.0, Rated torque = 8751 N-m, $\delta = 41.6°$, Rated armature current = 99 A, Rated $I'_f = 144.9$ A, Rated speed = 120 f/p rpm = $120 \times 60/6 = 1200$ rpm or 125.7 rad/sec.

1. Half rated speed = 600 rpm.

Since the ratio (V/f) is constant, I'_m is constant. According to equation (10.19) δ will remain the same. From equations (10.22) and (10.23), for constant values of δ and I'_m, the in-phase and quadrature components of armature current remain unchanged. Hence, I_s and the power factor also remain unchanged. This is true for any speed. Thus,

$I_s = 99$ A, $\quad \delta = 41.6°$ (electrical) and power factor = 1.0

2. Half the rated torque = 4375.5 N-m

From equation (10.19)

$$4375.5 = \frac{3(0.5 \times 36)}{0.5 \times 125.7} \times 105.8 \times 144.9 \sin \delta$$

or $\sin \delta = 0.33$ and $\cos \delta = 0.94$.

Since equations (10.22) and (10.23) are independent of frequency, they are applicable here also; hence,

$$I_s \cos \phi = -105.8 + 144.9 \times 0.94 = 30.4$$
$$I_s \sin \phi = 144.9 \times 0.33 = 47.8$$

which gives

$$I_s = \sqrt{30.4^2 + 47.8^2} = 56.6 \text{ A}$$
$$\cos \phi = 0.54 \text{ (leading)}$$

3. Since the operation is taking place at the rated (constant) terminal voltage, rated current, and unity power factor, developed power is

$$P_m = 1100 \times 10^3 \text{ W}$$

$$\omega_m = \frac{1500 \times 2\pi}{60} = 157.1 \text{ rad/sec}.$$

$$T = \frac{1100 \times 10^3}{157.1} = 7003 \text{ N-m}$$

At 1500 rpm,

$$X_s = \frac{1500}{1200} \times 36 = 45\ \Omega$$

$$I'_m = \frac{V}{X_s} = \frac{6600/\sqrt{3}}{45} = 84.68 \text{ A}$$

Since $\phi = 0°$, from equations (10.22) and (10.23),

$$I'_f \cos \delta = 84.68$$

$$I'_f \sin \delta = 99$$

Hence, $I'_f = 130$ A.

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PROBLEMS

10.1 A 10 kW, 3-phase, 440 V, 60 Hz, 4 pole, Y-connected permanent magnet synchronous motor has the following parameters:
$X_m = 15\ \Omega$, $X_{s\ell} = 1\ \Omega$, negligible R_s, Rated power factor = 0.85 (lagging)
Calculate

1. The equivalent field current I_f' and pull-out torque.
2. The torque angle at full load.
3. The power factor and the armature current at 60 percent of the rated torque.
4. The torque when operating at a unity power factor.

Neglect friction, windage, and core loss.

10.2 A 10 MW, 3-phase, 6600 V, 60 Hz, 4 pole, Y-connected wound-field motor has the following parameters:
$X_m = 8\ \Omega$, $X_{s\ell} = 0.5\ \Omega$, $R_s = 0.01\ \Omega$, Rated power factor = 1.0, Rated field current = 200 A, Field winding resistance = 1.2 Ω, Core, friction, and windage losses = 10 kW. Calculate

1. The constant n and the torque angle at full load.
2. The power factor, armature current, and efficiency at half the rated torque and rated field current. Assume core, friction, and windage losses remain constant.
3. The field current to get the unity power factor at 60 percent of the rated torque.

10.3 The motor of problem 10.1 is operating in regenerative braking. Calculate the armature current and power factor at the rated braking torque.

10.4 The motor of problem 10.2 is operating under regenerative braking. Calculate the braking torque when the machine operates at the rated armature current and unity power factor. Assume the core, friction, and windage losses remain constant at 10 kW. What is the field current?

10.5 A 15 kW, 3-phase, 440 V, 60 Hz, 4 pole, Y-connected, permanent magnet synchronous motor has the following parameters:
$X_m = 12\ \Omega$, $X_{s\ell} = 1.2\ \Omega$, negligible R_s, Rated power factor = 0.8 (lagging)
The motor is controlled by variable frequency control with a constant (V/f) ratio up to the base speed and at a constant (= rated) terminal voltage above the base speed.

Calculate and plot T, P_m, V, I_m', and the power factor against speed for the motor operation at the rated armature current. An application requires that the machine should supply 80 percent of full-load power for all speeds above base speed. What will be the maximum available speed for such an application?

Neglect friction, windage, and core loss.

10.6 For the drive of problem 10.5, calculate the torque which gives unity power factor operation at base speed. Will the power factor at this torque be unity for all speeds below base speed?

10.7 Calculate the speed at which the drive of problem 10.5 operates at the unity power factor and 0.8 of the rated current. Calculate the torque at this operating point. Will the power factor at this torque be unity for all speeds above base speed?

10.8 A 500 kW, 3-phase, 6600 V, 60 Hz, 6-pole, Y-connected wound-field synchronous motor has the following parameters:
$X_m = 78$, $X_{s\ell} = 3$, Rated power factor = 1.0, n = 5, R_s = negligible
The motor speed is controlled by variable frequency control with a constant (V/f) ratio up to base speed and a constant (= rated) terminal voltage above base speed.

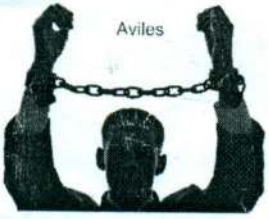
Calculate and plot T, P_m, V, I_m', and I_F versus speed for the motor operation at the rated armature current and unity power factor. What is the range of constant power operation? Neglect friction, windage, and core loss.

10.9 What value of the field current will be required in the drive of problem 10.8 to obtain unity power factor operation at 200 kW and 1200 rpm?

10.10 Derive an expression for torque for a salient pole synchronous motor fed from a fixed frequency current source.

11

Self-Controlled Synchronous Motor Drives (Brushless dc and ac Motor Drives)



The self-controlled variable speed synchronous motor drives have a number of advantages which make them either superior to or competitive with induction motor or dc motor variable speed drives.

1. The operation of a synchronous motor in the self-controlled mode eliminates hunting and stability problems, and permits the realization of versatile control characteristics of a dc motor without the limitations associated with commutator and brushes, such as limits on maximum speed, voltage, and power, frequent maintenance, inability to operate in contaminated and explosive environments, and so on. The self-controlled synchronous motor drives have been built for power ratings of tens of megawatts and speeds approaching 6000 rpm, which are beyond the capability of dc and induction motor drives. They have good dynamic response and smooth starting and braking operation with a high torque-to-current ratio.
2. The power factor of a wound-field synchronous motor can be controlled by controlling its field current. The operation of the drive at unity power factor minimizes the KVA rating, cost, and losses of variable frequency supplies, and maximizes the motor power output while reducing its losses. By operating the machine at a leading power factor, the inverter thyristors can be commutated by the armature induced voltages. Use of this commutation—known as load commutation—eliminates the need for thyristor commutation circuits, thus permitting substantial savings in cost, weight, volume, and losses in a thyristor inverter. Further, the load commutation increases the frequency range of a current source inverter and a cycloconverter. The permanent magnet motor can also operate with a leading power factor by the adjustment of its torque angle.

Hence, the load commutation and the associated benefits are available with the permanent magnet drives also.

Because of these advantages, the self-controlled synchronous motor drives are employed in the following variable speed applications:

1. The self-controlled synchronous motor drives fed from a load commutated current source inverter or a cycloconverter have been used in medium to very high power (tens of megawatts) or high-speed drives such as compressors, extruders, induced and forced draft fans, blowers, conveyers, aircraft test facilities, main line traction, steel rolling mills, large ship propulsion, flywheel energy storage, and so on.
2. The self-controlled synchronous motor drives fed from a line commutated cycloconverter are employed in low-speed gearless drives for ball mills in cement plants, mine hoists, rolling mills, and so on.
3. Recently, self-controlled permanent magnet synchronous motor drives are finding applications in servo drives which so far have been dominated by dc motors.
4. The self-control is also employed for starting large synchronous machines in gas turbine and pumped storage power plants. A common inverter is time-shared by a number of machines.

The applications of self-controlled synchronous motor drives are expected to increase, particularly in the areas dominated by dc drives. The only disadvantage of the self-controlled synchronous motor drives, compared to dc drives, is their complex control. However, this is not a major problem due to the recent progress in logic gates and microcomputers.

Figure 11.1 shows the phasor diagram for the variable frequency operation of a synchronous motor. It is obtained from the phasor diagram of figure 10.3. The cur-

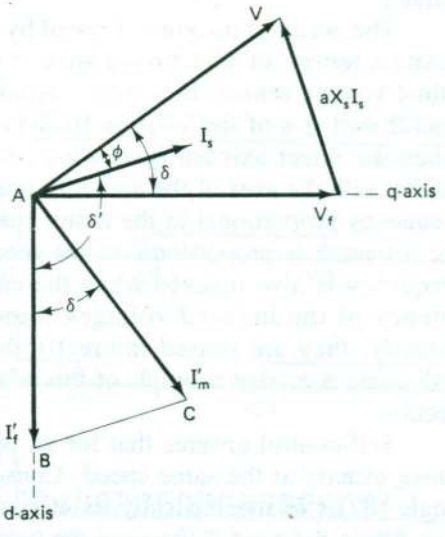


Figure 11.1 Synchronous motor phasor diagram.

rent phasors are independent of frequency. The induced voltages are given by the following equations [equations (10.8) and (10.9)]:

$$V = aX_s I'_m \tag{11.1}$$

$$V_f = aX_s I'_f \tag{11.2}$$

where the per-unit frequency "a" is the ratio of the frequency of operation f to the rated frequency f_{rated}—that is,

$$a = f/f_{rated} \tag{11.3}$$

and X_s is the synchronous reactance at the rated frequency f_{rated}.

When operating at given values of I_s, I'_f, and δ' (or δ), the parameters I'_m, flux, and torque have fixed values. The induced voltages V and V_f, and the reactance drop increase linearly with frequency but their phase relationship remains independent of frequency.

Since the equivalent circuit is based on the assumption of a negligible drop across R_s, V represents both the terminal voltage and the induced voltage (voltage E_t in fig. 10.2) due to the flux linking the armature (that is, the sum of the air-gap flux and armature leakage flux).

11.1 SELF-CONTROL

In self-control, as the rotor speed changes, the armature supply frequency is also changed proportionately so the armature field always moves at the same speed as the rotor. This ensures that the armature and rotor fields move in synchronism for all operating points. Consequently, a self-controlled synchronous motor does not pull out of step and does not suffer from the hunting oscillations and instability associated with a step change in torque or frequency when controlled from an independent oscillator.

The accurate tracking of speed by frequency is realized with the help of a rotor position sensor, of which there are two types: rotor position encoder or armature terminal voltage sensor. In a rotor position encoder, the firing pulses to the semiconductor switches of the variable frequency converter feeding the motor are delivered when the direct axis (or quadrature axis) of the rotor makes certain predetermined angles with the axes of the armature phases. Consequently the switches are fired at a frequency proportional to the motor speed. The frequency of the voltage induced in the armature is proportional to the speed. Hence, the desired tracking of speed by frequency is also realized when the converter frequency is made to track the frequency of the induced voltages. Since the induced voltages cannot be sensed directly, they are sensed indirectly through the armature terminal voltages. We will come across an example of this when we consider load commutation in the next section.

Self-control ensures that for all operating points the armature and rotor fields move exactly at the same speed. Consequently, the motor cannot adjust the torque angle (δ' or δ) mechanically as in conventional operation (see section 10.2.1). In fact we do not need it, because the torque angle can now be adjusted electronically. This feature provides an additional controllable parameter and thus, greater control

of the motor behavior. Consider a specific example involving a particular rotor position encoder and the current source inverter of figure 8.14 as a variable frequency source, mainly to explain how a rotor position encoder is employed to generate reference signals for the firing of the semiconductor switches of an inverter and how the torque angle δ' (or δ) is set electronically.

Figure 11.2 shows an optical rotor position encoder for a 4 pole synchronous machine. It consists of a circular disc mounted on the rotor shaft. The disc has two 90° (180° electrical) slots S′ and S″ on an inner radius, with a separation of 90° between them. The outer periphery has a large number of slots. There are four stationary optical sensors P_1 to P_4. Three of them P_1 to P_3 are placed 60° (or 120° electrical) apart on the inner radius. The fourth one P_4 is placed on the periphery. Each sensor has a light-emitting diode and a photo transistor. Consequently it gives an output whenever it faces a slot. The waveforms produced by the sensors are also shown in the figure. They are drawn such that time is measured from the instant the disc occupies the position shown in the figure. The square pulses P_1 to P_3 are of 180° duration with a phase shift of 120°, and their frequency is proportional to the motor

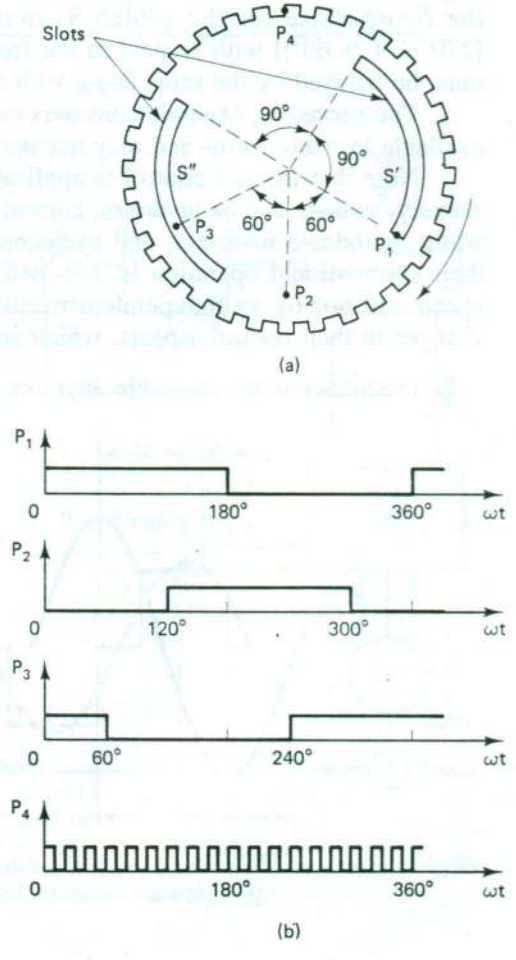


Figure 11.2 Rotor position encoder and its output waveforms (Le-Huy et al, 'Microprocessor control of a current-fed synchronous motor drive', IEEE IAS Annual Meeting, 1979, © IEEE 1979).

speed. The edges of the square pulse P_1 are used as reference points for the generation of firing pulses for semiconductor switches S_1 and S_4 of the inverter (fig. 8.14). Similarly, square pulses P_2 and P_3 are used to generate firing pulses for the other two switch pairs (S_3, S_6) and (S_5, S_2). The pulse train P_4 also has a frequency proportional to speed. It is used to sense speed and the setting of inverter firing angles. At all speeds, P_4 has the same number of pulses in the 180° duration of a square pulse. Hence, the firing angle can be measured independent of speed simply by counting the pulses starting from the reference points.

Figure 11.3 shows how switches S_1 and S_4 of the inverter are fired to obtain a desired value of the torque angle δ' for phase A. According to the phasor diagram of figure 11.1, the fundamental component of phase A current leads I_f' by an angle δ', and I_f' always lies along the direct axis which rotates at synchronous speed. Hence, switches S_1 and S_4 must be fired such that the peak of the fundamental component I_s occurs when the revolving direct axis is δ' electrical degrees behind the phase A axis. This is achieved by firing S_1 when the direct axis is $(\delta' + 60°)$ electrical behind the axis of phase A. S_4 is then fired 180° electrical later than S_1. If the slots are aligned with the direct axis such that the front edge of pulse P_1 leads I_f' by 270°, then the firing pulse for the switch S_1 must be delayed by an angle of $210° - \delta'$ $[270 - (\delta' + 60°)]$ with respect to the front edge of P_1, and the firing pulse for S_4 must be delayed by the same angle with respect to the trailing edge of P_1.

The preceding example considers one type of rotor position encoder. They are available in many forms and may use optical or magnetic sensors.

Note that the self-control is applicable to all variable frequency converters—namely, voltage source inverters, current source inverters, current controlled pulse-width modulated inverters, and cycloconverters. The only difference compared to their conventional operation is that their frequency is determined by the machine speed and not by an independent oscillator. This, however, leads to significant changes in their control aspects, which are listed here:

1. Frequency is not available anymore as an independent control variable.

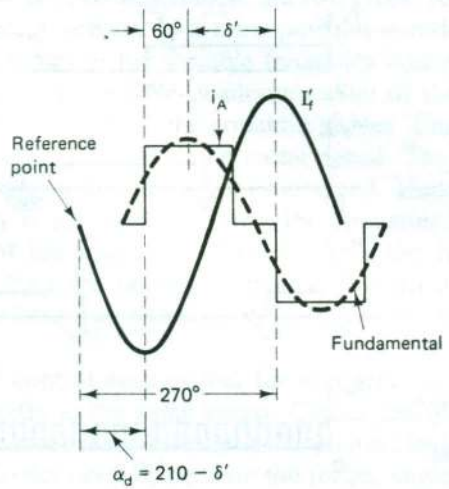
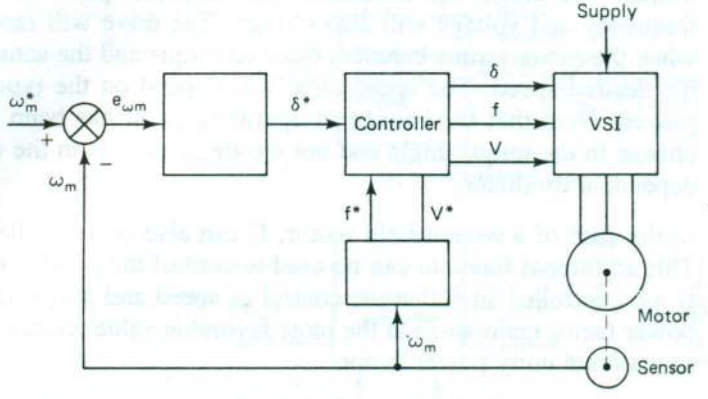
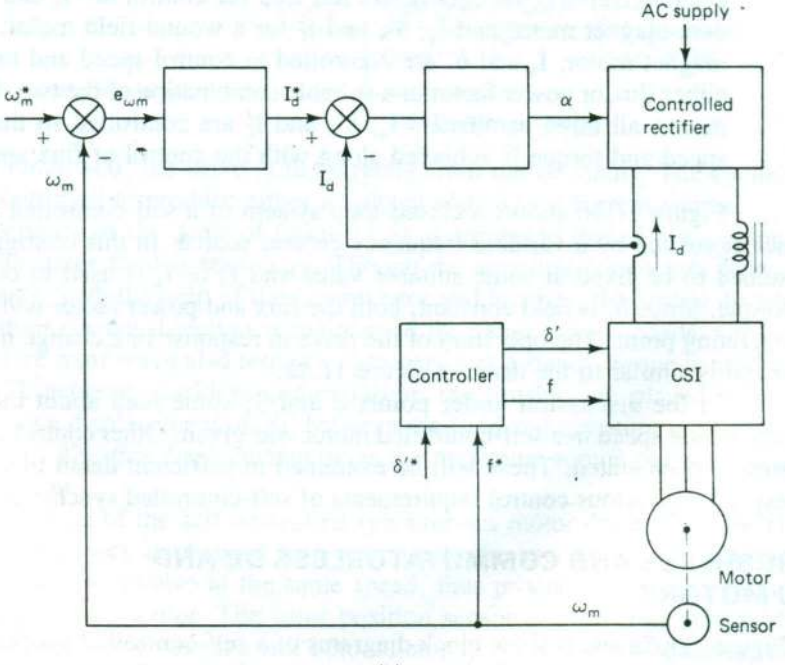


Figure 11.3 Control of torque angle δ'.

2. When the motor is fed from a voltage source inverter or cycloconverter, the voltage is changed as a function of frequency to obtain operation at a constant flux below base speed and at a constant terminal voltage above base speed. Hence, in the case of a permanent magnet motor, the only parameter left free for the control of torque and speed is the torque angle δ [equation (10.18)]. To illustrate this, figure 11.4a shows a closed-loop speed control system employ-



(a)



(b)

Figure 11.4 Closed loop speed control of self-controlled synchronous motor drives fed from (a) voltage source inverter and (b) current source inverter.

ing a voltage source inverter-fed self-controlled synchronous motor. The block diagram of the self-controlled machine shows that the frequency and output voltage of the inverter are changed as a function of speed. When operating in the steady state, an increase in speed command will produce a speed error, increasing the torque angle and causing the machine torque to exceed the load torque. The motor will accelerate. As the rotor speed increases, the inverter frequency and voltage will also change. The drive will reach the steady state when the motor torque balances the load torque and the actual speed is close to the desired speed. The speed error will depend on the type of controller employed. Note that the change in operating point has been brought on by the change in the torque angle and not the frequency, as in the control from an independent oscillator.

In the case of a wound-field motor, I_f' can also be controlled in addition to δ. This additional freedom can be used to control the power factor. Hence, δ and I_f' are controlled such that the control of speed and torque is achieved with the power factor maintained at the most favorable value. Generally, it is preferable to maintain unity power factor.

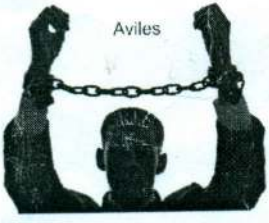
3. When the motor is fed from a variable frequency current source (current source inverter, current controlled pulse-width modulated inverter and current source cycloconverter), the parameters left free for control are I_s and δ' for a permanent magnet motor and I_s, δ', and I_f' for a wound-field motor. In a permanent magnet motor, I_s and δ' are controlled to control speed and torque along with either flux or power factor or a suitable combination of the two. In a wound-field motor, all three parameters I_s, δ', and I_f' are controlled so that the control of speed and torque is achieved along with the control of flux and power factor.

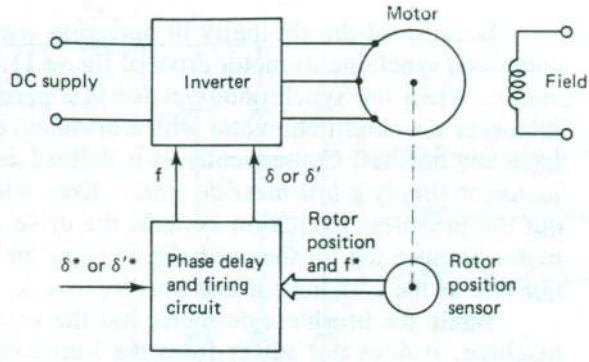
Figure 11.4b shows a closed-loop system of a self-controlled permanent magnet motor fed by a variable frequency current source. In this configuration, δ' is assumed to be fixed at some suitable value and I_d or I_s is used to control speed and torque. Since δ' is held constant, both the flux and power factor will change with the operating point. The operation of the drive in response to a change in the speed command is similar to the drive of figure 11.4a.

In the discussion under points 2 and 3, some idea about the control of the torque and speed in a self-controlled motor was given. Other control possibilities have simply been stated. These will be examined in sufficient detail in section 11.5 after explaining various control requirements of self-controlled synchronous motor drives.

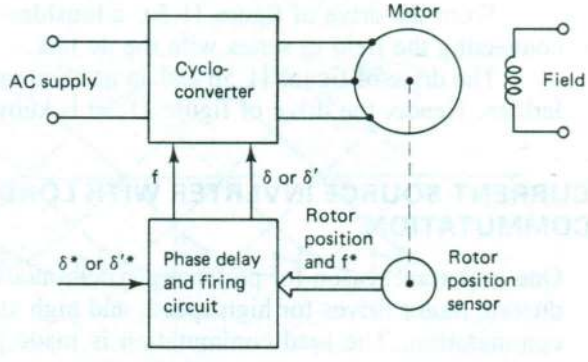
11.2 BRUSHLESS AND COMMUTATORLESS DC AND AC MOTORS

Figures 11.5a and b show block diagrams of a self-controlled synchronous motor fed from an inverter and a cycloconverter, respectively. When an inverter is employed, the drive is controlled from a dc source. The inverter may be a current source inverter, current controlled PWM inverter or a voltage source inverter. Depending on the type of inverter, the dc source may be a controllable current source or a constant or a controllable voltage source (figs. 8.2, 8.3, 8.17, and 8.26). When a cyclocon-





(a) Brushless DC motor



(b) Brushless AC motor

Figure 11.5 Self-controlled synchronous motors.

verter is employed, the drive is fed directly from the ac mains. The cycloconverter can be controlled to produce either a voltage source or a current source.

Operation of the drive of figure 11.5a is similar to that of a dc motor. In a dc motor, the stator field is stationary. Though the armature conductors carry alternating currents, with the help of the commutator and brushes, the spatial distribution of the armature current direction is maintained the same for all speeds. Consequently, the armature mmf wave also remains stationary and a steady torque is produced at all speeds. To achieve sparkless commutation, the brushes are placed in the neutral axis, giving a displacement of 90° between the field and armature mmf waves. From equation (10.27), this displacement gives the maximum torque per ampere of the armature current.

In the case of the self-controlled synchronous motor drive of figure 11.5a, the inverter's frequency is changed in proportion to the speed so that the armature and rotor mmf waves revolve at the same speed, thus producing a steady torque for all speeds, as in a dc motor. The rotor position sensor and inverter now perform the same function as the brushes and commutator in a dc motor. However, there is a slight difference between the two machines. Unlike a dc motor, the self-controlled drive does not have to operate at $\delta' = 90$, because it does not have a problem with sparking. Hence, δ' can be chosen at a suitable value to satisfy other performance requirements.

Because of the similarity in operation with a dc motor, the inverter-fed self-controlled synchronous motor drive of figure 11.5a is known as a *commutatorless dc motor.* When the synchronous motor is a permanent magnet motor, a reluctance motor, or a wound-field motor with a brushless excitation system, the drive does not have any brushes. Consequently, it is defined as a *brushless and commutatorless dc motor* or simply a *brushless dc motor.* Even when using a wound-field motor without the brushless excitation system, the drive is sometimes called a brushless dc motor because the brushes and slip rings of the excitation system are not as objectionable as the commutator and brushes in a dc motor.

While the brushless dc motor has the versatile control characteristics of a dc machine, it does not suffer from the limitations imposed by the commutator—namely, restrictions on the maximum speed and power ratings, frequent maintenance, inability to operate in explosive and contaminated environments, and so on.

From the drive of figure 11.5a, a brushless dc series motor can be obtained by connecting the field in series with the dc link.

The drive of figure 11.5b and an ac commutator motor also have the same similarities. Hence, the drive of figure 11.5b is known as *commutatorless ac motor.*

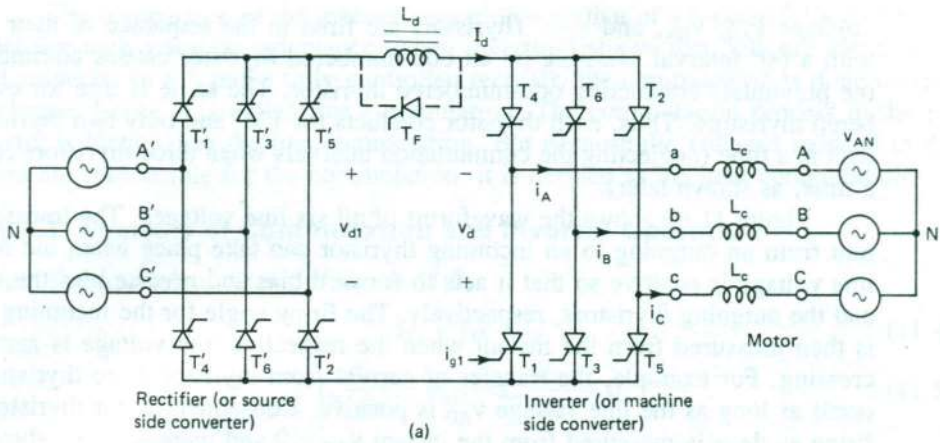
11.3 CURRENT SOURCE INVERTER WITH LOAD COMMUTATION

One important reason for preferring synchronous motor drives over squirrel-cage induction motor drives for high speed, and high and medium power ratings is the load commutation. The load commutation is made possible because of the ability of a synchronous motor to operate at a leading power factor. The load commutation has been employed for current source inverters and cycloconverters.

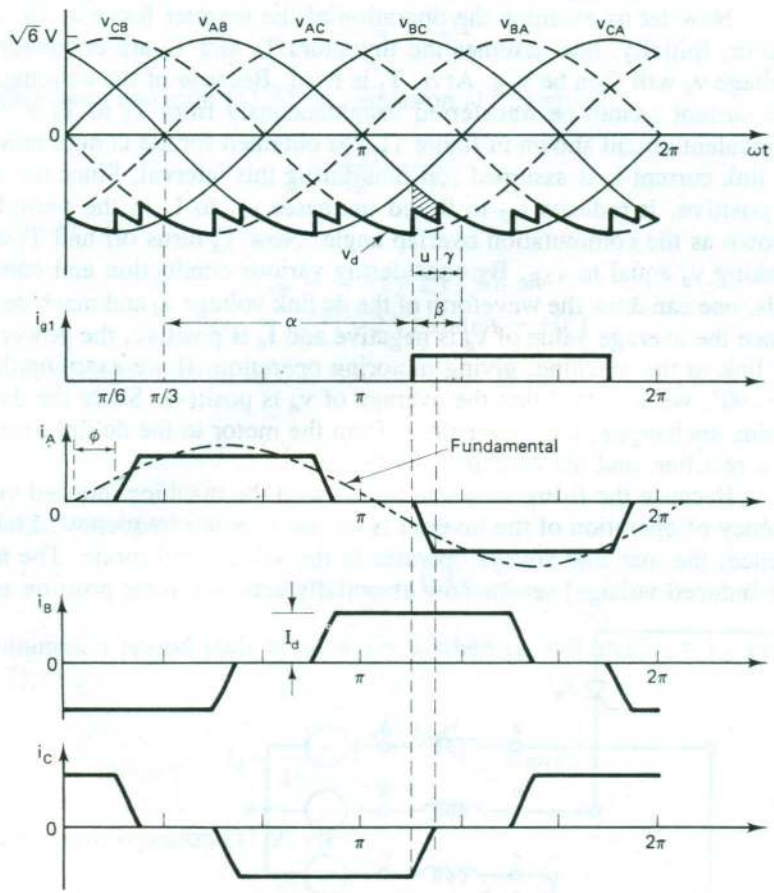
Figure 11.6 shows a current source inverter feeding a synchronous motor. The dc supply for the inverter is obtained from a 6-pulse fully controlled rectifier, which, together with the link inductor L_d, can be considered to constitute a current source for the inverter. The link inductor reduces the ripple in the link current I_d and prevents the inverter and rectifier from interfering with each other's operation.

The synchronous motor has been modeled as a three-phase set of sinusoidal voltages with rms value V behind a commutating inductance L_c per phase. Due to the very short duration of commutation transients, the commutating inductance is equal to the subtransient inductance of a synchronous motor.[11] The phase voltages v_{AN}, v_{BN}, and v_{CN} are the voltages induced in the armature due to the flux linking the armature (the air-gap flux and the armature leakage flux). Since the armature resistance drop has been neglected, these voltages are equal to the terminal voltages. Hence, they are represented by phasor V in the phasor diagram of figure 11.1 or figure 10.3b. Note that this model of the synchronous motor is valid only for operation during commutation intervals. For steady-state operation, the model of figure 10.3a—which consists of excitation voltage V_f behind the synchronous reactance X_s or its equivalent given in figure 10.3b—is applicable.

The inverter is similar to the current source inverter of figure 8.15, except that the commutation circuit, consisting of diodes D_1 to D_6 and capacitors C_1 to C_6, is absent. The function of the commutation circuit is now performed by induced



(a)



(b)

Figure 11.6 Load communication in a current source inverter.

voltages v_{AN}, v_{BN}, and v_{CN}. Thyristors are fired in the sequence of their numbers with a 60° interval. Turning on an odd-numbered thyristor causes commutation of the previously conducting odd-numbered thyristor. The same is true for even-numbered thyristors. Thus, each thyristor conducts for 120° and only two thyristors conduct at a time (neglecting the commutation intervals when three thyristors conduct at a time, as shown later).

Figure 11.6b shows the waveforms of all six line voltages. The transfer of current from an outgoing to an incoming thyristor can take place when the respective line voltage is positive so that it acts to forward bias and reverse bias the incoming and the outgoing thyristors, respectively. The firing angle for the incoming thyristor is then measured from the instant when the respective line voltage is zero and increasing. For example, the transfer of current from thyristor T_5 to thyristor T_1 can occur as long as the line voltage v_{AC} is positive. Consequently, for thyristor T_1, the firing angle α is measured from the instant $v_{AC} = 0$ and increasing, as shown in the figure. Each firing pulse has a duration of 120°.

Now let us examine the operation of the inverter for $\alpha > 90°$ as shown in the figure. Initially, one assumes the thyristors T_5 and T_6 are conducting. The dc link voltage v_d will then be v_{CB}. At α, T_1 is fired. Because of the commutating inductors, the current cannot be transferred instantaneously from T_5 to T_1. Consequently, the equivalent circuit shown in figure 11.7 is obtained for the commutation interval. The dc link current I_d is assumed constant during this interval. Since the line voltage v_{AC} is positive, it reduces i_{T5} to 0 and increases i_{T1} to I_d in the period u (fig. 11.6b) known as the commutation overlap angle. Now T_5 turns off and T_1 and T_6 conduct, making v_d equal to v_{AB}. By considering various conduction and commutation intervals, one can draw the waveform of the dc link voltage v_d and machine phase currents. Since the average value of v_d is negative and I_d is positive, the power flows from the dc link to the machine, giving motoring operation. If we examine the operation for $\alpha < 90°$, we will find that the average of v_d is positive. Since the direction of I_d remains unchanged, the power flows from the motor to the dc link, the inverter works as a rectifier, and the machine regenerates.

Because the firing is synchronized with the machine induced voltages, the frequency of operation of the inverter is the same as the frequency of induced voltages. Hence, the machine always operates in the self-control mode. The terminal voltage (or induced voltage) sensor now essentially acts as a rotor position sensor.

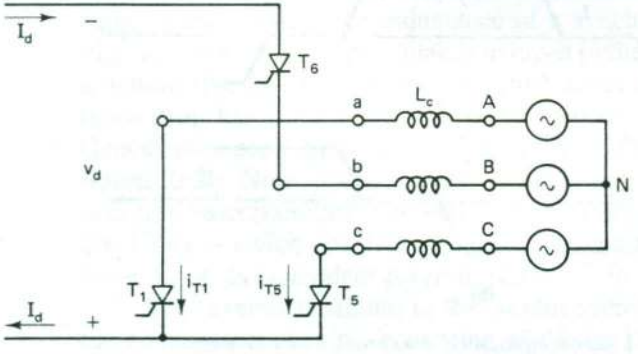


Figure 11.7 Equivalent circuit for the commutation of current from T_5 to T_1 in the inverter of Fig. 11.6.

The comparison of the inverter operation with that of a 6-pulse fully controlled rectifier with a source inductance (when inverting) shows that both are identical in all respects. In a 6-pulse fully controlled rectifier, the commutation is done by line voltages; hence it is called line commutation. The commutation process in the inverter is identical to the line commutation. But because the voltages induced in the load are responsible for the commutation, it is defined as the load commutation.

11.3.1 Analysis of Commutation and Inverter Operation

From the equivalent circuit of figure 11.7,

$$i_{T1} + i_{T5} = I_d \tag{11.4}$$

$$L_c \frac{di_{T1}}{dt} - v_{AC} - L_c \frac{di_{T5}}{dt} = 0 \tag{11.5}$$

Differentiating equation (11.4) yields

$$\frac{di_{T5}}{dt} = -\frac{di_{T1}}{dt} \tag{11.6}$$

Substituting from equation (11.6) into equation (11.5) gives

$$\frac{di_{T1}}{dt} = \frac{v_{AC}}{2L_c} \tag{11.7}$$

From figure 11.6b,

$$v_{AB} = \sqrt{3}(\sqrt{2}\,V) \sin \omega t \tag{11.8}$$

$$v_{AC} = \sqrt{3}(\sqrt{2}\,V) \sin(\omega t - 60°) \tag{11.9}$$

The dc link voltage is given by

$$v_d = -L_c \frac{di_{T1}}{dt} + v_{AB}$$

Substituting from equation (11.7) gives

$$v_d = (v_{AB} - 0.5 v_{AC}) \tag{11.10}$$

The commutation period ends at $\omega t = \alpha + u$ when $i_{T5} = 0$ and $i_{T1} = I_d$. From equation (11.7),

$$I_d = \frac{1}{2\omega L_c} \int_{\alpha+\pi/3}^{\alpha+\pi/3+u} v_{AC}\, d(\omega t)$$

Substituting from equation (11.9) yields

$$\cos \alpha - \cos(\alpha + u) = \frac{2\omega L_c I_d}{\sqrt{6}\,V} \tag{11.11}$$

Let us define an angle β as shown in figure 11.6b. Note that β is the angle of lead with respect to the instant when v_{AC} ceases to be positive. The angle β is called the *commutation lead angle* and is given by

$$\beta = 180° - \alpha \tag{11.12}$$

Substituting from equation (11.12) into equation (11.11) yields

$$\cos(\beta - u) - \cos\beta = \frac{2\omega L_c}{\sqrt{6}\,V} I_d \tag{11.13}$$

After turn-off, T_5 remains reverse biased until $v_{AC} = 0$ and is decreasing. If this duration is denoted by γ, then

$$\gamma = \beta - u \tag{11.14}$$

The angle γ is known as the *margin angle*. For a safe commutation,

$$\gamma > \omega t_q \tag{11.15}$$

where t_q is the turn-off time of thyristors. The waveforms for the commutation interval are shown in figure 11.6b. Figure 11.6b also shows the phase current waveforms. For phase A, the fundamental component is also shown in the figure. When overlap angle u is zero, the fundamental component leads v_{AN} by an angle β. The effect of the commutation overlap is to delay the fundamental component. The delay is approximately 0.5 u when the current waveform is assumed to be trapezoidal. Hence, in the presence of overlap, the fundamental component leads the induced voltage v_{AN} by $(\beta - 0.5\ u)$. By considering different intervals of operation, the waveforms shown in figure 11.8 can be drawn for the terminal voltage of phase A, v_{aN}. It differs from v_{AN} only during the overlap period as shown (assuming negligible drop across R_s). The fundamental component of the phase A current can also be assumed to lead v_{aN} by $(\beta - 0.5\ u)$. Hence the power factor angle of the machine ϕ is

$$\phi = \beta - 0.5\ u \text{ (leading)} \tag{11.16}$$

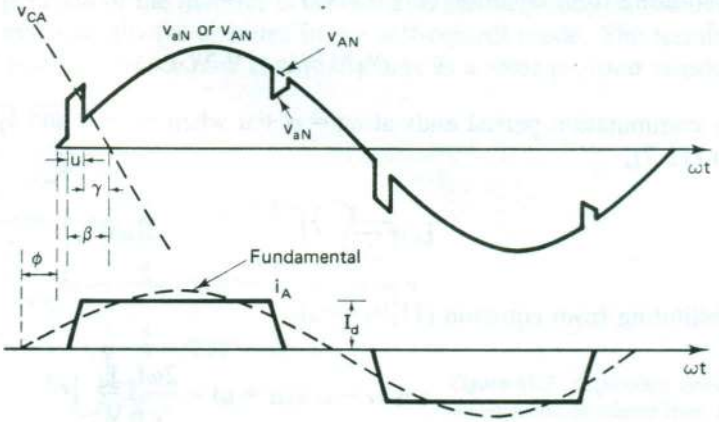


Figure 11.8 Motor terminal voltage and current.

For safe commutation, β must be greater than u (fig. 11.6b); hence, ϕ will always be positive. This shows that with load commutation, the machine always operates with a leading power factor.

The nature of variation of u, γ, and ϕ with I_s, when the machine operates at a fixed value of β equal to β_c and a constant flux (constant I'_m and variable I'_f) is shown in figure 11.10. At a constant flux the ratio (V/ω) is constant. From equation (11.13), when β is held constant, u depends only on I_d or I_s.

The phasor diagrams of the machine for motoring ($\beta < 90°$) and braking ($\beta > 90°$) operations with load commutation are shown in figure 11.9. Since the machine operates at a leading power factor, the torque angle δ' lies between 90° and 180° for motoring and between 180° and 270° for braking, as noted in section 10.3. In this range of δ', the conventional operation of a synchronous motor is steady-state unstable. But with self-control, the operation becomes stable. For example, an increase in δ' caused by a disturbance, with other parameters unchanged, will reduce the motor torque, causing a decrease in speed. Consequently, the induced voltage will decrease, increasing the motor current and torque, and restoring the balance between the motor and load torques.

In equation (11.13), ω depends on speed, and V depends on speed and I'_m. The current I'_m in turn depends on I_s and I'_f. Hence, for a given operating point characterized by given values of β, I_s (or I_d), I'_f, and speed, u has a fixed value. From

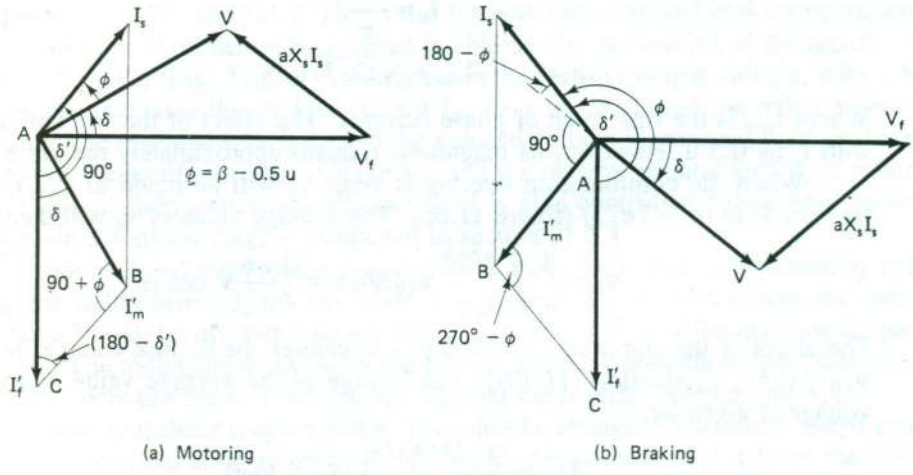


Figure 11.9 Phasor diagrams of synchronous motor fed by a load commutated current source inverter.

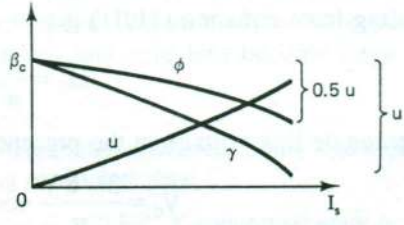


Figure 11.10 ϕ u and γ versus I_s curves for a constant β.

equation (11.16), the power factor angle ϕ has a unique value. In the phasor diagrams of figure 11.9, for given values of I_s, I_f', and ϕ, all sides and angles of triangle ABC have unique values. Therefore, the torque angle δ' (or δ) also has a unique value. Thus, for a given set of values of I_s, I_f', and speed, there is a unique value of δ' for a given β. This suggests that β can also be controlled indirectly by controlling δ'. Let us define a variable δ_o' as

$$\delta_o' = \delta' + 0.5 \text{ u} \tag{11.17}$$

When the commutation overlap is neglected, the armature current has a 6-step waveform shown in figure 11.3. When the firing pulses are generated using a rotor position encoder, as explained in section 11.1, then if the encoder is set to produce a constant torque angle δ_o', the torque angle will stay fixed at this value. When the commutation overlap cannot be neglected, as in a real situation, then the torque angle will be equal to δ_o' for a no-load operation ($I_s = 0$). As the load (or I_s) increases, the torque angle δ' will vary according to equation (11.17). The angle δ_o' is defined as the no-load torque angle. Note that the firing pulse to an odd-numbered thyristor of an inverter is delivered (fig. 8.14) when the direct axis makes an angle of $(\delta_o' + 60°)$ with respect to the axis of the respective phase (fig. 11.3).

For a 6-step current waveform, from equations (8.23) and (8.24)

$$I_s = \frac{\sqrt{6}}{\pi} I_d \tag{11.18}$$

$$I_{rms} = \sqrt{(2/3)}\, I_d \tag{11.19}$$

Where I_{rms} is the rms value of phase currents. The effect of the overlap is to phase shift I_s by 0.5 u. However, its magnitude remains approximately the same.

When the commutation overlap is zero, v_d will be equal to v_{AB} from $\omega t = (\alpha + \pi/3)$ to $(\alpha + 2\pi/3)$ (figure 11.6b). The average value of v_d will then be

$$V_d = \frac{3}{\pi} \int_{\alpha+\pi/3}^{\alpha+2\pi/3} v_{AB}\, d(\omega t) = \frac{3\sqrt{6}}{\pi} V \cos \alpha \tag{11.20}$$

The effect of the commutation overlap is to change the dc link voltage from v_{AB} to $v_{AB} - 0.5v_{AC}$ [equation (11.10)]. The change in the average value of the dc link voltage is given by

$$\begin{aligned} U_d &= \frac{3}{\pi} \int_{\alpha+\pi/3}^{\alpha+\pi/3+u} 0.5 v_{AC}\, d(\omega t) \\ &= \frac{3\sqrt{6}\, V}{2\pi} [\cos \alpha - \cos(\alpha + u)] \end{aligned} \tag{11.21}$$

Substituting from equation (11.11) gives

$$U_d = \frac{3}{\pi} (\omega L_c I_d) \tag{11.22}$$

The average dc link voltage in the presence of commutation overlap is

$$V_d = \frac{3\sqrt{6}\, V}{\pi} \cos \alpha - \frac{3}{\pi} (\omega L_c I_d) \tag{11.23}$$

The power input to the motor is

$$P_{in} = -V_d I_d \tag{11.24}$$

Power developed by the motor is

$$P_m = P_{in} - 3I_{rms}^2 R_s \tag{11.25}$$

Substituting from equations (11.24) and (11.19) gives

$$P_m = -V_d I_d - 2I_d^2 R_s$$

and developed torque is

$$T = \frac{P_m}{\omega_{ms}} = -\frac{1}{\omega_{ms}}[V_d I_d + 2I_d^2 R_s] \tag{11.26}$$

Also

$$P_m = 3VI_s \cos \phi - 3I_{rms}^2 R_s$$

or

$$T = \frac{3}{\omega_{ms}}[VI_s \cos \phi - I_{rms}^2 R_s) \tag{11.27}$$

Equations (11.26) and (11.27) are valid for both motoring and braking operations.

Speed control below base speed is obtained by the control of the rectifier output voltage V_{d1} (Fig. 11.6b). An increase in the rectifier output voltage, with β held constant, increases the dc link current I_d. As a result, I_s and the motor torque increase. The machine speeds up and the induced voltage V increases until the balance between the rectifier output voltage V_{d1} and the inverter counter emf V_d is restored. In a wound-field motor, usually the field is also controlled below base speed to maintain a constant flux, as explained in section 11.5.1.

For a wound-field motor, speed control above base speed is obtained by reducing the field current I_f'. A decrease in I_f' reduces I_m', which reduces the induced voltage V and the inverter counter emf V_d. Therefore, I_d and the machine torque increase. The motor speed increases until V has a value which allows the balance to be restored between the rectifier voltage V_{d1} and the inverter counter emf V_d.

In a permanent magnet motor, I_f' cannot be changed. Therefore, speed control above base speed is obtained by increasing β. An increase in β reduces the inverter counter emf V_d for a given V. The dc link current I_d increases, increasing torque. The motor speed increases until V is large enough to restore the balance between the inverter counter emf V_d and the rectifier output voltage V_{d1}. The increase in β has an adverse effect on the power factor [equation (11.16)]. Furthermore, the machine terminal voltage also increases with speed. Because of the poor power factor and high terminal voltage, both copper loss and core loss become large and efficiency decreases.

Example 11.1

The synchronous motor of a self-controlled drive, fed from a load-commutated current source inverter, has the following name plate data:

5 MW, 3-phase, 6600 V, 6-pole, 50 HZ, Y-connected, unity power factor

The parameters are $X_s = 10\ \Omega$, subtransient reactance $= 1.8\ \Omega$, and negligible R_s, core loss, friction, and windage. The field is controlled to maintain a constant flux below base speed and the rated terminal voltage above base speed. The machine is operated at a constant commutation lead angle of 50°.

1. Calculate the margin angle, power factor, developed power, and torque for machine operation at the rated armature current (rms) and speed.
2. Repeat 1, for a speed of 1600 rpm.
3. Calculate δ_o' for 1.
4. Calculate the dc link voltage for 1.

Solution:

$$V = \frac{6600}{\sqrt{3}} = 3810.5\ V$$

At the rated frequency, $\omega = 2\pi \times 50 = 314$ rad/sec.

$$\text{Rated speed} = \frac{120f}{p}\ \text{rpm} = \frac{120 \times 50}{6} = 1000\ \text{rpm}$$

$$= 104.7\ \text{rad/sec.}$$

$$\text{Rated armature current} = \frac{5 \times 10^6}{\sqrt{3} \times 6600} = 437.4\ A$$

When fed from an inverter, for an rms motor current of 437.7 A, from equations (11.18) and (11.19),

$$I_d = \sqrt{\frac{3}{2}} I_{rms} = \sqrt{\frac{3}{2}} \times 437.4 = 536\ A$$

$$I_s = \frac{\sqrt{6}}{\pi} I_d = \frac{\sqrt{6}}{\pi} \times 536 = 418\ A$$

1. Since the machine is operating at a constant flux, at rated speed the terminal voltage will have the rated value.

From equations (11.13) and (11.14)

$$\cos\gamma - \cos(50°) = \frac{2 \times 1.8}{\sqrt{6} \times 3810.5} \times 536$$

which gives $\gamma = 32°$.
Now $u = \beta - \gamma = 50° - 32° = 18°$

$$\phi = \beta - 0.5\ u = 50° - 9° = 41°$$

$$\text{Power factor} = \cos 41° = 0.75$$

$$\text{Developed power} = 3VI_s \cos\phi$$

$$= 3 \times 3810.5 \times 418 \times 0.75 = 3.58\ MW$$

$$\text{Torque} = \frac{3.58 \times 10^6}{104.7} = 3.4 \times 10^4\ \text{N-m}$$

2. Above the rated speed, the machine operates at the rated terminal voltage; hence

$$V = 3810.5 \text{ V}, \omega L_c = 1.6 \times 1.8 = 2.88\ \Omega, I_d = 536 \text{ A}$$

From equation (11.13),

$$\cos \gamma - \cos 50^\circ = \frac{2 \times 2.88}{\sqrt{6} \times 3810.5} \times 536$$

which gives $\gamma = 13^\circ$

$$u = \beta - \gamma = 50^\circ - 13^\circ = 37^\circ$$

$$\phi = \beta - 0.5\ u = 50^\circ - 18.5^\circ = 31.5^\circ$$

Power factor = $\cos 31.5^\circ = 0.85$

$$P_m = 3VI_s \cos \phi = 3 \times 3810.5 \times 418 \times 0.85 = 4.06 \text{ MW}$$

$$T = \frac{4.06 \times 10^6}{104.7 \times 1.6} = 2.43 \times 10^4 \text{ N-m}$$

3. $$I'_m = \frac{V}{X_s} = \frac{3810.5}{10} = 381 \text{ A}$$

From figure 11.9,

$$I'^2_f = I'^2_m + I^2_s - 2I'_m I_s \cos(90 + \phi)$$
$$= 381^2 + 418^2 + 2 \times 381 \times 418 \sin 41^\circ$$

or

$$I'_f = 727 \text{ A}$$

From triangle ABC,

$$\frac{\sin \delta}{I_s} = \frac{\sin(90 + \phi)}{I'_f} \quad \text{or} \quad \sin \delta = \frac{I_s}{I_f} \cos \phi = \frac{418}{727} \cos 41^\circ = 0.43 \quad \text{or} \quad \delta = 25.7^\circ$$

Now $\delta' = \delta + 90^\circ + \phi = 25.7^\circ + 90^\circ + 41^\circ = 156.7^\circ$ and from equation (11.17), $\delta'_o = \delta' + 0.5\ u = 156.7^\circ + 9^\circ = 165.7^\circ$

4. $\alpha = 180^\circ - \beta = 180^\circ - 50^\circ = 130^\circ$

From equation (11.23),

$$V_d = \frac{3\sqrt{6}}{\pi} \times 3810.5 \cos 130^\circ - \frac{3}{\pi}(1.8 \times 536)$$
$$= -6650 \text{ V}$$

11.3.2 Power Factor Optimization

In a motoring operation, the power factor is maximum when β is minimum or α is maximum. At a minimum β, the inverter output voltage is maximum. Consequently, the power transferred from the dc link to the machine for a given value of I_d is maximized. This in turn maximizes the machine torque for a given I_s.

A minimum value of β is obtained when the margin angle is chosen just sufficient (γ_{min}) to ensure safe commutation. Then equation (11.15) can be written as

$$\gamma_{min} = K_q \omega t_q \tag{11.28}$$

where K_q is a safety factor
Now the minimum value of β is

$$\beta_{min} = u + \gamma_{min} \tag{11.29}$$

and maximum power factor from equations (11.16) and (11.29) is

$$(PF)_{max} = \cos(\beta_{min} - 0.5\ u) = \cos(\gamma_{min} + 0.5\ u) \tag{11.30}$$

From equations (11.13) and (11.29),

$$\cos(u + \gamma_{min}) = \cos\ \gamma_{min} - \frac{2\omega L_c I_d}{\sqrt{6}\, V} \tag{11.31}$$

In equation (11.31), ω depends on speed, and V depends on speed and I'_m. The current I'_m in turn depends on I_s and I'_f. Hence, for a given operating point, determined by values of I_s (or I_d), I'_f, and speed, the overlap angle u can be evaluated from equation (11.31). Then β_{min} can be obtained from equation (11.29).

In the braking operation, the power factor, and consequently the regenerated power and braking torque for a given I_d are maximized when $\beta = 180°$ (or $\alpha = 0°$). Then the maximum power factor is

$$(PF)_{max} = |\cos(180 - 0.5\ u)| = \cos\ 0.5\ u \tag{11.32}$$

11.3.3 Generation of Inverter Firing Signals

The reference signals for the control of the commutation lead angle β can be generated by using either machine terminal voltages or a rotor position encoder.

Figure 11.8 shows the induced voltage v_{AN} and the terminal voltage v_{aN} for phase A. The induced voltage v_{AN} which is the sum of a sinusoidal voltage V_f and a nonsinusoidal drop across the synchronous reactance, is nonsinusoidal. The terminal voltage v_{aN} is more distorted because of commutation transients. Any attempt to filter it causes a frequency dependent phase shift, for which it is difficult to compensate. However, one can assume the zero crossing points of the terminal voltage are approximately the same as for the induced voltage. These zero crossing points can be used as reference points for generating inverter firing signals. This is achieved by converting terminal voltages into rectangular voltages using zero crossing detectors. A digital control circuit using counters then generates firing signals by using the zero crossing points of the rectangular waveforms as reference points.[14]

As explained in section 11.3.1, for given values of I_s, I'_f, and ω, there is a unique value of δ' or δ for each value of β. Consequently, β can be controlled indirectly by controlling δ' or δ using a rotor position encoder as explained in section 11.1 for a current source inverter without commutation overlap. The effect of commutation overlap is to delay the fundamental component I_s by 0.5 u. Hence, the odd-numbered thyristors of the inverter of figure 11.6 should be fired when the direct axis is $\delta' + 60° + 0.5$ u or $\delta'_o + 60°$ behind the respective armature phase axis.

The rotor position encoder arrangement is generally preferred because of simplicity. However, an encoder is not suitable for contaminated environments and high-speed drives.

11.3.4 Inverter Control Strategies

A number of approaches are possible for inverter control. A few commonly used control strategies are described here.

Constant Margin Angle Control: The operation of the inverter at the minimum safe value of the margin angle gives the highest power factor and the maximum torque per ampere of the armature current, thus allowing the most efficient use of both the inverter and motor. The exact implementation of this control strategy is not possible because the minimum value of the commutation lead angle cannot be accurately predicted. It is therefore implemented only approximately. Even then it requires complex control. The implementation of this control strategy using rotor position encoder and terminal voltage sensing methods is described in section 11.7.

Operation at a Constant Commutation Lead Angle: A simple control is obtained when the inverter operates at a constant β. From equations (11.29) and (11.31),

$$\cos \beta_{min} = \cos \gamma_{min} - K\left(\frac{I_d}{I'_m}\right) \tag{11.33}$$

where K is a constant.

Equation (11.33) shows that β_{min} has the highest value when the ratio (I_d/I'_m) has a maximum value. When the operation is limited to the base speed, the ratio (I_d/I'_m) is maximum when I_d or torque is maximum. If the inverter is operated at a constant commutation lead angle β_c, which is equal to the value of β_{min} at the maximum value of I_d, then commutation is ensured for all operating points. Now the machine will still operate at the highest power factor at the maximum torque; however, the power factor and efficiency will be lower for low torques (due to a decrease in u) compared to the control at a constant margin angle. A comparison of figures 11.9a and 11.11a, which are drawn for the same value of β but different value of I_s, confirms this behavior. Figures 11.11a and 11.11c compare the power factors for this scheme and the constant margin angle control for the same I_s.

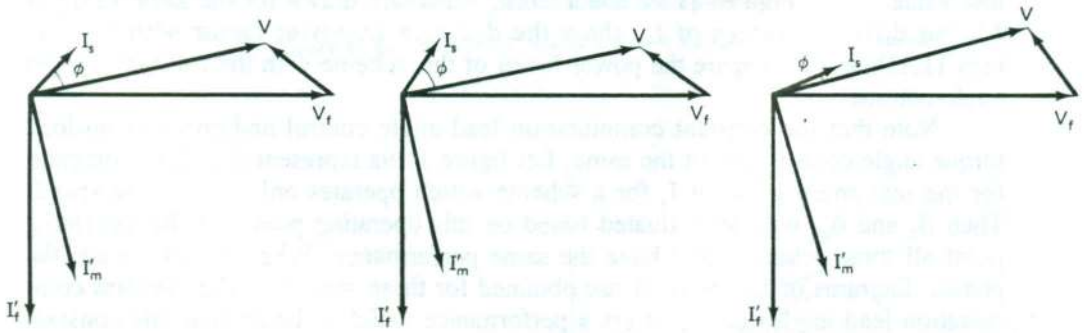


Figure 11.11 Variation of power factor for different control schemes.

When the machine also operates above base speed at a constant terminal voltage, β_{min} has a maximum value when the machine operates at maximum I_d and highest speed (equation (11.31)). Hence, β_c is chosen based on this point. Now for lower values of speed and I_d, the machine will operate at a lower power factor and efficiency compared to the constant margin angle control.

Operation at a Constant No-Load Torque Angle δ'_{oc}: In this scheme, the drive is operated at a constant no-load torque angle δ'_{oc}. The value of δ'_{oc} is chosen to provide enough commutation margin angle at the highest value of I_d. This will ensure safe commutation for lower values of I_d. As a specific example, let us consider how to evaluate δ'_{oc}. Assume that the machine operates at a constant flux. Then I'_m will be a known constant. Now β_{min} can be obtained for the highest value of I_d from equation (11.33). Since $u = \beta_{min} - \gamma_{min}$, ϕ can be evaluated from equation (11.16). Now from triangle ABC of figure 11.9a,

$$I'_f = \sqrt{I_s^2 + I_m'^2 - 2I_s I'_m \cos(90 + \phi)}$$
$$= \sqrt{I_s^2 + I_m'^2 + 2I_s I'_m \sin\phi} \tag{11.34}$$

Also

$$\frac{I'_m}{\sin(180 - \delta')} = \frac{I'_f}{\sin(90 + \phi)}$$

or

$$\sin\delta' = \left(\frac{I'_m}{I'_f}\right)\cos\phi \tag{11.35}$$

Since I'_m and ϕ are already known, I'_f for the maximum value of I_d (or I_s) is obtained from equation (11.34), $\sin\delta'$ is obtained from equation (11.35), and δ'_{oc} is then calculated from equation (11.17). When the drive also operates above base speed, at a constant terminal voltage, then I'_m (at the maximum I_d) will have a minimum value at the highest speed. Hence, δ'_{oc} must be evaluated for the highest values of I_d and speed. Also in this scheme the drive has a low power factor and efficiency for low values of I_d. Figures 11.9a and 11.11b, which are drawn for the same value of δ'_{oc} but different values of I_s, show the decrease in power factor with I_s. Figures 11.11b and c compare the power factor of this scheme with the constant margin angle control.

Note that the constant commutation lead angle control and constant no-load torque angle control are not the same. Let figure 11.9a represent the phasor diagram for the maximum value of I_s for a scheme which operates only up to base speed. Then β_c and δ'_{oc} will be evaluated based on this operating point. At this operating point all three schemes will have the same performance. When I_s is reduced, the phasor diagrams of figure 11.11 are obtained for these schemes. The constant commutation lead angle control offers a performance which is better than the constant no-load torque angle control, but inferior to constant margin angle control. The constant commutation lead angle control and constant no-load torque angle control are reported to give increased torque pulsations at low speeds.

Figures 11.12a, b, and c show the variation of β, ϕ, and δ' for the three control schemes for operation below base speed, when they are designed to provide the same margin angle at the highest current. Note that from equation (11.13) u is also a function of β. The u curve of figure 11.12a, which is drawn for the constant margin angle control, is not applicable to the other two control schemes.

There are a few other control schemes which essentially provide a compromise between the complexity and excellent performance of the constant margin angle control and the simplicity and poor performance of the constant commutation lead angle control or constant no-load torque angle control. Two of these are described next.

Commutation Lead Angle Varying Linearly with I_d: Here the value of the commutation lead angle is given by

$$\beta = \gamma_{min} + K_\beta I_d \tag{11.36}$$

K_β is suitably chosen to make enough margin angle available at all operating points, as shown in figure 11.13.

No-Load Torque Angle Varying Linearly with I_d: Here the no-load torque angle is given by

$$\delta'_o = \delta'_{om} + K_{\delta'} I_d \tag{11.37}$$

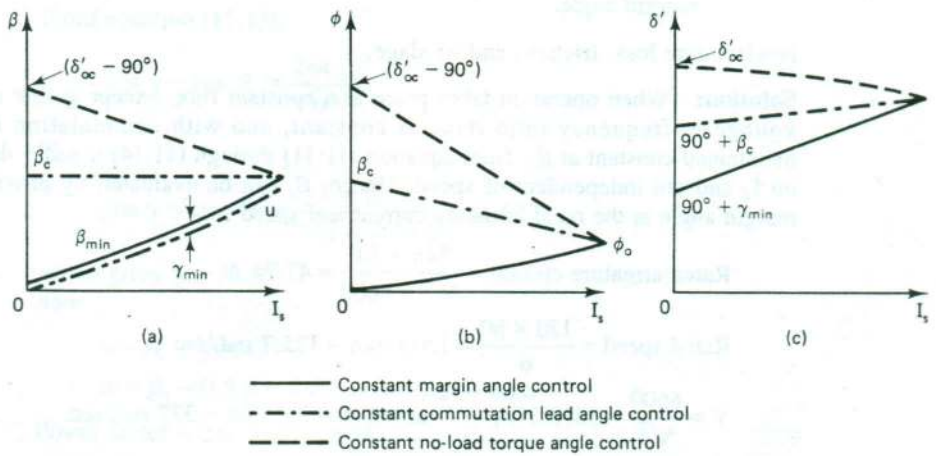


Figure 11.12 β, ϕ and δ' versus I_s curves for various control techniques.

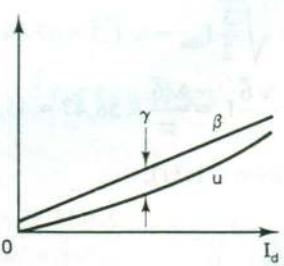


Figure 11.13 β varying linearly with I_d.

where δ'_{om} is the value of the torque angle which provides a margin of γ_{min} for $I_d = 0$. When $I_d = 0$ (or $I_s = 0$), V is in phase with V_f (figure 11.9a), $u = 0$ and from equations (11.16) and (11.29) $\phi = \gamma_{min}$. Hence from figure 11.9a,

$$\delta'_{om} = 90° + \gamma_{min} \tag{11.38}$$

The constant $K_{\delta'}$ in equation (11.37) is chosen to provide a sufficient margin angle for all operating points.

Example 11.2

A self-controlled load-commutated current source inverter drive employs a motor with the following name plate data: 546 kW, 3-phase, 6600 V, 60 Hz, 6 pole, Y-connected, $L_c = 50$ mH, $X_s = 81\ \Omega$, Power factor = 1.0, R_s = negligible

The field current is controlled to keep a constant flux up to base speed. Constant commutation lead angle control is employed. The drive is not required to operate above base speed. The maximum motor current (rms) is not to exceed its rated value.

1. Calculate β_c so that a minimum margin angle of 15° is available for all operating points.
2. With the value of β_c fixed as in 1, calculate the power factor, torque, and dc link voltage for the rated speed and the rated armature current.
3. Repeat 2 for 25 percent of the rated armature current. Also calculate the margin angle.

Neglect core loss, friction, and windage.

Solution: When operation takes place at a constant flux, except at low speeds, the voltage-to-frequency ratio remains constant, and with commutation lead angle maintained constant at β_c, from equation (11.11) through (11.14) u and γ depend only on I_d and are independent of speed. Hence, β_c can be evaluated by providing a 15° margin angle at the rated armature current and speed

$$\text{Rated armature current} = \frac{546 \times 10^3}{\sqrt{3} \times 6600} = 47.74 \text{ A}$$

$$\text{Rated speed} = \frac{120 \times 60}{6} = 1200 \text{ rpm} = 125.7 \text{ rad/sec.}$$

$$V = \frac{6600}{\sqrt{3}} = 3810.5 \text{ V}, \qquad \omega = 2\pi f = 2\pi \times 60 = 377 \text{ rad/sec.}$$

$$\omega L_c = 377 \times 50 \times 10^{-3} = 18.85\ \Omega$$

1. When fed from an inverter, $I_{rms} = 47.74$ A.

From equations (11.19) and (11.18),

$$I_d = \sqrt{\frac{3}{2}} I_{rms} = \sqrt{1.5} \times 47.74 = 58.47 \text{ A}$$

$$I_s = \frac{\sqrt{6}}{\pi} I_d = \frac{\sqrt{6}}{\pi} \times 58.47 = 45.6 \text{ A}$$

From equation (11.31),

$$\cos \beta_c = \cos 15° - \frac{2 \times 18.85 \times 58.47}{\sqrt{6} \times 3810.5}$$

or

$$\beta_c = 43°.$$

2. $u = \beta_c - \gamma_{min} = 43° - 15° = 28°$

$$\phi = \beta_c - 0.5\ u = 43° - 14° = 29°$$

Power factor = cos 29° = 0.87

$$P_m = 3VI_s \cos\phi = 3 \times 3810.5 \times 45.6 \times 0.87 = 453.5 \text{ kW}$$

$$T = \frac{455.5 \times 10^3}{125.7} = 3608 \text{ N-m}$$

$$\alpha = 180° - \beta_c = 180° - 43° = 137°$$

From equation (11.23),

$$V_d = \frac{3\sqrt{6}}{\pi} \times 3810.5 \cos 137° - \frac{3}{\pi}(18.85 \times 58.47)$$

$$= -7571 \text{ V}$$

3. $I_s = 0.25 \times 45.6 = 11.4$ A

$$I_d = 0.25 \times 58.47 = 14.62 \text{ A}$$

From equation (11.13),

$$\cos\gamma - \cos\beta_c = \frac{2\omega L_c}{\sqrt{6}V} I_d$$

or

$$\cos\gamma = \cos 43° + \frac{2 \times 18.85}{\sqrt{6} \times 3810.5} \times 14.62$$

which gives $\gamma = 37.8°$

now

$$u = \beta_c - \gamma = 43° - 37.8° = 5.2°$$

$$\phi = \beta_c - 0.5\ u = 43° - 0.5 \times 5.2° = 40.4°$$

Power factor = cos 40.4° = 0.76

$$P_m = 3 \times 3810.5 \times 11.4 \times 0.76 = 99 \text{ kW}$$

$$T = \frac{99 \times 10^3}{125.7} = 787.6 \text{ N-m}$$

From equation (11.23)

$$V_d = \frac{3\sqrt{6}}{\pi} \times 3810.5 \cos 137° - \frac{3}{\pi}(18.85 \times 14.62)$$

$$= -6782 \text{ V}$$

Example 11.3

The drive of example 11.2 now also operates above base speed. The maximum speed is 1800 rpm and the armature current is limited to the rated value.

1. Calculate β_c so that a minimum margin angle of 15° is available for all operating points.
2. With the value of β_c fixed as in 1, calculate the power factor, torque, and dc link voltage at half the rated speed and 25 percent of the rated current. What is the margin angle?

Solution: Now u has the highest value at the rated armature current and 1800 rpm. Hence, β_c should be chosen for these values of current and speed. From example 11.2, at the rated rms current (47.74 A):

$$I_s = 45.6 \text{ A} \quad \text{and} \quad I_d = 58.47 \text{ A}$$

At 1800 rpm, which is above base speed,

$$V = 3810.5 \text{ V}$$

$$\omega = \frac{1800}{1200} \times 377 = 565.5 \text{ rad/sec.}$$

$$\omega L_c = 1.5 \times 18.85 = 28.28 \text{ ohms}$$

From equation (11.31),

$$\cos \beta_c = \cos 15° - \frac{2 \times 28.28 \times 58.47}{\sqrt{6} \times 3810.5}$$

or

$$\beta_c = 52.3°$$

and

$$\alpha = 180° - \beta_c = 127.7°$$

2. $I_d = 0.25 \times 58.47 = 14.62$ A, $\quad I_s = 0.25 \times 45.6 = 11.4$ A

At half rated speed:

$$V = 0.5 \times 3810.5 = 1905 \text{ V}, \qquad \omega L_c = 0.5 \times 18.85 = 9.42 \ \Omega$$

From equation (11.13),

$$\cos \gamma - \cos \beta_c = \frac{2\omega L_c}{\sqrt{6}\, V} I_d$$

$$\cos \gamma = \cos 52.3° + \frac{2 \times 9.42}{\sqrt{6} \times 1905} \times 14.62$$

or

$$\gamma = 47.9°$$

$$u = \beta_c - \gamma = 52.3° - 47.9° = 4.4°$$

$$\phi = \beta_c - 0.5\, u = 50.1°$$

Power factor = 0.64

$$P_m = 3 \times 1905 \times 11.4 \times 0.64 = 41 \text{ kW}$$

$$T = \frac{41 \times 10^3}{0.5 \times 125.7} = 652 \text{ N-m}$$

From equation (11.23),

$$V_d = \frac{3\sqrt{6}}{\pi} \times 1905 \cos 127.7 - \frac{3}{\pi}(9.4 \times 14.62)$$

$$= -2856 \text{ V}$$

Example 11.4

The drive of example 11.2 is to be controlled by constant no-load torque angle control.

1. Calculate δ'_{oc} so that a minimum margin angle of 15° is available for all operating points.
2. Calculate the power factor, torque, and dc link voltage for rated speed and 25 percent of the rated armature current with δ'_{oc} set as in 1.
3. Calculate the power factor, torque, and dc link voltage for rated speed and 25 percent of the rated armature current, when constant margin angle control is employed.

Solution: From example 11.2 at the rated speed and armature current,

$$\phi = 29°, \qquad I_s = 45.6 \text{ A}, \qquad I_d = 58.47 \text{ A}, \qquad u = 28°$$

$$I'_m = \frac{V}{X_s} = \frac{3810.5}{81} = 47 \text{ A (constant)}$$

From equation (11.34),

$$I'^2_f = I_s^2 + I'^2_m + 2I_s I'_m \sin \phi$$

$$= 45.6^2 + 47^2 + 2 \times 45.6 \times 47 \times \sin 29°$$

or

$$I'_f = 79.8 \text{ A}$$

From equation (11.35),

$$\sin \delta' = \left(\frac{I'_m}{I'_f}\right) \cos \phi = \left(\frac{47}{79.8}\right) \cos 29° = 0.515$$

or

$$\delta' = 31° \quad \text{or} \quad 149°$$

Since the machine operates at a leading power factor, the feasible value of δ' is 149°. Therefore,

$$\delta'_{oc} = \delta' + 0.5 \text{ u}$$

$$= 149° + 14° = 163° \text{ (electrical) or } 54.3° \text{ (mechanical)}$$

2. This part can only be solved by iteration using the following sequence of steps:
   (a) Assume a value for β and obtain u from equation (11.13) and calculate ϕ.
   (b) Obtain I'_f from equation (11.34) and δ' from equation (11.35).
   (c) Calculate δ'_{oc} and compare it with its actual value. If different, go back to step (a) and try another value of β.

   The iteration for $\beta = 69°$, which yields δ'_{oc} — close to the actual value, is given here.

At 25 percent rated armature current: $I_d = 14.62$, $I_s = 11.4$.
From equation (11.13),

$$\cos(\beta - u) = \frac{2\omega L_c}{\sqrt{6}\,V} I_d + \cos\beta$$

or

$$\cos(69° - u) = \frac{2 \times 18.85}{\sqrt{6} \times 3810.5} \cdot 14.62 + \cos 69°$$

which gives $u = 3.67°$

$$\phi = \beta - 0.5\ u = 67.2°$$

From equation (11.34),

$$I_f'^2 = 11.4^2 + 47^2 + 2 \times 11.4 \times 47 \sin 67.2°$$

or

$$I_f' = 57.68 \text{ A}$$

From equation (11.35),

$$\sin\delta' = \frac{47}{57.68} \cos 67.2°$$

thus,

$$\delta' = 161.6°$$

$$\delta_{oc}' = \delta' + 0.5\ u = 163.4°$$

which is nearly equal to the actual value.
Now $P_m = 3 \times 3810.5 \times 11.4 \times \cos 67.2° = 50.5$ kW

$$T = \frac{50.5 \times 10^3}{125.7} = 401.8 \text{ N-m}$$

Power factor = $\cos 67.2° = 0.39$

From equation (11.23),

$$V_d = \frac{3\sqrt{6}}{\pi} \times 3810.5 \times \cos 111° - 3 \times 18.85 \times 14.62 \div \pi$$

$$= -3457 \text{ V}.$$

3. When constant margin angle control is used, for all operating points $\gamma = \gamma_{min} = 15°$

From equation (11.31),

$$\cos(u + 15°) = \cos 15° - \frac{2 \times 18.85 \times 14.62}{\sqrt{6} \times 3810.5}$$

or

$$u = 9.92°, \qquad \beta = 24.92°$$

and

$$\phi = 24.92° - 0.5 \times 9.92° = 19.96°$$

Power factor = cos 19.96° = 0.94

$$P_m = 3 \times 3810.5 \times 11.4 \times 0.94 = 122.50 \text{ kW}$$

$$T = \frac{122.5 \times 10^3}{125.7} = 974.5 \text{ N-m}$$

$$\alpha = 180° - 24.92° = 155°$$

From equation (11.23),

$$V_d = \frac{3\sqrt{6}}{\pi} \times 3810.5 \cos 155° - \frac{3 \times 18.85 \times 14.62}{\pi}$$

$$= -8341 \text{ V}$$

Next we compare the values of T and the power factor for the three control schemes for 25 percent of the rated armature current:

	Constant $\beta = \beta_c$	Constant $\delta'_o = \delta'_{oc}$	Constant Margin Angle
T(N-m)	787.6	401.8	974.5
Power factor	0.76	0.39	0.94

The superiority of constant margin angle control is demonstrated by these results.

11.3.5 Merits and Disadvantages of Load Commutation

The load commutated current source inverter has the following merits and disadvantages:

1. Compared to the forced commutated current source inverter of figure 8.15, the load commutated inverter of figure 11.6 uses 6 capacitors and 6 diodes less. Consequently, a load commutated inverter has lower cost, weight, volume, and losses. Hence, synchronous motor drives with load commutated inverters are widely used in high-power applications.
2. A forced commutated current source inverter has a low frequency range. The attainable frequency range of a load commutated inverter depends on the commutating inductance and the turn-off time of the inverter thyristors. With converter grade thyristors and even with motors with somewhat larger than the usual value of commutating inductances, operation at up to several hundred Hertz is possible.[8] When operating above base speed at a constant induced voltage V, the overlap angle increases with frequency, thus requiring operation at a larger β. But, because of the large value of u, the power factor does not become too low [equation (11.16)].
3. With permanent magnet motors, the operation above base speed is obtained at a large β; consequently, the motor operates at a low power factor and efficiency.

4. Load commutation is caused by induced voltages. At very low speeds, the induced voltages are too small to provide satisfactory commutation. Hence, for low speeds — typically below 10 percent of base speed — load commutation cannot be used.

11.3.6 Commutation at Low Speeds

As mentioned in the previous section, load commutation is employed only for speeds above 10 percent of base speed. Hence, for lower speeds, a pulsed mode of operation, also known as the link current interruption method, is employed.

In the inverter of figure 11.6, a cycle of operation consists of six 60° intervals. In these invervals, thyristors conduct dc link current in pairs of two in the sequence T_1T_6, T_1T_2, T_2T_3, T_3T_4, T_4T_5, and T_5T_6. Somewhat similar operation of the inverter is obtained when each time the current is to be transferred from one pair to another, I_d is forced to zero by making the firing angle of the line side converter close to 180°. Consequently, thyristors in the outgoing pair turn off due to the lack of current. Now the firing angle of the line side converter is brought back to the original value and the incoming pair of inverter thyristors is fired to establish the flow of I_d through it. Figure 11.14 shows the waveforms of the dc link current and machine phase currents and voltages. The six intervals of operation and thyristor pairs under

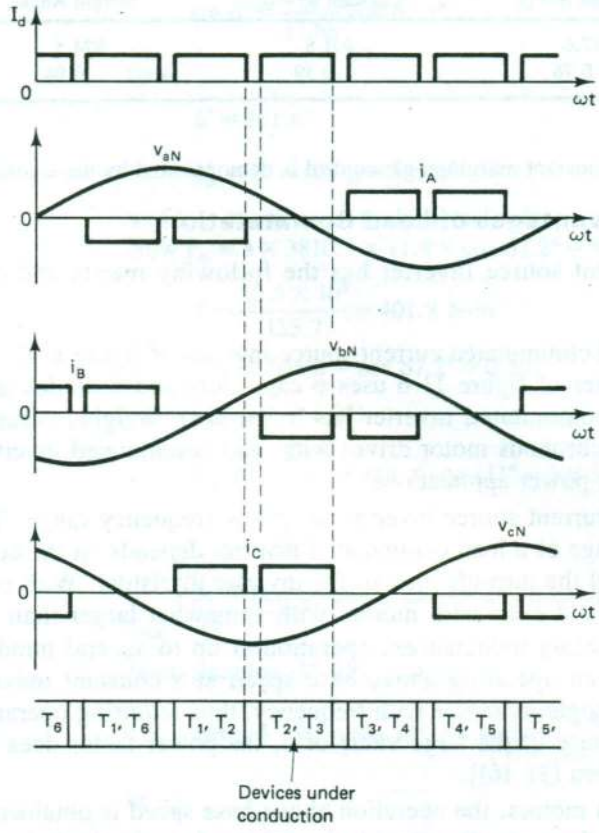


Figure 11.14 Pulsed mode of operation of the drive of Fig. 11.6.

conduction are also shown. The phase current pulse is less than 120° wide. The angular gap between two adjacent pulses depends on the time required by the line side converter to force I_d to 0 and to reestablish it. Because of a large value of L_d, a few cycles of the ac supply are required to accomplish this change in I_d. However, the angular gap remains small because of the low frequency of the inverter operation.

The effect of the angular gap is to reduce the motor current and torque. The gap can be substantially reduced by connecting a freewheeling thyristor across the dc link inductor, as shown by dotted lines in figure 11.6. At the time of commutation, the current I_d is transferred to the freewheeling thyristor, instead of being forced to zero. The operation is explained with the help of figure 11.15. At such a low speed, the line side converter output voltage is low. Consequently it operates at a firing angle slightly less than 90°. The dc link voltage v_{dl} is shown in the figure. When commutation is desired, a single firing pulse is diverted from the converter to the freewheeling thyristor. This causes v_{dl} to become negative for a duration of 90° (around 5 msec. for a 50 Hz supply). Consequently, the current is transferred to the freewheeling thyristor and the outgoing inverter thyristor pair is turned off. Now the incoming inverter thyristor pair is fired. The current I_d is transferred to this pair when the next firing pulse is given to the line side converter.

When the rotor position sensor consists of a rotor position encoder, the firing pulses required for the foregoing operation can be obtained from the encoder. At approximately 10 percent of base speed, the operation is shifted from the pulsed mode to normal operation. A terminal voltage sensor cannot be used to generate the firing pulses at standstill. Hence, when a terminal voltage sensor is employed, at standstill, the firing pulses are obtained from an independent oscillator. Soon after the motor starts, the independent oscillator is replaced by the terminal voltage sen-

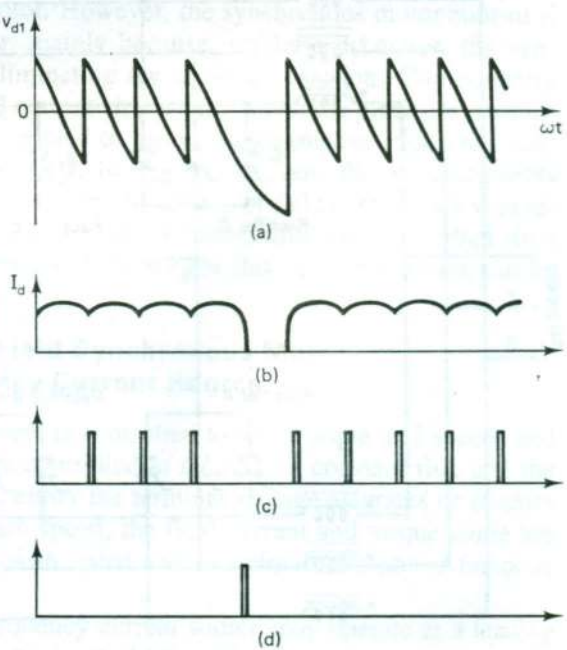


Figure 11.15 Pulsed mode of operation of the drive of Fig. 11.6 with a freewheeling thyristor. (a) v_{dl}, (b) I_d, (c) Firing pulses for converter thyristors, (d) Firing pulse for the freewheeling thyristor.

sor. When the speed reaches approximately 10 percent of base speed, the pulsed mode ceases and normal operation begins.

11.4 CYCLOCONVERTER WITH LOAD COMMUTATION

A cycloconverter feeding a synchronous motor is shown in figure 11.16. For commutation considerations, the synchronous motor is modeled in the same way as in figure 11.6. The cycloconverter consists of three dual converters A, B, and C, each consisting of two three-pulse controlled rectifiers. During the positive half-cycles of phase current i_A, i_B, and i_C, the machine draws currents from rectifiers A+, B+, and C+, respectively, and during the negative half-cycles, from rectifiers A−, B−, and C−, respectively. The rectifiers are controlled to produce the six-step current waveform of figure 11.6b. For this they conduct in pairs with a sequence (A+, B−), (A+, C−), (B+, C−), (B+, A−), (C+, A−), and (C+, B−). Load commutation is used mainly to obtain a changeover from one pair to another. For example, when pair (A+, C−) is conducting, phase A carries positive current and phase C carries negative current. These two pairs conduct together for 60° in a cycle of the machine induced voltage. During this conduction, the two rectifiers are controlled in the same way as a single rectifier. The next pair to conduct is (B+, C−). For this, current must transfer from rectifier A+ to rectifier B+. For this transfer, the induced voltage v_{AB} must be positive. Hence, the transfer is initiated at a suitable lead angle β. If at this instant thyristor A_1 of rectifier A+ has been in conduction, then thyris-

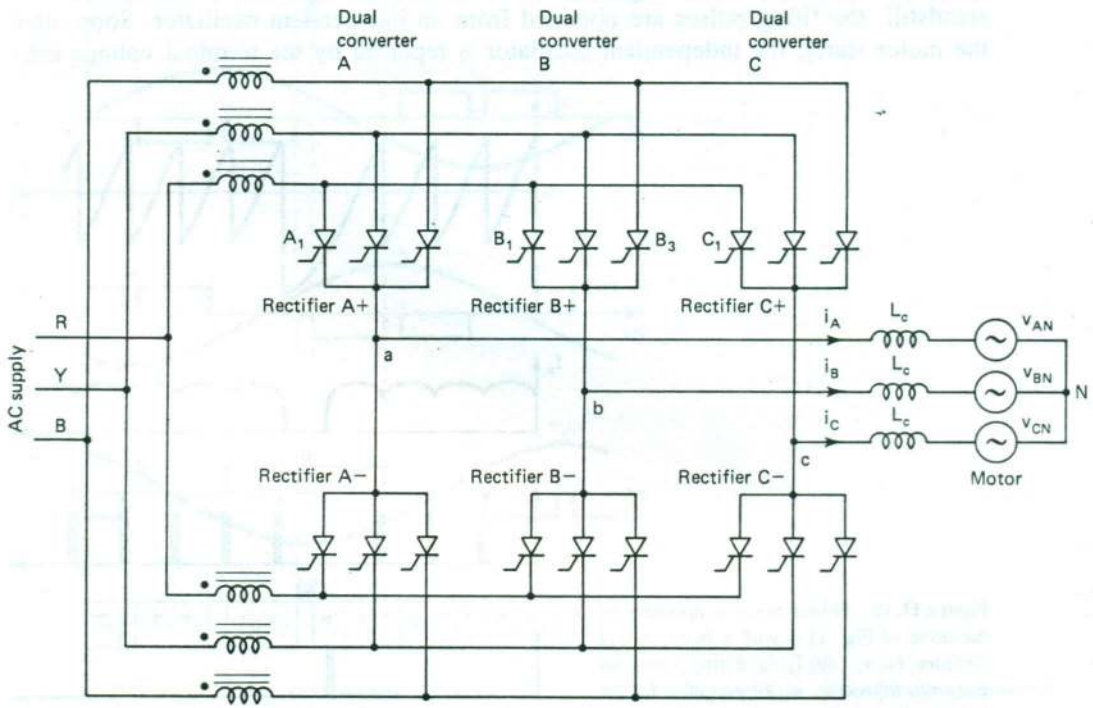


Figure 11.16 Load and source commutation in a cycloconverter.

tor B_1, which is connected to the same supply line as A_1, is fired at a lead angle β. The induced voltage v_{AB} transfers the current from A_1 to B_1 after an overlap angle u and subsequently subjects A_1 to a reverse bias of duration $(\beta - u)$ so that it can regain forward voltage blocking capability. The commutation process is identical to the current source inverter, and the analysis of commutation given in section 11.3.1 and description of sections 11.3.2 to 11.3.4 are applicable.

At low speeds, the machine induced voltages are too small to cause the transfer of current from one rectifier to another. Hence, the source voltage is used to effect this commutation. Let us again consider the transfer of current between the pairs $(A+, C-)$ and $(B+, C-)$, which requires transfer of conduction from rectifier A+ to rectifier B+. For this, the thyristor of rectifier B+, whose anode is at a higher potential than the conducting thyristor of rectifier A+ is fired. For example, if A_1 has been in conduction and V_{BR} is positive when the current transfer is desired, then B_3 is fired. Since voltage V_{BR} is positive, B_3 turns on and A_1 is commutated.

The main advantage of the load commutation in a cycloconverter is that the output frequency can be higher than the ac supply frequency, thus providing speed control over a wide range.

11.5 SYNCHRONOUS MOTOR CONTROL REQUIREMENTS

The basic control strategy is to operate the machine preferably at a constant flux below base speed and at reduced flux and constant terminal voltage above base speed. The same approach has been used for a rectifier controlled dc motor or an induction motor with variable frequency control. As explained in section 11.2, a variable frequency controlled synchronous motor with self-control essentially works as a commutatorless or a brushless dc motor. However, the synchronous motor control is not as simple as that of a dc motor, mainly because, unlike a dc motor, the synchronous motor operates without eliminating the armature reaction. Consequently the flux does not depend on the field current alone, but also on the armature current. Further, the additional requirements related to power factor impose additional constraints on the drive operation and control. In this section, how these requirements are achieved will be examined. Because of the flexibility of field control in a wound-field motor, the flux, power factor and speed can be simultaneously controlled. In a permanent magnet motor, along with speed, either the flux or power factor can be controlled.

11.5.1 Operation of a Wound-Field Synchronous Motor from a Variable Frequency Current Source

Below base speed, the armature current is controlled to vary torque and speed, and the field current and torque angle are controlled to maintain a constant flux and the desired power factor. At base speed, either the terminal voltage saturates or reaches the rated value. Therefore, above base speed, the field current and torque angle are controlled to vary the flux inversely with speed and to maintain the power factor at the desired value.

A drive fed from a variable frequency current source may operate at a leading power factor when load commutation is used otherwise it may operate at unity

power factor to maximize the use of the source and motor ratings and to minimize losses. Therefore, the motor will be examined for unity and leading power factor operations below and above base speed.

Operation at Unity Power Factor

Constant flux operation below base speed is obtained by keeping I'_m constant at the rated value. Thus, when I_s is increased to increase torque, I'_f and δ' are also controlled simultaneously to maintain I'_m constant and to obtain a unity power factor. Figure 11.17 shows the phasor diagrams for motoring and braking operations. From triangles ABC in these phasor diagrams,

$$I'_f = \sqrt{I_s^2 + I_M^2} \tag{11.39}$$

$$\delta = \pm\tan^{-1}(I_s/I_M)$$

$$\delta' = \pm[90 + |\delta|] = \pm[90 + \tan^{-1}(I_s/I_M)] \tag{11.40}$$

where a positive sign is for motoring, a negative sign is for braking, and I_M is the rated value of I'_m. I'_f and δ' as a function of I_s are shown in figure 11.17c. If I'_f and δ' are varied as a function of I_s as shown in figure 11.17c, then the machine will operate at a constant flux and unity power factor below base speed.

From equations (10.1) and (10.27),

$$T = \left(\frac{3}{2}pL_s\right)I_s I'_f \sin\delta' \tag{11.41}$$

where L_s is the synchronous inductance.

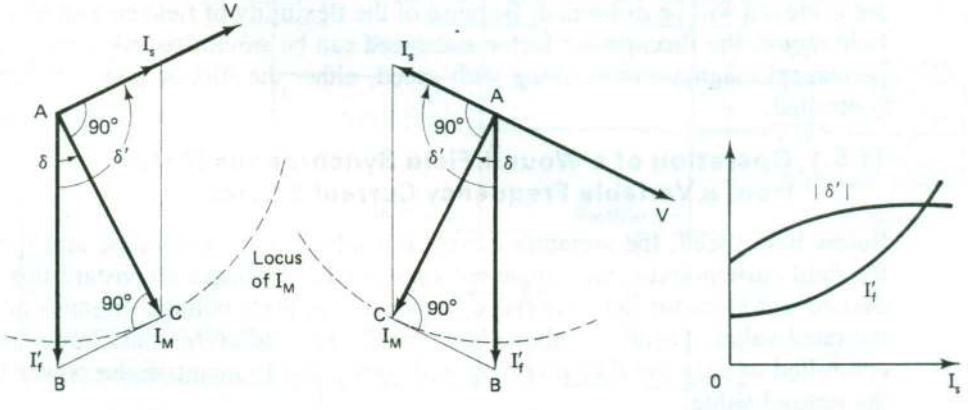
From triangles ABC,

$$I'_f \cos\delta = I_M$$

or

$$I'_f \sin\delta' = \pm I_M \tag{11.42}$$

where again a positive sign is for motoring and a negative sign is for braking.



(a) Motoring (b) Braking (c)

Figure 11.17 Unity power factor operation of a wound-field synchronous motor.

From equations (11.41) and (11.42),

$$T = \pm\left(\frac{3}{2} pL_s\right) I_s I_M \tag{11.43}$$

From equation (11.43), torque is proportional to armature current and it is independent of frequency or speed. When operating at the maximum permissible current, the torque has a constant value. Therefore, below the base speed, the machine operates in the constant torque mode.

Above base speed ($a > 1$), V is held constant. Hence, the magnetizing current must be reduced to (I_M/a). From equations (11.39) and (11.40),

$$I_f' = \sqrt{I_s^2 + (I_M/a)^2} \tag{11.44}$$

$$\delta' = \pm[90° + \tan^{-1}(aI_s/I_M)] \tag{11.45}$$

Now I_f' and δ' are functions of I_s and frequency. It is difficult to implement such a relationship. Now

$$P_m = 3VI_s \tag{11.46}$$

Since V is constant, the drive operates in a constant power mode.

The variation of T, V, I_f', and I_m' with per-unit speed "a" for the machine operation at a constant I_s is shown in figure 11.18.

Operation at a Leading Power Factor

The machine operates at a leading power factor when fed by a load commutated inverter or cycloconverter. Here, the machine operation is initially examined for a constant leading power factor. The results obtained are then extended to constant margin angle control, constant commutation lead angle control and constant no-load torque angle control. Note that constant margin angle control is required only during motoring, when the machine side converter operates as an inverter. For braking, β can be set at 180° for all operating points. Therefore, the analysis is considered only for motoring operation. The analysis for braking can be carried out in the same way.

From the triangle ABC of figure 11.9a,

$$I_f' = \sqrt{I_s^2 + I_M^2 - 2I_sI_M \cos(90 + \phi)}$$

$$= \sqrt{I_s^2 + I_M^2 + 2I_sI_M \sin\phi} \tag{11.47}$$

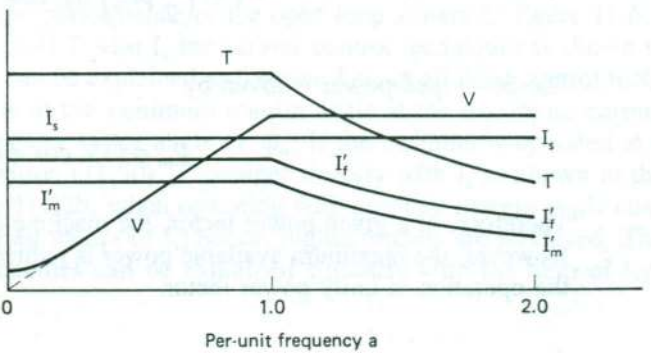


Figure 11.18 Unity power factor operation of a synchronous motor fed from a variable frequency source.

and

$$\frac{I_M}{\sin(180 - \delta')} = \frac{I_f'}{\sin(90 + \phi)}$$

or

$$I_f' \sin \delta' = I_M \cos \phi \tag{11.48}$$

or

$$\delta' = \sin^{-1}\left(\frac{I_M}{I_f'} \cos \phi\right) \tag{11.49}$$

For a given value of ϕ and constant I_M, I_f' is known as a function of I_s from equation (11.47). Then δ' is known as a function of I_f' from equation (11.49).

From equations (11.41) and (11.48),

$$T = \left(\frac{3}{2} p L_s\right) I_M I_s \cos \phi \tag{11.50}$$

At a constant power factor angle ϕ, torque is proportional to the armature current and power is proportional to speed. Consequently, the machine operates in a constant torque mode. The torque available at the maximum permissible current is reduced by a factor cos ϕ compared to the operation at unity power factor. This emphasizes the need for operation at the least value of ϕ—that is, with the constant margin angle control.

Expressions for the operation above base speed ($a > 1$) at a constant ϕ are obtained by substituting (I_M/a) for I_M in equations (11.47), (11.49), and (11.50) giving

$$I_f' = \sqrt{I_s^2 + (I_M/a)^2 + 2I_s(I_M/a) \sin \phi} \tag{11.51}$$

$$\delta' = \sin^{-1}\left(\frac{I_M}{aI_f'} \cos \phi\right) \tag{11.52}$$

$$T = \left(\frac{3}{2} p L_s\right) (I_M \cos \phi) \left(\frac{I_s}{a}\right) \tag{11.53}$$

The developed power is given by

$$P_m = (3V \cos \phi) I_s \tag{11.54}$$

Therefore, at a given power factor, the machine operates in a constant power mode. However, the maximum available power is reduced by a factor of cos ϕ compared to the operation at unity power factor.

Equations for the constant margin angle control below base speed are obtained by substituting from equation (11.30) into equations (11.47), (11.49), and (11.50), giving

$$I_f' = \sqrt{I_s^2 + I_M^2 + 2I_sI_M \sin(0.5u + \gamma_{min})} \tag{11.55}$$

$$\delta' = \sin^{-1}\left[\frac{I_M}{I_f'}\cos(0.5u + \gamma_{min})\right] \tag{11.56}$$

$$T = \left(\frac{3}{2}pL_s\right)I_MI_s\cos(0.5u + \gamma_{min}) \tag{11.57}$$

From equations (11.29) and (11.33),

$$\cos(u + \gamma_{min}) = \cos\gamma_{min} - K\left(\frac{I_d}{I_M}\right) \tag{11.58}$$

From equation (11.58), u is a function of I_d only. At low frequencies where flux is not constant u will also be a function of speed. Since ϕ is not constant, the torque does not vary linearly with T.

The equations for the constant margin angle control above base speed are obtained by substitution from equation (11.30) into equations (11.51), (11.52) and (11.54), giving

$$I_f' = \sqrt{I_s^2 + (I_M/a)^2 + 2I_s(I_M/a)\sin(0.5u + \gamma_{min})} \tag{11.59}$$

$$\delta' = \sin^{-1}\left[\frac{I_M}{aI_f'}\cos(0.5u + \gamma_{min})\right] \tag{11.60}$$

$$P_m = 3VI_s\cos(0.5u + \gamma_{min}) \tag{11.61}$$

Replacing I_M by (I_M/a) in equation (11.58) gives

$$\cos(u + \gamma_{min}) = \cos\gamma_{min} - K\left(\frac{aI_d}{I_M}\right) \tag{11.62}$$

According to equation (11.62), u increases with speed; consequently, the power factor and the power for a given I_s decrease with an increase in speed. Thus, maximum available power falls with an increase in speed.

The performance equations and control requirements for constant commutation lead angle control and constant no-load torque angle control can be similarly derived.

Now let us examine the performance of the open loop system of figure 11.6.

The nature of variation of T with I_s for various control techniques is shown in figure 11.19a. Their nature can be explained as follows. Let, in all three control techniques, the inverters operate at the minimum margin angle at the maximum current and let the corresponding power factor angle be ϕ_o. If the machine is operated at a constant ϕ_o, then from equation (11.50), T changes linearly with I_s as shown in the figure. According to figure 11.12b, when operating with constant margin angle control, ϕ is lower than ϕ_o for all values of I_s; hence, higher torques are produced. The curves for other control schemes can be explained similarly with the help of fig-

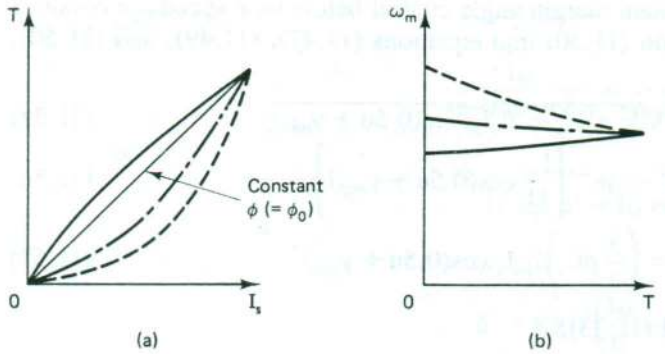


Figure 11.19 Performance curves for load commutated inverter-fed wound-field synchronous motor drive.

ure 11.12b. These characteristics are also applicable to the load commutated cycloconverter case. The nature of speed-torque characteristics for the three control schemes for a fixed value of the source side converter firing angle and for the operation below base speed is shown in figure 11.19b. Similar curves are obtained for a cycloconverter fed drive for a given value of the firing angle of the cycloconverter's dual converters. The difference between these curves at low torques occurs due to different values of β. In the case of constant margin angle control, β decreases as torque is reduced, but it remains constant for the constant commutation lead angle control (fig. 11.12a). For a larger β, a higher value of induced voltage and therefore of speed is required to obtain a balance between V_{d1} and V_d [(11.23)]. For the same reason the constant no-load torque angle control will give higher speeds than the constant commutation lead angle control at low torques. In the constant margin angle control, β increases with I_d; thus, the speed tends to increase with torque. On the other hand the voltage drops across the resistances of the dc link inductor and motor, and u increase with I_d and this tends to decrease the speed. Depending on the machine parameters, the speed may increase or decrease with an increase in torque.

Example 11.5

The drive of example 11.2 is now fed from a constant dc link voltage $V_d = -7571$ V. Calculate the speed and torque for $I_d = 58.47$ A, 43.85 A, 29.25 A, and 12.62 A for the constant margin angle control and the constant commutation lead angle control. For both the control schemes at $I_d = 58.47$ A, the drive operates at a minimum margin angle of 15°.

Solution: At $I_d = 58.47$ A, both the control schemes will have the same torque and speed. From example 11.2 these are 3608 N-m and 1200 rpm, respectively.

At the rated speed, V = 3810.5 V, $\omega = 377$ rad/sec.

$$I_M = \frac{V}{X_s} = \frac{3810.5}{81} = 47 \text{ A}, \qquad \omega L_c = 377 \times 50 \times 10^{-3} = 18.85\ \Omega$$

$$L_s = \frac{X_s}{\omega} = \frac{81}{377} = 0.215 \text{ H}$$

Since the motor operates at a constant flux, $(\omega L_c/V)$ and I'_m are constant at all speeds.

Constant Margin Angle Control: From equation (11.31),

$$\cos(u + 15°) = \cos 15° - \frac{2 \times 18.85}{\sqrt{6} \times 3810.5} I_d$$

$$= 0.966 - 0.004 I_d \qquad \text{(E11.1)}$$

From equation (11.23) for the per-unit speed a (actual speed/rated speed),

$$V_d = \frac{3\sqrt{6}}{\pi} Va \cos(180° - \beta) - \frac{3}{\pi} \omega L_c a I_d$$

$$= \left[\frac{3\sqrt{6}}{\pi} \times 3810.5 \cos(180° - \beta) - \frac{3}{\pi} \times 18.85 I_d\right] a$$

or

$$a = \frac{-7571}{8913 \cos(180° - \beta) - 18 I_d} \qquad \text{(E11.2)}$$

From equation (11.57),

$$T = \frac{3}{2} p L_s I_M I_s \cos \phi$$

$$= \frac{3}{2} \cdot 6 \times 0.215 \times 47 I_s \cos \phi$$

$$= 90.95 I_s \cos \phi \qquad \text{(E11.3)}$$

For I_d = 43.85 A,

$$I_s = \frac{\sqrt{6}}{\pi} \times I_d = \frac{\sqrt{6}}{\pi} \times 43.85 = 34.2 \text{ A}$$

From equation (E11.1),

$$\cos(u + 15°) = 0.966 - 0.004 \times 43.85$$

which gives $u = 22.76°$, $\beta = 22.76° + 15° = 37.76°$

Hence

$$\phi = \gamma_{min} + 0.5u = 15° + 0.5 \times 22.76° = 26.38°$$

From equation (E11.2),

$$a = \frac{-7571}{8913 \cos(180° - 37.76°) - 18 \times 43.85} = 0.966$$

$$\text{Speed} = a \times 1200 = 0.966 \times 1200 = 1159 \text{ rpm}$$

From equation (E11.3),

$$T = 90.95 \times 34.2 \times \cos 26.38° = 2786.6 \text{ N-m}$$

Following the foregoing steps, the speed and torque can also be calculated for other values of I_d. The results are listed in the table here:

I_d (A)	58.47	43.85	29.25	14.62
T (N-m)	3608	2786.6	1836	975
Speed (rpm)	1200	1159	1122	1087

Note that speed falls with a decrease in torque.

Constant Commutation Lead Angle Control: From example 11.2 for $I_d = 58.47$ A, speed and torque are 1200 rpm and 3608 N-m, respectively, and $\beta_c = 43°$. From equation (11.13),

$$\cos(43° - u) = \cos 43° + \frac{2 \times 18.85}{\sqrt{6} \times 3810.5} I_d$$

$$= 0.73 + 0.004 I_d \qquad \text{(E11.4)}$$

From equation (E11.2)

$$a = \frac{-7571}{8913 \cos 137° - 18 I_d} = \frac{-7571}{-6518.6 - 18 I_d} \qquad \text{(E11.5)}$$

For $I_d = 43.85$ A, From equation (E11.4),

$$\cos(43° - u) = 0.73 + 0.004 \times 43.85 = 0.9 \quad \text{or} \quad u = 18°$$

Then

$$\phi = \beta - 0.5\, u = 43° - 0.5 \times 18° = 34°$$

From equation (E11.5),

$$a = \frac{-7571}{-6518.6 - 18 \times 43.85} = 1.04$$

Hence, speed = $1200 \times 1.04 = 1248$ rpm

From equation (E11.3),

$$T = 90.95 \times 34.2 \times \cos 34° = 2579 \text{ N-m}$$

Following the preceding steps for different values of I_d gives the results tabulated here:

I_d (A)	58.47	43.85	29.25	14.62
T (N-m)	3608	2579	1688	789.5
speed (rpm)	1200	1248	1289	1339

11.5.2 Operation of a Permanent Magnet Synchronous Motor from a Variable Frequency Current Source

Since the operation of a permanent magnet motor takes place at a constant I_f', the analysis of this section is also applicable to the operation of a wound-field motor with a constant field current.

Below base speed, the speed is controlled by varying the armature current and above base speed it is controlled by weakening the flux. In a wound-field motor we varied I_s, I_f', and δ' (or β) to control T, flux, and the power factor for a given speed.

In a permanent magnet motor I_f' is constant. Consequently I_s and δ' can be varied to achieve torque control along with the control of either flux or the power factor for a given speed. The control of the power factor is important when load commutation is employed. On the other hand, flux control may be desirable when fast transient response is desired.

Operation at a Leading Power Factor

The analysis is presented for motoring operation only. We can use the phasor diagram of figure 11.9. Note that I_f' is constant and I_m' is a variable. From triangle ABC,

$$\frac{I_s}{\sin \delta} = \frac{I_f'}{\sin(90 + \phi)} = \frac{I_m'}{\sin(180 - \delta')} \tag{11.63}$$

which yields

$$\delta = \sin^{-1}\left(\frac{I_s}{I_f'} \cos \phi\right) \tag{11.64}$$

Now

$$\delta' = \delta + 90° + \phi \tag{11.65}$$

or

$$\delta' = (90 + \phi) + \sin^{-1}\left(\frac{I_s}{I_f'} \cos \phi\right) \tag{11.66}$$

The operation at a constant ϕ is obtained, if δ' is increased with I_s according to equation (11.66). Since δ' increases with I_s, according to equation (11.41), the torque increases less than linearly with I_s (note that $\delta' > 90°$).

When fast transient operation is required, the motor is allowed to carry several times the rated current. Consider the operation of the drive when maximum current allowed under transient operation is equal to twice the rated current. A permanent magnet motor is designed to have unity power factor at full load. For load commutation it is made to operate at a leading power factor by increasing the torque angle δ'. According to equation (11.41) this causes machine torque to reduce for the rated current. To achieve load commutation at twice the rated current requires a large increase in δ', consequently a proportionate increase in torque is much less than in current. Compare this behavior with that of a wound-field motor. In a wound-field motor, the leading power factor for twice the rated current is obtained partly by an increase in I_f' and partly by an increase in δ'. From equation (11.48) the decrease in the factor $\sin \delta'$ is compensated by an increase in I_f'; hence, torque increases in proportion to armature current.

It is useful to examine this behavior from another direction. From equation (11.63),

$$I_m' = I_f' \frac{\sin \delta'}{\cos \phi} \tag{11.67}$$

From equations (11.41) and (11.67),

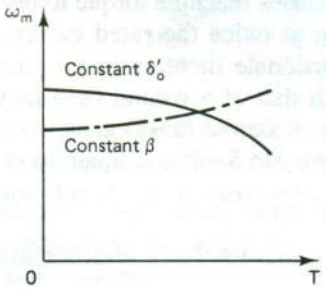
$$T = \left(\frac{3}{2} pL_s\right) I_s I'_m \cos \phi \tag{11.68}$$

As I_s is increased to increase torque, with ϕ held constant, δ' increases. Consequently I'_m decreases [equation (11.67)], decreasing flux and not allowing torque to increase in proportion to I_s [equation (11.68)]. Thus, unlike a wound-field motor, which operates at a constant flux, a permanent magnet motor operates at a reduced flux for a large I_s. This effect is similar to an armature reaction in a dc motor.

The operation above base speed is obtained as follows. At base speed, for given values of I_s and ϕ, δ' is given by equation (11.66). If for a given operating point, δ' is increased more than that given by equation (11.66), then I'_m and induced voltage will decrease (fig. 11.9a) and the motor speed will increase to obtain a balance between the terminal voltage and the induced voltage. Further, according to equation (11.66) for a given I_s, ϕ will increase with δ'. Hence, the motor power factor and power capability will reduce and losses will increase with an increase in speed.

Expressions for constant margin angle control are obtained by substituting from equation (11.30) into equations (11.66) and (11.68). According to equation (11.31), u now depends on I_d, I'_m, and speed. The performance equations and performance requirements for other control techniques can be similarly obtained.

Now examine the behavior of the open-loop drive of figure 11.6. Figure 11.20 shows the speed-torque curves for constant commutation lead angle control and constant no-load torque angle control for operation below base speed at a constant value of the source side converter firing angle. In the former control, speed increases with torque. With an increase in torque, I_s, δ', and u increase and flux decreases. The increase in I_s increases the voltage drops across various resistances, which has a tendency to decrease the speed. The effect of the commutation overlap is to decrease the speed for a given value of the inverter counter emf V_d. The field weakening tends to increase the speed. Since the latter effect dominates, the speed increases with torque. In constant no-load torque angle control, speed drops with torque. Here also, while the resistance drops tend to decrease speed with torque, the armature reaction tends to increase it with torque. Figure 11.12a shows that β decreases with an increase in I_s. This has the effect of reducing speed for a given V_d [equa-



Constant commutation lead angle control

Constant no-load torque lead angle control

Figure 11.20 Speed-torque curves for load commutated inverter-fed permanent magnet synchronous motor drive.

tion (11.23)]. The net result of these effects is to cause a decrease in speed with an increase in torque.

Next let us examine the behavior of the drive above base speed. When a rotor position encoder is employed, δ_o' is increased to increase speed. An increase in δ_o' causes β to increase and flux to reduce. Both changes cause speed to increase [equation (11.23)]. When a terminal voltage sensor is employed, operation above base speed is obtained by increasing β. Here again, an increase in speed is obtained due to an increase in β and a decrease in flux.

As explained earlier, the drive power factor, efficiency, and power capability decrease and machine terminal voltage increases with an increase in speed.

Operation at Unity Power Factor

The machine operates at unity power factor when δ' is varied with I_s. The relevant equations are obtained when $\phi = 0$ is substituted in equations (11.66) through (11.68).

Operation at a Constant δ' Below Base Speed

If a value of δ' can be chosen so that at the highest value of I_s or torque, $I_m' = I_f'$, then the flux will be the same as at no load. The locus of I_m' and terminal voltage for a given speed when I_s is increased from 0 to its maximum value at a constant δ' is shown in figure 11.21. The phasors are shown for $I_s = 0$ and the maximum value of I_s. I_m' is equal to I_f' for $I_s = 0$ and at the maximum value of I_s. The value of δ' can be obtained as follows.

From triangle ABC,

$$I_f'^2 + I_s^2 - 2I_f'I_s \cos(180 - \delta') = I_m'^2 = I_f'^2$$

or

$$\cos \delta' = -\frac{I_s}{2I_f'} \tag{11.69}$$

This operation allows high torque to be obtained at the maximum value of I_s. The power factor is also maintained high. However, because of the lagging power factor operation at high currents, forced commutation is required. This technique can be used for low-power drives where transistor inverters are employed.

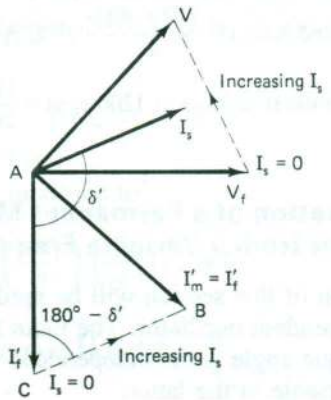


Figure 11.21 Phasor diagram for a constant δ' which make the value of I_m' at the maximum I_s equal to I_f'.

At a given torque, a constant power factor is maintained for all speeds. At a given speed, as the torque is reduced from its maximum value, the power factor increases from its minimum lagging value, reaches unity, becomes leading, and has a minimum value at the lowest armature current.

Example 11.6

A 10 kW, 3-phase, 440 V, 60 Hz, 4 pole, Y-connected permanent magnet synchronous motor has the following parameters:

$$X_s = 15\ \Omega, \qquad R_s \text{ is negligible}, \qquad \text{Rated power factor} = 1.0$$

The machine is controlled from a transistor current source inverter at a torque angle δ' such that at the rated current $I'_m = I'_f$. Calculate the motor torque at the rated current, and the terminal voltage at 1200 rpm and the rated current. Neglect harmonics, core loss, friction, and windage.

Solution:

$$V = 440/\sqrt{3} = 254 \text{ V}$$

$$\text{The rated } I_s = \frac{10 \times 10^3}{\sqrt{3} \times 440} = 13.1 \text{ A}$$

$$I'_m = \frac{V}{X_s} = \frac{254}{15} = 16.93 \text{ A}$$

From figure 11.17 for the unity power factor operation

$$I_f'^2 = I_s^2 + I_m'^2 = 13.1^2 + 16.93^2$$

or

$$I'_f = 21.4 \text{ A}$$

From equation (11.69), for $I'_m = I'_f$ at a given I_s,

$$\cos \delta' = -\frac{I_s}{2I'_f} = -\frac{13.1}{2 \times 21.4}$$

which gives $\delta' = 107.8°$.

From equation (11.41),

$$T = \left(\frac{3}{2} \times 4 \times \frac{15}{377}\right) \times 13.1 \times 21.4 \sin 107.8° = 63.72 \text{ N-m}$$

$$\text{Rated speed} = \frac{120 \times 60}{4} = 1800 \text{ rpm}$$

$$\text{Terminal voltage at 1200 rpm} = \frac{1200}{1800} \times \frac{21.4}{16.93} \times 440$$

$$= 371 \text{ V (line)}$$

11.5.3 Operation of a Permanent Magnet Synchronous Motor from a Variable Frequency Voltage Source

The discussion of this section will be applicable to self-control as well as to control from an independent oscillator. The main difference between the two controls is that while the torque angle is the independent variable in the former, frequency is the independent variable in the latter.

The basic control strategy is to operate the motor at a constant flux below base speed and at a constant terminal voltage above base speed. The constant flux operation below base speed is achieved by operating the machine at a constant (V/f) ratio, which is increased at low speeds to compensate for the armature resistance drop.

The phasor diagram of figure 11.1 is applicable, and torque and power are given by equations (10.16) to (10.19). Substituting for ω_{ms} from equation (10.1) and $X_s = \omega L_s$, gives

$$P_m = 3VI'_f \sin \delta \tag{11.70}$$

$$= 3L_s \omega I'_m I'_f \sin \delta \tag{11.71}$$

$$T = \frac{3p}{2}\left(\frac{V}{\omega}\right) I'_f \sin \delta \tag{11.72}$$

$$= \left(\frac{3pL_s}{2}\right) I'_m I'_f \sin \delta \tag{11.73}$$

Below base speed, the magnetizing current is constant and equal to its rated value I_M. Hence, from equations (10.22) and (10.23) or figure 11.1

$$I_s \sin \phi = I'_f \cos \delta - I_M \tag{11.74}$$

$$I_s \cos \phi = I'_f \sin \delta \tag{11.75}$$

Consider the operation of the motor below base speed for a constant armature current. Since both I'_f and I'_m are constant, all the sides of triangle ABC of figure 11.1 have fixed values; therefore, δ has a constant value. The torque is also constant from equation (11.73). This is true for any speed. Therefore the machine operates in the constant torque mode. Since δ is constant, from equations (11.74) and (11.75), the in-phase and quadrature components of armature current are also constant. Consequently, the power factor is also constant for all speeds.

Let us examine the operation of a motor which has been designed to operate at a lagging power factor. When it operates at the rated current, at any speed, it will operate at the designed lagging power factor. A decrease in I_s, irrespective of speed, reduces δ, T, and the quadrature component of armature current. At some value of I_s or δ, $(I'_f \cos \delta - I_M)$ becomes zero. Then the machine operates at a unity power factor. A further decrease in I_s causes the motor to operate at a leading power factor. The lowest leading power factor is obtained at the minimum value of I_s.

Next, let us consider the operation above base speed for which the per-unit frequency $a > 1$. Since V is constant

$$I'_m = \frac{I_M}{a} \tag{11.76}$$

Equations (11.73) and (11.74) are modified to

$$T = \left(\frac{3pL_s}{2}\right)\left(\frac{I_M}{a}\right) I'_f \sin \delta \tag{11.77}$$

$$I_s \sin \phi = I'_f \cos \delta - \frac{I_M}{a} \tag{11.78}$$

The in-phase component of the armature current is still given by equation (11.75). Now let us consider the operation at the rated armature current for which the machine operates at a lagging power factor below base speed. As shown in figure 11.22, for a constant I_s and variable I'_m ($= I_M/a$), point C of triangle ABC moves along a circular path with a radius equal to I_s. Consequently, as the speed increases, initially δ increases, reaches a maximum value, and then decreases again. From equations (11.70), (11.75) and (11.78), as the speed is increased above the base speed, with the armature current held at the rated value, the developed power and power factor increase initially. The power reaches the maximum value at a speed for which the power factor is unity and δ is maximum [equation (11.70), assuming that motor parameters are such that δ does not reach 90° before the power factor becomes unity]. A further increase in speed causes a decrease in power and the power factor, which is now leading. When the power factor becomes the same as at base speed, the motor produces the same power as at base speed. A further increase in speed reduces the power below the base speed value. This imposes a limit on the maximum speed for applications requiring essentially a constant power capability. Therefore, the speed range of a motor designed for a lagging power factor is generally higher than the one designed for a unity or leading power factor.

Now consider the variation of the power factor at a constant speed above base speed, when the current is reduced from the rated value. If the power factor at the rated current is lagging, it improves, reaches unity and becomes leading for low values of current. If the power factor has already become leading at the rated current, it becomes more leading as the current is reduced.

11.5.4 Operation of a Wound-Field Synchronous Motor from a Variable Frequency Voltage Source

Here also the machine operates at a constant flux below base speed and at a constant terminal voltage above base speed. Additional flexibility made available by the field current control is used to control the power factor.

It is preferable to maintain the power factor at unity. From equation (11.74), the motor operates at a unity power factor below base speed when

$$I'_f \cos \delta - I_M = 0$$

or

$$I'_f = \frac{I_M}{\cos \delta} = \frac{\text{constant}}{\cos \delta} \tag{11.79}$$

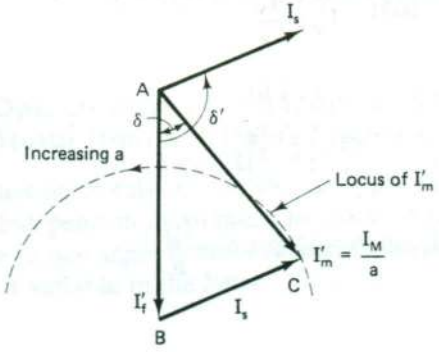


Figure 11.22 Field weakening by increasing δ'.

Thus I_f' should be varied with δ according to equation (11.79).

The power input to the machine is given by

$$P_m = 3VI_s \tag{11.80}$$

and

$$\begin{aligned} T &= \frac{3}{\omega_{ms}} VI_s \\ &= \left(\frac{3p}{2\omega}\right)(\omega L_s I_M) I_s \\ &= \frac{3pL_s}{2} I_M I_s \end{aligned} \tag{11.81}$$

Thus, at the unity power factor operation, the armature current varies linearly with torque. For a given armature current, the torque has a constant value at all speeds; hence the motor operates in the constant torque mode.

Above base speed, V is held constant; therefore

$$I_m' = \frac{I_M}{a} \tag{11.82}$$

From equation (11.79), for the unity power factor operation,

$$I_f' = \frac{I_M}{a \cos \delta} \tag{11.83}$$

Therefore, I_f' is to be changed as a function of δ and speed. Now

$$P_m = 3VI_s \tag{11.84}$$

which shows that the machine operates in the constant power mode above base speed.

At a given I_s, variables T, V, I_f', and I_m' vary with "a" in the same manner as shown in figure 11.18.

11.5.5 Operation of Permanent Magnet Motor at the Maximum Torque to Armature Current Ratio

From equation (10.27), which applies to current source operation, for a given I_f', the ratio (T/I_s) is maximized when motoring and braking operations take place at $\delta' = 90°$ and $\delta' = 270°$, respectively (that is, at pull-out points). The phasor diagram for the motoring operation at $\delta' = 90°$ is shown in figure 11.23a.

The maximum (T/I_s) ratio operation can also be identified in terms of the internal power factor, which is defined as the cosine of the angle between the armature current I_s and the excitation emf V_f. The ratio has a maximum value when the internal power factor is unity. Note that under this condition the terminal (actual) power factor is always less than unity; it is lagging for motoring and leading for braking.

In this operation, the (T/I_s) ratio is maximized essentially by boosting the air-gap flux. The machine may operate under heavy saturation and the core loss will be high but the copper loss will be low. When operating at $\delta' = 90°$ from a variable fre-

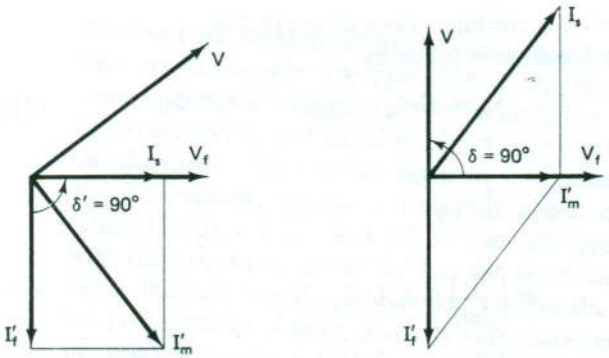


Figure 11.23 Operation at (a) maximum (T/I_s) ratio or $\delta' = 90°$ and (b) maximum (T/I'_m) ratio or $\delta = 90°$.

quency current source, for the same maximum dc link voltage, the base speed will have a lower value. For higher speeds, δ' will have to be increased to weaken the flux. This control technique of maximizing the (T/I_s) ratio by boosting the flux can be used in servo drives requiring fast transient response but only low-speed operation.

11.5.6 Operation of Permanent Magnet Motor at the Maximum Torque-to-Flux Ratio

From equation (10.19), applicable to voltage source operation, for a given I'_f, the (T/I'_m) ratio is maximized when motoring and braking operations take place at $\delta = 90°$ and $\delta = 270°$, respectively (that is, at pull-out operating points). The phasor diagram for the motoring operation at $\delta = 90°$ is shown in figure 11.23b. The machine operates at a lagging power factor.

In this operation the (T/I'_m) ratio is maximized by boosting the armature current. The machine will have high copper loss and low core loss. The base speed will have a higher value compared to the maximum (T/I_s) ratio operation. This operation can be used in servo applications requiring operation over a wide speed range.

11.5.7 True DC Motor Operation

Here the synchronous motor operation is termed true dc motor operation when it operates at $\delta' = 90°$ for motoring and at $\delta' = 270°$ for braking, thus yielding maximum torque per ampere of the armature current at a given field current. However, such an operation will have low base speed, and heavy saturation, high core loss, and a low lagging power factor when operating at and near the maximum permissible current. These drawbacks are caused by armature reaction. Armature reaction can be eliminated by providing a compensating field winding located in the quadrature axis and carrying the armature current. By suitably choosing the turns, the compensating winding mmf can be made equal to and opposite the armature winding mmf. Then the air-gap flux becomes independent of the armature current and depends on the value of field current only. Figure 11.24a shows the phasor diagram of a machine with a compensating winding. The current I_{fc} is the equivalent armature phase current, which produces the same revolving mmf as the compensating winding. It can be evaluated in the same way as I'_f (section 10.1.1). The current I_{fc} induces a voltage E_c

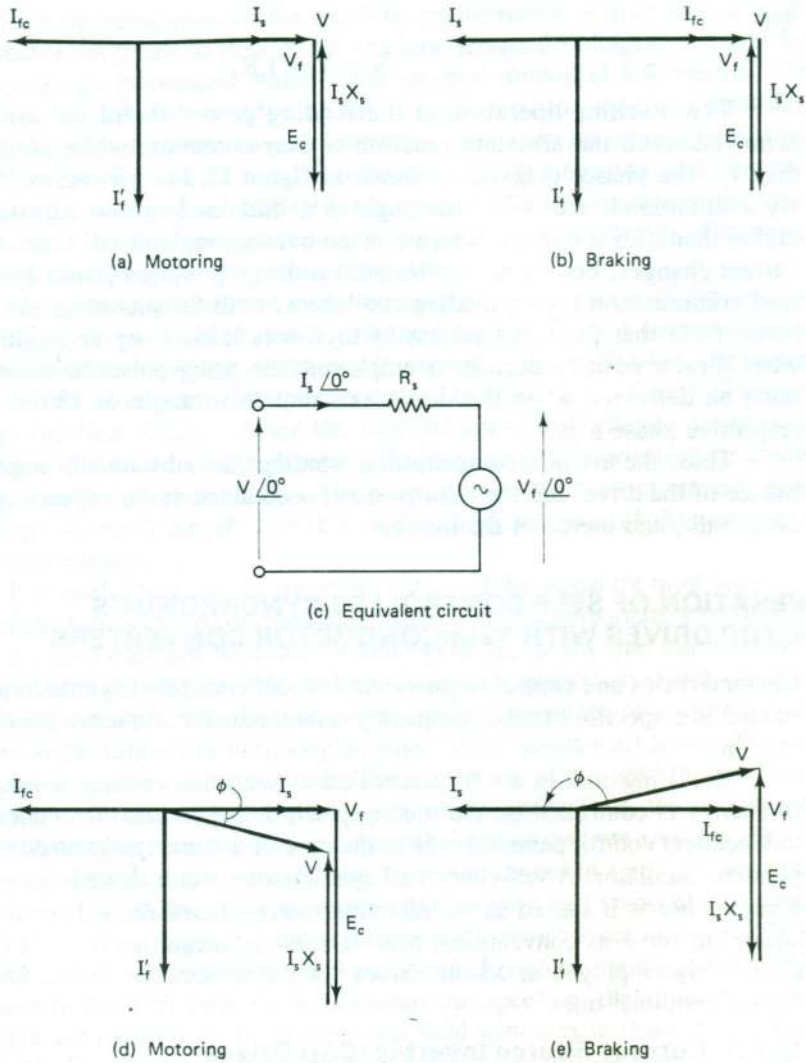


Figure 11.24 True dc motor operation with compensating winding: (a) to (c) are for the unity power factor operation and (d) and (e) are for the leading power factor operation.

which is equal and opposite to the I_sX_s drop. Hence, V, I_s, and V_f are in phase. The machine now operates at unity terminal, as well as at the unity internal power factor while retaining δ' at 90°. Figure 11.24b shows the corresponding diagram for the braking operation. This gives the optimum condition of operation for the drive in the sense that while the machine torque and power for given values of I_s and I'_f are maximized, the inverter rating for a given power output is minimized. The machine equivalent circuit under these operating conditions is shown in figure 11.24c, which is similar to a dc machine. From the phasor diagrams and the equivalent circuit,

$$V_f = X_s I'_f \tag{11.85}$$

and

$$V = V_f + I_s R_s \tag{11.86}$$

The machine operation at the leading power factor for all values of I_s is achieved when the armature reaction is over-compensated by choosing I_{fc} higher than I_s. The phasor diagram is shown in figure 11.24d for the motoring operation. By a suitable choice of I_{fc}, the angle ϕ at full load can be adjusted to be a little higher than $(0.5\ u + \gamma_{min})$, where u is the overlap angle at full load. As the armature current changes, both ϕ and u change, resulting in a good power factor and reliable load commutation for all loading conditions, both for motoring and braking operations. Note that the firing instant of thyristors leads I_s by an angle 0.5 u. Hence, when a rotor position encoder is employed, the firing pulses to respective thyristors must be delivered when the direct axis makes an angle of $150° + 0.5$ u with the respective phase axis.

Thus, the use of a compensating winding can substantially improve the performance of the drive. But the improvement is obtained at the expense of an increase in cost, bulk, and inertia of the motor.

11.6 OPERATION OF SELF-CONTROLLED SYNCHRONOUS MOTOR DRIVES WITH SEMICONDUCTOR CONVERTERS

Characteristics and control requirements of self-controlled synchronous motor drives related to a specific variable frequency semiconductor converter are described in this section.

Recalling that in a self-controlled synchronous motor drive, the converter frequency is controlled by the motor speed, and therefore, it is not available as an independent control parameter, as in the case of a converter controlled from an independent oscillator. A self-controlled synchronous motor drive is neither subjected to hunting, nor is it started as an induction motor; therefore, a damper winding is not needed to serve its conventional role. It does not mean that it is not needed at all. It is certainly employed in certain drives but the reasons for its use are different than the conventional ones.

11.6.1 Current Source Inverter (CSI) Drives

A CSI synchronous motor drive may employ a load commutated thyristor inverter (fig. 11.6), a forced commutated inverter (fig. 8.14), or a current controlled pulse-width modulated inverter (figs. 8.26 and 8.27). A forced commutated inverter may consist of a forced commutated thyristor inverter (fig. 8.15) or a self-commutated transistor or GTO inverter. The current controlled pulse-width modulated inverters usually employ a voltage source inverter (fig. 8.1) with transistor switches, though GTOs may also be used.

Inverter with Rectangular Current Waveforms

The load commutated (fig. 11.6) and forced commutated inverters (figs. 8.14 and 8.15) give approximately rectangular or trapezoidal current waveforms which are rich in harmonics. The harmonic content is given approximately by equation (8.22).

The effect of harmonics on the machine performance is discussed in section 10.9. The main effects are to distort the machine terminal voltages, increase losses, and produce torque pulsations which cause stepped motion at low speeds. The damper windings are helpful in reducing the distortion of the machine terminal voltage. They offer a low impedance path to the harmonic currents; consequently, the magnetizing current (fig. 10.15b) and flux become nearly sinusoidal, making the terminal voltage also nearly sinusoidal. The damper windings also reduce the commutating inductance, and thus help in reducing the commutation overlap. Therefore, damper windings are always employed in these drives.

At the time of commutation, the current in one phase of the armature jumps from 0 to I_d and in another phase it jumps from I_d to 0. This sudden change of phase currents produces voltage spikes in the terminal voltage. Further, the sudden change in the mmf's of two armature phases caused by the sudden change of current tends to change the flux linkage. Since the flux linkage cannot change abruptly, a burst of current is produced in the damper windings and field winding to counteract the changes in the armature mmf's. The magnitude of the burst of current in the damper windings is much larger than that in the field, because of the damper windings' lower impedance.

It is useful to examine the exact nature of the armature mmf wave. The operation of the motor can be divided into 6 intervals of approximately 60° duration, separated by commutation intervals. When operating in any one interval, two machine phases carry dc link current and produce a stationary armature mmf wave. The commutation shifts the dc link current from one phase to another; consequently the armature mmf wave jumps by an angle of 60° (electrical) around the air-gap. The mmf wave again remains stationary for an interval of 60° until the next commutation occurs and shifts it around the air-gap by another 60°. Thus, the armature mmf wave does not revolve continuously and smoothly but in discrete steps of 60°. As a result of this, at very low frequencies, the rotor also moves in steps. At high frequencies, the inertia causes the rotor to move continuously and smoothly. If now the rotor speed is assumed constant, the armature mmf wave moves at a variable speed with respect to the rotor, although its average speed remains the same as that of the rotor. Because of the difference in the instantaneous speeds of the armature mmf and rotor, currents are induced in the damper and field windings to maintain the flux linkages constant. This tendency to maintain a constant flux linkage smoothens out the effects of stepped motion of the armature mmf and produces a nearly sinusoidal air-gap flux wave. Hence, the machine terminal voltages are nearly sinusoidal, except at the instant of commutation where voltage spikes are produced by sudden changes of current.

The load commutated inverter or forced commutated inverter is fed from the closed-loop current source of figure 8.17b when the supply is ac and from the closed-loop current source of figure 8.17c when the supply is dc. When braking is required, a fully controlled rectifier is used in the scheme of figure 8.17b and a two-quadrant type D chopper is used in figure 8.17c.

In section 11.3, the load commutated inverter is discussed without a closed-loop current source because it is not essential for its operation. In actual practice, it is always employed with a closed-loop current source. The closed-loop current con-

trol causes the source side converter or chopper to track the inverter terminal voltage at a constant current. When the torque angle δ' (or angle β in a load commutated inverter using terminal voltage sensing) is changed to change the motor operation from motoring to regenerative braking, the inverter changes its operation from inversion to rectification and its terminal voltage reverses. Since the source side converter tracks the inverter terminal voltage at a constant dc link current, its operation automatically and smoothly shifts from rectification to inversion, causing the regenerated energy to be fed to the ac mains. Similarly, the change of motor operation back to motoring will cause the source side converter operation to change to rectification, automatically and smoothly. Similar operation is obtained for the chopper case.

For both load commutated and forced commutated inverter drives one-phase sequence gives operation in quadrants I and II. When the motor is stationary, the reversal of phase sequence, by interchanging the firing pulses between any two legs of the inverter, will drive the motor in the reverse direction and will provide operation in quadrants III and IV. The composite braking described for the induction motor for the scheme of figure 8.17c can also be employed. However, unlike an induction motor, dynamic braking can also be employed by deactivating the chopper, connecting a resistor across the dc link and causing the inverter to operate as a rectifier.

Only thyristor inverters can be built at high power levels. Therefore, the load commutated thyristor inverters are widely used in high power applications because of the absence of a commutation circuit. But then the inverter operates at a leading power factor. The power factor can be improved by the constant margin angle control but at the expense of complex control. The forced commutated thyristor inverter can provide unity power factor operation, but it is not employed because it is expensive and it is difficult to achieve reliable forced commutation at high power levels. The inverters at low power levels are usually built using power transistors. Since the comutation poses no problem, the drive can operate at unity or lagging power factor. While the unity power factor operation is required to minimize the inverter kVA rating and the drive losses, the operation at a lagging power factor is required when the motor is operated at $\delta' = 90°$ or $\delta = 90°$ to obtain a high peak torque to get fast transient response (section 11.5.5).

Current Controlled Pulse-Width Modulated Inverter

A current controlled pulse-width modulated inverter has very high frequency harmonics which have little effect on machine operation. Consequently, the torque pulsations are absent at all speeds, including very low speeds. The current controlled pulse-width modulated inverter is suitable for high-performance servo drives. A voltage source transistor inverter (inverter of figure 8.1a with transistor switches) is employed to obtain fast switching. The dc supply for the inverter is obtained using the circuits of figure 8.3, which are also employed for a pulse-width modulated voltage source inverter. Because of similar power circuits, the braking and multiquadrant operations described in section 8.1.5 for a pulse-width modulated voltage source inverter fed induction motor are applicable to a current controlled pulse-width modulated inverter fed synchronous motor with only one difference. While braking

in the former case is obtained by a change in frequency, in the latter case, it is obtained by a change in torque angle δ'.

11.6.2 Voltage Source Inverter Drives

The machine may be fed by a six-step inverter or a pulse-width modulated inverter. The output voltage of a 6-step inverter is shown in figure 8.1b and its harmonic content is given by equation (8.2). It was shown in section 10.9, that in the absence of damper windings, the machine offers a high impedance to harmonics; consequently, the harmonics are filtered out and the armature current has a sinusoidal waveform. Since in a self-controlled synchronous motor drive, damping action is not required, it is desirable not to use a damper winding in voltage source inverter drives. Since the machine is able to filter out the harmonics, a pulse-width modulated inverter may not be required.

The discussion of section 8.1.5 about the braking and multiquadrant operations of voltage source inverter fed induction motor drives is also applicable to voltage source inverter fed synchronous motor drives, except that the changeover from motoring to braking, or vice versa, in a synchronous motor drive is obtained by a change in torque angle δ rather than in frequency.

11.6.3 Cycloconverter Drives

A synchronous motor drive may employ a cycloconverter with line commutation or a cycloconverter with load commutation. A line commutated cycloconverter may operate with a current source characteristic or with a voltage source characteristic. In either case, the armature current is sinusoidal at low frequencies; hence, low-speed torque pulsations are absent. The changeover from motoring to braking, and vice versa, is done by the control of torque angle. The reversal of phase sequence allows operation in all four quadrants. To keep the harmonic content low in the machine current and voltage and in the source current, the frequency of operation is limited to 40 percent of the source frequence. Hence, the drive is suitable for applications requiring low speed range.

When load commutation is used, a cycloconverter can operate at a frequency higher than the source frequency. Hence, a wide speed range is obtained. At low speeds, a load commutated cycloconverter operates with line commutation, which eliminates low-speed torque pulsations. When operating in load commutation, the operation is somewhat similar to a load commutated current source inverter. Therefore, the description given in section 11.6.1 about harmonics and damper windings is applicable.

11.7 SELF-CONTROLLED SYNCHRONOUS MOTOR DRIVES (BRUSHLESS AND COMMUTATORLESS DC AND AC MOTOR DRIVES) AND THEIR APPLICATIONS

As explained earlier, self-controlled synchronous motor drives are popularly known as commutatorless dc and ac drives depending on whether the synchronous motor is fed from a dc supply through an inverter or from an ac supply though a cyclocon-

verter. When these drives employ a wound-field motor with a brushless excitation system or a permanent magnet motor then they are called brushless dc and ac drives, respectively.

Numerous control methods are possible. Here, a few methods will be described, basically to demonstrate the control principles described in section 11.5 for the cases which have practical applications.

11.7.1 Load Commutated Synchronous Motor Drives

In load commutation, the firing pulses may be derived either from the rotor position encoder or machine terminal voltage sensor and any one of the control strategies described in section 11.3.4 may be used. The machine may be fed from a load commutated current source inverter or a load commutated cycloconverter. The drive may accordingly be called a brushless (or commutatorless) dc or ac drive with load commutation.

Figure 11.25 shows a brushless dc motor drive employing a permanent magnet synchronous motor and a terminal voltage sensor. Figure 11.25a shows a drive with a constant commutation lead angle control. The drive employs an inner current control loop with an outer speed loop like the dc drive of figure 5.1b. The inner current control loop is nothing but a closed-loop current source. The terminal voltage sensor generates reference pulses of the same frequency as the machine induced voltage. The phase delay circuit shifts the reference pulses suitably to obtain control at a constant commutation lead angle β_c. Depending on the sign of the speed error $e_{\omega m}$, β_c is set to provide motoring or braking operation. Signals f and ω_m are obtained from the terminal voltage sensor. The speed and dc link current controllers are generally PI controllers.

An increase in speed command ω_m^* produces a speed error $e_{\omega m}$. The speed controller and current limiter set the dc link current command I_d^* at the maximum allowable value. The machine accelerates at the maximum available torque. When close to the desired speed, the current limiter desaturates and the drive settles at the desired speed and at the dc link current which balances the load torque. Similarly, a reduction in speed command produces a negative speed error. This sets β_c at 180° and the drive decelerates at the maximum torque. When the speed error changes sign, the operation shifts to motoring and the drive settles at the desired speed.

If the portion enclosed by dotted lines in figure 11.25a is replaced by the portion shown in figure 11.25b, the drive operates according to the control law of equation (11.36), which allows the machine to operate with improved performance.

Figure 11.26 shows the same drive when the terminal voltage sensor is replaced by a rotor position encoder. Figure 11.26a is for operation at a constant no-load torque angle, δ'_{oc}. When the circuit enclosed by dotted lines in figure 11.26a is replaced by the circuit shown in figure 11.26b, the drive operates according to the control law of equation (11.37), yielding better performance.

In the systems of figures 11.25 and 11.26, base speed is reached when the converter output voltage saturates. The speed above base speed is obtained by increasing the phase delay of the phase delay circuit with speed. This will increase β_c or δ'_{oc} and consequently increase the drive speed.

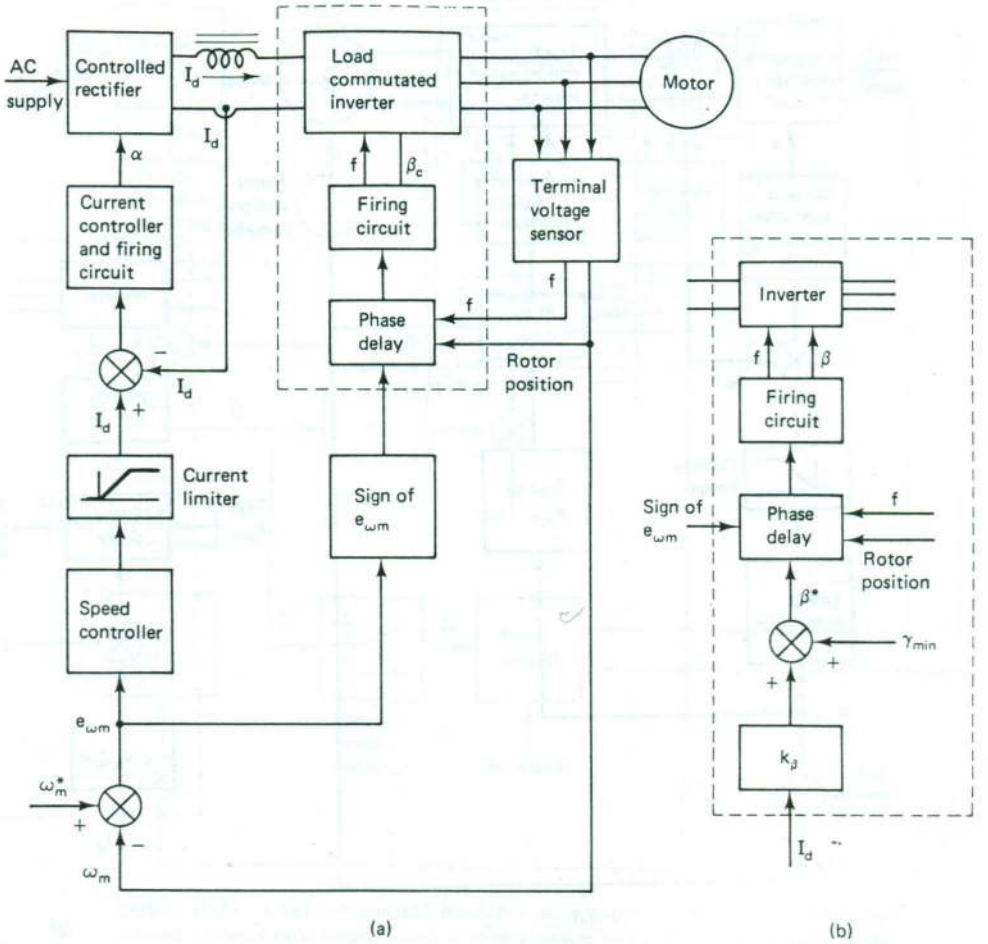


Figure 11.25 Load commutated brushless dc motor drive with terminal voltage sensor and permanent magnet motor.

Figure 11.27 shows a drive employing a wound-field synchronous motor, the terminal voltage sensor, and the control law of equation (11.36). The field current is controlled as a function of I_d to maintain a constant flux. The relationship between I_f^* and I_d can be worked out for the control law of equation (11.36) with the help of the analysis of section 11.5.1. The field current command I_f^* acts as a reference signal for the closed-loop control of field current; the details of which are not given in the figure. The drive is similar to the drive of figure 11.25 with the control law of equation (11.36). The only difference is the addition of field current control. The arrangement for the braking operation is not shown. It can be easily incorporated by causing β to become 180° whenever the speed error becomes negative. The speed control above base speed can be obtained by adding a variable negative offset signal to I_F^*. This will produce smaller and smaller values of field current as the offset

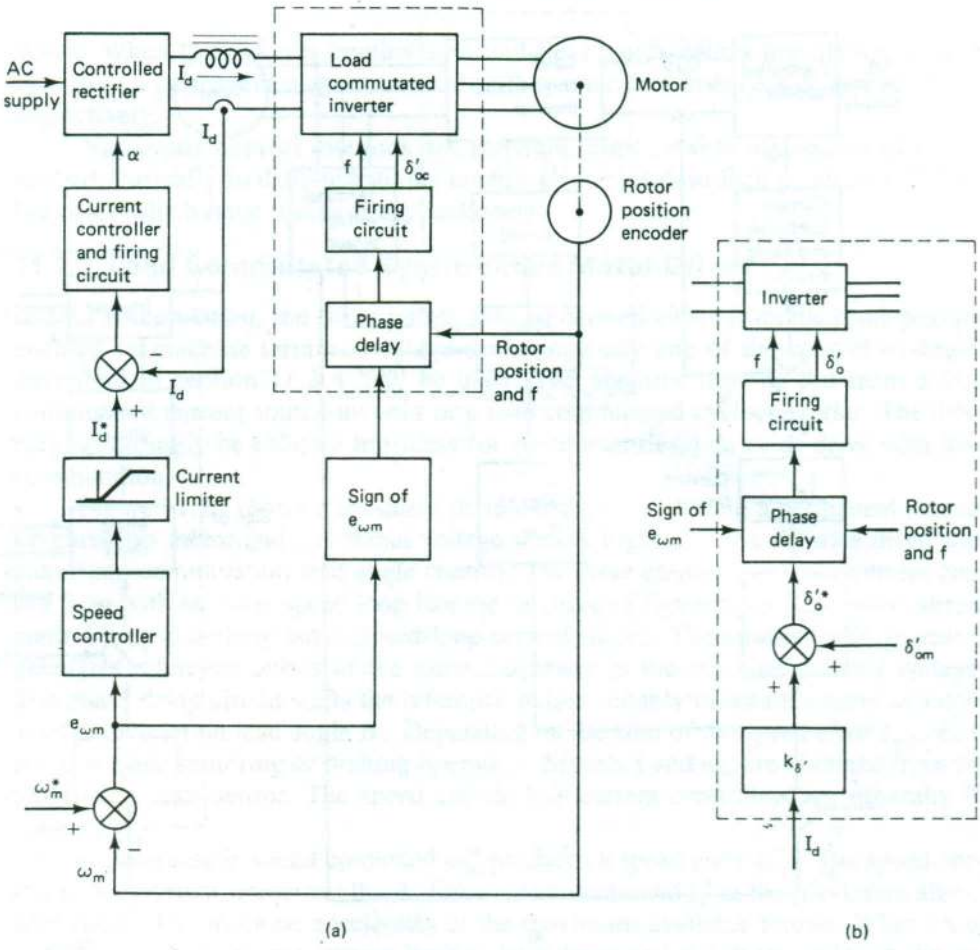


Figure 11.26 Load commutated brushless dc motor drive using rotor position encoder and permanent magnet motor.

signal is increased. It will also be necessary to increase K_β of the controller with the increase of the offset signal, or else commutation failure will occur.

Figure 11.28 shows the constant margin angle control for a wound-field motor drive employing a rotor position encoder. The scheme is based on the analysis of section 11.5.1. The drive has an outer speed loop and an inner current control loop as usual. In addition, it has an arrangement to produce constant flux operation and constant margin angle control. From the value of dc link current command I_d^*, I_s and 0.5 u are produced by blocks (1) and (2), respectively. The signal ϕ is generated from γ_{min} and 0.5 u [equations (11.16) and (11.29)] in adder (3). In block (4) I_f' is calculated from the known values of I_s, ϕ, and I_M [equation (11.47)]. Note that the magnetizing current I_m' is held constant at its rated value I_M to keep the flux constant. $I_f'^*$ sets a reference for the closed-loop control of the field current I_F. Block (5) estimates δ'^* from known values of ϕ and $I_f'^*$ [equation (11.49)]. The

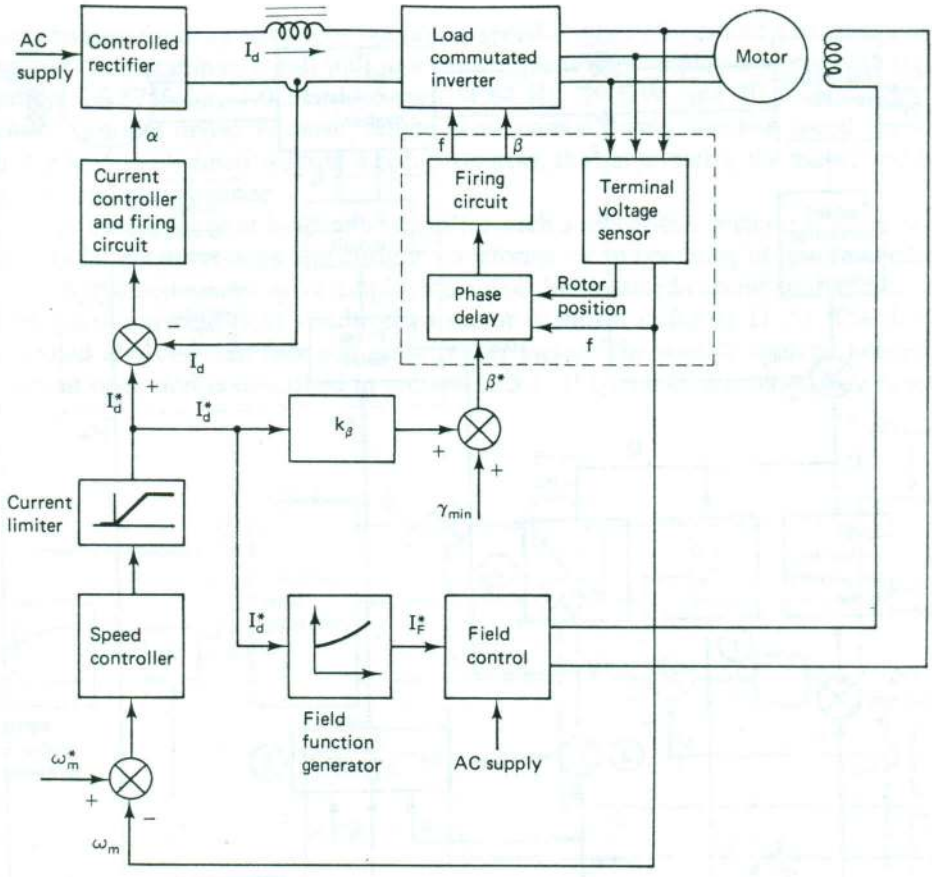


Figure 11.27 Load commutated brushless or commutatorless dc motor drive employing a wound field motor, terminal voltage sensor and *approximate* constant margin angle control.

phase delay circuit suitably shifts the pulses produced by the encoder to produce the desired value of δ_o' [$= (\delta' + 0.5\ u)$].

When the terminal voltage sensor is used, constant margin angle control can be obtained by a small modification to the scheme of figure 11.28. First, the portion enclosed in dotted lines is replaced by the portion of figure 11.27 enclosed by dotted lines. Second, block (5) of figure 11.28 is no longer needed. The command β^* for the phase delay circuit is obtained simply by adding u and γ_{min} [equation (11.29)].

In all the schemes discussed in this section, the circuit required for starting has been omitted for simplicity.

Because of the advantages described in section 11.3.4, the load commutated inverter drives are used in medium power, high-power and very high power (tens of megawatts) drives, and high-speed drives, such as compressors, extruders, induced and forced draft fans, blowers, conveyers, aircraft test facilities, steel rolling mills, large ship propulsion, main line traction, flywheel energy storage, and so on.[1-4,7,8,9]

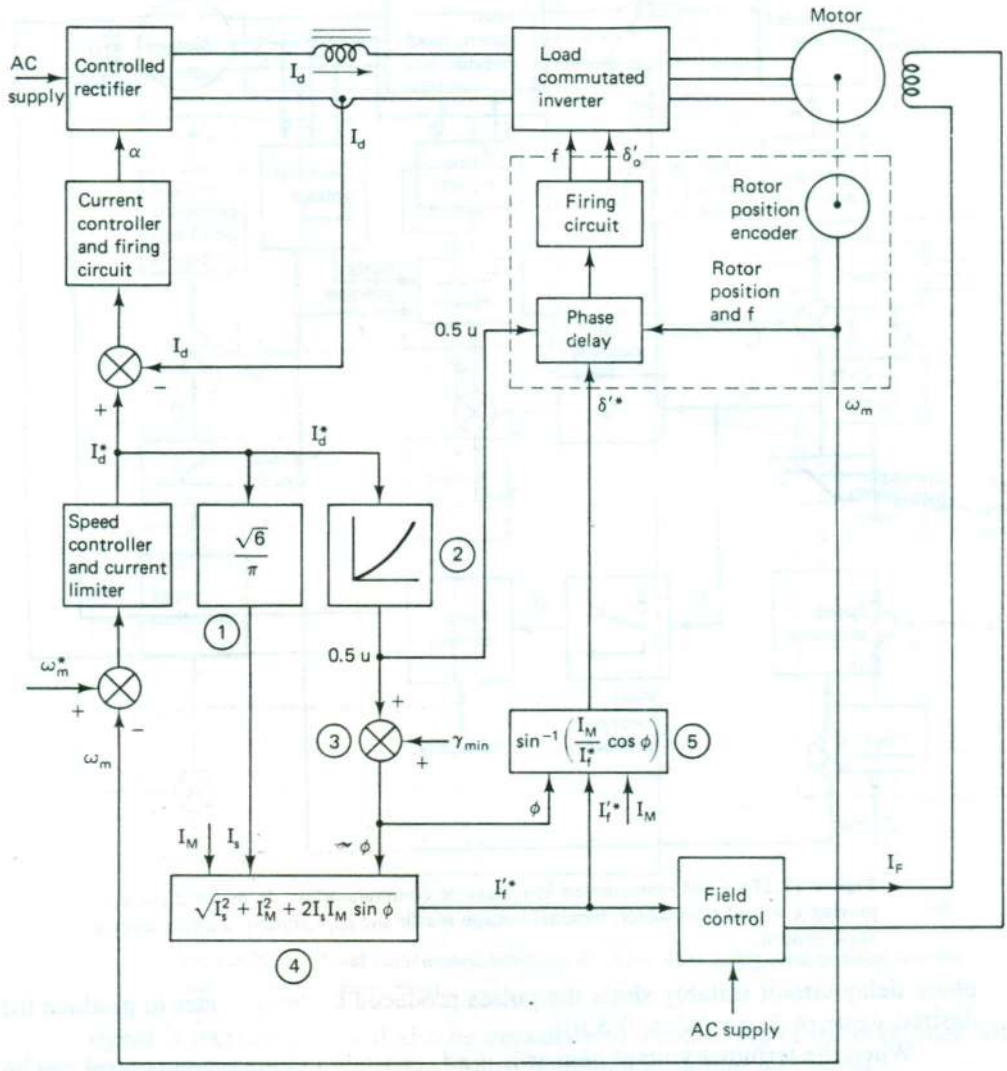


Figure 11.28 Load commutated brushless or commutatorless dc motor drive with constant margin angle control and using a wound-field motor and rotor position encoder.

They have also been used for the starting of large synchronous machines in gas turbine and pumped storage plants. High-power drives usually employ rectifiers with higher pulse numbers (12 or more), to minimize torque pulsations. The converter voltage ratings are also high so that efficient high voltage motors can be employed.

11.7.2 Line Commutated Cycloconverter Fed Synchronous Motor Drives

Line commutated cycloconverter fed wound-field synchronous motor drives (brushless or commutatorless ac drives) have been employed in low-speed gearless drives for rolling mills, mine hoists, ball mills in cement plants, and so on. These drives

are characterized by very low operating speeds, large power and fast transient response. For example, a ball mill in a cement plant may employ a motor with typical ratings of 8750 hp, 1.0 PF, 14.5 rpm, 4.84 Hz, 1900 V, and 40 poles.[5] They are called gearless drives because, unlike conventional drives, the low-speed operation of the load is obtained without a reduction gear, thus eliminating the associated cost, space, and maintenance.

A cycloconverter is ideally suited for such applications because it gives nearly sinusoidal output voltage and current waveforms when operating at low frequencies.

A cycloconverter drive employing a line commutated current source cycloconverter and a wound-field synchronous motor is shown in figure 11.29. The drive is operated at a constant flux and unity power factor. The control strategy needed for such an operation is described in section 11.5.1. If I_f' and δ' are varied as a function

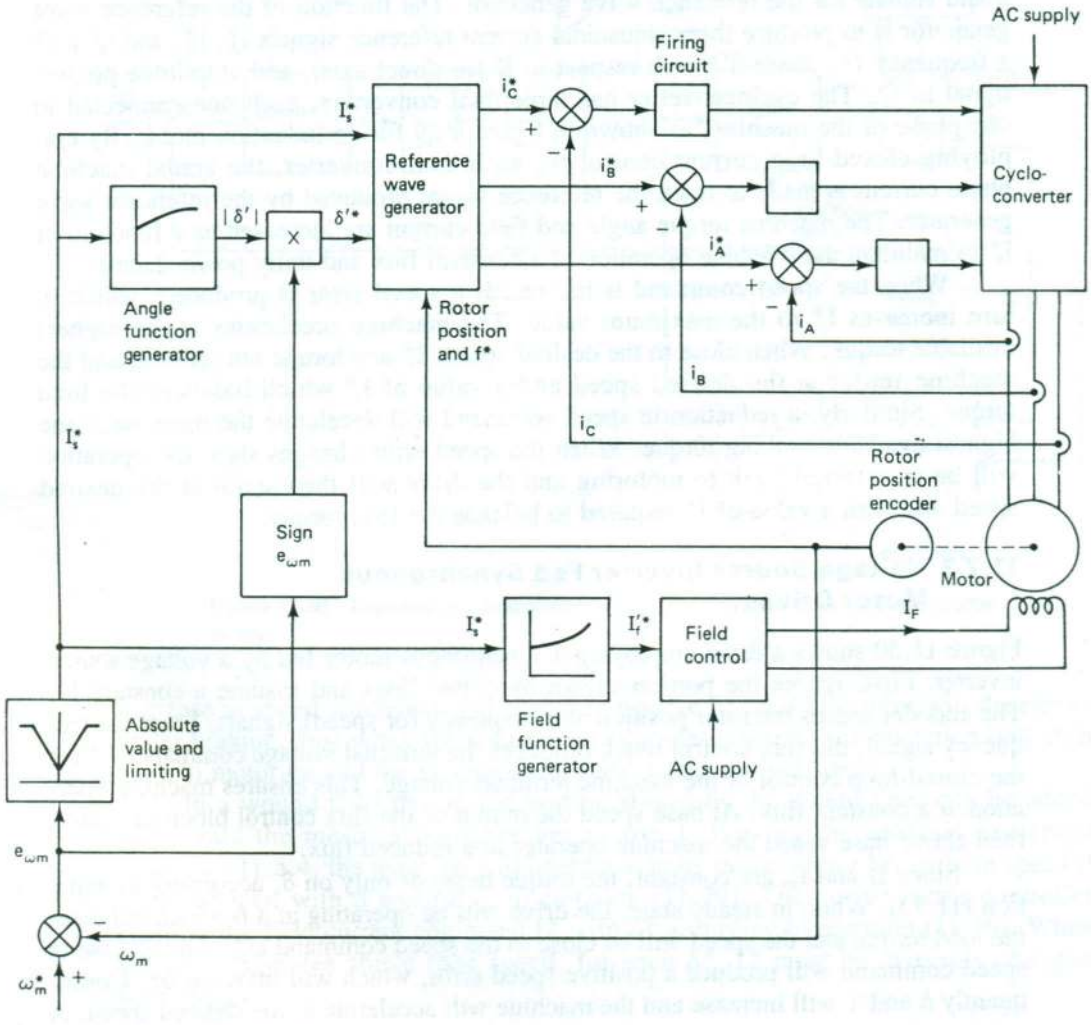


Figure 11.29 Brushless or commutatorless ac motor drive using line-commutated current source cycloconverter-fed wound-field machine.

of I_s, according to equations (11.39) and (11.40) (figure 11.17c), operation at unity power factor and constant flux is obtained. The required relationships between I_s and I_f' [equation (11.39)] and I_s and $|\delta'|$ [equation (11.40)] are implemented with the help of field and angle function generators. To obtain a motoring operation, the angle δ' is assigned a positive sign when the speed error is positive. When the speed error is negative, δ' is assigned a negative sign to obtain a braking operation. By incorporating an arrangement for the change of phase sequence at zero speed, a four-quadrant operation is obtained. The drive operates as follows.

Based on the speed error, the absolute value and limiting circuit sets a current reference I_s^*. Depending on the value of I_s^*, command signals $I_f'^*$ and δ'^* are produced by the field and angle function generators, respectively. $I_f'^*$ acts as a reference signal for the closed-loop control of the field current. I_s^*, δ'^*, and f^* act as command signals for the reference wave generator. The function of the reference wave generator is to produce three sinusoidal current reference signals i_A^*, i_B^*, and i_C^* with a frequency f^*, phase δ'^* with respect to $\bar{I}_f'$ (or direct axis), and amplitude proportional to I_s^*. The cycloconverter has three dual converters, each one connected to one phase of the machine, as shown in figure 8.29 for an induction motor. By employing closed-loop current control for each dual converter, the actual machine phase current is made to track the reference signal produced by the reference wave generator. The machine torque angle and field current are increased as a function of I_s^* to maintain the machine operation at a constant flux and unity power factor.

When the speed command is increased, a speed error is produced, which in turn increases I_s^* to the maximum value. The machine accelerates at the highest available torque. When close to the desired speed, I_s^* and torque are reduced and the machine settles at the desired speed and a value of I_s^* which balances the load torque. Similarly, a reduction in speed command will decelerate the machine at the highest available braking torque. When the speed error changes sign, the operation will be transferred back to motoring and the drive will then settle at the desired speed and with a value of I_s^* required to balance the load torque.

11.7.3 Voltage Source Inverter Fed Synchronous Motor Drives

Figure 11.30 shows a drive employing a synchronous motor fed by a voltage source inverter. First, ignore the portion shown by dotted lines and assume a constant I_f'. The encoder senses the rotor position and frequency (or speed) signals. From the frequency signal, the flux control block produces the terminal voltage command V^* for the closed-loop control of the machine terminal voltage. This ensures machine operation at a constant flux. At base speed the output of the flux control block saturates; then above base speed the machine operates at a reduced flux.

Since I_f' and I_m' are constant, the torque depends only on δ, according to equation (11.73). When in steady state, the drive will be operating at a δ which balances the load torque and the speed will be close to the speed command ω_m^*. An increase in speed command will produce a positive speed error, which will increase δ^*. Consequently δ and T will increase and the machine will accelerate to the desired speed. A

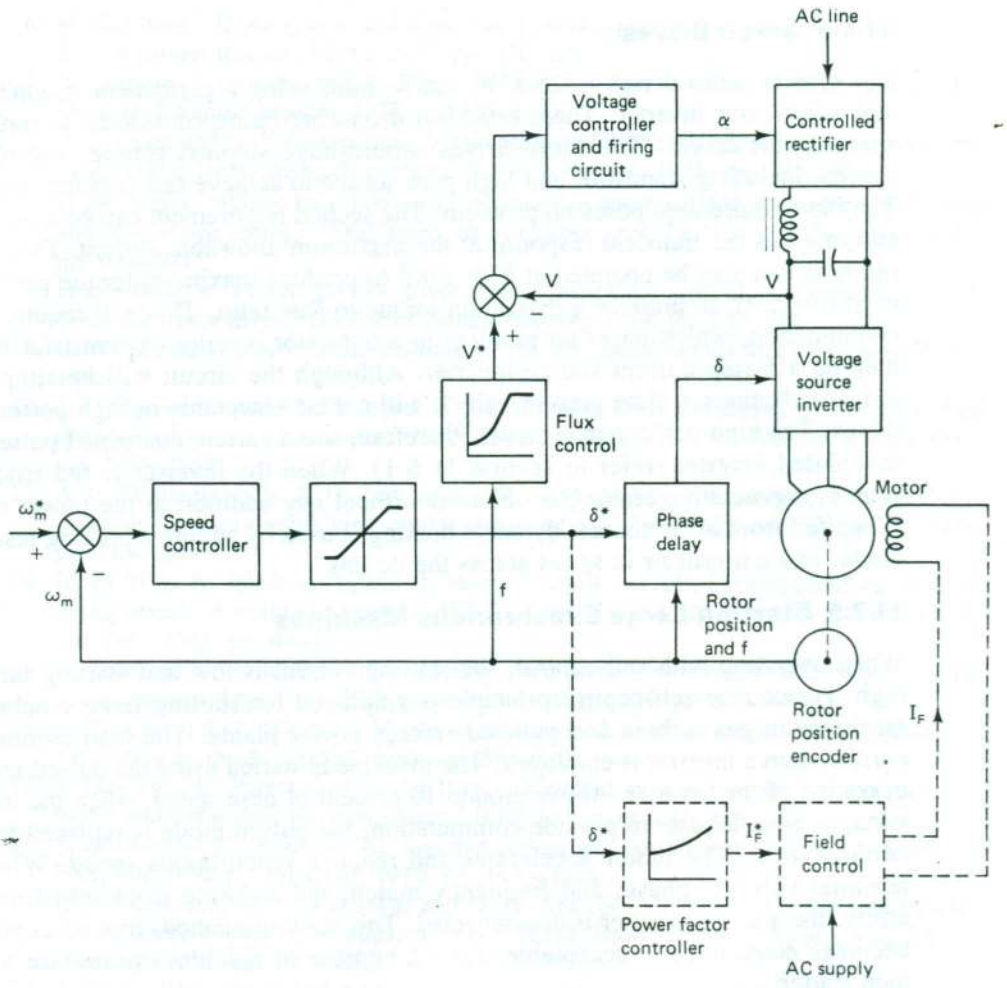


Figure 11.30 Brushless or commutatorless dc motor drive fed from a voltage source inverter.

decrease in speed command will produce negative δ^*, and the motor will decelerate under braking. Just when the speed error becomes positive, the operation will shift back to motoring and the machine will settle to the desired speed.

In a wound-field motor, the portion shown by the dotted lines can be added to operate the motor at unity power factor. Following the analysis presented in section 11.5.4, the machine will operate at unity power factor below base speed if I_f' is changed with δ according to equation (11.79). The power factor controller changes the field current command I_f^* with δ^* according to equation (11.79). When the drive operates above base speed, for each δ^*, I_F^* must be inversely changed with speed.

11.7.4 Servo Drives

Low-power servo drives (<25 kW) can be built using a permanent magnet motor and a transistor inverter. These brushless drives are being considered to replace dc servo motor drives. The servo drives should have stepless torque control at all speeds, including standstill, and high peak torque to achieve fast transient response. The first requirement poses no problem. The second requirement can be achieved by carrying out the transient response at the maximum allowable current. Further, the machine can also be operated at $\delta' = \pm 90°$ to produce maximum torque per ampere or at $\delta = \pm 90°$ to produce a maximum torque-to-flux ratio. This will require forced commutation, which poses no problem in a transistor inverter. A transistor inverter may be a 6-step current source inverter. Although the circuit will be simple, the torque pulsations will be present, which will not be acceptable in high-performance drives. The high-performance drives, therefore, use a current controlled pulse-width modulated inverter (refer to section 11.6.1). When the inverter is fed from a dc source, regenerative braking is obtained without any addition to the power circuit. When fed from an ac source, dynamic braking is used by adding a braking resistor, a diode, and a transistor in series across the dc link.

11.7.5 Starting Large Synchronous Machines

When operating with self-control, the starting current is low and starting torque is high. Hence, the self-control principle is employed for starting large synchronous machines in gas turbine and pumped storage power plants. The load commutated current source inverter is employed. The machine is started using the pulsed mode of operation of the inverter. Above around 10 percent of base speed, when the induced voltages are adequate to provide commutation, the pulsed mode is replaced by load commutation. The motor accelerates and reaches synchronous speed. When the terminal voltage, phase, and frequency match, the machine is switched into the utility line and the inverter is disconnected. This starting method, though expensive, becomes economically acceptable when a number of machines timeshare a common starter.

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PROBLEMS

11.1 A brushless dc motor drive with a load commutated current source inverter has a synchronous motor with the following name plate data: 10 MW, 3-phase, 11 kV, 60 Hz, 6 pole, Y-connected, unity power factor. The parameters are $X_s = 12\ \Omega$, subtransient reactance = 3 Ω, R_s = negligible. The field is controlled to maintain a constant flux below base speed and the rated terminal voltage above base speed. The drive operates at a constant commutation lead angle of 60°.

1. Calculate the margin angle, torque, and terminal voltage when the motor operates at the rated armature current (rms value) and 1200 rpm.

2. If the current is restricted to rated (rms) value and a minimum margin angle of 10° must be maintained for reliable commutation, calculate the highest speed at which the drive can operate.

3. Calculate δ_o' for 1.

Neglect core loss, friction, and windage.

11.2 A 1100 kW, 3-phase, 6600 V, 60 Hz, 6 pole, Y-connected, unity power factor synchronous motor has the following parameters: $X_s = 36\ \Omega$, $R_s = 1.2\ \Omega$, and subtransient reactance of 10 Ω.

This motor is employed in a load commutated current source inverter drive. The field current is controlled to keep a constant flux below base speed. The drive operation above base speed is not required. The rms value of the armature current is not allowed to exceed 1.5 times the rated current. Constant commutation lead angle control is employed.

1. Calculate β_c so that a minimum margin angle of 10° is available for all operating points.

2. With the value of β_c fixed as in 1, calculate the power factor, margin angle, torque, and dc link voltage for half the rated speed and half the rated current.

Neglect friction, windage, and core loss.

11.3 The drive of problem 11.2 is now controlled by constant no-load torque angle control.

1. Calculate δ'_{oc} so that a minimum margin angle of 10° is available for all operating points.
2. With δ'_{oc} fixed in 1, calculate the power factor, torque, dc link voltage, and margin angle for half the rated speed and half the rated current.

11.4 A brushless dc motor drive is fed by a load commutated current source inverter. The wound-field synchronous motor used in the drive has the following details: 5 MW, 3-phase, 6600 V, 6 pole, 60 Hz, Y-connected, 0.9 (lagging) power factor, $X_s = 10.8\ \Omega$, subtransient reactance = 3 Ω, and negligible R_s.

The field is controlled to maintain a constant flux up to base speed and the rated terminal voltage above base speed. The armature current is not allowed to exceed its rated (rms) value. The desired speed range is from 20 percent of base speed to 50 percent above base speed.

1. Calculate β_c so that a minimum margin angle of 15° is available for all operating points.
2. With the value of β_c fixed as in 1, calculate the power factor and torque at the rated current (rms) and the rated speed.
3. Repeat 2 for 20 percent of the rated current and the rated speed.

Neglect core loss, friction, and windage.

11.5 Repeat problem 11.4 when constant no-load torque angle control is employed.

11.6 The wound-field synchronous motor of a load commutated current source inverter fed brushless dc drive has the following details: 15 MW, 3-phase, 11 kV, 4 pole, 60 Hz, Y-connected, 0.85 (lagging) power factor, $X_s = 4\ \Omega$, subtransient reactance = 1.2 Ω, and negligible R_s.

The field is controlled to maintain constant flux. The maximum armature current (rms) allowed is twice its rated value.

1. If the machine is controlled by constant commutation lead angle control, calculate β_c to get a margin angle of 10° at the maximum allowable current. Calculate and plot ϕ, β, δ', and I'_f versus I_s curves.
2. Repeat 1 when the drive is controlled by constant no-load torque angle control and constant margin angle control.

Neglect friction, windage, and core loss.

11.7 A load commutated brushless ac motor drive employs the synchronous motor of problem 11.1. The drive is operated below base speed and at a constant flux. Calculate and plot, δ' versus I_s, and I'_f versus I_s curves for $\beta_c = 60°$.

11.8 Calculate and plot speed-torque curves for constant commutation lead angle control, constant no-load torque angle control, and constant margin angle control of the drive of problem 11.2 for a dc link voltage of 6000 Volts. At 1.5 times the rated current, all three control schemes are designed to give a minimum margin angle of 10°.

11.9 A 10 kW, 3-phase, 440 V, 60 Hz, 4 pole, Y-connected permanent magnet motor has the following parameters: $X_s = 16\ \Omega$, Rated power factor = 0.85 (lagging), and negligible R_s. Neglect core loss, friction, and windage. The motor is fed from a current source inverter to form a brushless dc motor drive. Calculate and plot δ' as a function of I_s to get the unity power factor operation for all operating points below base speed. Also obtain the T versus I_s plot and the motor terminal voltage versus I_s plot for the rated speed.

11.10 A 25 kW, 3-phase, 440 V, 50 Hz, 4 pole, Y-connected permanent magnet synchronous motor has the following parameters: $X_s = 15\ \Omega$, negligible R_s, and the rated power factor = 1.0.

The machine is controlled from a current source inverter at a torque angle δ' such that at the rated current $I'_m = I'_f$.

1. Calculate the torque at half the rated armature current.
2. What will be the motor terminal voltage at half the rated armature current and 1000 rpm?

Neglect harmonics, core loss, friction, and windage.

11.11 The drive of problem 11.10 operates above base speed by keeping the armature current constant at the rated value and increasing δ'. Calculate the torque, and power factor for a motor speed of 2000 rpm and rated terminal voltage.

11.12 A brushless dc motor drive employs a load commutated current source inverter-fed wound-field synchronous motor. The drive is controlled by constant commutation lead angle control. Derive expressions expressing I'_f as a function of I_s for the constant flux operation of the machine. Draw a block diagram of such a closed-loop speed control system.

11.13 Repeat problem 11.12 for a drive controlled by the approximate constant margin angle control given by equation (11.36).

11.14 A brushless dc motor drive employs a permanent magnet motor fed from a current source inverter. Draw the block diagram of a closed-loop speed control system which gives motor operation at unity power factor below base speed.

2

DC Motors

DC drives are widely used in applications requiring adjustable speed, good speed regulation and frequent starting, braking, and reversing. Some important applications are rolling mills, paper mills, mine winders, machine tools, and traction.

The present chapter describes the steady-state speed torque relations, methods of speed control, starting and braking, and the dynamics of dc motors.

Commonly used dc motors are shown in figure 2.1. In the case of a separately excited motor, the field and armature voltages can be controlled independent of each other. In a shunt motor, the field and armature are connected to a common source. Therefore, an independent control of the field current or armature voltage can be done only by inserting a resistance in the appropriate circuit; however, this is an inefficient method of control. In the case of a series motor, the field current is the same as the armature current, and, therefore, field flux is a function of armature current. In a cumulative compound motor, the magnetomotive force of a series field is a function of the armature current and is in the same direction as the mmf of the shunt field.

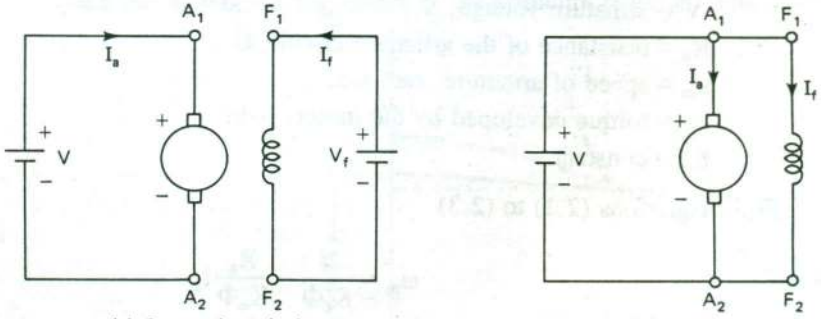
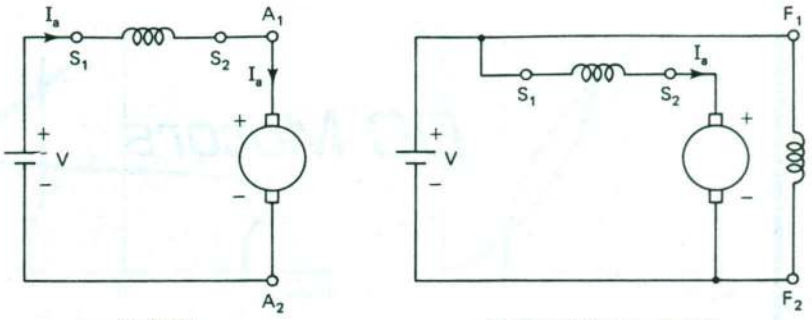


Figure 2.1 Commonly used dc motors *(continued on next page)*.



(c) Series (d) Cumulative compound

Figure 2.1 *(continued).*

2.1 STEADY-STATE SPEED TORQUE RELATIONS

The steady-state equivalent circuit of the armature of a dc machine is shown in figure 2.2. The resistor R_a is the resistance of the armature circuit. For separately excited and shunt motors, it is equal to the resistance of the armature winding; and for series and compound motors it is the sum of armature and series field winding resistances. Basic equations of a dc motor are

$$E = K_e \Phi \omega_m \tag{2.1}$$

$$V = E + R_a I_a \tag{2.2}$$

$$T = K_e \Phi I_a \tag{2.3}$$

where

Φ = flux per pole, Webers
I_a = armature current, A
V = armature voltage, V
R_a = resistance of the armature circuit, Ω
ω_m = speed of armature, rad/sec.
T = torque developed by the motor, N-m
K_e = constant

From equations (2.1) to (2.3)

$$\omega_m = \frac{V}{K_e \Phi} - \frac{R_a}{K_e \Phi} I_a \tag{2.4}$$

$$= \frac{V}{K_e \Phi} - \frac{R_a}{(K_e \Phi)^2} T \tag{2.5}$$

Equations (2.1) to (2.5) are applicable to all three types of dc motors: namely, separately (or shunt) excited, series, and compound motors.

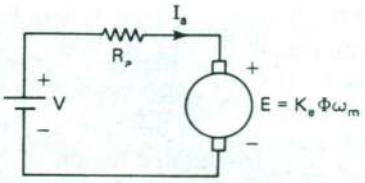


Figure 2.2 Steady-state equivalent circuit of the armature circuit of a dc motor.

In the case of separately excited motors, if the field voltage is maintained consant, one can assume the flux is practically constant as the torque changes. Let

$$K_e\Phi = K(\text{constant}) \tag{2.6}$$

Then from equations (2.1), (2.3), and (2.4) to (2.6)

$$T = KI_a \tag{2.7}$$

$$E = K\omega_m \tag{2.8}$$

$$\omega_m = \frac{V}{K} - \frac{R_a}{K} I_a \tag{2.9}$$

$$= \frac{V}{K} - \frac{R_a}{K^2} T \tag{2.10}$$

Thus, the speed torque characteristic of a separately excited motor is a straight line, as shown in figure 2.3. The no-load speed ω_{mo} is determined by the values of the armature voltage and the field excitation. Speed decreases as torque increases and speed regulation depends on the armature circuit resistance (equation (2.10)). In practice, due to armature reaction, the flux decreases as torque increases, even when the field current is kept constant. So the decrease in speed will be less than that given by equation (2.10). At high torque values, the field may be weakened to such an extent that the slope of the characteristic may become positive, leading to an unstable operation. In this situation, a relatively weak additional series field is employed to reduce the demagnetizing effect of the armature reaction. The usual drop

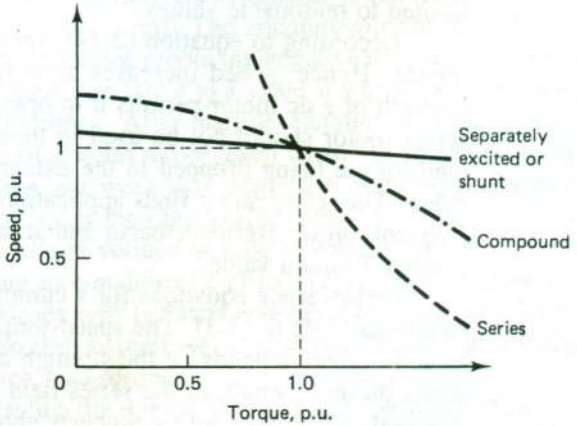


Figure 2.3 Speed-torque characteristics of dc motors.

in speed from no load to full load, in the case of a medium size motor, is of the order of five percent. Separately excited motors are employed in applications requiring good speed regulation and adjustable speed.

In the case of series motors, the flux is a function of armature current. In the unsaturated region of the magnetization characteristic, Φ can be assumed to be proportional to I_a. Thus,

$$\Phi = K_f I_a \tag{2.11}$$

Substituting in equations (2.1), (2.4), and (2.5) gives

$$T = K_e K_f I_a^2 \tag{2.12}$$

$$\omega_m = \frac{V}{K_e K_f I_a} - \frac{R_a}{K_e K_f} \tag{2.13}$$

$$= \frac{V}{\sqrt{K_e K_f}} \frac{1}{\sqrt{T}} - \frac{R_a}{K_e K_f} \tag{2.14}$$

where the armature circuit resistance R_a is now the sum of the armature and field winding resistances.

A speed-torque characteristic of a dc series motor is shown in figure 2.3. In the case of a series motor, any increase in torque is accompanied by an increase in the armature current and, therefore, an increase in flux. Because the flux increases with torque, the speed must drop to maintain a balance between the induced voltage and the supply voltage. The characteristic is, therefore, highly drooping. A motor of standard design works at the knee point of the magnetization characteristic at the rated torque. At heavy torque overloads, the magnetic circuit saturates and the speed-torque curve approaches a straight line.

Series motors are suitable for applications requiring high starting torque and heavy torque overloads. With an increase in torque, the flux also increases; therefore, for the same increase in torque, the increase in motor current is less compared to that in a separately excited motor. Thus, during heavy torque overloads, the power overload on the source and the thermal overloading of the motor are kept limited to reasonable values.

According to equation (2.14), speed varies inversely as the square root of the torque. Hence, speed increases as torque is reduced. Generally, the mechanical strength of a dc motor permits it to operate at about twice the rated speed. Thus the series motor should not be used in those drives where there is a possibility of the load torque being dropped to the extent that the speed may exceed twice the rated value. The series motor finds applications in those drives where frequent starting and frequent torque overloads occur and at the same time the torque does not drop below a safe minimum value.

Performance equations for a cumulative compound motor can be derived from equations (2.1) to (2.3). The speed-torque characteristic is drawn in figure 2.3. The no-load speed depends on the strength of the shunt field, and the drop-in speed depends on the strength of the series field. Suitable values of no-load speed and drop-in speed can be realized by properly choosing the relative strength of the two fields. Cumulative compound motors are used in those applications where a drooping char-

acteristic similar to that of a series motor is required and at the same time the no-load speed must be limited to a safe value; typical examples are lifts, winches, and so on. It is also used in intermittent load applications, where the load varies from almost no load to very heavy loads. In these applications, a fly-wheel is mounted on the motor shaft. During a heavy load period, the drooping characteristic allows the speed to drop, and, therefore, a large proportion of the total torque demanded by the load is supplied from the energy stored in the fly-wheel. The energy lost by the fly-wheel is replenished by the motor during a light load period. This permits the use of a smaller size motor and equalizes the load on the supply system. A pressing machine is a typical example of this type of application.

The characteristics shown in figure 2.3 are called the natural speed-torque characteristics because they are obtained when a motor is operated at the rated voltage and flux, and without any external resistance in series with the armature or field.

2.2 METHODS OF SPEED CONTROL

The speed-torque relation of dc motors, equation (2.5), shows that the speed can be controlled by any of the following three methods:

1. Armature voltage control.
2. Field flux control.
3. Armature resistance control.

2.2.1 Armature Voltage Control

If the armature voltage of a separate or series excited dc motor running at a steady-state speed is reduced (by a small amount), then, according to figure 2.2, the armature current, and, therefore, the motor torque will decrease. As the motor torque will be less than the load torque, the motor will decelerate, causing speed and back emf to decrease. It will finally settle at a lower speed at which its torque equals the load torque. If the armature voltage of a separately excited motor is reduced by a large amount, it may become less than the back emf. The armature current will then reverse and the motor will work as a generator producing negative torque. This operation will continue until the motor speed has fallen to a value at which the motor back emf becomes equal to the applied voltage. After that, deceleration will occur in the manner just explained. In the case of a series motor, even when the armature voltage is changed by a larger step, it does not work as a generator, and the deceleration occurs because the motor torque is lower than the load torque.

On the other hand, if the armature voltage of a dc motor running at a steady-state speed is increased, according to equations (2.2) and (2.3), the armature current, and, therefore, the motor torque will increase and the motor will accelerate, causing the motor speed and back emf to increase. It will finally settle at a higher speed at which its torque equals the load torque.

While increasing or decreasing the speed, the armature voltage should be changed only in small steps. A large change in the armature voltage causes a large

amount of current to flow through the armature, which may damage the commutator or reduce its life.

Steady-state speed-torque characteristics of dc separately and series excited motors for different armature voltages are shown by solid lines in figures 2.4a and b, respectively. By reducing the armature voltage, a motor can be operated for any speed-torque combination lying between the natural speed-torque curve and the torque axis. In the case of a separately excited motor, the no-load speed also changes and the speed-torque characteristics for different voltages are parallel straight lines. As the armature voltage cannot be increased beyond the rated value, this method of speed control is used only to get the motor operation below its natural speed-torque characteristics.

The important feature of this method of speed control is that the nature and the slope of the speed-torque characteristics do not change with the change in speed. It provides a constant torque drive because the maximum permissible armature current, and, therefore, the maximum torque capability, of the motor remains constant at all the speeds.

The variable dc voltage can be obtained by using either of the following semiconductor converters:

1. Controlled rectifier (or ac to dc converter).
2. Chopper (dc to dc converter).

Methods 1 and 2 are described in chapters 3 and 4, respectively.

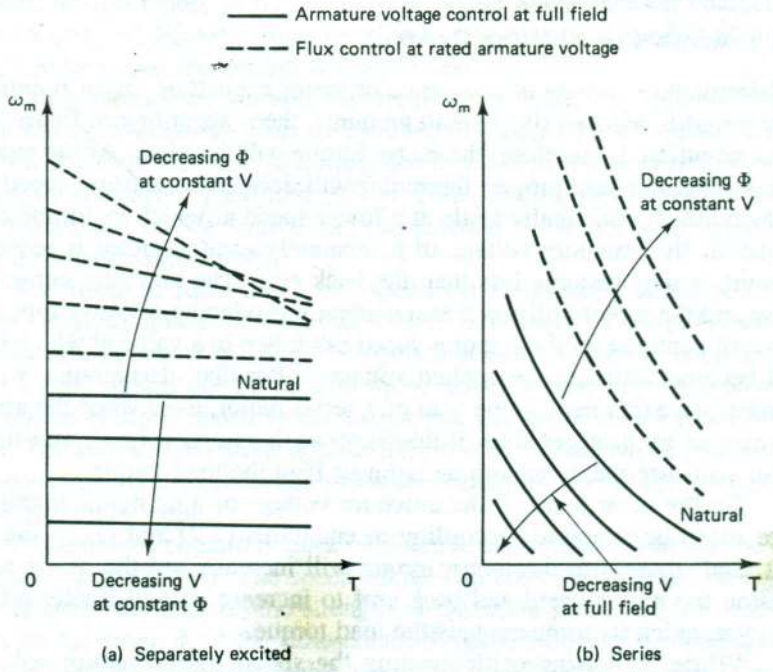


Figure 2.4 Speed-torque curves of separately excited and series motors.

2.2.2 Field Control

If the field of a separately or series excited motor running at a speed is weakened, its induced emf decreases. Because of low armature resistance, the current increases by an amount much larger than the decrease in the field flux. As a result, in spite of the weakened field, the torque is increased by a large amount, considerably exceeding the load torque. The surplus torque thus available causes the motor to accelerate and the back emf to rise. The motor will finally settle down to a new speed, higher than the previous one, at which the motor torque with the weakened field becomes equal to the load torque. Any attempt to weaken the field by a large amount will cause a dangerous inrush of current. Care should therefore be taken to weaken the field only slowly and gradually.

On the other hand, when the field of a separately excited motor is increased, the induced emf increases and often exceeds the supply voltage; thus, not only the armature current reduces but it often reverses. When this happens, the motor works as a generator and feeds energy to the supply system. This energy is obtained at the expense of the kinetic energy of the machine and the load. A rapid reduction of speed takes place and finally the motor settles at a new speed, lower than the previous one, at which the motor torque becomes equal to the load torque. In the case of a series motor, the increase in field reduces the armature current by a large amount (but does not reverse it). Because the motor torque is lower than the load torque, the motor decelerates to a lower speed at which the motor torque equals the load torque.

Steady-state speed-torque characteristics of separate and series excited motors at reduced flux are shown by dotted lines in figures 2.4a and 2.4b, respectively. At a reduced flux, for a given increase in torque, the armature current, and therefore, the armature drop, increases by a larger amount. Consequently, the back emf, and, therefore, the speed, drops by a larger amount. Thus, the lesser the flux, the greater is the slope of the speed-torque curves. At low values of flux, a decrease in flux may even lead to a decrease in speed if the torque demand is not low (fig. 2.4a).

In the case of a shunt motor, the lowest speed obtainable is that corresponding to full field with no external resistance in the field circuit. In the case of a separately excited motor, the lowest speed is limited by the heating of the field coils and the saturation of the magnetic circuit. Since at full excitation, modern machines work with an appreciable amount of saturation of the magnetic circuit, the speed can be decreased only by a small amount below the natural speed torque characteristic. The highest speed is limited by the instability of the motor due to the demagnetizing effect of armature reaction under the weak field and the mechanical strength of the motor. With a normally designed dc motor, a speed range of 1.5 to 2 times the rated speed can be obtained; and with specially designed motors, the range can be up to 6 times the rated speed. To prevent instability, separately excited motors are fitted with a relatively weak series field to assist the main field. Under the momentary heavy loads, a heavy current will strengthen the main field and tend to reduce the speed.

Field control of shunt and separately excited motors provides a constant power control because the maximum power capability of the motor remains nearly constant at all speeds. It is assumed that the maximum allowable armature current I_{amax} does not change as the field weakens. At the armature current I_{amax}, the counter emf E re-

mains constant for all speeds because the terminal voltage is held constant at V. Consequently, the allowable motor developed power E I_{amax} remains substantially constant over the speed range and the maximum allowable torque varies inversely with the speed.

The assumption that the maximum allowable armature current I_{amax} does not change with a reduction in flux is only approximate. The armature reaction becomes more effective as the main flux is reduced; therefore, the maximum current that the motor can carry without sparking at the commutator is decreased, reducing the maximum allowable developed power at high speeds.

In a separately excited motor, control of flux is obtained by the variation of the voltage across the field using a controlled rectifier or a chopper, depending on whether the main supply is ac or dc. Small size machines are connected as shunt motors, and the variation of flux is obtained by inserting a variable resistance in the field circuit. In a series motor, the control of flux is achieved by connecting a diverter resistance across the field winding. Some series motors have taps on the field winding. In these motors the flux can be controlled by changing the number of turns in the field winding.

2.2.3 Combined Armature Voltage and Field Flux Control

In drives requiring a wide range of speed control, armature voltage control is combined with field control. Armature voltage control has the advantage of retaining the maximum torque capability of the motor at all speeds. It is therefore employed wherever it can be, and field control is used only for getting speeds which cannot be obtained by armature control. In such drives, base speed is defined as the normal (rated) armature voltage full field speed. This is the speed at which the motor runs on the natural speed-torque characteristic. Speeds from standstill to base speed are obtained by the armature voltage control, and the field is maintained constant at the rated value. Speeds above the rated value cannot be obtained by armature control because the motor armature voltage must not be increased beyond the rated value. Therefore, speeds above base speed are obtained by field control, provided the load torque demand at these speeds can be met even with the reduced torque capability of the motor. Typical examples of such drives are rolling mills, coiler drives, traction, and so on. The variation of torque (T) and power (P_m) ratings of the drive with this dual control for speeds below and above base speed are shown in figure 2.5. The armature current rating is assumed to be constant for all speeds, as shown in the figure.

An exception to the practice of not using field control below base speed is made in multi-motor drives with a common armature supply. Paper mills and continuous rolling mills are examples of this type of a multi-motor drive. The combined speed variation of all the motors is done by armature voltage control. Finer control of relative speeds of various motors is achieved by field control.

Example 2.1

A 230 V, 500 rpm, 100 A separately excited dc motor has an armature resistance of 0.1 Ω. The motor is driving, under rated conditions, a load whose torque is constant and independent of speed. The speeds below the rated speed are obtained with armature

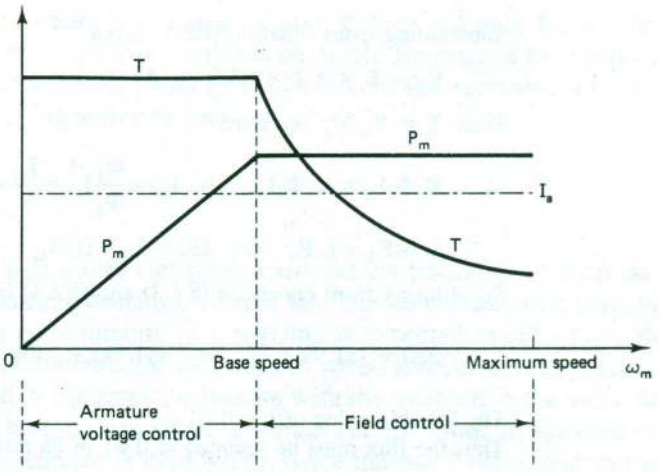


Figure 2.5 Torque and power limitations in combined armature voltage and field control.

voltage control (with full field) and the speeds above the rated speed are obtained by field control (with rated armature voltage).

1. Calculate the motor terminal voltage when the speed is 400 rpm.
2. By what amount should flux be reduced to get a motor speed of 800 rpm?

Neglect the motor's rotational losses.

Solution Back emf at 500 rpm.

$$E_1 = V - R_a I_{a1} = 230 - 10 = 220 \text{ V}$$

Rated speed $= \omega_{m1} = (500/60) \times 2\pi = 52.4$ rad/sec.
Let the flux at rated conditions $= \Phi_1$
Then $K_e\Phi_1 \times 52.4 = E_1 = 220$
or

$$K_e\Phi_1 = \frac{220}{52.4} = 4.2 \qquad \text{(E2.1)}$$

1. Back emf at 400 rpm,

$$E_2 = E_1 \times \frac{400}{500} = 220 \times \left(\frac{400}{500}\right) = 176 \text{ V}$$

Since the load torque is constant, $I_a = 100$ A
Now the motor terminal voltage

$$= E_2 + I_a R_a$$
$$= 176 + 10 = 186 \text{ V}$$

2. Let the new flux $\Phi_2 = k\Phi_1$ (E2.2)

Since $E = K_e\Phi\omega_m$

$$E_3 = K_e\Phi_2 \times \frac{800 \times 2\pi}{60}$$
$$= K_e k\Phi_1 \times 83.8$$

Substituting from equation (E2.1) gives

$$E_3 = k \times 4.2 \times 83.8 = 351.9k \tag{E2.3}$$

Since $T = K_e \Phi I_a$, we have

$$K_e \Phi_1 I_{a1} = K_e \Phi_2 I_{a2} \quad \text{or} \quad I_{a2} = \frac{\Phi_1}{\Phi_2} I_{a1} = \frac{I_{a1}}{k} = \frac{100}{k} \tag{E2.4}$$

$$V = E_3 + I_{a2} R_a \quad \text{or} \quad 230 = E_3 + 0.1 I_{a2} \tag{E2.5}$$

Substituting from equations (E2.3) and (E2.4) into equation (E2.5) gives

$$230 = 351.9k + \frac{10}{k} \quad \text{or} \quad 351.9k^2 - 230k + 10 = 0 \quad \text{or} \quad k = 0.61 \quad \text{or} \quad 0.05$$

The feasible value of $k = 0.61$.
Thus the flux must be reduced to 0.61 of its rated value.

2.2.4 Armature Resistance Control

Speed torque characteristics of separately excited (or shunt) and series motors for various values of external resistance R_e in series with the armature are shown in figures 2.6a and 2.6b, respectively.

The main drawback of this method of speed control is its poor efficiency. For example, for a constant torque load, the combined power input to the motor (both for separately excited and series motors) and the series resistance remains constant; while the power supplied to the load decreases in proportion to the speed. Thus the percentage efficiency of the motor is the same as the speed of the motor expressed as a percentage of the rated speed; at 10 percent of the rated speed, motor efficiency is just 10 percent.

Figure 2.6a shows that the armature resistance control changes the nature of the speed-torque characteristics of a separately excited (or shunt) motor from that of nearly a constant speed at all torques to variable speed characteristics. Because of this and because of the poor efficiency, this method is seldom used with separately excited motors, except for getting speeds which are required for very short times.

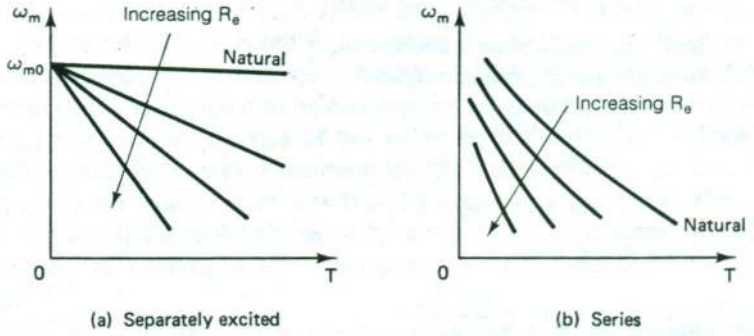


Figure 2.6 Speed-torque curves of dc motors with resistance control.

For intermittent drives where low speed running is required only for a short time, the decrease in the overall efficiency will not be much. Because of the simplicity and low initial cost, this method is found quite convenient and economical for intermittent duty drives employing series motors.

2.3 STARTING

The maximum current that a dc motor can safely carry during transients of short duration is limited by the maximum armature current that can be commutated without sparking. Theoretically, the commutating pole winding is expected to cancel out all the voltages opposing the commutation at all values of speed and current. In practice it is found that the cancellation becomes inadequate with the increase in the value of the current, and hence sparking occurs above a certain current limit. In the case of an uncompensated machine, current is kept within twice the rated value; and in the case of specially designed and compensated machines, it is allowed up to 3.5 times the rated value.

If started with full voltage across its terminals, a medium size motor will have to carry current of the order of 20 times the rated current. Such a high current will damage the motor due to heavy sparking at the commutator and heating of the winding. Therefore, it becomes necessary to limit the current to a safe value during starting. This is achieved by reducing the voltage across the motor terminals and progressively increasing it as the motor speed increases. The motor voltage is reduced either by decreasing the source voltage or by dropping a part of the source voltage across a series connected resistance.

In applications requiring adjustable speed, a controller is provided for the control of speed. The same controller can also be employed for limiting the current during starting. Where control of speed is not required, a starter is used to limit the current. In applications not requiring frequent starting, the most widely used method is to insert an extra sectionalized resistance in the armature circuit and to cut it gradually such that the motor current does not exceed a safe value and at the same time the torque developed by the motor remains higher than the load torque.

2.4 BRAKING

The reasons for using electric braking are given in section 1.2.5.

During braking, the motor essentially works as a generator. With the direction of current shown in the steady-state equivalent circuit of figure 2.2, the machine develops torque in the positive direction and converts electrical energy into mechanical energy, which is absorbed by the load. If by some means the motor armature current is reversed while maintaining the flux in the same direction, the motor torque will reverse and the machine will work as a generator, absorbing mechanical energy from the load and converting it into electrical energy. The mechanical energy is obtained from the load either from the energy stored in the inertia of the motor load system or from the active load torque on the motor shaft. The braking operation is classified in

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accordance with the manner in which the generated electrical energy is disposed of. There are three methods of braking a dc motor:

1. Regenerative braking.
2. Dynamic braking or rheostatic braking.
3. Plugging or reverse voltage braking.

2.4.1 Regenerative Braking

In regenerative braking, the energy generated is supplied to the source. Usually the source will not have the ability to store the energy. The energy supplied is diverted to other loads connected to the source, where it is usefully employed and the source is relieved from supplying this much energy. If the source does not have the ability to store energy nor are there other loads connected to the source, regenerative braking cannot be employed.

Separately Excited Motor

The steady-state equivalent circuit of a separately excited motor and source is given in figure 2.2. If by some method the induced emf E is made greater than the source voltage V, the current will reverse. The machine will work as a generator and the source will act as a sink of energy, thus giving regenerative braking. It may be emphasized that for regenerative braking to take place, the source-motor circuit should have the ability to carry current in either direction, and the source must have the ability to absorb energy. The induced emf E can be made greater than the source voltage V either by increasing E or decreasing V.

In those applications where the motor is supplied by its own unit source, V can be varied and kept below E for all speeds greater than zero. For speeds less than the rated no-load speed, the field current is maintained at the rated value. By appropriately varying V with E, the desired braking torque can be obtained. For speeds greater than the rated no-load speed, V is fixed at the rated value and the field current is varied to adjust E to get the desired braking torque. The field control also ensures that E does not exceed the rated motor voltage by a substantial amount. The speed-torque characteristics are shown in figure 2.7 for speeds below and above the rated no-load speed. They are obtained when the motoring characteristics of figure 2.4a are extended into the second quadrant. They can be calculated from equation (2.10) by using appropriate values of K.

In some applications, the source voltage V cannot be changed. For example, in electric traction, the same source caters to a number of loads which may be operating simultaneously and may require a constant voltage. In these applications, regenerative braking occurs only when E exceeds V. This happens when the speed increases beyond the rated no-load speed. For speeds substantially greater than the rated no-load speed, the field must be weakened to restrict the braking current and torque. For speeds lower than the rated no-load speeds, E can exceed V only by strengthening the field. Since the motor field is normally designed to operate at the knee point of the magnetization characteristic under rated conditions, the flux can be increased only by a small amount by increasing the field current. The heating of the field winding also does not permit the field current to be increased beyond a certain

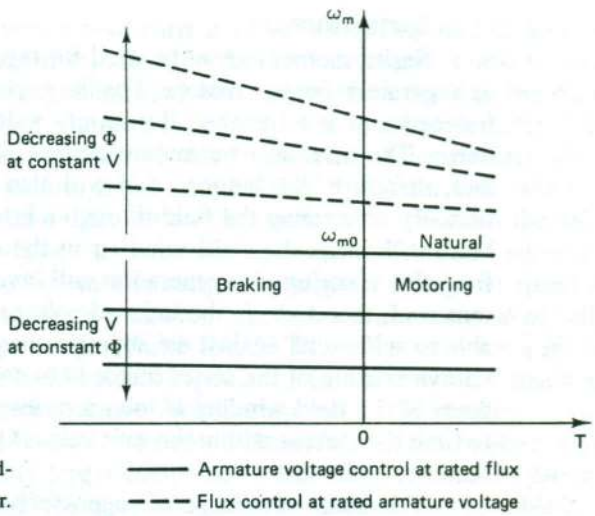


Figure 2.7 Regenerative braking speed-torque curves of a separately excited motor.

value. Therefore, when the machine is fed by a constant voltage source, regenerative braking can be employed only for speeds above and slightly below base speed.

When fed by a constant voltage source, regenerative braking below base speed can also be obtained by connecting a step-up chopper between the machine and the source. By the adjustment of the step-up ratio, the chopper output voltage can be made to exceed the source voltage even for very low values of the terminal voltage of the machine. This allows the regenerative braking operating down to very low speeds. This method of regenerative braking is described in section 4.4.

The variations of torque, power and motor current ratings shown in figure 2.5 for the motoring operation are also applicable for the regenerative braking operation.

Example 2.2

The separately excited motor of example 2.1 is now coupled to an overhauling load with a torque of 800 N-m. Determine the speed at which the motor can hold the load by regenerative braking. Source voltage is 230 V. Neglect the motor's rotational losses.

Solution: From example 2.1, $E_1 = 220$ V, $\omega_{m1} = 52.4$ rad/sec.

$$\text{Rated motor torque} = T_1 = \frac{E_1 \times I_{a1}}{\omega_{m1}} = \frac{220 \times 100}{52.4} = 420 \text{ N-m}$$

$$T = K_e \Phi I_a$$

$$\text{Thus} \quad 420 = K_e \Phi I_{a1} = K_e \Phi \cdot 100 \tag{E2.6}$$

$$\text{and} \quad 800 = K_e \Phi I_{a2} \tag{E2.7}$$

where I_{a2} is the current under regenerative braking. From equations (E2.6) and (E2.7)

$$I_{a2} = \frac{800 \times 100}{420} = 190 \text{ A}$$

back emf $E = V + I_{a2}R_a = 230 + 19 = 249$ V

The new speed

$$\omega_m = \frac{E}{E_1}\omega_{m1} = \frac{249}{220} \times 52.4 = 59.3 \text{ rad/sec} \quad \text{or} \quad 566 \text{ rpm}$$

Series motor

Series motors cannot be used for regenerative braking in the same simple way as separately excited motors. For the regenerative braking to take place, the motor induced emf must exceed the supply voltage and the armature current should reverse. The reversal of armature current will reverse the current through the field, and, therefore, the induced emf will also reverse, setting up a short-circuit condition. By connecting the field through a bridge rectifier, it is possible to maintain the current through the field winding in the same direction. However, a switch over from the motoring to regeneration will involve a change in the magnitude of the current, and, therefore, in the induced voltage. Because of this, the machine will not be able to self-excite against the supply voltage. One commonly used method of regenerative braking of the series motor is to connect it as a shunt motor. Since the resistance of the field winding is low, a series resistance is connected in the field circuit to limit the current within the safe value. One can also use a chopper, as explained in section 4.4.2.

The main advantage of regenerative braking is that the generated electrical energy is usefully employed instead of being wasted in rheostats as in the case of dynamic braking and plugging.

2.4.2 Dynamic Braking

The dynamic braking of a dc motor is effected by disconnecting it from the source and closing the armature circuit through a suitable resistance (fig. 2.8). The motor now works as a generator, producing the braking torque.

For the braking operation, the separately excited (or shunt) motor can be connected either as a separately excited generator (fig. 2.8b), where the flux remains constant, or it can be connected as a self-excited shunt generator, with the field winding in parallel with the armature (fig. 2.8c). When working with separate excitation, the speed-torque curves can be calculated from equation (2.10) by substituting $V = 0$ and the appropriate value of the armature circuit resistance R_a (R_a is now the sum of the braking resistance and the armature winding resistance). For the calculation of the speed-torque characteristics with the self-excitation, the actual mag-

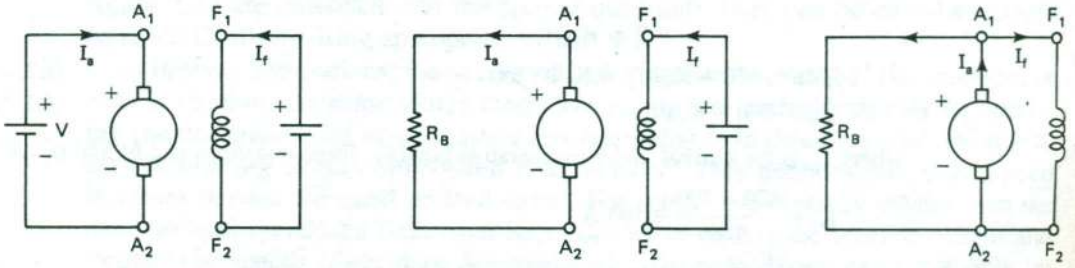


Figure 2.8 Dynamic braking of separately excited motor.

netization characteristics ($K_e\Phi$ versus field current curve) should be used to account for the saturation. When motoring at the rated speed, the induced emf is nearly equal to the supply voltage. If the braking is initiated now, a resistance equal to the starting resistance is required to limit the braking current within the safe limit. The speed-torque curves of a separately excited (or shunt) motor under the dynamic braking with separate and shunt excitations are shown in figure 2.10a, for two values of the braking resistance. In both cases, the torque decreases with speed. In the self-excitation case, the induced voltage, and, therefore, armature and field currents, decreases with speed. Thus, the decrease in torque for a given change in speed is larger compared to that for the separate excitation. Furthermore, in the case of self-excitation, the torque becomes zero at a finite speed. This is because for a given value of resistance, there is a critical value of speed ω_{mc} below which the machine fails to self-excite. After the braking torque falls to zero value, the machine coasts with only friction opposing the motion. Due to all these factors, the braking time with self-excitation is considerably larger compared to that with separate excitation. When used to hold an active load, the operation at lower speeds is not possible due to the absence of the braking torque. Therefore, self-excitation is used only for emergency stops in the event of failure of the supply.

When quick braking is required, variable resistance is used. The maximum value of the resistance is chosen such that the current at the initiation of braking at the highest speed will have the maximum permissible value. The resistance is reduced with the speed to maintain the braking torque at the highest value until $R_B = 0$.

For dynamic braking, the series motor is usually connected as a self-excited series generator. For the self-excitation, it is necessary that the current forced through the field winding by the induced emf aids the residual flux. This requirement is satisfied either by reversing the armature terminals or the field terminals (fig. 2.9). The speed torque characteristics are shown in figure 2.10b, for two values of the braking resistance. For a given value of the braking resistance, as the speed falls the torque decreases by a large amount and becomes zero at a finite speed. The speed-torque relation can be calculated from equations (2.4) and (2.5) by substituting $V = 0$ and

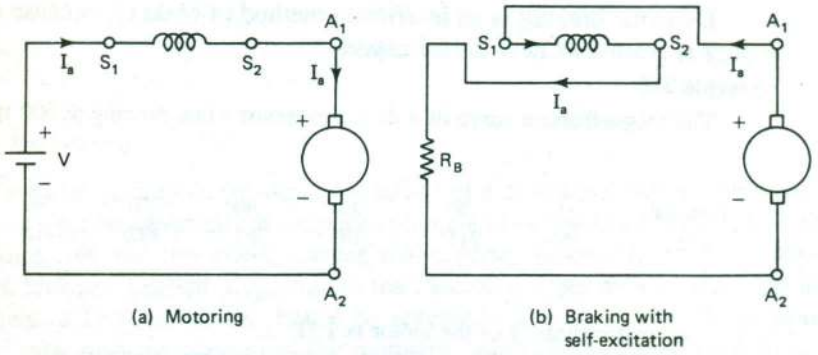


Figure 2.9 Dynamic braking of series motor.

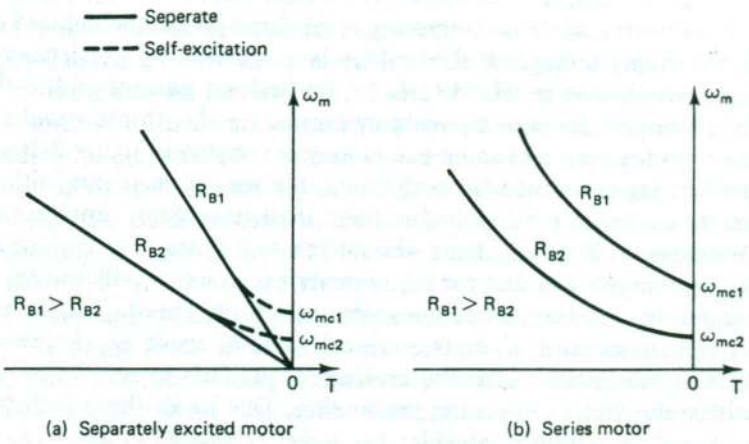


Figure 2.10 Speed-torque curves of dc motors under dynamic braking.

the appropriate value of the armature circuit resistance R_a (sum of the armature and field winding resistances and the braking resistance), and by using the actual magnetization characteristic (relation between I_f and $K_e\Phi$). According to equation (2.14), with $V = 0$, in the unsaturated region of the magnetization characteristic, the speed becomes independent of torque and is given by the following equation:

$$\omega_{mc} = \frac{R_a}{K_e K_f} \tag{2.15}$$

This is the critical speed, below which the machine fails to self-excite, and, therefore, the braking torque drops to zero.

As the braking operation is slow with self-excitation, when faster braking operation is required, the machine is connected for the separate excitation, and an appropriate resistance is inserted in series with the field to limit the current within the safe value.

Dynamic braking is an inefficient method of braking, because all the generated energy is wasted as heat in resistances.

Example 2.3

The magnetization curve of a dc series motor when running at 600 rpm is given by

current:	A	20	30	40	50	60	70	80
emf:	V	215	310	381	437	485	519	550

The total resistance of the motor is 1 Ω.

When connected for dynamic braking against a load giving a torque of 400 N-m, we wish to limit the motor speed to 500 rpm. What resistance must be connected across the motor terminals? Neglect the motor's rotational losses.

Solution

$$K_e\Phi = \frac{E}{\omega_m} \quad \text{and} \quad T = K_e\Phi I_a, \text{ N-m}.$$

From these relations and the magnetization characteristic

I_a	20	30	40	50	60	70	80
$K_e\Phi$	3.4	4.9	6.06	6.96	7.72	8.26	8.75
T	68	147	243	348	463	578	700

T and $K_e\Phi$ versus I_a plot is given in figure E2.1. For a torque of 400 N-m, $I_a = 54.3$ Amps and $K_e\Phi = 7.22$. At 400 N-m,

$$E = K_e\Phi\omega_m = 7.22 \times \frac{500}{60} \times 2\pi = 378 \text{ V}$$

Now $E = (R_B + R_a)I_a$ or $378 = (R_B + 1)54.3$ or $R_B = 6\Omega$

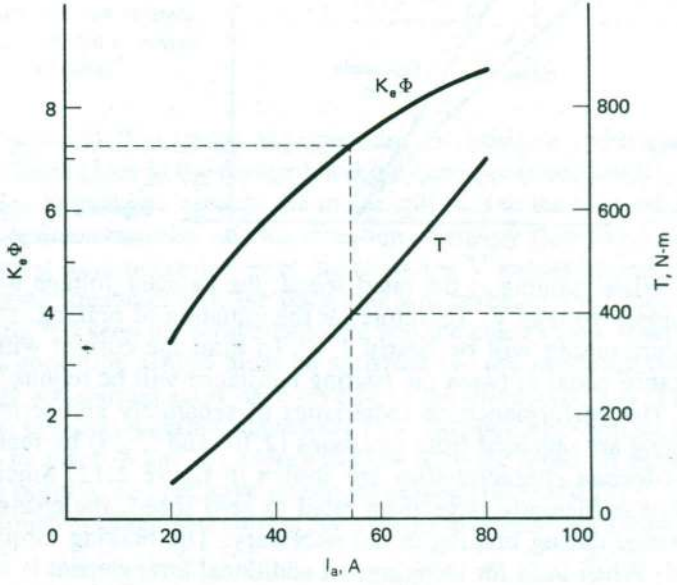


Figure E2.1

2.4.3 Plugging

If the armature terminals (or supply polarity) of a separately excited (or shunt) motor when running are reversed, the supply voltage and the induced voltage will act in the same direction and the motor current will reverse, producing braking torque. This type of braking is called plugging. In the case of a series motor, either the armature terminals or field terminals should be reversed. Reversing of both gives only the normal motoring operation. For the motoring connections of figures 2.8a and 2.9a, the plugging connections are shown in figure 2.11.

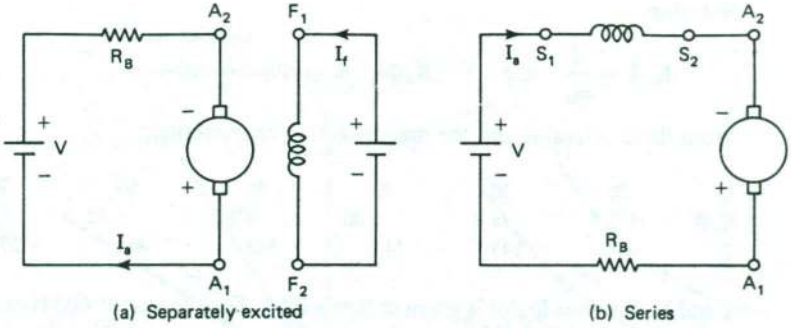


Figure 2.11 Plugging operation of dc motors.

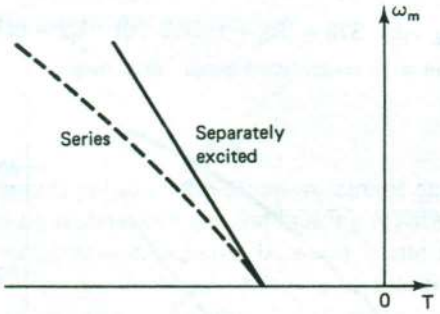


Figure 2.12 Speed-torque curves of dc motors under plugging.

When running at the rated speed, the induced voltage will be nearly equal to the supply voltage V. Therefore, at the initiation of braking, the total voltage in the armature circuit will be nearly 2 V. To limit the current within the safe value, a resistance equal to twice the starting resistance will be required.

The performance characteristics of separately and series excited motors for plugging are obtained from equations (2.10) and (2.14) by replacing V by $-V$. The speed-torque characteristics are shown in figure 2.12. Since the braking torque remains sufficiently large from rated to zero speed, the change in armature circuit resistance during braking is not necessary. The braking torque is not zero at zero speed. When used for stopping, an additional arrangement is required to disconnect the motor from the supply at or near zero speed, otherwise it will speed up in the reverse direction.

Plugging is a highly inefficient method of braking. Not only is power supplied by the load, but also the power taken from the source is wasted in resistances.

2.5 MULTIQUADRANT OPERATION OF SEPARATELY EXCITED DC MOTOR WITH REGENERATIVE BRAKING

Multiquadrant drives employing semiconductor converters mostly employ regenerative braking because of the saving in energy. Figure 2.13 shows the polarities of the source voltage, back emf, and armature current for the operation in different quadrants. These quantities—that is, V, E, and I_a— are taken positive for the forward

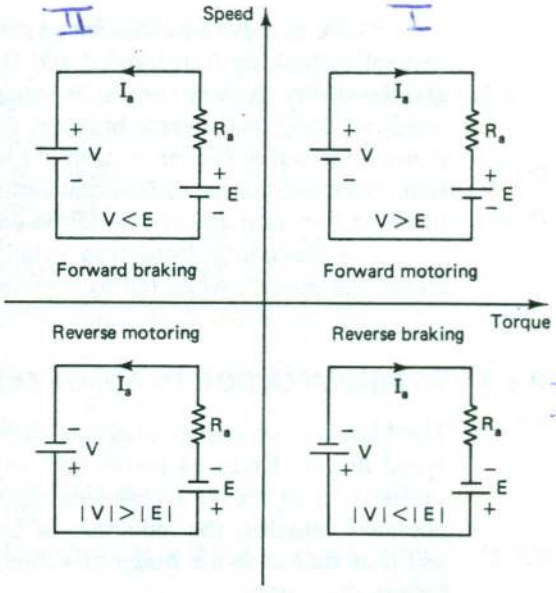


Figure 2.13 Sign of source voltage, motor current and back emf in various quadrants of motor operation.

motoring (quadrant I). The torque and speed are also positive in this quadrant. When the operation takes place in the forward braking quadrant (quadrant II), the back emf will continue to be positive because the motor still runs in the forward direction. For the torque to become negative and the direction of energy flow to reverse, the armature current must have opposite (−ve) direction and V should be less than E. When reverse motoring (quadrant III), the back emf will reverse (−ve) because of the reversal of speed. To keep the torque negative and the energy flow from the source to the motor (a condition which must be satisfied for the motoring operation), the source voltage and current must have reverse (−ve) directions, and $|V| > |E|$. When in the reverse braking quadrant (qauadrant IV), the back emf will be negative. For the torque to be positive and the energy to flow from the motor to the source, I_a and V must have positive and negative directions respectively, and $|V| < |E|$. The directions of the source voltage and current for various quadrants are listed in table 2.1.

TABLE 2.1

Quadrants of Operation	Polarity of Source Voltage	Direction of Source Current
Forward motoring (I quadrant)	+ve	+ve
Forward braking (II quadrant)	+ve	−ve
Reverse motoring (III quadrant)	−ve	−ve
Reverse braking (IV quadrant)	−ve	+ve

These entries show that for an operation consisting of forward motoring and regenerative braking (quadrants I and II), one needs a source with a positive voltage and the ability to carry current in either direction. For an operation consisting of forward motoring and reverse braking, the source should be able to provide voltage in either direction; however, it need not have the ability to carry current in either direction. For operation in all four quadrants, the source should be able to provide voltage in either direction and should allow the current to flow in either direction.

The preceding discussion is helpful in discovering the quadrants in which a motor can operate when fed by a given converter or chopper.

2.6 LOSS MINIMIZATION IN ADJUSTABLE SPEED DC DRIVES[1]

The high cost of energy provides an incentive to reduce energy losses in adjustable speed drives. Reduced losses not only reduce operating costs but also reduce the capital cost of the utility system supplying the adjustable speed drives. In battery operated vehicles, the reduction of losses allows more efficient use of the battery and thus increases the range of vehicle in terms of the distance traveled before the battery discharges.

At any operating point, characterized by a speed and a torque, a combination of the armature current and the field current can be found that meets the requirement of the operating point and minimizes the drive's losses.

A dc motor drive will have the following loss components:

1. Armature circuit loss P_a: This will be the sum of the loss in armature supply, the brush contact loss, and the armature winding loss. This loss can be expressed as $P_a = I_a^2 R_a$, where R_a is the armature circuit resistance, which accounts for losses in the armature supply, in the brush contact, and the armature winding.
2. Field circuit loss P_f, including the field circuit power supply: If the field circuit resistance is R_f, then $P_f = I_f^2 R_f$.
3. Armature core loss P_c which is a function of motor speed ω_m and air-gap flux density B_g: Let $P_c = K_c f(B_g, \omega_m)$.
4. Friction and windage loss P_w: Since P_w is a function of speed, $P_w = K_w f(\omega_m)$.
5. Stray loss P_s which is a function of the armature current I_a and the speed ω_m: Let $P_s = K_s f(I_a, \omega_m)$.

Loss due to the armature and the field supply harmonics are not considered here. When semiconductor supplies are used, the motor armature and/or the field supply voltage usually have a considerable amount of harmonics; therefore, this loss should be taken into account.

The total loss of the dc motor drive is

$$P = I_a^2 R_a + I_f^2 R_f + K_c f(B_g, \omega_m) + K_w f(\omega_m) + K_s f(I_a, \omega_m) \tag{2.16}$$

To a first order the loss component can be simplified to the following:

$$P = I_a^2 R_a + I_f^2 R_f + K_c B_g^2 \omega_m^2 + K_w \omega_m^3 + K_s I_a^2 \omega_m^2 \tag{2.17}$$

The effect of the armature reaction produced by I_a on the flux density is neglected here.

Due to the saturation of the magnetic circuit, the relation between B_g and I_f is nonlinear. Thus

$$B_g = K_f I_f \tag{2.18}$$

$$K_f = f(I_f) \tag{2.19}$$

Equation (2.19) indicates that K_f is a nonlinear function of I_f. Now,

$$T = K_e \Phi I_a = K_e' B_g I_a = K_e' K_f I_f I_a \tag{2.20}$$

or

$$I_a = \frac{T}{K_e' K_f I_f} \tag{2.21}$$

Substituting from equations (2.19) and (2.21) into equation (2.17) gives

$$P = \frac{R_a + K_s \omega_m^2}{(K_e' K_f)^2} \cdot \left(\frac{T}{I_f}\right)^2 + I_f^2 R_f + K_c K_f^2 I_f^2 \omega_m^2 + K_w \omega_m^3 \tag{2.22}$$

Equation (2.22) expresses the power loss P in terms of T, ω_m, and I_f. For a given operating point (a given torque T and a given speed ω_m), the loss will be minimal when the partial derivative of P with respect to I_f will be zero. Thus,

$$\left.\frac{\partial P}{\partial I_f}\right|_{T,\omega_m} = \frac{\partial}{\partial I_f}\left[\frac{(R_a + K_s \omega_m^2) T^2}{K_e'^2} \cdot \frac{1}{K_f^2 I_f^2} + I_f^2 (R_f + K_c K_f^2 \omega_m^2)\right] = 0$$

or

$$-\frac{2}{I_f^3} \frac{(R_a + K_s \omega_m^2) T^2}{K_e'^2 K_f^2} - \frac{2(R_a + K_s \omega_m^2) T^2}{K_e'^2} \cdot \frac{1}{K_f^3 I_f^2} \frac{\partial K_f}{\partial I_f}$$

$$+ 2 I_f (R_f + K_c K_f^2 \omega_m^2) + I_f^2 \cdot 2 K_f K_c \omega_m^2 \frac{\partial K_f}{\partial I_f} = 0$$

or

$$-\frac{(R_a + K_s \omega_m^2)}{I_f}\left(\frac{T}{K_e' K_f I_f}\right)^2 - \frac{(R_a + K_s \omega_m^2)}{K_f} \cdot \frac{\partial K_f}{\partial I_f}\left(\frac{T}{K_e' K_f I_f}\right)^2$$

$$+ I_f R_f + K_c \omega_m^2 K_f^2 I_f \left(1 + \frac{I_f}{K_f} \frac{\partial K_f}{\partial I_f}\right) = 0 \tag{2.23}$$

Substituting from equation (2.21) gives

$$-\frac{R_a + K_s \omega_m^2}{I_f} I_a^2 - \frac{R_a + K_s \omega_m^2}{K_f} \cdot \frac{\partial K_f}{\partial I_f} I_a^2 + I_f R_f + K_c \omega_m^2 K_f^2 I_f \left(1 + \frac{I_f}{K_f} \frac{\partial K_f}{\partial K_f}\right) = 0$$

or

$$(R_a + K_s \omega_m^2) I_a^2 \left(1 + \frac{I_f}{K_f} \frac{\partial K_f}{\partial I_f}\right) = I_f^2 R_f + K_c \omega_m^2 K_f^2 I_f^2 \left(1 + \frac{I_f}{K_f} \frac{\partial K_f}{\partial I_f}\right)$$

or

$$(R_a + K_s\omega_m^2)I_a^2 = I_f^2\left[\frac{R_f}{\left(1 + \frac{I_f}{K_f}\frac{\partial K_f}{\partial I_f}\right)} + K_c K_f^2 \omega_m^2\right] \tag{2.24}$$

If the magnetic circuit is assumed linear then K_f will be a constant according to equations (2.18) and (2.19). Then $\partial K_f/\partial I_f$ will be zero. Substituting this in equation (2.24) gives

$$(R_a + K_s\omega_m^2)I_a^2 = I_f^2[R_f + K_c K_f^2 \omega_m^2] \tag{2.25}$$

Equations (2.24) and (2.25) show that at a given speed and torque, the total drive loss is minimal when the armature current dependent loss is equal to the field current dependent loss. A given operating point P, characterized by the specific values of speed and torque, can be realized by a number of combinations of I_a and I_f, as shown in figure 2.14. However, there would be one combination which would satisfy equation (2.24) or equation (2.25) and thus minimize the total drive loss. The optimum solution is obtained as follows.

The speed, ω_m is taken as the independent variable. For a given ω_m, equation (2.24) or equation (2.25) is solved for I_a, and I_f, subjected to the constraint imposed by equation (2.20). Equation (2.24), which accounts for saturation, is a nonlinear algebraic equation. Some numerical technique, such as the Newton-Raphson method will have to be used to solve this equation.

In a practical system, the optimum solution must be subjected to additional constraints such as the maximum value of I_a and the minimum value of I_f from the consideration of commutation.

The implementation of the minimum loss control is shown in figure 2.15.[1] The speed command acting on the armature voltage controller sets the desired speed and maintains it constant. The torque and speed are sensed by a torque-sensor and a tachogenerator, respectively. It may be noted that T in equation (2.20) is the developed torque. It can be sensed electrically by sensing I_a and I_f. The signals T and ω_m are now fed to the minimum loss controller, whose function is to solve equation (2.24) or equation (2.25) for a given ω_m, subject to the constraint on T, and then output the optimum value of I_f. The optimum value of I_f acts as a command for the field current controller.

Equations (2.24) and (2.25) also suggest that at light loads the drive should be operated at reduced flux even below the base speed. In section 2.2.3 it was sug-

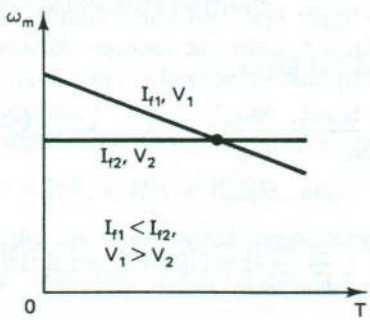


Figure 2.14

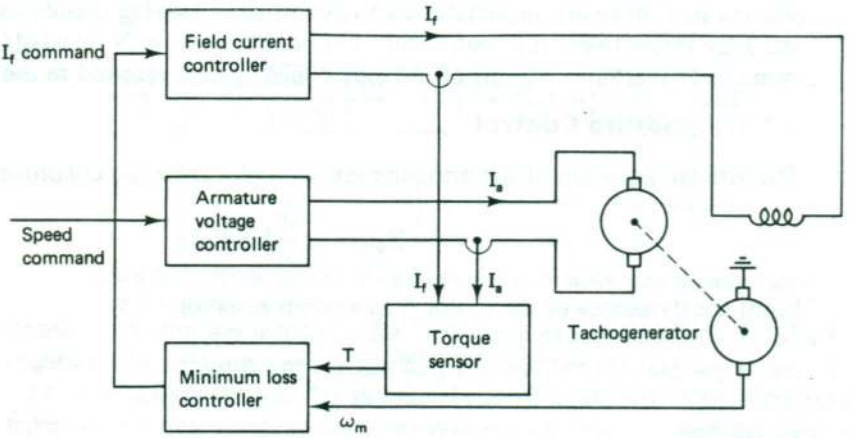


Figure 2.15 Minimum loss control of a dc drive.

gested that from the consideration of torque capability, below base speed the drive should always be operated at the rated flux. In the absence of the minimum loss control, this seems to be the most appropriate strategy. Because of the complex control, the minimum loss control is employed only when the energy saving is large enough to justify the additional cost and complexity. This will happen only when the drive runs at light loads for prolonged periods of time.

2.7 TRANSFER FUNCTIONS OF SEPARATELY EXCITED DC MOTOR

Transfer functions in an appropriate form are needed for the stability analysis and design of closed-loop drives. Closed-loop drives may be employed for speed control or position control, and the motor may be operated with armature control or field control. In this section, transfer functions of a separately excited motor with armature control and field control are derived in the form suitable for stability analysis and design of closed-loop drives.

The dynamic model of a dc separately excited motor is shown in figure 2.16. The source voltage, armature current, back emf and torque required to do the useful mechanical work are denoted by v, i_a, e, and T_M, respectively. The lower case letters have been employed for the source voltage, armature current, and back emf to em-

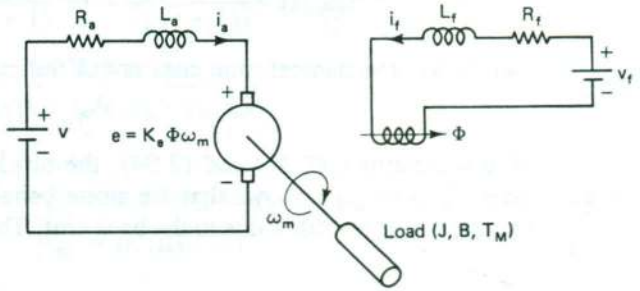


Figure 2.16 Dynamic equivalent circuit of dc separately excited motor.

phasize that these are instantaneous values of time varying quantities. The terms B and J are respectively the coefficient of viscous friction in N–m/(rad/sec.) and polar moment of inertia in Kg–m^2 of the motor load system referred to the motor shaft.

2.7.1 Armature Control

The voltage equation of the armature circuit under dynamic conditions is given by

$$v = R_a i_a + L_a \frac{di_a}{dt} + K\omega_m \tag{2.26}$$

From the dynamics of the motor load system equation (1.5),

$$J\frac{d\omega_m}{dt} = T - T_M - B\omega_m \tag{2.27}$$

Further,

$$T = Ki_a \tag{2.28}$$

Substitution in equation (2.27) yields

$$J\frac{d\omega_m}{dt} = Ki_a - T_M - B\omega_m \tag{2.29}$$

Taking the Laplace transform of equations (2.26) and (2.29), assuming zero initial conditions, gives

$$sL_aI_a(s) + R_aI_a(s) + K\omega_m(s) = V(s) \tag{2.30}$$

$$sJ\omega_m(s) + B\omega_m(s) + T_M(s) = KI_a(s) \tag{2.31}$$

where $I_a(s)$, $V(s)$, $\omega_m(s)$, and $T_M(s)$ are Laplace transforms of the variables i_a, v, ω_m, and T_M, respectively.

From equation (2.30),

$$I_a(s) = \frac{V(s)}{R_a(1 + s\tau_a)} - \frac{K}{R_a}\frac{\omega_m(s)}{(1 + s\tau_a)} = \frac{V(s) - K\omega_m(s)}{R_a(1 + s\tau_a)} \tag{2.32}$$

where the armature circuit time constant is

$$\tau_a = L_a/R_a \tag{2.33}$$

From equation (2.31)

$$\omega_m(s) = \frac{(K/B)I_a(s)}{(1 + s\tau_m)} - \frac{1}{B}\cdot\frac{T_M(s)}{(1 + s\tau_m)} = \frac{1}{B}\cdot\frac{T(s) - T_M(s)}{(1 + s\tau_m)} \tag{2.34}$$

where the mechanical time constant of the motor load system is

$$\tau_m = J/B \tag{2.35}$$

From equations (2.32) and (2.34), the block diagram shown in figure 2.17 is obtained. The figure shows that the motor behaves as a closed-loop system with an inherent speed feedback due to the back emf. There are two excitations, $V(s)$ and $T_M(s)$.

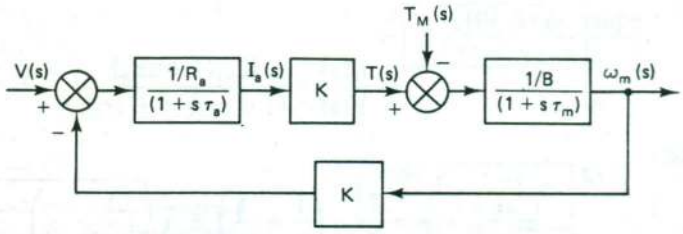


Figure 2.17 Block diagram of separately excited dc motor with armature control.

To be able to obtain the responses for both these excitations, two transfer functions are required: one relating $\omega_m(s)$ with $V(s)$ and another relating $\omega_m(s)$ with $T_M(s)$.

Let us consider the transfer function relating $\omega_m(s)$ with $V(s)$. The closed-loop dc drives employing semiconductor converters are usually operated with an outer speed control loop and an inner current control loop, as will be explained in chapter 5. The transfer function should be realized in a form appropriate for this configuration of the closed-loop drives. A suitable form of the transfer function is obtained when it is realized in two parts: one relating $I_a(s)$ with $V(s)$ and another relating $\omega_m(s)$ with $I_a(s)$. Substituting $T_M = 0$ in equation (2.34) gives

$$\omega_m(s) = \frac{K_m I_a(s)}{(1 + s\tau_m)} \tag{2.36}$$

where

$$K_m = K/B \tag{2.37}$$

Substituting from equation (2.36) into equation (2.32) yields

$$\left[1 + \frac{K^2}{R_a B} \cdot \frac{1}{(1 + s\tau_a)(1 + s\tau_m)}\right] I_a(s) = \frac{V(s)}{R_a(1 + s\tau_a)} \tag{2.38}$$

or

$$\begin{aligned}\frac{I_a(s)}{V(s)} &= \frac{B(1 + s\tau_m)}{(K^2 + R_a B) + R_a B(\tau_a + \tau_m)s + R_a B\tau_a\tau_m s^2} \\ &= \frac{B}{(K^2 + R_a B)}\left[\frac{(1 + s\tau_m)}{1 + \dfrac{R_a B(\tau_a + \tau_m)}{K^2 + R_a B}s + \dfrac{R_a B\tau_a\tau_m}{K^2 + R_a B}s^2}\right] \\ &= \frac{K_{m1}(1 + s\tau_m)}{1 + \tau_{m1}(1 + \tau_a/\tau_m)s + \tau_{m1}\tau_a s^2}\end{aligned} \tag{2.39}$$

where

$$\tau_{m1} = JR_a/(BR_a + K^2) \tag{2.40}$$

$$K_{m1} = B/(BR_a + K^2) \tag{2.41}$$

From equation (2.39),

$$\frac{I_a(s)}{V(s)} = \frac{K_{m1}(1 + s\tau_m)}{(1 + s\tau_1)(1 + s\tau_2)} \tag{2.42}$$

where

$$-\frac{1}{\tau_1}, -\frac{1}{\tau_2} = \frac{1}{2}\left[-\left(\frac{1}{\tau_a} + \frac{1}{\tau_m}\right) \pm \sqrt{\left(\frac{1}{\tau_a} + \frac{1}{\tau_m}\right)^2 - \frac{4}{\tau_{m1}\tau_a}}\right] \tag{2.43}$$

From equations (2.36) and (2.42), the block diagram shown in figure 2.18a is obtained.

The time constants τ_1 and τ_2 may be complex conjugate. This is true for large motors and converter drives with a filter inductor in series with the armature. In that case, equation (2.39) can be rewritten in a more appropriate form:

$$\frac{I_a(s)}{V(s)} = \frac{K_{m2}(1 + s\tau_m)}{s^2 + \left(\frac{1}{\tau_a} + \frac{1}{\tau_m}\right)s + \frac{1}{\tau_a\tau_{m1}}}$$

$$= \frac{K_{m2}(1 + s\tau_m)}{s^2 + 2\xi\omega_n s + \omega_n^2} \tag{2.44}$$

where

$$\omega_n^2 = \frac{1}{\tau_a\tau_{m1}} \tag{2.45}$$

$$K_{m2} = \frac{1}{\tau_a\tau_m R_a} \tag{2.46}$$

$$\xi = \frac{1}{2}(1 + \tau_a/\tau_m)\sqrt{\tau_{m1}/\tau_a} \tag{2.47}$$

Usually $\tau_a \ll \tau_m$, then from equation (2.38),

$$\left[1 + \frac{K^2}{R_a B}\left(\frac{1}{1 + s\tau_m}\right)\right] I_a(s) = \frac{V(s)}{R_a}$$

or

$$\frac{I_a(s)}{V(s)} = \frac{K_{m1}(1 + s\tau_m)}{(1 + s\tau_{m1})} \tag{2.48}$$

The simplified block diagram based on equations (2.36) and (2.48) is shown in figure 2.18b.

There are a number of applications where the load torque is proportional to the speed. In these applications the effect of the load torque on the drive performance can be taken into account by simply combining it with the viscous friction term and upgrading the value of B, equation (2.29). Then all the equations just derived will also account for the effect of the load torque on the drive performance. When the load torque is not proportional to the speed, a separate transfer function is required

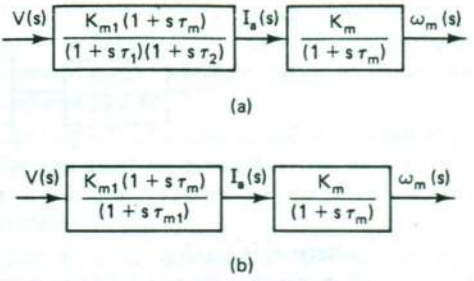


Figure 2.18 Simplified block diagrams of separately excited motor with armature control: (a) Exact, (b) Approximate.

between $\omega_m(s)$ and $T_M(s)$. This transfer function is obtained by combining the block diagram of figure 2.17 with other blocks of the system and setting the reference speed to zero.

2.7.2 Field Control

Some dc drives are operated with field control and with a constant current in the armature circuit. Usually, the armature current is maintained constant using a closed-loop system. Since the armature time constant is very small compared to the field time constant, the response time of the closed-loop system controlling the armature current can be considered zero, and thus the change in the armature current due to the variation of field current and motor speed can be neglected.

From the dynamic equivalent circuit of figure 2.16

$$V_f = R_f i_f + L_f \frac{di_f}{dt} \tag{2.49}$$

Assuming a linear magnetic circuit and noting that the armature current is constant,

$$T = K_a I_f \tag{2.50}$$

where K_a is a constant.

From the dynamics of the motor load system (equation (2.27)) and equation (2.50)

$$J\frac{d\omega_m}{dt} = K_a i_f - T_M - B\omega_m \tag{2.51}$$

Taking the Laplace transform of equations (2.49) and (2.51), assuming zero initial conditions, gives

$$V_f(s) = R_f I_f(s) + L_f s I_f(s) \tag{2.52}$$

$$Js\omega_m(s) = K_a I_f(s) - T_M(s) - B\omega_m(s) \tag{2.53}$$

From equations (2.52) and (2.53), by rearranging the terms,

$$I_f(s) = \frac{V_f(s)}{R_f(1+s\tau_f)} \tag{2.54}$$

$$\omega_m(s) = \frac{(K_a/B)I_f(s)}{(1+s\tau_m)} - \frac{T_M(s)}{B(1+s\tau_m)} \tag{2.55}$$

$$= \frac{T(s) - T_M(s)}{B(1+s\tau_m)} \tag{2.56}$$

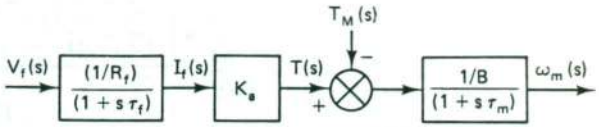


Figure 2.19 Block diagram of separately excited motor with field control.

where $\tau_f = L_f/R_f$.
Substituting from (2.54) in equation (2.55) gives

$$\omega_m(s) = \frac{K_{m3}}{(1 + s\tau_f)(1 + s\tau_m)} V_f(s) - \frac{T_M(s)}{B(1 + s\tau_m)} \tag{2.57}$$

where $$K_{m3} = \frac{K_a}{R_f B}.$$

From equations (2.50), (2.54), and (2.56) the block diagram shown in figure 2.19 is obtained. When the load torque is proportional to the speed, it can be combined with the viscous friction by upgrading the value of B. The transfer function for such a case is obtained by letting $T_M(s) = 0$ in equation (2.57).

REFERENCES

1. Alexander Kusko and Donald Galler,"Control means for minimization of losses in ac and dc motor drives," IEEE Trans. on Ind. Applications, vol. IA-19, July/Aug. 1983, pp. 561–570.

PROBLEMS

2.1 A separately excited dc motor is running at 1000 rpm, driving a load whose torque is proportional to the square of the shaft speed. The armature current is 100 A. The armature resistance drop and the rotational losses of the motor are negligible and the magnetic circuit can be assumed linear. The motor armature voltage is reduced from 200 V to 100 V. Mark and explain the correct answer.

(a) The motor speed will increase to 2000 rpm/decrease to 500 rpm/decrease to $(1000/\sqrt{2})$ rpm/remain constant.

(b) The armature current will increase to 200 A/decrease to 50 A/decrease to 25 A.

2.2 A separately excited dc motor is running at 1000 rpm, driving a load whose torque is constant. The motor armature current is 200 A, and the armature resistance drop and the rotational losses are negligible. The magnetic circuit can be assumed linear. The field current is reduced to half.
Mark and explain the correct answer.

(a) The motor speed will increase to 2000 rpm/decrease to 500 rpm/remain constant.

(b) The motor current will decrease to 100 A/increase to 400 A/remain constant.

2.3 A separately excited dc motor is running at 500 rpm, driving a load whose torque is proportional to the speed. The motor armature voltage is 220 V and the armature current is 20 A. What resistance should be inserted in the armature circuit to reduce the motor speed to 250 rpm? The armature resistance is 1 Ω.

2.4 A dc shunt motor is running at 1000 rpm, driving a load whose torque is constant at all speeds. The armature current is 100 A. The armature resistance drop can be neglected and the field circuit can be assumed linear. If the source voltage is reduced to half, calculate the motor speed and the armature current.

2.5 A 220 V, 960 rpm, 90 A separately excited dc motor has an armature resistance of 0.06 Ω. Under rated conditions the motor is driving a load whose torque is constant and independent of speed. The speeds below the rated speed are obtained with armature voltage control (with full field), and the speeds above the rated speed are obtained by field control (with rated armature voltage).

(a) Calculate the motor terminal voltage when the speed is 600 rpm.

(b) Obtain the value of flux as a percent of rated flux if the motor speed is 1200 rpm. Neglect the motor's rotational losses.

2.6 A 200 V dc series motor takes 20 A and runs at 500 rpm with a certain load on its shaft. The field winding and armature resistances are of 1 Ω each. A resistance of 9 Ω is now connected in parallel with the armature. Find the motor speed if the load on the motor shaft is removed. Neglect friction and windage.

2.7 A dc series motor is driving a load whose torque is proportional to the square of the speed. When supplied with 200 V it takes 100 A and runs at 1000 rpm. The total resistance of the armature and field is 0.1 Ω. What voltage should be applied to the motor to reduce its speed to 500 rpm? Assume the magnetic circuit of the motor is linear, and neglect friction and windage.

2.8 A dc series motor is driving a load whose torque is constant and independent of speed. The motor speed is increased from 1000 rpm to 1200 rpm by eonnecting a diverter resistance across its field. Find the ratio of the armature to the field winding current. Assume a negligible drop across the field winding and the armature resistance, and a linear magnetic circuit.

2.9 A dc series motor is driving a load whose torque is constant. The motor is running at 1000 rpm (clockwise) and the armature current is 200 A. Find the magnitude and the direction of the motor speed and armature current if the motor terminal voltage is reversed and the number of turns in the field winding are reduced to 80 percent. Neglect the voltage drop across the armature and field, and assume a linear magnetic circuit.

2.10 A motor is to be selected for driving a load having a large torque of short duration followed by a long no-load period. A fly-wheel of suitable inertia is already mounted on the load shaft. Out of the various types of dc motors (separately or shunt excited, series and compound) which one will you recommend for this application and why?

2.11 A 220-V, 960 rpm, 90 A dc separately excited motor has an armature resistance of 0.06 Ω. It is coupled to an overhauling load with a torque of 300 N-m. Determine the speed at which the motor can hold the load by regenerative braking.

2.12 The motor-load system of problem 2.11 is now braked by dynamic braking instead of regenerative braking. It is required that the machine should hold the load at 600 rpm. Calculate the value of external resistance to be connected across the armature.

2.13 The following figures give the magnetization curve of a dc series motor when running at 960 rpm:

field current	A	20	40	60	80	100
e.m.f.	V	261	540	738	882	945

The total resistance of the armature and field is 0.3 Ω. When connected for dynamic braking against a load whose torque is 600 N-m, it is desired to limit the motor speed to 750 rpm. What resistance must be connected across the motor terminals? Neglect the motor's rotational loss.

2.14 The motor of problem 2.11 is braked by plugging from an initial speed of 800 rpm. Calculate (a) the resistance to be placed in the armature circuit to limit the initial braking current to twice the full load value, (b) the initial braking torque, and (c) the torque when the speed has fallen to zero.

2.15 A dc separately excited motor is supplied by a dc source which can carry current in either direction. The motor is running on no load with a weak field. Now the field current is increased. State and explain the various operations (braking, motoring) the motor will have before it settles at a new steady-state speed.

3

Rectifier Control of DC Motors

Controlled rectifier fed dc drives are widely used in applications requiring a wide range of speed control and/or frequent starting, braking, and reversing. Some prominent applications are in rolling mills, paper mills, printing presses, mine winders, machine tools.

The line diagram of a controlled rectifier-fed separately excited dc motor drive is shown in figure 3.1. The maximum dc output voltage of the rectifier under continuous conduction should be equal to the rated armature voltage of the motor. If the ac

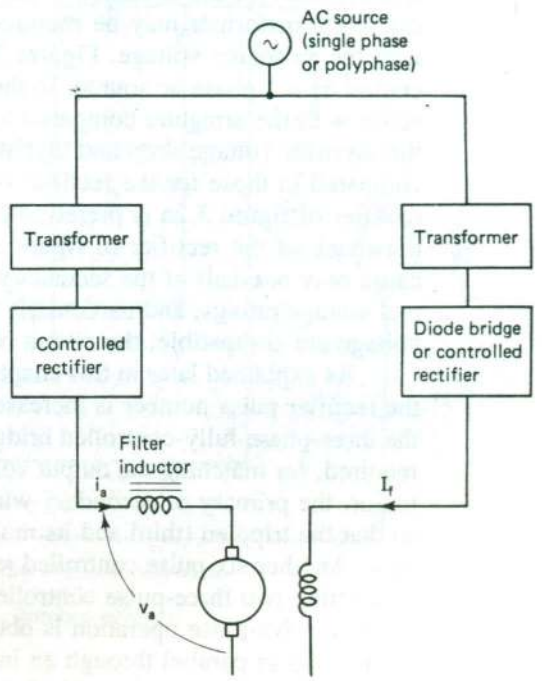


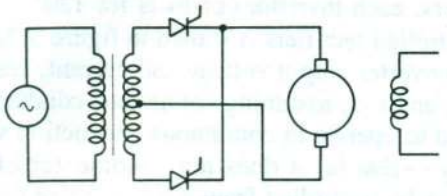
Figure 3.1 Line diagram of a controlled-rectifier fed dc motor drive.

source voltage magnitude is such that this requirement is satisfied, then some rectifiers can be directly connected to the ac source; otherwise a transformer with a suitable turns ratio is inserted between the ac source and the rectifier. A filter inductor is sometimes connected between the rectifier and the motor armature to reduce ripple in the motor current. This improves the motor performance. Usually, the field is supplied through a transformer and a diode bridge from the same ac source which supplies the armature. The transformer turns ratio is chosen to make the field voltage equal to the rated value. If field control is also required, the diode bridge is replaced by a controlled rectifier.

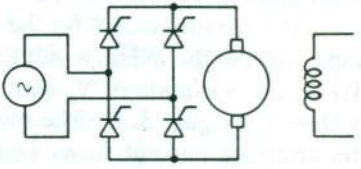
3.1 CONTROLLED RECTIFIER CIRCUITS

There are a number of controlled rectifier circuits, some fed from a 1-phase supply and others from a 3-phase supply. For the motor control, controlled rectifier circuits are classified as fully-controlled and half-controlled rectifiers. Some fully-controlled and half-controlled rectifiers are shown in figures 3.2 and 3.4 respectively. Single phase controlled rectifiers are employed up to a rating of 10 kW and in some special cases up to 50 kW. For higher power ratings 3-phase controlled rectifiers are employed. In those applications where only the 1-phase supply is available, such as in main line traction, 1-phase controlled rectifiers are also employed for ratings up to a few thousand kilowatts. In figures 3.2 and 3.4, the transformer is shown only when it is essential for the operation of the controlled rectifier. For other controlled rectifiers, the transformer may be required if the motor voltage rating is not compatible with the ac source voltage. Figures 3.2a and b show 2-pulse fully-controlled rectifiers fed by a 1-phase ac source. In the rectifier of figure 3.2a, only one thyristor is in series with the armature compared to two for the rectifier of figure 3.2b. Therefore, the thyristor voltage drop and thyristor losses are half for the rectifier of figure 3.2a compared to those for the rectifier of figure 3.2b. Because of these advantages, the rectifier of figure 3.2a is preferred for the control of low voltage motors. The main drawback of the rectifier of figure 3.2a is that a bulky transformer is required because only one-half of the secondary winding carries current at any instant. For normal voltage ratings, and particularly when the motor voltage rating and the ac source voltage are compatible, the circuit of figure 3.2b is employed.

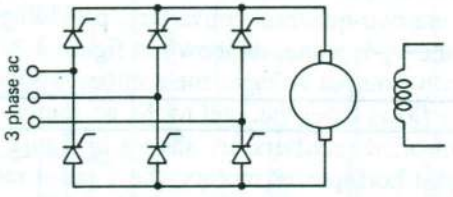
As explained later in this chapter, the performance of a drive is improved when the rectifier pulse number is increased. Six-pulse operation is realized by employing the three-phase fully-controlled bridge rectifier of figure 3.2c. When a transformer is required, for matching the output voltage of the rectifier and the voltage rating of the motor, the primary or secondary windings of the transformer are connected in delta so that the tripplen (third and its multiple) harmonics of the magnetizing current can flow. Another six-pulse controlled rectifier is shown in figure 3.2d. It is obtained by connecting two three-pulse controlled rectifiers in parallel through an interphase reactor. Twelve-pulse operation is obtained by connecting two controlled rectifiers of figure 3.2d in parallel through an interphase reactor. The two rectifiers are supplied by two transformer banks with their primaries connected in star and delta, respectively.[1,2] Twelve-pulse operation can also be obtained by connecting two six-pulse bridge controlled rectifiers of figure 3.2c in series and supplying them with a trans-



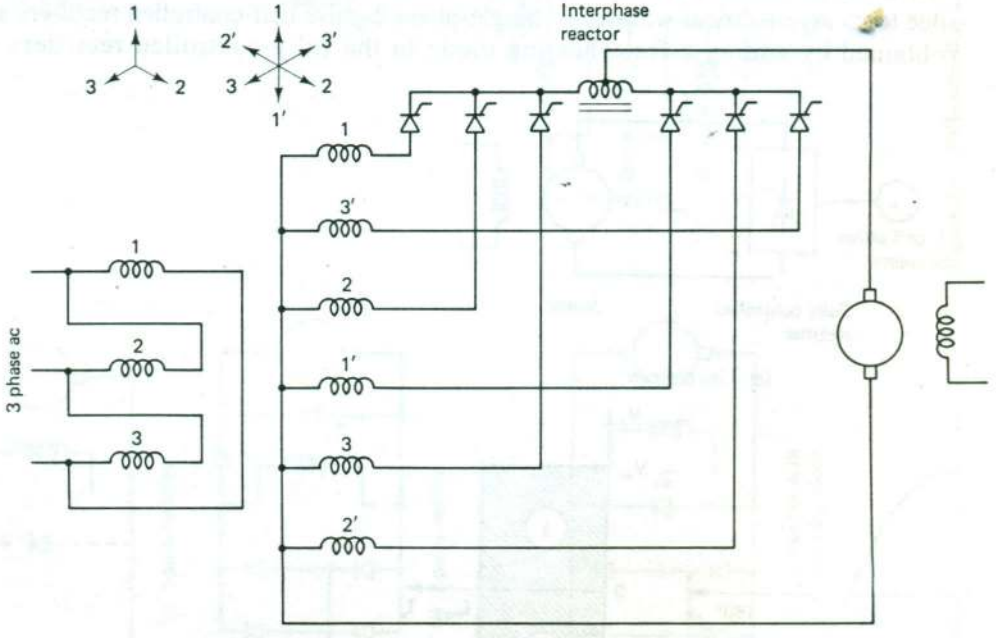
(a) 2-pulse midpoint rectifier



(b) 2-pulse bridge rectifier



(c) 6-pulse bridge rectifier



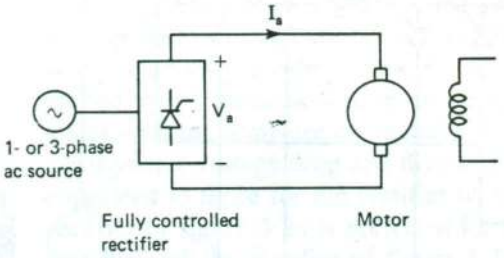
(d) 6-pulse midpoint rectifier with interphase reactor

Figure 3.2 Fully-controlled rectifiers.

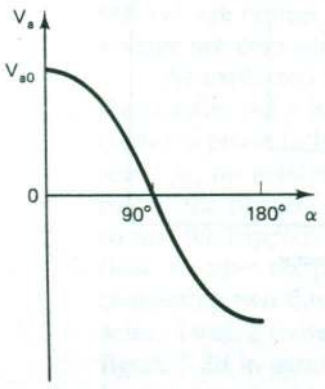
former having two sets of secondaries — one connected in star and another in delta.[1,2] In all these three-phase controlled rectifiers, each thyristor conducts for 120°.

The circuit symbol for the fully-controlled rectifiers is shown in figure 3.3a. V_a and I_a denote the average values of the converter output voltage and current, respectively. The variation of V_a with the firing angle α, assuming continuous conduction, is shown in figure 3.3b. The motor is said to operate in continuous conduction when the armature current flows continuously — that is, it does not become zero for a finite time interval. The output voltage can be controlled from a full-positive ($+V_{ao}$) to a full-negative value ($-V_{ao}$) by controlling the firing angle from 0° to 180°. In practice, the maximum value of α is restricted to 170° to avoid commutation failure of thyristors. Since the output voltage can be controlled in either direction, the fully-controlled rectifiers are two-quadrant converters, providing operation in the first and fourth quadrants of the V_a-I_a plane, as shown in figure 3.3c. I_{max} is the rated rectifier current. With a negative output voltage, the rectifier works as a line-commutated inverter and the power flows from the load to the ac source.

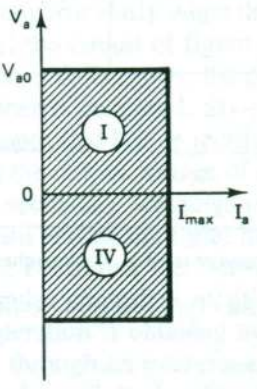
Some half-controlled rectifiers are shown in figure 3.4. For control of fractional and low integral horsepower motors, the 1-pulse rectifier of figure 3.4a may be employed, with and without a freewheeling diode. The cost of such a drive is low due to the lower number of devices in the rectifier. The major drawback of this rectifier is the presence of a direct component and even harmonics in the source current due to its asymmetrical waveform. Single-phase 2-pulse half-controlled rectifiers are obtained by adding a freewheeling diode to the fully-controlled rectifiers of



(a) Line diagram

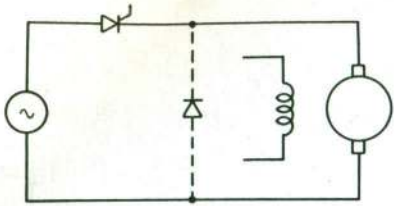


(b) Output voltage versus firing angle curve

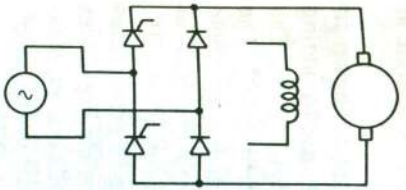


(c) Quadrants of operation

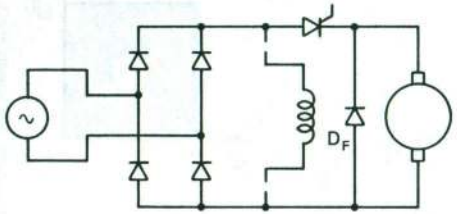
Figure 3.3 Fully-controlled rectifier characteristics.



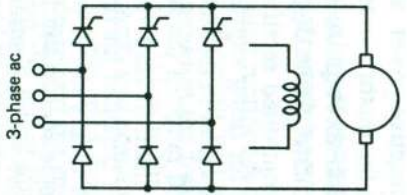
(a) 1-pulse rectifier



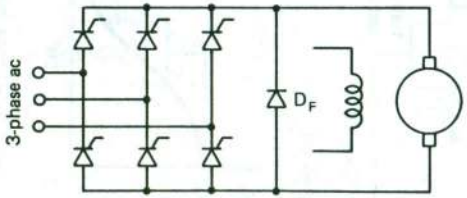
(b) 2-pulse rectifier



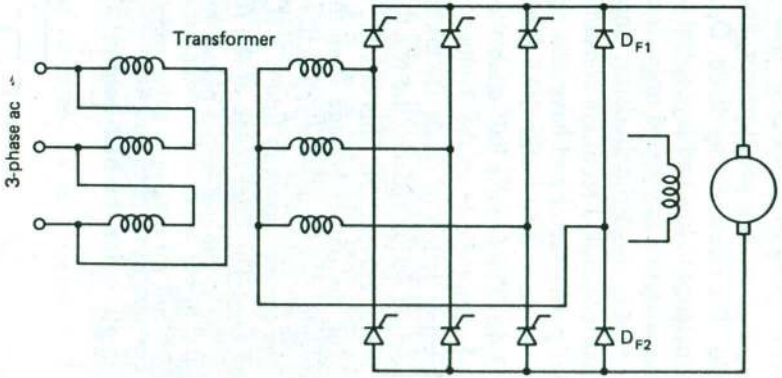
(c) 2-pulse rectifier



(d) 3-pulse rectifier



(e) 6-pulse rectifier with a freewheeling diode



(f) 6-pulse rectifier with two freewheeling diodes

Figure 3.4 Half-controlled rectifiers.

figures 3.2a and b. Alternatively, the half-controlled rectifier of figure 3.4b may be used. The circuit of figure 3.4c is sometimes used to reduce the cost of the drive. The circuit uses only one thyristor, and a common diode bridge feeds both armature and field. The freewheeling diode, D_F, can be dispensed with when the armature circuit inductance is low and low-speed operation is not required. In the case of a large inductance and/or low-speed operation, the drop across the freewheeling diode provides a bias to block the thyristor. Compared to 2-pulse fully-controlled rectifiers, 2-pulse half-controlled rectifiers consume less reactive power, and therefore operate at a higher power factor and have less ripple in the motor current.

A 3-phase half-controlled rectifier with three-pulse operation is shown in figure 3.4d. The 6-pulse half-controlled rectifier of figure 3.4e is obtained by adding a freewheeling diode to the 3-phase fully-controlled rectifier of figure 3.2c. The freewheeling action of diode D_F takes place for firing angles greater than 60°. The freewheeling action reduces the reactive component in the line current and the ripple in the motor current. For this circuit, the total range of the firing angle required for controlling the output voltage from the maximum to zero value is from 0° to 120°. The advantages of freewheeling action can be obtained for firing angles greater than 30° in the circuit shown in figure 3.4f, which makes use of two freewheeling diodes D_{F1} and D_{F2}. The total range of the firing angle required is 150°. This circuit is expensive due to the use of a 3-phase transformer with a neutral connection and an additional diode.

The circuit symbol and the variation of the average output voltage V_a with the firing angle α for the half-controlled rectifiers are shown in figures 3.5a and b,

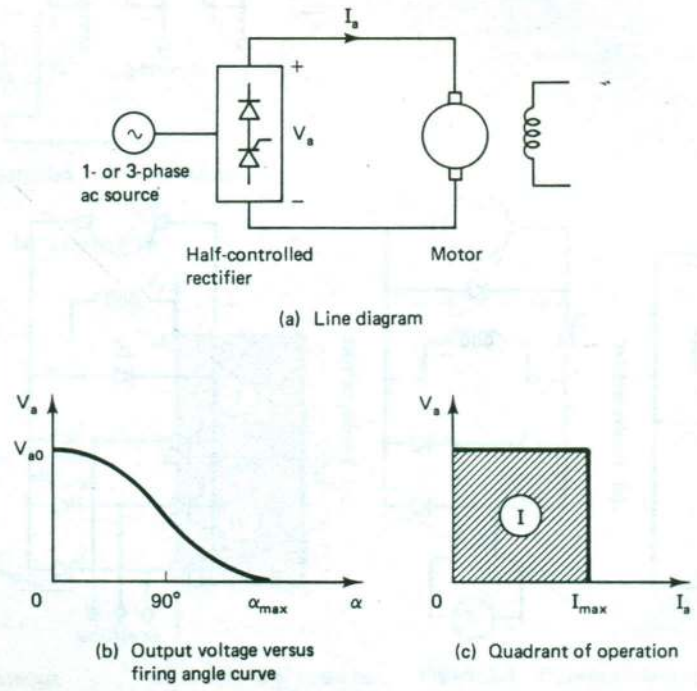


Figure 3.5 Half-controlled rectifier characteristics.

respectively. Various notations have the same meaning as stated for the fully-controlled rectifiers. α_{max} is the value of α for which V_a is zero. As just stated, it has different values for different circuits. Since only positive values of V_a can be obtained, the half-controlled rectifiers operate only in the first quadrant of the V_a-I_a plane, as shown in figure 3.5c.

The analysis and performance of drives will be considered only for those 1-phase and 3-phase rectifier-drives which are widely used. However, the knowledge gained will enable the reader to analyze any rectifier-drive.

3.2 BRAKING OPERATION OF RECTIFIER CONTROLLED SEPARATELY EXCITED MOTOR

A fully-controlled rectifier-fed dc separately excited motor is shown in figure 3.6a. The polarities of output voltage, back emf, and armature current shown are for the motoring operation in the forward direction. The rectifier output voltage is positive and the firing angle lies in the range $0 \leq \alpha \leq 90°$. The polarities of rectifier output voltage, back emf, and armature current show that the rectifier supplies power to the motor which is converted into mechanical power. With these polarities of the rectifier output voltage and the motor back emf, the direction of power flow can be reversed and thus the motor can be made to work under regenerative braking if the armature current can reverse. This is not possible because the rectifier can carry current only in one direction. The only alternative available for the reversal of the flow of power is to reverse both the rectifier output voltage V_a and the motor back emf E with respect to the rectifier terminals and make $|E| > |V_a|$ as shown in figure 3.6b. The rectifier output voltage can be reversed by making $\alpha > 90$, as shown in

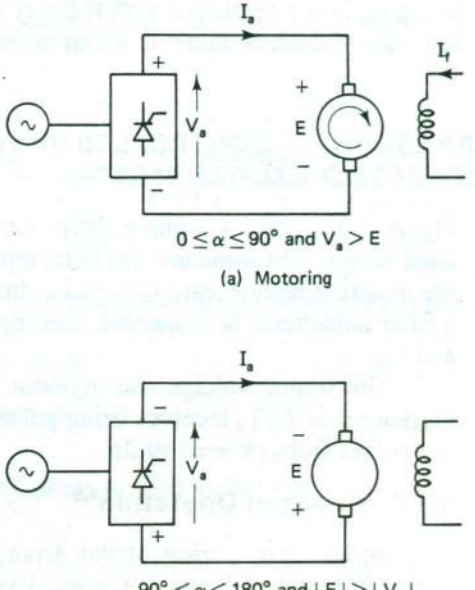


Figure 3.6 Two quadrant operation of fully-controlled rectifier-fed separately excited motor.

figure 3.3b. Under this condition, the rectifier works as a line commutated inverter, transferring power from the dc side to the ac mains. The condition $|E| > |V_a|$ can be satisfied for any motor speed by choosing an appropriate value of α in the range $90 < \alpha < 180°$. The reversal of the motor emf with respect to the rectifier terminals can be done by any of the following changes:

1. An active load coupled to the motor shaft may drive it in the reverse direction. This gives reverse regeneration (that is, operation in quadrant IV of the speed-torque plane). In this case no changes are required in the armature connection with respect to the rectifier terminals.
2. The field current may be reversed, with the motor running in the forward direction. This gives forward regeneration. In this case also no changes are required in the armature connection.
3. The motor armature connections may be reversed with respect to the rectifier output terminals, with the motor still running in the forward direction. This will give forward regeneration.

If the drive shown in figure 3.6a runs only in the forward direction and if there is no arrangement for the reversal of either field or armature, regenerative braking cannot be obtained. The drive then works essentially as a single quadrant drive.

Regenerative braking cannot be obtained with a half-controlled rectifier because the output voltage cannot be reversed. The plugging operation can be obtained both with half-controlled and fully-controlled rectifiers by reversing the back emf by any of the three methods just stated and keeping the rectifier voltage still positive. An external resistance must then be included to limit the current. Because of the poor efficiency and the need for external resistance to limit the armature current, plugging is not employed with rectifier drives. While operating in regenerative braking, care should be taken to avoid accidental plugging.

3.3 1-PHASE FULLY-CONTROLLED RECTIFIER-FED SEPARATELY EXCITED MOTOR

Figure 3.7a shows a 1-phase fully-controlled rectifier supplying a dc separately excited motor. The armature has been replaced by its equivalent circuit. R_a and L_a are the armature circuit resistance and inductance respectively, and E is the back emf. If a filter inductance is connected, then its resistance and inductance are included in R_a and L_a.

The source voltage and thyristor firing pulses are shown in figure 3.7b. The thyristor pair T_1, T_3 receives firing pulses from α to π and the pair T_2, T_4 receives firing pulses from $(\pi + \alpha)$ to 2π.

3.3.1 Modes of Operation[4,5]

The modes of operation of the drive for motoring and regenerative braking are shown in figures 3.8 and 3.9, respectively. The steady-state waveforms of the motor terminal voltage v_a and the armature current i_a, and the devices under conduction during different intervals of a cycle of the input ac voltage v_s are shown in these fig-

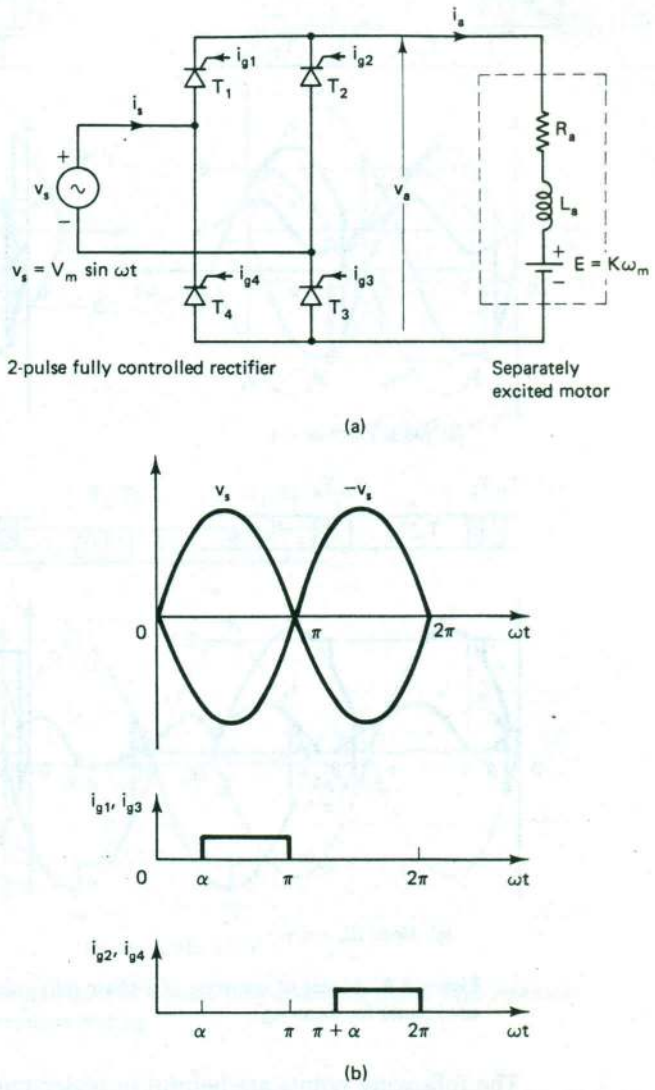


Figure 3.7 1-phase fully-controlled rectifier-fed dc separately excited motor.

ures. The drive is said to operate in discontinuous conduction when i_a becomes zero for a finite interval of time in each cycle.

The following notations have been used:

V_m = the peak value of the supply voltage, V

ω = supply frequency, rad/sec.

β = angle at which the armature current drops to zero value, rad

$\beta' = \beta - \pi$

$\gamma = \sin^{-1}(E/V_m)$ — that is, angle at which the source voltage v_s is equal to the back emf E, rad

$\gamma' = \pi - \sin^{-1}(|E|/V_m)$

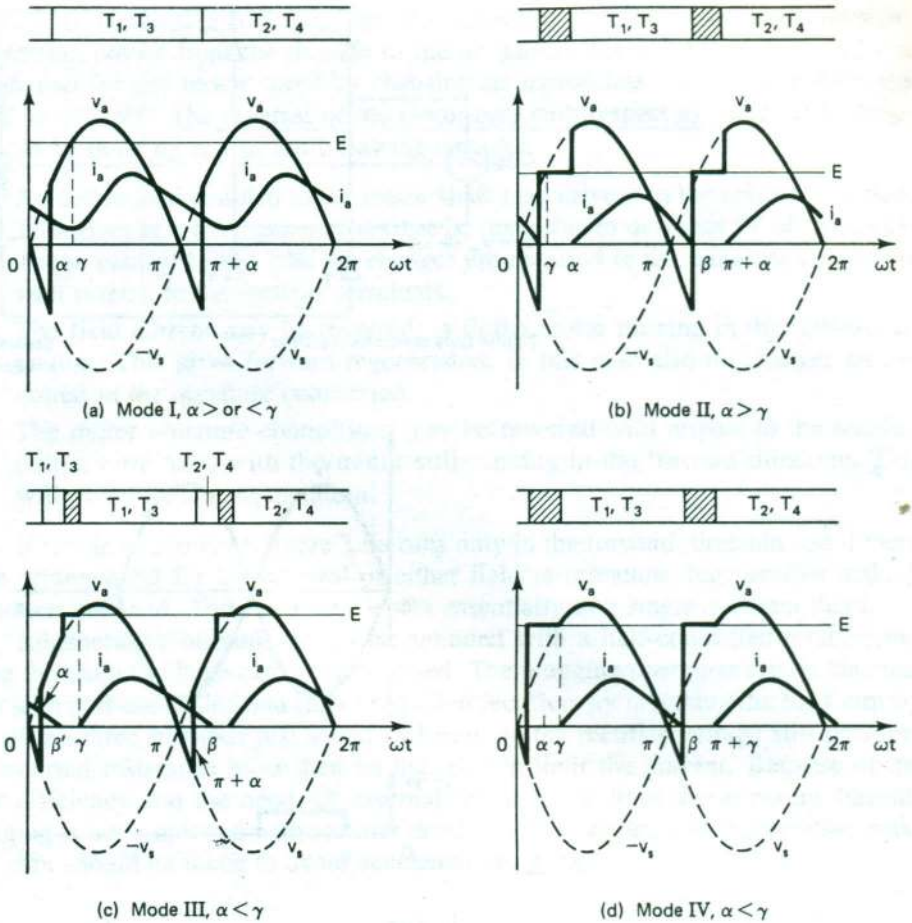


Figure 3.8 Modes of operation of 1-phase fully-controlled rectifier-fed separately excited motor for motoring.

The following points are helpful in understanding the modes of operation:

1. The motor armature current flows through the source, and either through the thyristor pair T_1, T_3 or through the pair T_2, T_4. When the pair T_1, T_3 conducts, $v_a = v_s$, and when the pair T_2, T_4 conducts, $v_a = -v_s$. When none of the thyristor pairs conducts, $i_a = 0$ and $v_a = E$.
2. When $i_a > 0$ at the instant of firing a thyristor pair, then the biasing on thyristors of this pair will be decided by the source voltage only. If the source voltage provides a positive bias, thyristors will turn on even when the source voltage is less than E. For example, if at the instant of firing of T_1 and T_3—that is, $\omega t = \alpha$, $i_a > 0$, then i_a must be carried by T_2 and T_4. Since T_2 and T_4 are on, the voltage across T_1 and T_3 will be v_s. T_1 and T_3 will turn on because v_s is positive. The turn on will occur regardless of whether v_s is greater or less than E.

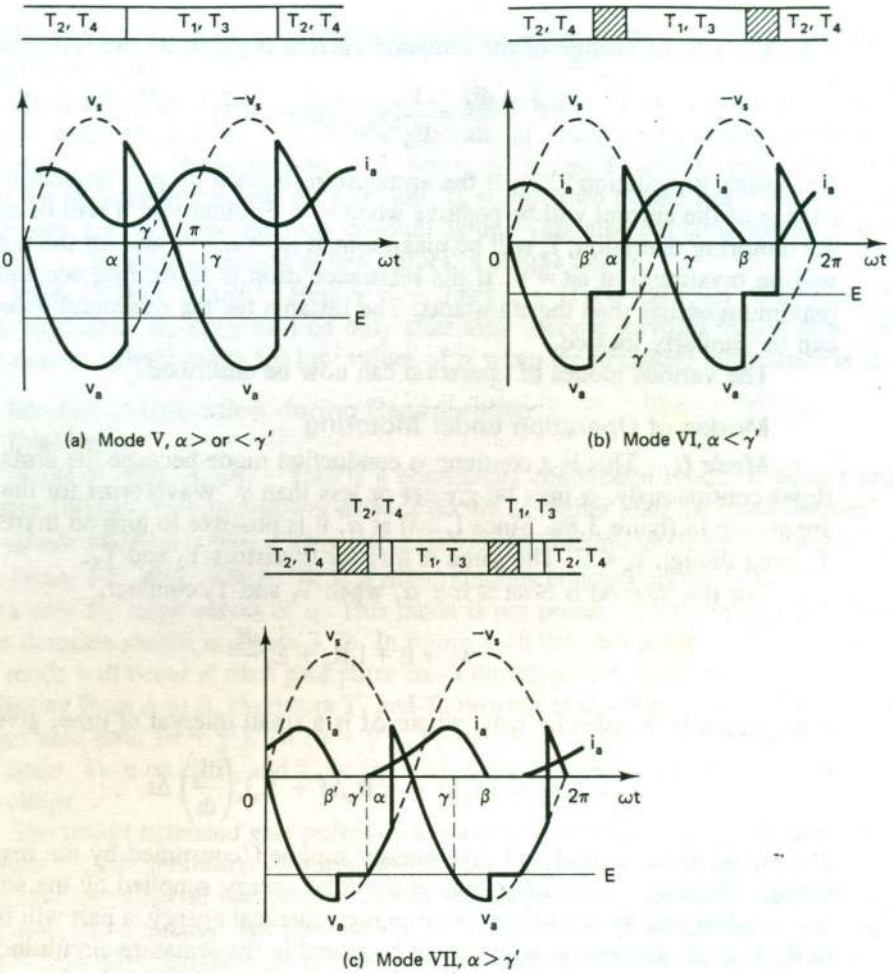


Figure 3.9 Modes of operation of 1-phase fully-controlled rectifier-fed separately excited motor for regenerative braking.

3. When $i_a = 0$ at the instant of firing a thyristor pair, then the biasing on thyristors of this pair will be decided by the difference of the source voltage and the back emf. The thyristors of this pair will conduct if the source voltage has the appropriate polarity and its magnitude is higher than E. For example, if $i_a = 0$ at $\omega t = \alpha$, then T_2 and T_4 must have already turned off due to the want of current. Then the voltage across T_1 and T_3 is $(v_s - E)$. T_1 and T_3 will conduct only if $v_s > E$, — that is, when $\alpha > \gamma$. Similarly, if $i_a = 0$ at $\omega t = (\pi + \alpha)$, T_2 and T_4 will turn on only if $(-v_s) > E$ or $(\pi + \alpha) > (\pi + \gamma)$.
4. The regenerative braking operation is obtained by reversing E by any of the three methods stated in section 3.2 and adjusting α so that the average terminal voltage of the converter, V_a, is negative. Thus, the modes of operation drawn for $+E$ are for motoring, and those drawn for $-E$ are for regeneration.

5. The rate of change of the armature current is given by the following equation:

$$\frac{di_a}{dt} = \frac{1}{L_a}[v_a - (E + i_a R_a)] \tag{3.1}$$

According to equation (3.1), if the armature resistance drop is neglected, the rate of change of the current will be positive when $v_a > E$; otherwise it will be negative. For the motoring operation, i_a will be maximum at $\omega t = \pi - \gamma$ and for the regeneration it will be maximum at $\omega t = \gamma$. If the resistance drop is taken into account, i_a will be maximum earlier than these instants. The instants for the minimum value of current can be similarly located.

The various modes of operation can now be described.

Modes of Operation under Motoring

Mode I: This is a continuous conduction mode because the armature current flows continuously. α may be greater or less than γ. Waveforms for the case $\alpha < \gamma$ are shown in figure 3.8a. Since $i_a > 0$ at α, it is possible to turn on thyristors T_1 and T_3 even though $v_s < E$. The same is true for thyristors T_2 and T_4.

For the interval $\alpha \leq \omega t \leq \pi + \alpha$, when T_1 and T_3 conduct,

$$v_s = E + i_a R_a + L_a \frac{di_a}{dt} \tag{3.2}$$

Multiplying both sides by $i_a \Delta t$, where Δt is a small interval of time, gives

$$v_s i_a \Delta t = E i_a \Delta t + i_a^2 R_a \Delta t + L_a i_a \left(\frac{di_a}{dt}\right) \Delta t \tag{3.3}$$

The terms of equation (3.3) give energy supplied/consumed by the respective elements. When $v_s > (E + i_a R_a)$, out of the total energy supplied by the source, a part will be absorbed by E and converted into mechanical energy, a part will be dissipated in R_a as heat, and the remainder will be stored in the armature circuit inductance L_a. On the other hand, when $v_s < (E + i_a R_a)$, energy consumed by E and R_a will be more than that supplied by the source v_s, and, therefore, the rest of the energy will be obtained from the energy stored in the inductance. Further, when v_s is negative ($\omega t > \pi$), v_s will also act as a sink of energy in addition to E and R_a and all this energy shall have to come from the energy stored in the inductance L_a. When L_a is small and/or i_a is low and/or α is large, the inductance will not be able to sustain the flow of armature current until (di_a/dt) becomes positive again either at $\omega t = \pi + \gamma$ when $\alpha < \gamma$ or at $(\pi + \alpha)$ when $\alpha > \gamma$. The armature current will therefore fall to zero and stay zero until the conditions become appropriate for it to flow again, and thus, discontinuous conduction is obtained.

In the absence of a filter inductance, a low-power drive — which is characterized by a low-armature circuit time constant — operates in the continuous conduction Mode I only for small values of α and current values larger than the rated motor current.

Three modes of motoring operation with discontinuous conduction are shown in figures 3.8b to d.

Mode II [Fig. 3.8b]: Here $\alpha > \gamma$, and i_a flow from α to β and stays zero from β to $(\pi + \alpha)$.

Mode III [Fig. 3.8c]: Here $\alpha < \gamma$. Since $i_a > 0$ at α, thyristors T_1 and T_3 turn on, even though $v_s < E$. Since the rate of change of current is negative [equation (3.1)], i_a drops to zero at β' where $\beta' < \gamma$. T_1 and T_3 get commutated due to the absence of current. At γ, T_1 and T_3 are forward biased again and since the gate pulses are still present, they conduct again. This also explains the need for a wide pulse. This mode occurs at low values of torque and α.

Mode IV [Fig. 3.8d]: Here $\alpha < \gamma$. Since $i_a = 0$ at α and $v_s(\alpha) < E$, T_1 and T_3 do not turn on at α. They turn on only after they become forward biased at γ. This mode occurs at light loads for low values of α when the armature inductance is low.

Modes of Operation during Regenerative Braking

Mode V [Fig. 3.9a]: This is a continuous conduction mode. It occurs only for large torques. In a low-power drive it occurs at higher than the rated torques.

Mode VI [Fig. 3.9b]: This discontinuous conduction mode occurs for $\alpha < \gamma'$.

Mode VII [Fig. 3.9c]: It is a discontinuous conduction mode for $\alpha > \gamma'$. It occurs only for large values of α. This mode is not present when the gate pulses are of the duration shown in figure 3.7b. In figure 3.7b the gate pulses end at π and 2π. This mode will occur if each gate pulse has a duration $\geq \pi$ for all values of α. After conducting from α to β, thyristors T_1 and T_3 turn off at β when i_a falls to zero value. i_a stays zero until $(\pi + \gamma')$. At $(\pi + \gamma')$, T_1 and T_3 become forward biased and conduct again. Turn on of T_2 and T_4 at $(\pi + \alpha)$ starts the next cycle of the rectifier output voltage.

The use of extended gate pulses of duration π, increases the conduction period of thyristors, and therefore, increases the regenerated power. Furthermore, it also reduces the difference in the average output voltage between the continuous and discontinuous conductions for the same α. Thus the change in the average output voltage from continuous to discontinuous conduction is not as abrupt as in the absence of the extended gate pulses.[5] In view of these advantages, it may be useful to use gate pulses of a duration greater than π instead of a duration of $(\pi - \alpha)$ as shown in figure 3.7b. However, the use of extended gate pulses results in the application of a gate signal when thyristors are reverse biased. This increases the reverse leakage current through thyristors.

3.3.2 Steady-State Motor Performance Equations

For the purpose of analysis, the following assumptions are made:

1. Thyristors are ideal switches—that is, they have no voltage drop when conducting and no leakage current when blocking. The main implication of this assumption is that the rectifier voltage drop and losses are neglected. This assumption should not be used with low-voltage motors.
2. The armature resistance and inductance are constant. The skin effect, which is present due to a ripple in the motor current, does alter the value of the resis-

tance. It is difficult to account for this variation. Since the variation is small, it is neglected here.

3. During a given steady-state operation, the motor speed is constant. The motor torque does fluctuate due to the ripple in the motor current. Because the mechanical time constant is very large compared to the period of current ripple, the fluctuation in speed is in fact negligible. At constant speed, one can assume the back emf E is an ideal direct voltage for a given steady-state operation.
4. Source inductance is negligible.

For all the modes described in the previous section, each period of the rectifier output voltage consists of one or two of the following intervals:

1. Duty Interval: When any one pair of thyristors is on and the ac source is connected to the motor. In this case the motor terminal voltage is v_s when T_1 and T_3 conduct and $-v_s$ when T_2 and T_4 conduct.
2. Zero-current Interval: During this interval, the armature current is zero and the motor terminal voltage is E.

The following equations describe the motor operation for the duty and zero-current intervals:

Duty Interval. When T_1 and T_3 conduct

$$v_a = L_a \frac{di_a}{dt} + R_a i_a + K\omega_m = V_m \sin \omega t \tag{3.4}$$

When T_2 and T_4 conduct

$$v_a = L_a \frac{di_a}{dt} + R_a i_a + K\omega_m = -V_m \sin \omega t \tag{3.5}$$

Zero-current Interval.

$$i_a = 0 \quad \text{and} \tag{3.6}$$

$$v_a = K\omega_m \tag{3.7}$$

where K is the motor back emf constant given by equation (2.6).

Mode I [Fig. 3.8a]: Each cycle of the output voltage consists of only a duty interval. For the output voltage cycle from α to $(\pi + \alpha)$, equation (3.4) is applicable. Solution of equation (3.4) may be considered to have two components, one due to the ac source, $(V_m/Z) \sin(\omega t - \psi)$ and the other due to the back emf, $(-K\omega_m/R_a)$. Each of these components has in turn a transient component. Let these be represented by a single exponent term $K_1 \exp(-t/\tau_a)$, then,

$$i_a(\omega t) = \frac{V_m}{Z} \sin(\omega t - \psi) - \frac{K\omega_m}{R_a} + k_1 \exp(-t/\tau_a), \quad \text{for } \alpha \le \omega t \le (\pi + \alpha) \tag{3.8}$$

where

$$Z = [R_a^2 + (\omega L_a)^2]^{1/2} \tag{3.9}$$

$$\tau_a = L_a/R_a \tag{3.10}$$

$$\psi = \tan^{-1}(\omega L_a/R_a) \tag{3.11}$$

and k_1 is a constant.

In the steady-state

$$i_a(\alpha) = i_a(\pi + \alpha) \tag{3.12}$$

The solution of equation (3.8), subject to the constraint (3.12), gives the following steady-state expression of current:

$$i_a(\omega t) = \frac{V_m}{Z}\left[\sin(\omega t - \psi) - \frac{2\sin(\alpha - \psi)\exp\{(\alpha - \omega t)\cot\psi\}}{1 - \exp(-\pi\cot\psi)}\right] - \frac{K\omega_m}{R_a} \quad \text{for} \quad \alpha \le \omega t \le (\pi + \alpha) \tag{3.13}$$

Since the flux is constant, the average motor torque depends only on the dc component (average value) of armature current, I_a. The ac components produce only pulsating torques with a zero average value. Therefore, the motor torque T_a is given by

$$T_a = KI_a \tag{3.14}$$

I_a can be obtained from the following expression:

$$I_a = \frac{1}{\pi}\int_{\alpha}^{\pi+\alpha} i_a(\omega t)\, d(\omega t)$$

This equation yields a long expression for I_a. A simple expression can be obtained using the following equation:

Average motor terminal voltage V_a = average voltage drop across R_a + average voltage drop across L_a + back emf (3.15)

$$\text{Now } V_a = \frac{1}{\pi}\int_{\alpha}^{\pi+\alpha} V_m \sin\omega t\, d(\omega t) = \frac{2V_m}{\pi}\cos\alpha = V_{ao}\cos\alpha \tag{3.16}$$

where $V_{ao} = 2V_m/\pi$

$$\text{Average drop across } R_a = \frac{1}{\pi}\int_{\alpha}^{\pi+\alpha} R_a i_a(\omega t)\, d(\omega t) = R_a I_a \tag{3.17}$$

$$\text{Average drop across } L_a = \frac{1}{\pi}\int_{\alpha}^{\pi+\alpha} L_a\left(\frac{di_a}{dt}\right) d(\omega t) = \frac{\omega}{\pi}\int_{i_a(\alpha)}^{i_a(\pi+\alpha)} L_a\, di_a = \frac{\omega L_a}{\pi}[i_a(\pi + \alpha) - i_a(\alpha)] = 0 \tag{3.18}$$

Thus the steady-state average voltage drop across the inductance is always zero. Substituting from equations (3.17) and (3.18) into equation (3.15) gives

$$V_a = I_a R_a + K\omega_m \tag{3.19}$$

Equation (3.19) is valid for the steady-state operation of a dc motor fed by any converter.
From equations (3.16) and (3.19),

$$I_a = \frac{(2V_m/\pi)\cos\alpha - K\omega_m}{R_a} \tag{3.20}$$

Substituting from equation (3.20) into equation (3.14) and rearranging the terms gives the relation between speed and torque,

$$\omega_m = \frac{2V_m}{\pi K}\cos\alpha - \frac{R_a}{K^2}T_a \tag{3.21}$$

Mode II [Fig. 3.8b]: Each cycle of the output voltage consists of a duty interval and a zero-current interval. For the output voltage cycle from α to $(\pi + \alpha)$, the duty interval is from α to β and the zero-current interval from β to $(\pi + \alpha)$. The current expression for the duty interval is obtained by subjecting equation (3.8) to the initial condition $i_a(\alpha) = 0$. Thus we get

$$i_a(\omega t) = \frac{V_m}{Z}[\sin(\omega t - \psi) - \sin(\alpha - \psi)\exp\{(\alpha - \omega t)\cot\psi\}] - \frac{K\omega_m}{R_a}[1 - \exp\{(\alpha - \omega t)\cot\psi\}] \quad \text{for } \alpha \le \omega t \le \beta \tag{3.22}$$

Since $i_a(\beta) = 0$, we have from equation (3.22)

$$\frac{V_m}{Z}\sin(\beta - \psi) - \frac{K\omega_m}{R_a} + \left[\frac{K\omega_m}{R_a} - \frac{V_m}{Z}\sin(\alpha - \psi)\right]\exp\{(\alpha - \beta)\cot\psi\} = 0 \tag{3.23}$$

β can be evaluated by the solution of the transcedental equation (3.23).
Now

$$V_a = \frac{1}{\pi}\left[\int_\alpha^\beta V_m \sin\omega t\, d(\omega t) + \int_\beta^{\pi+\alpha} E\, d(\omega t)\right] = \frac{V_m(\cos\alpha - \cos\beta) + (\pi + \alpha - \beta)K\omega_m}{\pi} \tag{3.24}$$

From equations (3.19) and (3.24),

$$I_a = \frac{V_m(\cos\alpha - \cos\beta) - (\beta - \alpha)K\omega_m}{\pi R_a} \tag{3.25}$$

For given α and ω_m, β is obtained from equation (3.23); I_a and then T_a are obtained from equations (3.25) and (3.14), respectively.

Substitution from equation (3.25) into equation (3.14) yields the speed-torque relation

$$\omega_m = \frac{V_m(\cos\alpha - \cos\beta)}{K(\beta - \alpha)} - \frac{\pi R_a}{K^2(\beta - \alpha)} T_a \tag{3.26}$$

Mode III [Fig. 3.8c]: Each cycle of the output voltage consists of two duty intervals and a zero-current interval. The output voltage cycle from γ to $(\pi + \gamma)$ consists of a duty interval from γ to $\pi + \alpha$ with $v_a = v_s$, another duty interval from $(\pi + \alpha)$ to β with $v_a = -v_s$, and a zero-current interval from β to $(\pi + \gamma)$. The current expression for the interval $\gamma \le \omega t \le (\pi + \alpha)$ is obtained by subjecting equation (3.8) to the initial condition $i(\gamma) = 0$. Thus we get

$$i_a(\omega t) = \frac{V_m}{Z}[\sin(\omega t - \psi) - \sin(\gamma - \psi)\exp\{(\gamma - \omega t)\cot\psi\}] - \frac{K\omega_m}{R_a}[1 - \exp\{(\gamma - \omega t)\cot\psi\}] \quad \text{for } \gamma \le \omega t \le \pi + \alpha \tag{3.27}$$

From equation (3.27)

$$i_a(\pi + \alpha) = -\frac{V_m}{Z}[\sin(\alpha - \psi) + \sin(\gamma - \psi)\exp\{-(\pi + \alpha - \gamma)\cot\psi\}] - \frac{K\omega_m}{R_a}[1 - \exp\{-(\pi + \alpha - \gamma)\cot\psi\}] \tag{3.28}$$

Solving equation (3.5), subjected to the initial condition given in equation (3.28), gives

$$i_a(\omega t) = -\frac{V_m}{Z}[\sin(\omega t - \psi) + 2\sin(\alpha - \psi)\exp\{(\pi + \alpha - \omega t)\cot\psi\} + \sin(\gamma - \psi)\exp\{(\gamma - \omega t)\cot\psi\}] - \frac{K\omega_m}{R_a}[1 - \exp\{(\gamma - \omega t)\cot\psi\}], \quad \text{for } (\pi + \alpha) \le \omega t \le \beta \tag{3.29}$$

Since $i_a = 0$ at β, equation (3.29) yields

$$\frac{V_m}{Z}[\sin(\beta - \psi) + 2\sin(\alpha - \psi)\exp\{(\pi + \alpha - \beta)\cot\psi\} + \sin(\gamma - \psi)\exp\{(\gamma - \beta)\cot\psi\}] + \frac{K\omega_m}{R_a}[1 - \exp\{(\gamma - \beta)\cot\psi\}] = 0 \tag{3.30}$$

β can be evaluated by the solution of equation (3.30).
Now,

$$V_a = \frac{1}{\pi}\left[\int_{\gamma}^{\pi+\alpha} V_m \sin\omega t\, d(\omega t) + \int_{\pi+\alpha}^{\beta} - V_m \sin\omega t\, d(\omega t) + \int_{\beta}^{\pi+\gamma} E\, d(\omega t)\right]$$
$$= \frac{V_m(2\cos\alpha + \cos\gamma + \cos\beta) + (\pi + \gamma - \beta)K\omega_m}{\pi} \tag{3.31}$$

From equations (3.19) and (3.31)

$$I_a = \frac{V_m(2\cos\alpha + \cos\gamma + \cos\beta) + (\gamma - \beta)K\omega_m}{\pi R_a} \tag{3.32}$$

For given α and ω_m, β can be obtained from equation (3.30). I_a and then T_a can be obtained from equations (3.32) and (3.14), respectively.

Substitution from equation (3.32) into equation (3.14) yields the speed torque relation

$$\omega_m = \frac{V_m}{(\beta - \gamma)K}(2\cos\alpha + \cos\gamma + \cos\beta) - \frac{\pi R_a}{K^2(\beta - \gamma)}T_a \tag{3.33}$$

Mode IV [Fig. 3.8d]: This mode is identical to mode II. In this mode current begins to flow at γ instead of at α as in mode II. The relevant equations for this mode are obtained by replacing α by γ in equations (3.23), (3.25), and (3.26). This yields

$$\frac{V_m}{Z}\sin(\beta - \psi) - \frac{K\omega_m}{R_a} + \left[\frac{K\omega_m}{R_a} - \frac{V_m}{Z}\sin(\gamma - \psi)\right]\exp\{(\gamma - \beta)\cot\psi\} = 0 \tag{3.34}$$

$$I_a = \frac{V_m(\cos\gamma - \cos\beta) - (\beta - \gamma)K\omega_m}{\pi R_a} \tag{3.35}$$

$$\omega_m = \frac{V_m(\cos\gamma - \cos\beta)}{K(\beta - \gamma)} - \frac{\pi R_a}{K^2(\beta - \gamma)}T_a \tag{3.36}$$

β can be evaluated from equation (3.34), and I_a and T_a can be calculated from equations (3.35) and (3.14), respectively.

Mode V [Fig. 3.9a] and Mode VI [Fig. 3.9b]: Modes V and I and modes VI and II are identical except that the back emf has a negative sign for modes V and VI. When the braking operation is obtained without a change in the armature connection, the negative E is obtained either due to K being negative or ω_m being negative. The negative K is obtained by the field reversal, which gives operation in the second quadrant; and the negative ω_m is obtained by the speed reversal, which gives operation in the fourth quadrant. When braking is obtained by the reversal of the armature connection, the net effect is the same as field reversal, and therefore, K can be considered negative. Equations (3.13), (3.14), (3.20), and (3.21) are valid for mode V, and equations (3.14) and (3.22) to (3.26) are valid for mode VI when appropriate signs are used for K, ω_m, I_a, and T_a.

Mode VII [Fig. 3.9c]: This mode is obtained only when extended gate pulses are present. The cycle of the output voltage from γ' to $(\pi + \gamma')$ consists of a duty interval from γ' to α with $v_a = -v_s$, another duty interval from α to β with $v_a = v_s$, and a zero-current interval from β to $(\pi + \gamma')$. The operation in the duty intervals is described by the following equations:

$$L_a\frac{di_a}{dt} + R_a i_a + K\omega_m = -V_m\sin\omega t \quad \text{for } \gamma' \le \omega t \le \alpha \tag{3.37}$$

$$L_a \frac{di_a}{dt} + R_a i_a + K\omega_m = V_m \sin \omega t \quad \text{for } \alpha \le \omega t \le \beta \tag{3.38}$$

In equations (3.37) and (3.38), K or ω_m will have a negative value depending on whether the braking operation is in the second or the fourth quadrant.

$i_a(\alpha)$ can be obtained from the solution of equation (3.37) with the initial condition $i_a(\gamma') = 0$. Equation (3.38) can then be solved with $i_a(\alpha)$ as the initial condition. Substituting $\omega t = \beta$ in this solution and noting that $i_a(\beta) = 0$, gives the following equation from which β can be evaluated:

$$\frac{V_m}{Z}[\sin(\beta - \psi) + \sin(\gamma' - \psi) \exp\{(\gamma' - \beta) \cot \psi\} - 2 \sin(\alpha - \psi) \times \exp\{(\alpha - \beta) \cot \psi\}] - \frac{K\omega_m}{R_a}[1 - \exp\{(\gamma' - \beta) \cot \psi\}] = 0 \tag{3.39}$$

Now,

$$V_a = \frac{1}{\pi}\left[\int_{\gamma'}^{\alpha} (-V_m \sin \omega t)\, d(\omega t) + \int_{\alpha}^{\beta} V_m \sin \omega t\, d(\omega t) + \int_{\beta}^{\pi+\gamma'} K\omega_m\, d(\omega t)\right]$$

$$= \frac{1}{\pi}[V_m(2 \cos \alpha - \cos \gamma' - \cos \beta) + K\omega_m(\pi + \gamma' - \beta)] \tag{3.40}$$

From equations (3.19) and (3.40),

$$I_a = \frac{1}{\pi R_a}[V_m(2 \cos \alpha - \cos \gamma' - \cos \beta) + K\omega_m(\gamma' - \beta)] \tag{3.41}$$

From equations (3.41) and (3.14)

$$\omega_m = -\frac{V_m}{K}\frac{(2 \cos \alpha - \cos \gamma' - \cos \beta)}{(\gamma' - \beta)} + \frac{\pi R_a}{K^2(\gamma' - \beta)} T_a \tag{3.42}$$

3.3.3 Mode Identification

For the given values of α and ω_m, the torque can be calculated by the approach described in section 3.3.2 if the mode of operation is known. This is done using the following logic for the motoring and braking operations.

Motoring

1. Check whether α is greater or less than γ.
2. If $\alpha > \gamma$, assume $i_a(\alpha) = 0$ and calculate $i_a(\pi + \alpha)$ from equation (3.22). A negative value of $i_a(\pi + \alpha)$ indicates that the current has ceased to flow before $(\pi + \alpha)$. Thus, if $i_a(\pi + \alpha) < 0$, it is mode II; otherwise, mode I.
3. If $\alpha < \gamma$, assume $i_a(\gamma) = 0$ and calculate $i_a(\pi + \alpha)$ from equation (3.27). If $i_a(\pi + \alpha) \le 0$, it is mode IV. If $i_a(\pi + \alpha) > 0$, calculate $i_a(\pi + \gamma)$ from equation (3.29). If $i_a(\pi + \gamma) < 0$, it is mode III; otherwise, mode I.

Regenerative Braking

1. Check whether α is greater or less than γ'.
2. If $\alpha < \gamma'$, assume $i_a(\alpha) = 0$ and calculate $i_a(\pi + \alpha)$ from equation (3.22). If $i_a(\pi + \alpha) < 0$, it is mode VI, otherwise, mode V.
3. If $\alpha > \gamma'$, assume $i_a(\gamma') = 0$ and calculate $i_a(\pi + \gamma')$ from equations (3.37) and (3.38). If $i_a(\pi + \gamma') < 0$, it is mode VII; otherwise, mode V.

3.3.4 Speed-Torque Characteristics

The speed-torque curves of a 2.2 kW, 1500 rpm dc motor fed by a 1-phase fully controlled rectifier with an ac source voltage of 230 V, 50 Hz are shown in figure 3.10 for quadrants I and IV. The regions of continuous and discontinuous conductions and the modes of operation have been marked.

The ideal no-load speed is obtained when $I_a = 0$. For firing angles from 0 to $\pi/2$, I_a becomes zero when the back emf E becomes equal to the peak of the source voltage, V_m. For firing angles $> \pi/2$, I_a becomes zero when $E = V_m \sin\alpha$. Thus the ideal no-load speed ω_{mo} is given by the following equations:

$$\omega_{mo} = \frac{V_m}{K}, \qquad 0 \le \alpha \le \pi/2 \tag{3.43}$$

$$= \frac{V_m \sin\alpha}{K}, \qquad \pi/2 \le \alpha \le \pi \tag{3.44}$$

The maximum average terminal voltage, $2V_m/\pi$, is chosen equal to the rated motor voltage. The ideal no-load speed of the motor when fed by a perfect direct voltage equal to the rated value will then be $2V_m/\pi K$. It is interesting to note that the maximum no-load speed with rectifier control is $\pi/2$ times this value.

The boundary between continuous and discontinuous conductions is shown by a dotted line (fig. 3.10). For torques less than the rated value, a low-power drive operates predominantly in the discontinuous conduction. In continuous conduction, the speed-torque characteristics are parallel straight lines, whose slope, according to equation (3.21), depends on the armature circuit resistance R_a. The effect of discontinuous conduction is to make the speed regulation poor. This behavior can be explained from the waveforms of figures 3.8 and 3.9. In continuous conduction, for a given α, any increase in load causes E and ω_m to drop so that I_a and T_a can increase. The average terminal voltage V_a remains constant. On the other hand, in discontinuous conduction, any increase in load, and the accompanied increase in I_a causes β to increase. Consequently V_a reduces, and the speed drops by a larger amount than in the case of continuous conduction. Other disadvantages of discontinuous conduction are the nonlinear transfer characteristics of the converter and the slower transient response of the drive.

The boundary between continuous and discontinuous conductions is obtained as detailed in the following section.

At the boundary, the minimum value of instantaneous current and the duration of the zero-current interval are zero. When operating in mode II, an increase in load

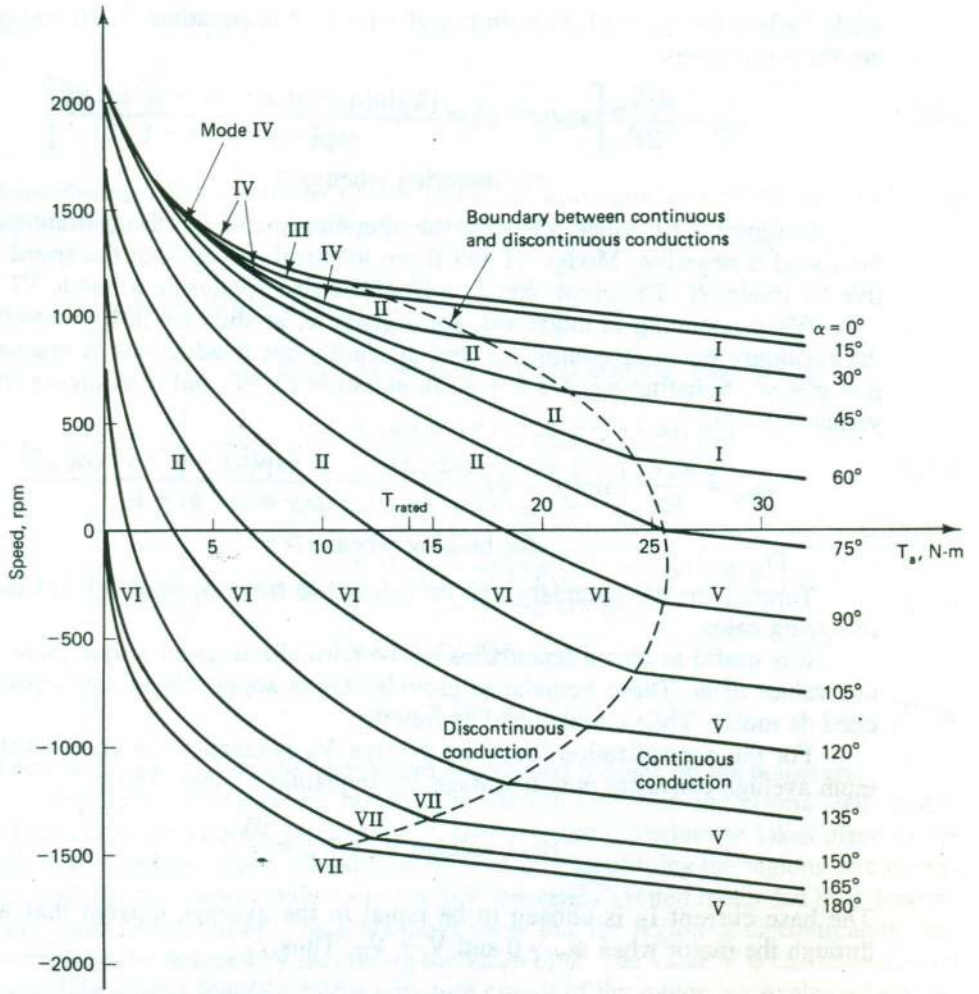


Figure 3.10 Speed-torque characteristics for a 1-phase fully-controlled rectifier drive, I-VI denote mode of operation.

increases I_a and angle β. The drive enters the continuous conduction mode I when $\beta = (\pi + \alpha)$. Substituting $\beta = (\pi + \alpha)$ in equation (3.23) and rearranging the terms gives

$$\omega_{mc} = \frac{R_a V_m}{ZK} \sin(\alpha - \psi) \left[\frac{1 + \exp(-\pi \cot \psi)}{\exp(-\pi \cot \psi) - 1} \right] \quad \text{for motoring when } \alpha > \gamma \quad (3.45)$$

where ω_{mc} is the critical speed (speed on the boundary).

When working in mode IV, an increase in load shifts the operation to mode III. Further increase in load increases β. The drive enters the continuous conduction

mode I when $\beta = (\pi + \gamma)$. Substituting $\beta = (\pi + \gamma)$ in equation (3.30) and rearranging the terms gives

$$\omega_{mc} = \frac{R_a V_m}{ZK}\left[\sin(\gamma - \psi) + \frac{\{2\sin(\alpha - \psi)\exp\{(\alpha - \gamma)\cot\psi\}}{\exp(-\pi\cot\psi) - 1}\right] \tag{3.46}$$

for motoring when $\alpha < \gamma$

In figure 3.10, mode VI is for the operation in the fourth quadrant for which the speed is negative. Modes VI and II are identical, except that the speed is negative for mode VI. Therefore, equation (3.45) is also applicable to mode VI.

When operating in mode VII, for a given α, an increase in I_a increases β and the boundary between continuous and discontinuous conductions is reached when $\beta = \pi + \gamma'$. Substituting $\beta = \pi + \gamma'$ in equation (3.39) and rearranging the terms yields

$$\omega_{mc} = \frac{R_a V_m}{KZ}\left[\sin(\gamma' - \psi) - \frac{2\sin(\alpha - \psi)\exp\{(\alpha - \pi - \gamma)\cot\psi\}}{\exp(-\pi\cot\psi) - 1}\right] \tag{3.47}$$

for braking when $\alpha > \gamma'$

Torques on the boundary can be calculated from equation (3.21) for all the foregoing cases.

It is useful to obtain boundaries on the normalized speed-torque plane for various values of ψ. These boundaries provide results applicable to any separately excited dc motor. They are obtained as follows:

For the normalization, the base voltage V_B is taken to be equal to the maximum average converter output voltage V_{ao} (equation (3.16)). Thus,

$$\text{Base voltage } V_B = V_{ao} = \frac{2V_m}{\pi} \tag{3.48a}$$

The base current I_B is chosen to be equal to the average current that will flow through the motor when $\omega_m = 0$ and $V_a = V_B$. Thus,

$$\text{Base current } I_B = \frac{V_{ao}}{R_a} = \frac{2V_m}{\pi R_a} \tag{3.48b}$$

Now, the normalized speed ω_{mn} is given by

$$\omega_{mn} = \frac{E}{V_B} = \frac{E}{V_{ao}} = \frac{\pi E}{2V_m} \tag{3.49}$$

And the normalized torque T_{an} is given by

$$T_{an} = I_{an} = \frac{I_a}{I_B} = \frac{I_a}{(V_{ao}/R_a)} = \frac{\pi R_a}{2V_m}(I_a) \tag{3.50}$$

Further

$$\gamma = \sin^{-1}[E/V_m] = \sin^{-1}\left[\frac{2}{\pi}\left(\frac{\pi}{2}\frac{E}{V_m}\right)\right] = \sin^{-1}\left(\frac{2}{\pi}\omega_{mn}\right) \tag{3.51a}$$

and

$$\gamma' = \pi - \sin^{-1}(|E|/V_m) = \pi - \sin^{-1}\left(\frac{2}{\pi}|\omega_{mn}|\right) \tag{3.51b}$$

Substituting from equations (3.48) to (3.50) into equations (3.45) to (3.47) and equation (3.21), and rearranging the terms gives,

$$\omega_{mcn} = \frac{\pi}{2}\cos\psi\sin(\alpha-\psi)\left[\frac{1-\exp(-\pi\cot\psi)}{\exp(-\pi\cot\psi)-1}\right] \tag{3.52}$$

for motoring when $\alpha > \gamma$ and braking when $\alpha < \gamma'$

$$\omega_{mcn} = \frac{\pi}{2}\cos\psi\left[\sin(\gamma-\psi) + \frac{\{2\sin(\alpha-\psi)\exp\{(\alpha-\gamma)\cot\psi\}}{\exp(-\pi\cot\psi)-1}\right] \tag{3.53}$$

for motoring when $\alpha < \gamma$.

$$\omega_{mcn} = \frac{\pi}{2}\cos\psi\left[\sin(\gamma'-\psi) - \frac{2\sin(\alpha-\psi)\exp\{(\alpha-\pi-\gamma)\cot\psi\}}{\exp(-\pi\cot\psi)-1}\right] \tag{3.54}$$

for braking with $\alpha > \gamma'$.

$$\omega_{mcn} = \cos\alpha - T_{cn} \tag{3.55}$$

where ω_{mcn} and T_{cn} denote normalized speeds and torques on the boundary.

Figure 3.15 shows the boundaries (dotted lines) on the normalized speed-torque plane for various values of ψ[10]. Discontinuous conduction takes place to the left of a boundary. These boundaries are useful in identifying the regions of continuous and discontinuous conductions for any separately excited motor fed by a 1-phase fully controlled rectifier. The boundaries show that the region of discontinuous conduction can be reduced by increasing the value of ψ. The value of ψ can be increased by adding a filter inductor in the armature circuit of the motor. As explained earlier, discontinuous conduction has a number of disadvantages such as poor speed regulation, nonlinear transfer characteristic of the rectifier, and poor transient response. Therefore, a filter inductor is sometimes added to reduce the zone of discontinuous conduction. However, the addition of the filter inductance increases the losses, armature circuit time constant, noise, and cost, weight, and volume of the drive.

Example 3.1

A 220 V, 1500 rpm, 11.6 A separately excited motor is controlled by a 1-phase fully-controlled rectifier with an ac source voltage of 230 V, 50 Hz. Enough filter inductance is added to ensure continuous conduction for any torque greater than 25 percent of rated torque, $R_a = 2\Omega$.

1. What should be the value of the firing angle to get the rated torque at 1000 rpm?
2. Calculate the firing angle for the rated braking torque and −1500 rpm.
3. Calculate the motor speed at the rated torque and $\alpha = 160°$ for the regenerative braking in the second quadrant.

Solution:

$$V_m = 230\sqrt{2} = 325.27 \text{ V}$$

$$E = 220 - 11.6 \times 2 = 196.8 \text{ V}$$

$$\omega_m = \frac{1500 \times 2\pi}{60} = 157 \text{ rad/sec}.$$

$$K = E/\omega_m = 196.8/157 = 1.25$$

For continuous conduction, from equation (3.20),

$$\frac{2V_m}{\pi} \cos \alpha = I_a R_a + E \qquad \text{(E3.1)}$$

1. At the rated torque $I_a = 11.6$ A
Back emf at 1000 rpm $= E_1 = (1000/1500) \times 196.8 = 131.2$ V
From equation (E3.1)

$$\frac{2 \times 325.27}{\pi} \cos \alpha = 11.6 \times 2 + 131.2 = 154.4$$

$$\text{or} \quad \cos \alpha = 0.75 \quad \text{or} \quad \alpha = 41.8°.$$

2. For the speed of -1500 rpm, $E = -196.8$ V.
From equation (E3.1)

$$\frac{2 \times 325.27}{\pi} \cos \alpha = 11.6 \times 2 - 196.8 = -173.6$$

$$\text{or} \quad \cos \alpha = -0.84 \quad \text{or} \quad \alpha = 147°$$

3. From equation (E3.1) for $\alpha = 160°$ and $I_a = 11.6$ A

$$\frac{2 \times 325.27}{\pi} \cos 160° = 11.6 \times 2 + E \quad \text{or} \quad E = -217.78 \text{ V}$$

Forward regeneration is obtained either by the field reversal or the armature reversal for which

$$K = -1.25$$

$$\text{Now} \quad \omega_m = \frac{E}{K} = \frac{-217.78}{-1.25} = 174.2 \text{ rad/sec}.$$

$$= 1663.8 \text{ rpm}$$

Example 3.2

Let the motor of example 3.1 have $L_a = 28.36$ mH. Calculate motor torques for the following cases:

1. $\alpha = 60°$ and speed = 400 rpm
2. $\alpha = 60°$ and speed = 995 rpm
3. $\alpha = 150°$ and speed = -640 rpm
4. $\alpha = 130°$ and speed = -1600 rpm

Solution: From example 3.1,
Rated speed = 1500 rpm = 157 rad/sec., E at 1500 rpm = 196.8 V

$$K = 1.25$$

$$\omega = 2\pi f = 2\pi \times 50 = 100\pi \text{ rad/sec}.$$

$$\psi = \tan^{-1}\frac{\omega L_a}{R_a} = \tan^{-1}\frac{100\pi \times 28.36 \times 10^{-3}}{2} = 1.35 \text{ rad}.$$

$$= 77.35°$$

$$\cot\psi = 0.22$$

$$Z = [R_a^2 + (\omega L_a)^2]^{1/2} = \left[(2)^2 + \left(\frac{100\pi \times 28.36}{1000}\right)^2\right]^{1/2} = 9.13\Omega$$

$$\frac{V_m}{Z} = \frac{230\sqrt{2}}{9.13} = 35.63$$

1. $E = (400/1500) \times 196.8 = 52.5$ V, $\omega_m = 400 \times 2\pi/60 = 41.9$ rad/sec.

$$\frac{K\omega_m}{R_a} = \frac{1.25 \times 41.9}{2} = 26.2$$

$$\gamma = \sin^{-1}\frac{E}{V_m} = \sin^{-1}\frac{52.5}{230\sqrt{2}} = 9.3°$$

Since $\alpha > \gamma$, the system may operate in mode I or mode II.
From equation (3.22)

$$i_a(\pi + \alpha) = 35.63[0.3 + 0.3 \times 0.5] - 26.2[1 - 0.5] = 2.9\text{A}$$

Since $i_a(\pi + \alpha) > 0$, the motor operates in the continuous conduction mode I.
From equation (3.21)

$$41.9 = \frac{2 \times 230\sqrt{2}}{\pi \times 1.25}\cos 60° - \frac{2}{(1.25)^2}T_a$$

which gives $T_a = 32$ N-m.

2. $E = (995/1500) \times 196.8 = 130.5$ V, $\omega_m = 995 \times 2\pi/60 = 104.2$ rad/sec.

$$\frac{K\omega_m}{R_a} = \frac{1.25 \times 104.2}{2} = 65.1$$

$$\gamma = \sin^{-1}\frac{E}{V_m} = \sin^{-1}\frac{130.5}{230\sqrt{2}} = 23.7°$$

Since $\alpha > \gamma$ the drive may operate in mode I or mode II.
From equation (3.22),

$$i_a(\pi + \alpha) = 35.63[0.3 + 0.3 \times 0.5] - 65.1[1 - 0.5] = -16.5\text{A}$$

Since $i_a(\pi + \alpha) < 0$, the motor operates in the discontinuous conduction mode II. The torque which is given by equation (3.26) can be calculated if β is known.
From equation (3.23),

$$35.63 \sin(\beta° - 77.35°) - 65.1 + [65.1 - 35.63 \times (-0.3)] \exp\left\{\left(\frac{\pi}{3} - \beta\right) \times 0.22\right\} = 0$$

$$\text{or} \quad 35.63 \sin(\beta° - 77.35°) + 75.8 \exp\left\{\left(\frac{\pi}{3} - \beta\right) \times 0.22\right\} = 65.1$$

Solution of this equation by trial and error gives $\beta = 215°$ or 3.75 rad.
From equation (3.26)

$$104.2 = \frac{230\sqrt{2}(\cos 60° - \cos 215°)}{1.25(3.75 - \pi/3)} - \frac{\pi \times 2}{(1.25)^2(3.75 - \pi/3)} T_a$$

which gives $T_a = 15.4$ N-m.
3. Since ω_m is $-$ve, and $\alpha > 90°$, the motor is operating in the fourth quadrant, in one of the regenerative braking modes V to VII, and K is positive.

$$E = -\frac{640}{1500} \times 196.8 = -84V; \quad \omega_m = -\frac{640 \times 2\pi}{60} = -67 \text{ rad/sec}.$$

$$\frac{K\omega_m}{R_a} = \frac{1.25 \times -67}{2} = -41.9$$

$$\gamma' = \pi - \sin^{-1}(|E|/V_m) = \pi - \sin^{-1}\left(\frac{84}{230\sqrt{2}}\right) = 165°$$

Since $\alpha < \gamma'$ the drive may operate either in mode V or VI.
From equation (3.22),

$$i_a(\pi + \alpha) = 35.63[-0.95 - 0.95 \times 0.5] + (41.9)(1 - 0.5) = -30A$$

Since $i_a(\pi + \alpha) < 0$, the motor operates in the dicontinuous conduction mode VI.
From equation (3.23),

$$35.63 \sin(\beta° - 77.35°) + 41.9 + [-41.9 - 35.63 \times 0.95] \exp\left[\left(\frac{5\pi}{6} - \beta\right) \times 0.22\right] = 0$$

$$\text{or} \quad 35.63 \sin(\beta° - 77.35°) - 75.8 \exp\left[\left(\frac{5\pi}{6} - \beta\right) \times 0.22\right] = -41.9$$

Solution of this equation by trial and error gives $\beta = 238°$ or 4.15 rad.
From equation (3.26),

$$-67 = \frac{230\sqrt{2}(\cos 150° - \cos 238°)}{1.25(4.15 - 2.62)} - \frac{\pi \times 2}{(1.25)^2(4.15 - 2.62)} T_a$$

which gives $T_a = 3.74$ N-m.

4. As ω_m is negative and $\alpha > 90°$, the motor is operating in the fourth quadrant and K is positive.

$$E = -\frac{1600}{1500} \times 196.8 = -209.9 \text{ V}, \qquad \omega_m = -\frac{1600 \times \pi}{30} = -167.55 \text{ rad/sec.}$$

$$\gamma' = \pi - \sin^{-1}(|E|/V_m) = \pi - \sin^{-1}(209.9/230\sqrt{2}) = 139.8°.$$

Since $\alpha < \gamma'$, possible modes are V and VI

$$\frac{K\omega_m}{R_a} = 1.25 \times (-167.55)/2 = -104.7\,.$$

From equation (3.22)

$$i_a(\pi + \alpha) = 35.63[-0.795 - 0.795 \times 0.5] + 104.7[1 - 0.5] = 9.86\text{A}$$

Since $i_a(\pi + \alpha) > 0$ this is a continuous conduction mode V.
From equation (3.21)

$$-167.55 = \frac{2 \times 230\sqrt{2}}{\pi \times 1.25} \cos 130° - \frac{2}{(1.25)^2} T_a$$

which gives $T_a = 47.7$ N-m.

Example 3.3

For the motor of example 3.1, calculate the speed and torque on the boundary between continuous and discontinuous conductions for $\alpha = 60°$.

Solution: From equation (3.45)

$$\omega_{mc} = \frac{2 \times 35.63}{1.25} \sin(60° - 77.35°) \left[\frac{1 + \exp(-\pi \times 0.22)}{\exp(-\pi \times 0.22) - 1}\right]$$

$$= 51.3 \text{ rad/sec.} = 490 \text{ rpm}$$

Since equation (3.45) is valid only when $\alpha > \gamma$, we should check this condition

$$E = K\omega_{mc} = 1.25 \times 51.3 = 64.1 \text{ V}$$

$$\gamma = \sin^{-1} \frac{64.1}{230\sqrt{2}} = 11.4°$$

thus $\alpha > \gamma$
Now from equation (3.21),

$$51.3 = \frac{2 \times 230\sqrt{2}}{\pi \times 1.25} \cos 60° - \frac{2}{(1.25)^2} T_{ac}$$

which gives $T_{ac} = 24.6$ N-m.

Example 3.4

For the converter motor system of example 3.1, calculate the firing angles for the following points:

1. $T_a = 30$ N-m and $N = 424$ rpm
2. $T_a = 32$ N-m and $N = -1178$ rpm
3. $T_a = 20$ N-m and $N = 733$ rpm

where N is the speed in rpm. L_a is 28.36 mH.

Solution: From examples 3.1 and 3.2, E at 1500 rpm = 196.8, K = 1.25, $\omega = 100\pi$ rad/sec. $V_m/Z = 35.6$ A, $\psi = 1.35$ rad or 77.35°, cot $\psi = 0.22$.

Before α can be calculated, the modes of operation must be identified. This can be done from figure 3.15 as follows:
From equations (3.49) and (3.50)

$$\omega_{mn} = \frac{\pi K \omega_m}{2V_m} = \frac{\pi K}{2V_m} \cdot \frac{\pi N}{30} = \frac{\pi^2 \times 1.25N}{60 \times 230\sqrt{2}} = 6.32 \times 10^{-4} N$$

$$T_{an} = \frac{\pi R_a}{2V_m} \cdot \frac{T_a}{K} = \frac{\pi \times 2}{2 \times 230\sqrt{2} \times 1.25} T_a = 7.73 \times 10^{-3} T_a$$

For the foregoing points, the normalized values are

1. $T_{an} = 0.23$ $\omega_{mn} = 0.27$
2. $T_{an} = 0.247$ $\omega_{mn} = -0.74$
3. $T_{an} = 0.15$ $\omega_{mn} = 0.46$

Figure 3.15a shows that the points 1 and 2 lie on the right of the boundary between continuous and discontinuous conductions for $\psi = 1.35$, and, therefore, the drive operates in continuous conduction at these points. The drive operates in discontinuous conduction at point 3 because it is on the left of the boundary. Now consider the calculation for each point separately.

1. From (3.21),

$$\frac{424 \times \pi}{30} = \frac{2 \times 230\sqrt{2}}{\pi \times 1.25} \cos \alpha - \frac{2}{(1.25)^2} \times 30$$

which gives $\cos \alpha = 0.50$ or $\alpha = 60°$.

2. Again from equation (3.21)

$$-\frac{1178 \times \pi}{30} = \frac{2 \times 230\sqrt{2}}{\pi \times 1.25} \cos \alpha - \frac{2 \times 32}{(1.25)^2}$$

which gives $\cos \alpha = -0.5$ or $\alpha = 120°$.

3. Equation (3.19) is valid both for continuous and discontinuous conductions. From this equation

$$V_a = I_a R_a + E = \frac{T_a}{K} R_a + E = \frac{20}{1.25} \times 2 + 96.2 = 128.2 \text{ V}.$$

In the present case the conduction is discontinuous. The drive may operate in modes II or IV. As a first approximation, α is calculated assuming continuous conduction. Thus,

$$\frac{2V_m}{\pi} \cos \alpha = 128.2$$

or

$$\cos \alpha = \frac{128.2 \times \pi}{2 \times 230\sqrt{2}} = 0.62 \quad \text{or} \quad \alpha = 51.7°$$

Now

$$\gamma = \sin^{-1}(E/V_m) = \sin^{-1}\left(\frac{96.2}{230\sqrt{2}}\right) = 17.2°.$$

Thus the approximate value of α, calculated assuming continuous conduction, is greater than γ. The effect of discontinuous conduction is to increase V_a for a given α or increase α for a given V_a. Therefore, the actual value of α must be greater than 51.7°. And hence the drive will be operating in mode II.

Substituting known values in equations (3.23) and (3.26) gives

$$35.63 \sin(\beta° - 77.35°) + [48 - 35.63 \sin(\alpha° - 77.35°)] \exp\{(\alpha - \beta) \times 0.22\} = 48 \quad \text{(E3.2)}$$

$$325.27(\cos \alpha° - \cos \beta°) - 96.2(\beta - \alpha) = 100.53 \quad \text{(E3.3)}$$

Simultaneous solution of equations (E3.2) and (E3.3) gives

$$\alpha = 60° \quad \text{and} \quad \beta° = 228°.$$

3.3.5 Rectifier with Controlled Flywheeling[5–10]

The output voltage waveforms of a single-phase fully-controlled rectifier have both positive and negative excursions. Since the current in the rectifier is always positive, the energy flows from the source to the load during a positive excursion and from the load to the source during a negative excursion. Thus, a negative excursion of the voltage during rectification produces energy which flows back and forth between the source and the load. It is nothing but reactive energy. Similarly, a positive excursion of the voltage during inversion produces the reactive energy. Hence, negative excursions of voltage during rectification and positive excursions of voltage during inversion cause a large amount of reactive power to be drawn from the source, particularly at low output voltages. In controlled flywheeling, negative excursions of the output voltage during rectification and positive excursions of the output voltage during inversion are eliminated by diverting the load current from the source to the freewheeling paths formed by the series connected thyristor pairs T_1, T_4 and T_2, T_3. This gives rise to a considerable improvement in the rectifier power factor and the motor performance. The rectifier power circuit remains the same as shown in figure 3.7.

The principle of controlled flywheeling is explained with the help of the waveforms of figure 3.11. The thyristor gate pulses are shown in figure 3.11a and the load voltage waveform under the assumption of continuous conduction is shown in figure 3.11b. The thyristor pair T_1, T_3 is turned on at α, connecting the source to the load. At an angle $\pi + \alpha_n$, T_4 is gated, which turns on T_4 and turns off T_3. The load current now freewheels through the thyristor pair T_1, T_4. In rectification, α_n is made zero and the output voltage variation is obtained by varying α from 0 to π. In inversion, α is made equal to $180 - \delta$, where δ is a small angle required for the safe commutation of thyristors. The variation of output voltage is obtained by controlling α_n from 0 to π. The load voltage waveforms for rectification and inversion are shown in figures 3.12a and 3.13a, respectively.

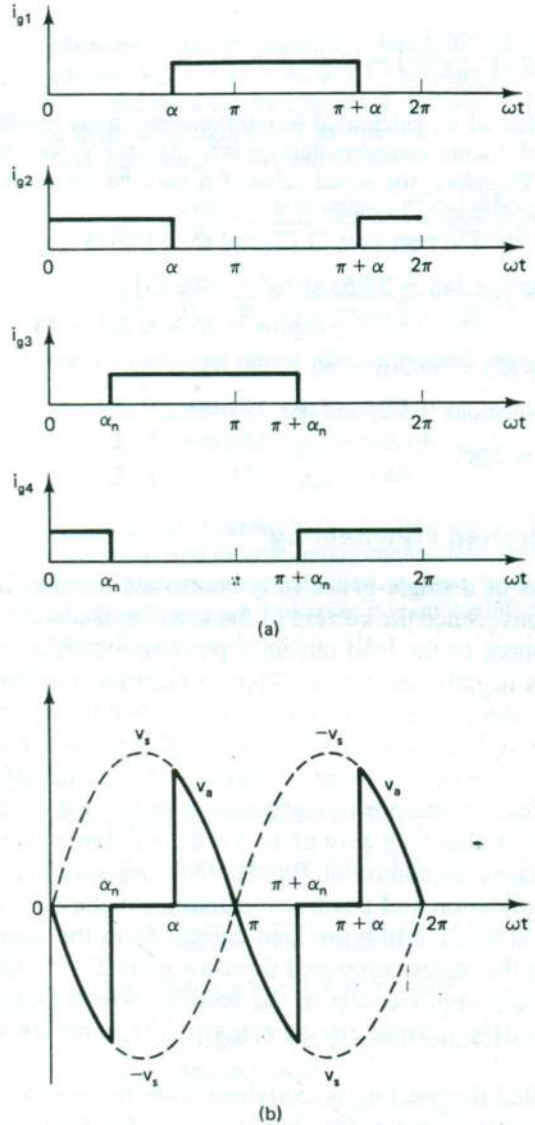


Figure 3.11 Principle of controlled flywheeling.

The modes of operation of a separately excited motor fed by a rectifier with the controlled flywheeling are shown in figures 3.12 and 3.13 for motoring and regenerative braking, respectively. The same notations have been used as listed in section 3.3.1. These modes can be explained following the arguments mentioned for the conventional operation of the rectifier in section 3.3.1. Each period of the motor terminal voltage may consist of any two or three from among the duty, freewheeling, and zero-current intervals. The motor operation for the duty and zero-current inter-

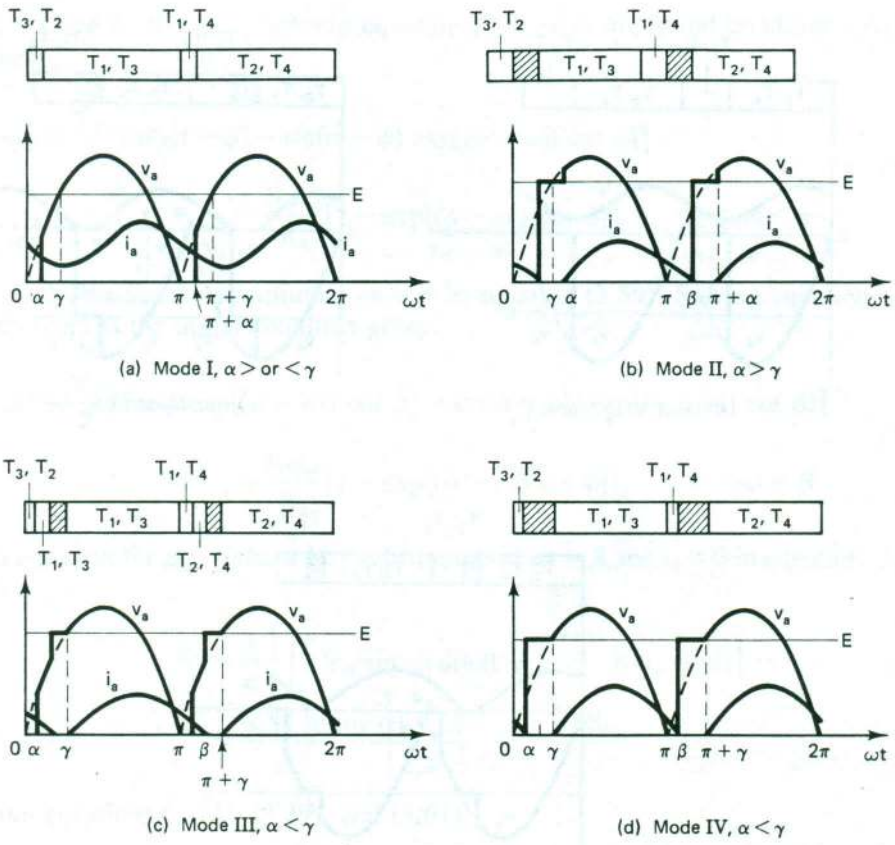


Figure 3.12 Modes of operation for motoring of dc separately excited motor-fed by 1-phase rectifier with controlled flywheeling with $\alpha_n = \pi$.

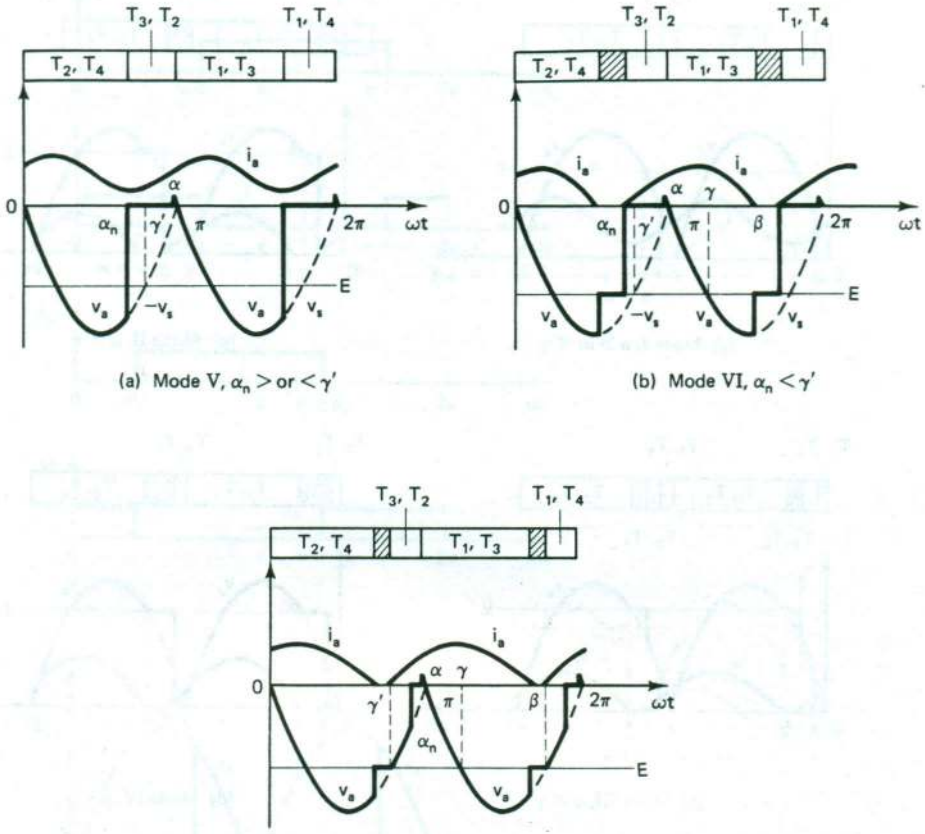
vals is described by equations (3.4) to (3.7). The following equation describes the operation for the freewheeling interval:

$$L_a \frac{di_a}{dt} + R_a i_a + K\omega_m = v_a = 0 \tag{3.56}$$

The performance equations for various modes of operation can be derived in the same way as for the conventional operation of the rectifier in section 3.3.2. The important equations are given next.

Motoring Operation

The current is continuous in mode I, and α can be greater or less than γ. The current is discontinuous in modes II to IV, mode II is for $\alpha > \gamma$, and modes III and IV are for $\alpha < \gamma$.



(a) Mode V, $\alpha_n > \text{or} < \gamma'$

(b) Mode VI, $\alpha_n < \gamma'$

(c) Mode VII, $\alpha_n > \gamma'$

Figure 3.13 Modes of operation for regenerative braking of separately excited motor-fed by 1-phase rectifier with controlled flywheeling with $\alpha = (\pi - \delta)$.

Mode I [Fig. 3.12a]:

$$V_a = \frac{1}{\pi}\int_{\alpha}^{\pi} V_m \sin \omega t \, d(\omega t) = \frac{V_m}{\pi}(1 + \cos \alpha) = \frac{V_{ao}}{2}(1 + \cos \alpha) \qquad (3.57)$$

From equations (3.14), (3.19), and (3.57),

$$\omega_m = \frac{V_m}{\pi K}(1 + \cos \alpha) - \frac{R_a}{K^2} T_a \qquad (3.58)$$

Mode II [Fig. 3.12b]: Here the current may become zero in the freewheeling interval ($\beta > \pi$) or in the duty interval ($\beta < \pi$). Figure 3.12b has been drawn for the former case.

Case 1. $\beta > \pi$: Solving equation (3.4) with the initial condition $i_a(\alpha) = 0$ gives

$$i_a(\omega t) = \frac{V_m}{Z}[\sin(\omega t - \psi) - \sin(\alpha - \psi)\exp\{(\alpha - \omega t)\cot\psi\}] - \frac{K\omega_m}{R_a}[1 - \exp\{(\alpha - \omega t)\cot\psi\}], \qquad \alpha \le \omega t \le \pi \tag{3.59}$$

$i_a(\pi)$ is obtained by substituting $\omega t = \pi$ in equation (3.59). Solving equation (3.56) with $i_a(\pi)$ as the initial condition gives

$$i_a(\omega t) = \frac{V_m}{Z}[\sin\psi\exp\{(\pi - \omega t)\cot\psi\} - \sin(\alpha - \psi)\cdot\exp\{(\alpha - \omega t)\cot\psi\}] - \frac{K\omega_m}{R_a}[1 - \exp\{(\alpha - \omega t)\cot\psi\}], \qquad \pi \le \omega t \le \beta \tag{3.60}$$

An equation for β is obtained by substitution of $\omega t = \beta$ and $i_a = 0$ in equation (3.60). Now

$$V_a = \frac{1}{\pi}\left[\int_\alpha^\pi V_m \sin\omega t\, d(\omega t) + \int_\beta^{(\pi+\alpha)} K\omega_m\, d(\omega t)\right] = \frac{V_m(1 + \cos\alpha) + (\pi + \alpha - \beta)K\omega_m}{\pi} \tag{3.61}$$

From equations (3.14), (3.19), and (3.61),

$$\omega_m = \frac{V_m(1 + \cos\alpha)}{K(\beta - \alpha)} - \frac{\pi R_a}{K^2(\beta - \alpha)}T_a \tag{3.62}$$

Case 2. $\beta < \pi$: An equation for β is obtained by the substitution of $\omega t = \beta$ and $i_a = 0$ in equation (3.59). Now,

$$V_a = \frac{1}{\pi}\left[\int_\alpha^\beta V_m \sin\omega t\, d(\omega t) + \int_\beta^{(\pi+\alpha)} K\omega_m\, d(\omega t)\right] = \frac{V_m(\cos\alpha - \cos\beta) + (\pi + \alpha - \beta)K\omega_m}{\pi} \tag{3.63}$$

From equations (3.14), (3.19), and (3.63)

$$\omega_m = \frac{V_m(\cos\alpha - \cos\beta)}{K(\beta - \alpha)} - \frac{\pi R_a}{K^2(\beta - \alpha)}T_a \tag{3.64}$$

Mode III [Fig. 3.12c]: Here i_a starts flowing at γ compared to at α in mode II. Therefore, the expressions for i_a for the intervals $\gamma \le \omega t \le \pi$ and $\pi \le \omega t \le (\pi + \alpha)$ are obtained by replacing α with γ in equations (3.59) and (3.60), respec-

tively. Solving equation (3.5) with the value of $i_a(\pi + \alpha)$, obtained from the later equation, as the initial condition gives

$$i_a(\omega t) = \frac{V_m}{Z}[\sin \psi \exp\{(\pi - \omega t) \cot \psi\} - \sin(\alpha - \psi) \exp\{(\pi + \alpha - \omega t) \cot \psi\} - \sin(\gamma - \psi) \exp\{(\psi - \omega t) \cot \psi\} - \sin(\omega t - \psi)] \quad (3.65)$$

$$- \frac{K\omega_m}{R_a}[1 - \exp\{(\gamma - \omega t) \cot \psi\}], \qquad (\pi + \alpha) \leq \omega t \leq \beta$$

An equation for β is obtained from equation (3.65) by substituting $\omega t = \beta$ and $i_a = 0$. Now

$$V_a = \frac{1}{\pi}\left[\int_{\gamma}^{\pi} V_m \sin \omega t \, d(\omega t) + \int_{\pi+\alpha}^{\beta} - V_m \sin \omega t \, d(\omega t) + \int_{\beta}^{\pi+\gamma} K\omega_m \, d(\omega t)\right]$$

$$= \frac{V_m(1 + \cos \gamma + \cos \beta + \cos \alpha) + K\omega_m(\pi + \gamma - \beta)}{\pi} \quad (3.66)$$

From equations (3.14), (3.15), and (3.66)

$$\omega_m = \frac{V_m(1 + \cos \alpha + \cos \beta + \cos \gamma)}{K(\beta - \gamma)} - \frac{\pi R_a}{K^2(\beta - \gamma)} T_a \quad (3.67)$$

Mode IV [Fig. 3.12d]: As in mode II, the current may become zero either before π or after π. Equations for this mode are obtained by replacing α with γ in equations (3.59) to (3.64).

Regenerative Braking

Here either ω_m will be negative (fourth quadrant operation) or K will be negative (second quadrant operation with the field or armature reversal). For the analysis, δ is assumed to be zero and therefore $\alpha = \pi$. While the current is continuous in mode V, it is discontinuous in modes VI and VII. For mode VI, $\alpha_n < \gamma'$, and for mode VII, $\alpha_n > \gamma'$.

Mode V [Fig. 3.13a]: A cycle of the output voltage from α_n to $\pi + \alpha_n$ consists of a freewheeling interval from α_n to π and a duty interval from π to $\pi + \alpha_n$ with $v_a = v_s$. Therefore,

$$V_a = \frac{1}{\pi}\int_{\pi}^{\pi+\alpha_n} V_m \sin \omega t \, d(\omega t) = -\frac{V_m}{\pi}(1 - \cos \alpha_n) \quad (3.68)$$

From equations (3.14), (3.19), and (3.68),

$$\omega_m = -\frac{V_m}{\pi K}(1 - \cos \alpha_n) - \frac{T_a}{K^2} R_a \quad (3.69)$$

Mode VI [Fig. 3.13b]: A cycle of the output voltage from α_n to $\pi + \alpha_n$ consists of a freewheeling interval from α_n to π, a duty interval from π to β with $v_a = v_s$, and a zero-current interval from β to $\pi + \alpha_n$. An expression for i_a for the interval $\alpha_n \leq \omega t \leq \pi$ is obtained from the solution of equation (3.56) with the initial

condition $i_a(\alpha_n) = 0$. $i_a(\pi)$ can be obtained from this expression. Then the solution of equation (3.4) with $i_a(\pi)$ as the initial current gives

$$i_a(\omega t) = \frac{V_m}{Z}[\sin(\omega t - \psi) - \sin\psi \exp\{(\pi - \omega t)\cot\psi\}] - \frac{K\omega_m}{R_a}[1 - \exp\{(\alpha_n - \omega t)\cot\psi\}], \qquad \pi \le \omega t \le \beta \tag{3.70}$$

An equation for β can be obtained by the substitution of $\omega t = \beta$ and $i_a = 0$ in equation (3.70). Now

$$V_a = \frac{1}{\pi}\left[\int_{\pi}^{\beta} V_m \sin\omega t\, d(\omega t) + \int_{\beta}^{\pi+\alpha_n} K\omega_m\, d(\omega t)\right] = \frac{-V_m(1 + \cos\beta) + (\pi + \alpha_n - \beta)K\omega_m}{\pi} \tag{3.71}$$

From equations (3.14), (3.19), and (3.71)

$$\omega_m = -\frac{V_m(1 + \cos\beta)}{(\beta - \alpha_n)K} - \frac{\pi R_a}{K^2(\beta - \alpha_n)}T_a \tag{3.72}$$

Mode VII [Fig. 3.13c]: A cycle of the output voltage from γ' to $\pi + \gamma'$ consists of a duty interval from γ' to α_n with $v_a = -v_s$, a freewheeling interval from α_n to π, a duty interval from π to β with $v_a = v_s$, and a zero-current interval from β to $\pi + \gamma'$. Solution of equations (3.5), (3.56), and (3.4) for the intervals $\gamma' \le \omega t \le \alpha_n$, $\alpha_n \le \omega t \le \pi$ and $\pi \le \omega t \le \beta$, respectively, with the initial condition $i_a(\gamma') = 0$ gives the following expression for the current:

$$i_a(\omega t) = \frac{V_m}{Z}[\sin(\omega t - \psi) + \sin(\gamma' - \psi)\exp\{(\gamma' - \omega t)\cot\psi\} - \sin(\alpha_n - \psi)\exp\{(\alpha_n - \omega t)\cot\psi\} - \sin\psi\exp\{(\pi - \omega t)\cot\psi\}] - \frac{K\omega_m}{R_a}[1 - \exp\{(\gamma' - \omega t)\cot\psi\}], \qquad \pi \le \omega t \le \beta \tag{3.73}$$

An equation for β can be obtained by the substitution of $\omega t = \beta$ and $i_a = 0$ in equation (3.73). Now

$$V_a = \frac{1}{\pi}\left[\int_{\gamma'}^{\alpha_n} -V_m \sin\omega t\, d(\omega t) + \int_{\pi}^{\beta} V_m \sin\omega t\, d(\omega t) + \int_{\beta}^{\pi+\gamma'} K\omega_m\, d(\omega t)\right] = \frac{-V_m(1 + \cos\gamma' + \cos\beta - \cos\alpha_n) + K\omega_m(\pi + \gamma' - \beta)}{\pi} \tag{3.74}$$

From equations (3.14), (3.19), and (3.74)

$$\omega_m = -\frac{V_m(1 + \cos\gamma' + \cos\beta - \cos\alpha_n)}{K(\beta - \gamma')} - \frac{\pi R_a}{K^2(\beta - \gamma')}T_a \tag{3.75}$$

The nature of speed-torque characteristics is shown in figure 3.14 for various values of α and α_n. For the motoring operation, no-load speeds are given by equations (3.43) and (3.44); and for the braking operation, the no-load speed is 0 for all α_n.

The boundaries between continuous and discontinuous conductions on the normalized speed-torque plane are shown in figure 3.15b for different values of ψ.[10] Normalization has been done using the same base values as in equations (3.48) to (3.51). A comparison of these boundaries with those of figure 3.15a for the conventional operation (without flywheeling) shows a considerable reduction in the zone of discontinuous conduction for the controlled flywheeling.

Example 3.5

Repeat example 3.4 for a rectifier with controlled flywheeling.

Solution: The normalized values of torque and speed have been obtained in example 3.4. From the boundary for $\psi = 1.35$ rad. in figure 3.15b, it is found that all the three points provide continuous conduction. Now consider the calculation for each point separately.

1. From equation (3.58)

$$\frac{424 \times \pi}{30} = \frac{230\sqrt{2}}{\pi \times 1.25}(1 + \cos \alpha) - \frac{2}{(1.25)^2} \times 30$$

which gives $\cos \alpha = 0$ or $\alpha = 90°$ and always $\alpha_n = 0$ for motoring.

2. At this point the drive works in mode V. From equation (3.69),

$$-\frac{1178 \times \pi}{30} = -\frac{230\sqrt{2}}{\pi \times 1.25}(1 + \cos \alpha_n) - \frac{2 \times 32}{(1.25)^2}$$

which gives $\alpha_n = 90°$ and always $\alpha = 180°$ for braking.

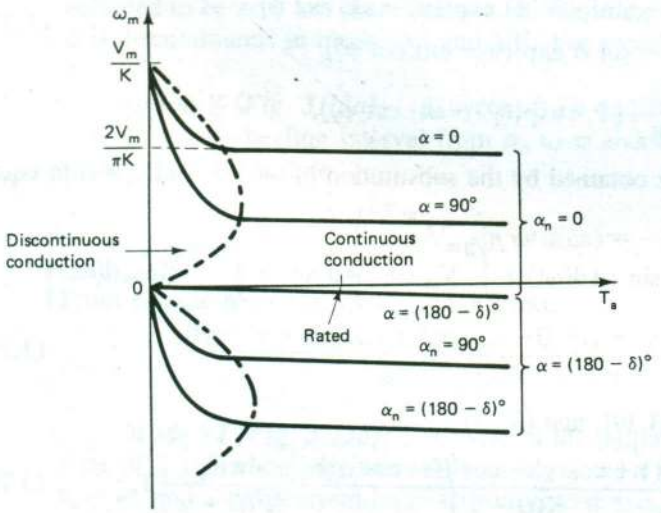


Figure 3.14 Speed-torque characteristics of 1-phase rectifier drive with controlled flywheeling.

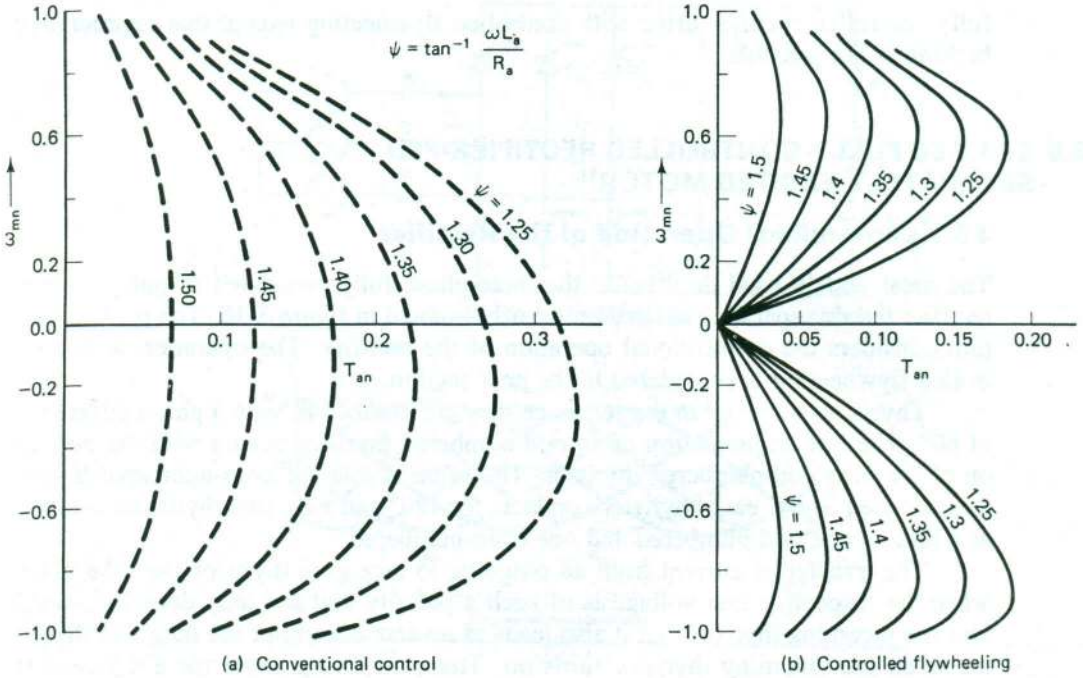


Figure 3.15 Boundaries between continuous and discontinuous conductions for 1-phase fully-controlled rectifier on normalised speed (ω_{mn}) and torque (T_n) plane; $\omega_{mn} = E/(2V_m/\pi)$ and $T_{an} = I_a/(2V_m/\pi R_a)$.

3. Again from equation (3.58)

$$\frac{733 \times \pi}{30} = \frac{230\sqrt{2}}{\pi \times 1.25}(1 + \cos \alpha) - \frac{2}{(1.25)^2} \times 20$$

which gives $\cos \alpha = 0.24$ or $\alpha = 76°$ and for motoring α_n is always 0.

3.4 1-PHASE HALF-CONTROLLED RECTIFIER-FED SEPARATELY EXCITED MOTOR

When regenerative braking is not required, the motor is fed by a half-controlled rectifier (fig. 3.4b), which is cheaper than a fully-controlled rectifier. The modes of operation and various waveforms of a half-controlled rectifier drive are identical to modes I to IV of figure 3.12, which are for the motoring operation of the fully-controlled rectifier drive with controlled flywheeling, except that different devices conduct during the various intervals. The equations derived in section 3.3.5 are also applicable. The speed-torque characteristics are the same as shown in the first quadrant of figure 3.14. The half-controlled rectifier drive has all the advantages of the

fully-controlled rectifier drive with controlled flywheeling except that regenerative braking is not possible.

3.5 3-PHASE FULLY-CONTROLLED RECTIFIER-FED SEPARATELY EXCITED MOTOR[11]

3.5.1 Conventional Operation of the Rectifier

The most widely used dc drive is the three-phase fully-controlled (6-pulse) bridge rectifier fed dc separately excited motor drive shown in figure 3.16. The present section considers the conventional operation of the rectifier. The operation with controlled flywheeling is considered in the next section.

Thyristors are fired in the sequence they are numbered, with a phase difference of 60°. The line commutation of an odd-numbered thyristor occurs with the turning on of the next odd-numbered thyristor. The same is true for even-numbered thyristors. Consequently, each thyristor conducts for 120° and only two thyristors conduct at a time — one odd-numbered and one even-numbered.

The transfer of current from an outgoing to incoming thyristor can take place when the respective line voltage is of such a polarity that not only does it forward bias the incoming thyristor but it also leads to reverse biasing of the outgoing thyristor when the incoming thyristor turns on. Thus, the firing angle for a thyristor is measured from the instant when the respective line voltage is zero and increasing. For example, the transfer of current from thyristor T_5 to thyristor T_1 can occur as long as the line voltage v_{AC} is positive. Hence, for thyristor T_1, the firing angle α is measured from the instant $v_{AC} = 0$ and increasing as shown in figure 3.16. Each firing pulse is of 120° duration and shifted to vary the firing angle from 0 to $(180° - \delta)$, where δ is a small positive angle required for the commutation of thyristors. If the line voltage v_{AB} is taken as the reference voltage, then

$$v_{AB} = V_m \sin \omega t \tag{3.76}$$

and

$$\alpha = \omega t - \pi/3 \tag{3.77}$$

The drive has the same number of and similar modes of operation as the 1-phase fully-controlled rectifier-fed drive with the conventional operation (figs. 3.8 and 3.9). The discontinuous conduction modes for the motoring operation with $\alpha < \gamma$ (corresponding to modes III and IV of fig. 3.8) and for the regenerative braking with $\alpha > \gamma'$ (corresponding to mode VII of fig. 3.9) occur in a very narrow range of operation and can be ignored without any appreciable loss of accuracy in the calculation of the drive performance. The remaining four modes of operation are shown in figure 3.17. Modes I and III are continuous conduction modes and modes II and IV are discontinuous conduction modes for the motoring and braking operations, respectively.

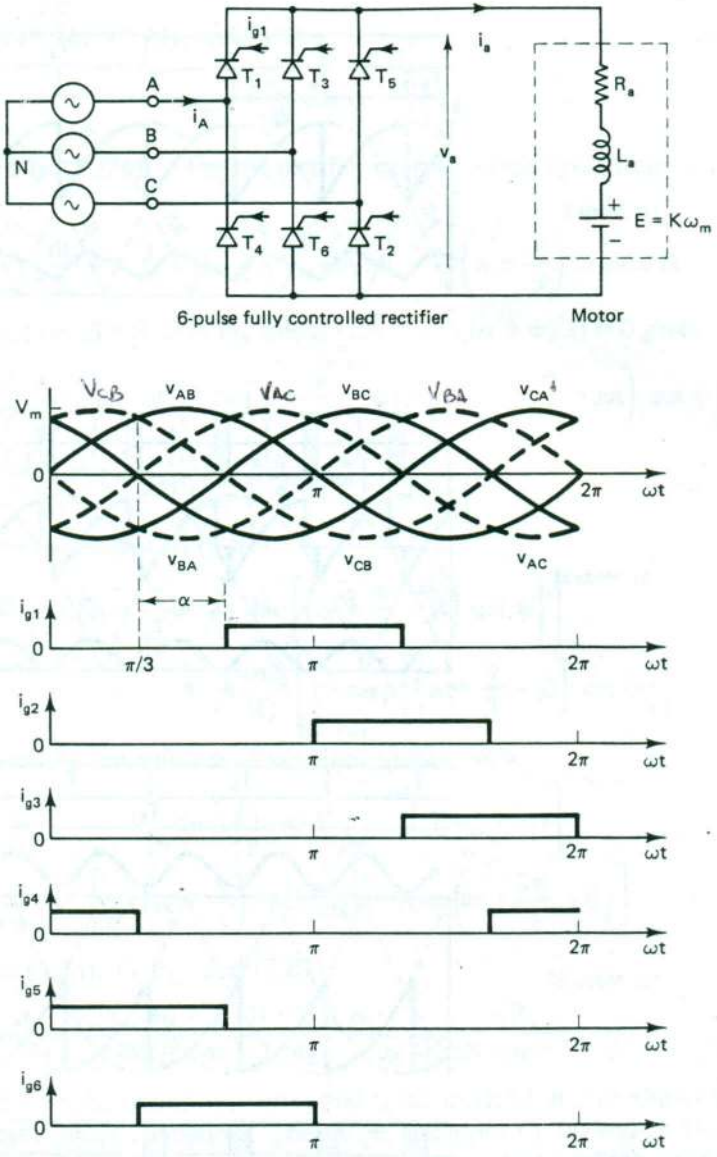


Figure 3.16 3-phase fully-controlled rectifier-fed dc motor.

Mode I [Fig. 3.17a]: For the converter output voltage cycle from $\omega t = \alpha + \pi/3$ to $\omega t = \alpha + 2\pi/3$,

$$V_a = \frac{3}{\pi}\int_{\alpha+\pi/3}^{\alpha+2\pi/3} V_m \sin \omega t \, d(\omega t) = \frac{3}{\pi} V_m \cos \alpha = V_{ao} \cos \alpha \qquad (3.78)$$

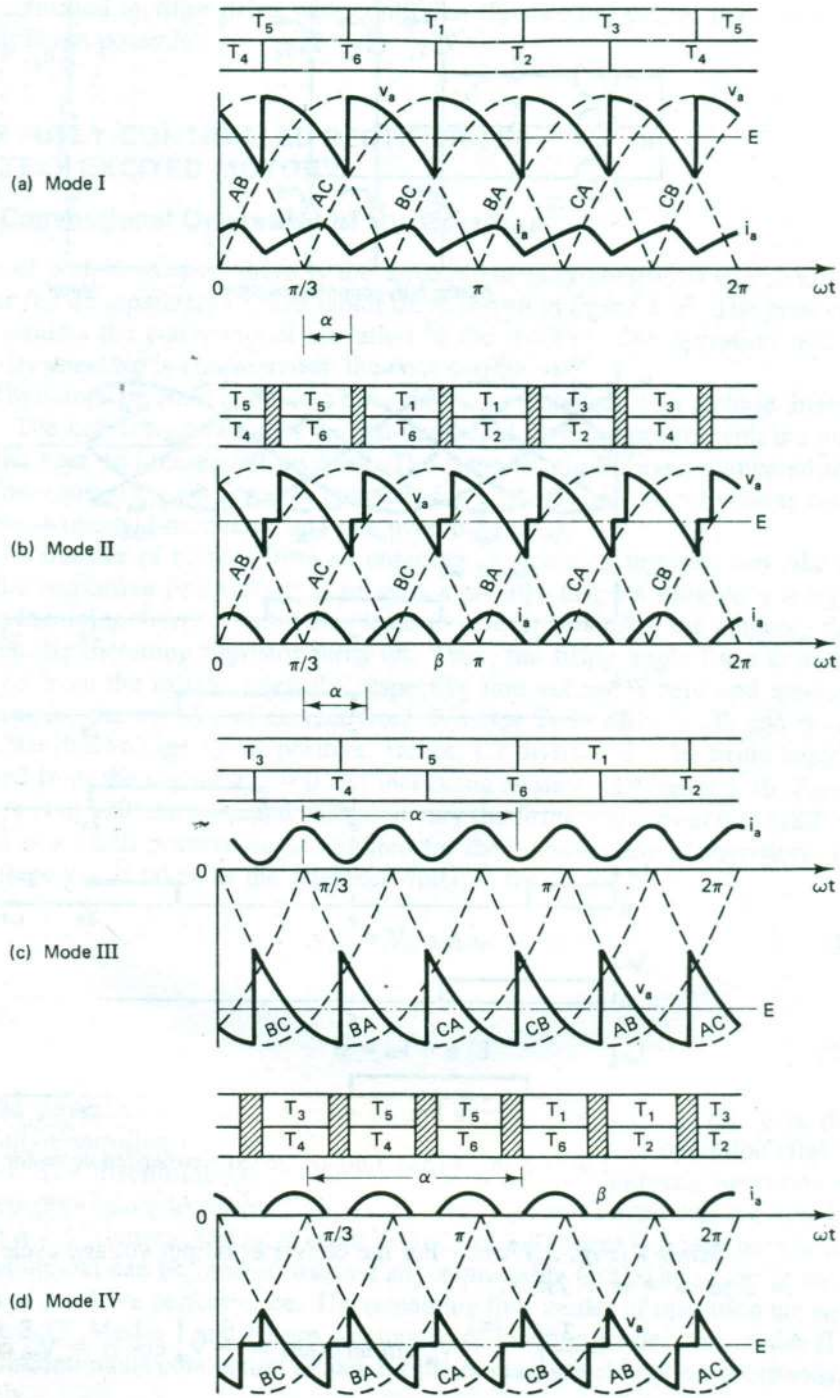


Figure 3.17 Modes of operation of 3-phase fully-controlled rectifier-fed dc motor.

where $V_{ao} = 3V_m/\pi$
From equations (3.14), (3.19), and (3.78),

$$\omega_m = \frac{3V_m}{\pi K}\cos\alpha - \frac{R_a}{K^2}T_a \tag{3.79}$$

Mode II [Fig. 3.17b]: For the rectifier output voltage cycle from $\alpha + \pi/3$ to $\alpha + 2\pi/3$

$$v_a = L_a\frac{di_a}{dt} + R_a i_a + K\omega_m = V_m \sin\omega t, \qquad \left(\alpha + \frac{\pi}{3}\right) \le \omega t \le \beta \tag{3.80}$$

Solution of equation (3.80) with the initial condition $i_a(\alpha + \pi/3) = 0$ gives

$$i_a(\omega t) = \frac{V_m}{Z}\left[\sin(\omega t - \psi) - \sin\left(\alpha + \frac{\pi}{3} - \psi\right)\exp\left\{\left(\alpha + \frac{\pi}{3} - \omega t\right)\cot\psi\right\}\right] - \frac{K\omega_m}{R_a}\left[1 - \exp\left\{\left(\alpha + \frac{\pi}{3} - \omega t\right)\cot\psi\right\}\right] \tag{3.81}$$

Since $i_a(\beta) = 0$, from equation (3.81)

$$\frac{V_m}{Z}\left[\sin(\beta - \psi) - \sin\left(\alpha + \frac{\pi}{3} - \psi\right)\exp\left\{\left(\alpha + \frac{\pi}{3} - \beta\right)\cot\psi\right\}\right] = \frac{K\omega_m}{R_a}\left[1 - \exp\left\{\left(\alpha + \frac{\pi}{3} - \beta\right)\cot\psi\right\}\right] \tag{3.82}$$

β can be obtained by the solution of equation (3.82). Now

$$V_a = \frac{3}{\pi}\left[\int_{\alpha+\pi/3}^{\beta} V_m \sin\omega t\, d(\omega t) + \int_{\beta}^{\alpha+2\pi/3} K\omega_m\, d(\omega t)\right] = \frac{3}{\pi}\left[V_m\left\{\cos\left(\alpha + \frac{\pi}{3}\right) - \cos\beta\right\} + K\omega_m\left(\alpha + \frac{2\pi}{3} - \beta\right)\right] \tag{3.83}$$

From equations (3.14), (3.19), and (3.83)

$$\omega_m = \frac{V_m\{\cos(\alpha + \pi/3) - \cos\beta\}}{K(\beta - \alpha - \pi/3)} - \frac{\pi R_a}{3K^2(\beta - \alpha - \pi/3)}T_a \tag{3.84}$$

The examination of the waveforms of v_a and i_a for modes I and III shows that they are governed by the same equations. Therefore, equations (3.78) and (3.79) are also applicable to mode III. Since mode III is for regenerative braking operation for which E is negative, speed should be taken negative for the fourth-quadrant operation and K should be taken negative for the second-quadrant operation, as explained in section 3.3.2.

What has been said about modes I and III is also true for modes II and IV. Therefore equations (3.80) to (3.84) are applicable to mode IV when appropriate signs are used for ω_m and K.

The speed-torque characteristics of a 2.2 kW, 1500 rpm dc motor fed by a 3-phase fully-controlled rectifier with an ac source voltage of 170.3 V (line), 50 Hz is shown in figure 3.18. The regions of continuous and discontinuous conductions and

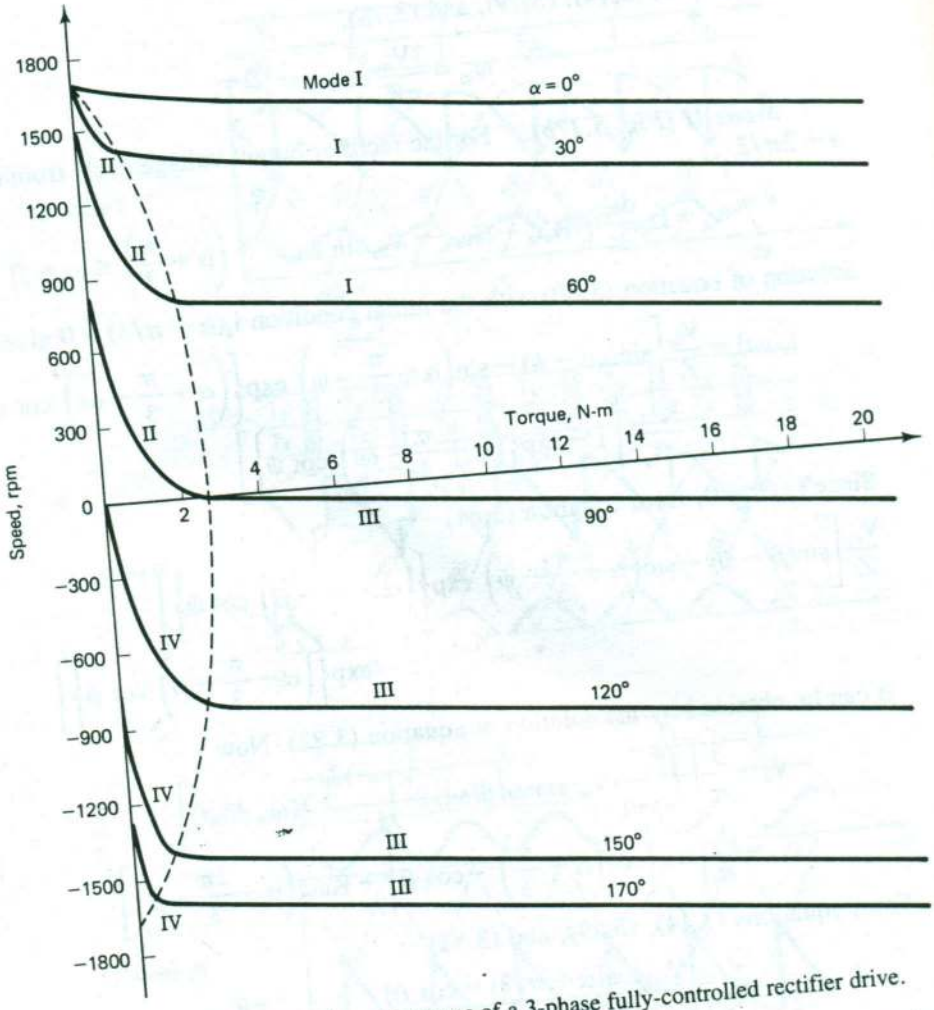


Figure 3.18 Speed-torque curves of a 3-phase fully-controlled rectifier drive.

the modes of operation have been marked. Comparison of these characteristics with those of figure 3.10 shows a considerable reduction in the zone of discontinuous conduction.

The ideal no-load operation is obtained when $E = V_m$ for $0 \le \alpha \le \pi/6$ rad. and $E = V_m \sin(\alpha + \pi/3)$ for $\pi/6 \le \alpha \le \pi$ radians. Thus, the no-load speeds are given by the following equations:

$$\omega_{mo} = \frac{V_m}{K}, \qquad 0 \le \alpha \le \pi/6 \tag{3.85}$$

$$\omega_{mo} = \frac{V_m \sin(\alpha + \pi/3)}{K}, \qquad \pi/6 \le \alpha \le \pi \tag{3.86}$$

A comparison of equations (3.44) and (3.86) and figures 3.10 and 3.18 shows that, unlike in the single-phase case, no-load speeds can be negative also (for $2\pi/3 < \alpha < \pi$).

When operating in mode II, the motor torque increases β. The boundary between discontinuous and continuous conductions is reached when $\beta = \alpha + 2\pi/3$. Substitution of this value of β in equation (3.82) gives an expression for the critical speed ω_{mc}. The critical torque T_{ac} is obtained from equation (3.79).

Figure 3.21 shows the boundaries between continuous and discontinuous conductions on the normalized speed-torque plane for different values of ψ.[11] In drawing these boundaries all the modes have been taken into account.

For normalization, the base voltage V_B is taken to be equal to the maximum (average) converter output voltage V_{ao} [equation (3.78)]. The base current I_B is chosen to be equal to the average motor current that will flow when $\omega_m = 0$ and $V_a = V_B$. Thus,

$$V_B = V_{ao} = \frac{3V_m}{\pi} \quad \text{and} \quad I_B = \frac{V_B}{R_a} = \frac{V_{ao}}{R_a} = \frac{3V_m}{\pi R_a} \tag{3.87}$$

Thus the normalized speed ω_{mn} and the normalized torque T_{an} are given by the following equations:

$$\omega_{mn} = \frac{E}{V_B} = \frac{E}{V_{ao}} = \frac{\pi E}{3V_m}, \qquad T_{an} = I_{an} = \frac{I_a}{I_B} = \frac{\pi R_a}{3V_m}(I_a) \tag{3.88}$$

These boundaries are helpful in calculating the motor performance and selecting a value of the filter inductance which will allow the discontinuous conduction to be eliminated for the drive's steady-state operation.

Example 3.6

A 220 V, 1500 rpm dc motor has the armature resistance and inductance of 2 Ω and 28.36 mH, respectively. It is controlled by a 3-phase fully-controlled rectifier from an ac source operating at 50 Hz.

1. Calculate the ac source voltage required to get the rated voltage across the motor terminals when operating in continuous conduction.
2. For the ac source voltage obtained in 1, calculate the motor speed for the following points:

(a) $\alpha = 60°$, $T_a = 1.0$ N-m
(b) $\alpha = 60°$, $T_a = 20.0$ N-m
(c) $\alpha = 120°$, $T_a = 20.0$ N-m

Solution: **1.** From equation (3.78) for $\alpha = 0$

$$220 = \frac{3x\sqrt{2}\,V}{\pi} \quad \text{or} \quad V = 162.9 \text{ V (line)}$$

and $V_m = 230.38$ V.

2. Before the speed can be obtained for a given α and T_a, one must know the mode of operation. For this the critical torque is evaluated. Then if the given torque is greater than the critical, the conduction will be continuous; otherwise, discontinuous.

Substitution of $\beta = \alpha + 2\pi/3$ in equation (3.82) and rearranging the terms gives

$$\omega_{mc} = \frac{R_a}{K}\frac{V_m}{Z}\left[\frac{\sin\left(\alpha + \frac{2\pi}{3} - \psi\right) - \sin\left(\alpha + \frac{\pi}{3} - \psi\right)\exp\left(-\frac{\pi}{3}\cot\psi\right)}{1 - \exp\left(-\frac{\pi}{3}\cot\psi\right)}\right] \quad \text{(E3.4)}$$

From examples 3.1 and 3.2

$$K = 1.25, \quad Z = 9.13\Omega, \quad \psi = 1.35 \text{ rad} \quad \text{or} \quad 77.35° \quad \text{and} \quad \cot\psi = 0.22.$$

$\alpha = 60°$: Substituting the known values in equation (E3.4) gives

$$\omega_{mc} = \frac{2 \times 162.9\sqrt{2}}{1.25 \times 9.13}\left[\frac{\sin(102.65°) - \sin(42.65°) \times 0.794}{1 - 0.794}\right)$$

$$= 85.8 \text{ rad/sec}.$$

$$I_{ac} = \frac{V_a - K\omega_{mc}}{R_a} = \frac{\frac{3}{\pi}V_m \cos\alpha - K\omega_{mc}}{R_a}$$

$$= \frac{\frac{3}{\pi}162.9\sqrt{2}\cos 60° - 1.25 \times 85.8}{2}$$

$$= 1.37\text{A}$$

$$T_{ac} = K \times 1.37 = 1.25 \times 1.37 = 1.71 \text{ N-m}.$$

Comparison of torques at points (a) and (b) with T_{ac} shows that the drive operates in discontinuous conduction at (a) and in continuous conduction at (b).

$\alpha = 120°$:

$$\omega_{mc} = \frac{2 \times 162.9\sqrt{2}}{1.25 \times 9.13}\left[\frac{\sin(162.65°) - \sin(102.65°) \times 0.794}{1 - 0.794}\right]$$

$$= -93.4 \text{ rad/sec}.$$

$$I_{ac} = \frac{\frac{3}{\pi}162.9\sqrt{2}\cos 120° - 1.25 \times (-93.4)}{2} = 3.376\text{A}$$

$$T_{ac} = 4.22 \text{ N-m}.$$

Since the torque at point (c) is greater than T_{ac}, the drive operates in continuous conduction. Having identified the modes of operation, torques are calculated as follows: (a) The drive operates in the discontinuous conduction mode II. From equation (3.82)

$$\frac{230.38}{9.13}\left[\sin(\beta° - 77.35°) - \sin 42.65° \exp\left\{\left(\frac{2\pi}{3} - \beta\right) \times 0.22\right\}\right]$$

$$- \frac{1.25\omega_m}{2}[1 - \exp\{(2\pi/3 - \beta) \times 0.22\}] = 0$$

or

$$25.23[\sin(\beta° - 77.35°) - 1.074\exp(-0.22\beta)]$$

$$- 0.625\omega_m[1 - 1.5853\exp(-0.22\beta)] = 0 \quad \text{(E3.5)}$$

From equation (3.84)

$$\omega_m = \frac{230.38}{1.25}\frac{(-0.5-\cos\beta^\circ)}{(\beta-2\pi/3)} - \frac{\pi\times 2}{3\times(1.25)^2}\frac{1}{(\beta-2\pi/3)}$$

$$\text{or}\quad \omega_m = 184.3\frac{(\cos\beta^\circ+0.5)}{(2\pi/3-\beta)} + \frac{1.34}{(2\pi/3-\beta)} \qquad \text{(E3.6)}$$

Iterative solution of equations (E3.5) and (E3.6) gives

$$\beta^\circ = 165.26^\circ \quad \text{and} \quad \omega_m = 107.55 \text{ rad/sec} \quad \text{or} \quad 1027 \text{ rpm}.$$

(b) From equation (3.79)

$$\omega_m = \frac{3\times 230.38\cos 60^\circ}{\pi\times 1.25} - \frac{2}{(1.25)^2}\times 20$$

$$= 62.4 \text{ rad/sec.} \quad \text{or} \quad 596 \text{ rpm}$$

(c) Again from (3.79)

$$\omega_m = \frac{3\times 230.38}{\pi\times 1.25}\cos 120^\circ - \frac{2\times 20}{(1.25)^2}$$

$$= -113.6 \text{ rad/sec.} \quad \text{or} \quad -1085 \text{ rpm}$$

Example 3.7

The motor of example 3.6 now drives a load whose torque is constant and independent of speed for a given setting. The minimum value of the load torque is 1.0 N-m. What inductance must be added to the armature circuit to get continuous conduction for all operating points?

Solution: From equation (3.88) the normalized value of torque corresponding to 1 N-m

$$T_{an} = \frac{\pi R_a}{3V_m}\left(\frac{T_a}{K}\right) = \frac{\pi\times 2}{3\times 230.38}\times\frac{1}{1.25} = 0.0073$$

For the elimination of discontinuous conduction, the straight line $T_{an} = 0.0073$ should be on the right of the boundary between continuous and discontinuous conductions. According to figure 3.21 this is achieved when $\psi = 1.5$ rad or 85.94°. Since

$$\psi = \tan^{-1}\frac{\omega L_a}{R_a}$$

$$L_a = \frac{R_a}{\omega}\tan\psi = \frac{2}{2\pi\times 50}\tan 85.94^\circ = 89.77 \text{ mH}$$

External inductance required $= 89.77 - 28.36 = 61.4$ mH.

3.5.2 Operation with Controlled Flywheeling[8, 12, 13]

The use of controlled flywheeling is beneficial in improving the power factor, and in reducing armature current ripple and the region of discontinuous conduction. The power circuit remains the same as shown in figure 3.16. As in the case of a single-phase fully-controlled rectifier, the controlled flywheeling in a 3-phase fully-controlled rectifier is implemented by eliminating negative excursions of the output voltage in rectification and positive excursions of the output voltage in inversion by

diverting the armature current to one of the three freewheeling paths formed by thyristor pairs T_1 T_4, T_3 T_6 and T_5 T_2; for equal loading they are used alternately. The transfer of the armature current to a freewheeling path is obtained by generating suitably timed additional gate pulses.

For firing angles less than 60°, the instantaneous output voltage of the rectifier is always positive, and for firing angles greater than 120°, it is always negative; therefore, controlled flywheeling cannot be used. For the firing angle range from 60°

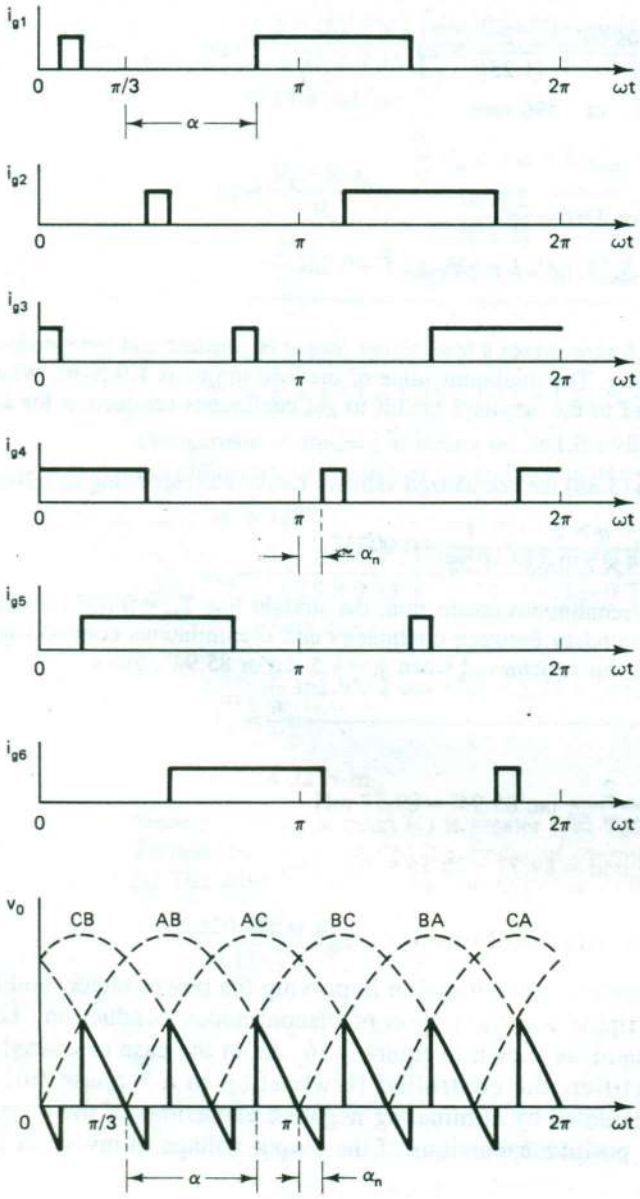


Figure 3.19 Firing schemes for regions II and III.

to 120° the instantaneous output voltage has both positive and negative excursions, and therefore, controlled flywheeling can be used.

Controlled flywheeling can be implemented using the approach shown in figure 3.19. For $\omega t < \alpha$, the motor current is freewheeling through the thyristor pair T_3, T_6. At α, T_1 is gated. Since v_{AB} is positive, T_1 turns on and T_3 turns off, and a voltage v_{AB} is applied across the motor. At angle α_n, T_4 is gated. Since v_{BA} is positive, T_4 turns on and T_6 turns off, and the motor current now flows through the freewheeling path formed by the thyristor pair T_1, T_4. For the motoring operation, α_n is fixed at 0 and variation in the output voltage is obtained by controlling α from $\pi/3$ to $(2\pi/3 - \delta)$, where δ is a small angle required for the commutation of thyristors. For regenerative braking, α is fixed at $(2\pi/3 - \delta)$ and α_n is controlled from 0 to $\pi/3$. The output voltage and current waveforms under continuous conduction for the motoring and braking operations are shown in figures 3.20a and 3.20c, respectively. The wide pulses shown in figure 3.19 have a duration of $2\pi/3$.

The drive operates in four distinct regions, which are listed in table 3.1. Regions I and II are for the motoring operation and regions III and IV are for the braking operation. Region I is for the range of α from 0 to 60° for which the controlled flywheeling cannot be used. The narrow pulses which control α_n can either be blanked or can be retained with $\alpha_n = 0$. Their presence has no effect on the converter operation. In the continuous conduction, the average output voltage varies from 1 per unit to 0.5 per unit. In region II, the output voltage varies from 0.5 per unit to 0. Here the narrow pulses are set to make $\alpha_n = 0$, and α is controlled by shifting the wide pulses. In region III, the wide pulses are set to make $\alpha = (2\pi/3 - \delta)$, and α_n is controlled from 0 to $\pi/3$ to vary the output voltage from 0 to -0.5 per unit. In region IV, the output voltage is controlled from -0.5 per unit to -1 per unit by blanking the narrow pulses and controlling α from 120° to $(180° - \delta)$. The same result is obtained by keeping $\alpha = (2\pi/3 - \delta)$ and varying α_n from $\pi/3$ to $2\pi/3$.

The drive has 12 modes of operation.[13] If the modes which occur in the narrow range of operation are ignored, one is left with 8 modes of operation. For regions I and IV, the modes are the same as modes I to IV shown in figure 3.17 for the conventional operation of the converter, and equations (3.76) to (3.84) are applicable. The remaining four modes, for operations in Regions II and III, are shown in figure 3.20. The performance equations for these modes can be derived in the same way as for modes I, II, V, and VI (figs. 3.12 and 3.13) of the single-phase converter with controlled flywheeling. Some important equations are derived next.

Mode V [Fig. 3.20a]: for $\pi/3 \le \alpha \le 2\pi/3$

$$V_a = \frac{3}{\pi}\int_{\alpha+\pi/3}^{\pi} V_m \sin \omega t \, d(\omega t)$$
$$= \frac{3V_m}{\pi}\left[1 + \cos\left(\alpha + \frac{\pi}{3}\right)\right] = V_{ao}\left[1 + \cos\left(\alpha + \frac{\pi}{3}\right)\right] \tag{3.89}$$

where $V_{ao} = 3V_m/\pi$

From equations (3.14), (3.19), and (3.89),

$$\omega_m = \frac{3V_m}{\pi K}\left[1 + \cos\left(\alpha + \frac{\pi}{3}\right)\right] - \frac{R_a}{K^2}T_a \tag{3.90}$$

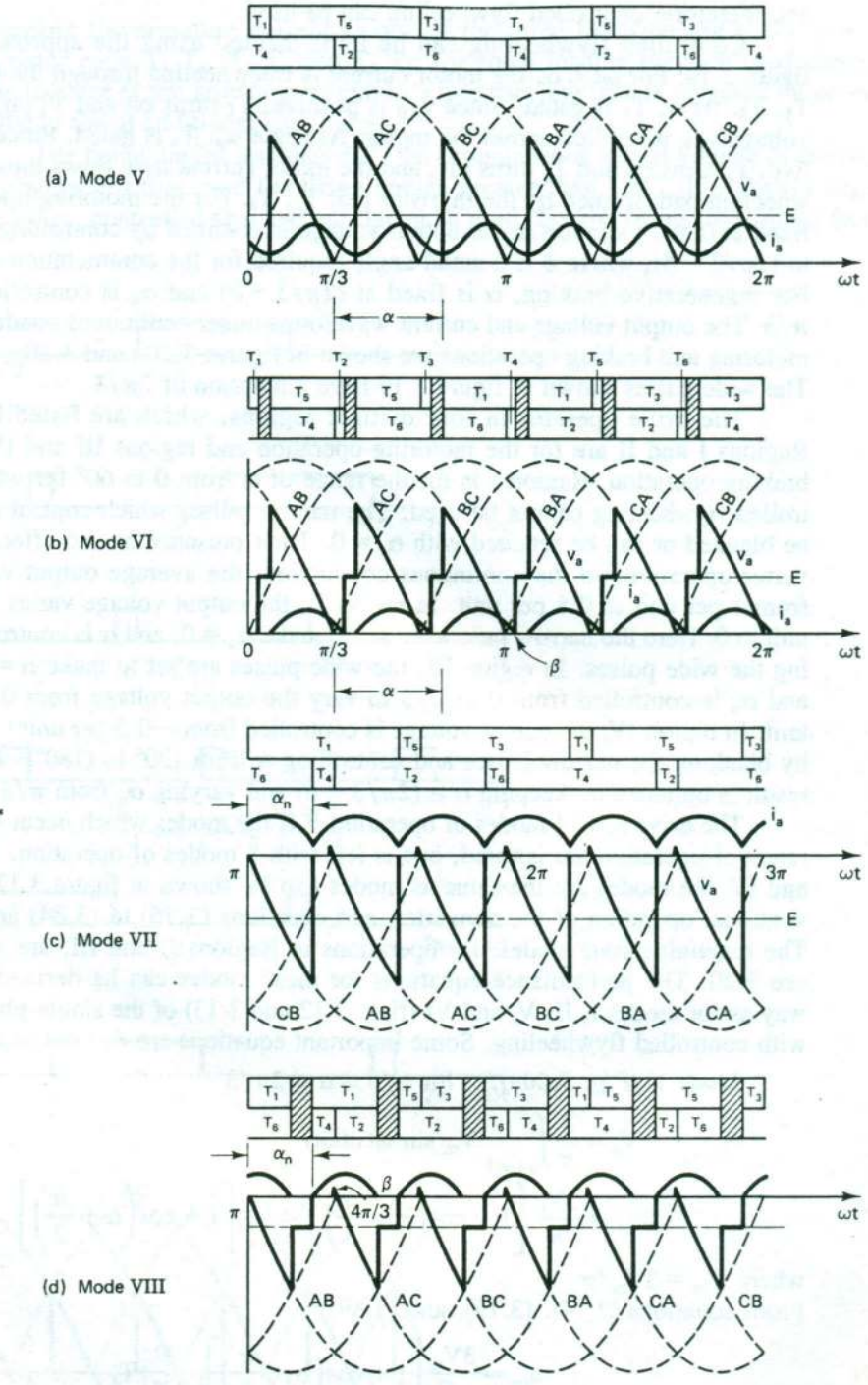


Figure 3.20 Modes of operation of dc drive fed by 3-phase rectifier with controlled flywheeling.

TABLE 3.1

Region of Operation	Drive Operation	Range of α	Range of α_n
I	Motoring	$0 < \alpha < \pi/3$	—
II	Motoring	$\pi/3 < \alpha < 2\pi/3$	$\alpha_n = 0$
III	Regeneration	$\left(\frac{2\pi}{3} - \delta\right)$	$0 < \alpha_n < \pi/3$
IV	Regeneration	either $2\pi/3 < \alpha < \pi$	or $\pi/3 < \alpha_n < 2\pi/3$

Mode VI [Fig. 3.20b]: for $\pi/3 \le \alpha \le 2\pi/3$

Replacing α by $\alpha + \pi/3$ in eqn. (3.60) yields

$$i_a(\omega t) = \frac{V_m}{Z}\left[\sin\psi \exp\{(\pi - \omega t)\cot\psi\} - \sin(\alpha + \pi/3 - \psi) \times \exp\left\{\left(\alpha + \frac{\pi}{3} - \omega t\right)\cot\psi\right\}\right] - \frac{K\omega_m}{R_a}\left[1 - \exp\left\{\left(\alpha + \frac{\pi}{3} - \omega t\right)\cot\psi\right\}\right], \quad \pi \le \omega t \le \beta \tag{3.91}$$

An equation for β is obtained by the substitution of $\omega t = \beta$ and $i_a = 0$ in equation (3.91). Now

$$V_a = \frac{3}{\pi}\left[\int_{\alpha+\pi/3}^{\pi} V_m \sin\omega t\, d(\omega t) + \int_{\beta}^{\alpha+2\pi/3} K\omega_m\, d(\omega t)\right] = \frac{3}{\pi}\left[V_m\left\{1 + \cos\left(\alpha + \frac{\pi}{3}\right)\right\} + K\omega_m\left(\alpha + \frac{2\pi}{3} - \beta\right)\right] \tag{3.92}$$

From equations (3.14), (3.19) and (3.92),

$$\omega_m = \frac{V_m}{K}\left[\frac{1 + \cos(\alpha + \pi/3)}{\beta - \alpha - \pi/3}\right] - \frac{\pi R_a T_a}{3K^2(\beta - \alpha - \pi/3)} \tag{3.93}$$

Mode VII [Fig. 3.20c]: for $0 \le \alpha_n \le \pi/3$

For an output voltage cycle from π to $4\pi/3$,

$$V_a = \frac{3}{\pi}\int_{\pi}^{\pi+\alpha_n} V_m \sin\omega t\, d(\omega t) = -\frac{3V_m}{\pi}(1 - \cos\alpha_n) \tag{3.94}$$

Thus

$$\omega_m = -\frac{3V_m}{\pi K}(1 - \cos\alpha_n) - \frac{R_a}{K^2}T_a \tag{3.95}$$

Mode VIII [Fig. 3.20d]: for $0 \le \alpha_n \le \pi/3$

Equation (3.56) is applicable from $\pi + \alpha_n$ to $4\pi/3$. The operation in the interval $4\pi/3$ to β is described by the following equation:

$$v_a = v_{AC} = V_m \sin(\omega t - \pi/3) = L_a \frac{di_a}{dt} + R_a i_a + K\omega_m, \qquad \frac{4\pi}{3} \le \omega t \le \beta \tag{3.96}$$

$i_a(4\pi/3)$ is obtained from the solution of equation (3.56) with the initial condition $i_a(\pi + \alpha_n) = 0$. Solution of equation (3.96) with $i_a(4\pi/3)$ as the initial condition gives

$$\begin{aligned} i_a(\omega t) = \frac{V_m}{Z}\left[\sin\left(\omega t - \frac{\pi}{3} - \psi\right) - \sin\psi \exp\left\{\left(\frac{4\pi}{3} - \omega t\right)\cot\psi\right\}\right] \\ - \frac{K\omega_m}{R_a}[1 - \exp\{(\alpha_n + \pi - \omega t)\cot\psi\}] \end{aligned} \tag{3.97}$$

An equation for β is obtained by the substitution of $\omega t = \beta$ and $i_a = 0$ in equation (3.97). For an output voltage cycle from $4\pi/3$ to $5\pi/3$,

$$\begin{aligned} V_a &= \frac{3}{\pi}\left[\int_{4\pi/3}^{\beta} V_m \sin(\omega t - \pi/3)\, d(\omega t) + \int_{\beta}^{\alpha_n + 4\pi/3} K\omega_m\, d(\omega t)\right] \\ &= \frac{3}{\pi}\left[-V_m\{1 + \cos(\beta - \pi/3)\} + K\omega_m\left(\alpha_n + \frac{4\pi}{3} - \beta\right)\right] \end{aligned} \tag{3.98}$$

From equations (3.14), (3.19), and (3.98)

$$\omega_m = -\frac{V_m}{K}\left[\frac{1 + \cos(\beta - \pi/3)}{(\beta - \pi - \alpha_n)}\right] + \frac{\pi R_a}{3K^2(\alpha_n - \beta + \pi)} T_a \tag{3.99}$$

It should be noted that modes VII and VIII are for regenerative braking and therefore K should be taken negative for the operation in the second quadrant, and ω_m should be taken negative for the operation in the fourth quadrant.

The speed-torque characteristics have essentially the same nature as for the 1-phase fully-controlled rectifier with controlled flywheeling, except that the discontinuous conduction occurs in a very narrow region and for $\pi/3 < \alpha_n < 2\pi/3$ ideal no-load speeds are negative.

The boundaries between discontinuous and continuous conductions on the normalized speed-torque plane were obtained taking into account all the 12 modes of operation.[13] These are plotted in figure 3.21. Speeds and torques have been normalized using equations (3.87) and (3.88). Comparison of these boundaries with those for the conventional operation shows that discontinuous conduction is considerably reduced by controlled flywheeling.

3.5.3 Operation with a Freewheeling Diode

When regenerative braking is not required, a freewheeling diode is added to the fully-controlled rectifier giving a circuit as shown in figure 3.4e. The operation of this circuit is identical to the motoring operation of the fully-controlled rectifier with the controlled flywheeling. Therefore, the performance equations derived for the

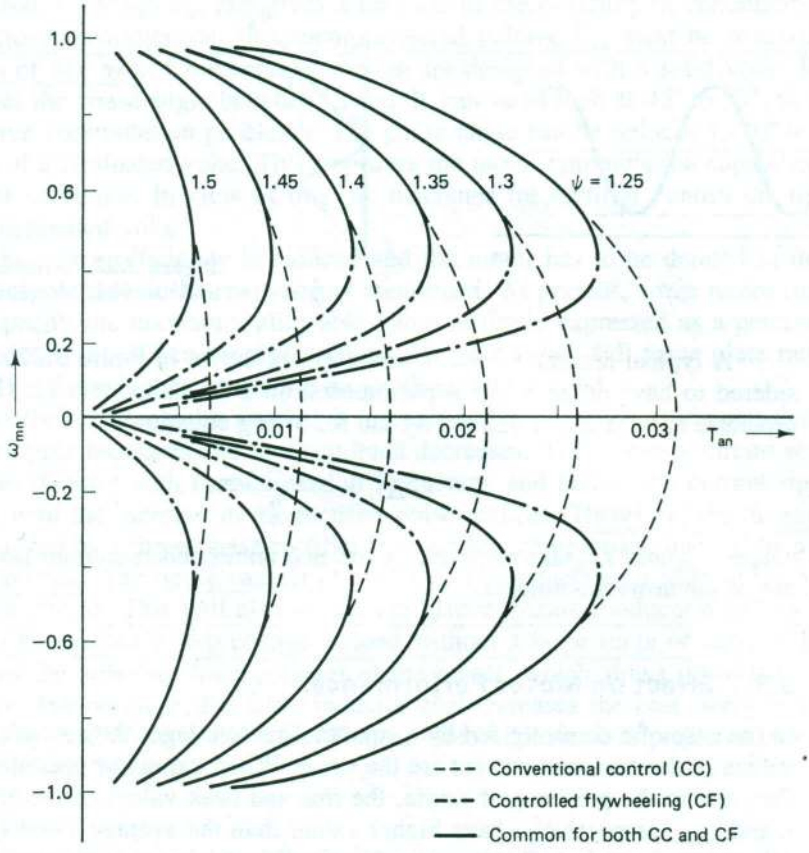


Figure 3.21 Boundary curves separating the regions of continuous and discontinuous conductions for 3-phase rectifier drives, $\omega_{mn} = E/(3V_m/\pi)$, $T_{an} = I_a/(3V_m/\pi R_a)$.

motoring operation in section 3.5.2 are valid for this circuit also. The only advantage this circuit has over the fully-controlled rectifier with the controlled flywheeling is that the control circuit is simpler.

3.6 ARMATURE CURRENT RIPPLE AND ITS EFFECT ON MOTOR PERFORMANCE

The performance of a converter-controlled dc motor is significantly different from the dc motor fed by a source of smooth direct voltage. Two major differences are the presence of discontinuous conduction and the ripple in the armature current. Discontinuous conduction has been discussed in the previous sections. The present section considers the effect of ripple on motor performance.

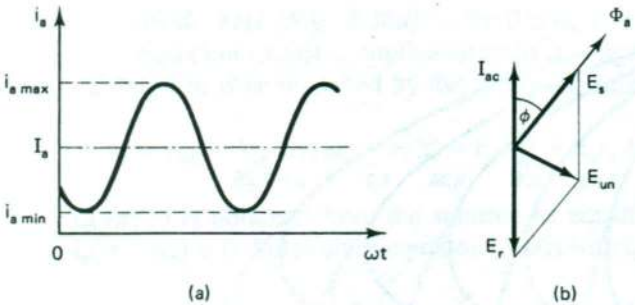


Figure 3.22 Armature current ripple and its effect on commutation.

A typical armature current waveform is shown in figure 3.22a. It can be considered to have an ac ripple superimposed on a dc component I_a. The ripple in the armature current Δi_a is defined by the following equation:

$$\Delta i_a = \frac{i_{amax} - i_{amin}}{2} \tag{3.100}$$

where i_{amax} and i_{amin} are respectively the maximum and minimum instantaneous values of the armature current.

3.6.1 Effect on Motor Performance

In the case of a dc motor fed by a smooth direct voltage, the average, rms, and peak values of the armature current are the same. When the motor is controlled by a rectifier, due to the presence of ripple, the rms and peak values of the armature current, I_r and i_{amax}, respectively, have higher values than the average value I_a. While I_a contributes to the torque, I_r is responsible for heat loss in the armature resistance, and i_{amax} is a measure of the burden on the motor commutation. Furthermore, due to the skin effect, the armature resistance at ripple frequency is higher than its dc value. Due to the larger values of I_r and the armature resistance, the armature copper loss is considerably increased. There is also a slight increase in the core loss due to the presence of ripple.

Another important point to be noted is about the interpole winding heating. The interpole winding is in series with the armature winding and hence the increase in the losses in the interpole winding and the consequent temperature rise of the interpole winding can become an important problem because, unlike the armature winding, the interpole winding is stationary.

The motor commutation problem becomes severe not only because the peak current i_{amax} is higher than I_a but also because the pulsating interpole flux produces eddy currents in the solid iron yoke, which damp the commutating flux and cause it to be phase displaced in time after the armature current. The armature and the interpole ripple current are represented by a vector I_{ac} in figure 3.22b. The flux Φ_a lags the current vector by an angle ϕ determined by the eddy currents. The speed emf E_s is in phase with Φ_a. The reactance voltage which is equal and opposite to E_s in the absence of eddy currents, is shown as a vector E_r. The result of E_s and E_r is the un-

compensated voltage E_{un} and gives a measure of the difficulty in commutation. For satisfactory commutation, the uncompensated voltage E_{un} must be restricted to a fraction of one volt. Conventional motors are designed with a solid yoke. In these machines the phase angle between I_{ac} and Φ_a can be as high as 45° to 50°, thus causing severe commutation problems. The phase angle can be reduced to 10° to 15° by the use of a laminated yoke. This improves the motor-commutation capability in the presence of ripple. In view of this, dc machines for rectifier control are designed with a laminated yoke.

The motor efficiency is reduced and the motor has to be derated in the presence of ripple due to the reasons just mentioned. At present, some motor manufacturers specify the maximum allowable value of ripple expressed as a percentage of the rated current. When the ripple is less than this value, full name plate rating can be used, and for higher values the motor should be derated.

As the rectifier pulse number is increased, the frequency of the rectifier output voltage ripple increases and its magnitude decreases. The armature circuit reactance increases directly with the increase in frequency, and hence, the current ripple decreases with the increase in the rectifier pulse number. Therefore, the derating of a dc motor fed by a three-phase rectifier is much less compared to one fed by a single-phase rectifier. The ripple can also be reduced by connecting a filter inductor in the armature circuit. This will also reduce the discontinuous conduction and enable the drive to experience a step change in load without a large surge of current. But this increases the armature time constant of the motor, which slows down the transient response. Furthermore, the filter inductor also increases the cost, weight, volume, power loss, and noise in the drive. Hence, often high ripple is preferred to the presence of a filter inductor.

3.6.2 Calculation of the Maximum Current Ripple and the Selection of Filter Inductance

The discussion in section 3.6.1 suggests that the armature current ripple should be restricted below a permissible limit. When the ripple is more than the prescribed limit, it should be reduced either by increasing the rectifier pulse number or by adding a filter inductor.

As long as the armature current is discontinuous, the current ripple, for a given firing angle, depends on the back emf. When the armature current is continuous, any change in the motor back emf causes change only in the dc component of the armature current, and therefore, the ripple becomes independent of back emf and speed. For a particular firing angle, the ripple becomes maximum and remains constant when the current is continuous. Therefore, it is adequate to know the ripple for the continuous conduction modes. For a given α, one can identify the regions in which the current will have the maximum and minimum values. The minimum and maximum values can be obtained by the differentiation of the relevant armature current expressions, and then the ripple can be calculated.

The variation of ripple with the converter output voltage for 1-phase drives, with and without controlled flywheeling, is shown in figure 3.23; where the normal-

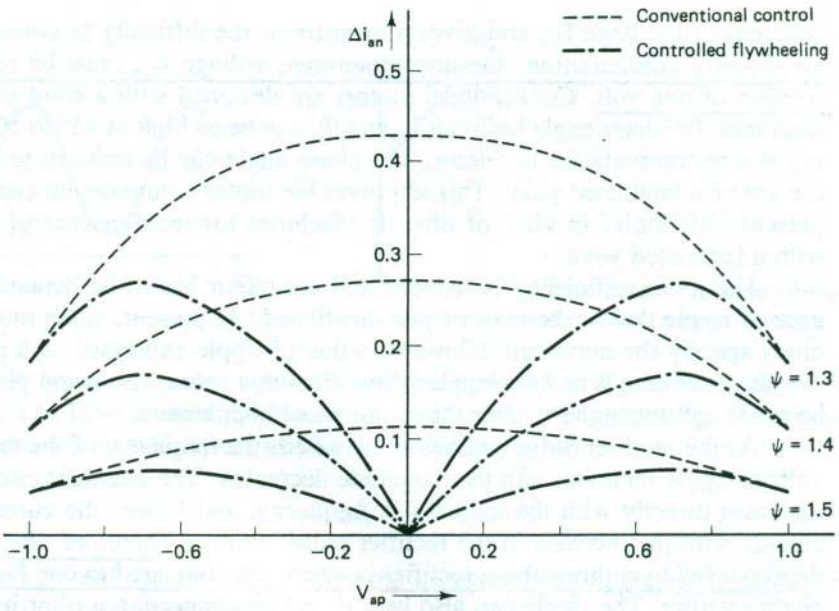


Figure 3.23 Normalised armature current ripple Δi_{an} versus per unit output voltage V_{ap} for 1-phase rectifier drives, $\Delta i_{an} = \Delta i_a/(V_m/\pi R_a)$ and $V_{ap} = V_a/(2V_m/\pi)$.

ized current ripple has been plotted against the per-unit output voltage for various values of ψ.[10] The normalized ripple Δi_{an} is defined by the following equation:

$$\Delta i_{an} = \frac{\Delta i_a}{V_m/(\pi R_a)} = \frac{\pi R_a (i_{amax} - i_{amin})}{2V_m} \tag{3.101}$$

The per-unit rectifier output voltage V_{ap} is obtained by dividing the rectifier output voltage by its maximum value $2V_m/\pi$, giving

$$V_{ap} = \frac{V_a}{V_{ao}} = \frac{V_a}{2V_m/\pi}$$

Thus, from equation (3.16) for the conventional control

$$V_{ap} = \cos \alpha \tag{3.102}$$

Also from equation (3.57) for the motoring operation with controlled flywheeling

$$V_{ap} = \frac{V_a}{V_{ao}} = \frac{1}{2}(1 + \cos \alpha) \tag{3.103}$$

Figure 3.23 shows that the controlled flywheeling reduces the ripple by a large amount at low speeds. The maximum ripple is also reduced substantially.

The normalized ripple versus per-unit output voltage plots for 3-phase rectifier drives are shown in figure 3.24.[11,13] The normalized current ripple is given by equation (3.101). The per-unit output voltage is given by the following equation:

$$V_{ap} = \frac{V_a}{V_{ao}} = \frac{V_a}{3V_m/\pi}$$

Thus, from equation (3.78) for the conventional control,

$$V_{ap} = \cos \alpha \tag{3.104}$$

And from equations (3.78) and (3.89) for the motoring operation with the flywheeling

$$V_{ap} = \cos \alpha \quad \text{for} \quad 0 \le \alpha \le \pi/3 \tag{3.105}$$

$$= 1 + \cos(\alpha + \pi/3) \quad \text{for} \quad \pi/3 \le \alpha \le 2\pi/3 \tag{3.106}$$

The curves of figures 3.23 and 3.24 are useful for calculating the derating of the motor and determining the filter inductance required to keep the ripple within permissible values. The curves are plotted only for three values of ψ. The intermediate values can be obtained by noting that the ripple changes linearly with ψ.[10,11,13]

Example 3.7

A 7.5 kW, 230 V, 850 rpm, 40 A dc separately excited motor has the armature resistance and inductance of 0.77 Ω and 11.8 mH, respectively. It is fed by a 3-phase fully-controlled rectifier with an ac source voltage of 162.9 V (line), 60 Hz. Obtain the

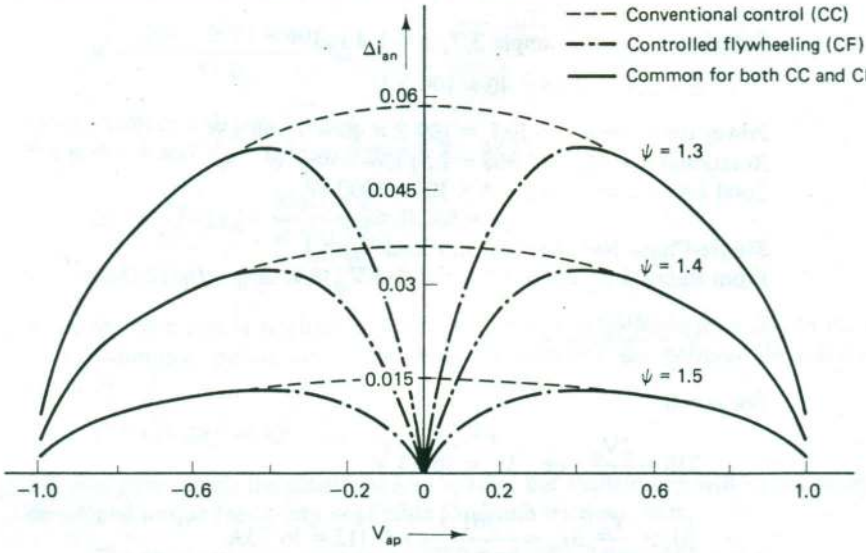


Figure 3.24 Normalised armature current ripple Δi_{an} versus per unit output voltage for 3-phase rectifier drives; $\Delta i_{an} = \Delta i_a(V_m/\pi R_a)$ and $V_{ap} = V_a/(3V_m/\pi)$.

maximum current ripple. What inductance must be added to the armature circuit to restrict the ripple to 5 percent of the rated current?

Solution:

$$\psi = \tan^{-1}\frac{\omega L_a}{R_a} = \tan^{-1}\frac{2\pi \times 60 \times 11.8 \times 10^{-3}}{0.77}$$

$$= 1.4 \text{ rad}$$

From figure 3.24, for $\psi = 1.4$, the maximum normalized ripple is 0.036:

$$\Delta i_a = \frac{V_m}{\pi R_a}\Delta i_{an} = \frac{162.9\sqrt{2}}{\pi \times 0.77} \times 0.036 = 3.43\text{A}$$

For 5 percent ripple $\Delta i_a = 0.05 \times 40 = 2\text{A}$

$$\Delta i_{an} = \frac{\pi R_a}{V_m}\Delta i_a = \frac{\pi \times 0.77}{162.9\sqrt{2}} \times 2 = 0.021$$

From figure 3.24, for $\Delta i_{an} = 0.021$, $\psi = 1.47$ rad or 84.2°. Thus

$$L_a = \frac{R_a}{\omega}\tan\psi^\circ = \frac{0.77}{2 \times 60\pi}\tan 84.2^\circ = 20.1 \text{ mH}$$

Inductance to be added $= 20.1 - 11.8 = 8.3$ mH.

Example 3.8

Find out the thermal derating of the motor of example 3.7 when fed by 1-phase and 3-phase fully-controlled rectifiers. For both the rectifiers the input ac voltage is adjusted to give 230 V across the motor for $\alpha = 0$. Assume rotational losses to remain fixed.

Solution: From example 3.7, $\psi = 1.4$ rad

$$E = 230 - 0.77 \times 40 = 199.2 \text{ V}$$

Power developed $P_d = EI_a = 199.2 \times 40 = 7.968$ kW
Rotational loss $P_w = (7.968 - 7.5)$ kW $= 468$ W
Total loss $= 230 \times 40 - 7.5 \times 10^3 = 1700$ W

Single-Phase Rectifier: At full load $V_{ap} = 1$
From figure 3.23 for $\psi = 1.4$ rad and $V_{ap} = 1$, $\Delta i_{an} = 0.112$. Now

$$V_a = \frac{2V_m}{\pi}\cos\alpha$$

For $\alpha = 0$

$$230 = \frac{2V_m}{\pi} \quad \text{or} \quad V_m = 361.3 \text{ V}$$

$$\Delta i_a = \frac{V_m}{\pi R_a}\Delta i_{an} = \frac{361.3}{\pi \times 0.77} \times 0.112 = 16.73\text{A}$$

RMS value of $\Delta i_a = 11.83\text{A}$.

To have the same thermal loading, the rms armature current should remain constant. If I_a is the permissible average current then,

$$I_a^2 + (11.83)^2 = 40^2 \quad \text{or} \quad I_a = 38.21\text{A}$$

$$E = 230 - 38.21 \times 0.77 = 200.58 \text{ V}$$

Developed power, $P_d = 200.58 \times 38.21 = 7664$ W
Output power $= P_d - P_w = (7664 - 468)$W $= 7.196$ kW
Percent derating $= (7.196/7.5) \times 100 = 96\%$

Three-Phase Rectifier: From figure 3.24 for $\psi = 1.4$ rad and $V_{ap} = 1$, $\Delta i_{an} = 0.0054$
For $\alpha = 0$, $V_a = 3V_m/\pi$ or $230 = 3V_m/\pi$ or $V_m = 241$ V

$$\Delta i_a = \frac{V_m}{\pi R_a}\Delta i_{an} = \frac{241}{\pi \times 0.77} \times 0.0054 = 0.54\text{A} = 0.38\text{A(rms)}$$

Now $I_a^2 + (0.38)^2 = 40^2$ or $I_a \simeq 40$ A
Thus with a three-phase rectifier one is able to get nearly full output.

It may be noted that in this example we have not considered derating due to commutation.

Example 3.9

Find out the torque derating at zero speed for the motor of example 3.7 when fed by a 1-phase fully-controlled rectifier with ac source voltage of 255.5 V, 60 Hz. Neglect core loss and the armature reaction.

Solution: Speed in rad/sec. $= 850 \times \pi/30 = 89.01$
Rated shaft torque $= 7500/83.78 = 84.26$ N-m

$$K = \frac{230 - 0.77 \times 40}{89.01} = 2.24$$

From example 3.7, $\psi = 1.4$ rad
For $\psi = 1.4$ and $V_{ap} = 0$ from figure 3.23, $\Delta i_{an} = 0.268$

$$\Delta i_a = \frac{V_m}{\pi R_a}\Delta i_{an} = \frac{255.5\sqrt{2}}{\pi \times 0.77} \times 0.268 = 40\text{A}$$

RMS ripple $= 40/\sqrt{2} = 28.28$A

Since the core loss is negligible, the motor heating takes place only due to the copper loss. Assuming cooling conditions remain unaffected, the permissible value of I_a is given by

$$I_a^2 + (28.28)^2 = 40^2, \quad \text{or} \quad I_a = 28.29\text{A}$$

Since at zero speed, the rotational loss is zero, the shaft-torque will be the same as the developed torque (assuming negligible Coulomb friction); thus,

The shaft torque at zero speed $= KI_a = 2.24 \times 28.29 = 63.4$ N-m

Derating of the torque at zero speed $= \dfrac{63.4}{84.26} \times 100 = 75.24$ percent.

3.7 RECTIFIER-SOURCE INTERACTION

Rectifiers have a poor power factor and draw nonsinusoidal current from the ac source. In a weak source, with high internal impedance, the line current harmonics cause fluctuations of the source voltage. Furthermore, the temporary short-circuit of the lines during the commutation of thyristors, causes sharp current pulses which further distort the line voltage. The voltage fluctuations adversely affect other loads connected to the source. The line current harmonics, particularly the sharp current pulses, cause electromagnetic interference of nearby telephone lines and communication equipment. A feedback of voltage fluctuations to the rectifier through the synchronizing signals may cause the drive to become unstable. The instability problem is overcome by the use of firing schemes immune to voltage fluctuations.

For the calculation of the rectifier power factor, it is customary to assume that the supply voltage is sinusoidal. This assumption is based on the fact that the total capacity of the source is usually much larger than the loads drawing nonsinusoidal current (including the rectifier) and that the lines connecting the rectifier with the source have a low impedance.

The power factor (P.F.) of a 1-phase rectifier is defined as

$$\text{P.F.} = \frac{\text{Real power}}{\text{Apparent power}} = \frac{VI_1 \cos \phi_1}{VI_{rms}} \tag{3.107}$$

where

V = rms source voltage, V

I_{rms} = rms source current, A

I_1 = fundamental component of source current, A

ϕ_1 = Phase difference between V and I_1

Therefore,

$$\text{P.F.} = (I_1/I_{rms}) \cos \phi_1 = \mu \cos \phi_1 \tag{3.108}$$

where μ is called the distortion factor and $\cos\phi_1$ is called the displacement factor or fundamental power factor. The distortion factor is a measure of the harmonic content in the source current.

An expression similar to equation (3.108) can be obtained for a three-phase rectifier.

The source current i_s can be described by the following Fourier series:

$$i_s = \sum_{n=1}^{\infty} a_n \sin n\omega t + \sum_{n=1}^{\infty} b_n \cos n\omega t \tag{3.109}$$

where

$$a_n = \frac{1}{\pi} \int_0^{2\pi} i_s \sin n\omega t \, d(\omega t) \tag{3.110}$$

$$b_n = \frac{1}{\pi} \int_0^{2\pi} i_s \cos n\omega t \, d(\omega t) \tag{3.111}$$

The rms value of the nth harmonic in the source current is given by

$$I_n = \frac{1}{\sqrt{2}}[a_n^2 + b_n^2]^{1/2} \tag{3.112}$$

The phase displacement of the nth harmonic $\emptyset_n$ is given by

$$\emptyset_n = \tan^{-1}\frac{b_n}{a_n} \tag{3.113}$$

The individual harmonics, including the fundamental component I_1, are obtained from equations (3.110) through (3.113). The rms value of the current is obtained by the following expression:

$$I_{rms} = \left[\frac{1}{2\pi}\int_0^{2\pi} i_s^2\, d(\omega t)\right]^{1/2} \tag{3.114}$$

In our analysis, thyristors and diodes have been assumed to be ideal switching elements and this implies a lossless rectifier. Therefore, the input power to the motor should be equal to the input power of the rectifier. Thus,

$$\text{Real power} = I_r^2 R_a + EI_a \tag{3.115}$$

where I_r is the rms value of the motor current.

In a dc drive, the rectifier power factor and the source current harmonics depend on speed, torque, and firing angle. For the comparison of various rectifiers, it is common to assume a ripple-less armature current. With this assumption, the

TABLE 3.2

Converter	Control Parameters	P.F.	Fundamental P.F.	I_n/I_a	Per-Unit Output Voltage
1-phase conventional	$0 \le \alpha \le \pi$	$0.9 \cos \alpha$	$\cos \alpha$	$\frac{0.9}{n}$ for n odd	$\cos \alpha$
1-phase with controlled flywheeling	$0 \le \alpha \le \pi$ $\alpha_n = 0$	$\frac{0.8(1+\cos\alpha)}{\sqrt{\pi - \alpha}}$	$\cos(\alpha/2)$	$\frac{0.9}{n}\sqrt{\frac{1+\cos n\alpha}{2}}$ for n odd	$\frac{1}{2}(1+\cos\alpha)$
3-phase conventional	$0 \le \alpha \le \pi$	$0.955 \cos \alpha$	$\cos \alpha$	$\frac{0.78}{n}$ for n = 1, 5, 7, 11	$\cos \alpha$
3-phase with controlled flywheeling	$0 \le \alpha \le \pi/3$	$0.955 \cos \alpha$	$\cos \alpha$	$\frac{0.78}{n}$ for n = 1, 5, 7, 11	$\cos \alpha$
	$\frac{\pi}{3} \le \alpha \le \frac{2\pi}{3}$ and $\alpha_n = 0$	$0.98\left[1+\cos\left(\alpha+\frac{\pi}{3}\right)\right]\Big/\left(\sqrt{\left(\frac{2\pi}{3}-\alpha\right)}\right)$	$\cos(\alpha/2)$	$\frac{1.56}{n}\sqrt{\frac{1+\cos n(\alpha+\pi/3)}{2}}$ for n = 1, 5, 7, 11	$1+\cos(\alpha+\pi/3)$

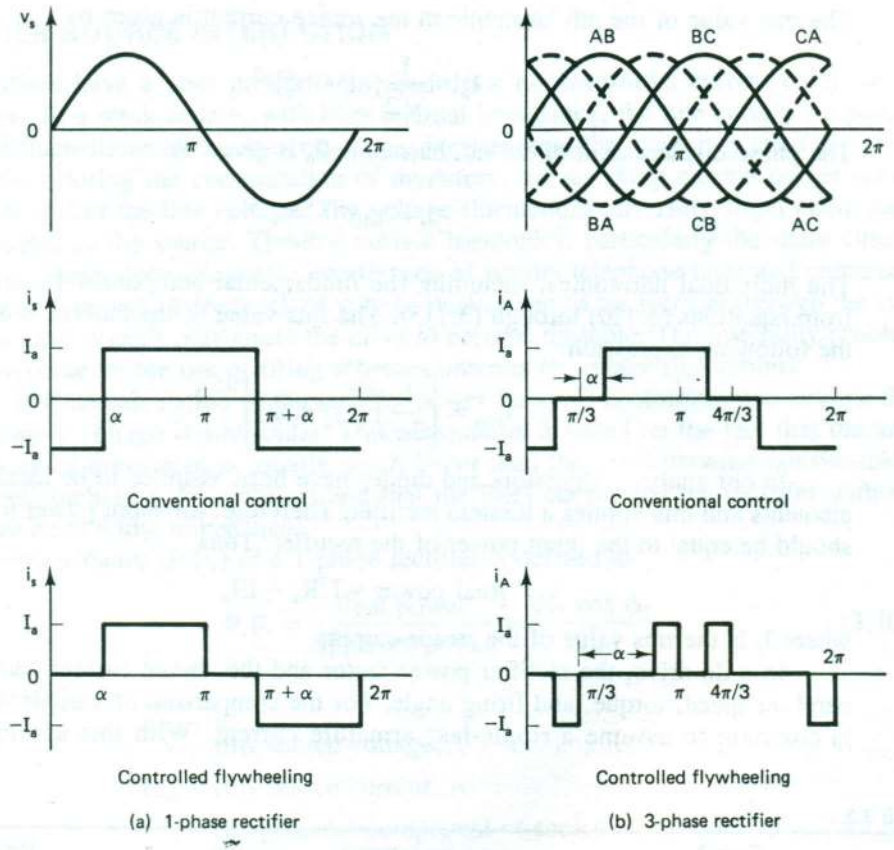


Figure 3.25 Source current waveforms for controlled rectifiers.

source current waveforms for some rectifiers are shown in figure 3.25. The time origins have been chosen in the same way as in figure 3.7b and figure 3.16b or equation (3.76). The expressions for the power factor, fundamental power factor, and nth harmonic expressed as a ratio of the dc component and per-unit output voltage, as defined in equations (3.102) to (3.112), are listed in table 3.2. The variation of power factor and I_n/I_s with the per-unit output voltage is shown in figure 3.26. The single-phase rectifiers have all the odd harmonics. Three-phase rectifiers have all the odd harmonics except the third and its multiples. For all the rectifiers, the power factor is low at low-output voltages. Three-phase (6-pulse) rectifiers have lower amplitude of harmonics than the single-phase (2-pulse) rectifiers. The harmonics decrease with the increase of the rectifier pulse number. Therefore, 12-pulse rectifiers are employed for high-power drives.

At high-power levels, the lagging reactive power and the line current harmonics pose a serious problem. The static reactive power compensators with associated filters are often employed to counteract this problem. Alternatively, rectifiers can be operated with improved control techniques, such as sequence control and pulse-width modulation.[15]

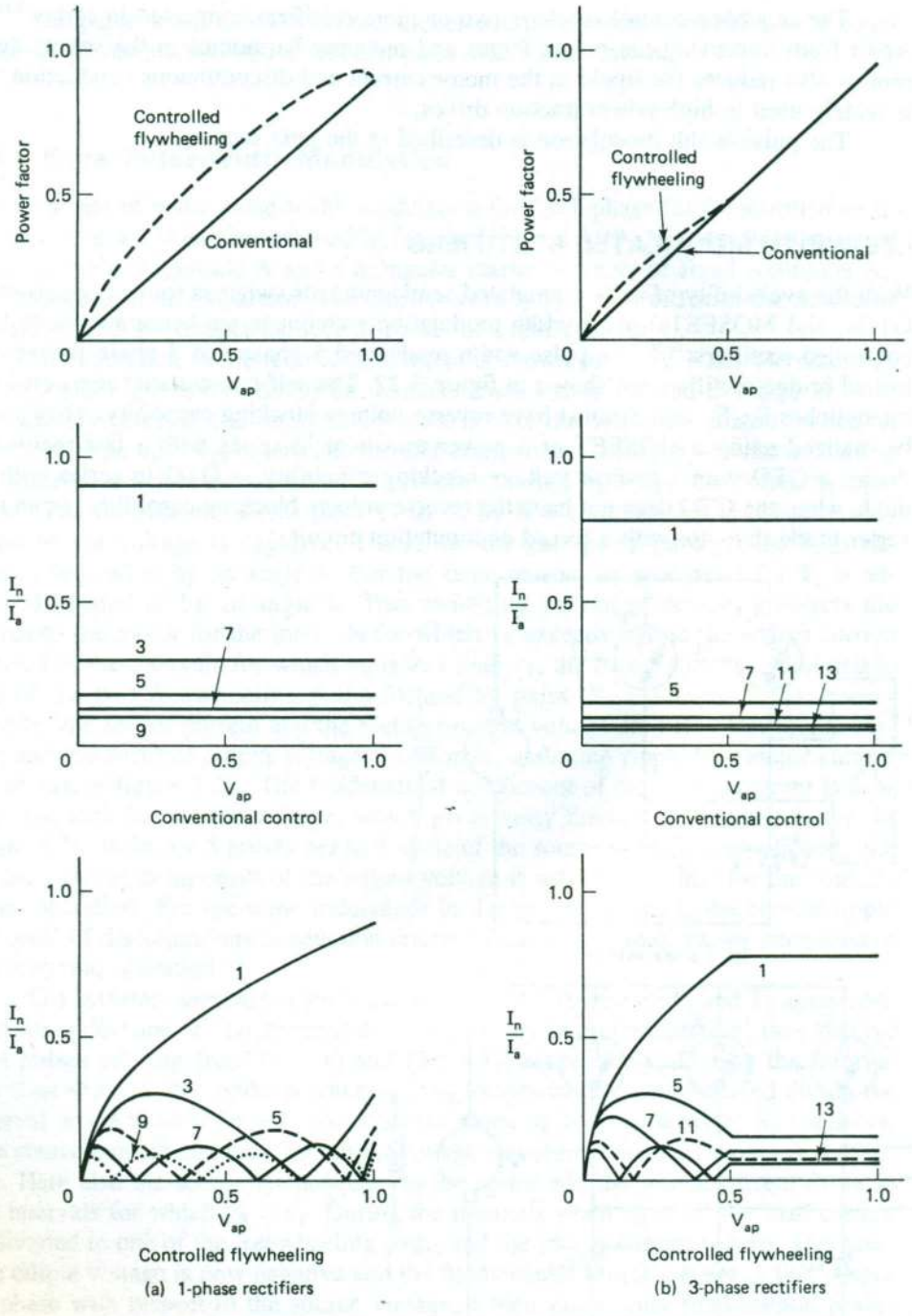


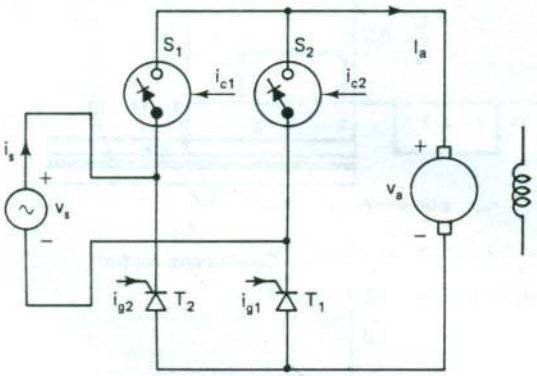
Figure 3.26 Power factor and harmonics versus per unit output voltage curves for controlled rectifiers.
(Previously published by the Institution of Electrical Engineers in IEE Proc. Part B, vol. 127, no. 4, July 1980).

The sequence control employs two or more rectifiers connected in series.[3.15-18] Apart from improving the power factor and reducing harmonics in the source current, it also reduces the ripple in the motor current and discontinuous conduction. It is widely used in high-power traction drives.

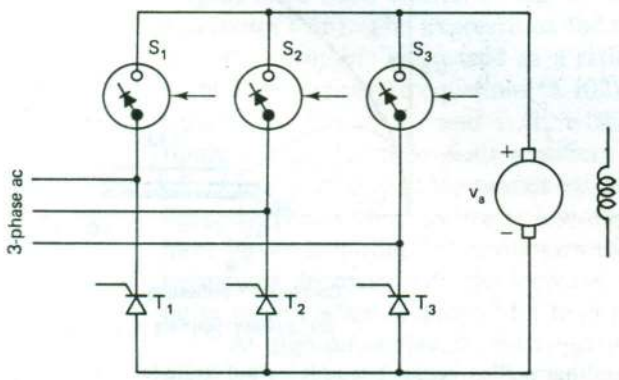
The pulse-width modulation is described in the next section.

3.8 PULSE-WIDTH MODULATED RECTIFIERS

With the availability of self-commutated semiconductor switches (power transistors, GTOs, and MOSFETs) pulse-width modulation techniques are being employed for controlled rectifiers.[19-24] The pulse-width modulated 1-phase and 3-phase fully-controlled bridge rectifiers are shown in figure 3.27. The self-commutated semiconductor switches S_1, S_2, and S_3 must have reverse voltage blocking capability. They may be realized using a MOSFET or a power transistor in series with a fast recovery diode, a GTO with a reverse voltage blocking capability, a GTO in series with a diode when the GTO does not have the reverse voltage blocking capability, or an inverter grade thyristor with a forced commutation circuit.



(a) 1-phase rectifier



(b) 3-phase rectifier

Figure 3.27 Pulse width modulated fully-controlled rectifiers.

The pulse-width modulation techniques commonly used in rectifiers are equal pulse-width modulation and sinusoidal pulse-width modulation.

3.8.1 Equal Pulse-Width Modulation

The principle of equal pulse-width modulation for the 1-phase fully-controlled rectifier of figure 3.27a is demonstrated in figures 3.28 and 3.29. A dc modulating signal v_R of variable magnitude A and a triangular carrier wave v_T of fixed amplitude A_m are compared in a comparator. The carrier wave is synchronized with the ac source voltage v_s and has integer number of cycles in a half-cycle of v_s.

The operation in the rectification mode is shown in figure 3.28. Thyristors T_1 and T_2, which are line commutated, conduct from λ to $\pi + \lambda$ and $\pi + \lambda$ to $2\pi + \lambda$, respectively. During the interval $\lambda \leq \omega t \leq \pi + \lambda$, the self-commutated switch S_1 conducts when $v_R > v_T$; otherwise, the self-commutated switch S_2 conducts. During the interval $\pi + \lambda \leq \omega t \leq 2\pi + \lambda$, S_2 conducts when $v_R > v_T$; otherwise, S_1 conducts. The transfer of current from T_1 to T_2 by line commutation is possible only when source voltage is negative. Therefore, the gate pulse for thyristor T_2 is advanced beyond π by an angle λ. For the same reason the gate pulse for T_1 is advanced beyond 0° by an angle λ. This switching pattern of devices connects the source to the motor for the intervals for which v_R exceeds v_T and the source current flows. For the intervals for which v_R is less than v_T, the motor current is diverted to one of the two freewheeling paths formed by pairs (S_1,T_2) and (S_2,T_1); consequently, the source current and the rectifier output voltage are zero. The source current and the rectifier output voltage waveforms, assuming ripple-less motor current are shown in figure 3.28. The fundamental component of the source current is now in phase with the source voltage, which gives unity fundamental power factor. In figure 3.28, there are 5 pulses per half-cycle of the source voltage. Consequently the fundamental ac component of the output voltage is smaller than that for the conventional operation. For the same inductance in the armature circuit, the current ripple and zone of discontinuous conduction operation have very small values compared to conventional operation.

The inverter operation is shown in figure 3.29. Thyristors T_1 and T_2 again conduct for a duration π. To commutate T_1 and T_2 by line commutation, they receive gate pulses starting from $(\pi - \lambda)$ and $(2\pi - \lambda)$, respectively. During the interval $-\lambda \leq \omega t \leq \pi - \lambda$, S_2 conducts when $v_R > v_T$; otherwise, S_1 conducts; and during the interval $\pi - \lambda \leq \omega t \leq 2\pi - \lambda$, S_1 conducts when $v_R > v_T$; otherwise, S_2 conducts. The source current and rectifier output voltage waveforms are also shown in the figure. Here also the source is connected to the motor and the source current flows in the intervals for which $v_R > v_T$. During the intervals when $v_R < v_T$, the load current is diverted to one of the freewheeling paths and the source current is zero. The average output voltage is now negative and the fundamental source current is 180° output of phase with respect to the source voltage, which gives unity fundamental power factor. The ripple in the motor current and zone of discontinuous conduction will again be small.

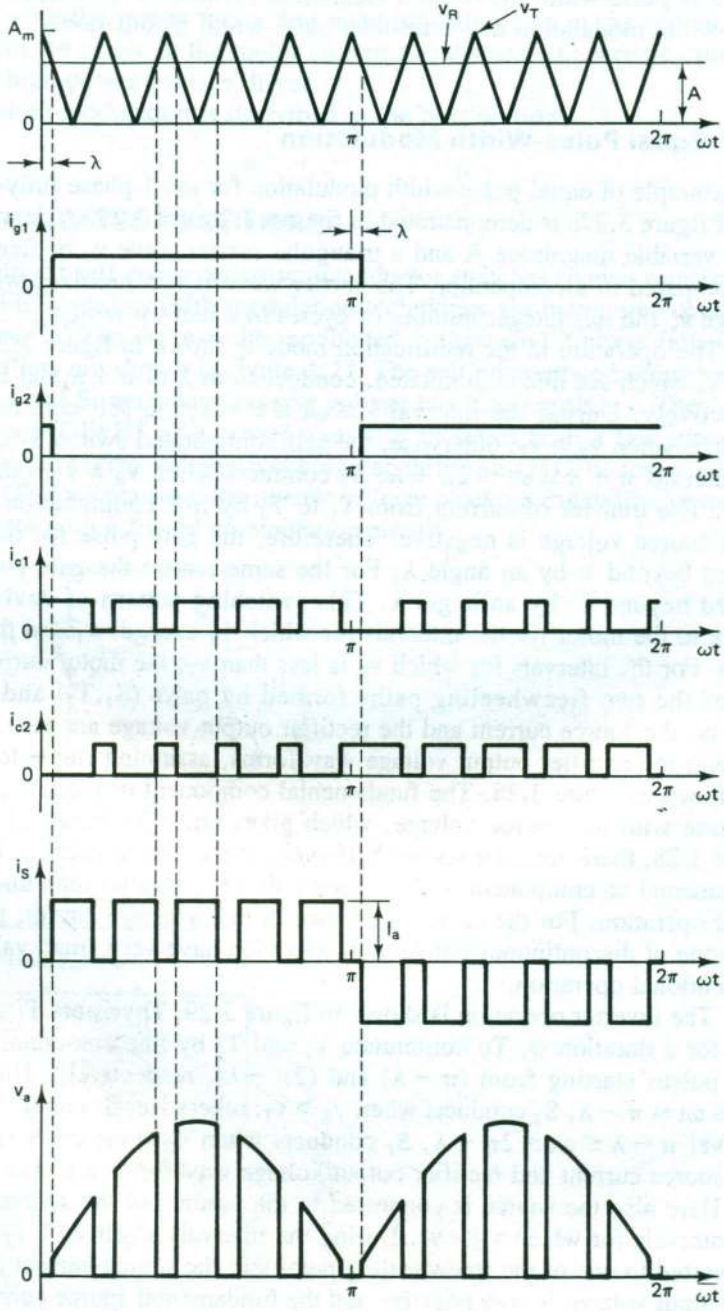


Figure 3.28 Waveforms for rectification mode of a 1-phase bridge rectifier with equal pulse width modulation.

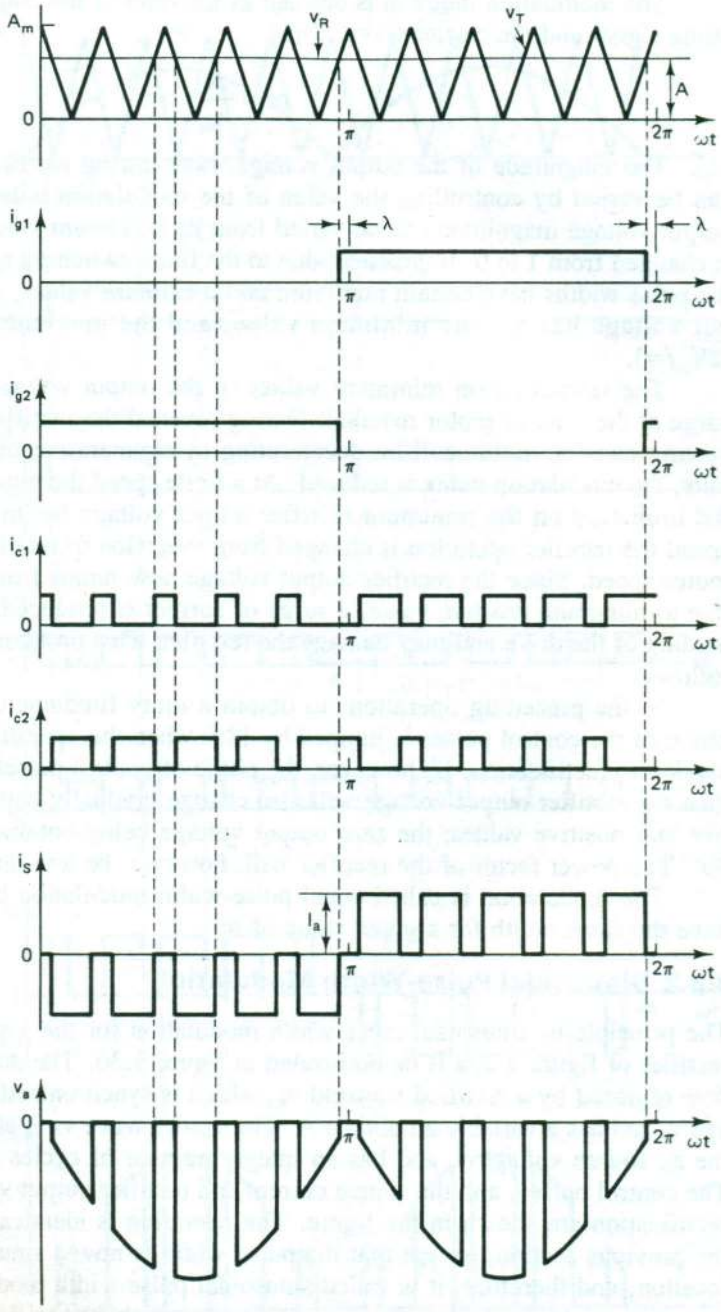


Figure 3.29 Waveforms for inversion mode of a 1-phase fully-controlled bridge rectifier with equal pulse width modulation.

The modulation index m is defined as the ratio of the amplitudes of the modulating signal and the carrier wave. Thus

$$m = \frac{A}{A_m} \tag{3.116}$$

The magnitude of the output voltage, both during rectification and inversion, can be varied by controlling the value of the modulation index. Theoretically, the output voltage magnitude can be varied from its maximum value $(2V_m/\pi)$ to 0 as m is changed from 1 to 0. In practice, due to the finite switching times of S_1 and S_2, the the pulse widths have certain minimum and maximum values; consequently the output voltage has a finite minimum value, and the maximum value is less than $(2V_m/\pi)$.

The restriction on minimum values of the output voltage produces a current surge at the time of motor reversal. During reversal the rectifier will be initially inverting and the motor will be decelerating in regenerative braking. As the speed falls, the modulation index is reduced. At a finite speed the regeneration stops due to the limitation on the minimum rectifier output voltage for inversion. Now at this speed the rectifier operation is changed from inversion to rectification to reverse the motor speed. Since the rectifier output voltage now jumps from its minimum negative to minimum positive value, a surge of current is produced, which causes shock loading of the drive and may damage the rectifier. This problem can be overcome as follows.

In the preceding operation, to obtain a unity fundamental power factor, the phase of the control pulses is jumped by 180° when the operation is shifted from inversion to rectification. If, however, the phase of control pulses is shifted gradually, then the rectifier output voltage will also change gradually between minimum negative and positive values; the zero output voltage being obtained when the phase is 90°. The power factor of the rectifier will, however, be low during this transition.

The modulation is called equal pulse-width modulation because all the pulses have the same width for a given value of m.

3.8.2 Sinusoidal Pulse-Width Modulation

The principle of sinusoidal pulse-width modulation for the 1-phase fully-controlled rectifier of figure 3.27a is demonstrated in figure 3.30. The dc modulating signal is now replaced by a rectified sinusoid v_R, which is synchronized with the source voltage v_s and has a variable amplitude A. The carrier wave v_T is also synchronized with the ac source voltage v_s and has an integer number of cycles in a half cycle of v_s. The control pulses, and the source current and rectifier output voltage waveforms for rectification are shown in the figure. The operation is identical to that described in the previous section, except that the pulse width is now a sinusoidal function of its location, and therefore, it is called sinusoidal pulse-width modulation. The inverter operation is obtained when the control signals are shifted by 180°.

Compared to the equal pulse-width modulation, the sinusoidal pulse-width modulation has higher power factor and lower harmonic content in the source current but higher ripple in the motor current. The modulation ceases to be sinusoidal

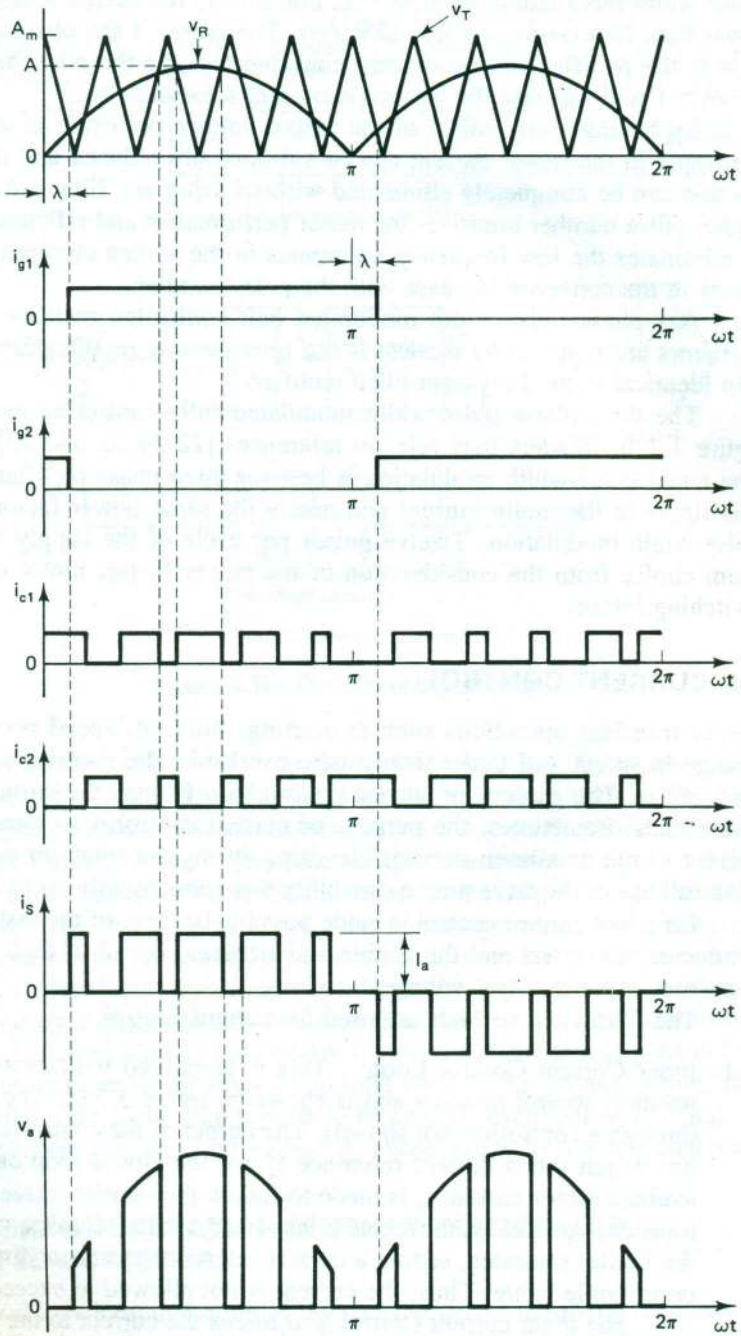


Figure 3.30 Waveforms for rectification mode of a 1-phase rectifier with sinusoidal pulse width modulation.

pulse-width modulation when m = 1. For m = 1, the output voltage is substantially lower than its maximum value ($2V_m/\pi$). Therefore, if the operation is restricted to m = 1, the rectifier will be derated considerably. On the other hand, the operation for m > 1 will increase the harmonic content substantially.

By having many pulses of the output voltage per cycle of the source voltage, the ripple in the motor current can be substantially reduced and discontinuous conduction can be completely eliminated without using any filter inductance. Thus, the higher pulse number improves the motor performance and efficiency. It also reduces or eliminates the low-frequency harmonics in the source current. But the switching losses in the converter increase with the pulse number.

A 1-phase pulse-width modulated half-controlled rectifier is obtained when thyristors are replaced by diodes. It can provide only rectification, giving an operation identical to the fully-controlled rectifier.

The three-phase pulse-width modulated fully-controlled rectifier is shown in figure 3.27b. Readers may refer to references [22-24, 32 and 33] for its operation. The equal pulse-width modulation is best for three-phase rectifiers because it gives less ripple in the motor current and nearly the same power factor as the sinusoidal pulse-width modulation. Twelve pulses per cycle of the supply voltage is an optimum choice from the consideration of the power factor, motor current ripple, and switching losses.

3.9 CURRENT CONTROL

Under transient operations such as starting, braking, speed reversal, and sudden change in speed, and under steady-state overloads, the rectifier current may exceed safe values. The purpose of current control is to prevent the current from exceeding safe values. Sometimes, the purpose of current control is to intentionally force the current to the maximum permissible value during the transient operations. This allows full use of the drive torque capability and consequently gives very fast response.

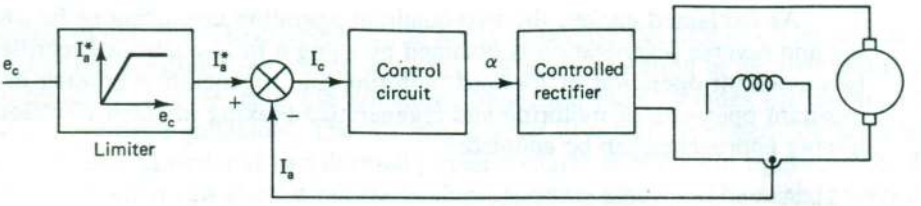
Effective current control is made possible because of the fast response of semiconductor converters and the simple and stepless control of their firing angles, and therefore, of their output voltages.

The following methods are used for current control:

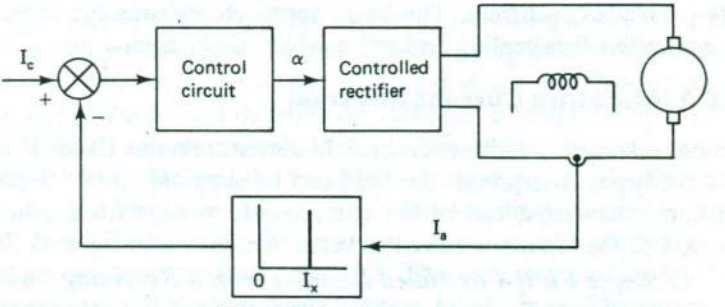
1. Inner Current Control Loop: This is employed with closed-loop speed and position control systems and is shown in figure 3.31a. The error is processed through a controller (not shown). The output of the controller e_c is fed to a limiter which sets a current reference I_a^* for the closed-loop current control. The average motor current I_a is made to follow the current reference I_a^*. During the transient operations, the signal e_c has a large value. Consequently, the output of the limiter saturates, setting a current reference corresponding to the maximum permissible value. Thus, the current is not allowed to exceed the safe limit.

   The inner current control also forces the current to the maximum permissible value during the greater part of the transient operation, consequently giving fast response.

   Open-loop drives are sometimes connected with closed-loop current control during starting, braking, and speed reversal. The block diagram of the drive



(a) Inner current control



(b) Current limit control

Figure 3.31 Current control schemes.

will be similar to that shown in figure 3.31a with the limiter and e_c removed. The reference signal I_a^* will be set for the maximum permissible value. The drive will operate at the maximum permissible current, giving fast response.

2. Current Limit Control: The block diagram is shown in figure 3.31b. If I_a is less than the maximum permissible value I_x, the output of the threshold circuit remains zero. As long as $I_a < I_x$, the motor operation is not affected by the threshold circuit. However, if I_a exceeds I_x, even by a small amount, a large signal is developed by the threshold circuit, and the rectifier firing angle is changed by a large amount in a direction to force the current to decrease fast below I_x. Soon after I_a falls below I_x, the threshold circuit goes out of action and the rectifier firing angle is brought back to the original value. If again I_a exceeds I_x, the same step repeats to bring the current below I_x. Thus, the transient process is completed without I_a exceeding I_x by a substantial amount.

3.10 MULTIQUADRANT OPERATION OF FULLY-CONTROLLED RECTIFIER-FED DC MOTOR

Here, the multiquadrant operation of the rectifier drives involving regenerative braking is considered. As explained in the previous section, the current control forms an integral part of such drives. During transient operations, it limits the current within safe values and sometimes forces it to stay at the maximum permissible value for the greater part of the transient operation to get fast response.

As explained earlier, the two-quadrant operation consisting of forward motoring and reverse regeneration is obtained by using a fully-controlled rectifier. For the two-quadrant operation of forward motoring and regenerative braking or the four-quadrant operation of motoring and regenerative braking in either direction, the following approaches can be adopted:

1. Armature current reversal.
2. Field current reversal.

These operations will now be described for the conventional operation of the fully-controlled rectifiers. The basic approach remains the same for rectifiers with the controlled flywheeling and pulse-width modulation.

3.10.1 Armature Current Reversal

In these schemes, the direction of field current remains fixed. If speed control above the base speed is required, the field can be supplied from a 1-phase half-controlled rectifier; otherwise it can be fed at a fixed dc voltage from a diode bridge. Various schemes of the armature current reversal are shown in figure 3.32.

1. Single Fully-Controlled Rectifier with a Reversing Switch: This scheme is shown in figure 3.32a. A fully-controlled rectifier feeds the motor through a re-

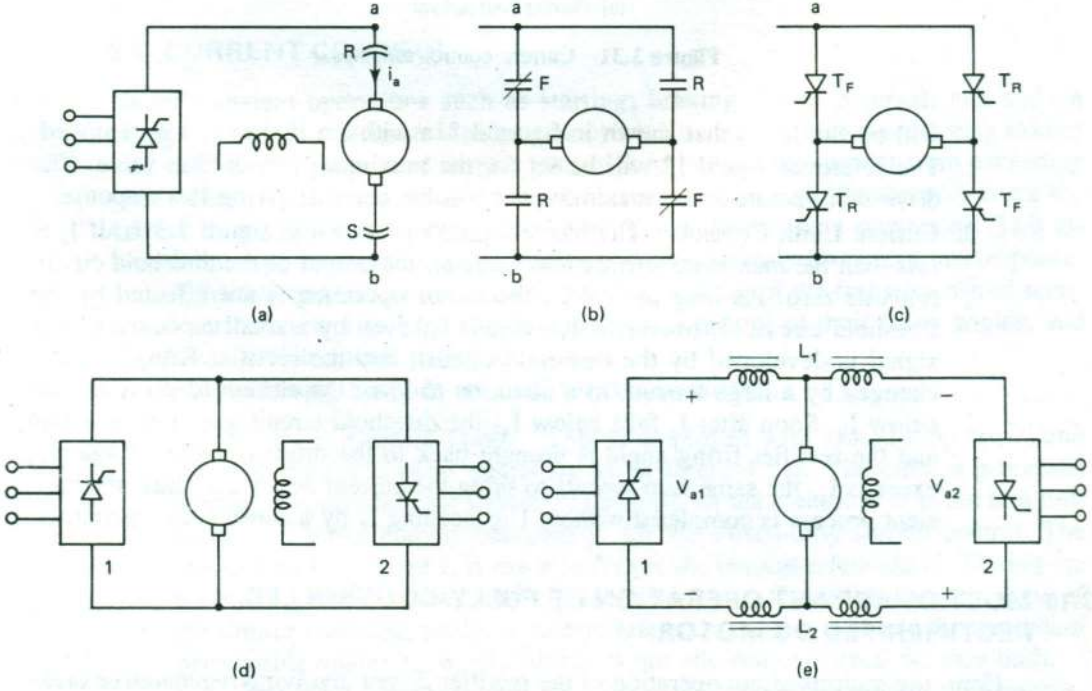


Figure 3.32 Armature current reversal: (a) One fully-controlled rectifier with a reversing switch, (b) and (c) reversing switch, (d) Dual converter with non-simultaneous control, (e) Dual converter with simultaneous control.

versing switch RS. The purpose of the RS is to reverse the motor armature with respect to the rectifier. One setting of switch RS gives operation in the first and fourth quadrants. The reversal of the armature connection then provides operation in the second and third quadrants. The RS may consist of a relay operated contactor with two normally closed and two normally open contacts as shown in figure 3.32b. The reversing switch can also be realized using a drum controller. The speed reversal is carried out as detailed in the following section.

To avoid inductive voltage surges and to reduce the size of the reversing switch, it is necessary to perform the switching at zero armature current. The armature current i_a is forced to zero quickly by setting the firing angle at the highest permissible value. When the current value is zero, firing pulses are stopped. As the instant of zero current cannot be sensed accurately due to the fluctuations in the current and also due to the current through the snubbers, a dead time of 2 to 10 milliseconds is provided to ensure that the current has in fact become zero. Now the armature is reversed by RS and the firing pulses are released. The firing angle is already set at the highest value. In a 3-phase fully-controlled rectifier, according to figure 3.17d, at this firing angle the armature current will be either zero or close to zero. The firing angle is gradually reduced. The armature current builds up and a smooth transition into the regenerative braking takes place. The connection of the rectifier for braking at the highest firing angle is called the advanced firing scheme. It permits a transfer to braking without any surge of current and torque. However, it introduces a considerable amount of delay because the braking torque remains low for a considerable period.

As the firing angle is reduced, the armature current tends to exceed the safe limit but this is prevented by the current control loop. The motor decelerates to zero speed under regenerative braking and then accelerates in the reverse direction with nearly maximum torque under current control. When the speed reaches close to the steady-state value, the current reduces and the motor settles at a speed in the reverse direction.

The slow speed of response is the major limitation of this scheme. The contactor reversal time alone can be 50 to 100 mS. It needs zero-current sensing and frequent maintenance due to the moving contacts. As the cost is low, it is used in low-power drives where fast response is not required.

Considerable improvement in performance at the expense of an increase in cost is obtained if reversing is done using four thyristors as shown in figure 3.32c. The pair T_F is kept on continuously while the pair T_R is blocked. The reversal is obtained by blocking the pair T_F at zero current and firing the pair T_R.

2. Dual Converters: A dual converter consists of two fully-controlled rectifiers connected in antiparallel across the motor armature, as shown in figures 3.32d and e. If rectifier 1 provides operation in the first and fourth quadrants, rectifier 2 provides operation in the second and third quadrants. The dual converters may operate in simultaneous or nonsimultaneous control. In the simultaneous control, commonly known as circulating current control, both rectifiers operate simultaneously. In the nonsimultaneous control, also known as circulating-current-free control, only one rectifier operates at any given time and another is blocked.

In nonsimultaneous control, figure 3.32d, speed reversal is carried out as follows:

Initially let the drive be in operation in the first quadrant. Then rectifier 1 will be in operation and the firing pulses to rectifier 2 will be blocked. For speed reversal, the motor must initially operate in the second quadrant and then in the third quadrant. For this, the operation must transfer to rectifier 2 from rectifier 1. Before rectifier 2 can be activated all thyristors in rectifier 1 must turn off; otherwise, a line side short-circuit of rectifier 2 through the conducting thyristors of rectifier 1 will take place; the resulting current cannot be regulated by the current control and must be cleared by a high-speed breaker or expensive fuse links. The following steps take care of these requirements.

The armature current is forced to zero by setting the firing angle of rectifier 1 at the highest value. After zero current is sensed, a dead time of 2 to 10 mS is provided to ensure the turn-off of all the thyristors of rectifier 1. Now the firing pulses are withdrawn from rectifier 1 and released to rectifier 2. The motor speed will not change appreciably during this period owing to inertia. The firing angle of rectifier 2, α_2, can be set in accordance with the advanced firing just described. Because of the delay involved, it is not usually employed in high-performance drives. The emf matching method is used in high-performance drives. In this method α_2 is initially set to make the armature terminal voltage under continuous conduction equal to the back emf. It is then reduced to make the maximum allowable current to flow. The motor torque builds up fast and the speed reversal is achieved at the maximum torque under the influence of current control.

The speed of response of a dual converter with nonsimultaneous control is considerably affected by dead time. In high-performance drives, particularly in low-inertia fast-servo drives, it should be reduced to a minimum. The value of dead time depends on the accuracy with which the current zero is sensed. Thyristors need only 50 to 100 microseconds to turn off after the current has ceased to flow. If zero current can be sensed exactly, a delay of few hundred microseconds would suffice. Because of the fluctuations in current and also due to the current through snubbers, it is not possible to sense the current zero accurately. In a converter rated for a few thousand amperes, a current of a few amperes can be considered zero for all practical purposes; but not in the present case, because even at this current thyristors of the outgoing rectifier may still be conducting. Adequate dead time is therefore provided to take care of the uncertainty involved in sensing the current zero.

Since the main purpose of sensing zero current and then providing an adequate time delay is to ensure that the thyristors of the outgoing rectifier turn off before the incoming rectifier is activated, it will be more appropriate to sense the state of the thyristors of the outgoing rectifier instead of zero current. A thyristor turns off completely (that is, develops forward voltage blocking capability) if it is kept reverse biased for a duration greater than the gate recovery time after it has started blocking the reverse voltage. Since the gate recovery time is of the order of a few microseconds, a delay of a few hundred microseconds after the thyristor has started blocking the reverse voltage should be adequate. In a 3-phase fully-controlled rectifier, thyristors are fired and conduct in pairs in series; it will, therefore, be adequate to monitor the states of the lower (or upper) three thyristors.

The dual converter for simultaneous control is shown in figure 3.32e. The two rectifiers are controlled simultaneously in such a manner that the sum of their average terminal voltages is zero so that no dc current circulates in the loop formed by the two rectifiers. Thus,

$$V_{a1} + V_{a2} = 0$$

or $$V_{a0} \cos \alpha_1 + V_{a0} \cos \alpha_2 = 0$$

or $$\cos \alpha_1 = -\cos \alpha_2$$

or $$\alpha_1 + \alpha_2 = 180° \qquad (3.117)$$

Equation (3.117) shows that when one rectifier rectifies another inverts. Since the two rectifiers work in different modes, one in rectification and another in inversion, their instantaneous voltages are not equal. This causes ac circulating current to flow in the loop formed by the two rectifiers. The inductors L_1 and L_2 are connected to restrict the ac circulating current.

Although both rectifiers remain in operation simultaneously, the motor control in the first and fourth quadrants is still provided by rectifier 1. Rectifier 2 carries only the circulating current and remains ready to take over whenever the need arises. The two rectifiers reverse their roles when the operation takes place in the second and third quadrants. Speed reversal from the first to third quadrant is carried out as described next.

When operating in quadrant one, rectifier 1 will be rectifying ($0 < \alpha_1 < 90°$) and rectifier 2 will be inverting ($90° < \alpha_2 < 180°$). For the speed reversal α_1 is increased and α_2 is decreased to satisfy equation (3.117). The motor back emf exceeds $|V_{a1}|$ and $|V_{a2}|$. The armature current shifts to rectifier 2 and the motor operates in the second quadrant. As α_2 is decreased gradually, the motor decelerates under regenerative braking. When zero speed is reached $\alpha_1 \simeq \alpha_2 \simeq 90°$. Reduction of α_2 below 90° will make it work as a rectifier and the motor will accelerate to a speed in the reverse direction. During these operations, the armature current is regulated below the safe value by a current control loop.

The circulating current increases the losses, and reduces the efficiency and power factor of the drive; therefore, it must be kept to a minimum. On the other hand, the circulating current must be continuous so that the rectifier carrying this current remains ready to take over the control from the other rectifier without any delay. The relation (3.117) does not take into account the voltage drops across various elements and the dissimilarity in the firing of the two rectifiers. Therefore, when controlled according to the relation (3.117), a large circulating current may flow. This can be limited by operating the two rectifiers with separate current control loops as shown in figure 3.33. For a positive e_c, rectifier 1 carries the armature current and circulating current, and rectifier 2 carries the circulating current. For a negative e_c, the rectifiers reverse their roles. For a positive e_c, current limiter 2 sets a small (fixed) current reference I_2^*, thus allowing only the necessary amount of current to circulate. The current control loop for rectifier 1 operates in the same way as the inner current control loop of figure 3.31a. For a negative e_c, current limiter 1 sets a small (fixed) current reference I_1^* thus allowing only the necessary amount of current

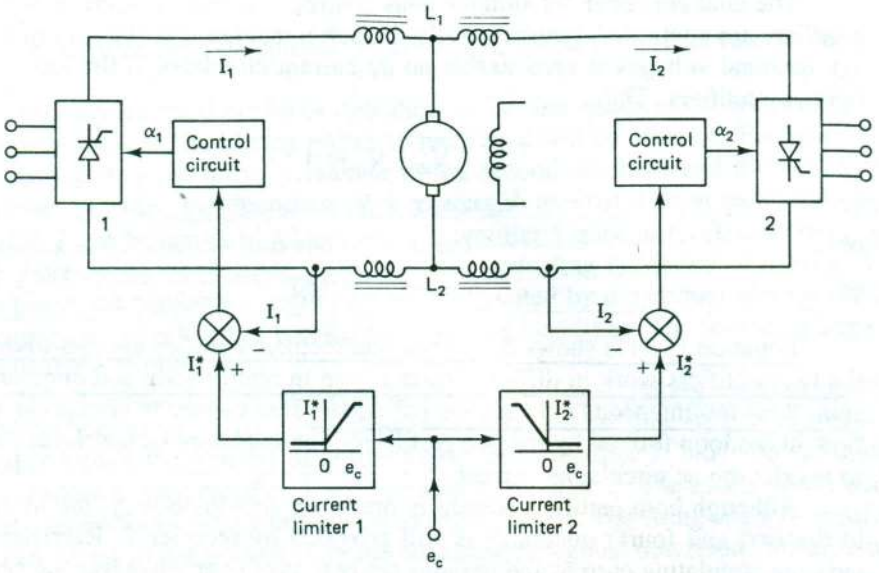


Figure 3.33 Control of circulating current in a dual converter with simultaneous control.

to circulate, and the current control loop for rectifier 2 now operates in the same way as the inner current control loop of figure 3.33a.

Simultaneous control has a number of advantages. The control is simple. Continuous conduction is guaranteed because of the natural freedom of the motor current to flow in either direction; hence, a constant gain transfer characteristic of the dual converter is ensured and the drive has good speed regulation. It has a number of disadvantages, too. The presence of reactors L_1 and L_2 increases the cost, weight, volume, noise, and power loss. The transient response becomes slow due to the increase in the time constant. Power factor and efficiency are low due to the circulating current. The circulating current increases the reactive power and consequently the power factor deteriorates for all firing angles, and progressively at lower motor currents, since the circulating current is almost unaffected by the motor current. When working at a speed, a large drop in the source voltage due to a fault may create a large difference in the motor back emf and the inverter terminal voltage. A large current will flow through the motor and inverter. The current cannot be regulated by the current control loop and must be cleared by a breaker or fuse link.

Nonsimultaneous control has the advantages of increased efficiency and a higher power factor due to the absence of the circulating current. It does not need reactors and saves the associated cost, weight, space, noise, and power loss and provides quicker response to the changes in the firing angle. The main drawbacks are discontinuous conduction at light loads and the consequent poor speed regulation and nonlinear transfer characteristics of the converter, the presence of a dead zone during the reversal of the armature current, and the additional logic requirement to detect current zero and implement dead time.

Because of the need for a complex control circuitry and the presence of dead time, nonsimultaneous control was not used in high-performance drives in the past. Due to the availability of integrated circuits and better methods of detecting current zero, nonsimultaneous control can be easily implemented and in fact provides faster response (due to a very small delay time of 2 to 5 mS). Hence nonsimultaneous control is very widely used now. Simultaneous control is employed only in the low-inertia high-performance servo drives where a dead zone is not acceptable. Recently a pulse-width modulated GTO dual converter with simultaneous control has been developed where circulating current is completely eliminated.[32,33]

3.10.2 Field Current Reversal

Schemes employing field current reversal are shown in figure 3.34. The field current can be reversed either by using a dual converter or a single fully-controlled rectifier with a reversing switch, which can either be a reversing contactor or a drum controller. If one direction of the field current gives operation in the first and fourth quadrants, another direction of the field current will give operation in the second and third quadrants. Since the field current is much smaller compared to the armature current, the ratings of the rectifiers of the field circuit will be much smaller compared to those of the armature circuit. This makes this drive the cheapest four-quadrant dc drive. Because of the large field time constant, the field reversal takes a long time. Often field forcing, involving the use of three to five times the normal voltage during the field reversal, is used to reduce the reversing time. The voltage induced in the armature winding by the transformer action during field reversal adversely affects commutation, making it necessary to keep the armature current at zero during field reversal. Because of the large field reversal time in which the torque remains essentially zero, this drive is rarely used in spite of the low cost. It is employed in high-power drives with large inertia, where the delay in field reversal forms only a fraction of the mechanical time constant, and fast response is not mandatory.

The following steps are used for the reversal of speed when the field is supplied by a dual converter. The armature rectifier firing angle is set at the highest value to force the armature current to zero. The firing angle of the rectifier supplying the field is now set for the highest value. It inverts and the field current is forced to

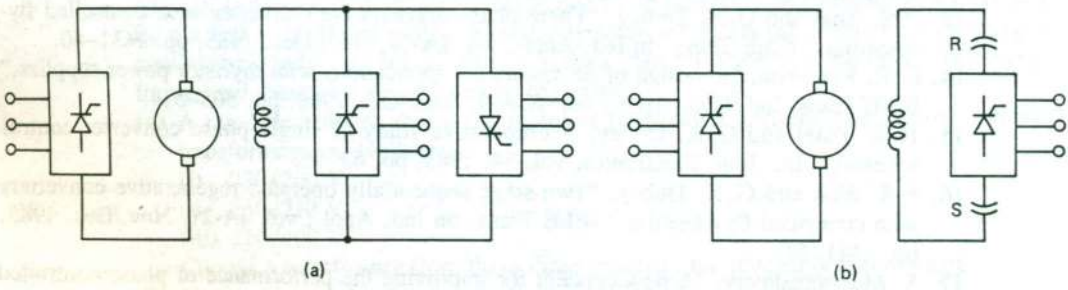


Figure 3.34 Field current reversal: (a) Field control with a dual converter, (b) Field control with a single rectifier and a reversing switch.

zero. After a suitable dead time, the second rectifier is activated at the lowest firing angle. When the field current has nearly settled, and the motor back emf has reversed, the armature rectifier is activated again using the advanced firing scheme explained earlier. The firing angle is progressively changed to first brake and then accelerate the machine in the reverse direction.

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PROBLEMS

3.1 A 7.46 kW, 230 V, 500 rpm, 40 A separately excited dc motor has an armature resistance of 0.478 Ω. This motor is controlled by a 1-phase fully-controlled rectifier fed from a 230 V, 60 Hz ac supply through a transformer. It has sufficient inductance to get continuous conduction for all torques greater than 50 percent of the rated. The transformer and source impedance can be neglected.

(a) A rated dc voltage across the motor at full load is desired. The following three transformers are available:

(i) 230/260 V
(ii) 230/210 V
(iii) 230/400 V

Choose a transformer from these three to satisfy the preceding requirement.

(b) Having chosen the transformer find the following for the drive:

(i) The rectifier firing angle for the rated torque and half the rated speed.
(ii) The rectifier firing angle for the rated braking torque and the speed of 400 rpm in the reverse direction.

(iii) The motor connections are now reversed to get the regenerative braking in the forward direction. What should be the firing angle to develop rated torque at 300 rpm? Neglect friction and windage.

3.2 Repeat problem 3.1 for a 1-phase fully-controlled rectifier with controlled flywheeling.

3.3 A 220 V, 1500 rpm, 11.6 A separately excited motor has the armature resistance and inductance of 2 Ω and 28.36 mH, respectively. This motor is controlled by a 1-phase fully-controlled rectifier with an ac source voltage of 230 V, 50 Hz. Identify modes and calculate developed torques for the following conditions of operation:

(a) $\alpha = 30°$ and speed = 1480 rpm
(b) $\alpha = 30°$ and speed = 1000 rpm
(c) $\alpha = 120°$ and speed = −1178 rpm
(d) $\alpha = 120°$ and speed = −640 rpm

3.4 The motor of problem 3.3 is now controlled by a 1-phase rectifier with controlled flywheeling. The ac source voltage is 230 V, 50 Hz. Identify modes and calculate developed torques for the following conditions of operation:

(a) $\alpha_n = 0$ and $\alpha = 60°$, and speed = 1050 rpm
(b) $\alpha_n = 0$ and $\alpha = 60°$, and speed = 891 rpm
(c) $\alpha_n = 120°$ and $\alpha = 180°$, and speed = −1171 rpm
(d) $\alpha_n = 120°$ and $\alpha = 180°$, and speed = −1475 rpm

3.5 Let the armature inductance of the rectifier drive of problem 3.1 be 13 mH. Calculate no-load speeds, and speeds and developed torques on the boundary between continuous and discontinuous conductions for $\alpha = 30°$ and $\alpha = 120°$.

3.6 For the rectifier drive of problem 3.4, calculate no-load speeds, and the speeds and developed torques on the boundary between continuous and discontinuous conductions for

(a) $\alpha = 60°$, $\alpha_n = 0$
(b) $\alpha = 180°$, $\alpha_n = 120°$.

3.7 Let the armature inductance of the rectifier drive of problem 3.1 be 18.83 mH. Identify modes and calculate firing angles for the following points:

(a) $T_a = 260$ N-m, N = 100 rpm
(b) $T_a = 140$ N-m, N = 200 rpm
(c) $T_a = 170$ N-m, N = −250 rpm

N is the motor speed in rpm.

3.8 Repeat problem 3.7 for a rectifier with controlled flywheeling.

3.9 A 2.4 kW, 220 V, 480 rpm, 12.8 A dc motor has the armature resistance and inductance of 2.2 Ω and 40.0 mH, respectively. It is fed by a 1-phase fully-controlled rectifier with an ac source voltage of 240 V, 60 Hz. Identify modes and calculate speeds for the following points:

(a) $\alpha = 60°$, $T_a = 80$ N-m
(b) $\alpha = 60°$, $T_a = 60$ N-m
(c) $\alpha = 120°$, $T_a = 60$ N-m

3.10 The drive of problem 3.9 is now operated with the controlled flywheeling. Calculate speeds for the following points:

(a) $\alpha = 180°$, $\alpha_n = 30°$, $T_a = 30$ N-m
(b) $\alpha = 60°$, $\alpha_n = 0$, $T_a = 30$ N-m

3.11 A 12.2 kW, 230 V, 850 rpm, 56 A dc separately excited motor is controlled by a 3-phase fully-controlled rectifier fed from 460 V, 60 Hz ac supply through a transformer. It has an armature resistance of 0.284 Ω and sufficient inductance to assure

continuous conduction for all operating points with torques greater than 20 percent of the rated. The transformer and the source impedance can be neglected.

(a) A rated dc voltage across the motor at full load is desired. Choose a suitable transformer from the following three available:
- **(i)** 460/460 V
- **(ii)** 460/230 V
- **(iii)** 460/180 V

(b) Having chosen the transformer find the following:
- **(i)** The rectifier firing angle for the rated torque and speed.
- **(ii)** The rectifier firing angle for the rated braking torque and the speed of 600 rpm in the reverse direction.
- **(iii)** The motor field is now reversed to get regenerative braking in the forward direction. What should be the firing angle to develop rated torque at 500 rpm? Neglect friction and windage.

3.12 Repeat problem 3.11 for a 3-phase fully-controlled rectifier with controlled flywheeling.

3.13 A 30 kW, 230 V, 860 rpm, 144 A dc motor has an armature resistance of 0.07 Ω. It is fed by a 3-phase fully-controlled rectifier from an ac source of 170.3 V (line), 60 Hz. Assuming continuous conduction, calculate motor speeds for the following cases:

(a) $\alpha = 60°$, $T_a = 300$ N-m

(b) $\alpha = 150°$, $T_a = 400$ N-m

(c) $\alpha = 120°$, $T_a = -400$ N-m (obtained by the field current reversal)

3.14 If the rectifier in problem 3.13 is now operated with controlled flywheeling, calculate speeds for the following points, assuming continuous conduction:

(a) $\alpha = 90°$, $\alpha_n = 0$, $T_a = 300$ N-m

(b) $\alpha = 120°$, $\alpha_n = 60°$, $T_a = -300$ N-m (obtained by the field current reversal)

3.15 A 3.74 kW, 1000 rpm, 230 V, 20 A dc motor has an armature resistance and inductance of 1.4 Ω and 16.5 mH, respectively. The motor is fed by a 3-phase fully-controlled rectifier with an ac source voltage of 170.3 (line), 60 Hz. Identify modes and calculate speeds for the following points:

(a) $\alpha = 60°$, $T_a = 1.0$ N-m

(b) $\alpha = 60°$, $T_a = 35$ N-m

(c) $\alpha = 150°$, $T_a = -35$ N-m (obtained by the reversal of the field current)

3.16 For the motor of problem 3.15 identify modes and calculate firing angles for the following:

(a) $T_a = 11.0$ N-m, $N = 500$ rpm

(b) $T_a = 30.0$ N-m, $N = 750$ rpm

(c) $T_a = 30.0$ N-m, $N = -500$ rpm

(d) $T_a = -30.0$ N-m, $N = 500$ rpm (obtained by the field current reversal)

3.17 A 10 kW, 1000 rpm, 230 V, 49 A dc motor has an armature resistance of 0.2 Ω. It is controlled by a 3-phase fully-controlled rectifier with an ac source voltage of 230 V (line), 60 Hz. When operating at full load with a motor terminal voltage of 230 V, the armature current ripple was found to be 5 A (rms). Assuming a 10 percent increase in rotational loss, calculate the power factor and efficiency.

If the thermal loading on the motor is mainly dependent on the copper loss, calculate the derating of the motor.

3.18 The drive of problem 3.15 is now used to drive a load whose torque remains constant for a given setting. What should be the minimum setting of the load torque for the drive to operate always in continuous conduction?

The minimum load torque setting is now reduced to half of the value just calculated. Calculate the value of the external inductance to be connected in the armature circuit to get continuous conduction for all operating points.

3.19 A 1.5 kW, 230 V, 1000 rpm, 7.8 A dc motor has an armature resistance and inductance of 2.5 Ω and 16 mH, respectively. It is fed by a 1-phase fully-controlled rectifier with a 230 V, 50 Hz ac supply. What external inductance must be inserted in the armature circuit to reduce the maximum ripple under continuous conduction to 60 percent of the rated current? At what speed will the ripple be maximum? For this speed, calculate the approximate value of the torque on the boundary between continuous and discontinuous conductions (hint: On the boundary $\Delta i_a = I_a$).

3.20 A 7.5 kW, 230 V, 1000 rpm, 40 A dc motor has an armature resistance and inductance of 0.5 Ω and 5 mH, respectively. It is supplied by a 3-phase fully-controlled rectifier from an ac source of 170.3 V (line) and 60 Hz. Find out the maximum ripple as a percentage of the rated current. How much inductance must be added to keep the maximum ripple to 10 percent of the rated current?

3.21 The motor of problem 3.20 is controlled by a 3-phase fully-controlled rectifier fed from an ac source of 208 V (line), 60 Hz. The rotational losses can be assumed to be 10 percent higher with the rectifier supply.

(a) Determine the ripple at full load.

(b) Calculate the power factor and efficiency.

(c) If the thermal loading of the motor is mainly dependent on the copper loss, calculate

(i) the derating in the output power and

(ii) the derating of torque at zero speed, assuming forced cooling.

3.22 List the sequence of steps required for identifying the modes of operation of a separately excited dc motor fed by a 1-phase fully-controlled rectifier with controlled flywheeling.

3.23 Describe the sequence of steps for identifying the modes of operation and calculating the speed-torque curves for a 3-phase fully-controlled rectifier-fed separately excited dc motor.

3.24 Repeat problem 3.23 for a 3-phase fully-controlled rectifier with controlled flywheeling.

3.25 A 3-phase dc drive is to be selected for the four-quadrant operation of a large-power high-inertia load. Suggest a suitable drive and explain reasons for your choice.

3.26 When a dc motor is fed by a 3-phase controlled rectifier, a step-down transformer can be interposed between the ac source and the converter to get the rated motor voltage for $\alpha = 0$. Alternatively, the rectifier can be connected directly to the ac source. Then the rated voltage is obtained for $\alpha > 0$. Compare these two alternatives from the consideration of the power factor, efficiency, motor derating, and cost.

4

Chopper Control of DC Motors

Choppers are used for the control of dc motors because of a number of advantages such as high efficiency, flexibility in control, light weight, small size, quick response, and regeneration down to very low speeds. Chopper controlled dc drives have applications in servos and traction. In traction, they have been used in underground transit, in battery operated vehicles such as forklift trucks, trolleys, and so on, and in 1500 V dc traction to replace resistance controllers.

In servo applications, separately excited dc motors or permanent magnet field dc motors are used because of their flexible control characteristics. In the past, the series motor was mainly used in traction. Presently, the separately excited motor is also employed in traction. The main reason for using a series motor was the high starting torque. The series motor, however, has a number of limitations. The field of a series motor cannot be easily controlled by static means. If field control is not employed, the series motor must be designed with its base speed equal to the highest desired speed of the drive. The higher base speeds are obtained by using fewer turns in the field windings. This, however, reduces the torque per ampere at zero and low speeds. Further, there are a number of problems with regenerative braking of a series motor, as shown later in this chapter. On the other hand, regenerative braking of a separately excited motor is fairly simple and can be carried out down to very low speeds. Because of the limitations of series motors, separately excited motors are now preferred even for traction applications. In view of the reduced importance of series motors in recent years, series motor drives will be described only briefly.

For a dc motor control in open-loop and closed-loop configurations, the chopper offers a number of advantages over controlled rectifiers. Because of the higher frequency of the output voltage ripple, the ripple in the motor armature current is less and the region of discontinuous conduction in the speed-torque plane is smaller.

As explained in the last chapter, a reduction in the armature current ripple reduces the machine losses and its derating. A reduction or elimination of discontinuous conduction region improves speed regulation and transient response of a drive. To realize a higher frequency of output voltage ripple, it is customary to use a rectifier with a higher pulse number. Use of a rectifier with a higher pulse number results in a low utility factor for thyristors and a relatively high cost. On the other hand, a chopper can be operated at comparatively high frequencies. For example, it is possible to operate a chopper at 300 Hz even with converter grade thyristors. Frequency can be increased to 600 Hz with inverter grade thyristors. If the output voltage range can be lowered, corresponding frequencies can be increased to 400 Hz and 800 Hz, respectively. When power transistors are employed, frequency can be higher than 2.5 kHz. For low power applications, power MOSFETs can be used, and the frequency can be higher than 200 kHz. The rectifier output voltage and current have a much lower frequency — 100 Hz in the case of a single-phase rectifier and 300 Hz in the case of a three-phase fully-controlled rectifier — when the ac source frequency is 50 Hz.

Even when the supply is ac, a chopper drive consisting of a diode bridge followed by a chopper is sometimes preferred. The operation of a chopper in synchronism with the ac source voltage allows an improvement in the line power factor and a reduction in the armature current ripple.

4.1 PRINCIPLE OF OPERATION AND CONTROL TECHNIQUES

The circuit diagram and the steady-state waveforms of a chopper are shown in figure 4.1. A source of direct voltage V supplies an inductive load through a self-commutated semiconductor switch S. The symbol of a self-commutated semiconductor switch has been used because a chopper can be built using any device from among thyristor with a forced commutation circuit, GTO, power transistor, and MOSFET. This symbol was explained in section 1.6.6. The diode shows the direction in which the device can carry current. A diode D_F is connected in parallel with the load. The semiconductor switch S is operated periodically with a period T and remains closed for a time $t_{on} = \delta T$ with $0 < \delta < 1$. The variable $\delta (= t_{on}/T)$ is called the duty ratio or duty cycle of a chopper. Figure 4.1 also shows the waveform of control signal i_c. Control signal i_c will be a base current for a transistor chopper, and a gate current for the GTO of a GTO chopper or the main thyristor of a thyristor chopper. If a power MOSFET is used, it will be a gate to source voltage. When the control signal is present, the semiconductor switch S will conduct if forward biased. It is assumed that the circuit operation has been arranged such that the removal of i_c will turn off the switch.

During the on interval of the switch, $(0 \leq t \leq \delta T)$, the load is subjected to a voltage V and the load current increases from i_{a1} and i_{a2}. The switch is opened at $t = \delta T$. During the off period of the switch, $(\delta T \leq t \leq T)$, the load inductance maintains the flow of current through diode D_F. The load terminal voltage stays zero (if the diode drop is neglected in comparison to V) and the current decreases from i_{a2} to i_{a1}. The interval $0 \leq t \leq \delta T$ is called the duty interval and the interval $\delta T \leq t \leq T$ is known as the freewheeling interval. Diode D_F provides a path for the load current to

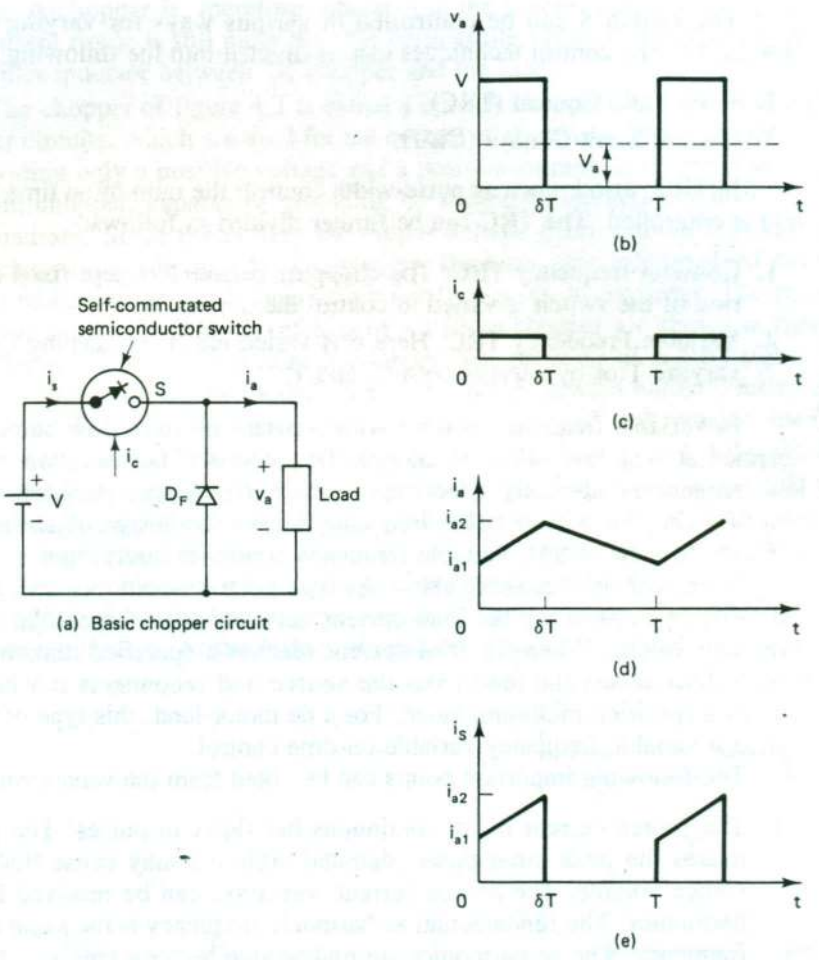


Figure 4.1 Principle of operation of a step-down (or class A) chopper: (a) Basic chopper circuit, (b) to (e) Waveforms.

flow when switch S is off and thus improves the load current waveform. Furthermore, by maintaining the continuity of the load current at turn-off, it prevents transient voltage from appearing across switch S, due to the sudden change of the load current. The source current waveform is also shown in figure 4.1e. The source current flows only during the duty interval and is equal to the load current.

The direct component or average value of the load voltage V_a is given by

$$V_a = \frac{1}{T}\int_0^T v_a\,dt = \frac{1}{T}\int_0^{\delta T} V\,dt = \delta V \qquad (4.1)$$

By controlling δ between 0 and 1, the load voltage can be varied from 0 to V. Thus a chopper allows a variable dc voltage to be obtained from a fixed voltage dc source.

The switch S can be controlled in various ways for varying the duty ratio $\delta(=t_{on}/T)$. The control techniques can be divided into the following categories:

1. Time Ratio Control (TRC).
2. Current Limit Control (CLC).

In TRC, also known as pulse-width control, the ratio of on time to chopper period is controlled. The TRC can be further divided as follows:

1. Constant frequency TRC: The chopping period T is kept fixed and the on period of the switch is varied to control the duty ratio, δ.
2. Variable Frequency TRC: Here δ is varied either by keeping t_{on} constant and varying T or by varying both t_{on} and T.

In variable frequency control with constant on time, low-output voltages are obtained at very low values of chopper frequencies. The operation of a chopper at low frequencies adversely affects the motor performance. Furthermore, the operation of a chopper with variable frequency makes the design of an input filter very difficult. In view of this, variable frequency control is rarely used.

In current limit control, also known as point-by-point control, δ is controlled indirectly by controlling the load current between certain specified maximum and minimum values. When the load current reaches a specified maximum value, the switch disconnects the load from the source and reconnects it when the current reaches a specified minimum value. For a dc motor load, this type of control, in effect, is a variable frequency variable on-time control.

The following important points can be noted from the waveform of figure 4.1.

1. The source current is not continuous but flows in pulses. The pulsed current makes the peak input power demand high and may cause fluctuation in the source voltage. The source current waveform can be resolved into dc and ac harmonics. The fundamental ac harmonic frequency is the same as the chopper frequency. The ac harmonics are undesirable because they interfere with other loads connected to the dc source and cause radio frequency interference through conduction and electromagnetic radiation. Therefore, an L-C filter is usually incorporated between the chopper and the dc source. The filter keeps the harmonic content in the source current due to the chopper within permissible limits. At higher chopper frequencies, harmonics can be reduced to a tolerable level by a cheaper filter. From this aspect, a chopper should be operated at the highest possible frequency.
2. The load terminal voltage is not a perfect direct voltage. In addition to a direct component, it has harmonics of the chopping frequency and its multiples. The load current also has an ac ripple; its adverse effects on the dc motor performance were described in section 3.6.1.

For a given duty ratio, the harmonics in the load voltage have fixed magnitudes. The harmonic current in the load, and therefore, the ripple in the load current depend on the chopper frequency and the load inductance. The ripple in the load current decreases as the chopping frequency is increased or the load inductance is in-

creased. A chopper is, therefore, operated at the highest possible frequency. If the load current ripple is still more than the permissible value, it is reduced by introducing a filter inductor between the chopper and the load.

The chopper of figure 4.1 is called a class A chopper. It is one of a number of chopper circuits, which are used for the control of dc drives. This chopper is capable of providing only a positive voltage and a positive current. It is, therefore, a single-quadrant chopper, capable of providing dc separately excited motor control in the first quadrant. Since it can vary the output voltage from V to 0, it is also a step-down chopper or a dc to dc buck converter. The basic principle involved can also be used to realize a step-up chopper or a boost type dc to dc converter. The circuit diagram and the steady-state waveforms of a step-up chopper are shown in figure 4.2. This chopper is known as a class B chopper. The presence of control signal i_c indicates the duration for which switch S can conduct if forward biased. During a chopping period T, it remains closed for an interval $0 \leq t \leq \delta T$ and remains open for an interval $\delta T \leq t \leq T$. During the on period, i_s increases from i_{s1} to i_{s2}, thus increasing the magnetic energy stored in inductance L. When the switch is opened, current flows through the parallel combination of the load and C. Since the current is forced against a higher voltage, the rate of change of the current is negative. It decreases from i_{s2} to i_{s1} in the switch's off period. The energy stored in the inductance and the energy supplied by the low voltage source are given to the load. The capacitor C serves two purposes. At the instant of opening of switch S, the source current i_s and

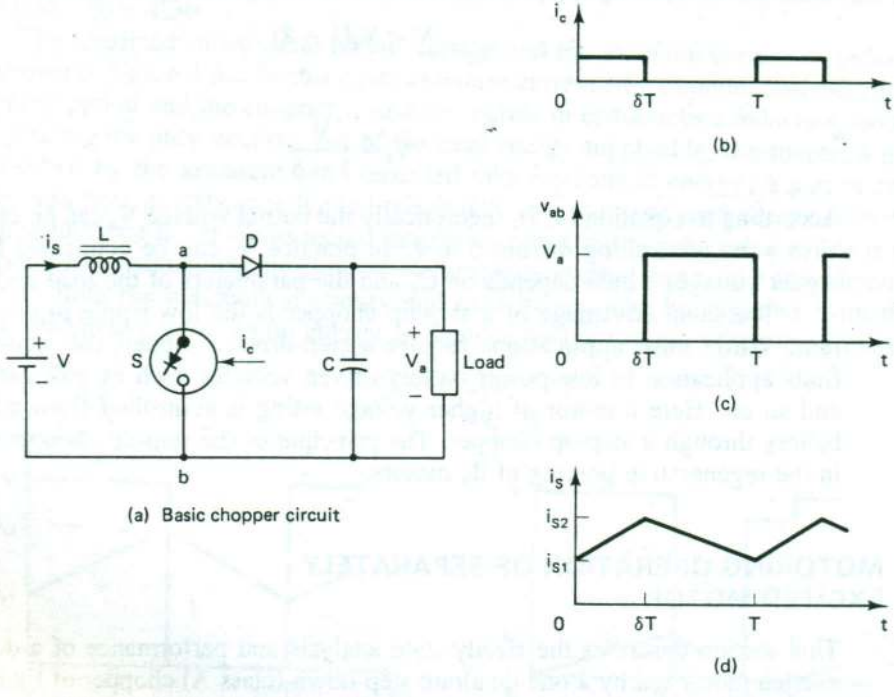


Figure 4.2 Principle of operation of a step-up (or class B) chopper: (a) Basic chopper circuit, (b) to (d) Waveforms.

load current i_a are not the same. In the absence of C, the turn off of S will force the two currents to have the same values. This will cause high induced voltages in L and the load inductance. Another use of C is to reduce the load voltage ripple. The purpose of the diode D is to prevent any flow of current from the load into switch S or source V.

For an understanding of the step-up action, C is assumed large enough to maintain a constant voltage V_a across the load. The average voltage across the terminals a, b is given by

$$V_{ab} = \frac{1}{T}\int_{\delta T}^{T} v_{ab}\,dt = V_a(1-\delta) \tag{4.2}$$

Average voltage across the inductance

$$\begin{aligned} V_L &= \frac{1}{T}\int_0^T \left(L\frac{di}{dt}\right) dt \\ &= \frac{1}{T}\int_{i_{s1}}^{i_{s1}} L\,di = 0 \end{aligned} \tag{4.3}$$

The source voltage

$$V = V_L + V_{ab} \tag{4.4}$$

Substituting from equations (4.2) and (4.3) into equation (4.4) gives

$$V = V_a(1-\delta)$$

or

$$V_a = \frac{V}{1-\delta} \tag{4.5}$$

According to equation (4.5), theoretically the output voltage V_a can be changed from V to ∞ by controlling δ from 0 to 1. In practice V_a can be controlled from V to a higher voltage, which depends on C, and the parameters of the load and chopper.

The main advantage of a step-up chopper is the low ripple in the source current. While most applications require a step-down chopper, the step-up chopper finds application in low-power battery-driven vehicles such as golf carts, trolleys, and so on. Here a motor of higher voltage rating is controlled from a low-voltage battery through a step-up chopper. The principle of the step-up chopper is also used in the regenerative braking of dc motors.

4.2 MOTORING OPERATION OF SEPARATELY EXCITED MOTOR

This section describes the steady-state analysis and performance of a dc separately excited motor fed by a one-quadrant step-down (class A) chopper of figure 4.1. The basic scheme of the drive is shown in figure 4.3a. An L-C filter is connected between the source and the chopper to reduce fluctuations in the source current and voltage. When the filter inductor is assumed lossless and the capacitor C is suffi-

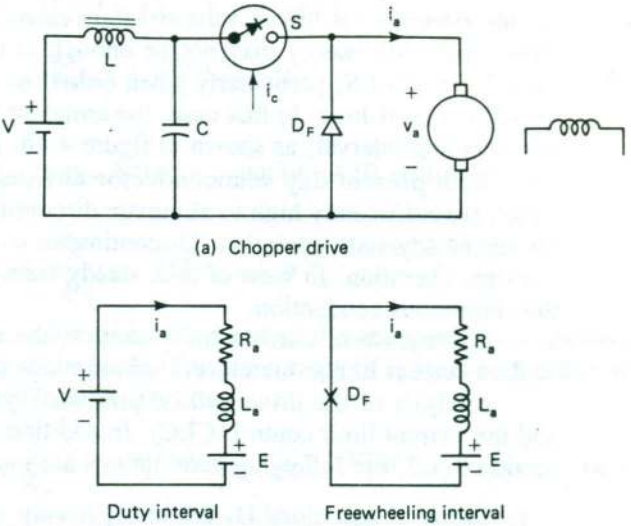


Figure 4.3 Chopper controlled dc separately excited motor.

ciently large, then the chopper input voltage will be equal to the source voltage V. The equivalent circuits of the motor for the duty and freewheeling intervals are shown in figure 4.3b.

The idealized steady-state output voltage and the armature current waveforms are shown in figure 4.4a. In this case, armature current flows continuously during a chopping period and the chopper is said to operate in continuous conduction mode.

During the duty interval, out of the total energy supplied by the source, a part is absorbed by the armature and converted into mechanical energy, a part is converted into heat in resistance R_a and the switch, and the remaining energy is stored in the inductance L_a. It is this stored magnetic energy in the inductance which is responsible for maintaining the flow of the armature current during the freewheeling interval; both the mechanical energy and heat losses must be supplied from this stored magnetic energy, as no energy is supplied by the source. When the armature

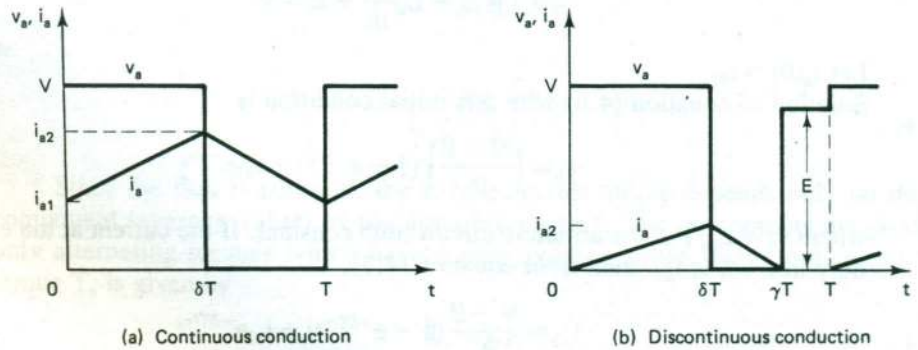


Figure 4.4 Motor terminal voltage and current waveforms.

circuit inductance is low and the armature current is small (at low motor torque), the stored magnetic energy may not be enough to maintain the flow of current during the off period of S, particularly when either the back emf is large or the duration of the off period is long. In that case, the armature current may become zero during the freewheeling interval, as shown in figure 4.4b, giving discontinuous conduction.

With present day semiconductor devices, choppers operate at frequencies which are sufficiently high to eliminate discontinuous conduction during the motor's normal steady-state operation. Discontinuous conduction takes place only under the transient operation. In view of this, steady-state analysis will be considered only for the continuous conduction.

In the case of current limit control, the chopper operates between the prescribed current limits; therefore, discontinuous conduction does not occur.

Analysis of the drive will be presented both for the time ratio control (TRC) and the current limit control (CLC). In addition to assumptions 2 and 3 described in section 3.3.2, the following assumptions are made:

1. Switch S and diode D_F are ideal, having zero drop when conducting and zero leakage current when not conducting.
2. The turn-off of the switch does not have any effect on the motor terminal voltage. This assumption is generally valid except in the case of some thyristor choppers.
3. The discontinuous conduction is ignored due to the reasons just given.
4. The chopper terminal voltage is constant and equal to V as explained earlier in this section.

4.2.1 Steady-State Analysis for Time Ratio Control (TRC)

The analysis aims at the calculation of the motor speed-torque curves and the armature current ripple. The motor performance equations for the duty and freewheeling intervals are as follows:

Duty Interval ($0 \leq t \leq \delta T$): From figure 4.3b

$$R_a i_a + L_a \frac{di_a}{dt} + E = V \tag{4.6}$$

Let $i_a(0) = i_{a1}$

Solution of equation (4.6) with this initial condition is

$$i_a = \left(\frac{V - E}{R_a}\right)(1 - e^{-t/\tau_a}) + i_{a1} e^{-t/\tau_a} \tag{4.7}$$

where $\tau_a = L_a/R_a$, the armature circuit time constant. If the current at the end of the duty interval is i_{a2}, then from equation (4.7),

$$i_{a2} = \frac{V - E}{R_a}(1 - e^{-\delta T/\tau_a}) + i_{a1} e^{-\delta T/\tau_a} \tag{4.8}$$

Freewheeling Interval ($\delta T \leq t \leq T$): From figure 4.3b,

$$R_a i_a + L_a \frac{di_a}{dt'} + E = 0 \tag{4.9}$$

where $t' = t - \delta T$.
The initial current (at $t' = 0$) is i_{a2}. Solving equation (4.9) with this initial condition gives

$$i_a = -\frac{E}{R_a}(1 - e^{-t'/\tau_a}) + i_{a2}e^{-t'/\tau_a} \tag{4.10}$$

In the steady state, the value of i_a at the end of the chopping cycle should be the same as at the beginning of the cycle. Thus, the value of i_a for $t' = (1 - \delta)T$ will be i_{a1}. Substituting this in equation (4.10) gives

$$i_{a1} = -\frac{E}{R_a}(1 - e^{-(1-\delta)T/\tau_a}) + i_{a2}e^{-(1-\delta)T/\tau_a} \tag{4.11}$$

Solving equations (4.8) and (4.11) for i_{a1} and i_{a2} gives

$$i_{a1} = \frac{V}{R_a}\left(\frac{e^{\delta T/\tau_a} - 1}{e^{T/\tau_a} - 1}\right) - \frac{E}{R_a} \tag{4.12}$$

$$i_{a2} = \frac{V}{R_a}\left(\frac{1 - e^{-\delta T/\tau_a}}{1 - e^{-T/\tau_a}}\right) - \frac{E}{R_a} \tag{4.13}$$

The current ripple Δi_a is given by the following equation:

$$\Delta i_a = \frac{i_{a2} - i_{a1}}{2} = \frac{V}{2R_a}\left[\frac{1 + e^{T/\tau_a} - e^{\delta T/\tau_a} - e^{(1-\delta)T/\tau_a}}{e^{T/\tau_a} - 1}\right] \tag{4.14}$$

It is explained in section 3.3.2 that the steady-state average voltage drop across the inductance is zero, and, therefore,

$$V_a = E + R_a I_a \tag{4.15}$$

where V_a and I_a are the average values of the armature terminal voltage and current, respectively. Substituting from equation (4.1) gives

$$\delta V = E + I_a R_a$$

or

$$I_a = \frac{\delta V - E}{R_a} \tag{4.16}$$

Since the flux is constant, the average motor torque depends only on the dc component (average value) of the armature current. The ac components produce only alternating torques which have a zero average value. Therefore, the motor torque T_a is given by

$$T_a = KI_a \tag{4.17}$$

From equations (4.16) and (4.17) and noting that $E = K\omega_m$, where ω_m is the motor speed in radians/sec.,

$$\omega_m = \frac{\delta V}{K} - \frac{R_a}{K^2} T_a \tag{4.18}$$

It should be noted that equation (4.18) may not be valid for very light loads (approaching ideal no load), where discontinuous conduction may be present.

The speed-torque curves with δ as a parameter are similar to those shown by the continuous lines in figure 2.4a for the ideal direct voltage. The natural characteristic ideally corresponds to $\delta = 1$ (assuming a lossless chopper). The lower characteristics are obtained for the smaller values of δ. If speed control above the base speed is required, the field may also be supplied by a chopper. In this case the duty ratio of the armature chopper is either set at 1 or the motor is directly connected to the source. The latter alternative is preferred as it eliminates the chopper losses. The duty ratio of the field chopper is reduced to get higher speeds. The speed torque curves for various values of the duty ratio of the field chopper will be similar to those shown by the dotted lines in figure 2.4a.

When the load torque changes, the average value of the armature current I_a changes to make the motor torque equal to the load torque. At a given δ, the necessary change in I_a is obtained due to a change in E. This, however, does not affect the value of the ripple in the armature current, which is independent of both E and I_a according to equation (4.14). The current ripple is a function of δ and the T/τ_a ratio. For a given T/τ_a ratio, the ripple has a maximum value at $\delta = 0.5$ (see example 4.1).

At present, some motor manufacturers prescribe the maximum ripple that can be permitted without adversely affecting the motor commutation. For the maximum ripple and $\delta = 0.5$, the T/τ_a ratio can be calculated from equation (4.14). The chopper frequency is decided based on the capability of the device used to realize switch S and the desired minimum output voltage.[2] The armature time constant τ_a and the desired armature circuit inductance L_a^* are calculated. If the armature circuit inductance is less than L_a^*, an external inductance is added.

Example 4.1

A 250-V separately excited motor dc has an armature resistance of 2.5 Ω. When driving a load at 600 rpm with constant torque, the armature takes 20 A. This motor is controlled by a chopper circuit with a frequency of 400 Hz and an input voltage of 250 V.

1. What should be the value of the duty ratio if one desires to reduce the speed from 600 to 400 rpm, with the load torque maintained constant?
2. What should be the minimum value of the armature inductance, if the maximum armature current ripple expressed as a percentage of the rated current is not to exceed 10 percent?

Solution: With an input voltage of 250 V and at a constant torque, the motor will run at 600 rpm when $\delta = 1$.

1. At 600 rpm

$$E = V_a - I_a R_a = 250 - 20 \times 2.5 = 200 \text{ V}$$

At 400 rpm, the back emf

$$E_1 = 200 \times \frac{400}{600} = 133 \text{ V}$$

The average chopper output voltage

$$V_{a1} = E_1 + I_a R_a = 133 + 20 \times 2.5 = 183 \text{ V}.$$

Now $\delta V = V_{a1}$ or $\delta = V_{a1}/V = 183/250 = 0.73$.

2.

$$\Delta i_a = \frac{V}{2R_a}\left[\frac{1 + e^{T/\tau_a} - e^{\delta T/\tau_a} - e^{(1-\delta)T/\tau_a}}{e^{T/\tau_a} - 1}\right] \tag{4.14}$$

$$\text{Per-unit current ripple} = (\Delta i_a)_p = \frac{\Delta i_a}{I_{rated}}$$

$$= \frac{V}{2R_a I_{rated}}\left[\frac{1 + e^{T/\tau_a} - e^{\delta T/\tau_a} - e^{(1-\delta)T/\tau_a}}{e^{T/\tau_a} - 1}\right] \tag{E4.1}$$

For the maximum value of the per-unit ripple

$$\frac{d(\Delta i_a)p}{d\delta} = 0,$$

therefore from equation (E4.1)

$$-\frac{T}{\tau_a}e^{\delta T/\tau_a} + \frac{T}{\tau_a}e^{(1-\delta)T/\tau_a} = 0 \quad \text{or} \quad \delta = 1 - \delta \quad \text{or} \quad \delta = 0.5.$$

Substituting in equation (E4.1), the maximum value of the per-unit ripple (Δi_a)pm is given by the following equation:

$$(\Delta i_a)\text{pm} = \frac{V}{2R_a I_{rated}}\left[\frac{e^{0.5T/\tau_a} - 1}{e^{0.5T/\tau_a} + 1}\right] \tag{E4.2}$$

For (Δi_a)pm = 0.1

$$\frac{e^{0.5T/\tau_a} - 1}{e^{0.5T/\tau_a} + 1} = \frac{0.2R_a I_{rated}}{V} = \frac{0.2 \times 2.5 \times 20}{250} = 0.04$$

$$\text{or} \quad e^{0.5T/\tau_a} = 1.08 \quad \text{or} \quad 0.5T/\tau_a = \ell n(1.08) = 0.08 \quad \text{or} \quad \tau_a = \frac{T}{0.16}$$

$$\text{or} \quad L_a = \frac{R_a T}{0.16} = \frac{2.5}{400 \times 0.16} = 39.1 \text{ mH}.$$

4.2.2 Steady-State Analysis for Current Limit Control (CLC)

In CLC, a chopper operates to control the armature current between the prescribed limits i_{a1} and i_{a2}. The chopper adjusts the values of δ and T such that the current fluctuates between these prescribed limits. In CLC, the analysis aims at calculating δ, T, and speed-torque curves.

Equations (4.6) to (4.8) are valid for the duty interval, and equations (4.9) to (4.11) are valid for the freewheeling interval. From equation (4.8),

$$\delta T = \tau_a \log_e \left(\frac{V - E - R_a i_{a1}}{V - E - R_a i_{a2}} \right) \tag{4.19}$$

and from equation (4.11),

$$(1 - \delta)T = -\tau_a \log_e \left(\frac{E + R_a i_{a1}}{E + R_a i_{a2}} \right) \tag{4.20}$$

Adding equations (4.19) and (4.20) yields

$$T = \tau_a \log_e \left[\left(\frac{V - E - R_a i_{a1}}{V - E - R_a i_{a2}} \right) \left(\frac{E + R_a i_{a2}}{E + R_a i_{a1}} \right) \right] \tag{4.21}$$

Once T is obtained from equation (4.21), δ can be calculated from equation (4.19). Then I_a and T_a are obtained from equations (4.16) and (4.17), respectively.

For given values of i_{a1} and i_{a2}, the average current and torque are nearly constant for all speeds. This can be shown as follows:

Since the durations δT and $(1 - \delta)T$ are small compared to the armature circuit time constant, the variation of i_a between i_{a1} and i_{a2} takes place along the initial parts of exponential curves, which can be approximated by straight lines. Thus,

$$I_a = \frac{1}{T} \left[\int_0^{\delta T} \left(i_{a1} + \frac{i_{a2} - i_{a1}}{\delta T} t \right) dt + \int_0^{(1-\delta)T} \left(i_{a2} + \frac{i_{a1} - i_{a2}}{(1 - \delta)T} t' \right) dt' \right]$$

$$= \frac{i_{a1} + i_{a2}}{2} = \text{Constant} \tag{4.22}$$

Thus, the speed-torque curves are parallel to the speed axis. The CLC changes the motor speed-torque characteristic from a constant speed to a constant torque characteristic. This type of characteristic can be used for constant-current starting of a motor or for torque control of a motor driving a battery-operated vehicle. This type of characteristic is not suitable for driving most loads due to the problem of instability, unless an additional closed loop is incorporated to control the motor speed.

4.3 MOTORING CONTROL OF SERIES MOTOR

The main problem in the analysis of a chopper controlled series motor arises due to the nonlinear relationship between the induced voltage E and the armature current i_a because of the saturation in the magnetization characteristic. At a given motor speed, the instantaneous back emf e changes between E_1 and E_2 as i_a changes between i_{a1} and i_{a2}, as shown in figure 4.5. The average effect of this changing emf can be accounted for by a fixed emf E_a which is given by the following equation:

$$E_a = K_e \Phi \omega_m = f(I_a)\omega_m = K\omega_m \tag{4.23}$$

Thus, the motor back emf constant is assumed to be a function of the average value of armature current, I_a. For a given I_a, K is obtained from the magnetization characteristic of the motor. The magnetization characteristic is obtained by driving the se-

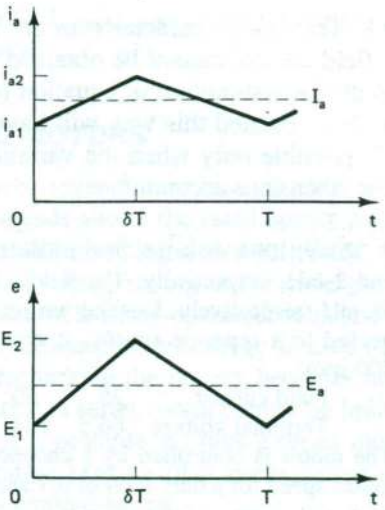


Figure 4.5 Approximation of time varying back emf by a fixed emf.

ries machine by a prime mover at a fixed speed and exciting the field by a separate source. For different values of the field winding current I_a, the armature induced voltages are obtained. Dividing the induced voltages by ω_m allows a K versus I_a curve to be obtained.

In addition to the preceding simplifying assumption, it is also assumed that the motor's field inductance and resistance are constant. The field inductance does vary considerably due to the saturation of the magnetic circuit and to the eddy currents. However, the variation of the inductance affects the steady-state performance curves of the motor only by a small amount.[4]

4.3.1 Steady-State Analysis for Time Ratio Control (TRC)

The analysis of the series motor for TRC is now carried out using the foregoing two assumptions and the assumptions described in section 4.2 for the separately excited motor. From equations (4.15) and (4.23),

$$\delta V = K\omega_m + I_a R_a$$

or

$$\omega_m = \frac{\delta V - I_a R_a}{K} \tag{4.24}$$

For the calculation of the motor speed-torque characteristics, the following sequence of steps is followed. A value of I_a is chosen; the corresponding value of K is obtained from the magnetization characteristic; ω_m is calculated from equation (4.24); and the torque T_a is obtained from equation (4.17). The speed torque curves with δ as a parameter have the same nature as the curves shown by the continuous lines in figure 2.4b for the ideal direct voltage. The natural characteristic ideally corresponds

to $\delta = 1$. The lower characteristics are obtained for smaller values of δ. In a series motor, field control cannot be obtained by a chopper.

With the assumption of equation (4.23), the ripple is given by equation (4.14). The ripple calculated this way will have a large error. Satisfactory estimation of the ripple is possible only when the variation of the field inductance and the eddy currents are taken into account.[4]

Example 4.2

A 220-V, 100A dc series motor has an armature resistance and an inductance of 0.06 Ω and 2 mH, respectively. The field winding resistance and inductance are 0.04 Ω and 18 mH, respectively. Running on no load as a generator, with the field winding connected to a separate source, it gives the following magnetization characteristic at 700 rpm:

Field current	25	50	75	100	125	150	175	A
Terminal voltage	66.5	124	158.5	181	198.5	211	221.5	V

The motor is controlled by a chopper operating at 400 Hz and 220 V. Calculate the motor speed for a duty ratio of 0.7 and a load torque equal to 1.5 times the rated torque.

Solution: The speed at which the magnetization characteristic was measured = $700 \times 2\pi/60 = 73.3$ rad/sec.

$$\text{voltage induced } E = K_e \Phi \omega_m$$

$$K_e \Phi = K = \frac{E}{\omega_m}$$

$$\text{Torque } T_a = K I_a = \frac{E I_a}{\omega_m} \tag{E4.3}$$

From equation (E4.3) and the magnetization characteristic

I_a	25	50	75	100	125	150	175	A
T_a	22.7	84.6	162.2	246.9	338.5	431.8	528.8	N-m

The rated torque (torque at 100A) = 247 N-m
$1.5 \times$ Rated torque $= 1.5 \times 247 = 370.5$ N-m
From the above T_a/I_a table the current at $1.5 \times$ Rated torque = 133 A
Also K at 133 A $= 370.5/133 = 2.79$

$$R_a = 0.06 + 0.04 = 0.1\ \Omega$$

$$\text{Now } \omega_m = \frac{\delta V - I_a R_a}{K} = \frac{0.7 \times 220 - 100 \times 0.1}{2.79} = 51.6 \text{ rad/sec.} = 492.7 \text{ rpm}$$

4.3.2 Steady-State Analysis with Current Limit Control (CLC)

As with equation (4.22) and the related arguments, for a series motor we have

$$I_a = \frac{i_{a1} + i_{a2}}{2}$$

Since i_{a1} and i_{a2} are known values for the CLC, I_a can be calculated from the foregoing equation. K is obtained from the magnetization characteristic and T_a is calculated from equation (4.17). For given values of i_{a1} and i_{a2} the torque is constant and

independent of speed. Thus, with CLC, the speed-torque curves are the same as those for the separately excited motor.

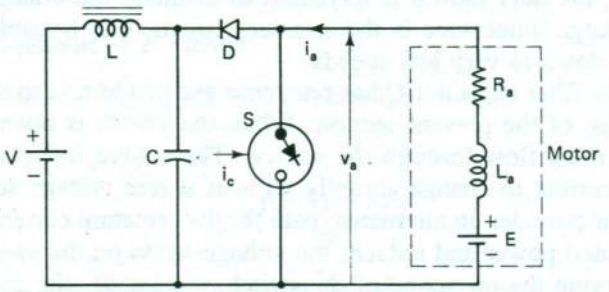
4.4 REGENERATIVE BRAKING OF DC MOTORS

When fed by a fixed voltage source, regenerative braking of a separately excited motor can be carried out only for speeds above the rated speed. With chopper control, it is possible to obtain regenerative braking down to nearly zero speed. This feature has allowed a large amount of energy saving in the underground traction and battery-operated vehicles. In the case of battery-operated vehicles, the regenerated power can be stored in the battery. Consequently, energy is saved and the vehicles can travel longer distances before recharging the battery becomes necessary.

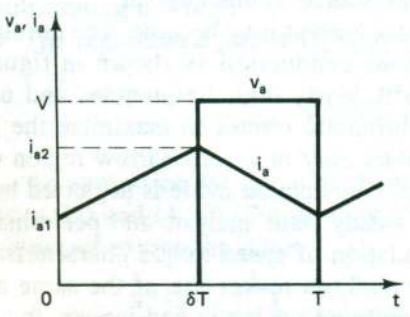
In the absence of chopper control, a series motor cannot be braked by regenerative braking. With chopper control it is possible to brake a series motor using regenerative braking. However, regenerative braking of a series motor is not as simple and effective as that of a separately excited motor.

4.4.1 Separately Excited Motor

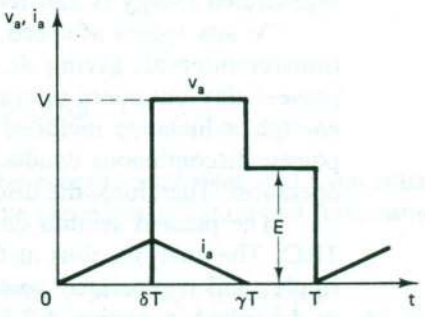
The regenerative braking circuit is shown in figure 4.6a. It uses essentially the step-up (class B) chopper of figure 4.2a, with the motor (working as a generator) forming the low voltage side and the source the high voltage side. The function of inductance



(a) Chopper circuit



(b) Continuous conduction



(c) Discontinuous conduction

Figure 4.6 Regenerative braking of separately excited motor.

L in figure 4.2a is now performed by the armature circuit inductance L_a. A filter is also connected, between the source and the chopper, for reducing the fluctuations in the source current and voltage.

The steady-state waveforms of the motor terminal voltage, v_a, and the armature current, i_a, for continuous and discontinuous conductions are shown in figure 4.6b and c respectively.

The semiconductor switch S is operated periodically with a period T. It remains closed for an interval $0 \leq t \leq \delta T$ and remains open for an interval $\delta T \leq t \leq T$. During the on period of the switch, the motor terminal voltage remains zero and due to the back emf E the armature current increases from i_{a1} to i_{a2}. The mechanical energy supplied by the load and the inertia of the motor load system (only if the speed is changing) is converted by the machine into the electrical energy. This energy is partly used in increasing the stored magnetic energy in the armature circuit inductance, and the remainder is dissipated in the armature circuit resistance and the switch. When the switch is interrupted at $t = \delta T$, the armature current flows through diode D against the source voltage V. During the interval $\delta T \leq t \leq T$, the sum of the energy generated by the machine and the energy stored in the inductor during the on period of the switch is partly dissipated in resistance R_a and diode D and the remaining energy is fed to the source, giving regenerative braking. Thus, with the help of a step-up chopper, it is possible to transfer energy from a back emf of lower voltage to a dc source V of constant and higher potential. As the speed falls and the back emf reduces, the duty ratio δ is increased to maintain the braking torque. With a sufficiently large inductance in the armature circuit, it is possible to obtain regenerative braking down to very low speeds.

The filter capacitor C has one more use in addition to that described in the first paragraph of the present section. When the switch is opened, the motor armature current must flow through the source. The source inductance does not allow the source current to change abruptly without severe voltage stress on the switch. The capacitor provides an alternative path for the armature current, and thus increases the regenerated power and reduces the voltage-stress on the switch.

During the on period of the switch, $0 \leq t \leq \delta T$, the energy is stored in the armature inductance. Therefore, the on interval is also called the energy storage interval. The off interval $\delta T \leq t \leq T$ is called the energy transfer interval, because the regenerated energy is transferred to the source in this interval.

At low values of speed, the motor current may become zero during the energy transfer interval, giving discontinuous conduction as shown in figure 4.6c. As present-day choppers operate at sufficiently high frequencies and usually with enough inductance included in the armature circuit to maximize the regenerated power, discontinuous conduction occurs only in a very narrow region of the drive operation. Therefore, the discontinuous conduction mode is neglected here.

The present section describes steady-state analysis and performance for the TRC. The analysis aims at the calculation of speed-torque characteristics, current ripple, and regenerated power. The analysis makes use of the same assumptions as described in section 4.2 for the motoring operation and ignores the presence of the filter.

Energy Storage Interval ($0 \leq t \leq \delta T$). The machine terminals are shorted by the closed switch S. Thus

$$R_a i_a + L_a \frac{di_a}{dt} = E \tag{4.25}$$

and let

$$i_a(0) = i_{a1}$$

Energy Transfer Interval ($\delta T \leq t \leq T$). The armature current now flows through the source. Therefore,

$$R_a i_a + L_a \frac{di_a}{dt} + V = E \tag{4.26}$$

and let

$$i_a(\delta T) = i_{a2}$$

Since the machine is working as a generator,

$$V_a = E - I_a R_a \tag{4.27}$$

From figure 4.6b,

$$V_a = (1 - \delta)V \tag{4.28}$$

Substituting in equation (4.27) gives

$$(1 - \delta)V = E - I_a R_a$$

or

$$I_a = \frac{E - (1 - \delta)V}{R_a} \tag{4.29}$$

Torque is still given by equation (4.17), but it has a negative sign due to the reversal of the armature current I_a. The derivation of the current ripple gives an expression similar to equation (4.14). The maximum ripple can be obtained from equation (4.14) by substituting $\delta = 0.5$.

The regenerated power P_{rg} is given by

$$P_{rg} = \frac{1}{T} \int_0^{(1-\delta)T} (V \cdot i_a)\, dt$$

An expression for i_a can be derived for the energy transfer interval from equations (4.25) and (4.26). Substituting for i_a in the preceding equation and integrating the resultant expression gives

$$P_{rg} = \frac{V^2}{R_a}\left[\left(\frac{E}{V} - 1\right) \cdot (1 - \delta) + \frac{\tau_a}{T}\left\{\frac{e^{(1-\delta)T/\tau_a} + e^{\delta T/\tau_a} - e^{T/\tau_a} - 1}{1 - e^{T/\tau_a}}\right\}\right] \tag{4.30}$$

The speed-torque curves with δ as a parameter are similar to those shown by the continuous lines in figure 2.7 for the ideal direct voltage. Assuming a lossless chopper, the natural characteristic corresponds to $\delta = 0$. The lower characteristics are obtained for larger values of δ. If regenerative braking above the rated speed is desired, the field is supplied through a step-down chopper for reducing the field current. The armature chopper is set for $\delta = 0$ or preferably is bypassed to eliminate chopper losses. The duty ratio of the field chopper is controlled to get the regenerative braking for speeds higher than the rated no-load speed. This gives the speed torque curves shown by the dotted lines in figure 2.7. The regenerated power versus speed curves with δ as a parameter are shown in figure 4.7. For a given δ the regenerated power increases linearly with speed.

In the absence of field control, the armature current may become excessive under certain operating conditions. At any operating speed, the armature current is prevented from exceeding the permissible value by a reduction in the value of δ. There is always a limitation on the minimum time for which the switch can be on. Thus, there is always a limitation on the minimum value of δ. According to equation (4.29), when operating with the minimum value of δ, there is a critical speed beyond which the armature current cannot be prevented from exceeding a safe value. For the minimum values of δ encountered in thyristor choppers, the critical speed may be close to the rated speed and sometimes even less than the rated speed. The critical speed may be easily exceeded in drives involving active loads. For example, when a battery-operated vehicle or an electric train is moving down a gradient, the critical speed may be crossed if the driver fails to apply the brakes (regenerative braking) in time. In such a situation, the armature current can be limited either by weakening the field or raising the value of R_a by inserting an external resistance in the armature circuit. While the former has the drawback of reducing the torque capability of the motor, the latter increases the loss.

The analysis and performance with CLC is not considered here. The necessary expressions can be derived from equations (4.25) and (4.26). According to equation (4.22) and the related explanation, the torque remains constant and independent of speed for given settings of i_{a1} and i_{a2}. Naturally, the braking power increases linearly with speed. The CLC is employed in battery-operated vehicles.

Example 4.3

A 230 V, 500 rpm, 90 A separately excited dc motor has the armature resistance and inductance of 0.115 Ω and 11 mH respectively. The motor is controlled by a chopper operating at 400 Hz. If the motor is regenerating,

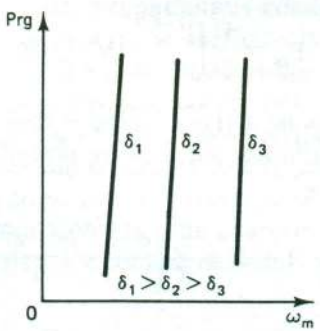


Figure 4.7 Regenerative braking performance curves of TRC chopper-fed dc separately excited motor.

1. Find the motor speed and the regenerated power at the rated current and a duty ratio of 0.5.
2. Calculate the maximum safe speed if the minimum value of the duty ratio is 0.1.

Solution: At rated conditions of operation,

$$E_r = V - I_{ar}R_a = 230 - 90 \times 0.115 = 219.7 \text{ V}$$

1. In regenerative braking

$$(1-\delta)V = E - I_aR_a \quad \text{or} \quad E = (1-\delta)V + I_aR_a \tag{E4.4}$$

At $\delta = 0.5$ and $I_a = 90$ A

$$E = 0.5 \times 230 + 90 \times 0.115 = 125 \text{ V}$$

$$\text{Since } \frac{N}{N_r} = \frac{E}{E_r}$$

where N_r = rated speed in rpm and N = speed to be calculated

Thus

$$N = \frac{N_rE}{E_r} = \frac{500 \times 125}{219.7} = 284.5 \text{ rpm}$$

$$\tau_a = \frac{11 \times 10^{-3}}{0.115} = 95.65 \text{ mS}, \qquad T = \frac{1}{400} = 2.5 \text{ mS}$$

$$T/\tau_a = \frac{2.5 \times 10^{-3}}{95.65 \times 10^{-3}} = 0.026 \quad \text{and} \quad \tau_a/T = 38.3$$

Equation (4.30) is repeated here:

$$P_{rg} = \frac{V^2}{R_a}\left[\left(\frac{E}{V} - 1\right) \cdot (1-\delta) + \frac{\tau_a}{T}\left\{\frac{e^{(1-\delta)T/\tau_a} + e^{\delta T/\tau_a} - e^{T/\tau_a} - 1}{1 - e^{T/\tau_a}}\right\}\right] \tag{4.30}$$

Now

$$\frac{e^{(1-\delta)T/\tau_a} + e^{\delta T/\tau_a} - e^{T/\tau_a} - 1}{1 - e^{T/\tau_a}} = x = \frac{e^{0.5T/\tau_a} + e^{0.5T/\tau_a} - e^{T/\tau_a} - 1}{1 - e^{T/\tau_a}}$$

$$= \frac{e^{0.5T/\tau_a} - 1}{e^{0.5T/\tau_a} + 1} = \frac{0.013}{2.013} = 0.0065$$

From equation (4.30),

$$P_{rg} = \frac{V^2}{R_a}\left[\left(\frac{E}{V} - 1\right) \cdot (1-\delta) + \frac{\tau_a}{T}x\right]$$

$$= \frac{230^2}{0.115}\left[\left(\frac{125}{230} - 1\right)(1 - 0.5) + 38.3 \times 0.0065\right]$$

$$= 9.52 \text{ kW}$$

2. The maximum safe speed will be obtained at the minimum value of δ and the rated armature current. For higher speeds, the armature current will exceed

the rated motor current and this operation will not be safe for the motor. At the maximum safe speed N_m, the back emf E_m is given by

$$E_m = (1 - \delta_{min})V + I_{ar}R_a = 0.9 \times 230 + 90 \times 0.115 = 217$$

$$N_m = \frac{N_r}{E_r} \times E_m = \frac{500}{219.7} \times 217 = 494 \text{ rpm}$$

4.4.2 Series Motor

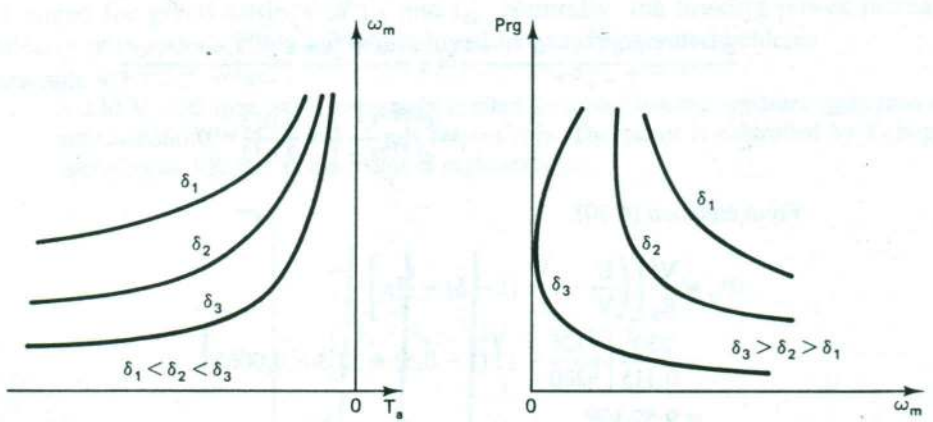
The circuit employed for regenerative braking of a series motor is the same as shown in figure 4.6. In regenerative braking, the series motor is made to work as a self-excited dc generator, which requires that the field be reversed with respect to the armature, compared to their connection in the motoring operation. The principle of operation remains the same as described for a separately excited motor.

Using the approximation of equation (4.23) and following the method described for the separately excited motor in section 4.4.1, the following equation can be obtained for the TRC control:

$$\omega_m = \frac{(1 - \delta)V + R_a I_a}{K} \tag{4.31}$$

The regenerated power is given by equation (4.30).

For a given δ, the speed-torque and the speed-regenerated power characteristics are calculated using the following sequence of steps. A value of I_a is chosen; K is obtained from the motor magnetization characteristic; and ω_m, T_a, and P_{rg} are then obtained from equations (4.31), (4.17), and (4.30), respectively. The nature of the speed-torque and the speed-regenerated power characteristics is shown in figure 4.8a and b. For a given δ, the torque increases with a decrease in speed and the regenerated power increases with an increase in speed. While the speed-torque curves are suitable for stopping a motor, they are not suitable for holding an active load, because of the positive slope the drive may become unstable.



(a) Speed-torque curves (b) Regenerated power-speed curves

Figure 4.8 Regenerative braking performance curves of TRC chopper-fed dc series motor.

As in the case of the motoring operation, section 4.3.2, it can be shown that in the CLC, the torque remains constant and independent of speed, and the regenerated power increases linearly with speed for a given setting of current limits i_{a1} and i_{a2}.

As explained in the case of the separately excited motor in section 4.4.1, the armature current may become excessive at large speeds due to the restriction on the minimum value of δ. This is prevented either by inserting an external resistance in the armature or weakening the field using a special circuit.[6] The regenerative brak- of a dc series motor is not as simple and smooth as that of a separately excited motor. The main problems are the difficulty in the initial build-up of the back emf and the poor stability.[7]

4.5 DYNAMIC AND COMPOSITE BRAKING OF DC MOTORS

4.5.1 Dynamic Braking

The basic circuit is shown in figure 4.9. The only difference between this circuit and the dynamic braking circuits of figure 2.9 is the addition of a self-commutated semiconductor switch S, in parallel with the braking resistance R_B. The switch is operated periodically with a period T and remains closed for the duration δT. This allows a stepless variation in the effective value of the resistance between the terminals a, b.

If the ripple in the armature current is neglected, then the energy consumed by the resistance R_B in one cycle of the chopper is given by

$$E_B = I_a^2 R_B (1 - \delta) T$$

The average power consumed by R_B

$$P_a = \frac{E_B}{T} = I_a^2 R_B (1 - \delta)$$

The effective value of the resistance between the terminals a, b

$$R_e = \frac{P_a}{I_a^2} = (1 - \delta) R_B \tag{4.32}$$

This equation suggests that the effective value of the braking resistance can be steplessly changed from R_B to 0 when the duty ratio of the switch, δ, is controlled from 0 to 1. With a reduction in speed, δ can be increased to reduce R_e. Thus the braking

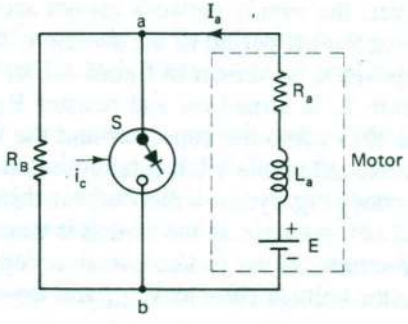


Figure 4.9 Dynamic braking with chopper control.

can be carried out at a constant torque until $\delta = 1$. Then the motor is braked along the speed-torque curve corresponding to $R_B = 0$ (fig. 2.10). The inclusion of the semiconductor switch allows the braking operation at the highest torque from full to very close to zero speed.

For dynamic braking, the series motor must self-excite, which requires that the field be reversed.

4.5.2 Composite Braking

When electrical energy is generated in regenerative braking, it should either be stored in the source or supplied to the loads connected to the source. With the exception of batteries, a source cannot store energy; hence, there should be some loads available to use this energy. If the loads are not available or they are not adequate to consume all the regenerated energy, then this energy must be dissipated using dynamic braking. The combination of regenerative and dynamic brakings is called composite braking. This type of braking is used in dc traction.

The dc supply for traction is usually obtained from an ac supply. In general, ac is converted into dc by uncontrolled rectifiers, which permit energy to flow in one direction only — that is, from ac to dc. When a traction vehicle is returning its braking energy, it should be taken up by other vehicles since it cannot be fed back to the ac supply. This may not happen, because, firstly, there may be no vehicles running at that time, and, secondly, the peak power fed back is always greater than the power that can be accepted by one other vehicle. There are two possible ways of absorbing this energy:

1. By feeding it to the ac supply using a line commutated inverter. This, however, increases the cost of conversion equipment.
2. By using composite braking.

A scheme of composite braking is shown in figure 4.10a. A braking resistor R_B and a thyristor T_1 are added to the regenerative braking circuit of figure 4.6a. The waveforms of the motor armature current and the filter capacitor voltage are shown in figure 4.10b and c. Thyristor T_1 is not given the gate pulse as long as the capacitor voltage is below the tolerance limit of the supply voltage V_{cmax}. If, however, the capacitor voltage crosses this limit, the gate pulse is applied to T_1 to turn it on.

As long as the dc network is able to absorb all the regenerated power, the capacitor voltage remains below the tolerance limit V_{cmax} and regenerative braking takes place. The waveform of v_c and i_a for this case are shown in figure 4.10b. If, however, the supply network cannot accept the regenerated power, which flows into it during the off period of the switch S, the energy gets stored in the capacitor and its voltage rises, as shown in figure 4.10c. When it exceeds the prescribed limit V_{cmax}, thyristor T_1 is turned on and resistor R_B is switched in. Now the motor current no longer flows into the capacitor and the source but into the series resistor R_B. When the semiconductor switch S is turned on again at T — that is, at the beginning of the next chopping cycle — the current through resistor R_B ceases and thyristor T_1 is turned off. As soon as the switch is turned off, the current first flows to offer energy to the source. If the source cannot accept energy, it flows through the capacitor. The capacitor voltage rises to V_{cmax} and thyristor T_1 switches in the braking resistor R_B.

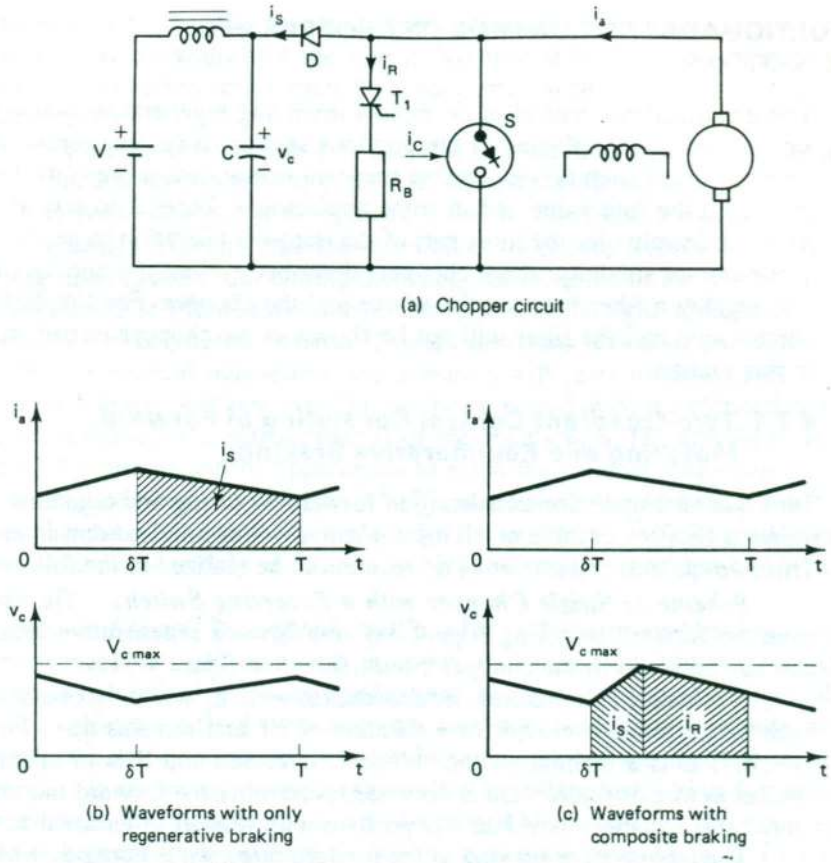


Figure 4.10 Composite braking.

The main advantage of this circuit is that the source capacity to accept the energy is checked in every chopping cycle. Energy is supplied to the source as long as it is able to absorb it. The energy is wasted in the resistance only when the source cannot accept it. Another advantage is that the power circuit is simple. A thyristor without its own commutation circuit switches the braking resistor in and out.

4.6 CURRENT CONTROL

The purpose of current control and methods of its implementation are described in section 3.9 for the case of rectifier control. They are also applicable to chopper control.

Because of the ability of a self-commutated semiconductor switch to turn off at any instant, the current-limit control is implemented differently. The instantaneous value (instead of the average value for rectifiers) of the armature current is compared with the maximum permissible (instantaneous) value. If the current exceeds the maximum permissible value, switch S is turned off. This automatically adjusts the duty ratio such that the current is maintained within safe limits.

4.7 MULTIQUADRANT CONTROL OF CHOPPER-FED DC MOTORS

The multiquadrant control of dc motors involving regenerative braking will be considered here. As explained in the previous section, a current control loop forms an integral part of such drives. During transient operations, it prevents the current from exceeding the safe value and in some applications forces it to stay at the maximum permissible value for the most part of the transient operation to get fast response. As in the case of single-quadrant choppers described earlier, the multiquadrant choppers also employ a filter between the source and the chopper. For simplicity, the current control loop and the filter will not be shown in the chopper circuits to be described in this section.

4.7.1 Two-Quadrant Control Consisting of Forward Motoring and Regenerative Braking

Two-quadrant operation consisting of forward motoring and regenerative braking requires a chopper capable of giving a positive voltage and current in either direction. This two-quadrant operation of dc motors can be realized in the following two ways.

Scheme 1: Single Chopper with a Reversing Switch. The chopper circuits used for forward motoring [fig. 4.3a] and forward regenerative braking [fig. 4.6] can be combined in the chopper circuit shown in figure 4.11.

S is a self-commutated semiconductor switch, which is operated periodically such that it remains closed for a duration of δT and remains open for a duration of $(1 - \delta)T$. C is a manual switch. When C is closed and S is in operation, a circuit similar to that of figure 4.3a is obtained, permitting the forward motoring operation. Under this situation, terminal a is positive with respect to terminal b.

The regenerative braking in the forward direction is obtained when C is opened and the armature connection is reversed with the help of the reversing switch RS, making terminal b positive with respect to terminal a. During the on period of switch S, the motor current flows through a path consisting of the motor armature, switch S and diode D_1, and increases the energy stored in the armature circuit inductance. When S is opened, the current flows through the armature, diode D_2, source V, diode D_1, and back to the armature, thus feeding energy to the source.

When working in motoring, the changeover to regeneration is done in the following steps. Switch S is deactivated and C is opened. This forces the armature current to flow through diode D_2, source V, and diode D_1. The energy stored in the

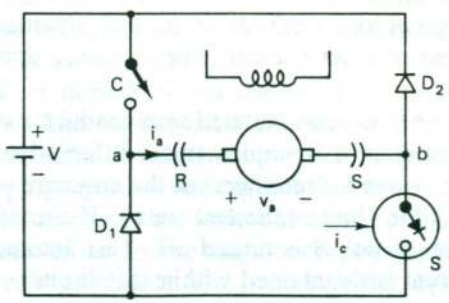


Figure 4.11 Forward motoring and regenerative braking control with a single chopper.

armature circuit is fed back to the source and the armature current falls to zero. After an adequate delay to ensure that the current has indeed become zero, the armature connections are reversed and switch S is reactivated with a suitable value of δ to start regeneration.

The chopper circuit of figure 4.11 can also be used for a series motor. Its armature is connected as shown and the field is connected outside the reversing switch—that is, either between a and R or between S and b. The field is connected outside the reversing switch to ensure that the direction of its current remains the same both for motoring and regeneration. During motoring, the direction of the induced voltage will be to make the left terminal positive with respect to the right terminal. Just after the switch-over to regeneration (with C open, armature reversed, and zero motor current), due to residual magnetism, the armature will have some induced voltage with the right side terminal positive. During the on period of S, the induced emf will force a current through the path consisting of the armature, S, D_1, and the field. The direction of the current will be appropriate to assist the residual magnetism. Consequently, the induced emf will build up by self-excitation. It should be noted that field reversal cannot be used in this case, otherwise the machine will fail to self-excite.

This scheme is widely used in underground traction and battery-operated vehicles.

Scheme 2: Class C Two-Quadrant Chopper. In some applications, such as servo drives, machine tools, and so on, a smooth transition from motoring to braking and vice versa is required. For such applications, the class C chopper of figure 4.12a is used. The self-commutated semiconductor switch S_1, and diode D_1 constitute one chopper and, the self-commutated semiconductor switch S_2, and diode D_2 form another chopper. Both the choppers are controlled simultaneously, both for motoring and regeneration. The switches S_1 and S_2 are closed alternately. In the chopping period T, S_1 is kept on for a duration δT, and S_2 is kept on from δT to T. To avoid a direct short-circuit across the source, care is taken to ensure that S_1 and S_2 do not conduct at the same time. This is generally achieved by providing some delay between the turn-off of one switch and the turn-on of another switch.

The waveforms of the control signals, v_a, i_a, and i_s, and the devices under conduction during different intervals of a chopping period are shown in figure 4.12b. In drawing these waveforms, the delay between the turn-off of one switch and turn-on of another switch has been ignored because it is usually very small. The control signals for the switches S_1 and S_2 are denoted by i_{c1} and i_{c2}, respectively. It is assumed that a switch conducts only if the control signal is present and the switch is forward biased.

The following points are helpful in understanding the operation of this circuit.

1. In this circuit, discontinuous conduction does not occur, irrespective of its frequency of operation. It may be recalled that discontinuous conduction occurs when the armature current falls to zero and stays at the zero value for a finite interval of time. As explained in sections 4.2 and 4.4.1, the current may become zero either during the freewheeling interval or in the energy transfer interval. In the present circuit, freewheeling will occur when S_1 is off and the current is flowing through D_1. This will happen in interval $\delta T \leq t \leq T$, which is also the interval for which S_2 receives the control signal. If i_a falls to zero in the freewheeling interval, the back emf will immediately drive a current

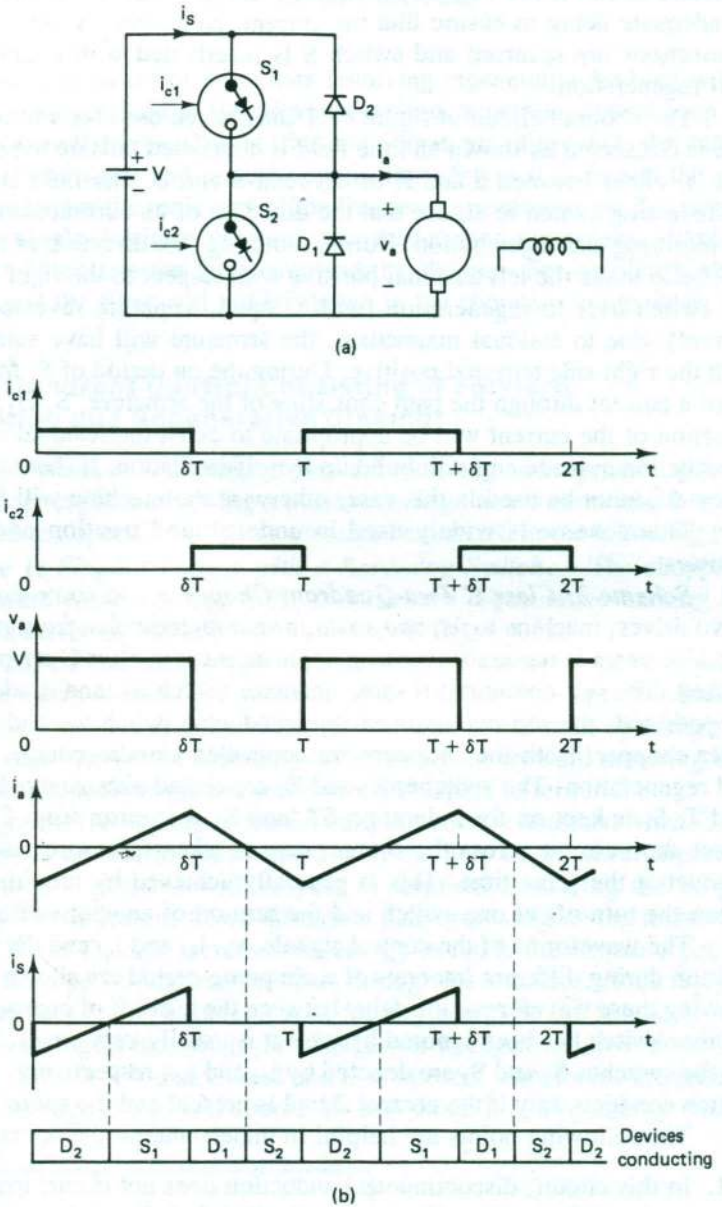


Figure 4.12 Forward motoring and braking control using class C two quadrant chopper: (a) Chopper circuit, (b) Waveforms.

through S_2 in the reverse direction, thus preventing the armature current to stay zero for a finite interval of time. Similarly, the energy transfer interval will be present when S_2 is off and D_2 is conducting — that is, during the interval $0 \leq t \leq \delta T$. If the current falls to zero during this interval, S_1 will conduct im-

mediately because i_{c1} is present and $V > E$. The armature current will flow, preventing discontinuous conduction.

2. Since discontinuous conduction is absent, the motor current will be flowing all the time. Thus, during the interval $0 \leq t \leq \delta T$, the motor armature will be connected to the source either through S_1 or D_2. Consequently, the motor terminal voltage will be V and the rate of change of i_a will be positive because $V > E$. Similarly, during the interval $\delta T \leq t \leq T$, the motor armature will be shorted either through D_1 or S_2. Consequently, the motor terminal voltage will be zero and the rate of change of i_a will be negative. This explains the nature of the waveforms of v_a and i_a.
3. During the interval $0 \leq t \leq \delta T$, the positive armature current is carried by S_1 and the negative armature current is carried by D_2. The source current flows only during this interval and it is equal to i_a. During the interval $\delta T \leq t \leq T$, the positive current is carried by D_1 and the negative current by S_2. This gives the explanation about the conduction of the devices in various intervals of the chopper cycle.
4. From the motor terminal voltage waveform of figure 4.12b,

$$V_a = \delta V$$

Hence,

$$I_a = \frac{\delta V - E}{R_a} \tag{4.33}$$

Equation (4.33) suggests that the motoring operation (+ve I_a) takes place when $\delta > (E/V)$, and that regenerative braking (−ve I_a) occurs when $\delta < (E/V)$. The no-load operation is obtained when $\delta = (E/V)$.

It can be shown that for a given δ, the ripple in the armature current is given by equation (4.14). This equation shows that the ripple is independent of motor speed. A change in speed at a given δ only changes the average value of the current but not the ripple.

Let us examine the operation for a given δ and different speeds. Let the drive be initially working at no load. Then I_a will be zero and the waveform of i_a will be symmetrical about the time axis [fig. 4.12b]. A decrease in speed will increase I_a, and the entire waveform of i_a will shift upward. When I_a becomes greater than the magnitude of ripple, i_a will always be positive. Now only S_1 and D_1 will conduct. Though S_2 will be receiving the control signal, it will not have a chance to conduct due to the reverse bias applied by the conducting diode D_1. Since the ripple may be at the most 5 percent of the rated current, S_2 and D_2 will have to conduct only when the motor is very lightly loaded. Similarly, an increase in speed above the no-load speed will produce a negative I_a and the waveform of i_a will shift downward. When i_a is always negative, the current will be carried only by S_2 and D_2, and S_1 and D_1 will not have any chance to conduct.

The foregoing discussion shows that the motoring current is carried by S_1 and D_1 and the braking current by S_2 and D_2, except at very low torques when all the four devices have to conduct.

Example 4.4

The motor of example 4.3 is controlled by a class C two-quadrant chopper operating with a source voltage of 230 V and a frequency of 400 Hz.

1. Calculate the motor speed for a motoring operation at $\delta = 0.5$ and half of rated torque.
2. What will be the motor speed when regenerating at $\delta = 0.5$ and rated torque?

Solution: At the rated conditions of operation,

$$E_r = V - I_{ar}R_a = 230 - 90 \times 0.115 = 219.7 \text{ V}$$

1. From equation (4.33),

$$\delta V = E + I_a R_a \qquad \text{(E4.5)}$$

At half the rated torque, $I_a = 45$ A
At $\delta = 0.5$

$$E = \delta V - I_a R_a = 0.5 \times 230 - 45 \times 0.115 = 109.8 \text{ V}$$

$$N = \frac{N_r E}{E_r} = \frac{500 \times 109.8}{219.7} = 250 \text{ rpm}$$

2. In the regenerative braking at the rated torque, $I_a = -90$ A
From equation (E4.5),

$$E = \delta V - I_a R_a = 0.5 \times 230 + 90 \times 0.115 = 125.4$$

$$N = \frac{N_r E}{E_r} = \frac{500 \times 125.4}{219.7} = 285 \text{ rpm}$$

4.7.2 Two-Quadrant Control Consisting of Forward Motoring and Reverse Regenerative Braking

Two-quadrant operation consisting of forward motoring and reverse regenerative braking requires a chopper capable of giving a positive current and voltage in either direction. Such a two-quadrant operation is rarely used. But the choppers involved have applications in current source inverters and the discussion involved provides a background necessary for understanding four-quadrant control presented in the next section.

This two-quadrant operation can be obtained in the following two ways:

Scheme 1: Single Chopper Control. The circuit of figure 4.11 can be used *without* the reversing switch. For the motoring operation, manual switch C is closed and S is controlled giving the class A chopper. When running in the reverse direction, terminal b will be positive with respect to a. Now if C is opened and S is controlled, the regenerative braking is obtained.

Scheme 2: Class D Two-Quadrant Chopper. In this scheme the two-quadrant chopper of figure 4.13 is used. This chopper can be controlled using a number of schemes. Here a superior scheme[8] is presented.

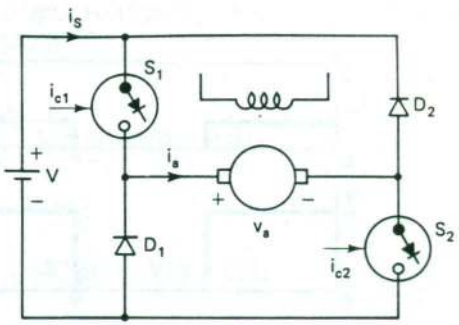


Figure 4.13 Forward motoring and reverse braking control using class D two quadrant chopper.

The control signals and the waveforms of v_a, i_a, and i_s for the forward motoring and the reverse regenerative braking are shown in figures 4.14a and b, respectively.

The self-commutated semiconductor switches S_1 and S_2 are turned on with a phase difference of T sec. Switch S_1 is turned on at $t = 0$ and turned off at $t = 2\delta T$. Switch S_2 is turned on at $t = T$ and turned off at $t = T + 2\delta T$. The period of operation of each switch is 2T and $\delta = t_{on}/2T$, where t_{on} is the duration for which each switch is closed. It should be noted that the instants of turn-on are fixed for both the switches.

The pattern of the control signals suggests that under continuous conduction, the drive will operate in four different modes. These modes of operation and corresponding values of the instantaneous output voltage of the chopper are as follows:

Mode I occurs when S_1 and S_2 conduct. The armature current flows through the path consisting of V, S_1, the armature, and S_2. Here

$$v_a = +V \tag{4.34}$$

Mode II is present when S_1 is on and S_2 is open. The armature current flows through D_2 and S_1 and

$$v_a = 0 \tag{4.35}$$

Mode III occurs when S_1 is open and S_2 is closed. The armature current flows through S_2 and D_1, and

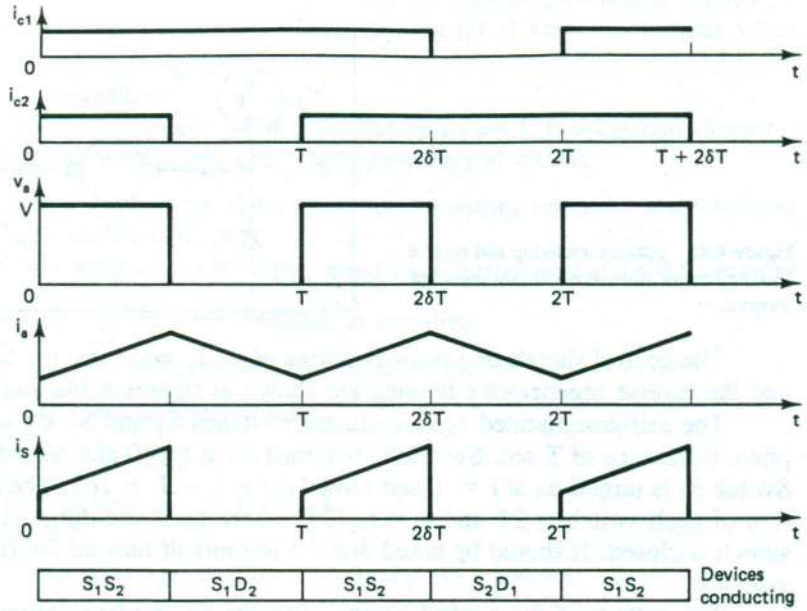
$$v_a = 0 \tag{4.36}$$

Mode IV is present when both S_1 and S_2 are open. The armature current flows through D_2, V, D_1. Here

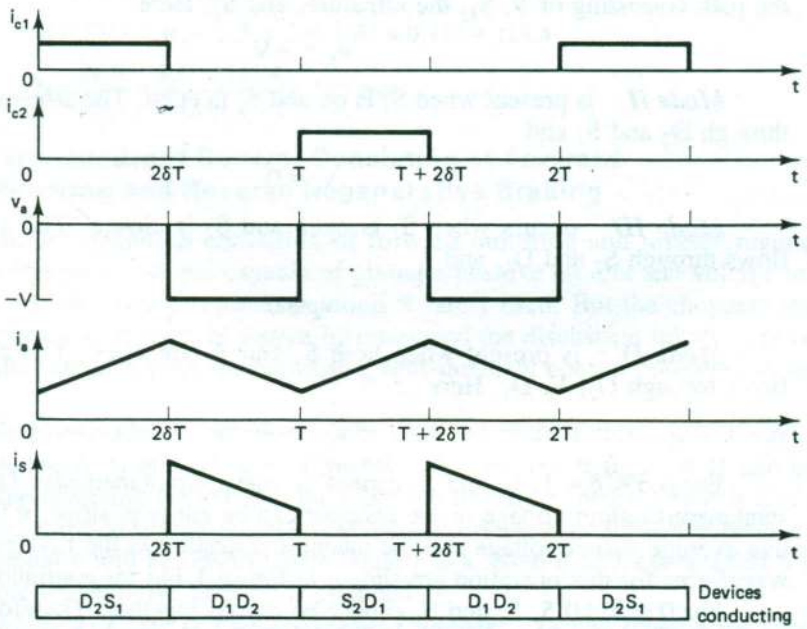
$$v_a = -V \tag{4.37}$$

For $0.5 < \delta < 1$, S_1 and S_2 cannot be open simultaneously. Therefore, the instantaneous output voltage of the chopper can be either positive or 0, giving a positive average output voltage and the machine operation in the first quadrant. Various waveforms for this operation are shown in figure 4.14a for continuous conduction.

For $0 < \delta < 0.5$, S_1 and S_2 cannot be closed together. Therefore, the instantaneous chopper output voltage can be either 0 or $-V$, giving a negative average output voltage and machine operation in the fourth quadrant. The waveforms for this operation are shown in figure 4.14b for continuous conduction.



(a) Forward motoring, $0.5 < \delta \leq 1$



(b) Reverse regeneration, $0 \leq \delta \leq 0.5$

Figure 4.14 Waveforms of the two quadrant chopper of Fig. 4.13.

An expression for the average output voltage V_a can be derived from the waveforms of v_a shown in figures 4.14a and b.
For $0.5 < \delta < 1$

$$V_a = \frac{1}{T}\int_{T}^{2\delta T} V\,dt = 2V(\delta - 0.5) \tag{4.38}$$

and for $0 < \delta < 0.5$

$$V_a = \frac{1}{T}\int_{2\delta T}^{T} (-V)\,dt = 2V(\delta - 0.5)$$

Notice that the same expressions are obtained for both ranges of δ.
Now

$$I_a = \frac{V_a - E}{R_a} = \frac{2V(\delta - 0.5) - E}{R_a} \; . \tag{4.39}$$

4.7.3 Four-Quadrant Control

The four-quadrant operation can be obtained by using the class E chopper shown in 4.15. The chopper can be controlled using the following methods.

Method I. If S_2 is kept closed continuously and S_1 and S_4 are controlled, one gets a two-quadrant chopper as shown in figure 4.12a. This provides a variable positive terminal voltage and the armature current in either direction, giving the motor control in quadrants I and II.

Now if S_3 is kept closed continuously and S_1 and S_4 are controlled, a two-quadrant chopper is obtained, which can supply a variable negative terminal voltage and the armature current in either direction, giving motor control in quadrants III and IV.

For the changeover from forward motoring to reverse motoring, the following sequence of steps is followed.

In the first quadrant S_2 is on continuously, and S_1 and S_4 are being controlled. For the changeover, δ is reduced to its minimum value. The motor current reverses [equation (4.33)] and reaches the maximum permissible value. The current control

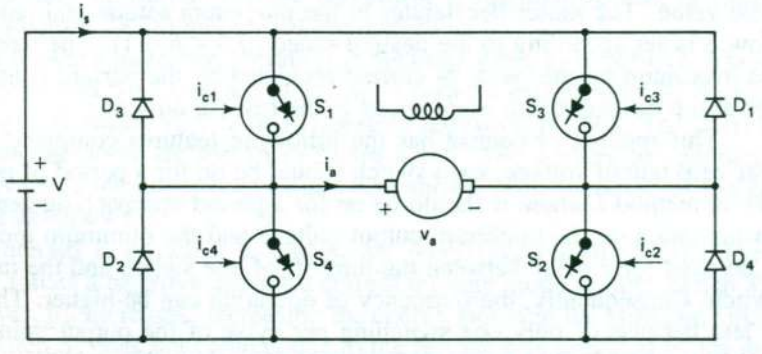


Figure 4.15 Class E four-quadrant chopper.

loop restricts it from exceeding the maximum permissible value. The motor decelerates at the maximum torque and reaches zero speed. Now S_2 is opened, S_3 is continuously closed and δ for the pair S_1, S_4 is adjusted corresponding to the desired speed. The motor now accelerates at the maximum torque in the reverse direction and its current is regulated by the current-control loop. Finally it settles at the desired speed.

This method of control has the following features: The utilization factor of the switches is low due to the asymmetry in the circuit operation. Switches S_3 and S_2 should remain on for a long period. This can create commutation problems when the switches are realized using thyristors. The minimum output voltage depends directly on the minimum time for which the switch can be closed. Since there is always a restriction on the minimum time for which the switch can be closed, particularly in thyristor choppers, the minimum available output voltage, and, therefore, the minimum available motor speed, is restricted.

To ensure that the switches S_1 and S_4, and S_3 and S_4 are not on at the same time, some fixed time interval must elapse between the turn-off of one switch and the turn-on of another switch. This restricts the maximum permissible frequency of operation. It also requires two switching operations during a cycle of the output voltage.

Method II. Switches S_1 and S_2 with diodes D_1 and D_2 provide a circuit identical to the chopper of figure 4.13. This chopper can provide a positive current and a variable voltage in either direction, thus allowing motor control in quadrants I and IV. Switches S_3 and S_4 with diodes D_3 and D_4 form another chopper, which can provide a negative current and a variable voltage in either direction, thus allowing the motor control in quadrants II and III.

The switch-over from quadrant I to quadrant III can be carried out using the following sequence of steps. In quadrant I, the switches S_1 and S_2 are controlled with $0.5 < \delta < 1.0$. The armature current has the direction shown in figure 4.15. For the changeover, S_1 and S_2 are turned off. The armature current now flows through diode D_1, source V, and diode D_2, and quickly falls to zero. The motor back emf has the polarity with the left terminal positive. Now the switches S_3 and S_4 are controlled with δ in the range $0 < \delta < 0.5$, but approaching 0.5. The motor current flows in the reverse direction and reaches the maximum value [equation (4.39)]. The current-control loop regulates δ to keep the current from exceeding the maximum permissible value. The motor decelerates at the maximum torque and reaches zero speed. Now δ is set according to the desired speed ($0.5 < \delta < 1$). The motor accelerates at the maximum torque, with its current regulated by the current-control loop and settles at the desired steady-state speed in the reverse direction.

This method of control has the following features compared to method I: At near-zero output voltage, each switch should be on for a period of nearly T sec., unlike in method I where it should be on for a period approaching zero. Thus, there is no limitation on the minimum output voltage and the minimum motor speed. There is no need for a delay between the turn-off of one switch and the turn-on of another switch. Consequently, the frequency of operation can be higher. The switching loss is less because of only one switching per cycle of the output voltage compared to two in method I. Due to the symmetrical operation, the switches have a better utilization factor.

Method III. This method is a modification of method II. In method II, switches S_1 and S_2 with diodes D_1 and D_2 form one chopper, which allows motor control in quadrants I and IV. The second chopper, providing operation in quadrants II and III is formed by switches S_3 and S_4, and diodes D_3 and D_4. In method II, these choppers are controlled separately. In the present method, these choppers are controlled simultaneously as follows.[9]

The control signals for the switches S_1–S_4 are denoted by i_{c1}, i_{c2}, i_{c3}, and i_{c4}, respectively. As with the convention adopted, a switch conducts if its control signal is present and it is forward biased; otherwise it remains open. The control signal i_{c1} to i_{c4}, and the waveform of v_a, i_a, and i_s for forward motoring and forward regeneration are shown in figure 4.16a and b, respectively. Switches S_1 and S_2 are given control signals with a phase difference of T secs. Switch S_1 receives a control signal from $t = 0$ to $t = 2\delta T$, where $\delta = t_{on}/2T$. The control signal for switch S_2 is present from $t = T$ to $t = T + 2\delta T$. Switches S_1 and S_4, and S_2 and S_3 form complementary pairs in the sense that the switches of the same pair receive control signals alternately. Usually some interval must elapse between the turn-off of one switch and the turn-on of another switch of the same pair to ensure that they are not on at the same time. This interval has been neglected in drawing the waveforms of figure 4.16.

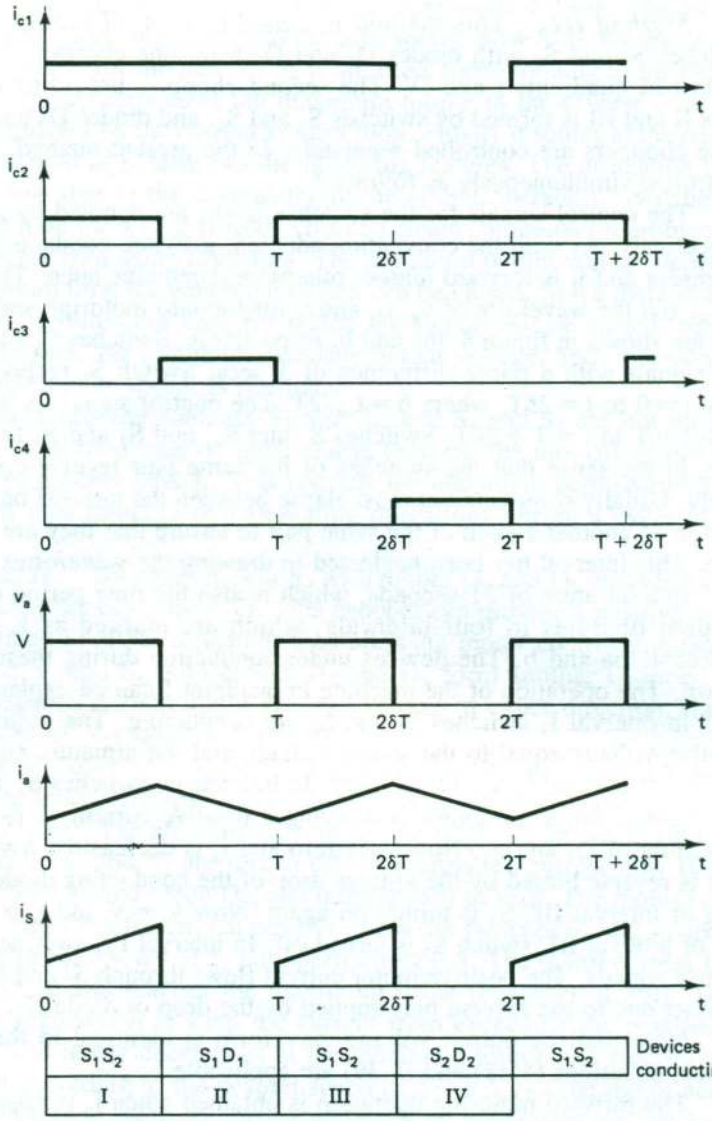
In a duration of 2T seconds, which is also the time period of each switch, the chopper operates in four intervals, which are marked as I, II, III, and IV in figures 4.16a and b. The devices under conduction during these intervals are also shown. The operation of the machine in quadrant I can be explained as follows.

In interval I, switches S_1 and S_2 are conducting. The motor is subjected to a positive voltage equal to the source voltage and the armature current increases. At the end of interval I, S_2 is turned off. In interval II, switches S_1 and S_3 receive control signals. Since the motor is carrying a positive current, it flows through a path consisting of D_1 and S_1. Now v_a is zero and i_a is decreasing. Switch S_3 remains off as it is reverse biased by the voltage drop of the conducting diode D_1. At the beginning of interval III, S_2 is turned on again. Now $v_a = V$ and i_a is increasing. At the end of interval III, switch S_1 is turned off. In interval IV, switches S_2 and S_4 receive control signals. The positive motor current flows through S_2 and D_2, and S_4 does not conduct due to the reverse bias applied by the drop of diode D_2.

Note that the output voltage waveform is identical to that of figure 4.14a. Hence, equations (4.38) and (4.39) are applicable.

The forward motoring operation is obtained when I_a is positive. The operation can be transferred from forward motoring to forward regeneration by decreasing δ or increasing E to make $V_a < E$ or I_a negative [equation (4.39)]. The waveforms for forward regeneration are shown in figure 4.16b. The devices in conduction in the four intervals of the chopper cycle are also shown. The operation of the chopper is explained as follows.

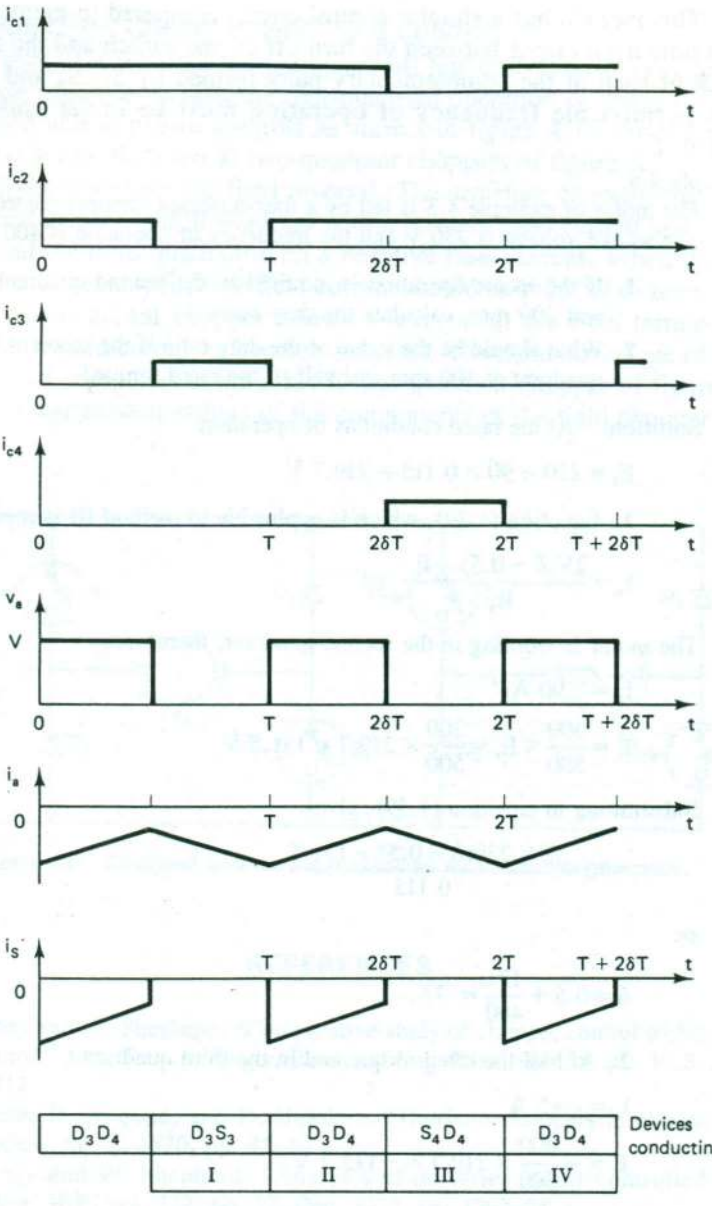
In interval I, switches S_1 and S_3 are receiving control signals. The positive back emf forces a negative armature current through diode D_3 and switch S_3. During this interval, $|i_a|$ increases, increasing the energy stored in the armature circuit inductance. Switch S_1 does not conduct due to the reverse bias provided by the drop of the conducting diode D_3. Switch S_3 is opened at the end of interval I. The armature current is forced through diode D_3, source V, and diode D_4, and the energy is fed to the



(a) Forward motoring, $0.5 \leq \delta \leq 1.0$ and $V_a > E$

Figure 4.16 Waveforms of the four quadrant chopper of Fig. 4.15 using *method III* *(continued on next page)*.

source. Although switches S_1 and S_2 are receiving the control signals, they remain open due to the reverse bias provided by the voltage drops of diodes D_3 and D_4. The motor terminal voltage is now V and $|i_a|$ is decreasing. S_4 is turned on in interval III. The armature current now flows through switch S_4 and diode D_4. Switch S_2 also receives a control signal; however, it does not conduct due to the reverse bias applied by diode D_4. The armature current magnitude again builds up. S_4 is turned off at the



(b) Forward regeneration, $0.5 \leq \delta \leq 1.0$ and $V_a < E$

Figure 4.16 (continued).

end of interval III. The armature current is forced again through diode D_3, the source, and diode D_4, and the energy is fed to the source.

The motoring and regenerative braking operations in the reverse direction are obtained when $0 < \delta < 0.5$, for which V_a is negative. Reverse motoring is obtained by setting δ such that $|V_a| > |E|$ and reverse regeneration is realized when $|E| > |V_a|$.

This method has a simpler control circuit compared to methods I and II. Since some time must elapse between the turn-off of one switch and the turn-on of another switch of each of the complementary pairs formed by S_1, S_4 and S_2, S_3, the maximum permissible frequency of operation must be lower compared to that of method II.

Example 4.5

The motor of example 4.3 is fed by a four-quadrant chopper controlled by method III. The source voltage is 230 V and the frequency of operation is 400 Hz.

1. If the motor operation is required in the second quadrant at the rated torque and 300 rpm, calculate the duty ratio.
2. What should be the value of the duty ratio if the motor is working in the third quadrant at 400 rpm and half of the rated torque?

Solution: At the rated conditions of operation

$$E_r = 230 - 90 \times 0.115 = 219.7 \text{ V}$$

1. Equation (4.39), which is applicable to method III is reproduced here:

$$I_a = \frac{2V(\delta - 0.5) - E}{R_a} \tag{4.39}$$

The motor is working in the second quadrant, therefore,

$$I_a = -90 \text{ A}$$

$$E = \frac{300}{500} \times E_r = \frac{300}{500} \times 219.7 = 131.8 \text{ V}$$

Substituting in equation (4.39), gives

$$-90 = \frac{2 \times 230(\delta - 0.5) - 131.8}{0.115}$$

or

$$\delta = 0.5 + \frac{121}{460} = .76 .$$

2. At half the rated torque and in the third quadrant

$$I_a = -45 \text{ A}$$

$$E = -\frac{400}{500} \times 219.7 = -175.7 \text{ V}$$

Substituting in equation (4.39), gives

$$-45 = \frac{2 \times 230(\delta - 0.5) + 175.7}{0.115}$$

or

$$\delta = 0.5 - \frac{181}{460} = 0.11 .$$

Four-quadrant Operation with Field Control

When field control is required for getting speeds higher than base speed and the transient response need not be fast, the four-quadrant operation is obtained by a combination of field and armature controls as shown in figure 4.17. Both armature and field are supplied by the class D two-quadrant choppers of figure 4.13. The reversal switch RS is employed for the field reversal. The armature chopper provides operation in the first and the fourth quadrant with a positive field current and operation in the second and the third quadrant with a negative field current. When the field connection is to be reversed, first the field current should be reduced to zero. The use of the class D two-quadrant chopper allows a reversal of the field terminal voltage, which forces the field current to become zero fast. The main advantage of this circuit is the lower cost compared to the class E four-quadrant chopper of figure 4.13, because of the lower current ratings of the components of the field chopper.

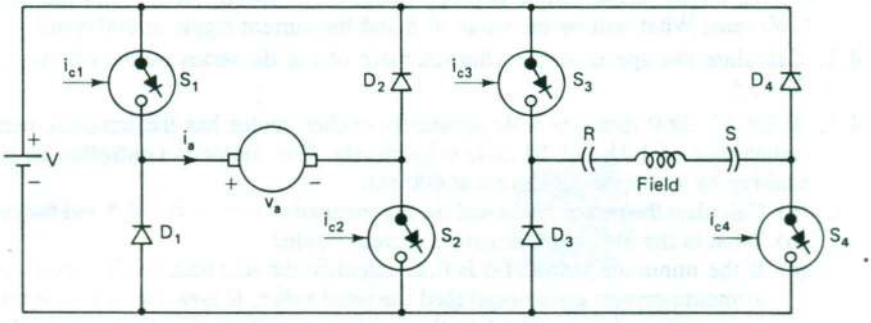


Figure 4.17 Combined armature and field control for four-quadrant operation.

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PROBLEMS

4.1. A 2.2 kW, 220 V, 11.6 A, 1500 rpm dc separately excited motor has an armature resistance and inductance of 2 Ω and 32.5 mH, respectively. This motor is controlled by a chopper with a frequency of 500 Hz and the input voltage of 220 V. The motor is driving a load whose torque is proportional to the speed. At $\delta = 0.9$ the motor runs at 1260 rpm. What will be the value of δ and the current ripple at 800 rpm?

4.2. Calculate the speed-torque characteristic of the dc series motor of example 4.2 for $\delta = 0.5$.

4.3. A 230 V, 1000 rpm, 20 A dc separately excited motor has the armature resistance and inductance of 1 Ω and 50 mH, respectively. The motor is controlled in regenerative braking by a chopper operating at 600 Hz.
(a) Calculate the motor speed and the regenerated power for $\delta = 0.5$ and the rated torque.
(b) What is the maximum armature current ripple?
(c) If the minimum value of δ is 0.2, calculate the maximum safe speed for which the armature current does not exceed the rated value. If now the field current is reduced by a factor of 0.8, calculate the maximum safe speed assuming a linear relation between the flux and the field current.

4.4. The dc series motor of example 4.2 is controlled by a chopper in regenerative braking. The source voltage is 220 V.
(a) Calculate the motor speed for $\delta = 0.5$ and the driving torque equal to twice the rated motor torque. Neglect friction and windage losses.
(b) Calculate δ if the machine is running at 500 rpm and the driving torque is equal to the rated torque.

4.5. **(a)** Derive expressions for the armature current ripple and the average torque for the dynamic braking of a chopper controlled separately excited motor.
(b) The separately excited motor of example 4.3 is controlled by a chopper in dynamic braking. The braking resistance R_B (fig. 4.9) is 5 ohms, and the chopper operating frequency is 450 Hz. Calculate the motor speed-torque characteristic for $\delta = 0.5$.

4.6. A 7.46 kW, 230 V, 500 rpm, 40 A separately excited dc motor has the armature resistance and inductance of 0.478 Ω and 13 mH, respectively. The motor is controlled by the Class C two-quadrant chopper of figure 4.12a operating at 400 Hz. The source voltage is 230 V.
(a) The motor is braked from its initial speed of 500 rpm to zero by regenerative braking. The current control loop adjusts δ automatically to keep the motor current at the rated value. The braking is slow, and, therefore, the chopper and motor can be assumed to be operating in steady state for all the speeds. Calculate and plot the variation of δ with speed.

(b) The motor is now working in the first quadrant at 300 rpm. Calculate the value of δ when $I_a = 5A$. Obtain the instantaneous armature current waveform and identify the devices conducting during different intervals of chopper operation.

4.7. Draw the waveforms of control signals, motor terminal voltage, motor armature current, and source current for the class E four-quadrant chopper of figure 4.15 controlled by method III for the motor operation in the third and fourth quadrants.

4.8. A 12.2 kW, 230 V, 850 rpm, 56 A separately excited dc motor is controlled by the four-quadrant chopper of figure 4.17. The armature resistance and inductance are 0.284 Ω and 4.4 mH, respectively. The chopper operating frequency is large enough to ensure continuous conduction.

(a) The motor is operating in the first quadrant at 700 rpm at the rated torque. Find δ.

(b) Now the motor operation is transferred to the second quadrant by reversing the field current. Assuming the motor speed to be still 700 rpm and the motor to be operating in steady state, calculate the value of δ for the rated torque.

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5

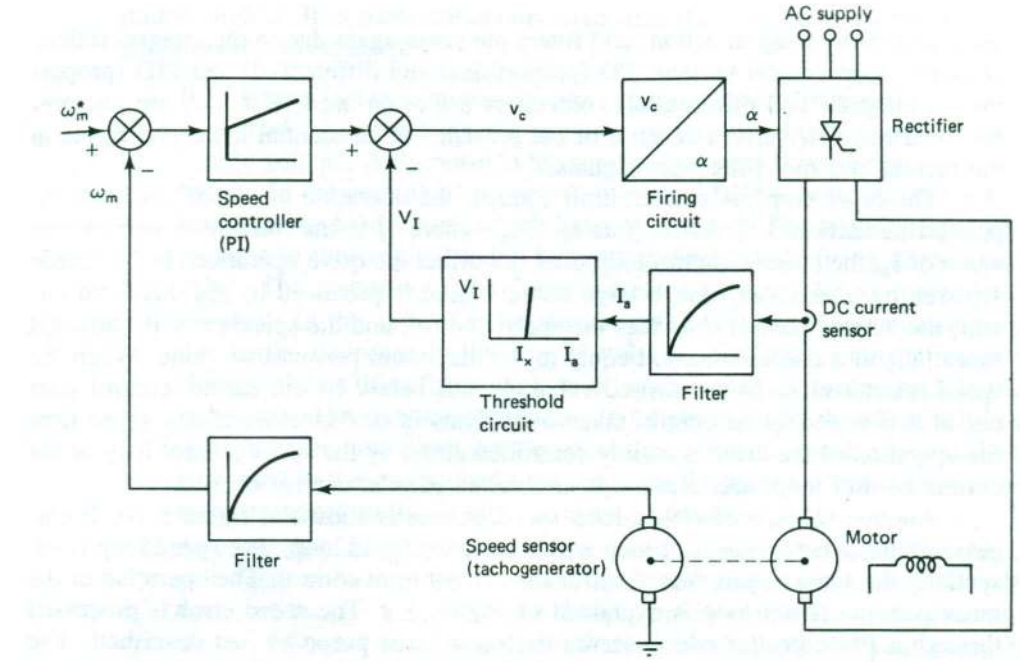
Closed-Loop Control of DC Drives

When the steady-state accuracy requirement cannot be satisfied in an open-loop configuration, the drive is operated in a closed-loop system. Additional feedback loops are provided to limit the parameters to safe or acceptable limits and to improve the dynamic performance. Here we are mainly concerned with closed-loop variable speed drives which are widely used in industry. The ratings of such drives range from as low as fractional kW to 10000 kW and more. Closed-loop rectifier drives are more widely used than chopper drives. In view of this, mainly rectifier drives will be described here. The same schemes are used in chopper drives.

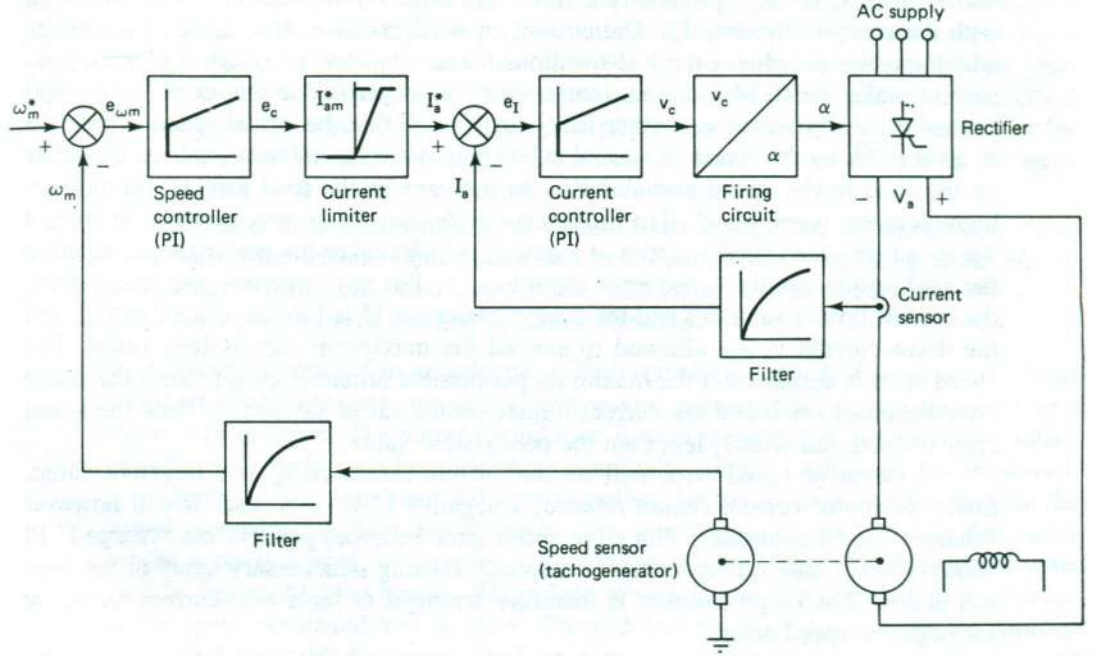
5.1 SINGLE-QUADRANT VARIABLE-SPEED DRIVES

5.1.1 Armature Voltage Control at Constant Field

A basic scheme of the closed-loop speed control system employing current limit control, also known as parallel current control, is shown in figure 5.1a. ω_m^* sets the speed reference. A signal proportional to the motor speed is obtained from the speed sensor. The speed sensor output is filtered to remove the ac ripple and compared with the speed reference. The speed error is processed through a speed controller. The output of the speed controller v_c adjusts the rectifier firing angle α to make the actual speed close to the reference speed. The speed controller is usually a PI (proportional and integral) controller and serves three purposes — stabilizes the drive and adjusts the damping ratio at the desired value, makes the steady-state speed-error



(a) Drive with current limit control



(b) Drive with inner current control loop

Figure 5.1 One quadrant closed-loop speed control.

close to zero by integral action, and filters out noise again due to the integral action. In closed-loop control systems PD (proportional and differential) and PID (proportional, integral, and differential) controllers are often used. But they are not preferred in converter drives because of the presence of substantial noise and ripple in the current and speed feedback signals.

The drive employs current limit control, the operation of which has been explained in section 3.9. As long as $I_a < I_x$, where I_x is the maximum permissible value of I_a, the current control loop does not affect the drive operation. If I_a exceeds I_x, even by a small amount, a large output signal is produced by the threshold circuit, the current control overrides the speed control, and the speed error is corrected essentially at a constant current equal to the maximum permissible value. When the speed reaches close to the desired value, I_a falls below I_x, the current control goes out of action and speed control takes over. Thus in this scheme, at any given time the operation of the drive is mainly controlled either by the speed control loop or the current control loop, and hence it is also called parallel current control.

Another scheme of closed-loop speed control is shown in figure 5.1b. It employs an inner current control loop within an outer speed loop. The speed loop is essentially the same as just described for the current limit control. The operation of the inner current control loop is explained in section 3.9. The speed error is processed through a PI controller which serves the same three purposes just described. The output of the speed controller e_c is applied to a current limiter which sets the current reference I_a^* for the inner current control loop. The armature current I_a is sensed by a current sensor, filtered, preferably by an active filter to remove ripple, and compared with the current reference I_a^*. The current error is processed through a PI controller which enables to achieve the just-mentioned three objectives, though it is not necessary to make the steady-state current error close to zero. The output of the current controller v_c adjusts the converter firing angle such that the actual speed is brought to a value set by the speed command ω_m^*. Any positive speed error, caused by either an increase in the speed command or an increase in the load torque, produces a higher current reference I_a^*. The motor accelerates due to an increase in I_a, to correct the speed error and finally settles at a new I_a^* which makes the motor torque equal to the load torque and the speed error close to zero. For any large positive speed error, the current limiter saturates and the current reference I_a^* is limited to a value I_{am}^*, and the drive current is not allowed to exceed the maximum permissible value. The speed error is corrected at the maximum permissible armature current until the speed error becomes small and the current limiter comes out of saturation. Now the speed error is corrected with I_a less than the permissible value.

A negative speed error will set the current reference I_a^* at a negative value. Since the motor current cannot reverse, a negative I_a^* is of no use. It will however "charge" the PI controller. When the speed error becomes positive the "charged" PI controller will take a longer time to respond, causing unnecessary delay in the control action. The current limiter is therefore arranged to set a zero-current reference for negative speed errors.

Since the speed control loop and the current control loop are in cascade, the inner current control is also known as cascade control. It is also called current

guided control. It is more commonly used than the current-limit control because of the following advantages:

1. It provides faster response to any supply voltage disturbance. This can be explained by considering the response of the two drives to a decrease in the supply voltage. A decrease in the supply voltage reduces the motor current and torque. In the current-limit control, the speed falls because the motor torque is less than the load torque which has not changed. The resulting speed error is brought to the original value by setting the rectifier firing angle at a lower value. The response of the drive is mainly governed by its mechanical time constant. In the case of inner current control, the decrease in motor current, due to the decrease in the supply voltage, produces a current error which changes the rectifier firing angle to bring the armature current back to the original value. The transient response is now governed by the electrical time constant of the motor. Since the electrical time constant of a drive is much smaller compared to the mechanical time constant, the inner current control provides a faster response to the supply voltage disturbances.
2. As explained later, for certain firing schemes, the rectifier and the control circuit together have a constant gain under continuous conduction. The drive is designed for this gain to set the damping ratio at 0.707, which gives an overshoot of 5 percent. Under discontinuous conduction, the gain reduces. The higher the reduction is in the conduction angle, the greater the reduction is in the gain. The drive response becomes sluggish in discontinuous conduction and progressively deteriorates as the conduction angle reduces. If an attempt is made to design the drive for discontinuous conduction operation, the drive is likely to be oscillatory or even unstable for continuous conduction. The inner current control loop provides a closed loop around the rectifier and the control circuit, and therefore, the variation of their gain has much less affect on the drive performance. Hence, the transient response of the drive with the inner current loop is superior to that with the current-limit control.
3. In the current-limit control, the current must first exceed the permissible value before the current-limit action can be initiated. Since the firing angle can be changed only at discrete intervals, substantial current overshoot can occur before the current limiting becomes effective.

Small motors are more tolerant to high transient currents. Therefore, to obtain a fast transient response, much higher transient currents are allowed by selecting a larger size rectifier. The current regulation is then needed only for abnormal values of current. In such cases because of the simplicity, current-limit control is employed.

Both the schemes have different responses for the increase and decrease in the speed command. A decrease in speed command at the most can make the motor torque zero; it cannot be reversed as braking is not possible. The drive decelerates mainly due to the load torque. When load torque is low, the response to a decrease in the speed command will be slow. These drives are therefore suitable for applications with large load torques, such as paper and printing machines, pumps, and blowers.

5.1.2 Field Weakening

The schemes of figure 5.1 can provide speed control up to base speed. For speed control above base speed, field control must be combined with aramature voltage control. Preferably, speed control from zero to base speed should be done at the maximum field by armature voltage control, and control above base speed should be done by field weakening at the rated armature voltage. This strategy can be approximately implemented using the scheme shown in figure 5.2. This is an inner current control scheme with an additional loop for the field control. The field current is controlled by a controlled rectifier.

In the field control loop, the back emf E ($= V_a - I_aR_a$) is compared with a reference voltage E* which is chosen to be between 0.85 to 0.95 of the rated armature voltage. The higher value is used for motors with a low armature circuit resistance. For speeds below base speed, the field controller saturates due to a large value of

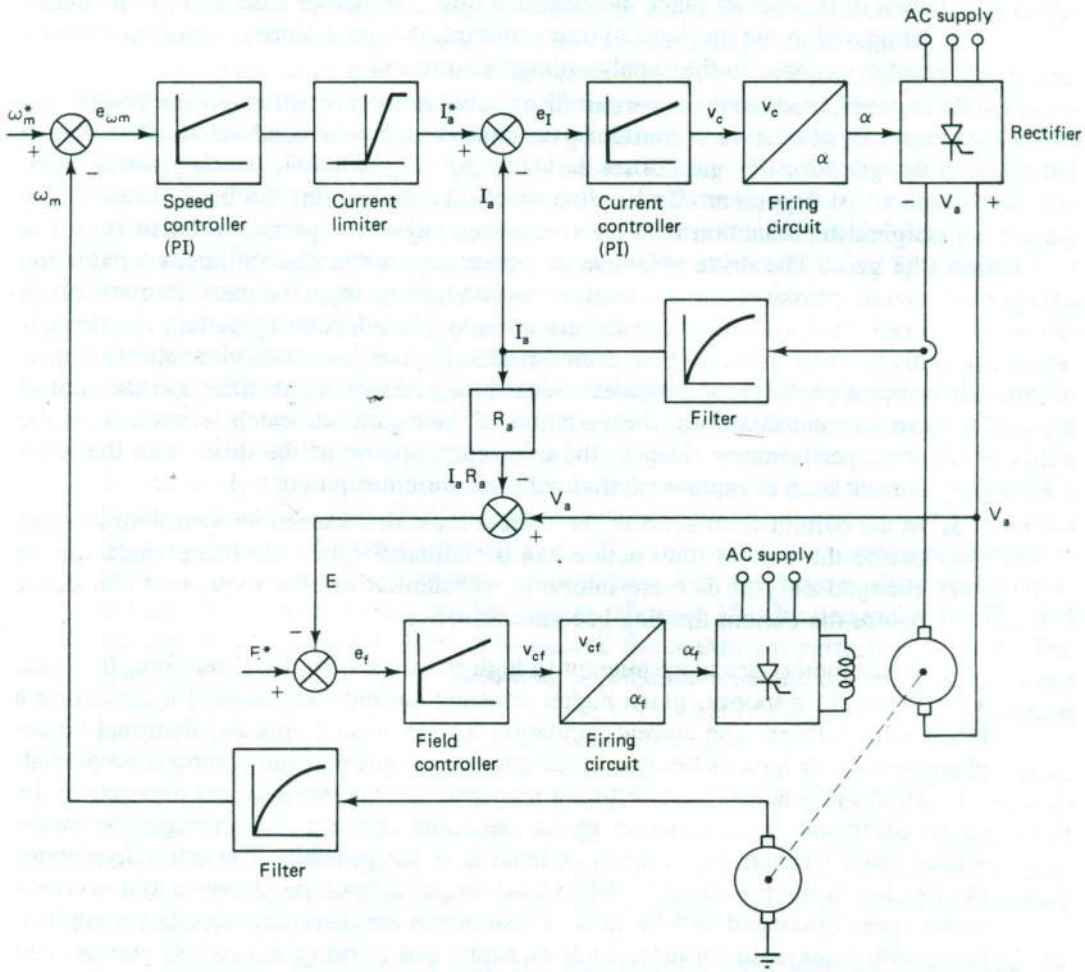


Figure 5.2 Closed loop armature control with field weaking.

error e_f, applying the rated voltage across the field. This ensures the maximum field current for motor operation below base speed. When close to base speed, the field controller comes out of saturation. Now if the reference speed ω_m^* is set for a speed above base speed, a positive speed error $e_{\omega m}$ is produced and the current reference I_a^* is set for a higher value. The firing angle of the armature rectifier is reduced to initially increase V_a. The motor accelerates, the back emf E increases, and the field control loop error e_f decreases, decreasing the field current. The motor speed continues to increase, in the process decreasing the field current until the motor speed is set at the value demanded by ω_m^*. Since the speed error $e_{\omega m}$ will now be small, V_a will return to a value close to the original value. Thus, speed control above base speed will be obtained by the field weakening with the armature terminal voltage maintained near the rated value.

In the field weakening region, the drive responds very slowly due to the large field time constant. Field forcing is sometimes used to improve the transient response, but then the control becomes complex. One can use a half-controlled rectifier but a fully controlled rectifier is usually preferred. Due to the ability to reverse the voltage, a fully controlled rectifier can reduce the field current much faster than a half-controlled rectifier.

5.1.3 Details of Various Blocks of Closed-Loop Drives

Details of various blocks of figures 5.1 and 5.2 are described next.

Speed Sensing

Two methods are used for speed sensing: induced voltage sensing and the use of tachometers. Speed is proportional to the back emf at a constant field. Therefore, if field control is not used, speed can be sensed by measuring the back emf ($= V_a - I_a R_a$). The accuracy of measurement is affected by the difficulty in sensing I_a accurately due to the presence of ripple, the variation of flux due to the field supply disturbance, and the variation of temperatures of the field and armature windings. The method is inexpensive and provides speed measurement with an accuracy of ± 2 percent of base speed.

More accurate speed regulation is achieved by using a tachometer driven from the motor shaft. A tachometer is an ac or dc generator with a high order of linearity between its speed and output voltage. For dc drives, dc tachometers are usually used. A dc tachometer is built with a permanent magnetic field and sometimes with silver brushes to reduce the contact drop between the brush and commutator. Typical voltage outputs are 10 V per 1000 rpm. The tachometer output voltage consists of a ripple whose frequency depends on its speed. At low speeds, adequate filtering can only be done by a filter with a large enough time constant to affect the dynamics of the drive. Special large diameter tachometers with a large number of commutator segments are sometimes built to overcome this problem. Tachometers are available to measure speed with an accuracy of ± 0.1 percent. The tachometer should be coupled to the motor with a torsionally stiff coupling so that the natural frequency of the system consisting of the armatures of the motor and tachometer lies well beyond the bandwidth of the speed control loop.[5,6] When very high speed accuracies are required, as in computer peripherals, paper mills and so on, digital tachometers are

used. A digital tachometer employs a shaft encoder which gives a frequency proportional to the motor speed. The encoder consists of a transparent plastic or aluminum disc mechanically coupled to the motor shaft. The transparent plastic disc is alternately painted black on its periphery to provide alternately transparent and nontransparent parts. In an aluminum disc, a number of holes or slots are uniformly made around its periphery. An opto-coupler unit, consisting of a light source and a light sensor, is so mounted that the disc will run between the light source and the sensor. The sensor senses the light source whenever a transparent part/slot/hole crosses the opto-coupler and a voltage pulse is produced. The frequency of the pluse train is proportional to the speed of the shaft.

Current Sensing

To avoid interaction between the control circuit, carrying low voltage and current, and the power circuit, involving high voltage, high current, and a substantial amount of harmonics, isolation must be provided between the two circuits. Therefore, except in low-voltage converters, the current sensor should also provide isolation.

The currents in the ac lines of a rectifier carry information on the dc side armature current when freewheeling is not present. The rectified output of the current transformers, with their primaries connected in the ac lines, then yields a signal proportional to the armature current. One single-phase current tranformer is needed for a single-phase rectifier. A three-phase current transformer is preferred for a three-phase rectifier, though one single-phase transformer can also be used. The frequency of ripple in the rectified current of a three-phase transformer is three times that of a single transformer. Because of the higher frequency of the ripple, the filter time constant can be lower, providing fast response of the current control loop. A circuit using a three-phase current transformer is shown in figure 5.3. The major limitations of this scheme are that it cannot sense the current direction and cannot be used for rectifiers employing free-wheeling. The scheme is widely used due to its low cost, simplicity, and reliability.

A number of methods are available for the direct sensing of armature current. Two commonly used methods are described here. The first method involves the use of a current sensor employing Hall effect. It also has the ability to sense the current direction. It is commercially available for a wide range of currents (a few amperes to

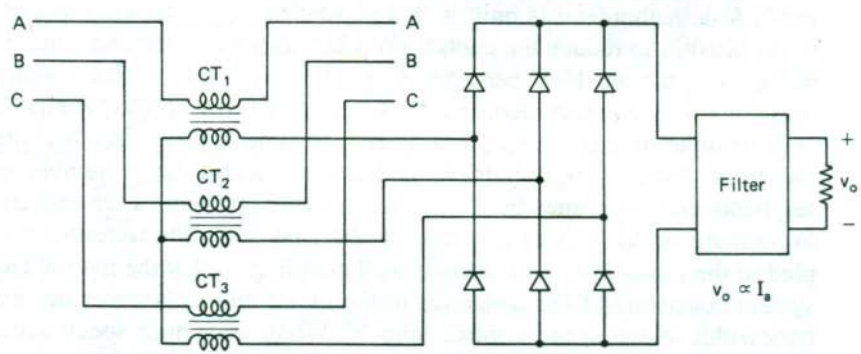


Figure 5.3 Current sensing in a 3-phase fully-controlled rectifier.

several hundred amperes) with a typical accuracy of one percent up to 400 Hz. The second method involves the use of a noninductive resistance shunt in conjunction with an isolation amplifier which has an arrangement for amplification and isolation between the power and the control circuits. The main limitation of the shunt is that it provides only a small output voltage of the order of 7.5 mV to 75 mV at the rated current. The use of shunts of higher resistance results in the increased power dissipation and drift of resistance with temperature. In the current control loop of a variable speed drive, accurate sensing of current is not necessary, and, therefore, the drop across the interpole winding is often used for current sensing. The isolation amplifier may consist of any one of the following circuits.

The voltage drop across the shunt is filtered, amplified, modulated, and then applied to the primary of an isolation transformer. The output of the transformer is demodulated by a phase sensitive demodulator, filtered, buffered, and applied to output terminals. The method also allows the sensing of the current direction. In the alternative scheme, the shunt voltage drop is filtered, amplified, and then processed through an opto-isolator. The opto-isolator output is buffered and then brought to the output terminals. Since the opto-isolator gain is temperature dependent and nonlinear, two identical opto-isolators are employed in a feedback loop to compensate for these nonlinearities.[4,7]

The direct sensing of the armature current using a shunt is fast compared to the indirect sensing involving current transformers. However, it is more expensive.

PI Controller

The error detector, PI controller, and limiter are combined in a single-circuit as shown in figure 5.4. Diode D_1 and zener diode D_{z1} provide limitation on the maximum positive voltage, and diode D_2 and zener diode D_{z2} provide limitation on the maximum negative voltage. When this circuit forms a part of the inner current control loop, these limitations are used to restrict the magnitudes of the control voltage v_c and thus provide restriction on the range of firing angle as explained in the next section. When employed in the speed loop, they restrict the maximum positive and negative values of current reference I_a^*. In a single-quadrant drive negative current reference is not required and hence diode D_2 alone may be used instead of D_{z2} and D_2.

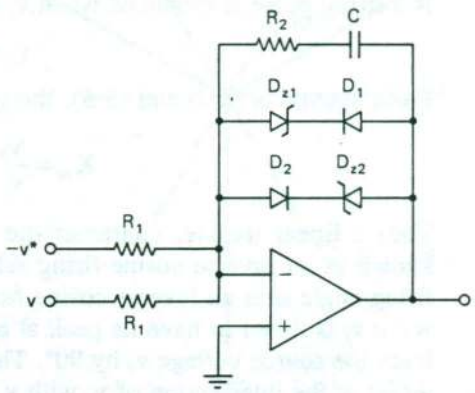


Figure 5.4 PI controller with error detector and limiter; v* and v are reference and feedback signals respectively.

The transfer function of the circuit in the linear region of its operation is given by

$$G_c(s) = \frac{K_c(1 + s\tau_c)}{s} \tag{5.1}$$

where

$$\tau_c = CR_2 \quad \text{and} \quad K_c = \frac{1}{R_1 C} \tag{5.2}$$

Transfer Characteristic of Rectifier and Control Circuit

The firing of a rectifier is a discrete process. After the need for the change of the rectifier firing angle has been assessed, a 3-phase fully controlled rectifier fed by a 50 Hz source may take from 0 to 3.33 ms (the time interval between two consecutive firing instants) before the firing angle can be changed. Since the mechanical time constant of the motor is much larger compared to this delay, the delay is ignored and the firing angle change is considered instantaneous. With this approximation the rectifier can be modeled simply as a gain element. This approximate model is found adequate when the aim is to design an adequately damped system. However, it is not valid close to the stability limit. An improved, but again approximate model is obtained by adding a term $1/(1 + s\tau_d)$ to the gain, where τ_d is the average delay which is 1.67 ms for a 3-phase fully controlled rectifier and 5 ms for a 1-phase rectifier when they are fed by a 50 Hz source.

The transfer characteristic of a control unit is often selected to match the transfer characteristic of the converter. It is therefore useful to consider the transfer characteristic of the combination.

For the conventional operation of 3-phase and 1-phase rectifiers under continuous conduction, from equations (3.16) and (3.78),

$$V_a = V_{ao} \cos \alpha \tag{5.3}$$

Let us generate a reference timing wave given by the following equation:

$$v_r = V_{rm} \cos \alpha \tag{5.4}$$

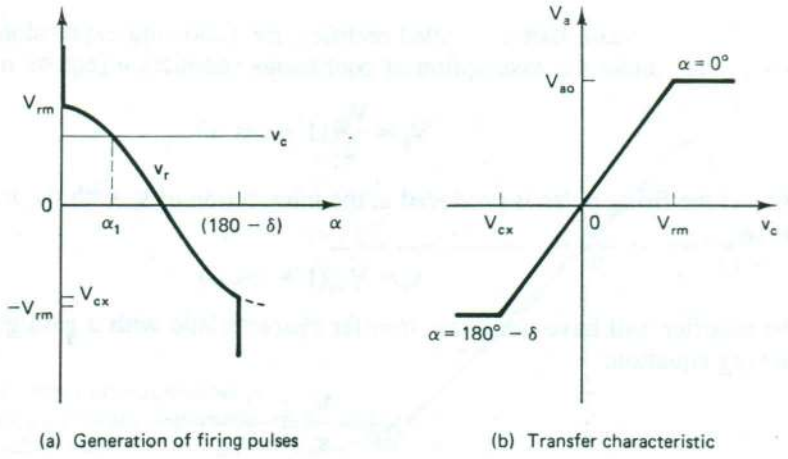
If a firing pulse is produced when $v_c = v_r$ (fig. 5.5a) then,

$$v_c = V_{rm} \cos \alpha \tag{5.5}$$

From equations (5.3) and (5.5), the gain of the combination K_A is given by

$$K_A = \frac{V_a}{v_c} = \frac{V_{ao}}{V_{rm}} = \text{constant} \tag{5.6}$$

Thus a linear transfer characteristic as shown in figure 5.5b is obtained. This is known as an inverse cosine firing scheme because according to equation (5.5), the firing angle α is an inverse cosine function of the control voltage v_c. The reference wave v_r is timed to have its peak at $\alpha = 0$. For a single-phase rectifier (fig. 3.7), v_r leads the source voltage v_s by 90°. The firing pulses for thyristors T_1 and T_3 are produced at the intersection of v_r with v_c, and the firing pulses for thyristors T_2 and T_4



(a) Generation of firing pulses (b) Transfer characteristic

Figure 5.5 Inverse cosine firing.

are produced at the intersection of $-v_r$ with v_c. For a three-phase converter, (fig. 3.16) v_r is timed to have its peak at $\pi/3$—that is, at the instant for which $\alpha = 0$. The phasor diagram of figure 5.6 shows that the phase voltage $(-V_B)$ has the required phase. v_r can therefore be obtained from $(-V_B)$. Thyristor T_1 is then fired at the intersection of this v_r and v_c. v_r's for thyristors T_2, T_3, T_4, T_5, and T_6, which are fired in the sequence of their numbers with a phase difference of 60°, can be obtained from the phase voltages V_A, $(-V_C)$, V_B, $(-V_A)$ and V_C, respectively.

To ensure the firing of thyristors, v_c should always be less than V_{rm}. Further, the maximum value of α should be restricted to some suitable value 180-δ, where δ is a positive angle required for the commutation. These restrictions are implemented by restricting the output voltage of the PI controller of figure 5.4 with the help of the zener diodes D_{z1} and D_{z2}. Alternatively, one can superimpose sharp narrow pulses (fig. 5.5) on v_r to satisfy these restrictions. These pulses, commonly known as "end-stop" pulses, also ensure firing under the supply voltage dips.

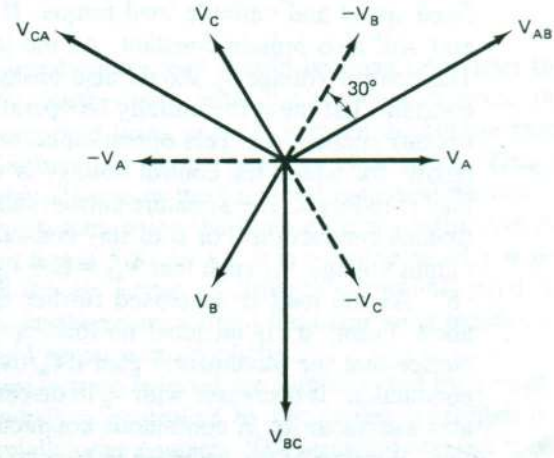


Figure 5.6 Phasor diagram of 3-phase source voltages.

For a 1-phase half-controlled rectifier, the following expression gives the output voltage under the assumption of continuous conduction [equation (3.57)]

$$V_a = \frac{V_{ao}}{2}(1 + \cos \alpha) \tag{5.7}$$

Now if the firing pulse is produced at the intersection of v_c with the following timing wave

$$v_r = V_{rm}(1 + \cos \alpha) \tag{5.8}$$

the rectifier will have the linear transfer characteristic with a gain given by the following equation:

$$K'_A = \frac{V_a}{v_c} = \frac{V_{ao}}{2V_{rm}} \tag{5.9}$$

For the 1-phase fully controlled rectifier with the controlled flywheeling and 3-phase fully controlled rectifier with the freewheeling diodes or with the controlled flywheeling, the linear transfer characteristics cannot be obtained because of different relations between V_a and α for the different ranges of α. In such cases the α versus V_a relations can be approximated by a suitable straight line.[3] The slope of the straight line will then be the gain of the rectifier. Such an approximation is acceptable for designing the drive with adequate damping.

Sometimes firing angles are generated by comparing v_c with a triangular ramp synchronized with the supply voltage. Here α is proportional to v_c, and, therefore, the output voltage is a cosine function of v_c. The incremental gain of the rectifier dV_a/dv_c is then a sine function of v_c. In such situations the rectifier gain is assumed equal to the average of its maximum and minimum values.[8]

Effect of Discontinuous Conduction on the Transfer Characteristics of Rectifiers. When a fully controlled rectifier is operated with inverse cosine firing, a linear transfer characteristic shown in figure 5.5b is obtained under the continuous conduction. This characteristic is modified considerably by discontinuous conduction as explained in figure 5.7. The figure shows the operation of the drive for a fixed speed and variable load torque. If the field is maintained constant, the back emf will also remain constant. As the load on the drive changes, I_a must change. The control voltage v_c should also change, to change V_a such that $V_a - I_aR_a = E =$ constant. Let the drive initially be operating with a load large enough to ensure continuous conduction. This operation corresponds to point "a" on the transfer characteristic for which the control voltage is v_{c1} and the armature current is I_{a1}. Now the load is reduced. The armature current falls to I_{a2} and the drive operates under discontinuous conduction. For E to stay constant, v_c must change to v_{c2} to get the rectifier output voltage V_{a2} such that $V_{a2} = E + I_{a2}R_a$. The operation now takes place at point "b". As the load is decreased further the operating point moves along the curve abcd. Point "d" is an ideal no-load operating point for which $I_a = 0$ and $V_a = E$. Notice that the incremental gain dV_a/dv_c is constant and the highest in continuous conduction. It decreases with v_c in discontinuous conduction. The drive which operates satisfactorily in continuous conduction fails to do so in discontinuous conduction. Because of the decrease in incremental gain, the transient response to a change

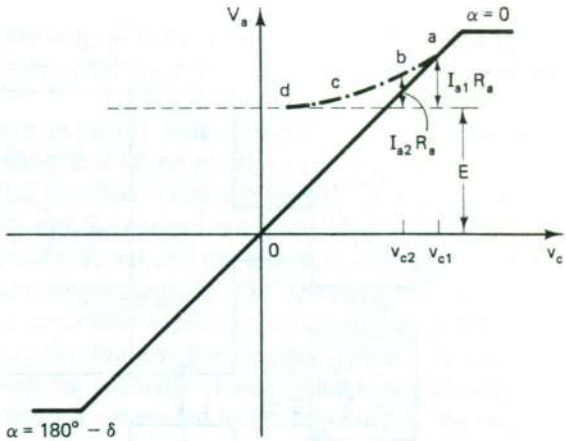


Figure 5.7 Effect of discontinuous conduction on the transfer characteristic of fully-controlled rectifier.

in the speed command or the load disturbance becomes sluggish and the steady-state error becomes large. The effect of discontinuous conduction on the drive performance can be reduced by designing the current control loop with a high gain; however, due to the presence of substantial noise, the gain cannot be made too high without impairing the stability and the dynamic response. Alternatively, an additional loop with the rectifier output voltage feedback can be introduced within the current control loop.[9,10] A number of other methods can also be used, such as the nonlinear controller,[5] dual-mode current controller,[11] feed forward current control,[12] and adaptive current control.[2]

5.2 FOUR-QUADRANT VARIABLE-SPEED DRIVES

Various methods of multiquadrant operation of fully controlled rectifier-fed dc drives are described in section 3.10. The reader may wish to review section 3.10 before the present one.

5.2.1 Drive Employing Armature Reversal by a Contactor

As explained in section 3.10, the armature reversal should be done only after the armature current has ceased to flow. Furthermore, after the armature reversal, the rectifier must be activated at an appropriate firing angle to prevent the current from shooting up to a large value and subjecting the drive to shock loading. This is achieved by either the advanced firing scheme or the back emf matching method.

A four-quandrant speed control system using the inner-current control and the advanced firing scheme is shown in figure 5.8. As the fast response usually is not the primary consideration for such drives, advanced firing is commonly used, in spite of the resulting slow response. Furthermore, current limit control is more commonly used than inner-current control because of simplicity.

The steps required for armature current reversal are implemented by a master controller. A reversing contactor, which is controlled by the master controller, has three normally closed and three normally open contacts. The armature reversal is ini-

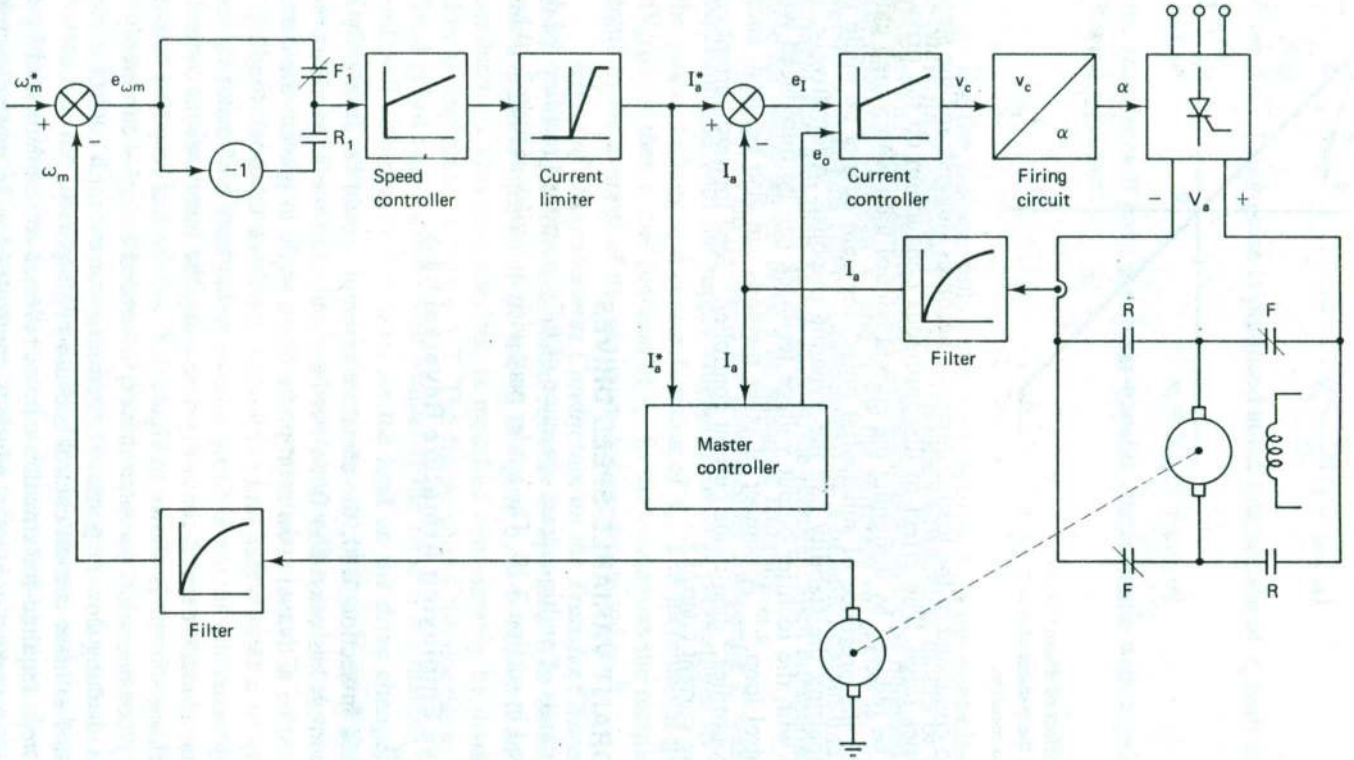


Figure 5.8 Four quadrant closed-loop speed control with armature reversal.

tiated by the master controller with the help of the contactor whenever I_a^* and I_a become zero simultaneously. The master controller also implements the advanced firing scheme.

Let the drive be initially running in steady state in the forward direction with the contactor in the off position. Both I_a^* and I_a are positive, and the errors $e_{\omega m}$ and e_I are close to zero because of the PI controllers. Now the speed command ω_m^* is reduced, which makes $e_{\omega m}$ negative and sets the current reference $I_a^* = 0$. As e_I is negative now, the rectifier output voltage is reduced and the armature current is reduced to zero. Now both I_a^* and I_a are zero simultaneously. On detecting this condition (that is, $I_a^* = 0$ and $I_a = 0$) the master controller applies a sufficiently large signal e_0 to set the converter firing angle α at the highest permissible value and simultaneously operates the contactor, opening the normally closed contacts and closing the normally open contacts. The converter gets connected to the armature at the highest firing angle. The signal e_0 is now slowly reduced to zero. The armature current builds up slowly and a smooth transition into braking occurs. The transfer of connection from F_1 to R_1 sets I_a^* at a positive value. The drive decelerates under the current control. When ω_m becomes less than ω_m^*, I_a^* becomes zero again and I_a is forced to zero. Since I_a^* and I_a are zero simultaneously, the master controller applies e_0 again to set α at the highest value. It also opens the contactor, thus, reconnecting the rectifier to the armature through the normally closed contacts F and setting I_a^* to a positive value. e_0 is now slowly reduced to zero. The motor current builds up and the drive settles at the speed for which $e_{\omega m} = 0$.

One can anticipate three types of reference speed settings when considering the armature reversal. One kind of setting is the reduction of speed in the same direction. The operation of the drive has already been considered for this kind of speed setting. It requires two armature reversals—one for shifting the operation to braking and another back to motoring. In the second type of setting, the reference speed is set for speed reversal. In this case only one armature reversal is required; after the reversal the drive is initially braked and then accelerated in the reverse direction under motoring to a new speed. The third possible setting is when the reference speed is set for a higher value in the same direction. In this case no armature reversal is necessary.

5.2.2 Drive Employing a Dual-Converter with Nonsimultaneous Control

A number of closed-loop schemes employing a dual-converter with nonsimultaneous control are possible. They may differ in the following respects:

1. Each rectifier may have a separate firing circuit, or a single firing circuit may be used for both rectifiers.
2. The current control may be done by the inner-current control loop or the current-limit control.
3. The rectifier switch-in can be done either by the advanced firing scheme or the back emf matching method.
4. The state of the outgoing rectifier respecting turn-off may be sensed either indirectly by sensing zero current or directly by sensing the state of the thyristors.

A scheme for a particular application is selected depending on its requirements related to the speed of response. A high-performance drive will employ the inner-current control loop, the back emf matching method, and the direct sensing of the state of the thyristors. For slower schemes other combinations may be chosen to reduce the complexity.

A scheme employing separate firing circuits, inner-current control loop, back emf matching method, and zero-current sensing is described here (fig. 5.9).

The inverse cosine firing, described in section 5.1.3, is used. The firing pulses for the thyristors of rectifier 1 are produced at the intersection of the control voltage v_c with the reference voltages v_{r1} — v_{r6} (solid lines) obtained from the phase voltages $(-V_B)$, V_A, $(-V_C)$, V_B, $(-V_A)$ and V_C as shown in figure 5.10. The firing pulses for the thyristors of rectifier 2 are obtained at the intersection of v_c with reference voltages v'_{r1}. . . . v'_{r6} (dotted lines), which are obtained from the phase voltages which are 180° out of phase from the phase voltages used for rectifier 1 — that is, they are obtained from the phase voltages V_B, $(-V_A)$, V_C, $(-V_B)$, V_A, and $(-V_C)$. The transfer characteristics of the two rectifiers are shown in figure 5.11.

The master controller (fig. 5.9) performs the following functions:

1. Through the logic variables F and V_{BLOCK} it decides which of the two rectifiers must receive the firing pulses. When both F and V_{BLOCK} are 1, rectifier 1 receives the firing pulses. When both $\overline{F}$ and V_{BLOCK} are 1, rectifier 2 receives the firing pulses. When V_{BLOCK} is 0, none of the rectifiers receives the firing pulses.
2. It senses the polarity of I_a^* and current zero ($I_a = 0$). If I_a^* is negative and $I_a = 0$, it initiates the process of switch-over from rectifier 1 to rectifier 2. On the other hand, if I_a^* is positive and $I_a = 0$, it initiates the process of switch-over from rectifier 2 to rectifier 1.
3. It also implements the process of switch-over. By setting V_{BLOCK} to 0, it withdraws the firing pulses from the outgoing rectifier. After a fixed delay τ (2 to 10 ms), it releases the firing pulses to the incoming rectifier by setting V_{BLOCK} back to 1 and F to the appropriate value.

v_{co}-control and the associated connections, shown by the dotted lines, are mainly to implement the back emf matching method. It is called v_{co}-control because it sets the initial value of v_c for the firing circuit of the incoming rectifier to make its output voltage under continuous conduction equal to the back emf. It is also called the initialization of the PI controller. It is controlled by the master controller through the logic variables F and $\overline{F}$. v_{co}-control operates on the current controller 2 when $\overline{F}$ is 1 and on the current controller 1 when F is 1. Examination of function 1 of the master controller, just stated, shows that v_{co}-control always operates on the idle rectifier. A reference signal v_R — obtained from the tachometer and related to E by the same proportionality constant as v_c with V_a under continuous conduction — is compared with the actual output voltage of the current controller of the idle rectifier. The error is amplified and fed to the input of the current controller to force its output voltage to v_R. Thus the output voltage of the current controller of the idle rectifier continuously tracks the motor back emf. When the idle rectifier is switched in, its terminal voltage under continuous conduction will be equal to the back emf. This gives fast switch-over without a current surge.

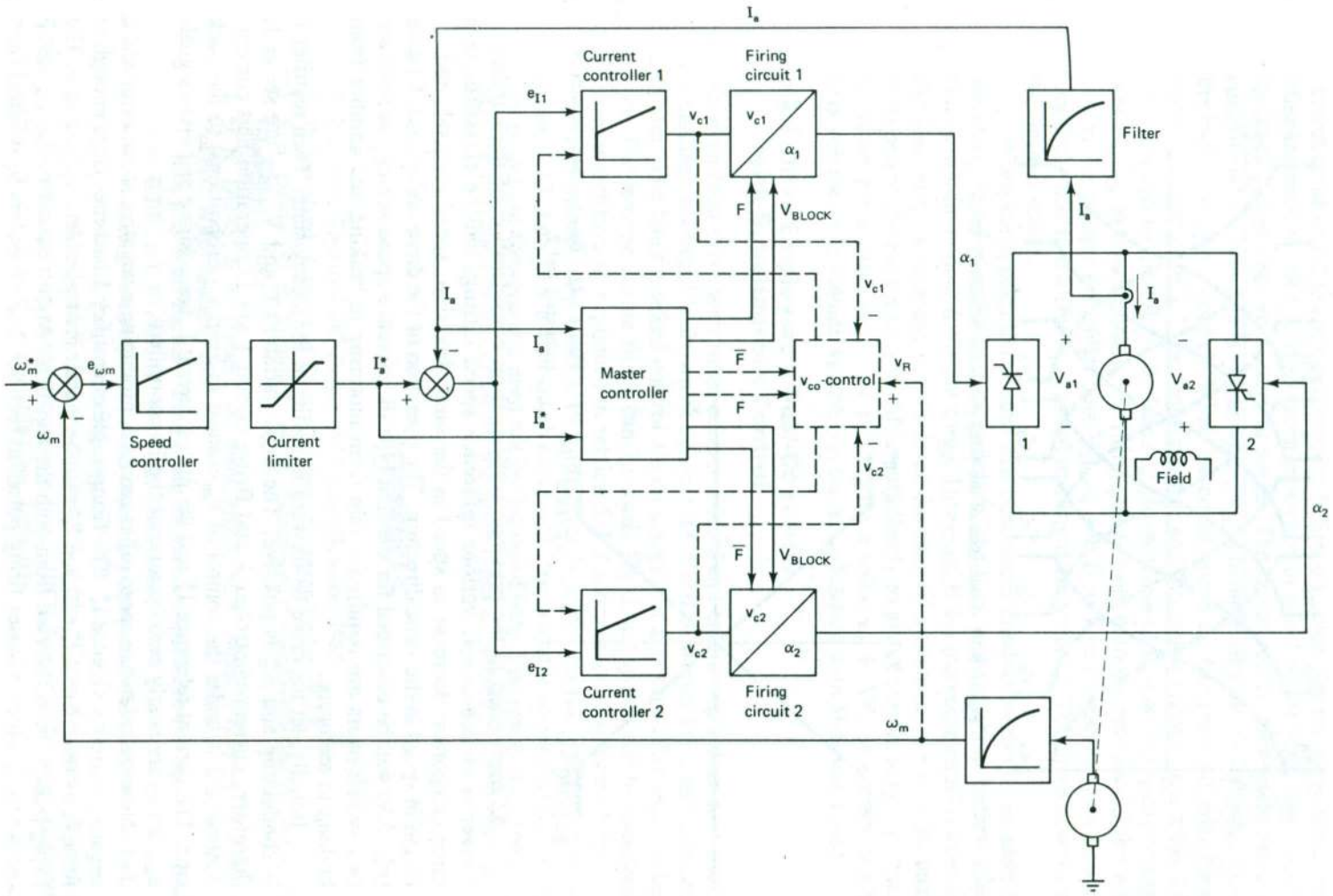


Figure 5.9 Four quadrant closed-loop speed control by dual converter with non-simultaneous control.

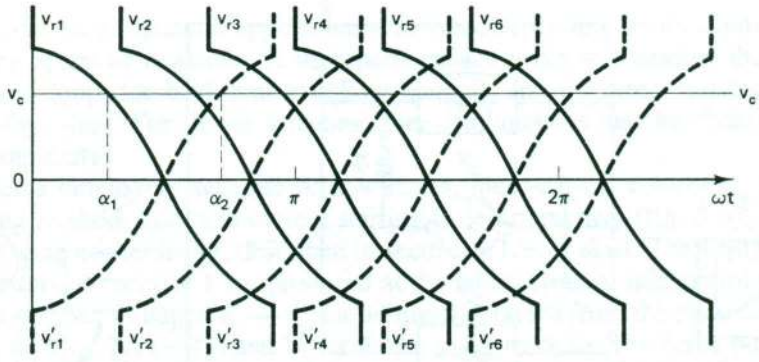


Figure 5.10 Generation of the firing pulses for rectifiers 1 and 2.

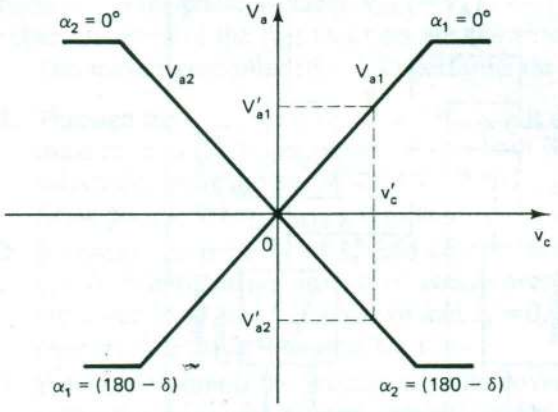


Figure 5.11 Transfer characteristics of rectifiers 1 and 2.

As mentioned in the previous section, from consideration of the number of converter switch-overs, various reference speed settings can be classified into three categories: decrease in speed in the same direction, speed reversal, and increase in speed in the same direction. The operation of the drive under consideration (fig. 5.9) will be described for the first type of reference speed setting. In this case two switch-overs are required—one from motoring to braking and another from braking to motoring.

Initially, let the drive be forward motoring in the steady state. Then rectifier 1 is conducting and ω_m^* is positive. The logic variables F and V_{BLOCK} are set at 1; therefore, current-controller 1 and firing circuit 1 are in operation and current-controller 2 is under the control of v_{co}-control to keep v_{c2} proportional to the back emf. The current reference I_a^* and the motor current I_a are positive and error signals $e_{\omega m}$ are e_I are nearly zero because of the PI controllers.

Now the reference speed ω_m^* is reduced, producing a negative speed error and a negative current reference I_a^*. The firing angle of rectifier 1 becomes large enough to force I_a to zero. Since $I_a^* < 0$ and $I_a = 0$, the master controller sets V_{BLOCK} to 0. The firing pulses are withdrawn from both the rectifiers. After a period τ, V_{BLOCK} and $\overline{F}$ are set to 1, which releases firing pulses to rectifier 2 and transfers v_{co}-control from

current controller 2 to current controller 1. Since v_{c2} was already set to make the rectifier terminal voltage at the instant of switch-in equal to the back emf, rectifier 2 switches in fast without a currrent surge. The motor regenerates under the current control and the speed falls. Meanwhile, v_{c1} is being continuously set proportional to the back emf by v_{co}-control. When the actual speed ω_m becomes less than the reference speed setting ω_m^*, I_a^* becomes positive and the output voltage of rectifier 2 is set large enough to force $I_a = 0$. Since I_a^* is positive and $I_a = 0$, the master controller sets V_{BLOCK} to 0 to withdraw the firing pulses from both the rectifiers. After a duration τ, the master controller sets V_{BLOCK} and F to 1 to release the firing pulses to rectifier 1 and to transfer v_{co}-control to current controller 2. The drive now settles at the desired speed.

When the application is such that continuous conduction can be assured in the incoming rectifier at the time of switch-in, the back emf matching method allows a fast switch-in without a current surge. However, if the incoming rectifier operates in the discontinuous conduction mode at the time of switch-in, the rectifier terminal voltage and the back emf may differ sufficiently to give a current surge. In this case v_{c1} is set for a value $v_R - \Delta V$ and v_{c2} for a value $v_R + \Delta V$, where ΔV is a fixed bias voltage. The voltage v_R still has the same relation with E as just stated.

5.2.3 Drive Employing a Dual-Converter with Simultaneous Control

An open-loop drive with inner-current control and employing a dual-converter with simultaneous control is shown in figure 3.33. Each converter has a separate current-limiter. The block marked control circuit consists of a PI controller and a firing circuit. The inverse cosine firing can be used. The firing pulses can be generated for the two rectifiers as explained in section 5.2.2 and as shown in figure 5.10.

If a speed loop with a PI controller is added to the open-loop drive of figure 3.33, one gets a four-quadrant closed-loop variable-speed drive. The operation is straight-forward and need not be discussed here. The reader may wish to consider the operation of this drive for the three types of reference speed settings.

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6

Induction Motors

Induction motors, particularly squirrel-cage type induction motors, have a number of advantages when compared with dc motors. Some of these are ruggedness; lower maintenance requirements; better reliability; lower cost, weight, volume, and inertia; higher efficiency; and the ability to operate in dirty and explosive environments. The major drawback of dc motors is the presence of commutators and brushes, which require frequent maintenance and make the motor unsuitable for explosive and dirty environments. Because of their advantages, induction motors are more widely used than all the rest of the electric motors put together. However, until recently, they have been mainly used in applications requiring constant speed. Variable speed applications have been dominated by dc motors. This can be attributed to the fact that conventional methods of speed control of induction motors have been either expensive or highly inefficient.

With the improvement in capabilities and the reduction in the cost of thyristors and more recently of power transistors and gate-turn-off thyristors, it has become possible to build variable speed induction motor drives which can match and in some cases even surpass dc drives in performance and cost. As a result of this development, induction motor drives have succeeded in replacing dc drives in a number of variable speed applications. It is projected that in the future induction motors will be widely used in variable speed drives.

6.1 PERFORMANCE OF 3-PHASE INDUCTION MOTORS

A 3-phase induction motor consists of a balanced three-phase winding on the stator. The rotor of a wound-rotor type induction motor consists of a balanced three-phase winding of the same number of poles as that of the stator winding. The rotor of a

squirrel-cage type induction motor has conductors shorted by end rings. By induction, the same number of phases and poles are produced by the squirrel-cage rotor as in the stator winding. When the stator is supplied by a balanced three-phase ac source of frequency ω radians per second (or f Hz), a rotating field moving at a synchronous speed ω_{ms} radians per sec. is produced. Where

$$\omega_{ms} = \frac{2}{p}\omega = \frac{4\pi f}{p} \text{ rad/sec.} \quad \text{and} \quad p = \text{number of poles.} \tag{6.1}$$

If the rotor speed is ω_m rad/sec then the relative speed between the stator rotating field and the rotor is given by

$$\omega_{s\ell} = \omega_{ms} - \omega_m = s\omega_{ms} \tag{6.2}$$

where $\omega_{s\ell}$ is called the slip speed. The parameter s is known as slip and is given by

$$s = \frac{\omega_{ms} - \omega_m}{\omega_{ms}} \tag{6.3}$$

Also

$$\omega_m = (1 - s)\omega_{ms} \tag{6.4}$$

Because of the relative speed between the stator field and the rotor, balanced three-phase voltages are induced in the rotor windings. The frequency of these voltages is proportional to the slip speed. Hence

$$\omega_r = \frac{\omega_{s\ell}}{\omega_{ms}}(\omega) = s\omega \text{ rad/sec.} \tag{6.5}$$

where ω_r is the rotor frequency in rad/sec.

For $\omega_m < \omega_{ms}$, the relative speed is positive. Consequently, the rotor induced voltages have the same phase sequence as the stator voltages. The three-phase currents flowing through the rotor produce a field which moves with respect to the rotor at the slip speed in the same direction as the rotor speed. Consequently, the rotor field moves in space at the same speed as the stator field and a steady torque is produced. The torque is given by the following equation:[1]

$$T = -\frac{\pi}{2}\left(\frac{p}{2}\right)^2 \Phi_{ma} F_{mr} \sin \delta_r \tag{6.6}$$

where Φ_{ma} = air-gap flux per pole, Webers

F_{mr} = peak rotor mmf, Ampere-turns

and δ_r = torque angle or phase angle between the rotor and air-gap mmf's

For any speed $\omega_m < \omega_{ms}$, the rotor and stator fields remain stationary, and a steady torque is produced. For $\omega_m = \omega_{ms}$, the relative speed between the rotor and stator field becomes zero. Consequently, no voltages are induced in the rotor and no torque is produced by the motor.

For $\omega_m > \omega_{ms}$, the relative speed between the stator field and the rotor reverses. Consequently, the rotor induced voltages and currents also reverse and have a phase sequence opposite to that of the stator. The 3-phase rotor currents produce a field

which moves with respect to the rotor at the slip speed in the direction opposite to the rotor speed. Hence, again the rotor field moves in space at the same speed as the stator field and a steady torque is produced. Since the direction of rotor currents has reversed, the developed torque has a negative sign suggesting generator operation. The generator operation is employed to produce regenerative braking.

6.1.1 Steady-State Analysis

A per-phase equivalent circuit of an induction motor is shown in figure 6.1a. a_{T1} is the ratio of the stator turns n_s to rotor turns n_r. The equivalent circuit can be simplified by referring the rotor quantities to the stator frequency and number of turns. The resultant equivalent circuit is shown in figure 6.1b. R_r' and X_r' are the rotor resistance and reactance referred to the stator and are given by the following equations:

$$R_r' = a_{T1}^2 R_r \quad \text{and} \quad X_r' = a_{T1}^2 X_r \tag{6.7}$$

The portion on the left of the dotted line AB in figure 6.1b can be replaced by its Thevenin's equivalent circuit, giving a simplified equivalent circuit shown in figure 6.1c. $V_t\angle\theta_t$, R_t, and X_t are Thevenin's equivalent voltage source, resistance, and reactance, respectively. They are given by the following equations:

$$V_t = \frac{VX_m}{\sqrt{R_s^2 + (X_s + X_m)^2}} \tag{6.8}$$

$$\theta_t = \frac{\pi}{2} - \tan^{-1}\left(\frac{X_s + X_m}{R_s}\right) \tag{6.9}$$

$$R_t + jX_t = \frac{jX_m(R_s + jX_s)}{R_s + j(X_s + X_m)} \tag{6.10}$$

In the equivalent circuit of figure 6.1b, the stator current I_s is related to the equivalent rotor current I_r' by the following equation:

$$\bar{I}_s = \frac{[(R_r'/s) + j(X_r' + X_m)]\bar{I}_r'}{jX_m} \tag{6.11}$$

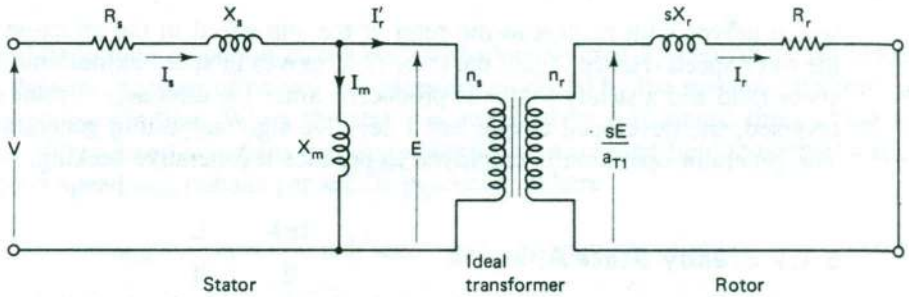
The approximate equivalent circuit of figure 6.1d is often used to simplify calculations. It is based on the assumption of negligible stator impedance drop compared to the terminal voltage.

From the equivalent circuit of figure 6.1c,

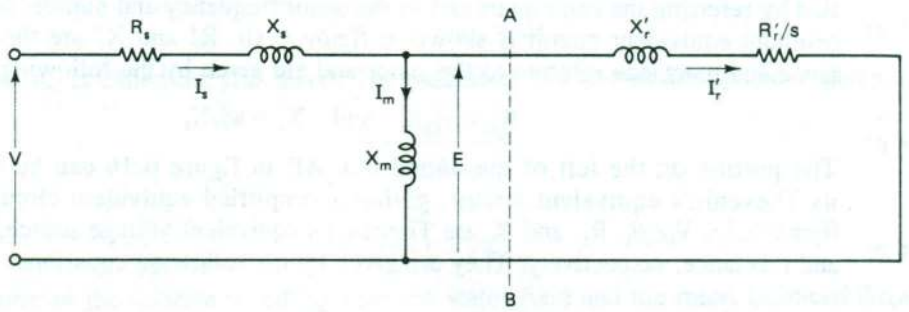
$$\bar{I}_r' = \frac{V_t\angle\theta_t}{\left(R_t + \frac{R_r'}{s}\right) + j(X_t + X_r')} \tag{6.12}$$

The power transferred across the air-gap P_g is equal to the power absorbed in the rotor equivalent resistance R_r'/s. Hence,

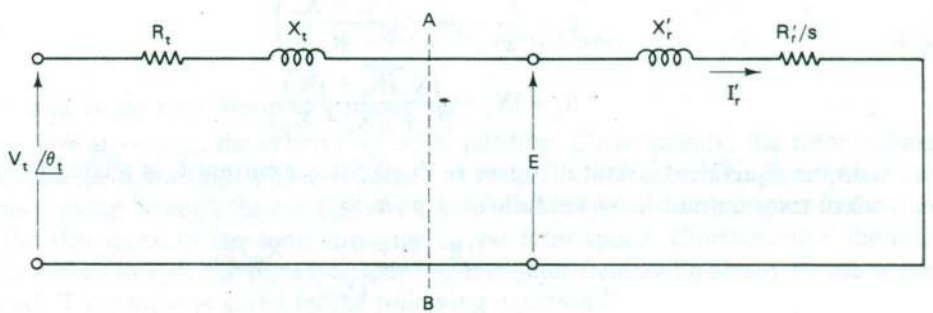
$$P_g = 3I_r'^2\left(\frac{R_r'}{s}\right) \tag{6.13}$$



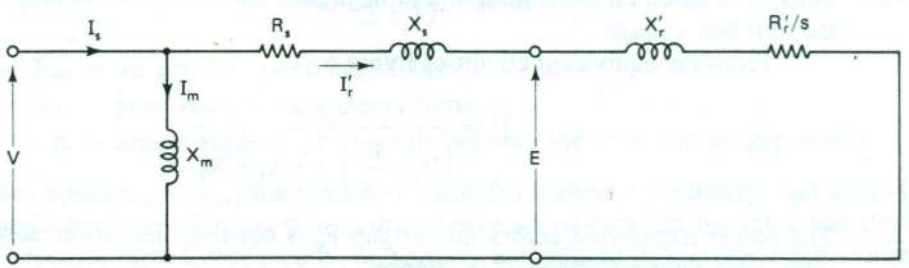
(a) Per-phase equivalent circuit



(b) Per-phase equivalent circuit referred to the stator



(c) Per-phase simplified equivalent circuit referred to the stator



(d) Per-phase approximate equivalent circuit referred to the stator

Figure 6.1 Per phase equivalent circuits of an induction motor.

P_g, which is known as the air-gap power, is also given by the following equation:

$$P_g = 3EI_r' \cos \Theta_r \tag{6.14}$$

where Θ_r is the phase difference between $\bar{E}$ and $\bar{I}_r'$. A portion of the air-gap power is wasted as rotor copper loss, and the remaining power is converted into mechanical power. Hence, mechanical power is given by

$$P_m = P_g - 3I_r'^2 R_r'$$

Substituting from equation (6.13) gives

$$P_m = 3I_r'^2 R_r' \left(\frac{1-s}{s}\right) = P_g(1-s) \tag{6.15}$$

Now the torque developed by the motor is

$$T = \frac{P_m}{\omega_m}$$

Substituting from equations (6.4) and (6.15) gives

$$T = \frac{P_g}{\omega_{ms}} \tag{6.16}$$

Substituting from equation (6.13) for P_g yields

$$T = \frac{3}{\omega_{ms}} I_r'^2 \frac{R_r'}{s} \text{ N-m} \tag{6.17}$$

Substituting for I_r' from equation (6.12) gives

$$T = \frac{3}{\omega_{ms}} \left[\frac{V_t^2 R_r'/s}{\left(R_t + \frac{R_r'}{s}\right)^2 + (X_s + X_r')^2} \right] \text{ N-m} \tag{6.18}$$

The motor output torque at the shaft is obtained by deducting friction, windage, and core loss torques from T. Further, from equations (6.16) and (6.17),

$$sP_g = 3I_r'^2 R_r' = \text{rotor copper loss} \tag{6.19}$$

The term sP_g is known as the slip power, because it is proportional to slip for a given value of P_g. It is the portion of the air-gap power P_g which is not converted into mechanical power. From equation (6.16)

$$T = \frac{\text{slip power}}{s\omega_{ms}} \tag{6.20}$$

Equation (6.20) is useful in deriving an expression for torque in a doubly fed wound-rotor induction motor. A motor is called doubly fed when it is supplied both from stator and rotor. When the motor is supplied from the stator only, according to equation (6.19), the slip power is equal to the rotor copper loss. According to equation (6.18), for given values of the stator voltage and frequency, the developed torque is a function of slip. The slip for the maximum developed torque can be ob-

tained by differentiating T with respect to s and equating it to zero. Alternatively, one can use the following approach.

From equation (6.16), the torque is maximum when the air-gap power P_g is maximum. The air-gap power is the same as the power absorbed by the equivalent rotor resistance R_r'/s. One can consider the rest of the circuit to be a voltage source V_t with an internal impedance equal to $R_t + j(X_t + X_r')$. The power transferred across the air-gap will be maximum at the slip s_m for which R_r'/s_m is equal to the internal impedance. Thus

$$\frac{R_r'}{s_m} = \pm\sqrt{R_t^2 + (X_r' + X_t)^2} \tag{6.21}$$

The maximum torque T_{max} is obtained by substituting from equation (6.21) into equation (6.18). Thus,

$$T_{max} = \frac{3}{2\omega_{ms}} \cdot \frac{V_t^2}{R_t \pm \sqrt{R_t^2 + (X_t + X_r')^2}} \tag{6.22}$$

Equation (6.22) shows that the maximum torque is independent of rotor resistance. However, according to equation (6.21), the slip at the maximum torque changes in proportion to rotor resistance. From equations (6.12) and (6.21) it can be shown that the rotor current at the maximum torque has a unique value, which is independent of rotor resistance.

The phase of the stator current with respect to the source voltage is obtained from equations (6.9), (6.11), and (6.12),

$$\phi_s = -\tan^{-1}\left(\frac{X_s + X_m}{R_s}\right) + \tan^{-1}\left(\frac{X_r' + X_m}{R_r'/s}\right) - \tan^{-1}\left(\frac{X_t + X_r'}{R_t + \dfrac{R_r'}{s}}\right) \tag{6.23}$$

Now,

$$\text{power factor} = \cos \phi_s$$

Equations relevant to the equivalent circuit of figure 6.1d are obtained when V_t, R_t, X_t, and θ_t are replaced by V, R_s, X_s, and 0, respectively in equations (6.12), (6.18), (6.21), and (6.22). The power factor angle is given by the following equation:

$$\phi_s = -\frac{\pi}{2} + \tan^{-1}\left(\frac{X_s + X_r' + X_m}{R_s + \dfrac{R_r'}{s}}\right) - \tan^{-1}\left(\frac{X_s + X_r'}{R_s + \dfrac{R_r'}{s}}\right) \tag{6.24}$$

The nature of the speed-torque and speed-rotor current characteristics of induction motors for the speed range from synchronous to zero speed are shown in figure 6.2. The rotor current which is zero at the synchronous speed increases due to the decrease in R_r'/s as the speed is decreased. The torque also increases due to the increase of the component of the rotor current in phase with the back emf E (or flux), as shown by equations (6.14) and (6.17). The torque reaches the maximum value at the slip s_m. A further decrease in speed decreases the torque in spite of an

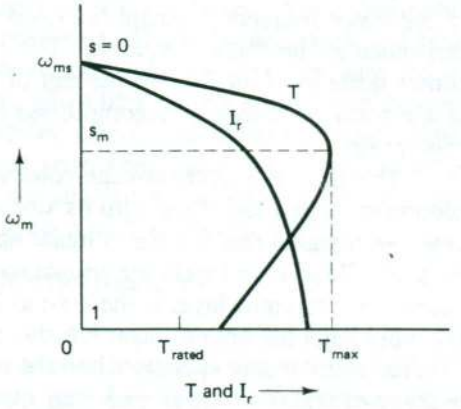


Figure 6.2 Speed-torque and speed-rotor current characteristics of an induction motor.

increase in the rotor current. This can be attributed to the low rotor power factor due to the low value of R_r'/s.

The motor operates stably on the portion of the speed-torque characteristic between $s = 0$ and s_m. With most loads, the operation is unstable for $s > s_m$. Therefore, the portion of the speed-torque curve with $s > s_m$ is often termed a statically unstable part of the speed-torque curves. When the rotor resistance is low, the change of speed from no load to rated torque is only a small percentage of the motor speed. Thus, the motor operates essentially at a constant speed. The short time overloading of the motor can be done up to the maximum torque. If the load torque exceeds the maximum torque — which is also called the breakdown torque — the motor stops and the overload protection must immediately disconnect the source to prevent damage due to overheating.

6.1.2 Squirrel-Cage Induction Motors

In a squirrel-cage motor, various parameters have to be chosen at the design stage. A low rotor resistance is preferred because it gives improved performance during normal running by providing high output power and efficiency and good speed regulation due to the low value of the rated slip. However, the starting performance of such a motor is poor due to the high starting current and low starting torque. An increase in the rotor resistance can improve the starting performance by reducing the starting current and increasing the starting torque, but then the normal running performance deteriorates. The breakdown torque is a measure of the short time overloading capability of the motor. A low rotor leakage reactance gives a high breakdown torque, which is, however, obtained at the expense of the increase in the starting current. Thus the design requirements for normal running and for starting performance are contradictory.

In a squirrel-cage induction motor, a good normal running performance combined with a good starting performance is obtained by using a rotor whose resistance automatically varies with speed, providing high resistance at standstill which progressively decreases as the speed increases and becomes quite small at normal speeds. Such a variation of rotor resistance is realized by making it frequency dependent. At standstill, the rotor frequency is high and equals the stator frequency. The

rotor frequency decreases as the speed increases, reaching a value of 2 to 10 percent of the stator frequency around the rated speed. The rotor is designed to have a high resistance at the stator frequency. The resistance decreases with frequency and becomes quite small at 2 to 10 percent of the stator frequency. This kind of variation of the rotor resistance is accomplished by using either deep-bar or double squirrel-cage rotors.

The deep-bar squirrel-cage rotor employs a deep and narrow bar. The cross section of such a bar along with its slot and the leakage flux is shown in figure 6.3a. One can imagine that the bar is made up of a number of narrow layers connected in parallel. Three such layers are shown crosshatched in the figure. The leakage inductance of the topmost layer is the least as it is linked to the minimum leakage flux. On the other hand the bottom layer has the highest leakage inductance due to the highest flux linkage. At low speeds, when the rotor frequency is high, the difference in the leakage reactance values cause an unequal distribution of currents between the layers — the highest amount of current carried by the top layer and the lowest by the bottom layer. Because of the unequal distribution of the current, the effective rotor resistance increases and the leakage reactance decreases.[1,2] When near the rated speed, the leakage reactances of the layers are low due to the low rotor frequency, and the distribution of current is mainly decided by their dc resistances. Since the dc resistance values are the same, the current gets uniformly distributed across the cross section of the bar, decreasing the effective value of the rotor resistance.

The double squirrel-cage rotor consists of two layers of bars short-circuited by the end rings. The cross section of a slot which consists of a bar of each layer, and the distribution of the leakage flux are shown in figure 6.3b. The top bar is of smaller cross-sectional area than the bottom bar and consequently has a higher resistance. On the other hand, due to the substantial amount of flux crossing the slot between the two bars, the lower bar has much higher leakage inductance than the upper bar. Thus the upper layer has high resistance and low leakage inductance and the lower layer has low resistance and high leakage inductance. At low speeds, for which the rotor frequency is high, the lower layer carries only a small portion of the rotor current. Therefore, the low-speed operation of the motor is governed by the upper layer. When close to the rated speed, the reactances of both the layers are small due to the low rotor frequency and can be neglected. The rotor resistance, which is approximately equal to that of the two layers in parallel, is now very small.

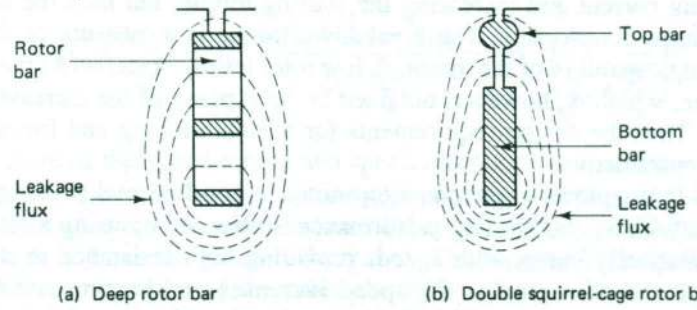


Figure 6.3 Squirrel cage rotors with variable rotor resistances.

The steady-state analysis of section 6.1.1 is based on the assumption of constant rotor resistance and leakage reactance. The rotor resistance and leakage reactance of the deep-bar rotor and double-cage rotor motors vary with frequency. The large size squirrel-cage motors, even when they are not specifically designed to have a deep-bar rotor, exhibit some deep-bar effect, and therefore, their rotor resistance and leakage reactance also change with frequency. The equivalent circuits of figure 6.1 are still applicable, but the analysis must account for the dependence of the rotor resistance and leakage reactance on the frequency.[1,2]

6.1.3 Squirrel-Cage Induction Motor Design Classification

The squirrel-cage induction motor is used in numerous applications with different requirements for the speed-torque and speed-current characteristics. By a suitable choice of rotor resistance and leakage reactance, and the degree of deep-bar or double-cage effects, a squirrel-cage induction motor can be designed to suit the requirements of a specific application. Large size motors are usually designed for a specific application. The medium and low size motors (around 150 kW and less) have been classified into standard designs whose speed-torque and speed-current characteristics have been found over a period of time to satisfy the majority of squirrel-cage induction motor needs. Application engineers choose the one whose characteristics are closest to their needs. Various countries have their own standard classifications. The American standard (NEMA) classification is briefly described next.

The NEMA has classified the squirrel-cage motor designs into five standard categories: A, B, C, D, and F. The nature of speed-torque characteristics of these standard designs are shown in figure 6.4. It may be noted that individual motors differ significantly from these curves.

A class A design motor uses a single low resistance and low leakage reactance cage winding. It has a low full-load slip, high running efficiency and power factor, high breakdown torque, normal starting torque, and high starting current. It is essentially an energy efficient motor designed to obtain a good running performance at the expense of the starting performance. The full-load slip may vary from 2 percent for a large size motor to 4 percent for a small motor. The breakdown torque is well over

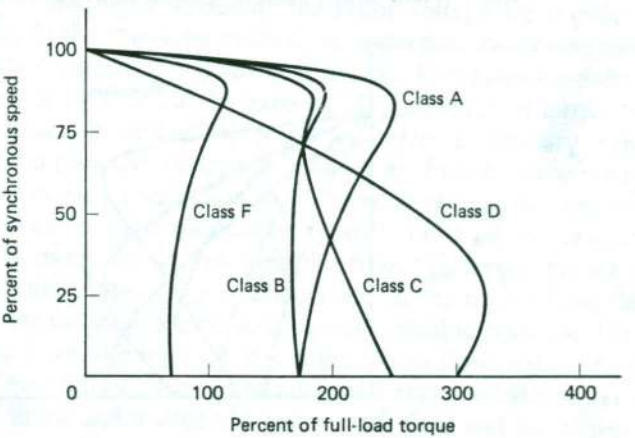


Figure 6.4 Standard squirrel-cage induction motor designs.

twice the full-load torque. The starting torque ranges from equal to full-load torque for large motors to twice full-load torque for small motors. The corresponding range of starting current is from 8 to 5 times the rated current.

Class B design also has a low full-load slip and a normal starting torque. The starting current is reduced to 70 to 80 percent of that of class A by designing for a relatively high leakage reactance, and the starting torque is maintained by employing a deep-bar or double-cage rotor. The use of higher leakage reactance reduces the breakdown torque to a little over twice the full-load torque and slightly reduces the full-load power factor.

Class C design employs a double-cage winding with a greater resistance in the high resistance rotor winding than the class B design. The result is a higher starting torque (about 2.5 times the full-load torque) with low starting current, lower breakdown torque (about 2 times the full-load torque), slightly lower running efficiency, and higher full-load slip (less than 5 percent) than the class B design.

Class D design uses a single-cage rotor with high resistance and low leakage reactance. It gives exceptionally high starting torque, about 3 times full-load torque, and low starting current. This excellent starting performance is obtained at the expense of a large full-load slip (5 to 50 percent) and low running efficiency.

Class F design combines a good running performance (full-load slip 2 to 4 percent) with a low starting current. This is obtained at the expense of starting and breakdown torques.

6.1.4 Wound-Rotor Induction Motors

One of the important features of the wound-rotor induction motor is that unlike the squirrel-cage motor it need not be designed to obtain a compromise between the normal running performance and the starting performance. The rotor winding is designed to have a low resistance so that the running efficiency is high and the full-load slip is low. The improved starting performance is obtained by connecting an external resistance in series with the rotor winding. The speed-torque and speed-rotor current characteristics of a wound-rotor induction motor with different values of external resistances are shown in figure 6.5. The increase in the rotor resistance does not affect the value of maximum torque but increases the slip at maximum torque. When high starting torque is needed, the rotor resistance can be chosen to

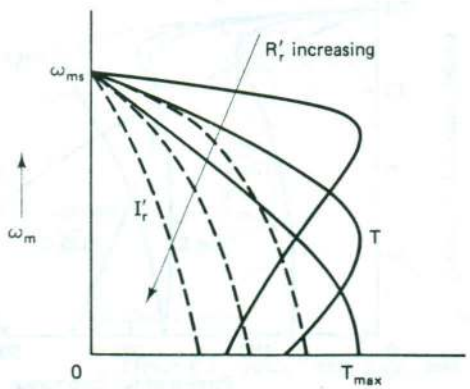


Figure 6.5 Speed-torque (——) and speed-rotor current (– – –) curves of a wound-rotor motor.

obtain the maximum torque at standstill. This also reduces the starting current substantially. As the rotor speeds up, external resistances can be decreased, making the maximum torque available throughout the accelerating range. Since most of the rotor copper loss occurs in the external resistors, the rotor temperature rise during starting is substantially lower than it would be if the resistance were incorporated in the rotor winding, as in the case of squirrel-cage motors. This allows optimum use of the motor torque capabilities. The wound-rotor motor is therefore widely used in applications requiring frequent starting and braking with large motor torques.

Because of the availability of rotor winding for changing the rotor resistance or injecting voltage in the rotor circuit, the wound-rotor motor offers greater flexibility for control. However, it has a number of disadvantages in comparison to the squirrel-cage motor, such as greater cost, need for occasional maintenance due to slip-rings and brushes, lack of sturdiness, and so on. In view of this, it is not as widely used as the squirrel-cage motor.

6.2 STARTING

When started directly on line, an induction motor draws large current. When supplied from a line with low capacity (or appreciable internal impedance), this can cause a dip in the line voltage which adversely affects other loads connected to the same line. Thus, some arrangement must be made to reduce motor current during starting. Various features introduced in the squirrel-cage motors at the design stage to reduce the starting current are discussed in section 6.1.3. The starting of wound-rotor motors using external resistors is explained in section 6.1.4. Other starting methods, applicable to both the squirrel-cage and wound-rotor motors, include reducing the stator voltage, varying the stator frequency and increasing the stator impedance. The wound rotor motor can also be started by injecting voltage in the rotor. These methods, except the one involving the increase of stator impedance, are also used for speed control, and therefore, they will be mainly discussed along with speed control in the present and later chapters.

Conventional methods of starting by reducing the stator voltage are star-delta and autotransformer methods. In the star-delta method, an induction motor designed to operate normally with delta connection is connected in star. This allows a reduction of motor voltage and current by $1/\sqrt{3}$ at the expense of a reduction of torque by $1/3$. In the autotransformer starting method, with the secondary to primary turns ratio a_T, the motor current and torque are reduced by a factor a_T^2. In both these methods, the changeover from the starting connection to the normal running connection may lead to severe current transients if the open circuit transition is used. In the open circuit transition, the motor is momentarily disconnected from the supply before it is reconnected directly across the supply. The disconnection of the motor from the supply causes the stator currents to become zero and the stator field to collapse. Due to a large time constant of the rotor, the rotor current continues to flow, maintaining the rotor field and causing the voltage to be induced in the stator windings. The phase of the voltage induced in the stator depends on the rotor field and is independent of the phase of the supply voltage. At the time of reconnection, the induced

voltage and the supply voltage may aid instead of opposing, resulting in a large current inrush.

Large size motors are often designed with two stator windings which are normally connected in parallel. During starting only one winding is used. This doubles the stator impedance. This method is also known as part winding starting.

6.3 BRAKING

The need for electric braking was explained in section 1.2.5. As in the case of dc motors, various methods of braking induction motors can be divided into the following categories:

1. Regenerative braking.
2. Plugging or reverse voltage braking.
3. Dynamic or rheostatic braking.

Unlike dc motors, dynamic braking of induction motors can be realized in a number of ways. Here only dc dynamic braking, which is the most widely used method among dynamic braking methods, will be discussed.

6.3.1 Regenerative Braking

Equations (6.18), (6.21), and (6.22) provide expressions for torque, s_m, and T_{max}. These expressions are valid for speeds below synchronous speed ($s > 0$), speeds above synchronous speed ($s < 0$), and also for negative speeds ($s > 1$). Figure 6.2 shows the speed-torque curves for the speed range from 0 to synchronous speed. Figure 6.6 shows the speed-torque curves for all the three ranges of speed by continuous lines. The operations for $\omega_m > \omega_{ms}$ (or $s < 0$) and $\omega_m < 0$ ($s > 1$) produce negative power and therefore correspond to the braking operation. The speed-torque curves obtained by the reversal of the phase sequence of the motor terminal voltages are also shown by dotted lines.

With a positive sequence voltage across the motor terminals, the operation above synchronous speed gives the regenerative braking operation (portion BAE). Similarly, with a negative sequence voltage across the motor terminals, regenerative braking is obtained for speeds above the synchronous speed in the reverse direction (portion bae). In regenerative braking, the motor works as an induction generator, converting mechanical energy supplied by the load to electrical energy, which is fed to the source. Thus the generated energy is usefully employed. It should be understood that if the source cannot accept energy then the regenerative braking cannot be used. The operation of the motor in regenerative braking can be explained as follows.

When the motor runs at a speed greater than the synchronous speed, the relative speed between the rotating field and the rotor is negative. The rotor-induced voltage and current have directions opposite to those under the motoring operation. The stator current which flows to balance the rotor ampere turns is also in the opposite direction. Thus, the power flows from the motor to the source and the motor works as an induction generator. The magnetizing current required to produce flux is

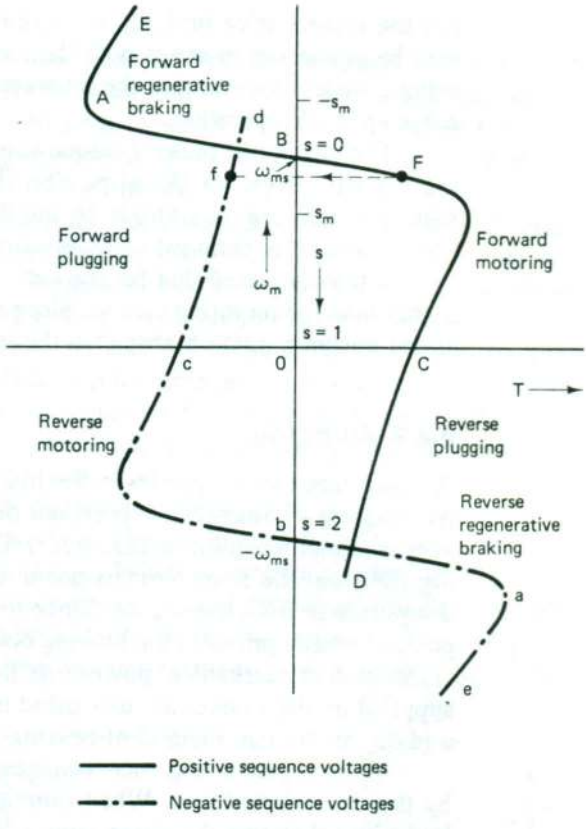


Figure 6.6 Speed-torque curves of an induction motor for speeds greater and less than synchronous speeds in either direction.

obtained from the source. It may be noted that the machine cannot regenerate unless it is connected to a source.

For regenerative braking to take place, the motor's speed should be greater than synchronous speed. When the motor is fed by a fixed frequency source, regenerative braking is possible only for speeds greater than synchronous speed. When the motor is fed by a variable frequency source, the source frequency can be adjusted to give a synchronous speed less than the motor speed for any motor speed; and therefore regenerative braking can be obtained up to nearly zero speed.

When regenerative braking is employed for holding the speed against an active load, care should be taken to restrict the operation in the region between the synchronous speed and the speed for which the braking torque is the maximum—that is, on the portion AB (or ab for the negative sequence voltages) for which $0 > s > -s_m$. For slips more negative than $-s_m$ (portion AE), the braking torque reduces drastically, leading to runaway speeds, because, the faster the motor runs, the lesser will be the braking torque. This restriction on the slip range must also be observed when braking against an active load by varying the supply frequency.

When holding an active load by regenerative braking, a short duration dip in the supply voltage or a momentary increase in the load torque may shift the operation to the unstable region. In such a situation mechanical brakes may be used to as-

sist the regenerative braking to prevent runaway speeds. Alternatively, capacitors may be connected in series with the motor to increase the braking torque. If one is using a wound-rotor motor, the rotor resistance may also be increased to increase the range of stable operation.

The developed braking torque can be calculated from equation (6.18) by using the negative sign for the slip. The shaft torque is obtained by adding friction windage and core loss torque to the developed torque. The maximum developed braking torque is obtained from equation (6.22) when the negative sign is used.

It may be noted that for the same absolute value of slip, the braking torque is higher than the motoring torque. Since the braking speeds are also higher, the regenerated power is much higher than the motoring power.

6.3.2 Plugging

An induction motor operates in the plugging mode for slips greater than 1. For positive sequence voltages, a slip greater than 1 is obtained when the rotor moves in the reverse direction (portion CD, fig. 6.6). Since the relative speed between the rotating field and the rotor remains positive, the motor torque is positive and the motor draws power from the source. Since the motor is running in the reverse direction, a positive torque provides the braking operation. The electrical power generated by the conversion of mechanical power supplied by the load and inertia, and also the power supplied by the source, are dissipated in the motor circuit's resistances. Thus, this is a highly inefficient method of braking.

With negative sequence voltages, plugging takes place on portion cd, shown by the chain-dotted line. When running in the forward direction, the motor can be braked by changing the phase sequence of the motor terminal voltages by simply interchanging the connections of any two motor terminals. This will transfer the operation from point F to f and braking will commence. The motor torque is not zero at zero speed. When braked for stopping, the motor should be disconnected from the supply at or near zero speed. An additional device will be required for detecting zero speed and disconnecting the motor from the supply. Therefore, plugging is not suitable for stopping. It is, however, quite suitable for reversing the motor. As the motor is already connected with the negative sequence voltages and the torque is finite at zero speed, it accelerates to a speed in the reverse direction.

Because of high values of slip (nearly 2 at point f), the equivalent rotor resistance R_r'/s has a low value. A high current flows, but the torque is low due to the low power factor of the rotor. In the case of a wound-rotor motor, external resistors are connected in the rotor to reduce the current and increase the braking torque. The value of the external resistor can be chosen to provide the maximum torque for $s = 2$. As s falls, the resistance can be varied to brake and reverse the motor at the maximum torque.

From the forward motoring (portion BC), the reverse plugging operation (portion CD) is obtained when an active load drives the motor in the reverse direction, as in crane and hoist applications. When operating this way, plugging is sometimes called counter-torque braking.

6.3.3 DC Dynamic Braking

In dc dynamic braking, the motor is disconnected from the ac supply and connected to a dc supply. The ways in which the motor can be connected to a dc supply are shown in figure 6.7. Connections c and f provide uniform loading for all the three phases but complicate the switching operation. Connections a, b, d, and e are generally used because of the simpler switching operations.

The flow of direct current through the stator windings sets up a stationary magnetic field. The relative speed between the stationary stator field and the moving rotor is now negative. Consequently, 3-phase voltages of reverse polarity and phase sequence (compared to the motoring in the same direction) are induced in the rotor. The resultant three-phase rotor currents produce a rotating field, moving at the rotor speed in the direction opposite to that of rotor, thus giving a stationary rotor field. Since both stator and rotor fields are stationary and rotor current flows in the reverse direction, a steady braking torque is produced at all speeds. It, however, becomes zero at standstill due to zero rotor currents.

Since the dc current flowing through the stator depends on its resistance which is low, a low voltage dc supply is required. This is obtained from the ac supply by a step-down transformer and a diode bridge. When controlled braking (braking with variable torque) is required, a thyristor bridge is used instead of the diode bridge. When quick braking is required, to produce large braking torque, one can allow the stator current to be as high as ten times the rated current. But then either the supply

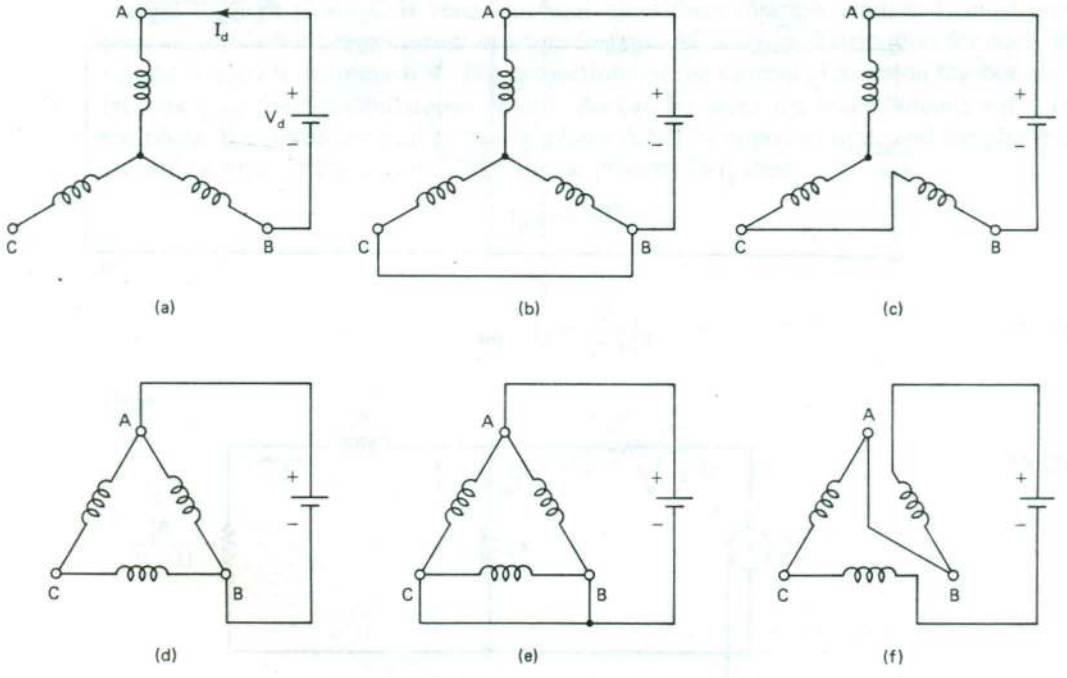
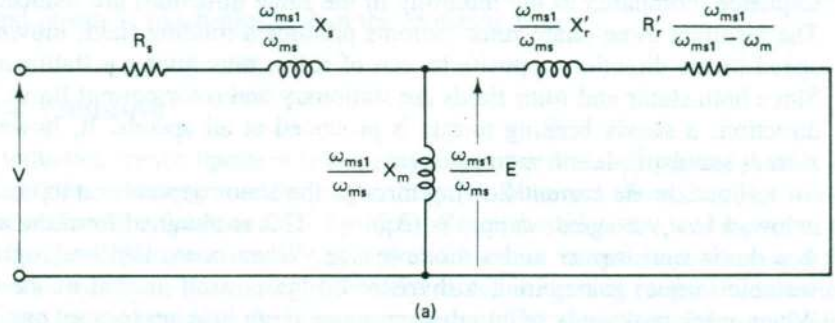


Figure 6.7 Stator connections for dc dynamic braking.

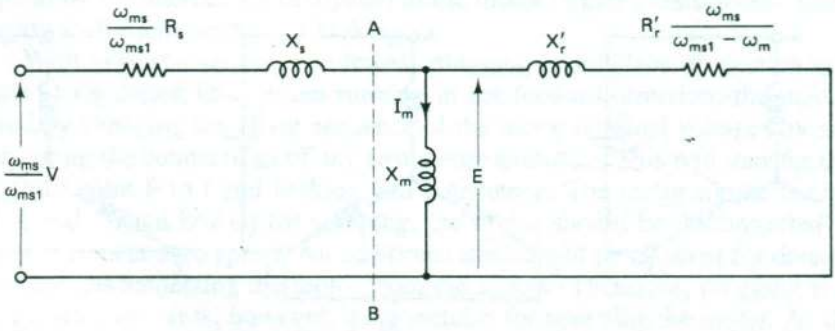
must be removed or the current must be reduced below the rated value soon after the motor stops, otherwise the motor will be overheated.

The dc dynamic braking equivalent circuit of an induction motor is obtained as follows.

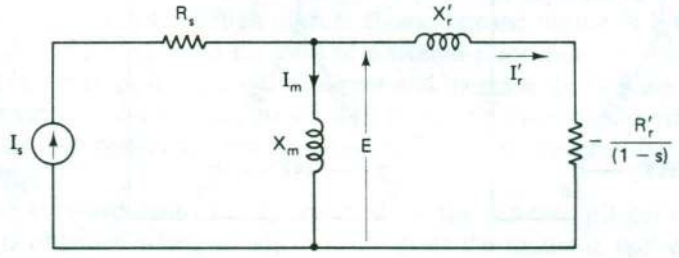
The equivalent circuit of figure 6.1b uses the values of various parameters at a frequency ω rad/sec. for which the synchronous speed is ω_{ms}. Figure 6.8a illustrates the equivalent circuit for a frequency ω_1, for which the synchronous speed is ω_{ms1}. Various parameters are given in terms of their values at the frequency ω. Dividing all the parameters by $(\omega_{ms1}/\omega_{ms})$ gives a circuit as shown in figure 6.8b. For dc dy-



(a)



(b)



(c)

Figure 6.8 DC dynamic braking equivalent circuits of an induction motor.

namic braking, $\omega_1 = 0$ and $\omega_{ms1} = 0$. Substitution of these values in figure 6.8b, gives the dc dynamic braking equivalent circuit shown in figure 6.8c.

The minus sign with the term $[-R_r'/(1 - s)]$ shows that the torque is negative and the machine is acting as a sink of mechanical power. As long as it is kept in mind that the torque is negative, it is not necessary to include the minus sign in the equivalent circuit. Here also the minus sign will be ignored in the derivations given later. The portion of figure 6.8b to the left of line AB has been replaced by a current source I_s in series with a resistance R_s. The explanation for this is as follows. The substitution of $\omega_{ms1} = 0$ gives an infinite voltage source in series with an infinite resistance. This can be replaced by a current source I_s. Any impedance in series with a current source will have no effect on the current value; hence, X_s can be dropped. However, R_s may be retained to account for the stator copper loss.

The value of current source I_s, which is an ac equivalent of the dc current through the stator, depends on the type of stator connection employed. It is obtained as follows.

The direct current I_d through the stator windings sets up a stationary field. The circuit of figure 6.8c is an ac equivalent circuit. An ac current I_s will be equivalent to the dc current I_d if it produces a rotating field of the same magnitude as that produced by I_d. If this condition is satisfied, then there would be an instant when the distribution of the ac currents through the three-phase stator windings will be the same as that due to the dc current. For example, in connection "a" of figure 6.7 only two phases are energized: phase A has a current I_d, phase B has a current $-I_d$, and the current through phase C is zero. An equivalent three-phase excitation I_s must produce this distribution of current at some instant. AC current distribution for such an instant is shown in figure 6.9. The projections of the current phasors on the horizontal axis give their instantaneous values. As can be seen, the instantaneous value of the phase B current is equal to that of phase A but of opposite sign, and the phase C current is zero. If the amplitude of the ac phasors is I_p then

$$I_p \cos 30° = I_d$$

or

$$I_p = \frac{2}{\sqrt{3}} I_d \tag{6.25}$$

Now

$$I_s = I_p/\sqrt{2} = \sqrt{\frac{2}{3}} I_d \tag{6.26}$$

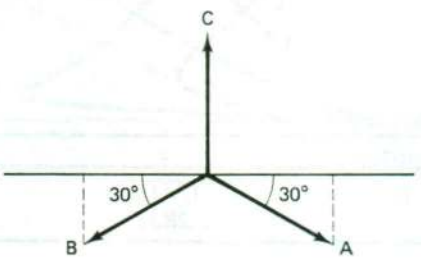


Figure 6.9

Further,

$$V_d = 2R_s I_d \tag{6.27}$$

The relationship between I_s and I_d, and V_d and I_d for connections a, b, d, and e are listed in table 6.1. The connections c and f require a different approach, which is not considered here.

Current I_s is shared between the rotor impedance and the magnetizing reactance X_m. For low values of $(1 - s)$, the rotor current will be small. Therefore, the magnetizing current will be nearly equal to I_s. It is common practice to use large values of I_s to get fast braking. Then for low values of $(1 - s)$, I_m will be large enough to cause heavy saturation in the magnetic circuit. Because of the saturation, the magnetization characteristic (the relation between E and I_m) will be nonlinear and X_m will vary with I_m.

The magnetization characteristic can be obtained experimentally. A wound-rotor motor is run at synchronous speed with the rotor open and with dc excitation on the stator. Since the rotor is open, the rotor current I_r is zero; thus $I_m = I_s$. The rotor induced phase voltage (E) can now be measured for various values of I_m. The magnetization characteristic of a squirrel-cage motor is obtained by running it at the synchronous speed with the ac excitation on the stator. Since I_r is zero, the input phase current and phase voltage are then equal to I_m and E, respectively. By changing the input ac excitation, the magnetization characteristic is obtained.

The analysis of the braking, taking into account the nonlinearity of the magnetization characteristic, is done as follows.

From the equivalent circuit of figure 6.8c,

$$I_s^2 X_m^2 = I_r'^2 \left[\left(\frac{R_r'}{1 - s} \right)^2 + (X_r' + X_m)^2 \right] \tag{6.28}$$

$$E^2 = I_r'^2 \left[\left(\frac{R_r'}{1 - s} \right)^2 + X_r'^2 \right] \tag{6.29}$$

and

$$E = I_m X_m \tag{6.30}$$

Substituting from equation (6.30) into (6.29), subtracting the resultant equation from equation (6.28) and rearranging the terms gives

$$I_r'^2 = \frac{I_s^2 - I_m^2}{1 + \dfrac{2X_r'}{X_m}} \tag{6.31}$$

TABLE 6.1

Connection	a	b	d	e
I_s	$I_d\sqrt{2/3}$	$I_d/\sqrt{2}$	$\sqrt{2}I_d/3$	$I_d/\sqrt{6}$
V_d	$2R_sI_d$	$1.5R_sI_d$	$2R_sI_d/3$	$R_sI_d/2$

Also from equation (6.29),

$$s = 1 - \frac{R'_r}{\sqrt{(E/I'_r)^2 - X'^2_r}} \tag{6.32}$$

The braking torque T is given by

$$T = \frac{3}{\omega_{ms}} \left(I'^2_r \frac{R'_r}{1-s} \right) \tag{6.33}$$

The shaft torque is obtained by adding friction, windage, and core loss torques to the developed torque T.

To avoid solution of nonlinear algebraic equations, the following sequence is used to calculate speed-torque characteristics. Assume a value for I_m, obtain corresponding E from the magnetization characteristic, calculate X_m from equation (6.30), obtain I'_r from equation (6.31), evaluate s from equation (6.32) and ω_m from equation (6.4), and then obtain T from equation (6.33).

The speed-torque characteristics for two values of I_d and R'_r are shown in figure 6.10. For a given I_d, an increase in R'_r increases the speed for which the torque is maximum, but the magnitude of the maximum torque remains constant. This behavior is similar to the motoring operation and it is explained next.

Substituting from equation (6.28) into equation (6.33) gives

$$T = \frac{3}{\omega_{ms}} \left[\frac{I_s^2 X_m^2 (R'_r/(1-s))}{(R'_r/(1-s))^2 + (X_m + X'_r)^2} \right] \tag{6.34}$$

Differentiation of equation (6.34) with respect to $(1 - s)$ gives the following expression for the slip at the maximum torque:

$$s_m = 1 - \frac{R'_r}{(X'_r + X_m)} \tag{6.35}$$

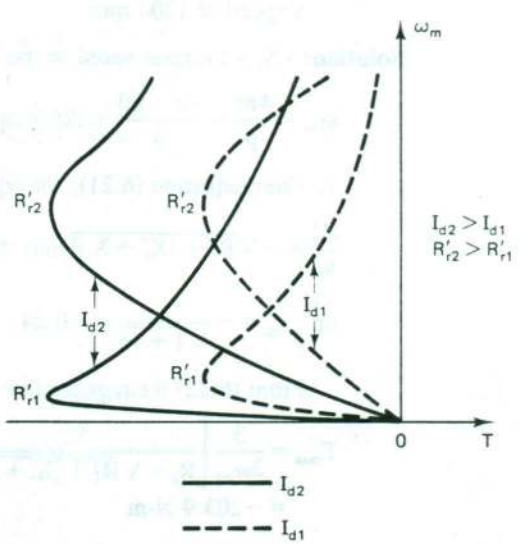


Figure 6.10 DC dynamic braking speed-torque curves.

Substitution from equation (6.35) into equation (6.34) yields

$$T_{max} = \frac{3}{2\omega_{ms}} \frac{I_s^2 X_m^2}{(X_r' + X_m)} \tag{6.36}$$

Thus, the maximum torque is independent of R_r'. In the linear region of operation, at a given speed, the torque is proportional to I_s^2. For low speeds, the torque is less than proportional to I_s^2 because of the decrease in X_m due to the saturation. From equations (6.31) – (6.33), (6.35), and (6.36) it can be shown that for a given I_s, I_r' has an unique value at the maximum torque.

External resistors are connected in the rotor of a wound-rotor induction motor for improving the braking performance at high speeds. When quick braking is desired, R_r' is chosen to get the maximum torque at the highest speed. As the speed falls, external resistors can be reduced to brake the motor at the maximum torque.

This method of braking is very effective for fast stopping of loads. Since the stator resistance is very small, a large I_d can be easily produced to get large braking torques. As the torque is zero at zero speed, no arrangement is required to disconnect the motor from the supply. Because of the lower consumption of energy compared to plugging, it is also used for holding speeds against active loads for prolonged periods, such as in mine winders.

Example 6.1

A 3-phase, Y-connected, 6-pole, 60 Hz induction motor has the following constants:

$$V_t = 231 \text{ V}, \qquad R_t = R_r' = 1\ \Omega, \qquad X_t = X_r' = 2\ \Omega$$

1. If the motor is used for regenerative braking,
   (a) Determine the range of active load torque it can hold and the corresponding range of speed.
   (b) Calculate the speed and current for an active load torque of 150 N-m.
2. If the motor is used for plugging, determine the braking torque and current for a speed of 1200 rpm.

Solution: Synchronous speed in rpm $N_s = 120f/p = 120 \times 60/6 = 1200$ rpm

$$\omega_{ms} = \frac{4\pi f}{p} = \frac{4\pi \times 60}{6} = 125.7 \text{ rad/sec.}$$

1. From equation (6.21), for regenerative braking

$$\frac{R_r'}{s_m} = -\sqrt{R_t^2 + (X_r' + X_t)^2}$$

$$\text{or} \quad s_m = -\frac{1}{\sqrt{1+16}} = -0.24$$

From (6.22) for regenerative braking,

$$T_{max} = \frac{3}{2\omega_{ms}} \left[\frac{V_t^2}{R_t - \sqrt{R_t^2 + (X_t + X_r')^2}} \right] = \frac{3}{2 \times 125.7} \left[\frac{231 \times 231}{1 - \sqrt{1+16}} \right]$$

$$= -203.9 \text{ N-m}$$

(a) The range of active load torque the motor can hold: 0 to 203.9 N-m
Speed in rpm at the maximum torque:

$$N = (1 - s_m)N_s = (1 - (-0.24)) \times 1200 = 1488 \text{ rpm}$$

Therefore the range of speed is 1200 to 1488 rpm.

(b) From equation (6.18)

$$T = \frac{3}{\omega_{ms}}\left[\frac{V_t^2 R_r'/s}{\left(R_t + \frac{R_r'}{s}\right)^2 + (X_t + X_r')^2}\right] \tag{6.18}$$

For regenerative braking, $T = -150$ N-m;
thus

$$\frac{3}{125.7} \cdot \frac{(231)^2(1/s)}{\left(1 + \frac{1}{s}\right)^2 + 16} = -150$$

Let $1/s = x$; then

$$(1 + x)^2 + 16 = -8.49x \quad \text{or} \quad x^2 + 10.49x + 17 = 0$$

$$\text{or} \quad x = \frac{-10.49 \pm \sqrt{(10.49)^2 - 4 \times 17}}{2} = \frac{-10.49 \pm 6.48}{2} = -2 \quad \text{or} \quad -8.5$$

$$\text{or} \quad s = \frac{1}{x} = -0.5 \quad \text{or} \quad -0.118$$

Since stable operation is usually obtained only up to s_m which is -0.24, the feasible value of s is -0.118.
Motor speed $N = (1 - s)\, N_s = 1.118 \times 1200 = 1342$ rpm

$$I_r' = \frac{V_t}{\sqrt{\left(R_t + \frac{R_r'}{s}\right)^2 + (X_t + X_r')^2}}$$

$$= \frac{231}{\sqrt{\left(1 + \frac{1}{-0.118}\right)^2 + 16}} = 27.2 \text{ A}$$

2. For the motor speed = N rpm

$$s = \frac{N_s - N}{N_s}$$

In the present example, for plugging

$$N_s = -1200 \text{ rpm} \qquad N = 1200 \text{ rpm}$$

$$s = \frac{-1200 - 1200}{-1200} = 2$$

$$I_r' = \frac{V_t}{\sqrt{\left(R_t + \frac{R_r'}{s}\right)^2 + (X_t + X_r')^2}} = -\frac{231}{\sqrt{(1+0.5)^2 + 16}} = 54.1 \text{ A}$$

$$T = \frac{3}{\omega_{ms}} \frac{I_r'^2 R_r'}{s} = -\frac{3}{-125.7} \cdot (54.1)^2 \cdot \frac{1}{2} = 34.9 \text{ N-m}$$

In spite of a large current, plugging torque is very low.

Example 6.2

A 2.8 kW, 3-phase, 50 Hz, 4-pole, Y-connected wound-rotor induction motor has the following parameters:

$$R_r' = 2\ \Omega, \qquad X_r' = 3\ \Omega$$

Following are the two points on the magnetization characteristic with dc dynamic braking connection b of figure 6.8:

Point	I_m, Amps	E, Volts
A	0.9	80
B	6.53	266

Point A is from the unsaturated part of the magnetization characteristic.

1. Calculate dc dynamic braking torque, speed, and rotor current for the foregoing two points for $I_d = 12$ A.
2. For the speed obtained for point B in question 1 calculate the torque, neglecting saturation. How much is the error in calculation when the saturation is neglected?

Solution: For connection b of figure 6.7 from table 6.1,

$$I_s = I_d/\sqrt{2} = 12/\sqrt{2} = 8.49 \text{ A}$$

$$\omega_{ms} = \frac{4\pi f}{p} = \frac{4\pi \times 50}{4} = 157.1 \text{ rad/sec.}$$

Synchronous speed in rpm $= N_s = 1500$ rpm.

1. For point A $X_m = 80/0.9 = 88.9\ \Omega$

From equation (6.31),

$$I_r'^2 = \frac{I_s^2 - I_m^2}{1 + \frac{2X_r'}{X_m}} = \frac{(8.49)^2 - (0.9)^2}{1 + \frac{2 \times 3}{88.9}}$$

$$= 66.8 \text{ A}$$

or $I_r' = 8.17$ A

From equation (6.32),

$$(1-s)=\frac{R_r'}{\sqrt{(E/I_r')^2-X_r'^2}}=-\frac{2}{\sqrt{(80/8.17)^2-9}}$$
$$=0.21$$

Motor speed = $(1-s)\,N_s = 0.21\times 1500 = 315$ rpm
From equation (6.33),

$$T=\frac{3}{\omega_{ms}}I_r'^2\frac{R_r'}{(1-s)}=\frac{3}{157.1}(8.17)^2\cdot\frac{2}{0.21}=12.15\text{ N-m}$$

For point B

$$X_m=\frac{266}{6.53}=40.74\ \Omega$$

$$I_r'^2=\frac{(8.49)^2-(6.53)^2}{1+\dfrac{2\times 3}{40.74}}=25.67$$

$$I_r'=5.07\text{ A}$$

$$1-s=-\frac{2}{\sqrt{(266/5.07)^2-9}}=0.038$$

Motor speed = $0.038\times 1500 = 57.3$ rpm.

$$T=\frac{3}{157.1}(5.07)^2\times\frac{2}{0.038}=25.8\text{ N-m.}$$

2. Speed = 57.3 rpm or 5.97 rad/sec.

If saturation is neglected, $X_m = 88.9\ \Omega$ (from point A). From equation (6.34),

$$T=\frac{3}{\omega_{ms}}\left[\frac{I_s^2X_m^2(R_r'/(1-s))}{(R_r'/(1-s))^2+(X_m+X_r')^2}\right]$$
$$=\frac{3}{157.1}\left[\frac{(8.49)^2\times(88.9)^2\times(2/0.038)}{(2/0.038)^2+(88.9+3)^2}\right]=51\text{ N-m}$$

Percent error = $((51-25.8)/25.8)\times 100 = 97.7\%$
Thus, a very large error is obtained if saturation is neglected.

6.4 SPEED CONTROL

The present section describes the basic principles of speed control methods employed in power semiconductor controlled induction motor drives. These methods are

1. Variable terminal voltage control.
2. Variable frequency control.

3. Rotor resistance control.
4. Injecting voltage in the rotor circuit.

Methods 1 and 2 are applicable to both squirrel-cage and wound-rotor motors, and methods 3 and 4 can be used only for wound-rotor motors.

6.4.1 Variable Terminal Voltage Control

The torque developed by an induction motor is proportional to the square of terminal voltage [see equations (6.18) and (6.8)]. The torque-speed curves, therefore, retain their shape but shrink or grow as the square of the terminal voltage. Speed control is achieved by varying the terminal voltage until the torque required by the load is developed at the desired speed. Since one cannot allow the terminal voltage to be more than the rated value, this method allows speed control only below the normal rated speed.

While the torque for a specific slip is proportional to the square of terminal voltage, the rotor current is directly proportional to the terminal voltage [equations (6.12) and (6.8)]. Hence the torque to current ratio decreases with the terminal voltage. Consequently, the torque available for a given thermal loading of the motor also decreases. Moreover, the breakdown torque decreases in proportion to the square of terminal voltage. Therefore, low-speed operation without the overheating of the machine is possible only if the load torque decreases with speed, as in the case of a fan load.

A set of motor speed-torque curves along with that of a fan load are shown in figure 6.11. For a wide variation of speed, a motor with a high normal full-load slip is required. Therefore, either a clsss D squirrel-cage motor with a normal full-load slip from 0.1 to 0.2 is used or a low slip wound-rotor motor with external rotor resistors is employed. The wound-rotor motor has the advantage that most of the rotor copper loss takes place outside the motor. Therefore, a smaller motor can be used. But this does not necessarily ensure a lower cost of the drive because of the higher cost and maintenance requirements of a wound-rotor motor, and the need for external resistors.

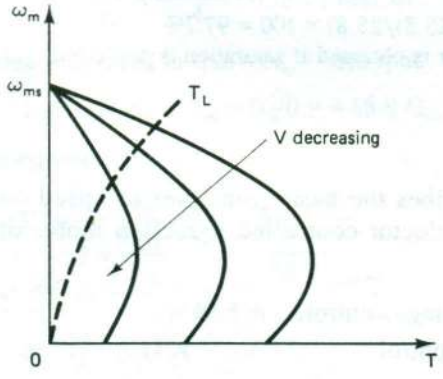


Figure 6.11 Speed control by variation of rotor resistance.

If the stator copper loss and the friction, windage, and core losses are ignored, from equation (6.15) the motor efficiency is given by

$$\eta_M \simeq \frac{P_m}{P_g} = (1 - s) \tag{6.37}$$

Equations (6.15) and (6.19) show that the developed power decreases but the rotor copper loss increases with the increase in slip. Consequently, the motor efficiency is very poor at low speeds.

The variable voltage is obtained by using ac voltage controllers. This method of speed control is described in chapter 7.

6.4.2 Variable Frequency Control

The synchronous speed is directly proportional to the supply frequency [equation (6.1)]. Hence, the synchronous speed and the motor speed can be controlled below and above the normal full-load speed by changing the supply frequency.

The voltage induced in the stator E is proportional to the product of the supply frequency and the air-gap flux. If the stator drop is neglected, the motor terminal voltage can be considered proportional to the product of the frequency and the flux. Any reduction in the supply frequency, without a change in the terminal voltage, causes an increase in the air-gap flux. Induction motors are designed to operate at the knee point of the magnetization characteristic to make full use of the magnetic material. Therefore, the increase in flux will saturate the motor. This will increase the magnetizing current, distort the line current and voltage, increase the core loss and the stator copper loss, and produce a high-pitch accoustic noise. While an increase in flux beyond the rated value is undesirable from the consideration of saturation effects, a decrease in flux is also avoided to retain the torque capability of the motor. Therefore, the variable frequency control below the rated frequency is generally carried out by reducing the machine phase voltage V along with the frequency f in such a manner that the flux is maintained constant. Above the rated frequency, the motor is operated at a constant voltage because of the limitation imposed by the stator insulation or by supply voltage limitations.

Let us define a variable "a" as

$$a = f/f_{rated} \tag{6.38}$$

where f is the operating frequency and f_{rated} is the rated frequency of the motor. The variable "a" is called the per-unit frequency.

Operation Below the Rated Frequency ($a < 1$)

As just stated, it is generally preferred to operate the motor at a constant flux. The motor will operate at a constant flux if I_m is maintained constant at all operating points. According to figure 6.1b, at the rated condition of motor operation

$$I_m = \frac{E_{rated}}{X_m} = \frac{E_{rated}}{f_{rated}} \cdot \frac{1}{2\pi L_m} \tag{6.39}$$

where L_m is the magnetizing inductance. When the motor is operated at a frequency f, then

$$I_m = \frac{E}{aX_m} = \frac{E}{a \cdot f_{rated}} \cdot \frac{1}{2\pi L_m} \tag{6.40}$$

Comparison of equation (6.40) with equation (6.39) shows that I_m will stay constant at a value equal to its rated value if

$$E = aE_{rated} \tag{6.41}$$

Equation (6.41) suggests that the flux will remain constant if the back emf changes in the same ratio as the frequency—in other words, when (E/f) ratio is maintained constant.

We next examine the motor operation for a constant (E/f) ratio. At a frequency f, from the equivalent circuit of figure 6.1b,

$$I_r' = \frac{aE_{rated}}{\sqrt{(R_r'/s)^2 + (aX_r')^2}} = \frac{E_{rated}}{\sqrt{R_r'^2/(as)^2 + X_r'^2}} \tag{6.42}$$

where

$$s = \frac{a\omega_{ms} - \omega_m}{a\omega_{ms}} \tag{6.43}$$

Note that ω_{ms} is the synchronous speed at the rated frequency.

$$\begin{aligned} T &= \frac{3}{a\omega_{ms}} I_r'^2 R_r'/s \\ &= \frac{3}{\omega_{ms}} \left[\frac{E_{rated}^2 R_r'/(as)}{R_r'^2/(as)^2 + X_r'^2} \right] \end{aligned} \tag{6.44}$$

Now for a given frequency, E is maintained constant. The power transferred across the air-gap will be maximum at a slip s_m for which

$$aX_r' = \pm R_r'/s_m \quad \text{or} \quad s_m = \pm \frac{R_r'}{aX_r'} \tag{6.45}$$

Substituting in equation (6.44) gives

$$T_{max} = \pm \frac{3}{2\omega_{ms}} \frac{E_{rated}^2}{X_r'} \tag{6.46}$$

Equation (6.46) shows that for a variable frequency control at a constant flux, the breakdown torque remains constant for all frequencies, both during motoring and regenerative braking. Further, the examination of equations (6.42) and (6.44) shows that for a constant (sa), the rotor current I_r' and torque T are constant. If $\bar{E}$ is taken as a reference vector, then the phase lag of $\bar{I}_r'$ is given by

$$\Theta_r = \tan^{-1}(as\, X_r'/R_r') \tag{6.47}$$

Since Θ_r is also constant for a given (sa), the stator current will also be constant. Thus the motor operates at constant values of torque, I_s and I_r' when the flux and (as)

are maintained constant. Let us now examine the physical significance of sa. From equation (6.43)

$$sa = \frac{a\omega_{ms} - \omega_m}{\omega_{ms}} = \frac{\omega_{s\ell}}{\omega_{ms}} \tag{6.48}$$

where

$$\omega_{s\ell} = a\omega_{ms} - \omega_m \tag{6.49}$$

Note that $\omega_{s\ell}$ is the slip speed, which is the difference in the field speed at the frequency f (or synchronous speed $a\omega_{ms}$) and the rotor speed ω_m. According to equation (6.48), a constant value of (as) implies the motor operation at a constant slip speed $\omega_{s\ell}$. Note that $\omega_{s\ell}$ is the drop in motor speed from its no-load speed ($a\omega_{ms}$), when the machine is loaded.

The foregoing discussion shows that for any value of T, the drop in the motor speed from its no-load speed ($a\omega_{ms}$) is the same for all frequencies. Hence the machine speed-torque characteristics for $0 < s < s_m$ are parallel curves. The nature of speed-torque curves for the variable frequency operation at a constant flux are shown in figure 6.12a, both for the motoring and braking operations.

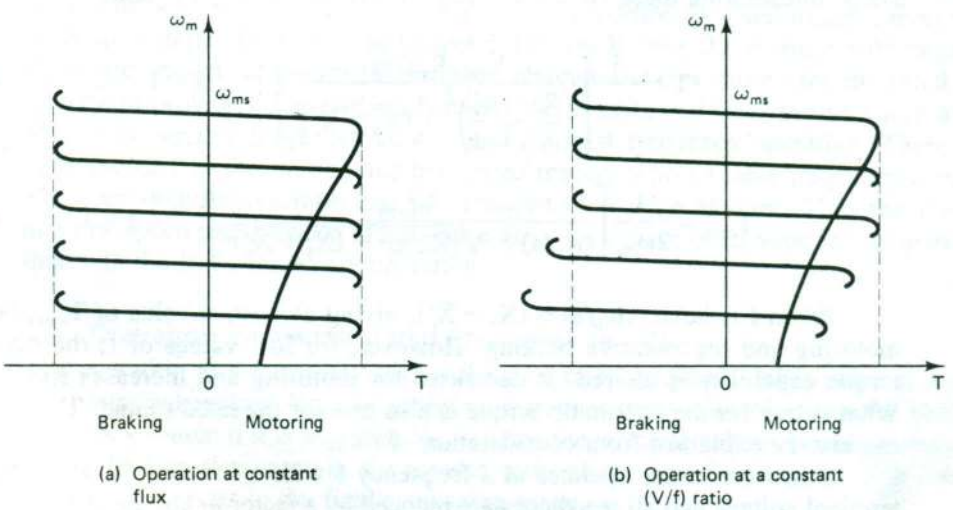
The operation of the machine at a constant slip speed also implies the operation at a constant rotor frequency as shown by equation (6.50):

$$sa = \frac{s \cdot f}{f_{rated}} = \frac{f_r}{f_{rated}} = \frac{\omega_r}{\omega_{rated}} \tag{6.50}$$

where ω_r and f_r are the rotor frequency in rad/sec. and Hz, respectively.

For $s < s_{max}$, $(R_r'/s) \gg (aX_r')$; hence from equation (6.44) and equation (6.48)

$$T = \frac{3E_{rated}^2}{\omega_{ms}R_r'}(as) = \text{constant} \times \omega_{s\ell} \tag{6.51}$$



(a) Operation at constant flux

(b) Operation at a constant (V/f) ratio

Figure 6.12 Speed-torque curves with variable frequency control.

Equation (6.51) suggests that for $s < s_m$, the speed-torque curves are nearly straight lines. Since they are also parallel, the speed-torque characteristics are approximately parallel straight lines for $s < s_m$, when flux is maintained constant.

The operation of the machine at a constant flux requires a closed-loop control of flux. When the operating point changes, the closed-loop control adjusts the motor voltage to maintain a constant flux. The closed-loop control becomes complicated because the measurement of flux is always difficult. Hence the flux is controlled indirectly by operating the machine at a constant (V/f) ratio for most of the frequency range, except at low frequencies, where the (V/f) ratio is increased to compensate for the stator resistance drop as explained next.

The (V/f) ratio is chosen equal to its value at the rated voltage and frequency. As the load on the machine is increased, the stator resistance drop increases and the back emf decreases and the flux reduces. Consequently, the machine does not operate exactly at a constant flux. We will now examine the motor operation when the (V/f) ratio is held constant.

For simplicity, the equivalent circuit of figure 6.1d is used here. Note that E is still defined in the same way as in figure 6.1b. E, and not I_m, will be taken as a measure of flux. From the equivalent circuit, at rated motor terminal voltage (V_{rated}) and frequency (ω_{rated})

$$T = \frac{3}{\omega_{ms}}\left[\frac{V_{rated}^2(R_r'/s)}{(R_s + R_r'/s)^2 + (X_s + X_r')^2}\right] \tag{6.52}$$

and

$$T_{max} = \frac{3}{2\omega_{ms}}\left[\frac{V_{rated}^2}{R_s \pm \sqrt{R_s^2 + (X_s + X_r')^2}}\right] \tag{6.53}$$

For a frequency f defined by equation (6.38), the synchronous speed, the terminal voltage, and any reactance X will have the values $a\omega_{ms}$, aV_{rated}, and aX, respectively. Substituting these values in equations (6.52) and (6.53) yields

$$T = \frac{3}{\omega_{ms}}\left[\frac{V_{rated}^2 R_r'/(as)}{\left(\frac{R_s}{a} + \frac{R_r'}{sa}\right)^2 + (X_s + X_r')^2}\right], \qquad a < 1 \tag{6.54}$$

$$T_{max} = \frac{3}{2\omega_{ms}}\left[\frac{V_{rated}^2}{(R_s/a) \pm \sqrt{(R_s/a)^2 + (X_s + X_r')^2}}\right], \qquad a < 1 \tag{6.55}$$

When f is large, $(R_s/a) \ll (X_s + X_r')$, giving a constant value of T_{max}, both for motoring and regenerative braking. However, for low values of f, the maximum torque capability is altered. It decreases for motoring and increases for braking. What is true for the maximum torque is also true for the rated torque. This behavior can also be explained from consideration of flux.

When the motor operates at a frequency f with a constant (V/f) control, the terminal voltage and all reactances are reduced by a factor a, but the stator resistance

remains fixed. The resistance drop, which is negligible for high values of f, becomes appreciable in comparison with the terminal voltage at low values of f. As a result, the (E/f) ratio decreases, decreasing flux and the motor torque capability. The lower the frequency, the greater the reduction in the torque capability.

When working in the regenerative braking mode, the rotor current direction is opposite to that in motoring. Hence, the stator resistance drop has an opposite effect—that is, the flux and braking torque have higher values at low frequencies.

The nature of the speed-torque characteristics for constant (V/f) control and for $f < f_{rated}$ are shown in figure 6.12b. The decrease in motoring torque and increase in braking torque at low frequencies have higher values for motors of low power rating.

To make full use of the motor's torque capability at the start and for low speeds, the (V/f) ratio is increased to compensate for the stator resistance drop at low frequencies, as shown in figure 6.14. This allows a constant maximum torque to be obtained for motoring operation at all frequencies.

The motoring speed-torque characteristics become similar to those shown in figure 6.12a. The braking torque which is already high at low frequencies is increased further. The large increase in braking torque may cause severe mechanical stress on the motor and the load.

For a given frequency, the exact compensation for the stator resistance drop can be done only for a particular operating point. This point is chosen either at the rated torque or the breakdown torque. Then for lower torques, the motor saturates due to a large flux, particularly at low frequencies. It may be saturated to such an extent that the no-load current may be greater than rated current. With the reduced cooling at low speeds, this may lead to overheating.

To obtain a linear relation between V and f, V may also be varied as $V = V_o + K \cdot f$, where V_o is chosen to produce the nominal flux at zero speed and the constant K is chosen to get the rated terminal voltage at the rated frequency.

To get a high torque to current ratio, and a high efficiency and power factor, the motor is operated for $s < s_m$—that is, on the portion of speed-torque curves with a negative slope. Therefore, in figures 6.12a and b, only the portions with negative slope are shown. However, a complete characteristic is shown for the rated frequency to provide a comparison between the starting and low-speed torques available with variable frequency control and constant frequency operation. There is a large increase in the starting and low-speed torques with variable frequency control. The corresponding currents are also reduced by a large amount. Thus the starting and low-speed performance of a variable frequency drive is far superior compared to that with the fixed frequency operation.

Operation above the Rated Frequency ($a > 1$)

The operation at a frequency higher than the rated frequency takes place at a constant terminal voltage V_{rated} or at the maximum voltage available from the variable frequency source if it is less than V_{rated}. Since the terminal voltage is maintained constant, the flux decreases in the inverse ratio of per-unit frequency a. The motor, therefore, operates in the field weakening mode.

The torque expressions for the operation in this frequency range are obtained by the substitution of $a\omega_{ms}$ for ω_{ms} and $a(X_s + X_r')$ for $(X_s + X_r')$ in equations (6.52) and (6.53), giving

$$T = \frac{3}{\omega_{ms}}\left[\frac{V_{rated}^2 R_r'/(as)}{(R_s + R_r'/s)^2 + a^2(X_s + X_r')^2}\right], \quad a > 1 \tag{6.56}$$

$$T_{max} = \frac{3}{2\omega_{ms}a}\left[\frac{V_{rated}^2}{R_s \pm \sqrt{R_s^2 + a^2(X_s + X_r')^2}}\right], \quad a > 1 \tag{6.57}$$

Since $a > 1$, the breakdown torque decreases with the increase in frequency and speed. The speed-torque curves for operation for field weakening mode of operation are shown in figure 6.13. Here also the motor is made to operate for $s < s_m$ to get high torque per ampere, high efficiency, and a good power factor.

Torque and Power Capabilities

The torque and power variations for a given stator current and for frequencies below and above the rated frequency are shown by dots in figure 6.13. When the stator current has the maximum permissible value, these will represent the maximum torque and power capabilities of the machine. The variations of various variables, such as developed torque and power (P_m), slip speed $\omega_{s\ell}$, and terminal voltage V with per-unit frequency a, for the motor operation at a given stator current are shown in fig. 6.14. When the stator current has the maximum permissible value, these curves will give the maximum torque and power capabilities of the machine. These variations can be explained as follows.

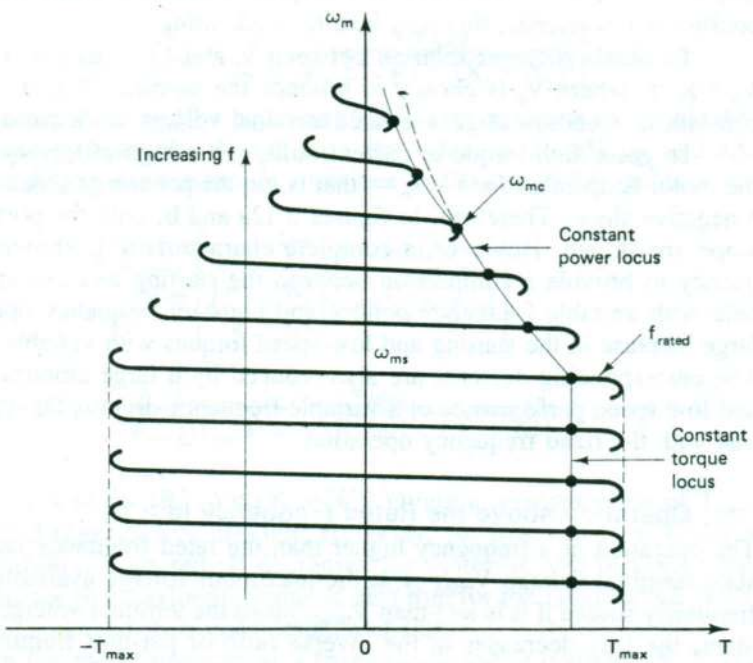


Figure 6.13 Speed-torque curves for variable frequency control of induction motor.

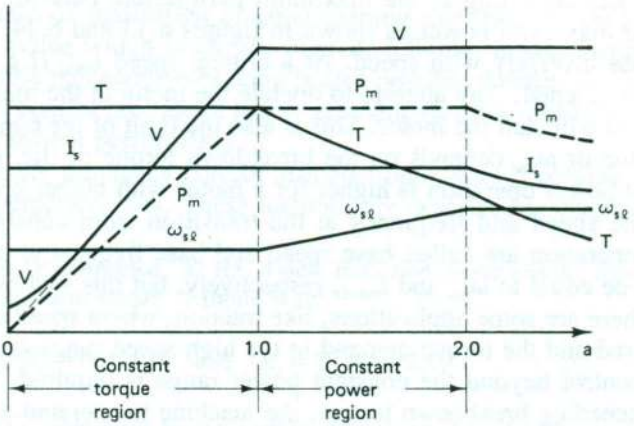


Figure 6.14 V, T, P_m, I_s and $\omega_{s\ell}$ versus per unit frequency, 'a', plots.

It is shown in the previous section that when the motor operates at a constant flux and a given stator current, the developed torque and slip speed have constant values at all frequencies. Thus for $a < 1$, the variable frequency control with a constant flux gives constant torque operation. When constant (V/f) control is used and the (V/f) ratio is increased at low frequencies to compensate for the stator resistance drop, at the maximum permissible current the drive operates essentially at a constant flux, providing constant torque operation.

For $a > 1$,

$$I_r' = \frac{V_{rated}}{\sqrt{\left(R_s + \frac{R_r'}{s}\right)^2 + a^2(X_s + X_r')^2}} \tag{6.58}$$

Since the slip is small,

$$I_r' = \frac{sV_{rated}}{R_r'} = \frac{V_{rated}}{R_r'}\left(\frac{a\omega_{ms} - \omega_m}{a\omega_{ms}}\right)$$

or

$$(a\omega_{ms} - \omega_m) = \omega_{s\ell} = \frac{R_r'\omega_{ms}}{V_{rated}}(aI_r')$$

Thus, at a given I_r' and hence at a given I_s, the slip speed $\omega_{s\ell}$ increases linearly with a.

Since the slip is small, I_r' is in phase with E. If the rotor resistance loss is neglected, the developed power P_m is given by

$$P_m = 3EI_r'.$$

If the stator drop is neglected, $E = V_{rated}$ and

$$P_m = 3V_{rated}I_r'$$

Consequently, P_m is constant for a given I_r', and therefore for a given I_s. Thus, for $a > 1$, the variable frequency control at a constant voltage gives constant power opera-

tion. When operating at the maximum permissible current, the motor develops a constant maximum power as shown in figures 6.13 and 6.14. The maximum torque decreases inversely with speed. At a critical speed ω_{mc} (fig. 6.13) the breakdown torque is reached. Any attempt to operate the motor at the maximum current beyond this speed will stall the motor. This is also the limit of the constant power operation. The value of ω_{mc} depends on the breakdown torque of the machine. The range of constant power operation is higher for a motor with higher breakdown torque.

The speed and frequency at the transition from constant torque to constant power operation are called base speed and base frequency, respectively. They will usually be equal to ω_{ms} and f_{rated}, respectively, but this will not always be so.

There are some applications, like traction, where speed control in a wide range is required and the torque demand in the high-speed range is low. For such applications, control beyond the constant power range is required. To prevent the torque from exceeding breakdown torque, the machine is operated at a constant slip speed and the machine current and power are allowed to decrease as shown in figure 6.14. Now, the motor current reduces inversely with speed; and the torque decreases inversely as the speed squared. This characteristic is often referred to as the series motor characteristic. The torque produced in this region is somewhat higher than that produced by a dc series motor.

Control and Advantages

To get the advantages of a high torque to current ratio, high efficiency, and a good power factor, the motor is always operated on the portion of the speed-torque curves with a negative slope, by limiting either the slip speed or the current.

Let us consider the operation of the drive for a change in speed command. When motoring, a decrease in the speed command decreases the supply frequency. This shifts the operation to regenerative braking (fig. 6.13). The drive decelerates under the influence of braking torque and load torque. For speeds below ω_{ms}, the voltage and frequency are reduced with speed to maintain the desired (V/f) ratio or constant flux, and to keep the operation on the portion of the speed-torque curves with a negative slope by limiting the slip speed. For speeds above ω_{ms}, the frequency alone is reduced with speed to maintain the operation on the portion of speed torque curves with a negative slope. When close to the desired speed, the operation shifts to the motoring operation and the drive settles at the desired speed. When regenerative braking is not possible due to the inability of the source to accept the energy, the motor torque can be made zero, at most. Then the deceleration occurs mainly due to the load torque.

When motoring, an increase in the speed command increases the supply frequency, The motor torque exceeds the load torque and the drive accelerates. Again the operation is maintained on the portion of the speed-torque curves with a negative slope by limiting the slip speed. The drive finally settles at the desired speed.

Usually a class B squirrel-cage motor is used. Some energy efficient applications may also employ the class A design. It may be recalled that these designs have low full-load slips, which result in high running efficiency and good speed regulation. Even with these designs, variable frequency control gives large torques with reduced currents for the complete range of speeds. Thus, variable frequency control

allows simultaneous realization of good running and transient performance from a squirrel-cage motor which is cheap, rugged, reliable, longer lasting, and maintenance free. Since regenerative braking is also possible down to zero speed, the variable frequency control provides a highly efficient variable speed drive with excellent running and transient performance.

The variable frequency control of induction motors is obtained by using either inverters or cycloconverters. These methods are described in chapter 8.

Example 6.3

A 3-phase, Y-connected, 60 Hz, 4-pole induction motor has the following parameters for the equivalent circuit of figure 6.1d:

$$R_s = R_r' = 0.024\ \Omega \quad \text{and} \quad X_s = X_r' = 0.12\ \Omega$$

The motor is controlled by the variable frequency control with a constant (V/f) ratio. For an operating frequency of 12 Hz, calculate

1. The breakdown torque as a ratio of its value at the rated frequency for both motoring and braking.
2. The starting torque and rotor current in terms of their values at the rated frequency.

Solution: From equation (6.38), $a = 12/60 = 0.2$

1. From equation (6.55), the ratio of breakdown torques for $a = 0.2$ and $a = 1$ is

$$\frac{T_{max}(a=0.2)}{T_{max}(a=1)} = \frac{0.024 \pm \sqrt{(0.024)^2 + (0.24)^2}}{\frac{0.024}{0.2} \pm \sqrt{\left(\frac{0.024}{0.2}\right)^2 + (0.24)^2}}$$

For motoring, $$\frac{T_{max}(a=0.2)}{T_{max}(a=1)} = 0.68$$

For braking, $$\frac{T_{max}(a=0.2)}{T_{max}(a=1)} = 1.46$$

2. Substitution of $s = 1$ in equation (6.54) gives an expression for the starting torque T_s. Thus

$$T_s = \frac{3}{\omega_{ms}} \left[\frac{V_{rated}^2 R_r'/a}{\left(\frac{R_s + R_r'}{a}\right)^2 + (X_s + X_r')^2} \right] \qquad \text{(E6.1)}$$

From equation (E6.1), the ratio of starting torques for $a = 0.2$ and $a = 1$ is

$$\frac{T_s(a=0.2)}{T_s(a=1)} = \frac{\frac{0.024/0.2}{(0.048/0.2)^2 + (0.24)^2}}{\frac{0.024}{(0.048)^2 + (0.24)^2}} = 2.6$$

The starting rotor current is given by

$$I'_{rs} = \frac{V_{rated}}{\sqrt{\left(\frac{R_s + R'_r}{a}\right)^2 + (X_s + X'_r)^2}} \tag{E6.2}$$

The ratio of starting currents for a = 0.2 and 1 is

$$\frac{I'_{rs}(a = 0.2)}{I'_{rs}(a = 1)} = \frac{\sqrt{(0.048)^2 + (0.24)^2}}{\sqrt{\left(\frac{0.048}{0.2}\right)^2 + (0.24)^2}} = 0.72$$

The preceding ratios of starting torques and starting rotor currents show that the constant (V/f) control provides a high starting torque with a reduced motor current.

Example 6.4

If the rated slip of the induction motor of problem 6.3 is 0.04, calculate the motor speed for rated torque and f = 30 Hz. The motor is controlled with a constant (V/f) ratio.

Solution:

$$a = 30/60 = 0.5$$

From equation (6.52) for rated torque and f = 60,

$$T_{rated} = \frac{3}{\omega_{ms}}\left[\frac{V^2_{rated}(0.024/0.04)}{\left(0.024 + \frac{0.024}{0.04}\right)^2 + (0.24)^2}\right] = \frac{3V^2_{rated}}{\omega_{ms}}(1.34) \tag{a}$$

and for 30 Hz from equation (6.54),

$$T_{rated} = \frac{3}{\omega_{ms}}\left[\frac{V^2_{rated}\left(\frac{0.024}{0.5\ s}\right)}{\left(\frac{0.024}{0.5} + \frac{0.024}{0.5\ s}\right)^2 + (0.24)^2}\right] \tag{b}$$

Equating equations (a) and (b) gives

$$\frac{\frac{0.024}{0.5\ s}}{\left(\frac{0.024}{0.5} + \frac{0.024}{0.5\ s}\right)^2 + (0.24)^2} = 1.34$$

or $26.04\ s^2 - 0.98\ s + 1 = 0$

or $s = \frac{14.62 \pm \sqrt{(14.62)^2 - 4 \times 26.04}}{2 \times 26.04} = 0.089$ or 0.43

The slip on the stable part of the speed torque curve will be 0.089.
Now synchronous speed for 30 Hz = 900 rpm.
Therefore, the motor speed = 900(1 − 0.089) = 820 rpm.

Example 6.5

A 400 V, 50 Hz, 6-pole, 960 rpm, Y-connected induction motor has the following parameters per phase referred to the stator:

$$R_s = 0.4\ \Omega, \qquad R_r' = 0.20\ \Omega, \qquad X_s = X_r' = 1.5\ \Omega, \qquad X_m = 30\ \Omega$$

The motor is controlled by variable frequency control at a constant flux of rated value.

1. Calculate the motor speed and the stator current at half the rated torque and 25 Hz.
2. Solve for part 1 assuming the speed-torque curves to be straight lines for $s < s_m$.
3. Calculate the frequency, the stator current, and voltage at the rated braking torque and 800 rpm.

Use the equivalent circuit of figure 6.1b.

Solution: From the rated conditions of operation,
Synchronous speed $N_s = 120f/p = 120 \times 50/6 = 1000$ rpm

$$\omega_{ms} = \frac{1000 \times 2\pi}{60} = 104.7 \text{ rad/sec.}$$

$$s = \frac{1000 - 960}{1000} = 0.04$$

Rotor impedance $Z_r' = \dfrac{R_r'}{s} + jX_r' = 5 + j1.5$

$$= 5.22\angle 16.7^\circ\ \Omega$$

Stator impedance $Z_s = 0.4 + j1.5 = 1.55\angle 75^\circ\ \Omega$

Machine impedance $Z_{in} = Z_s + \dfrac{Z_r' Z_m}{Z_r' + Z_m}$

$$= 4.82 + j3.63 = 6\angle 37^\circ\ \Omega$$

$$I_s = \frac{400/\sqrt{3}}{6} = 38.5 \text{ A}$$

$$I_r' = \left|\frac{Z_m}{Z_m + Z_r'}\right| I_s = \frac{30}{31.89} \times 38.5 = 36.22 \text{ A}$$

$$E = I_r' |Z_r'| = 36.22 \times 5.22 = 189 \text{ V}$$

$$\text{Rated torque} = \frac{3}{\omega_{ms}} I_r'^2 R_r'/s = \frac{3}{104.7} \times (36.22)^2 \cdot \frac{0.2}{0.04} = 188 \text{ N-m}.$$

1. At 25 Hz, $a = 25/50 = 0.5$

Substituting the known values in equation (6.44) gives

$$\frac{188}{2} = \frac{3}{104.7} \cdot \frac{(189)^2 \times 0.2/(0.5\ s)}{(0.2)^2/(0.5\ s)^2 + 1.5^2}$$

which gives $s = 0.0374$
From equation (6.43), $\omega_m = a\omega_{ms}(1 - s)$

$$\text{or} \quad N = aN_s(1 - s) = 0.5 \times 1000(1 - 0.0374)$$
$$= 481.3 \text{ rpm}$$

At 25 Hz, $E = 0.5 \times 189 = 94.5$ V

$$Z_r' = \frac{R_r'}{s} + jaX_r' = \frac{0.2}{0.0374} + j0.75 = 5.4\underline{/8°}\ \Omega$$

Taking E as a reference vector

$$I_r' = \frac{E}{Z_r'} = \frac{94.5}{5.4\underline{/-8°}} = 17.5\underline{/-8°}\ \text{A}$$

$$\bar{I}_m = \frac{E}{jaX_m} = \frac{94.5}{j0.5 \times 30} = 6.30\underline{/-90°}\ \text{A}$$

$$\bar{I}_s = \bar{I}_r' + \bar{I}_m = 17.5\underline{/-8°} + 6.30\underline{/-90°}$$

Hence $I_s = 19.85$ A.

2. Slip speed in rpm at the rated torque and frequency

$$N_{s\ell} = sN_s = 0.04 \times 1000 = 40 \text{ rpm}$$

Since the speed torque curve is a straight line, slip speed at half the rated torque $N_{s\ell 2} = 0.5 \times 40 = 20$ rpm. At 25 Hz, $N_s = 25/50 \times 1000 = 500$ rpm. Since the slip speed remains constant for a given torque,

$$\text{Motor speed } N = N_s - N_{s\ell 2} = 500 - 20 = 480 \text{ rpm}$$

For a constant flux, the (E/f) ratio must be constant.
Hence, at 25 Hz, $E = 0.5 \times 189 = 94.5$ V

$$s = \frac{N_{s\ell 2}}{N_s} = \frac{20}{500} = 0.04$$

$$Z_r' = \frac{R_r'}{s} + jaX_r' = 5 + j0.75 = 5.06\underline{/8.5°}\ \Omega$$

Taking E as a reference vector

$$\bar{I}_r' = \frac{E}{Z_r'} = \frac{94.5}{5.06\underline{/8.5°}} = 18.7\underline{/-8.5°}\ \text{A}$$

$$\bar{I}_m = \frac{E}{jaX_m} = \frac{94.5}{j15} = 6.30\underline{/-90°}\ \text{A}$$

$$\bar{I}_s = \bar{I}_r' + \bar{I}_m = 18.7\underline{/-8.5°} + 6.30\underline{/-90°}$$

Hence $I_s = 20.6$ A.

3. At the rated braking torque, the slip speed will be the negative of the slip speed at rated motoring torque.

Hence slip speed = $N_{sf3} = -40$ rpm
Synchronous speed = $N + N_{sf3} = 800 - 40 = 760$ rpm
Frequency = $(760/1000) \times 50 = 38$ Hz

$$a = 38/50 = 0.76$$

At 38 Hz, $E = (38/50) \times 189 = 143.64$ V, $s = -40/760 = -0.0526$

$$\bar{Z}'_r = \frac{R'_r}{s} + jaX'_r = -3.8 + j1.14 = 3.97\underline{/163.3°}\ \Omega$$

Taking E as a reference vector,

$$\bar{I}'_r = \frac{E}{\bar{Z}'_r} = \frac{143.64}{3.97\underline{/-163.3°}} = 36.2\underline{/-163.3°}\ A$$

$\bar{I}_m$ remains the same as the foregoing.

$$\bar{I}_s = 36.2\underline{/-163.3°} + 6.30\underline{/-90°}$$
$$= -34.62 - j16.88 = 38.52\underline{/-154°}$$
$$\bar{V} = \bar{E} + \bar{Z}_s\bar{I}_s = 143.64 + (0.4 + j0.76 \times 1.5) \times 38.52\underline{/-154°}$$

or $\bar{V} = 156\underline{/-17.3°}$ V

Note that the phase angle between $\bar{V}$ and $\bar{I}_s$ is more than 90°. Hence, power flows from the motor to the source.

6.4.3 Rotor Resistance Control

The speed-torque and speed-rotor current characteristics of a wound-rotor induction motor with rotor resistance control are shown in figure 6.5 and explained in section 6.1.4. For a given load torque, the motor speed is reduced as the rotor resistance is increased. The no-load speed, however, remains unaffected by the variation of the rotor resistance.

Equations (6.19) and (6.37) are also applicable to the rotor resistance control. These equations show that the motor efficiency drops and the rotor copper loss increases with the decrease in speed. Thus, rotor resistance control is an inefficient method of speed control like terminal voltage control. It has, however, a number of advantages over terminal voltage control. It provides a constant torque operation with a high torque to current ratio. Though the rotor copper loss increases with the decrease in speed, most of it is dissipated in the external resistors. For a fixed torque, the copper loss inside the motor in fact remains constant. Because of this, a motor of smaller size can be employed.

The rotor resistance control is implemented by a diode bridge and a chopper. It is described in chapter 9.

6.4.4 Injection of Voltage in the Rotor Circuit

Figure 6.15a shows the equivalent circuit of a wound-rotor induction motor with an injected voltage of $V_r\underline{/\phi_r}$ volts per phase. Consider the ideal no-load operation for

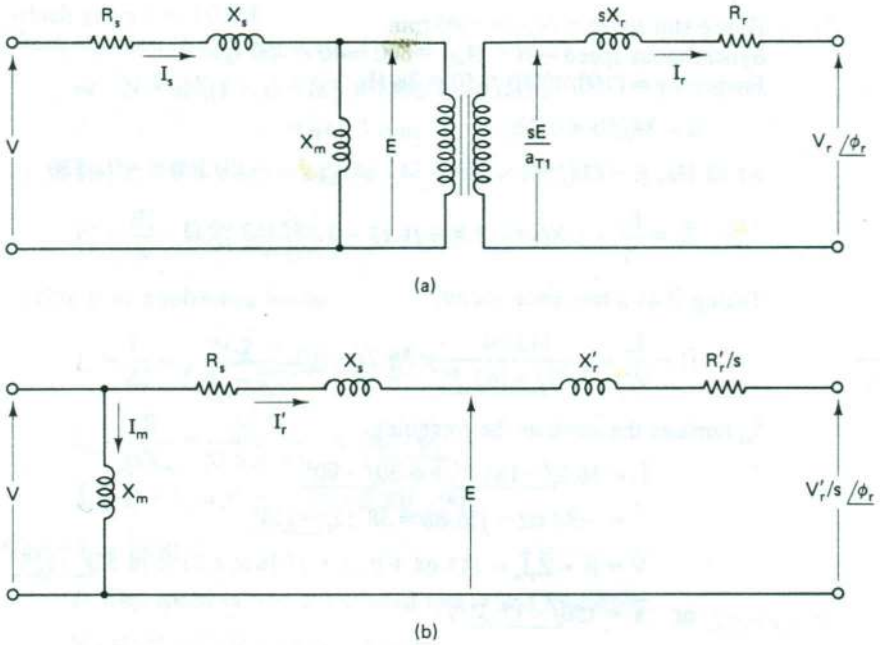


Figure 6.15 Induction motor equivalent circuits with rotor-injected voltage.

which I_r must be zero. In the absence of the injected voltage, I_r is zero when the motor speed is equal to the synchronous speed. When V_r is in phase with E, I_r is zero when $sE/a_{T1} = V_r$
or

$$s = \frac{a_{T1}V_r}{E} \tag{6.59a}$$

The no-load speed is

$$\omega_{mo} = \left(1 - \frac{a_{T1}V_r}{E}\right)\omega_{ms} \tag{6.59b}$$

According to equation (6.59b), the no-load speed can be changed from synchronous to standstill by varying V_r from 0 to (E/a_{T1}). Further, if V_r is reversed, s will be negative and the motor no-load speed will be higher than the synchronous speed. The relative speed between the stator field and the rotor will now be the opposite of that for speeds less than the synchronous speed. Hence, the phase sequence and direction of the rotor induced voltages will also be opposite. Thus, for operation above synchronous speed, both the polarity and the phase sequence of the injected voltage will have to be changed. Further, as the speed changes, the frequency of the rotor induced voltage changes. For the injected voltage to balance the induced voltage, the frequency of the injected voltage must track the frequency of the induced voltage.

Let us now consider the operation with a fixed V_r of positive polarity and phase sequence. The no-load speed of the motor will be less than the synchronous speed. An application of a positive load torque will reduce the motor speed from its no-load

speed, causing the slip and the rotor induced emf to increase. A positive rotor current will flow and motoring torque will be produced. As the load torque is increased, the motor speed falls to compensate for the increase in the machine impedance drop due to the increase in the rotor current. The higher the load torque, the greater the drop in speed from the no-load speed. This is the motor's subsynchronous motoring mode of operation. Let P_r denote the power absorbed by the source V_r. In this mode of operation, both P_r and P_m are positive. Also V_r has a positive polarity and positive phase sequence (figure 6.16a).

Let the load torque be removed. This will restore the operation to the no-load speed. Now let a negative load torque be applied. A negative load torque will increase the motor speed. Consequently, the rotor induced emf will decrease (because the motor is running at a speed less than the snychronous speed) and a negative rotor current will flow. Because of the negative rotor current, the machine will develop a negative torque and operate under regenerative braking. The motor will run at a speed higher than the no-load speed but less than the synchronous speed. The higher the magnitude of the negative load torque, the greater will the increase be in speed from the no-load speed. In this mode of operation, known as subsynchronous regenerative braking, both P_r and P_m are negative and V_r, which acted as a sink of power for subsynchronous motoring, acts as a source of power now (figure 6.16b).

Let us now consider the operation of the drive when the polarity and phase sequence of V_r are reversed. The no-load speed is obtained from equation (6.59b). It will be higher than the synchronous speed. The induced voltage will have negative polarity and negative phase sequence. Let a positive load torque be applied. The mo-

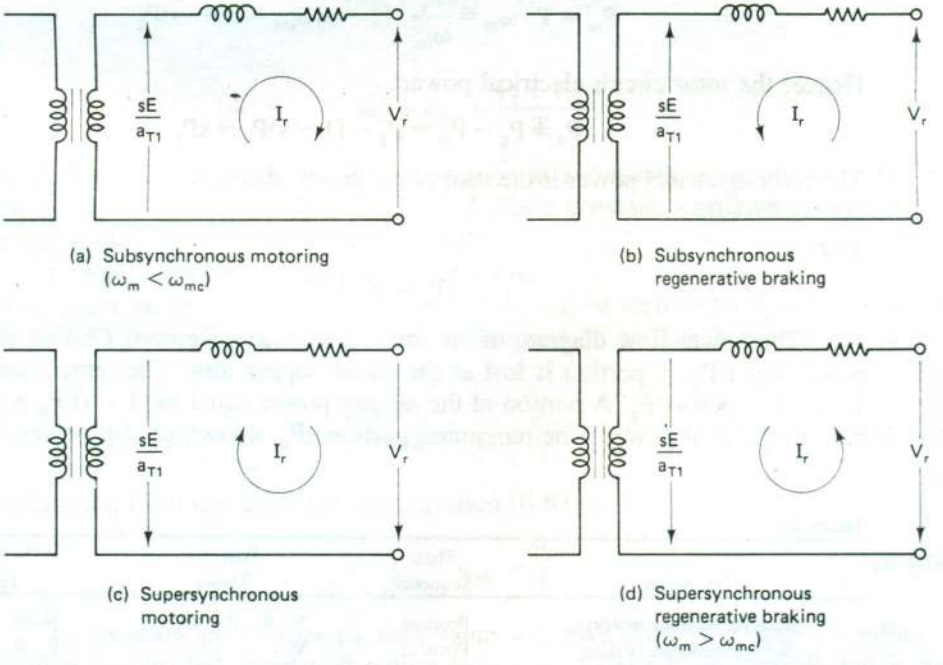


Figure 6.16 Polarities of rotor variables for different modes of drive operation for a constant magnitude of V_r and variable load torque.

tor will slow down. The induced voltage magnitude will decrease and a positive rotor current will flow, producing motoring torque. The motor will run at a speed less than the no-load speed but greater than the synchronous speed. The speed will fall with an increase in the load torque. This mode of operation is known as supersynchronous motoring. Here P_m is positive and P_r is negative. The voltage V_r has negative polarity and negative phase sequence (figure 6.16c).

Let the load torque be removed. This will restore the operation to the no-load speed which is higher than the synchronous speed. Now let a negative load torque be applied. This will increase the motor speed and the induced voltage magnitude will increase. A negative rotor current will flow, producing a negative torque and regenerative braking. The higher the load torque, the greater will the increase be of motor speed beyond the no-load speed. In this mode of operation, known as supersynchronous regenerative braking, P_m is negative and P_r is positive. The voltage V_r has negative polarity and negative phase sequence (figure 6.16d). V_r, which acted as a source of power for supersynchronous motoring now acts as a sink of power.

Table 6.2 summarizes the polarity and phase sequence of the injected voltage V_r for the drive's four modes of operation. It also gives the signs of the power terms P_r, P_m, and P_g (air-gap power) when the rotor copper loss P_{cr} is neglected. The speed-torque curves for three values of V_r are shown later in figure 6.18.

For a air-gap power P_g,

$$T = \frac{P_g}{\omega_{ms}} \tag{6.16}$$

$$P_m = T \cdot \omega_m = \frac{P_g}{\omega_{ms}} \cdot (1 - s)\omega_{ms} = (1 - s)P_g \tag{6.15}$$

Hence, the rotor circuit electrical power

$$P_e = P_g - P_m = P_g - (1 - s)P_g = sP_g \tag{6.60}$$

The rotor electrical power is the sum of the power absorbed by $V_r (= P_r)$ and the rotor copper loss P_{cr}.
Thus,

$$sP_g = P_r + P_{cr} \tag{6.61}$$

The power flow diagram of the drive is shown in figure 6.17. Out of the total power input P_{in}, a portion is lost as the stator copper loss. The remaining power is the air-gap power P_g. A portion of the air-gap power equal to $(1 - s)\,P_g$ is converted into mechanical power. The remaining portion sP_g, known as slip power, is used to

Table 6.2

Operation	Phase Sequence	Positive Terms	Negative Terms
1. Subsynchronous motoring	Positive	V_r, P_g, P_m, sP_g, P_r	None
2. Subsynchronous braking	Positive	V_r	P_g, P_m, sP_g, P_r
3. Supersynchonous motoring	Negative	P_g, P_m	V_r, sP_g, P_r
4. Supersynchronous braking	Negative	sP_g, P_r	V_r, P_m, P_g

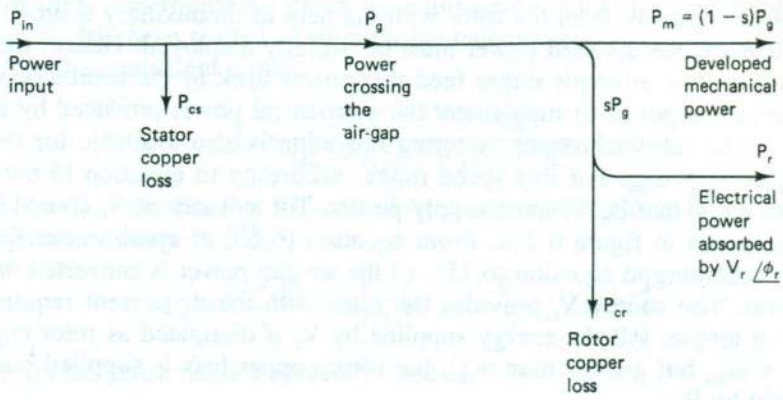


Figure 6.17 Power flow diagram for an induction motor with a rotor injected voltage.

supply rotor copper loss P_{cr} and the power absorbed by the auxiliary source V_r ($= P_r$). The power flow diagram of figure 6.17 is applicable to all the four modes of operation when the appropriate signs are assigned to the power terms. From figure 6.17,

$$P_m = P_g - P_r - P_{cr} \tag{6.62}$$

or

$$T\omega_m = T\omega_{ms} - P_r - P_{cr}$$

or

$$\omega_m = \omega_{ms} - \frac{P_r + P_{cr}}{T} \tag{6.63}$$

Let us now examine the drive operation at a constant value of torque for the four modes of operation when V_r is varied. For a constant magnitude of torque, the air-gap power P_g ($= T\omega_{ms}$) has a fixed value.

1. Subsynchronous Motoring ($1 \geq s \geq 0$):

When operating at a fixed value of torque T, P_r will be zero when $V_r = 0$. The motor speed for $V_r = 0$, ω_{mc} is obtained by the substitution of $P_r = 0$ in equation (6.63). Thus

$$\omega_{mc} = \omega_{ms} - \frac{P_{cr}}{T} \tag{6.64a}$$

Substituting from this equation into equation (6.63) yields

$$\omega_m = \omega_{mc} - \frac{P_r}{T} \tag{6.64b}$$

An increase in V_r from its zero value will give positive P_r and according to equation (6.64b) the speed will reduce below ω_{mc}. Since P_g is constant and P_r has increased, according to equation (6.62), P_m will reduce. Thus, in subsynchronous motoring the speed control is obtained essentially by diverting a portion of the air-

gap power away from the rotor with the help of an auxiliary source. To have good efficiency, the diverted power must be usefully employed. Hence, the drives which work on this principle either feed this power back to the source or convert it into mechanical power to supplement the mechanical power produced by the motor.

The subsynchronous motoring operation is also available for the speed range $\omega_{mc} \leq \omega_m \leq \omega_{ms}$. For this speed range, according to equation (6.64b), P_r must be negative — that is, V_r must supply power. The polarity of V_r should be opposite to that shown in figure 6.16a. From equation (6.63) at synchronous speed $P_r = -P_{cr}$ and according to equation (6.15), all the air-gap power is converted into mechanical power. The source V_r provides the rotor with the dc current required to produce given torque. All the energy supplied by V_r is dissipated as rotor copper loss. For $\omega_m < \omega_{ms}$ but greater than ω_{mc}, the rotor copper loss is supplied partly by V_r and partly by P_g.

2. Supersynchronous Motoring ($s < 0$):

Since for all operating points, $\omega_m > \omega_{mc}$ (because for a positive T, $\omega_{ms} > \omega_{mc}$ from equation (6.64a)), equation (6.64b) shows that P_r is negative throughout. In addition to the air-gap power ($= T\omega_{ms}$), an additional power $[T(\omega_m - \omega_{ms})]$ is converted into mechanical power, The additional power and the rotor copper loss are supplied by the source V_r (figure 6.16c). An increase in the magnitude of V_r increases P_m and the motor speed for a given torque.

3. Subsynchronous Braking ($1 \geq s \geq 0$):

For the braking operation, torque T, braking power P_m, and air-gap power P_g are negative. However, the magnitude of the air-gap power for a given torque is still constant. Therefore, the power supplied by the rotor to the stator is also constant. According to equation (6.64a), the speed ω_{mc} for which $P_r = 0$ is greater than synchronous speed because T is negative. From equation (6.64b) for a speed less than synchronous speed, P_r is negative. By increasing V_r the magnitude of P_r is increased (figure 6.16b). Since the magnitude of P_g is constant, the magnitude of P_m is decreased (equation 6.62). Consequently, the motor speed for a given torque is reduced. At the synchronous speed, V_r supplies rotor copper loss and provides a negative dc current to produce the given braking torque (because $E = 0$ in figure 6.16b).

4. Supersynchronous Braking ($s < 0$):

Again here the air-gap power is negative and constant. T and P_m are also negative. The speed ω_{mc} is greater than synchronous speed (equation (6.64a)). For $\omega_m > \omega_{mc}$, P_r is positive from equation (6.64b). From figure 6.16d V_r is negative. If the magnitude of V_r is increased to increase P_r, from equation (6.62) the magnitude of P_m must increase, because the magnitude of P_g is constant. Consequently, motor speed increases. For $\omega_{mc} > \omega_m > \omega_{ms}$, V_r will be positive. The rotor copper loss is supplied partly by V_r and partly by P_m (figure 6.16d). At $\omega_m = \omega_{ms}$, the rotor copper loss is entirely supplied by V_r.

In the foregoing discussion, V_r was considered in phase or in phase opposition to E. A more general case is when both the magnitude and phase of the injected voltage are controlled. Let the injected voltage be $\underline{V_r/\phi_r}$, where ϕ_r is the phase angle between $\overline{V_r}$ and the motor terminal phase voltage $\overline{V}$. Referring the rotor quantities to

the stator turns and frequency gives the equivalent circuit shown in figure 6.15b. Note that X_m has been shifted to simplify the calculations.

From the equivalent circuit,

$$\bar{I}'_r = \frac{V\underline{/0^\circ} - (V'_r/s)\underline{/\phi_r}}{(R_s + R'_r/s) + j(X_s + X'_r)} \tag{6.65}$$

Equation (6.65) indicates that by controlling ϕ_r, the motor power factor can be controlled. From equation (6.61)

$$P_g = \frac{P_r + P_{cr}}{s} = 3\left[I'^2_r R'_r/s + \frac{V'_r}{s} I'_r \cos\theta'_r\right] \tag{6.66}$$

where θ'_r is the phase angle between $\bar{V}'_r$ and $\bar{I}'_r$.

$$T = \frac{P_g}{\omega_{ms}} = \frac{3}{s\omega_{ms}}[I'^2_r R'_r + V'_r I'_r \cos\theta'_r] \tag{6.67}$$

Also from equations (6.60) and (6.66),

$$P_e = sP_g = 3[I'^2_r R'_r + V'_r I'_r \cos\theta'_r] \tag{6.68}$$

Speed torque curves of a 10 kW induction motor for V'_r equal to $\pm 0.5\ V\underline{/0^\circ}$ and 0 are shown in figure 6.18. Supersynchronous braking torque is very small, except for small values of $|s|$ and $|V_r|$. Note that the curve for $V'_r = 0$ is nothing but the locus of ω_{mc} for different values of T.

By making the injected voltage track the rotor induced voltage frequency and phase sequence, it has been ensured that for all speeds, the rotor mmf wave remains

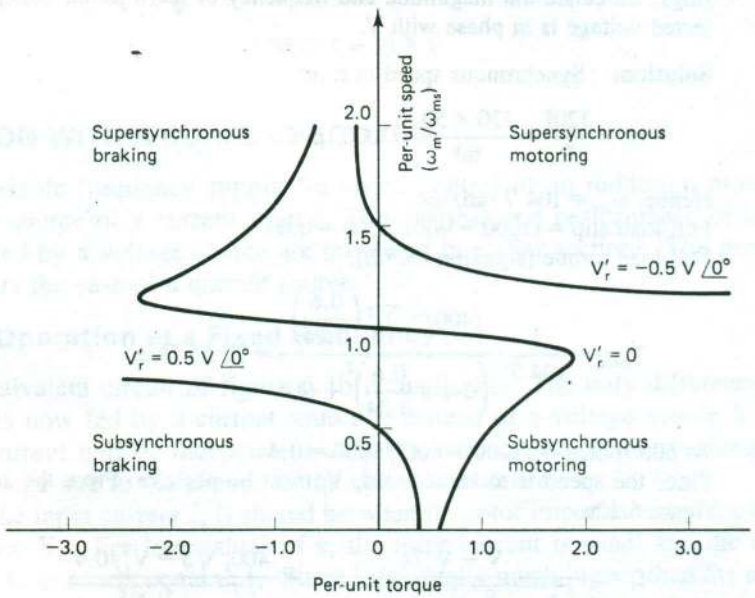


Figure 6.18 Speed control by injection of voltage in rotor.

stationary with respect to the air-gap flux wave. Hence, a steady torque is produced at all speeds. Thus, the main feature of the induction motor—that is, the ability to produce steady torque for all speeds—has been retained in this operation. This operation can be considered a true induction mode of operation to distinguish it from the synchronous mode of operation described next.

If the frequency of the injected voltage is made independent of the stator frequency, then the motor operates at a fixed speed for which the stator and rotor fields are stationary with respect to each other. For example, for an injected voltage frequency of 30 Hz, a 60 Hz induction motor will run either at 0.5 ω_{ms} or 1.5 ω_{ms} depending on whether the injected voltage has the same or opposite phase sequence compared to the stator supply. For any other speed, the motor will behave like a synchronous motor under pull-out. Hence, the operation of the induction motor with an independent control of the frequency of the injected voltage can be considered a synchronous mode of operation.

The true induction mode is always preferred because it does not suffer from the instability problems associated with the synchronous mode.

The implementation of speed control of the induction motor by voltage injection is considered in chapter 9.

Example 6.6

A 3-phase, 400 V, 50 Hz, 6-pole, 10 kW, 960 rpm, star-connected wound-rotor induction motor has the following parameters:

$$R_s = 0.4\ \Omega, \qquad R_r' = 0.6\ \Omega, \qquad X_s = X_r' = 2\ \Omega$$

Stator to rotor turns ratio is 2.5.

The speed at full-load torque is reduced to 600 rpm by injecting a voltage into the slip rings. Calculate the magnitude and frequency of the injected voltage. Assume the injected voltage is in phase with V.

Solution: Synchronous speed in rpm:

$$N_s = \frac{120f}{P} = \frac{120 \times 50}{60} = 1000 \text{ rpm}$$

Hence, $\omega_{ms} = 104.7$ rad/sec.

Full-load slip $= (1000 - 960)/1000 = 0.04$

Full-load torque [equation (6.52)],

$$T_{rated} = \frac{3}{104.7} \cdot \frac{(400/\sqrt{3})^2 \left(\frac{0.6}{0.04}\right)}{\left(0.4 + \frac{0.6}{0.04}\right)^2 + (4)^2} = 90.55 \text{ N-m}$$

At 600 rpm, $s = (1000 - 600)/1000 = 0.4$

Since the speed is to be reduced, V_r must be positive. From the equivalent circuit of figure 6.15b

$$I_r' = \frac{V - V_r'/s}{\sqrt{\left(R_s + \frac{R_r'}{s}\right)^2 + (X_1 + X_r')^2}} = \frac{400/\sqrt{3} - V_r'/0.4}{\sqrt{\left(0.4 + \frac{0.6}{0.4}\right)^2 + 16}}$$

$$\bar{I}_r' = \frac{400/\sqrt{3} - V_r'/0.4}{4.43} \angle -64.6°$$

From equation (6.67)

$$T = \frac{3}{s\omega_{ms}}[I_r'^2 R_r' + V_r' I_r' \cos(64.6°)]$$

$$= \frac{3}{0.4 \times 104.7}\left[\left(\frac{400/\sqrt{3} - V_r'/0.4}{4.43}\right)^2 \times 0.6 + V_r'\left(\frac{400/\sqrt{3} - V_r'/0.4}{4.43}\right) \times 0.428\right]$$

$$= 0.0716(-0.052V_r'^2 - 12.87V_r' + 1680.5)$$

Since a torque equal to full-load torque is required,

$$0.0716(-0.052V_r'^2 - 12.87V_r' + 1630.5) = 90.55$$

or

$$V_r'^2 + 247.5V_r' - 7036 = 0$$

which gives

$$V_r' = \frac{-247.5 \pm \sqrt{247.5^2 + 4 \times 7036}}{2}$$

$$= 25.75 \quad \text{or} \quad -273.25$$

Negative V_r' is for supersynchronous operation. Hence, for 600 rpm,

$$V_r' = 25.75 \text{ V}$$

$$V_r = 25.75/a_{TI} = 25.75/2.5 = 10.3 \text{ V}$$

6.5 OPERATION WITH A CURRENT SOURCE

The variable frequency supply for speed control of an induction motor can be a voltage source or a current source. The analysis and performance of an induction motor fed by a voltage source are presented in earlier sections. The present section considers the case of a current source.

6.5.1 Operation at a Fixed Frequency

The equivalent circuit of figure 6.1b is applicable. The only difference is that the motor is now fed by a current source I_s instead of a voltage source V. The motor input current will be independent of motor parameters and the terminal voltage V will change due to the change in the motor impedance.

The input current I_s is shared between the rotor impedance and the magnetizing reactance X_m. For low values of s, the rotor current is small and the magnetizing current I_m is nearly equal to I_s. Since I_s is usually much higher than the normal magnetizing current, the motor operates under saturation for low values of slip. Therefore, the motor should be analyzed taking the saturation into account. The nonlinear

relationship between E and I_m is obtained experimentally as explained in section 6.3.3. The following equations can be written from the equivalent circuit (fig. 6.1b):

$$\left[\left(\frac{R_r'}{s}\right)^2 + X_r'^2\right] I_r'^2 = E^2 \tag{6.69}$$

$$\left[\left(\frac{R_r'}{s}\right)^2 + (X_r' + X_m)^2\right] I_r'^2 = I_s^2 X_m^2 \tag{6.70}$$

$$E = I_m X_m \tag{6.71}$$

Subtracting equation (6.69) from (6.70) and then substituting from equation (6.71) gives

$$I_r'^2 = \frac{I_s^2 - I_m^2}{1 + \dfrac{2X_r'}{X_m}} \tag{6.72}$$

Also from equation (6.69),

$$s = \frac{R_r'}{\sqrt{(E/I_r')^2 - X_r'^2}} \tag{6.73}$$

Now,

$$T = \frac{3}{\omega_{ms}}\left(I_r'^2 \frac{R_r'}{s}\right) \tag{6.74}$$

Motor input impedance

$$\bar{Z}_{in} = (R_s + jX_s) + \frac{jX_m\left(\dfrac{R_r'}{s} + jX_r'\right)}{R_r'/s + j(X_m + X_r')} = R_{in} + jX_{in} \tag{6.75}$$

where

$$R_{in} = R_s + \frac{\dfrac{R_r'}{s} X_m^2}{(R_r'/s)^2 + (X_m + X_r')^2} \tag{6.76}$$

$$X_{in} = X_s + \frac{X_m[(R_r'/s)^2 + X_r'(X_m + X_r')]}{(R_r'/s)^2 + (X_m + X_r')^2} \tag{6.77}$$

$$V = I_s(R_{in}^2 + X_{in}^2)^{1/2} \tag{6.78}$$

Equations (6.69) to (6.75) are nonlinear algebraic equations due to the nonlinear relations of E and X_m with I_m. To avoid the need for a numerical solution, the calculations can be done in the following sequence.

A suitable value (less than I_s) is assumed for I_m for a given I_s; E and X_m are obtained from the magnetization characteristic; I_r' is calculated from equation (6.72); s is evaluated from equation (6.73), and then T and V are obtained from equations (6.74) and (6.78), respectively.

The speed-torque curves of a 2.8 kW, 400 V, 50 Hz, Y-connected induction motor at different per unit values of I_s and the rated frequency are shown in figure 6.19. The per-unit value of I_s is obtained by dividing the actual value by the rated stator current. The speed-torque and speed-current curves of the motor when fed by a voltage source of rated voltage and frequency are also shown by dash-dots for the sake of comparison. The variation of the terminal voltage for low values of slip and two values of I_s is also shown. When fed by a voltage source of rated voltage and frequency, the motor operates nearly at the nominal value of flux and I_m for all loads, where the nominal values are defined as the values attained at the rated voltage and frequency on no load.

With a constant current source, the starting torque is low due to the low values of flux (as I_m is low) and rotor current compared to their values at the rated voltage. The torque increases with speed due to the increase in flux. The flux and I_m have nominal values at the intersections with the rated voltage speed-torque curve. A further increase in speed increases the terminal voltage beyond the rated value. The

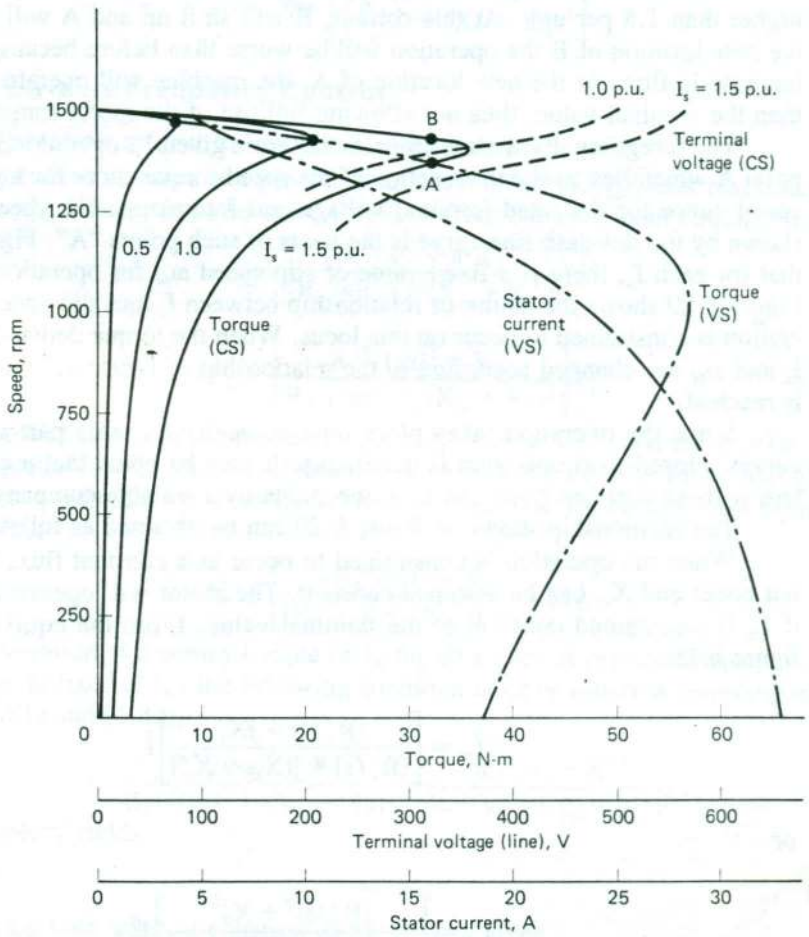


Figure 6.19 Induction motor operation with a current source.

flux and the magnetizing current are also increased beyond nominal values, and the motor saturates. Because of the saturation, the increase in terminal voltage and torque is much lower than what would be predicted if the saturation is neglected.

Let us examine points A and B on the speed-torque curve for a constant current $I_s = 1.5$ per unit. At both these points, the motor develops the same torque. Point A also lies on the motor speed-torque curve for the rated terminal voltage and frequency. Hence, at point A the motor operates at nominal flux and rated terminal voltage. At point B, the motor terminal voltage will be higher than the rated value and the flux will be more than the nominal value. The machine will operate under saturation and core loss will be higher than at point A. Point B will also provide stable operation with most loads. On the other hand, the rotor copper loss will be slightly higher at point A. The stator copper loss will be the same at both points. On the whole, losses will be less and saturation will be avoided if the machine is operated at point A. Because of these advantages, the operation at point A is preferred, even though it lies on the statically unstable part of the speed-torque curve.

Let us now consider the motor operation for the same developed torque but I_s higher than 1.5 per unit. At this current, B will shift up and A will shift down. At the new location of B the operation will be worse than before because of the further increase in flux. At the new location of A, the machine will operate at a flux lower than the nominal value, thus not allowing full use of the motor torque capability.

The foregoing discussion suggests that for a given I_s, operation is preferred at a point A which lies on the intersection of the speed-torque curve for a given I_s and the speed curve for the rated terminal voltage and frequency. The speed-torque curve shown by the dot-dash line curve is the locus of such points "A". Figure 6.19 shows that for each I_s, there is a fixed value of slip speed $\omega_{s\ell}$ for operation on this locus. Figure 6.20 shows the nature of relationship between I_s and slip speed when the operation is constrained to occur on this locus. When the torque demand changes, both I_s and $\omega_{s\ell}$ are changed according to the relationship of figure 6.20 until equilibrium is reached.

Since the operation takes place on a statically unstable part of speed-torque curves, closed-loop operation is mandatory. It may be noted that a closed-loop system with an unstable plant can be made stable by a suitable compensator.

The relationship shown in figure 6.20 can be obtained as follows.

When the operation is constrained to occur at a constant flux, saturation does not occur and X_m can be assumed constant. The motor will operate at nominal flux if I_m is maintained constant at the nominal value. From the equivalent circuit of figure 6.1b,

$$\bar{I}_m = \left[\frac{(R_r'/s) + jX_r'}{(R_r'/s) + j(X_m + X_r')}\right]\bar{I}_s$$

or

$$I_m^2 = \left[\frac{(R_r'/s)^2 + X_r'^2}{(R_r'/s)^2 + (X_m + X_r')^2}\right]I_s^2 \qquad (6.79)$$

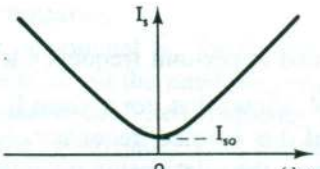


Figure 6.20

From equation (6.79), for a given I_s one can calculate s which will provide operation at a constant flux of the nominal value. Now,

$$T = \frac{3}{\omega_{ms}} I_r'^2 \frac{R_r'}{s}$$

Substituting for I_r' from equation (6.70) yields

$$T = \frac{3}{\omega_{ms}} \left[\frac{I_s^2 X_m^2 R_r'/s}{\left(\frac{R_r'}{s}\right)^2 + (X_r' + X_m)^2} \right] \tag{6.80}$$

6.5.2 Variable Frequency Control

Operation at and below Rated Frequency

When the drive operates at a per-unit frequency "a" ($= f/f_{rated}$), any reactance X will become aX. Hence, from equation (6.79) or from the equivalent circuit of 6.1b,

$$I_m^2 = \frac{(R_r'/s)^2 + a^2 X_r'^2}{(R_r'/s)^2 + a^2 (X_m + X_r')^2} I_s^2$$

or

$$I_m^2 = \left[\frac{R_r'^2/(sa)^2 + X_r'^2}{R_r'^2/(sa)^2 + (X_m + X_r')^2} \right] I_s^2 \tag{6.81}$$

Let us now consider the operation at a given I_s — say at I_{s1} — and variable frequency. From equation (6.79) one can obtain the value of slip for a = 1, which will give operation at a nominal value of I_m for $I_s = I_{s1}$. Let this slip be s_1. Then from equation (6.79),

$$I_m^2 = \left[\frac{(R_r'^2/s_1)^2 + X_r'^2}{(R_r'/s_1)^2 + (X_m + X_r')^2} \right] I_{s1}^2 \tag{6.82}$$

To get operation at a nominal value of I_m for all values of per-unit frequency a, with I_s remaining fixed at I_{s1}, the following condition must be satisfied according to equations (6.81) and (6.82):

$$\frac{R_r'^2/(sa)^2 + X_r'^2}{R_r'^2/(sa)^2 + (X_m + X_r')^2} = \frac{R_r'^2/s_1^2 + X_r'^2}{R_r'^2/s_1^2 + (X_m + X_r')^2}$$

This equation yields

$$sa = s_1 \tag{6.83}$$

Multiplying both sides of equation (6.83) by ω_{ms} gives

$$as\omega_{ms} = s_1 \omega_{ms}$$

or

$$\text{slip speed at per-unit frequency a} = \text{slip speed at rated frequency} \qquad (6.84)$$

Equation (6.84) shows that, for a given I_s, the slip speed which provides motor operation at nominal flux at rated frequency also gives operation at nominal flux at all frequencies. Hence, the relationship obtained between I_s and $\omega_{s\ell}$ at rated frequency for the operation at nominal flux is valid for all frequencies.

The relation of equation (6.83) can also be derived directly from the equivalent circuit of figure 6.1b. When the motor operates at a given I_s and variable frequency, I_m will be held constant if the ratio between the rotor impedance and the magnetizing reactance is kept fixed. Since the reactances change in proportion to a, slip must change in inverse proportion to a. This gives equation (6.83).

When operating at a per-unit frequency a, ω_{ms} and any reactance X in equation (6.80) should be replaced by $a\omega_{ms}$ and aX, respectively, giving

$$T = \frac{3}{\omega_{ms}} \left[\frac{I_s^2 X_m^2 R_r'/(sa)}{\left(\frac{R_r'}{sa}\right)^2 + (X_r' + X_m)^2} \right] \qquad (6.85)$$

For a given I_s, if slip speed is maintained constant, torque will also be constant for all frequencies.

By controlling slip speed as a function of I_s, the motor is made to operate at a constant flux. Hence, though it is fed by a current source, performance is identical to that on a variable frequency voltage source described in section 6.4.2. It can be shown that equations (6.44) and (6.85) are also identical.

The speed-torque curves for a constant flux operation are shown in figure 6.21. The figure also shows how these curves are obtained by selecting constant flux operating points for different I_s.

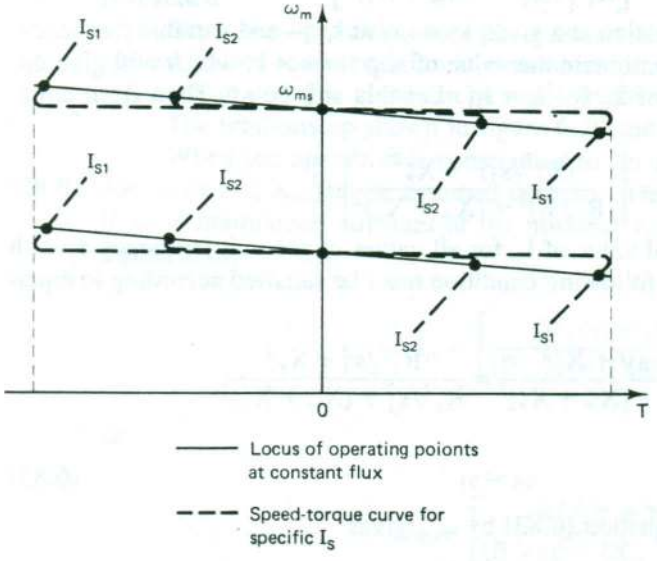


Figure 6.21 Speed-torque curves at a constant flux for induction motor fed by a variable frequency current source.

Operation above Rated Frequency

When operating at a nominal flux or nominal I_m, for $0 < a < 1$, the motor terminal voltage changes as shown in figure 6.14. At the rated frequency, it becomes equal to the rated value. Hence, operation above the rated frequency is carried out with the terminal voltage held constant at the rated value. With the operation constrained to occur at a fixed voltage, the machine behavior is identical to that when fed by a constant voltage variable frequency source, described in section 6.4.2 and shown in figures 6.13 and 6.14. The operation at the maximum permissible current gives operation at a constant maximum power. The maximum torque decreases inversely with speed. To get a constant terminal voltage, the machine impedance must be held constant as frequency is increased. This is achieved by increasing the slip speed to compensate for an increase in reactances. At a certain speed the breakdown torque is reached. The machine should now be operated at a constant slip speed and the current should decrease with the increase in frequency to keep the terminal voltage constant.

Variable frequency current sources are obtained by using either current source inverters or a cycloconverter. The induction motor operation with these current sources is described in chapter 8.

Example 6.7

The motor of example 6.5 is fed by a variable frequency current source. At all operating points, the motor is made to operate at the rated flux.

1. Calculate slip speed for $I_s = 60$ A.
2. Calculate the frequency and stator current for operation at 500 rpm for the following torque values:
   (a) 139 N-m
   (b) −188 N-m
3. Also obtain the solution of 2(a) assuming speed-torque curves to be straight lines in the region of interest.

Solution: From example 6.5 for 50 Hz operation $N_s = 1000$ rpm, $\omega_{ms} = 104.7$ rad/sec.
Rated torque = 188 N-m
Slip speed at rated torque = 40 rpm
E at rated conditions = 189 V

1. $I_m = E/X_m = 189/30 = 6.3$ A

Substituting the known values in equation (6.79) gives

$$(6.30)^2 = \left[\frac{(0.2/s)^2 + (1.5)^2}{(0.2/s)^2 + (31.5)^2}\right] \times (60)^2$$

This gives $s = 0.067$
Slip speed in rpm = sN_s rpm = $0.067 \times 1000 = 67$ rpm.

2.(a) Since the flux is constant for a given torque, the slip speed will also be constant for all frequencies. Thus the slip speed can be evaluated from the rated frequency operation.

Now

$$T = \frac{3}{\omega_{ms}}\left[\frac{E_{rated}^2 R_r'/s}{(R_r'/s)^2 + X_r'^2}\right]$$

Substituting the known values

$$139 = \frac{3}{104.7}\cdot\frac{(189)^2 \times (0.2/s)}{(0.2/s)^2 + (1.5)^2}$$

or

$$\left(\frac{0.2}{s}\right)^2 + (1.5)^2 = \left(\frac{1.472}{s}\right)$$

which gives s = 0.0284
Slip speed $N_{s\ell 1} = 0.0284 \times 1000 = 28.4$ rpm
Now for operation at 500 rpm:
Synchronous speed $N_s = N + N_{s\ell 1} = 500 + 28.4 = 528.4$ rpm
Frequency = $(528.4/1000) \times 50 = 26.42$ Hz

$$a = 26.42/50 = 0.528$$

$$s = \frac{N_{s\ell 1}}{N_s} = \frac{28.4}{500} = 0.0568$$

$$as = 0.528 \times 0.0568 = 0.03$$

Substituting the known values in equation (6.81) gives

$$I_s = 6.3\left[\frac{(0.2/0.03)^2 + (30 + 1.5)^2}{(0.2/0.03)^2 + (1.5)^2}\right]^{1/2} = 29.7 \text{ A}$$

2.(b) This is the rated braking torque. Hence the slip speed will be the same as for the rated motor torque, but of the opposite sign.

Hence, $N_{s\ell 2} = -40$ rpm

$$N_s = N + N_{s\ell 2} = 500 - 40 = 460 \text{ rpm}$$

frequency = $(460/1000) \times 50 = 23$ Hz

$$a = \frac{23}{50} = 0.46, \qquad s = \frac{N_{s\ell 2}}{N_s} = \frac{-40}{460} = -0.087$$

$$as = -0.04$$

Substituting the known values in equation (6.81), gives

$$I_s = 6.3\left[\frac{(0.2/0.04)^2 + (30 + 1.5)^2}{(0.2/0.04)^2 + (1.5)^2}\right]^{1/2}$$

$$= 38.5 \text{ A}$$

3. Since, at a constant flux, speed-torque curves for different frequencies are straight lines, slip speed

$$N_{s\ell 3} = \frac{139}{188} \times 40 = 29.56 \text{ rpm}.$$

Hence, synchronous speed $N_s = N + N_{s\ell 3} = 500 + 29.56$

$= 529.56$ rpm

Frequency $= (529.56/1000) \times 50 = 26.48$ Hz

$$s = \frac{N_{s\ell 3}}{N_s} = \frac{29.56}{529.56} = 0.056$$

$$a = \frac{26.48}{50} = 0.53, \qquad sa = 0.03$$

Substituting the known values in equation (6.81) gives

$$I_s = 6.30 \left[\frac{(0.2/0.03)^2 + (30 + 1.5)^2}{(0.2/0.03)^2 + (1.5)^2} \right]^{1/2} = 29.7 \text{ A}$$

6.6 LOSS MINIMIZATION

The high cost of energy provides an incentive to reduce energy losses. Reduced losses not only reduce operating cost but also reduce the capital cost of the utility system. In battery-powered vehicles, the reduction of losses allows more efficient use of the battery and thus increases the range of the vehicle in terms of the distance travelled before the battery discharges. In solar-powered drives, the solar cells constitute a major portion of the total cost of the drive. An efficient operation of the drive can lead to substantial saving in its capital cost.

The present section briefly describes ideas related to reducing losses in induction motor drives fed by fixed and variable frequency supplies.

6.6.1 Fixed Frequency Variable Voltage Operation

Nola[9] has shown that induction motor efficiency can be improved by means of reduced voltage operation at light loads. This can be explained by the equivalent circuit of figure 6.1b and the phasor diagrams of figure 6.22. At light loads, (R_r'/s) is large compared to X_r' due to a small s. Therefore, one can assume that I_r' is in phase with E and that the developed torque is proportional to the product EI_r'. It may also be noted that the core loss increases with flux at a given frequency.

Figure 6.22a shows the phasor diagram for the rated impressed voltage and a light load. A large flux is required to produce a back emf E to balance the terminal

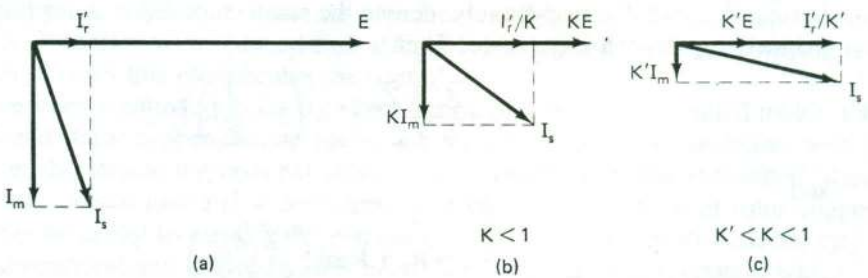


Figure 6.22 Reduced voltage operation.

voltage. The motor draws a large magnetizing current and the stator current I_s is large in spite of the low rotor current. The motor operates with a low power factor, high core loss, and high stator copper loss. Figure 6.22b is drawn for a reduced terminal voltage for which the induced emf is KE, where K is a constant less than 1. The rotor current must increase to (I_r'/K) and the slip must increase to (s/K^2) to produce the same torque as before. The magnetizing current is reduced to KI_m, and consequently I_s is also reduced. The flux is also reduced in the same proportion as I_m. Because of the decrease in flux and I_s, the core loss and the stator copper loss are reduced and the motor power factor is improved. Though the rotor copper loss is increased, the total loss is reduced and motor efficiency is improved. It may be emphasized that the reduction in losses and the improvement in efficiency are obtained mainly due to the operation at reduced flux for light loads.

There is an optimum value of terminal voltage (or flux) and slip for minimum loss. Decrease of voltage beyond this value increases losses and reduces efficiency. Figure 6.22c has been drawn mainly to illustrate this point. Further reduction in the induced emf gives large values of I_r' and I_s. Though the core loss will be less, the total loss will be higher due to the large values of stator and rotor copper losses. It may also be noted that the power factor is still improving. Thus the minimum loss operation occurs at a higher flux than the maximum power factor operation.

The load's torque requirement puts a constraint on the amount of voltage reduction; too great a reduction in voltage will create a high slip and lead to pull-out and motor stalling.

For the analysis of the motor for maximum efficiency at a given output, core loss must be taken into account. Hence, the equivalent circuit shown in figure 6.23 is employed. The resistance R_m accounts for the core loss. Now the developed power will be equal to the shaft power plus the friction and windage loss, because the core loss is already being accounted for due to R_m.

The developed power is given by equation (6.15). The corresponding electric power input to the motor is

$$P_{in} = 3I_s^2 R_{in} \tag{6.86}$$

where R_{in} is the resistive part of the input impedance. The electrical efficiency is given by

$$\eta = \frac{P_m}{P_{in}} = \frac{I_r'^2 R_r'(1-s)/s}{I_s^2 R_{in}} \tag{6.87}$$

Let $\bar{Z}_s$, $\bar{Z}_r'$, and $\bar{Z}_m$, respectively, denote the stator impedance, rotor impedance, and magnetizing branch impedance. Then

$$\bar{I}_s = \frac{\bar{Z}_r' + \bar{Z}_m}{\bar{Z}_m}\bar{I}_r' = \left(1 + \frac{\bar{Z}_r'}{\bar{Z}_m}\right)\bar{I}_r' \tag{6.88}$$

and

$$R_{in} = R_s + \text{Real}\left(\frac{\bar{Z}_r'\bar{Z}_m}{\bar{Z}_r' + \bar{Z}_m}\right) \tag{6.89}$$

Substituting from equations (6.88) and (6.89) into equation (6.87) yields

$$\eta = \frac{1 - s}{1 + \frac{2R_s}{R_m} + \frac{\alpha}{s} + s\left[\beta - \left(1 + \frac{2R_s}{R_m}\right)\right]} \tag{6.90}$$

where

$$\alpha = R_r'\left(\frac{1}{R_m} + \frac{R_s}{X_m^2}\right) + \frac{R_s R_r'}{R_m^2} \tag{6.91}$$

$$\beta = 1 + \frac{R_s}{R_r'}\left(1 + \frac{X_r'}{X_m}\right)^2 + \left(\frac{2R_s}{R_m} + \frac{X_r'^2}{R_r' R_m} + \frac{R_s X_r'^2}{R_r' R_m^2}\right) \tag{6.92}$$

Equation (6.90) shows that efficiency is independent of the terminal voltage and depends only on slip. Differentiating η with respect to s and equating to zero gives an expression for the optimum slip:

$$s_{op} = \frac{\alpha}{\beta}[1 - \sqrt{1 + \beta/\alpha}] \tag{6.93}$$

Thus, efficiency is maximized if the motor is operated at the constant slip s_{op}. The variation in torque and output power is then obtained by varying the terminal voltage. The flux then automatically changes with load to provide operation with a minimum loss for a given output power. It may also be noted that the same expression is obtained for the optimum slip when efficiency is maximized for a given torque.[11] Since the optimum operation is obtained at a fixed speed, optimization can be done only for drives where speed control is not required.

The efficiency of a motor fed by a current source can also be improved by operating it at reduced flux for light loads. It can be shown that there is an optimum slip which provides maximum efficiency for all values of I_s when saturation is neglected (see problem 6.20 and reference 15).

6.6.2 Variable Frequency Operation

In describing the variable frequency control of induction motor with a voltage source (see section 6.4.2) or a current source (section 6.5), the desirability of maintaining the flux at nominal value was emphasized mainly to make full use of the motor torque capability at all speeds. The previous section makes it obvious that this is not the right approach when considering efficiency and losses. A more appropriate strategy is to operate with reduced flux at light loads, during both motoring and regeneration, though this complicates the control.

It has been mentioned in the previous section that in a lightly loaded motor, the core loss and stator copper loss decrease, and the rotor copper loss increases with a decrease in flux below the nominal value. The optimum operation is reached when the decrease in core loss and stator copper loss equals the increase in rotor copper loss. It may be useful to identify the two components of stator copper loss: the magnetizing component and the load component. The magnetizing component, which is due to I_m, decreases with flux, and the load component due to I_r' changes inversely

with flux. When operating above full load, an increase in flux will decrease I_r', and, therefore, rotor copper loss and the load dependent stator copper loss. But the magnetizing component of the stator copper loss and the core loss will increase. As the load component of stator copper loss will dominate over the magnetizing component for large loads, the total loss will decrease with an increase in flux. Optimum operation may be attained at a flux higher than the nominal, provided that this does not lead to heavy saturation which will increase I_m substantially and ultimately reduce efficiency. Operation of the motor above nominal flux may not be allowed at the rated frequency, because then the motor terminal voltage will be higher than the rated value; however, such a restriction does not apply at low frequencies.

The core loss decreases with flux and frequency. When operating at low frequencies, the core loss forms a smaller proportion of the total loss. The optimum operation for a given torque is reached at a higher flux than that at the rated frequency.

The discussion in the preceding two paragraphs suggests that for large torque demands at low speeds, the optimum operation may be obtained at a flux greater than nominal. This emphasizes the need for taking saturation into account when accurate analysis is desired.

Since core loss is a function of both flux and frequency, there is a need to remodel R_m (fig. 6.23). Since the back emf is proportional to the product of flux and the supply frequency f,

$$E = K\Phi f \tag{6.94}$$

The stator core loss is given by the following equation:

$$P_{cs} = K_h f\Phi^2 + K_e f^2\Phi^2 \tag{6.95}$$

where K_h and K_e are hysteresis and eddy current coefficients, respectively.

The hysteresis loss obeys Steinmetz's formula, where the flux is raised to the power n, "n", which is known as the Steinmetz index, is determined experimentally for the material and configuration considered. Since n is in the range 1.5 to 2.5, it is assumed to be 2 in equation (6.95).

The rotor core loss is given by

$$P_{cr} = K_h sf\Phi^2 + K_e s^2 f^2\Phi^2 \tag{6.96}$$

Total core loss is given by

$$P_c = P_{cs} + P_{cr} = K_h f\Phi^2(1 + s) + K_e f^2\Phi^2(1 + s)^2 \tag{6.97}$$

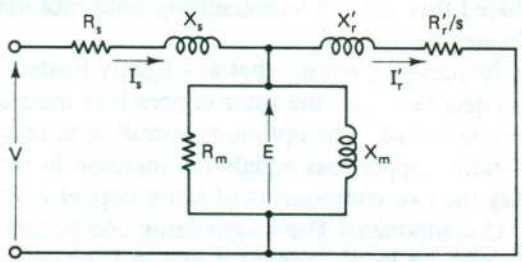


Figure 6.23 Equivalent circuit accounting for core loss.

It is difficult to measure K_h and K_e. A reasonable assumption is to measure core loss at the rated frequency and divide it equally between hysteresis and eddy current loss. Now,

$$R_m = \frac{E^2}{P_c} \tag{6.98}$$

Substituting from equation (6.94) and (6.97) gives

$$R_m = \frac{K^2}{\frac{K_h(1+s)}{f} + K_e(1+s)^2} \tag{6.99}$$

This gives the equivalent circuit of figure 6.24 for the variable frequency operation. Per-unit frequency a is defined by equation (6.38). Since $s = (a\omega_{ms} - \omega_m)/(a\omega_{ms})$

$$a = \frac{\omega_m}{\omega_{ms}(1-s)} \tag{6.100}$$

In variable frequency control employing a voltage source, we have two variables that can be controlled, V and f. An operating point P with given T and ω_m can be obtained with a number of combinations of the variables, as shown in figure 6.25a. However, the optimum operation with the highest efficiency (or with the lowest input power) will be provided only by one set of values of V and f. Similarly, with a current source, optimal operation for a given T and ω_m will be obtained with one set of values of I_s and f.

The optimum solution for a voltage source at the given values of torque (T_1) and speed (ω_{m1}), neglecting saturation, is obtained numerically as follows.[14]

The slip is considered as an independent variable. The voltage and the frequency therefore become dependent variables. For a chosen value of s and the given value of speed ω_{m1}, a is obtained from equation (6.100). This determines all the parameters of the equivalent circuit. Now, the torque for the rated terminal voltage is calculated. Let this be T_r. The terminal voltage required to produce the torque T_1 is given by

$$V_1 = V_r\sqrt{T_1/T_r} \tag{6.101}$$

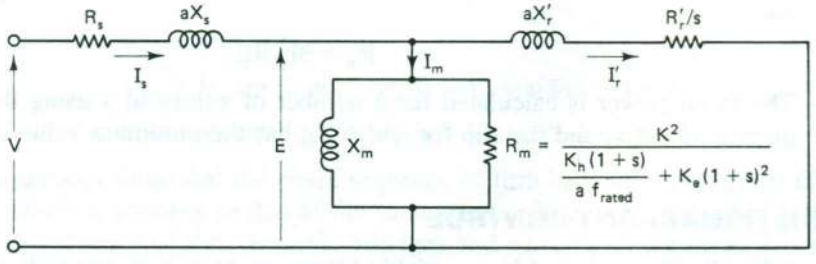


Figure 6.24 Variable frequency equivalent circuit accounting for core loss.

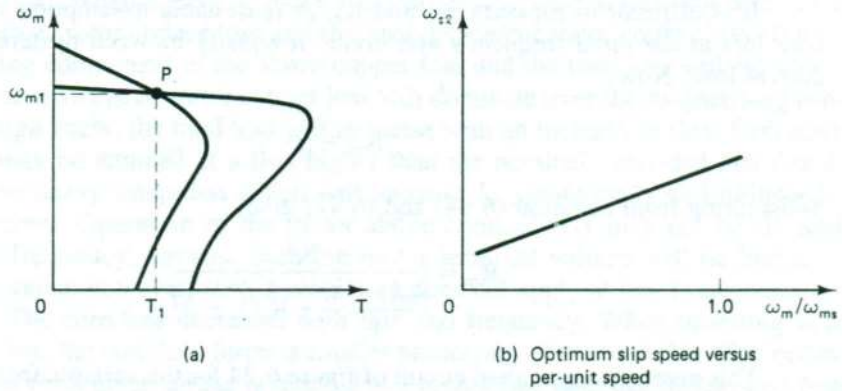


Figure 6.25 Optimum control of variable frequency drives.

The input power is obtained from the equivalent circuit

$$P_{in} = 3V_1^2 \frac{R_{in}}{Z_{in}^2} \tag{6.102}$$

where Z_{in} is the input impedance and R_{in} is the real part of this impedance.

The input power is calculated for a number of values of s following the foregoing procedure, and the optimum slip for which P_{in} is the minimum is identified. One can do these calculations for a number of operating points. The optimal solution for the entire range of speed has the nature shown in figure 6.25b, where the optimum slip speed has been plotted against the per unit speed. The readers are referred to reference 14 for the solution accounting for saturation.

The optimum solution for the motor fed by a variable frequency current source can also be obtained using the preceding approach. The condition for the minimum input power for a given torque (T_1) and speed (ω_{m1}), neglecting saturation, is obtained as follows.

For a chosen value of s and a given speed ω_{m1}, a is obtained from equation (6.100). This determines all the parameters of the equivalent circuit. The torque for the rated value of stator current I_{sr} is obtained. Let this be T_r. The stator current required to produce torque T_1 is given by

$$I_{s1} = I_{sr}\sqrt{T_1/T_r} \tag{6.103}$$

and

$$P_{in} = 3I_{s1}^2 R_{in} \tag{6.104}$$

The input power is calculated for a number of values of s using the foregoing sequence of steps, and the slip for which P_{in} has the minimum value is identified.

6.7 MULTIQUADRANT CONTROL

When the motor is fed by a variable frequency source, multiquadrant control is implemented using regenerative braking, provided that the source has the ability to absorb the energy generated. For speed reversal, the operation is shifted from motoring

to braking by reducing the supply frequency to make synchronous speed less than the motor speed. As the motor decelerates the frequency is continuously adjusted to keep the synchronous speed less than the motor speed. The phase sequence is reversed at zero speed and the frequency is slowly increased to bring the speed to the desired value in the reverse direction. A current control is usually employed to restrict the motor current during the speed reversal. When the generated energy cannot be accepted by the source, it is diverted to a resistor. Then the motor essentially works under dynamic braking.

When fed by a fixed frequency source, multiquadrant operation is obtained by plugging. The changeover from motoring to braking is done by changing the phase sequence. The motor brakes under plugging to zero speed and then reverses. When a sustained braking operation is required, a large resistance is included in the rotor of a wound-rotor motor to restrict the current and ensure stable operation. When used for speed reversal, then a suitable resistance is also included to increase the braking torque and decrease the current. In the case of a squirrel-cage motor, the terminal voltage may be reduced to reduce the braking current at the expense of a large reduction in the braking torque.

6.8 OPERATION WITH NONSINUSOIDAL SUPPLIES

The output voltages of variable voltage and variable frequency voltage sources employing semiconductor converters are nonsinusoidal. The output current of a variable frequency current source using semiconductor converters is also nonsinusoidal. A nonsinusoidal waveform can be resolved into fundamental and harmonic components using Fourier analysis [equations (3.109) to (3.113)]. Because of the half-wave symmetry only odd harmonics will be present.

6.8.1 Positive, Negative, and Zero Sequence Harmonics

Consider the fundamental phase voltage components

$$v_{AN} = V_1 \sin \omega t, \qquad v_{BN} = V_1 \sin(\omega t - 2\pi/3), \quad \text{and} \quad v_{CN} = V_1 \sin\left(\omega t - \frac{4\pi}{3}\right)$$

with the phase sequence ABC. They produce the main field wave in the air-gap which rotates at synchronous speed. The corresponding fifth harmonic phase voltages are

$$v_{AN} = V_5 \sin 5\omega t$$
$$v_{BN} = V_5 \sin 5(\omega t - 2\pi/3) = V_5 \sin(5\omega t - 4\pi/3)$$
$$v_{CN} = V_5 \sin 5(\omega t - 4\pi/3) = V_5 \sin(5\omega t - 2\pi/3)$$

These equations show that the phase sequence of fifth harmonic voltages is CBA (or ACB), which is opposite to that of the fundamental. In general it can be shown that the phase sequence of the harmonic voltages and currents of the order $k = 6n - 1$, where n is an integer, is opposite to that of the fundamental. These are known as negative sequence harmonics. Since they are carried by the same winding as the fundamental, they have the same number of poles. Because the phase sequence is oppo-

site and the frequency is k times that of the fundamental, they produce harmonic field waves in the air-gap, which rotates at k times the fundamental synchronous speed in the opposite direction to the main field.

Next consider the seventh harmonic phase voltages

$$v_{AN} = V_7 \sin 7\omega t$$
$$v_{BN} = V_7 \sin 7(\omega t - 2\pi/3) = V_7 \sin(7\omega t - 2\pi/3)$$
$$v_{CN} = V_7 \sin 7(\omega t - 4\pi/3) = V_7 \sin(7\omega t - 4\pi/3)$$

These equations show that the phase sequence of the seventh harmonic voltage is ABC, which is the same as that of the fundamental. This is true in general for the harmonic voltages and currents of the order $k = 6n + 1$. These are known as positive sequence harmonics. They have the same number of poles as the fundamental and produce harmonic field waves in the air-gap, which rotate at k times the fundamental synchronous speed in the same direction as the fundamental.

It can be shown that the phase voltages and currents of harmonics of the order $k = 3n$ are in phase. Consequently, these are called zero sequence harmonics. They do not produce rotating air-gap field waves. The zero sequence harmonic voltages are able to produce currents and affect motor performance only in the Y-connected stator winding with the neutral connection. Since their frequencies are in multiples of 3, they are also known as tripplen harmonics.

6.8.2 Harmonic Equivalent Circuits

The equivalent circuit of figure 6.1b is valid for the fundamental components of voltage and current. The equivalent circuit for the kth harmonic voltage and current can be derived from this equivalent circuit. All the reactances are increased by a factor k. The stator and rotor resistances are also increased due to the skin effect. As explained in the previous section, the kth harmonic field wave may rotate forward or backward at a speed $k\omega_{ms}$, and consequently the harmonic slip is given by

$$s_k = \frac{k\omega_{ms} \mp \omega_m}{k\omega_{ms}} \tag{6.105}$$

where the negative sign is applicable to positive sequence harmonics and the positive sign is valid for negative sequence harmonics. The kth harmonic equivalent circuit is shown in figure 6.26a.

The harmonic slip can be expressed in terms of the fundamental slip by substituting for ω_m from equation (6.4) into equation (6.105), giving

$$s_k = 1 \mp \frac{(1 - s)}{k} \tag{6.106}$$

If the motor speed varies from the synchronous speed to standstill, s varies from 0 to 1. From equation (6.106), the corresponding variations of s_5 and s_7 are from 1.2 to 1 and 0.857 to 1, respectively, which are close to unity. For higher harmonics, s_k is even closer to unity. Except for very low fundamental frequencies (around less than 5 Hz), the magnetizing reactance is much higher compared to the rotor impedance

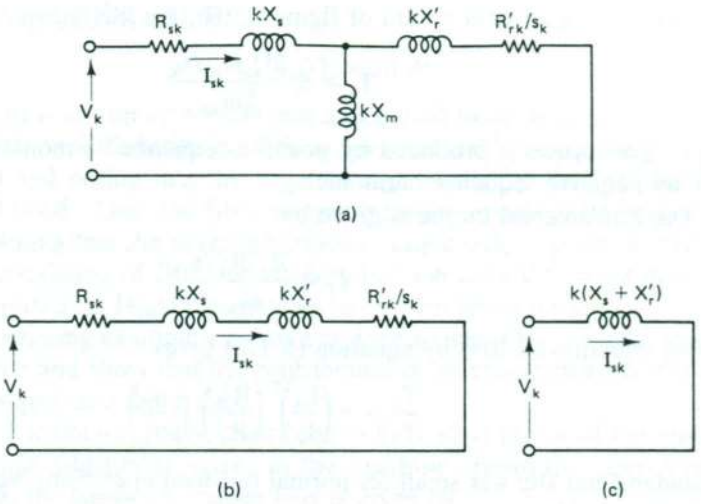


Figure 6.26 kth harmonic equivalent circuits of induction motor.

in parallel with it, and can be omitted (fig. 6.26b). The equivalent circuits of figure 6.26 are applicable to both voltage and current sources. For a voltage source, V_k is constant and I_{sk} varies; the reverse is true for a current source.

Since s_k is nearly unity at all motor speeds, the harmonic equivalent circuit is independent of the motor speed. Thus, harmonic currents are substantially constant and independent of the motor load and speed. From figure 6.26b,

$$I_{sk} = \frac{V_k}{\sqrt{\left(R_{sk} + \frac{R'_{rk}}{s_k}\right)^2 + k^2(X_s + X'_r)^2}} \tag{6.107}$$

When the fundamental frequency is more than about 20 Hz for small motors and more than about 10 Hz for large motors, X_s and X'_r have values comparable to R_s and R'_r. At harmonic frequencies their values are increased k times. The increase in the stator and rotor resistances due to skin effect is much less. Since s_k is close to unity, the resistances have negligible values compared to the reactances. Hence,

$$I_{sk} \cong \frac{V_k}{k(X_s + X'_r)} \tag{6.108}$$

This approximation gives the equivalent circuit as shown in figure 6.26c.

6.8.3 Efficiency and Derating

The rotating harmonic air-gap flux waves induce harmonic rotor currents of slip frequency. The slip frequency rotor currents create harmonic mmf waves which travel at slip speed with respect to the rotor. The mmf and the air-gap flux wave produced by the same harmonic are stationary relative to each other and a steady torque is produced due to their interaction. It can be readily shown that the harmonic torques are negligible compared to the fundamental.

From the equivalent circuit of figure 6.26b, the kth harmonic torque is

$$T_k = \pm \frac{3I_{sk}^2 R'_{rk}/s_k}{k\omega_{ms}} \tag{6.109}$$

The positive torque is produced by positive sequence harmonics and the negative torque by negative sequence harmonics.

The fundamental torque is given by

$$T_1 = \frac{3I_r'^2 R'_r/s}{\omega_{ms}} \tag{6.110}$$

Dividing equation (6.109) by equation (6.110) gives

$$\frac{T_k}{T_1} = \pm \left(\frac{I_{sk}}{I'_r}\right)^2 \left(\frac{R'_{rk}}{R'_r}\right) \left(\frac{s}{ks_k}\right) \tag{6.111}$$

The fundamental slip s is small for normal full-load operation. Neglecting s in equation (6.106) yields

$$s_k = \frac{k \mp 1}{k} \tag{6.112}$$

Substituting in equation (6.111) gives

$$\frac{T_k}{T_1} = \pm \left(\frac{I_{sk}}{I'_r}\right)^2 \left(\frac{R'_{rk}}{R'_r}\right) \left(\frac{s}{k \mp 1}\right) \tag{6.113}$$

A quantitative picture can be obtained by taking a specific case. Let us consider the operation of a typical induction motor with a starting current 5 times the rated and with a six-step voltage source inverter as described in chapter 8. At starting, resistances have negligible values compared to reactances; consequently,

$$\frac{V_1}{X_s + X'_r} = 5I'_r$$

Thus,

$$I'_r = \frac{V_1}{5(X_s + X'_r)} \tag{6.114}$$

where V_1 and I'_r, respectively, are the fundamental rated phase voltage and current of the motor.

For a six-step voltage source inverter (equation (8.2)),

$$V_k = \frac{V_1}{k}$$

Now consider the fifth harmonic torque. Substitution of $V_k = V_1/5$ and $k = 5$ in equation (6.108) yields

$$I_{s5} = \frac{V_1/5}{5(X_s + X'_r)} = \frac{V_1}{25(X_s + X'_r)} \tag{6.115}$$

From equations (6.114) and (6.115)

$$I_{s5}/I'_r = (0.2) \tag{6.116}$$

Assume a full-load slip $s_1 = 0.04$ and a threefold increase in the rotor resistance — that is, $R'_{r5}/R'_r = 3$. Substituting these values and that from equation (6.116) in equation (6.113) and noting that the negative sign applies to the fifth harmonic, gives $T_5/T_1 = -0.0008$. Thus the fifth harmonic torque is only 0.08 percent of the fundamental. Noting that the seventh harmonic torque will be positive, the contribution of the pair consisting of fifth and seventh harmonics to the motor developed torque will be insignificant. Higher harmonics have even lesser torques.

The foregoing example was for a voltage source. One can take an example of a current source and show that the contribution of harmonic currents to the developed torque is insignificant there also.

While harmonics do not contribute to the output power of the motor, they certainly produce additional losses in the machine. From the equivalent circuit of figure 6.26b, the harmonic copper loss is given by

$$P_h = \sum_{k=5,7} I_{sk}^2(R_{sk} + R'_{rk}) \tag{6.117}$$

The core loss is also increased somewhat in the presence of harmonics.

Harmonic losses reduce efficiency and increase thermal loading. The higher the harmonic content, the greater the reduction in efficiency and the increase in thermal loading. As explained earlier, the harmonic currents remain constant at all operating conditions of the motor; consequently, the harmonic losses are also independent of the motor load and speed. They cause a significant reduction in motor efficiency and a large increase in its heating at light loads. In general at full load, the reduction in efficiency is not very significant; however, the increase in thermal loading may lead to appreciable derating of the motor.

When fed by a voltage source, according to equations (6.108) and (6.117), the harmonic losses are less when the motor reactance $(X_s + X'_r)$ is high. The class B design has a higher value of reactance compared to the class A. Hence, the class B design is preferred when the voltage source has a high harmonic content. The class A design is used only when the harmonic content is negligible.

6.8.4 Torque Pulsations

The mmf and air-gap flux waves produced by different harmonics (including fundamental) are not stationary relative to each other. Consequently, they produce pulsating harmonic torques, which have zero average value. The prominent components are those arising from the interaction between the fundamental air-gap flux and rotor mmf waves produced by fifth and seventh harmonic air-gap fluxes. The negative sequence fifth harmonic air-gap flux wave produces a rotor mmf wave which moves backward at five times the fundamental synchronous speed. The relative speed between the fifth harmonic rotor mmf wave and the fundamental air-gap flux wave being six times the fundamental synchronous speed, their interaction produces a pulsating torque at six times the fundamental frequency. The positive sequence seventh harmonic air-gap flux wave produces a rotor mmf which rotates forward at

seven times the fundamental synchronous speed. Since the relative speed between the fundamental air-gap flux wave and the seventh harmonic rotor mmf wave is six times the fundamental synchronous speed, their interaction also produces a pulsating torque at six times the fundamental frequency. Similarly it can be shown that the eleventh and thirteenth harmonics produce a torque pulsation twelve times the fundamental frequency, but its amplitude is small.

The torque pulsations cause fluctuations in motor speed. When the fundamental frequency is sufficiently large, speed fluctuations are sufficiently low because of the motor inertia. When the fundamental frequency, and thus, the motor speed is low, large fluctuations in motor speed are obtained producing a jerky or stepped motion.

The amplitude of the torque pulsations depends on the magnitude of fifth and seventh harmonic currents, which in turn depend on the corresponding harmonic voltages and the motor reactance. Higher motor reactance reduces torque pulsations. This also points towards the suitability of the class B design for such applications. The torque pulsations also depend on the magnitude of the fundamental air-gap flux. The saturation increases this flux leading to higher magnitude of torque pulsations. This is also one of the reasons for operating a current source drive on the region of speed-torque curves with a positive slope.

Example 6.8

A 440-V, 50 Hz, 6-pole, 960 rpm, Y-connected induction motor has the following parameters per phase referred to the stator:

$$R_s = 0.6\ \Omega, \qquad R_r' = 0.3\ \Omega, \qquad X_s = X_r' = 1\ \Omega$$

X_m is very large and can be ignored.

The motor is fed by a nonsinusoidal voltage source. The fundamental component of the source voltage is 440 V. Fifth and seventh harmonics are 20 percent and 14 percent of the fundamental, respectively. Higher harmonics can be ignored. Skin effect causes the rotor resistance to increase three times for the fifth harmonic and four times for the seventh harmonic.

Calculate the derating of the machine due to nonsinusoidal supply. Neglect friction, windage, and core loss. Also calculate the rated motor torque with nonsinusoidal supply.

Solution: Let us first consider the motor operation with a sinusoidal supply.

$$\text{Synchronous speed } N_s = \frac{120f}{p} = \frac{120 \times 50}{6} = 1000 \text{ rpm}$$

$$\omega_{ms} = 104.7 \text{ rad/sec}.$$

$$\text{Rated slip} = \frac{1000 - 960}{1000} = 0.04$$

$$\text{Rated current } I_{rated} = \frac{440/\sqrt{3}}{\sqrt{\left(0.6 + \dfrac{0.3}{0.04}\right)^2 + (1+1)^2}} = 30.45 \text{ A}$$

$$\text{Rated power developed} = 3I_{rated}^2\left(\frac{R_r'}{s} - R_r'\right)$$

$$= 3 \times (30.45)^2\left(\frac{0.3}{0.04} - 0.3\right) = 20 \text{ kW}$$

$$\text{Motor heat loss} = 3I_{\text{rated}}^2(R_s + R_r') = 3 \times (30.45)^2 \times (0.6 + 0.3) = 2.5 \text{ kW}$$

Let us now consider the motor operation on a nonsinusoidal supply.

$$V_1 = \frac{440}{\sqrt{3}} = 254 \text{ V}$$

$$V_5 = 0.2 \times 254 = 50.8 \text{ V},$$

$$V_7 = 0.14 \times 254 = 35.56 \text{ V}.$$

From equation (6.108)

$$I_{s5} = \frac{50.8}{5(2)} = 5.08 \text{ A}$$

Copper loss due to the fifth harmonic

$$= 3 \times I_{s5}^2(R_s + 3R_r')$$
$$= 3 \times (5.08)^2(0.6 + 3 \times 0.3) = 116 \text{ W}$$

Again from equation (6.108)

$$I_{s7} = \frac{35.56}{7 \times 2} = 2.5 \text{ A}$$

Copper loss due to the seventh harmonic

$$= 3 \times I_{s7}^2(R_s + 4R_r')$$
$$= 3 \times (2.5)^2(0.6 + 4 \times 0.3) = 33.75 \text{ W}$$

Total harmonic copper loss = 116 + 33.75 = 0.15 kW
For the same heating of the machine as under rated conditions with a sinusoidal supply, the maximum fundamental copper loss allowed = 2.5 − 0.15 = 2.35 kW.
Hence, the maximum fundamental current allowed is

$$I = \left(\frac{2.35 \times 10^3}{3(R_s + R_r')}\right)^{1/2} = \left(\frac{2.35 \times 10^3}{3 \times 0.9}\right)^{1/2} = 29.5 \text{ A}.$$

$$29.5 = \frac{V_1}{\sqrt{\left(R_s + \frac{R_r'}{s}\right)^2 + (X_s + X_r')^2}}$$

$$= \frac{254}{\sqrt{\left(0.6 + \frac{0.3}{s}\right)^2 + 4}}$$

which gives s = 0.0386
Since only the fundamental contributes to the developed power,

$$\text{Rated developed power} = 3 \times (29.5)^2 \times \left(\frac{0.3}{0.0386} - 0.3\right)$$
$$= 19.5 \text{ kW}$$

$$\text{Percent derating} = \frac{19.5}{20} \times 100 = 97.5 \text{ percent}$$

Since only the fundamental contributes to the output torque,

$$\text{Rated torque} = \frac{3}{104.61}(29.5)^2 \times \frac{0.3}{0.0386} = 193.8 \text{ N-m}$$

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PROBLEMS

6.1 A 37.3 kW, 460 V, 60 Hz, 6-pole, 1180 rpm, Y-connected squirrel-cage induction motor has the following parameters per phase referred to the stator:
$R_s = 0.19\ \Omega$, $X_s = 0.75\ \Omega$, $X_m = 20\ \Omega$
$R_r' = 0.07\ \Omega$, $X_r' = 0.38\ \Omega$
Calculate
(a) The full-load current, torque, p.f., and efficiency.
(b) The starting torque and current as a ratio of their full-load values.
(c) The breakdown torque (developed) as a ratio of the full-load torque.
(d) The sum of the core and friction losses at full load.

6.2 Why is it essential to operate an induction motor between the synchronous speed and the breakdown speed when it is used to holding an active load by the regenerative braking?

6.3 The motor of problem 6.1 is used for regenerative braking. Calculate
(a) The range of active load torque it can hold and corresponding speed range.
(b) The maximum power it can generate.
(c) The speed at the developed torque of 300 N-m.

6.4 Plugging is used for the speed reversal of the induction motor of problem 6.1 from full-load speed. What changes must be made in its connection? Calculate the initial braking torque and current as a ratio of their full-load values.

Show the nature of the speed-torque characteristic during reversal from rated forward speed to rated reverse speed.

6.5 A 2.8 kW, 400 V, 50 Hz, 4-pole, 1370 rpm, Y-connected wound-rotor induction motor has the following parameters per phase referred to the stator:
$R_s = 1.9\ \Omega$, $R_r' = 4.575\ \Omega$, $X_s = X_r' = 3\ \Omega$
The magnetization characteristic with the two-lead connection (two phases in series with the third left open) is as follows:

I_m, A	0.1265	0.368	0.612	0.9	1.22	1.71
E, V	12.1	31.8	54	80	108	143
I_m, A	2.24	2.86	3.86	4.9	6.53	8.16
E, V	173	199	226	246	266	281

The motor is used for dc dynamic braking with a two-lead connection and $I_d = 12$ A. Calculate and plot the speed-torque characteristic.

6.6 The motor of problem 6.5 is initially running at full load. It is controlled by the terminal voltage control. The magnetizing current, and core, friction and windage losses can be neglected.
(a) If the load torque is directly proportional to speed, calculate the motor terminal voltage and current at half the rated speed. Can the motor be allowed to run continuously at this speed?
(b) Repeat (a) for a fan load.

6.7 The motor of problem 6.5 is controlled by the rotor resistance control. The stator to rotor turns ratio is 4.25. The magnetizing current, and core, friction and windage losses can be neglected.
(a) What resistance must be inserted in the rotor to run the motor at 400 rpm at the full-load torque?
(b) What resistance must be inserted in the rotor to get the maximum plugging torque at the rated speed?

6.8 A 7.5 kW, 220 V, 6-pole, 60 Hz, Y-connected induction motor has the following parameters per phase referred to the stator for the equivalent circuit of figure 6.1d:
$R_s = 0.29\ \Omega$, $X_s = 0.5\ \Omega$, $X_m = 13.3\ \Omega$
$R_r' = 0.145\ \Omega$, $X_r' = 0.21\ \Omega$
The motor is controlled by variable frequency control with a constant (V/f) ratio for frequencies below 60 Hz and with a constant terminal voltage for frequencies greater than 60 Hz.
(a) Calculate breakdown torques at different frequencies, both for motoring and regenerative braking, and plot them against frequency.
(b) Calculate the starting torque for different frequencies below 60 Hz, and plot them against frequency.

6.9 The induction motor of problem 6.8 has a full-load slip of 0.02. The motor is controlled by the variable frequency control with a constant (V/f) ratio. Calculate the rotor current as a percentage of its full-load value for a frequency of 15 Hz and at full-load torque. Can you run the motor continuously at this operating point? Explain your answer.
What change must be made so that the rotor draws full-load current at full-load torque when the supply frequency is 15 Hz?

6.10 The motor of problem 6.8 is controlled by variable frequency control. It is required that the rotor current at full-load torque remains constant for all frequencies below 60 Hz. Calculate the values of terminal voltage for different frequencies and plot them against frequency. Compare this plot with that for the constant (V/f) control.
Use the equivalent circuit of figure 6.1d.

6.11 The motor of problem 6.8 is controlled by variable frequency control. It is required that the breakdown torque should remain constant for all frequencies below 60 Hz. Calculate the values of terminal voltage for different frequencies and plot them against frequency. Compare this plot with that for the constant (V/f) control.
Obtain the (V/f) ratio for a frequency of 20 Hz for the preceding two cases. Calculate the regenerative braking breakdown torques for these two voltage ratios and the frequency of 20 Hz.
Use the equivalent circuit of figure 6.1d.

6.12 A 440 V, 50 Hz, 6-pole Y-connected induction motor has the following parameters per phase referred to the stator:
$R_s = 0.6\ \Omega$, $R_r' = 0.3\ \Omega$, $X_s = X_r' = 1\ \Omega$
X_m is very large and can be ignored. The normal full-load slip is 0.05.
Calculate the motor speed at the full-load developed torque when fed at 180 V and 20 Hz.

6.13 The motor of problem 6.12 is controlled by the variable frequency control with a constant (V/f) ratio. For an operating frequency of 10 Hz, calculate
(a) The breakdown torque as a ratio of its value at the rated frequency for both motoring and braking.
(b) The starting torque and current as a ratio of their values at the rated frequency.

6.14 A 400 V, 50 Hz, 4-pole, 1370 rpm, Y-connected induction motor fed by a variable frequency source has the following parameters per phase referred to the stator:
$R_s = 1.9\ \Omega$, $R_r' = 2\ \Omega$, $X_s = X_r' = 3\ \Omega$
X_m is very large and can be ignored.
Calculate the motor torque and speed at 60 Hz and the rated motor current.

6.15 A 460 V, 60 Hz, 4-pole, 1720 rpm, Y-connected induction motor has the following parameters per phase referred to the stator:
$R_s = 0.5\ \Omega$, $R_r' = 0.2\ \Omega$, $X_s = X_r' = 1\ \Omega$, $X_m = 30\ \Omega$

The motor is controlled by variable frequency control. The flux is kept constant at the rated value. Calculate

(a) The frequency at 1200 rpm and the rated torque.

(b) The motor speed for a frequency of 30 Hz and the rated torque.

(c) The frequency and the stator current at half the rated torque and 1500 rpm.

(d) And solve (c) assuming speed-torque curves, in the region of interest, to be straight lines at a given frequency and constant flux.

Use the equivalent circuit of figure 6.1b.

6.16 The motor of problem 6.15 is now braked by regenerative braking with variable frequency control at a constant rated flux. Calculate

(a) The frequency at 1200 rpm and rated braking torque

(b) The motor speed and stator current at half the rated braking torque and 30 Hz.

(c) And solve (b) assuming speed-torque curves to be straight lines, in the region of interest, at a given frequency and constant flux.

Use the equivalent circuit of figure 6.1d.

6.17 A 460 V, 60 Hz, 4-pole, 1720 rpm, Y-connected wound-rotor induction motor has the following parameters referred to the stator:

$R_r' = 0.3\ \Omega$, $X_r' = 2\ \Omega$, stator to rotor turns ratio = 2.

The stator impedance and the magnetizing reactance may be ignored.

The motor is running at full load with a fan load. What voltage should be injected in the rotor to reduce the motor speed to 1000 rpm? Neglect friction, windage, and core loss.

6.18 A 400 V, 4-pole, 50 Hz, Y-connected wound-rotor induction motor has the following constants referred to the stator:

$R_r' = 0.1\ \Omega$, $X_r' = 1.0\ \Omega$, stator to rotor turns ratio = 5.

The stator resistance and reactance and the magnetizing reactance may be ignored.

The motor speed when driving a load with constant torque is 1420 rpm. Calculate the magnitude and phase of the voltage to be impressed on slip rings to operate the motor at 1000 rpm and a unity power factor.

6.19 A 460 V, 60 Hz, 1185 rpm, 6-pole, Y-connected wound-rotor induction motor has the following parameters per phase referred to the stator:

$R_s = 0.05\ \Omega$, $X_s = 0.3\ \Omega$, $X_m = 10\ \Omega$

$R_r' = 0.041\ \Omega$, $X_r' = 0.38\ \Omega$

Stator to rotor turns ratio is 2.

The machine speed is controlled by injecting a voltage in the rotor circuit. The injected voltage is kept either in phase or in phase opposition to the terminal voltage.

(a) For an injected voltage of 46 V, calculate the speed for

(i) The torque equal to half the rated motor torque.

(ii) The braking torque equal to half the rated motor torque.

(b) For an injected voltage of -46 V, calculate the speed for

(i) The torque equal to the rated motor torque.

(ii) The braking torque equal to half the rated motor torque.

6.20 Derive an expression for the optimum slip which maximizes the efficiency of an induction motor fed by a current source of variable magnitude and fixed frequency.

6.21 The motor of problem 6.15 is controlled by a variable frequency current source. The flux is maintained constant at nominal value.

(a) Calculate and plot I_s versus $\omega_{s\ell}$, for both motoring and braking, for the range of I_s from 0 to 3 times the rated current.

(b) Calculate the motor speed and stator current for 40 Hz and the rated torque both for motoring and braking.

(c) Calculate the frequency and stator current for half the rated torque and a motor speed of 1000 rpm, both for motoring and braking.

6.22 The motor of problem 6.14 is fed by a nonsinusoidal supply. Calculate the motor torque, current, and efficiency at the rated speed if the fundamental, fifth and seventh harmonic phase voltages are 254 V, 100 V, and 40 V, respectively. Neglect higher harmonics, friction, windage, core loss, and skin effect.

7

Control of Induction Motors by AC Voltage Controllers

The speed control of induction motors by stator voltage control has been described in section 6.4.1. It has been shown that a class D design squirrel-cage induction motor should be used for wide variation of speed. Since the torque to current ratio decreases with voltage, the method is suitable for applications requiring a low torque at low speeds.

The variation of motor voltage is obtained by ac voltage controllers. It may be recalled that the function of ac voltage controllers is to allow a variable ac voltage of the same frequency to be obtained from a fixed ac voltage. However, this conversion is obtained at the expense of a low power factor and a considerable amount of harmonics in the output voltage of ac voltage controllers. The harmonic content increases and the power factor decreases with the decrease in output voltage. The harmonics increase the losses and require derating of the motor. The motor torque capability, which is already low at low voltages, is further reduced.

The induction motors controlled by ac voltage controllers find wide applications in fan, pump, and crane drives.

7.1 AC VOLTAGE CONTROLLER CIRCUITS

Figure 7.1 shows two commonly used symmetrical 3-phase ac voltage controller circuits for wye- and delta-connected stators, respectively. For small size motors, each antiparallel thyristor pair can be replaced by a triac. Connection C_1 can also be used with a delta-connected stator. With a delta connection, a third harmonic current may circulate, increasing the motor losses. Each thyristor pair of circuit C_1 carries the line current, whereas each thyristor pair of circuit C_2 carries only the phase current.

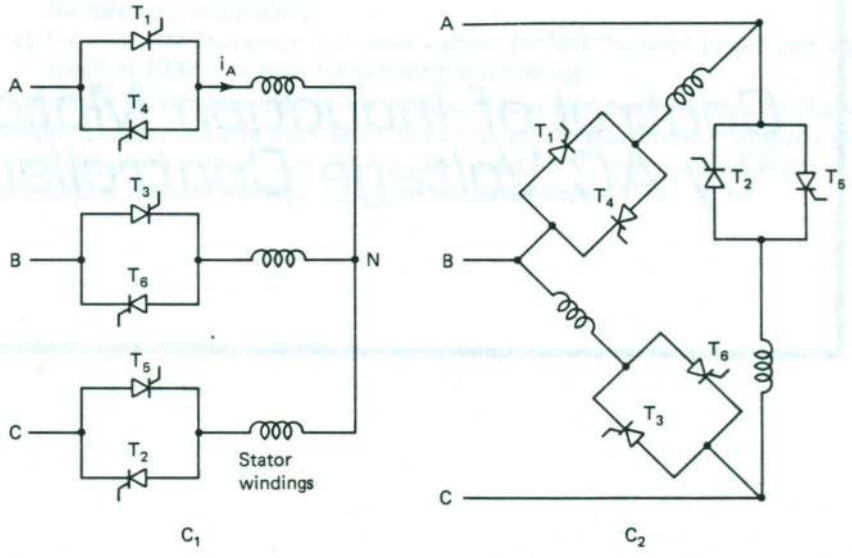


Figure 7.1 3-phase ac voltage controller circuits.

Thus, in C_2 the thyristor current rating is reduced by a factor of $\sqrt{3}$. Under normal operation, the maximum voltage to which the thyristors of circuit C_1 are subjected is $(\sqrt{3}/2)$ times smaller than that of the thyristors of circuit C_2.[2] However, under abnormal conditions, the maximum voltage is the same (that is, equal to the peak of the line voltage). Such an abnormal condition can arise if thyristors in one of the phases are rendered conductive by device failure or misfiring, when all phases are intended to be off. Since such abnormal conditions can always arise, the thyristor voltage rating is chosen to be higher than the peak of the line voltage. Thus voltage controller C_2 would cost less than circuit C_1. However, circuit C_2 can be used only when the machine is delta-connected and both ends of the phase windings are available. This situation may not exist in many commercially available motors.

A cheaper controller can be obtained by replacing one thyristor in each phase of circuit C_1 by a diode. This approach introduces even harmonics. The predominant harmonic is the second, compared to the fifth and third for the circuits C_1 and C_2, respectively. The presence of the low-frequency second harmonic increases the losses considerably, particularly at low motor speeds.

Asymmetrical circuits are obtained by removing one or two antiparallel thyristor pairs from circuit C_1 and connecting corresponding motor phases directly to the source. These asymmetrical circuits cause asymmetrical operation of the induction motor and increase harmonics. The asymmetrical operation and the large harmonic content substantially derate the motor. The torque capability, which is already low at low speeds, is greatly reduced. Hence these asymmetrical circuits are not used.

The thyristors of the controllers of figure 7.1 are fired in the sequence of their numbers with a phase difference of 60°. For circuit C_1, the firing angle α is measured from the instant the phase voltage V_{AN} has a zero value. For circuit C_2, the fir-

ing angle is measured from the instant the line voltage V_{AB} has a zero value. Let us define an angle

$$\phi = \tan^{-1}(X_{in}/R_{in}) \tag{7.1}$$

where $(R_{in} + jX_{in})$ is the input impedance of the induction motor.

For the firing angle $\alpha \leq \phi$, the motor terminal voltage remains constant and nearly equal to the supply voltage. Both motor voltage and current are sinusoidal. For higher values of α, the current flows discontinuously and the motor voltage decreases with an increase in α. The zero motor voltage and current are reached at $\alpha = 150°$ and $\alpha = 180°$ for the circuits C_1 and C_2, respectively. Because of the discontinuous conduction and dependence of R_{in} and X_{in} on the motor slip s, the analysis of the induction motor fed by an ac voltage controller becomes complicated. The speed-torque characteristics also deviate considerably from those obtained with the variable sinusoidal ac voltage (fig. 6.11).

The motor voltage and current waveforms depend on the firing angle. A typical current waveform for circuit C_1 at $\alpha = 60°$ is shown in figure 7.2. The delay from the instant the phase current reaches zero to the firing of the succeeding thyristor in that phase is called hold-off angle γ.

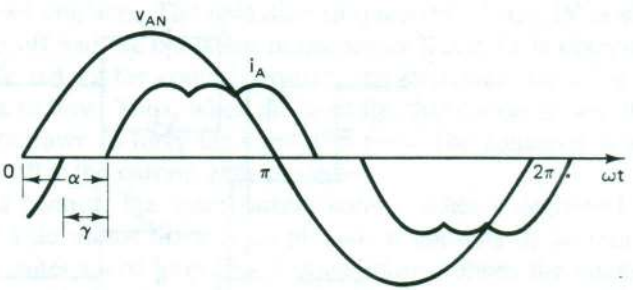
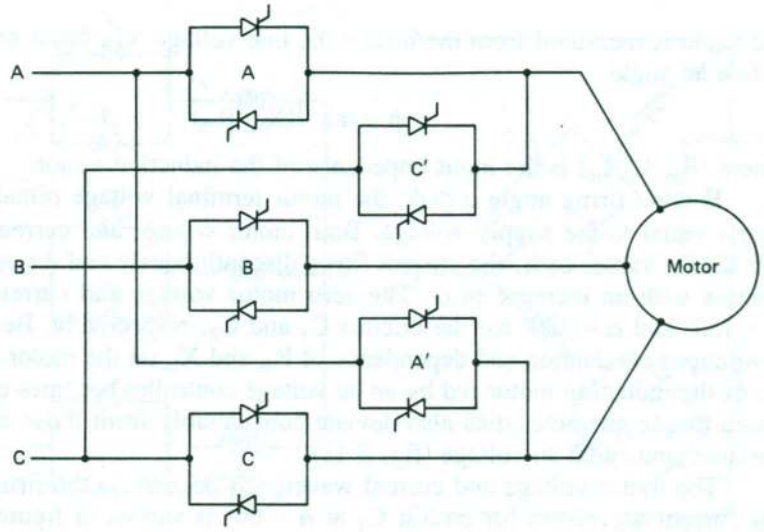


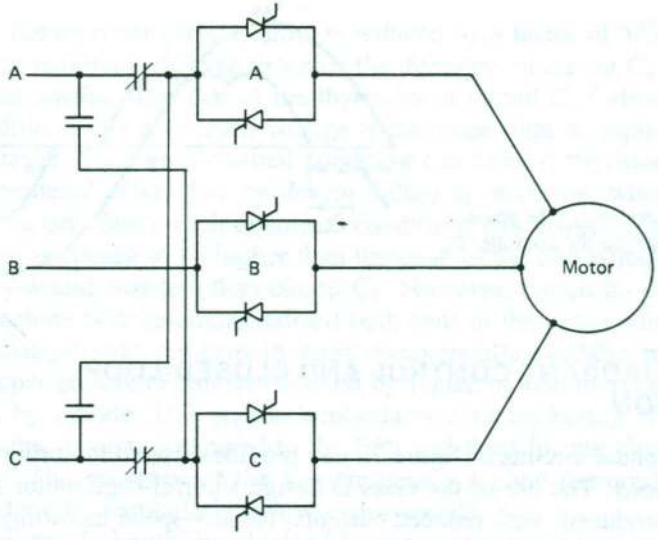
Figure 7.2 The waveform of the phase current i_A for $\alpha = 60°$ for the controller C_1 of Fig. 7.1.

7.2 FOUR-QUADRANT CONTROL AND CLOSED-LOOP OPERATION

The 3-phase circuits of figure 7.1 can provide forward motoring and reverse plugging operations. The use of the class D design squirrel-cage motor allows higher torques to be produced, with reduced currents, for low-speed motoring and high-speed braking operations. The four-quadrant operation with plugging is obtained by the use of the circuit of figure 7.3a. Thyristor pairs A, B, and C provide operation in quadrants I and IV. The speed-torque curve at a fixed stator voltage and for operation in quadrants I and IV is shown by a solid line in figure 7.4. Use of thyristor pairs A′, B, and C′ changes the phase sequence, and thus gives operation in quadrants II and III. The speed-torque curve for the same stator voltage and operation in quadrants II and III is shown in figure 7.4 by a dotted line. While changing from one set of thyristor pairs to another—that is, from ABC to A′BC′ and vice versa—care should be taken to ensure that the incoming pair is activated only after the outgoing



(a)



(b)

Figure 7.3 Four quadrant ac voltage controllers.

pair is fully turned off. Failure to satisfy this condition will cause short circuiting of the supply by the conducting thyristors of the two pairs. The protection against such a fault can be provided only by the fuse links and not by the current control. Therefore, when changing from one set of thyristor pairs to another, the firing pulses are withdrawn to force the current to zero. After the current zero is sensed by the zero-current sensor, a dead time of 5 to 10 ms is allowed to ensure that all the thyristors

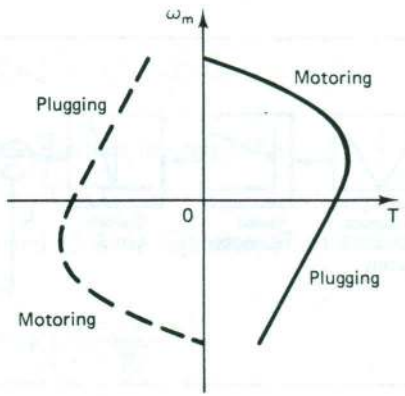


Figure 7.4 Speed-torque curves for a fixed stator voltage and +ve and −ve phase sequence.

of the outgoing pair have in fact turned off. Now the pulses are released to the incoming set of thyristor pairs. In the multiquadrant operation, the current control (section 3.9) is employed to restrict the motor current below a safe value.

The four-quadrant operation can also be obtained by the circuit of figure 7.3b. It consists of three pairs of thyristors A, B, and C and a contactor with two normally open and two normally closed contacts. The operation in quadrants I and IV is obtained when the contactor is off and the operation in quadrants II and III is obtained when the contactor is on. To reduce the contactor rating, the switching operation is done after the current ceases to flow. Thus, when the need for changeover arises, the thyristor gate pulses are withdrawn to force the current to zero. The contactor is allowed to operate some time after the current zero is sensed.

For closed-loop speed control, the inner-current control scheme described in section 5.1 (fig. 5.1b) for a dc motor drive is employed. It consists of an inner-current control loop and an outer speed loop. For 1-quadrant operation, the voltage controllers of figure 7.1 are used. For 4-quadrant operation, the voltage controllers of figure 7.3 are employed. A closed-loop scheme for the single-quadrant control is shown in figure 7.5. A four-quadrant closed-loop drive can be realized by using the

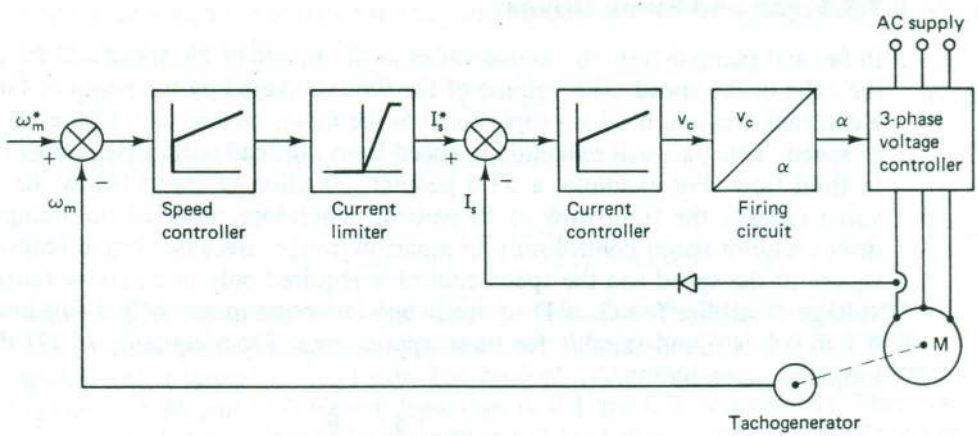


Figure 7.5 Single quadrant closed-loop speed control.

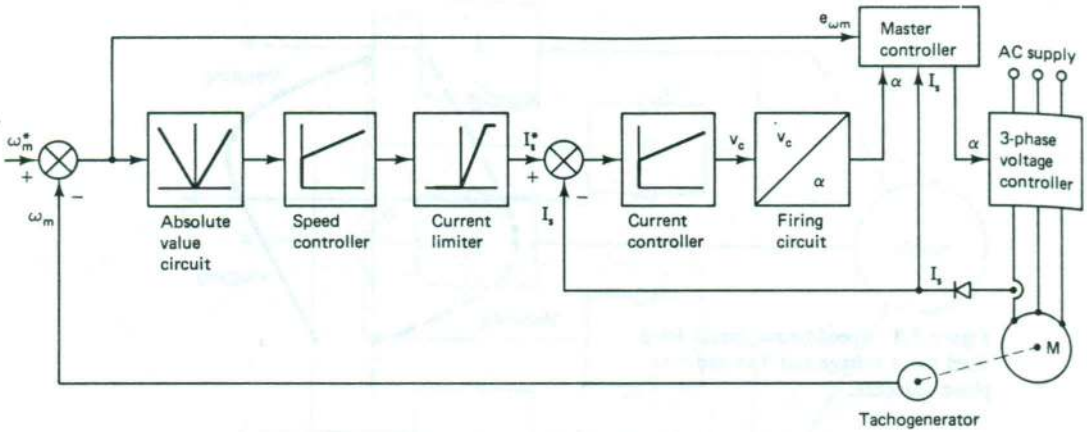


Figure 7.6 Four quadrant closed-loop speed control.

voltage controller of figure 7.3a and the drive of figure 7.6. Let us consider the operation of the drive for speed reversal. When the speed command is set for the reverse direction, the speed error $e_{\omega m}$ reverses and exceeds a prescribed limit. The master controller, on sensing this, withdraws the gate pulses to force the current to zero. After the zero current is sensed, the master controller provides a delay of 5 to 10 ms to ensure that the outgoing thyristors have turned off. Now the gate pulses are released to the other set of thyristors. The drive first decelerates and then accelerates in the reverse direction at a constant maximum allowable current and finally settles at the desired speed.

7.3 FAN OR PUMP AND CRANE HOIST DRIVES

Two major applications of induction motors fed by ac voltage controllers are fan or pump drives and the crane hoist drive.

7.3.1 Fan and Pump Drives

In fan and pump drives, the torque varies as the square of the speed and the power as the cube of the speed. The volume of the fluid delivered by the pump or fan against a constant pressure head is proportional to the motor power, and, hence, to the cube of speed. Thus, a small reduction in speed from full load will cause a large reduction in fluid flow. For example, a 53.6 percent reduction of speed below the full-load value reduces the fluid flow to 10 percent. Therefore, most of the pump and fan drives require speed control only in a narrow range. Because torque reduces as the square of the speed and the speed control is required only in a narrow range, the ac voltage controller fed class D squirrel-cage induction motor with a full-load slip of 0.1 to 0.2 is found suitable for these applications. From equation (6.17) the motor torque is given by

$$T = \frac{3}{\omega_{ms}} I_r'^2 \frac{R_r'}{s} \tag{7.2}$$

Also

$$T_L = C\omega_m^2 = C(1-s)^2\omega_{ms}^2 \tag{7.3}$$

where C is a constant.
If the friction, windage, and core loss torques are neglected,

$$T = T_L \tag{7.4}$$

Substituting from equations (7.2) and (7.3) in equation (7.4) and rearranging the term gives

$$I_r' = K\left[\frac{(1-s)\sqrt{s}}{\sqrt{R_r'}}\right] \tag{7.5}$$

where

$$K = \sqrt{(C\omega_{ms}^3/3)} \tag{7.6}$$

If the magnetizing branch of the equivalent circuit of figure 6.1d is ignored, then $I_s = I_r'$ and hence from equation (7.5)

$$I_s = K\left[\frac{(1-s)\sqrt{s}}{\sqrt{R_r'}}\right] \tag{7.7}$$

Equation (7.7) shows that for a given slip, the motor current is inversely proportional to the square root of the rotor resistance, R_r'. If the full-load stator current and the motor slip are denoted by I_{rated} and s_{rated}, respectively, then from equation (7.7)

$$I_{rated} = K\left[\frac{(1-s_{rated})\sqrt{s_{rated}}}{\sqrt{R_r'}}\right] \tag{7.8}$$

Dividing equation (7.7) by equating (7.8) gives

$$\frac{I_s}{I_{rated}} = \frac{(1-s)\sqrt{s}}{(1-s_{rated})\sqrt{s_{rated}}} \tag{7.9}$$

The slip at which the maximum value of I_s occurs is obtained by equating (dI_s/ds) in equation (7.7) to zero, giving

$$s_m = 1/3 \tag{7.10}$$

Substitution from equation (7.10) into equation (7.9) gives

$$\frac{I_{max}}{I_{rated}} = \frac{2}{3\sqrt{3}\,(1-s_{rated})\sqrt{s_{rated}}} \tag{7.11}$$

where I_{max} is the maximum value of I_s — greater than the rated motor current.

Equation (7.11) suggests that the maximum current has a lower value for a motor with a larger full-load slip. The ratio of maximum to rated motor current has values of 1.35 and 1.07 for full-load slips of 0.1 and 0.2, respectively. Therefore, if a motor with a power rating equal to the full-load power requirement of the load is chosen, it will be overloaded for speeds less than the rated speed.

When a speed range from synchronous speed to two-thirds of snychronous speed (or $s = 1/3$), or to a speed less than this, is required, then the motor current rating will have to be selected as equal to I_{max}. When this is done, the motor current at full load will be less than the rated current. Consequently, the full-load power delivered by the motor will be less than the motor power rating and thus the motor will be derated. The motor derating will be approximately equal to the ratio of rated to maximum current. Thus, for full-load slips of 0.1 and 0.2, the motor will be derated by factors 0.74 and 0.93, respectively. Except when forced cooling is used, the motor cooling will be less at $s = 1/3$ compared to a full-load speed. Hence, actual derating will be higher. These figures suggest that when considering derating, a higher value of full-load slip is preferred. But then the full-load efficiency is adversely affected. A compromise value of the full-load slip is chosen considering these aspects. It generally lies between 0.1 and 0.2.

The foregoing discussion does not account for the harmonics generated by the ac voltage controller. As the motor terminal voltage is decreased to reduce speed, harmonics in the stator and rotor currents increase. According to section 6.8.2 or equation (6.108), the harmonic motor current value depends approximately on the motor reactance alone. Hence the harmonic current remains nearly the same for motors with 0.1 and 0.2 slips if they have the same reactance. Thus, the harmonic current at $s = 1/3$ will not appreciably change by the increase in the full-load slip. When the additional heating of the motor due to harmonics is considered, the motor derating will be more than just mentioned.

In a fan or pump drive, braking is not required, because the fluid pressure provides adequate braking torque. To maintain the constant fluid flow with variations in pressure head and the nature of the pumped fluid, the drive is often operated with a closed-loop speed control.

7.3.2 Crane Hoist Drive

The hoist drive on a crane is required to provide motoring and braking operations in either direction, with a smooth switch-over from motoring to braking and vice versa. Therefore, the motor is fed by the four-quadrant controller of figure 7.3a. Furthermore, it must operate smoothly at controlled speeds over a wide speed range irrespective of the load. The drive may, therefore, be operated with a closed-loop speed control with inner-current control (fig. 7.6).

The use of a class D design squirrel-cage motor fed by an ac voltage controller allows good braking and low-speed motoring torques to be obtained. But prolonged operation under these conditions is not possible because of excessive current. Hence, a wound-rotor induction motor with variable resistances in the rotor is used. By having a sufficiently large rotor resistance, a high torque with reduced motor current is obtained. The motor heating is substantially reduced due to the reduction in current and due to the fact that part of the heat dissipation occurs in the external resistor. The variable resistance may consist of a few sections which are cut in and out by contactors. Alternatively, a chopper controlled resistance (section 9.1) may be used. By the use of combinations of ac voltage controller firing angles and rotor resistance values, the motor characteristics are shaped to satisfy varying torque and speed requirements during motoring and braking.

7.4 AC VOLTAGE CONTROLLER STARTERS

AC voltage controllers are finding applications as induction motor starters. Because of the stepless control of the motor voltage and the control flexibility provided by the low power control circuitry, the ac voltage controller starters have a number of advantages over conventional starters such as an autotransformer starter, a wye-delta starter, and so on. Some of the advantages are smooth acceleration and deceleration; ease in implementation of the current control; simple protection against single-phasing or unbalanced operation; reduced maintenance requirement in applications requiring automatic cyclical starting and stopping; absence of the current inrush which occurs in the open circuit transition to line voltage in the autotransformer and wye-delta starters as explained in section 6.2.

When the operating conditions are favorable, the ac voltage controller starter can allow energy savings by operating the motor at the optimum voltage as discussed in the next section.

7.5 LOSS MINIMIZATION

It was explained in section 6.6 that considerable savings in motor losses can be obtained by reduced voltage operation at light loads. In such applications, voltage control is used for the reduction of losses and not for speed control. The savings in losses primarily depends on three factors: the loading on the motor, the magnitude of the applied voltage, and the quality of the motor construction. The saving in losses is more with 1-phase motors than with three-phase motors.

The most significant factor affecting savings is the loading on the motor. The lighter the load, the greater the savings. The duty cycle of the motor operation is defined as the ratio of the full-load period to the sum of the no-load and full-load periods. The applications with low-duty cycles will allow greater savings in energy. Applications where a motor operates at no load for a significant period of time include drill presses, cutoff saws, gang ripsaw, woodhog, reciprocating air compressors, machine tools, and industrial sewing machines. In such applications, the motor is fed at a fixed voltage, as speed control is not required. Considerable savings in energy can be obtained by operating them at a variable voltage.

The energy savings also depend on the supply voltage. A motor operating near a distribution substation will have higher voltage than the one at the end of the distribution line. Therefore, the voltage reduction will allow greater savings. A badly designed motor will draw more magnetizing current and will have higher core losses from the poor quality of laminations and a larger air-gap. The reduced voltage operation under light loads will allow greater savings than for a well-designed motor.

The variation in the motor voltage can only be obtained by interposing an ac voltage controller between the source and the motor. The ac voltage controller loss and the additional motor loss from harmonic voltages substantially reduce the savings in energy. In most applications, the net savings in energy may not justify the additional cost of an ac voltage controller. However, if the ac voltage controller is also used for motor starting, then the same controller can be employed for loss reduction.

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PROBLEMS

7.1 Calculate and plot the (I_{max}/I_{rated}) versus s_{rated} relation for s_{rated} values from 0.05 to 0.25. Also plot the derating versus s_{rated} curve on the same axis.

7.2 In a pump drive, the fluid flow is to be varied from full to 50 percent. If the full-load slip is 0.15, calculate the maximum motor current to rated current ratio and the motor derating.

8

Frequency-Controlled Induction Motor Drives

The rectifier-fed dc motor is widely used in variable speed drives. It was shown in section 6.4.2 that variable frequency control allows good running and transient performance to be obtained from a squirrel-cage induction motor. The squirrel-cage motor has a number of advantages over a dc motor. It is cheap, rugged, reliable, and longer lasting. Because of the absence of a commutator and brushes, it requires practically no maintenance, it can be operated in an explosive and contaminated environment, and can be designed for higher speeds, voltage, and power ratings. It also has lower inertia, volume, and weight. However, in general, the cost of variable frequency supplies is higher than that of controlled rectifiers. Though the cost of a squirrel-cage motor is much lower compared to that of a dc motor of the same rating, the overall cost of a variable frequency ac drive is, in general, higher.

In special applications requiring maintenance-free operation, such as underground and underwater installations, and also in applications involving explosive and contaminated environments, such as in mines and the chemical industry, variable frequency induction motor drives have already gained popularity. Because of the advantages of squirrel-cage motors and variable frequency control, the variable frequency ac drives find applications in traction, mill runout tables, pumps, fans, blowers, compressors, spindle drives, conveyers, machine tools, and so on.

Due to the availability of power transistors with improved ratings and characteristics, general purpose low-power variable frequency drives are now available with a cost comparable to that of dc drives. The recent progress in GTOs may provide variable frequency drives which will compete very well and probably replace dc drives in medium and lower range high-power drives.

This chapter considers the implementation and performance of variable frequency drives employing power semiconductor converters. The following converters are employed:

1. Voltage source inverter
2. Current source inverter and
3. Cycloconverter

Inverters convert dc to variable frequency ac. An inverter belongs to the voltage source category if, viewed from the load side, the ac terminals of the inverter function as a voltage source. Similarly, an inverter which behaves as a current source at its ac terminals is called a current source inverter.

Because of a low internal impedance, the terminal voltage of a voltage source inverter remains substantially constant with variations in load. It is therefore equally suitable to single-motor and multi-motor drives. Any short-circuit across its terminals causes current to rise very fast, due to the low time constant of its internal impedance. The fault current cannot be regulated by current control and must be cleared by fast-acting fuse links.

Because of a large internal impedance, the terminal voltage of a current source inverter changes substantially with a change in load. Therefore, if used in a multi-motor drive, a change in load on any motor affects other motors. Hence, current source inverters are not suitable for multi-motor drives. Since the inverter current is independent of load impedance, it has inherent protection against short-circuits across its terminals.

The cycloconverter allows a variable frequency ac supply with voltage or current source characteristics to be obtained from a fixed frequency voltage source.

8.1 CONTROL OF INDUCTION MOTOR BY VOLTAGE SOURCE INVERTERS

The variable frequency control of an induction motor has been described in section 6.4.2 and the effects of harmonics on the motor performance have been presented in section 6.8. It is useful to summarize salient points before the operation of an induction motor fed from voltage source inverters is considered.

It was explained in section 6.4.2 that motor operation below the base speed is carried out at a constant (V/f) ratio and that operation above the base speed is done at a constant voltage. At low speeds, the (V/f) ratio may be increased to compensate for the stator resistance drop. The operation of the drive at the maximum permissible current allows constant torque operation from standstill to base speed and constant power operation from base speed to critical speed at which the breakdown torque is reached (figs. 6.13 and 6.14). Operation for speeds above critical speed is possible at reduced power. In this speed range, the drive is usually controlled at a constant slip speed to prevent pull-out. The range of constant power operation can be increased by the use of a motor with a higher breakdown torque. A motor has a higher breakdown torque when its leakage reactance has a low value. When the leakage reactance is low, the motor has higher derating and torque pulsations in the presence of harmonics, as explained next.

The points stated in the previous paragraph for motoring operation are also applicable to regenerative braking operation, with the exception that the constant torque operation is available up to a speed higher than the base speed.

When the (V/f) ratio is increased to compensate for the stator resistance drop at low speeds, during the motoring operation the motor saturates at light loads and may carry a current higher than the rated current.

Usually, a class B design squirrel-cage motor is employed. Some applications may use the energy efficient class A design.

The operation of an induction motor on a nonsinusoidal supply has been described in section 6.8. Because the magnetizing reactance has a large value (fig. 6.1b or d), the supply harmonics are able to produce a negligible magnetizing current, and hence, the flux remains essentially sinusoidal. The harmonic rotor current, which is independent of motor speed, depends on the harmonic voltage and the motor reactance [equation (6.108)]. The motor reactance acts as a low-pass filter and substantially reduces high-frequency harmonics in the rotor current. Consequently, a motor with a high reactance is preferred when its terminal voltage has a high harmonic content. Because the flux is essentially sinusoidal, the contribution of harmonics to the developed torque and power is negligible. However, harmonic currents increase the copper loss, leading to motor derating. The skin effect substantially increases the harmonic copper loss in deep-bar and double-cage rotors. They may not be used when the harmonic content is large. The interaction between the fundamental flux and the fifth and seventh harmonic rotor currents produces a pulsating torque six times the fundamental frequency. Similarly, eleventh and thirteenth harmonic rotor currents produce a pulsating torque 12 times the fundamental frequency, but its magnitude is comparatively much smaller. At low fundamental frequency, for which the motor speed is low, the pulsating torques cause large fluctuations in speed, producing a jerky or stepped motion and shaft cogging at the time of motor reversal. A high motor reactance helps in reducing machine derating and torque pulsations. Hence a class B design squirrel-cage motor with a large reactance must be employed when the terminal voltage has a large harmonic content. But then the breakdown torque and the speed range of constant power operation reduce. The class A design motor must be used only when the motor terminal voltage has a negligible harmonic content.

The present section describes 3-phase voltage source inverters and the variable frequency drives employing voltage source inverters.

8.1.1 Three-Phase Voltage Source Inverter

The power circuit of a three-phase voltage source inverter is shown in figure 8.1a. It consists of six self-commutated semiconductor switches (see section 1.6.6) S_1 to S_6 with the antiparallel diodes D_1 to D_6. The switches need not have reverse voltage blocking capability. They may be realized using power transistors, GTOs, MOSFETs, or inverter grade thyristors with forced commutation circuits. The relative merits of these devices are given in section 1.6. For diodes D_1 to D_6, fast recovery diodes are employed. A snubber (not shown) is required for each switch-diode pair. The motor, which is connected across terminals A, B, and C may have wye or delta connection. The inverter may be operated as a six-step inverter or as a pulse-width modulated (PWM) inverter as described in the following sections.

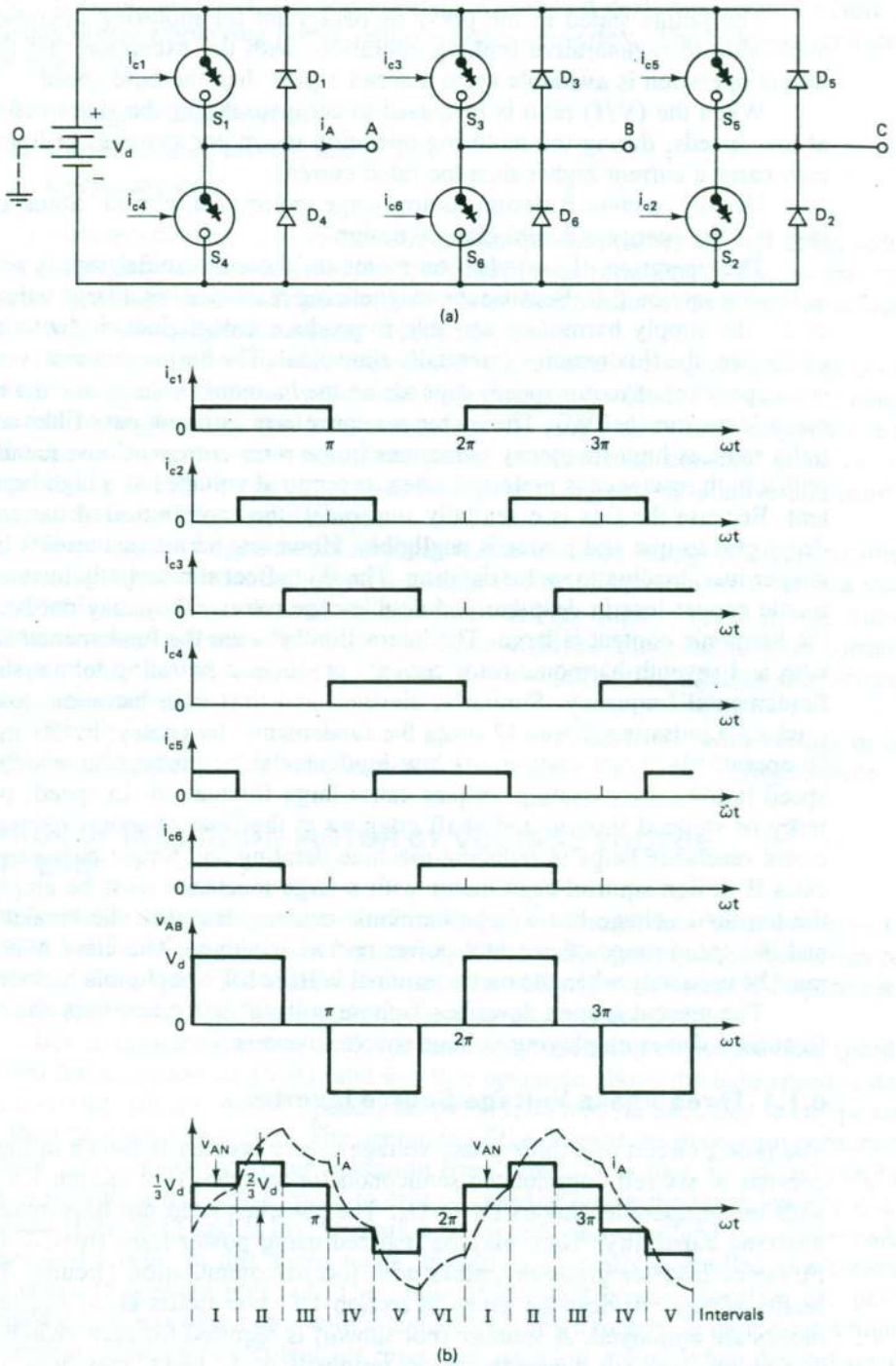


Figure 8.1 Three phase voltage source inverter.

8.1.2 Six-Step Inverter

The control signals for the six switches of the inverter of figure 8.1a, i_{c1} to i_{c6}, are shown in figure 8.1b. In the inverter output voltage period of 2π radians (or T sec.), each control signal has a duration of π radians. The control signals are applied to the switches in the sequence of their numbers, with a phase difference of $\pi/3$ radians. In figure 8.1b, the period T has been divided into six equal intervals. During each interval the following switches receive control signals:

Interval	Control Signals Applied To	Interval	Control Signals Applied To
I	1, 5, 6	IV	2, 3, 4
II	1, 2, 6	V	5, 4, 3
III	1, 2, 3	VI	5, 4, 6

A switch conducts and carries current in the direction of its diode when the control signal is present and the switch is forward biased. A switch will always be forward biased, except when its antiparallel diode conducts and thus reverse biases the switch by its voltage drop. In any case, either the switch or its antiparallel diode will be in conduction during the presence of the control signal, and the current will be free to flow in either direction. Using this information the waveforms of line voltages can be drawn. For example, during interval I, the switch-diode pairs 1, 5, and 6 are in conduction. Hence, terminals A and C are connected to the positive terminal of the dc source and terminal B is connected to the negative terminal of the dc source, giving the following values of line voltages:

$$v_{AB} = V_d, \qquad v_{BC} = -V_d \quad \text{and} \quad v_{CA} = 0$$

The line voltage v_{AB} is shown in figure 8.1b. The line voltages v_{BC} and v_{CA} will lag behind v_{AB} by 120° and 240°, respectively. For a wye-connected load, the phase voltages can be obtained in the same way. The phase voltage v_{AN} is also shown in figure 8.1b.

The switch pairs (S_1, S_4), (S_3, S_6), and (S_5, S_2) form three legs of an inverter. The switches in the same leg conduct alternately. Some time must elapse between the turn-off of one switch and turn-on of another switch in the same leg to ensure that both do not conduct simultaneously. Their simultaneous conduction will cause a short-circuit of the dc source. The resultant current will rise very fast. This fault, known as a shoot-through fault, can only be cleared by fast-acting fuse links.

In the operation under consideration, a cycle of the line voltage or the phase voltage is generated in six steps. Hence, when operating this way, the inverter is known as a six-step inverter. The voltages v_{AB} and v_{AN} are described by the following Fourier series:

$$v_{AB} = \frac{2\sqrt{3}}{\pi} V_d[\sin(\omega t + \pi/6) + \frac{1}{5}\sin(5\omega t - \pi/6) + \frac{1}{7}\sin(7\omega t + \pi/6)\ldots] \tag{8.1}$$

$$v_{AN} = \frac{2}{\pi} V_d \left[\sin \omega t + \frac{1}{5} \sin 5\omega t + \frac{1}{7} \sin 7\omega t \ldots \right] \tag{8.2}$$

The rms value of the fundamental phase voltage

$$V_1 = \frac{\sqrt{2}}{\pi} V_d \tag{8.3}$$

The rms value of the phase voltage is

$$V = \left[\frac{1}{\pi} \left\{ \int_0^{\pi/3} \left(\frac{1}{3} V_d \right)^2 d(\omega t) + \int_{\pi/3}^{2\pi/3} \left(\frac{2}{3} V_d \right)^2 d(\omega t) + \int_{2\pi/3}^{\pi} \left(\frac{1}{3} V_d \right)^2 d(\omega t) \right\} \right]^{1/2}$$
$$= \frac{\sqrt{2}}{3} V_d \tag{8.4}$$

The harmonic content of line and phase voltages is the same. The different waveforms are due to a different phase relationship between the harmonics and the fundamental. Only odd harmonics of the order $k = 6n \pm 1$ are present, where n is an integer. The tripplen (multiple of 3) harmonics are absent, and hence, there is no problem of circulating current in a delta-connected stator winding.

The nature of the phase-current waveform for a wye-connected stator is shown in figure 8.1b. This is also the output current i_A of the inverter. It is shared between the switch-diode pairs (S_1, D_1) and (S_4, D_4). During the interval $0 \leq \omega t \leq \pi$, when S_1 receives the control signal, the pair (S_1, D_1) carries current i_A; the positive i_A is carried by S_1, and the negative i_A flows through D_1. During the interval $\pi \leq \omega t \leq 2\pi$, when S_4 receives the control signal, i_A is carried by the pair (S_4, D_4); the positive i_A flows through D_4, and the negative i_A is carried by S_4. An identical waveform of i_A is obtained for a delta-connected stator.

Equations (8.1) and (8.2) give the fundamental and harmonics in the phase voltages of delta- and wye-connected stators, respectively. The analysis presented in section 6.8 can be employed for the calculation of the motor performance. As explained in section 8.1 (or section 6.8), the developed torque can be obtained by considering the fundamental alone. The equivalent circuit of figure 6.1d or equations (6.54) to (6.57) is used to calculate the developed torque and also the fundamental component of the current. The harmonic currents are obtained from equation (6.108).

Induction motor control requires the simultaneous variation of frequency and voltage. The frequency of the inverter output voltage can be controlled by varying the time period of one cycle. This is done simply by varying the duration of the control signals I_{c1} to i_{c6} (figure 8.1b). According to equations (8.1) and (8.2), the fundamental output voltage of this inverter is fixed. It can be varied by using the methods described in the next section.

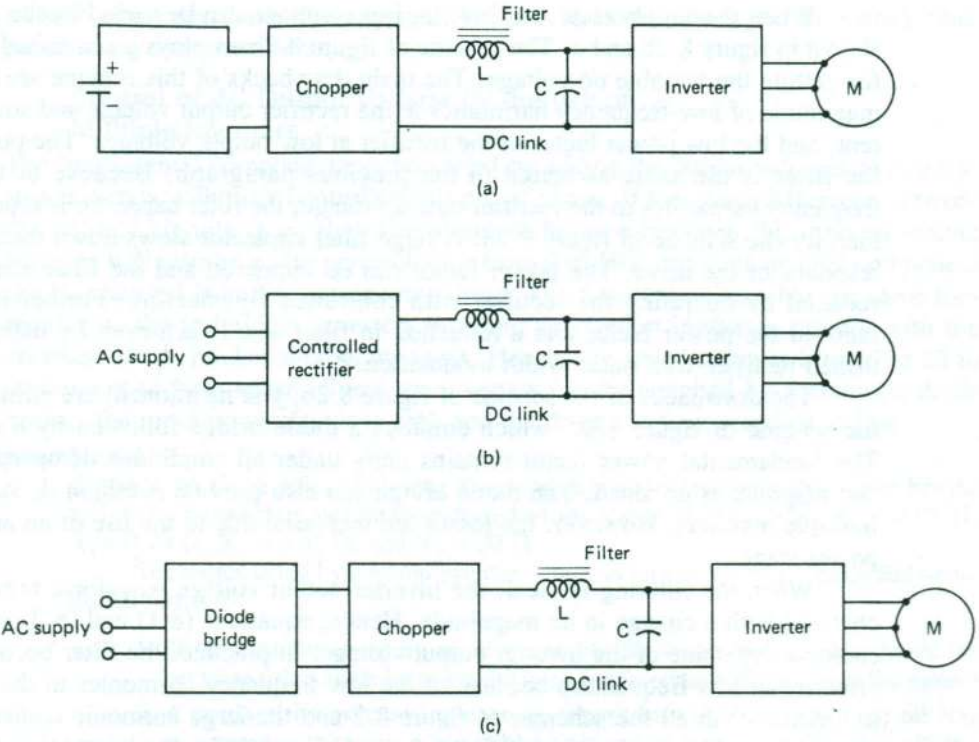


Figure 8.2 Voltage control in six-step inverter drive.

8.1.3 Six-Step Inverter Voltage Control

The output voltage of a six-step inverter can be controlled by either of the following methods:

1. Control of the dc input voltage.
2. Control of the ac output voltage by the use of multiple inverters.

Control of DC Input Voltage

The fundamental component of the output voltage of an inverter can be changed by controlling the inverter input voltage V_d. When the inverter is fed from a dc source, as in the case of underground traction, battery-operated vehicles, drives powered by solar cells, and so on, the dc input voltage of the inverter can be varied by connecting a chopper in between the source and the inverter, as shown in figure 8.2a. Depending on the application, the chopper can be a step-down or step-up. An L-C filter is interposed between the chopper and inverter to maintain a ripple-free dc voltage at the input of the inverter, and thus, prevent the harmonics in the chopper output voltage from interfering with the inverter, and the harmonics in the inverter input current from interfering with the chopper.

When the supply is ac, the inverter input voltage can be varied by the schemes shown in figure 8.2b and c. The scheme of figure 8.2b employs a controlled rectifier for getting the variable dc voltage. The main drawbacks of this scheme are the high magnitude of low-frequency harmonics in the rectifier output voltage and source current, and the low power factor of the rectifier at low-output voltages. The purpose of the filter is the same as stated in the previous paragraph. Because of the low-frequency harmonics in the rectifier output voltage, the filter capacitor is much larger than for the scheme of figure 8.2a. A large filter capacitor slows down the transient response of the drive. The power factor can be improved and the filter size can be reduced by operating the rectifier with controlled flywheeling. Further improvement in the power factor and a reduction in filter size is achieved by using a controlled rectifier with pulse-width modulation.[29]

The drawbacks of the scheme of figure 8.2b, just mentioned, are eliminated in the scheme of figure 8.2c, which employs a diode bridge followed by a chopper. The fundamental power factor remains unity under all conditions of operation and fast response is obtained. The diode bridge can also provide common dc supply for multiple inverters. However, the losses are increased due to the use of an additional power stage.

When the filtering is ideal, the inverter output voltage waveform remains unchanged with a change in its magnitude. Hence, equations (8.1) and (8.2) are applicable to any value of the inverter output voltage. In practice, the filter becomes less effective at low frequencies because of the low-frequency harmonics in the inverter input current in all the schemes of figure 8.2 and the large harmonic content in the output voltage of the rectifier of figure 8.2b. Consequently, the harmonic content of the output voltage increases with the decrease in the motor speed. However, for the analysis of the drive, the filtering is assumed to be ideal.

The schemes of figure 8.2a to c suffer from the following drawbacks:

1. Because of the presence of low-frequency harmonics, the motor losses are increased at all speeds causing derating of the motor.
2. Torque pulsations are present at low speeds owing to the presence of fifth, seventh, eleventh, and thirteenth harmonics.
3. The harmonic content increases at low speeds, increasing the losses. The increase in the (V/f) ratio at low speeds to compensate for the stator resistance drop may cause a higher motor current to flow at light loads due to saturation. These two effects may overheat the machine at low speeds.

Drawbacks 2 and 3 require the minimum speed to be restricted to 10 percent of base speed.

A thyristor inverter suffers from an additional drawback. The drive requires the same current commutation capability at all speeds. But the commutation capability of a thyristor inverter reduces with the reduction in the input dc voltage. This problem is overcome by the use of a separate fixed voltage supply for the commutation circuit. The inverters employing power transistors, MOSFETS, and GTOs do not suffer from this problem, because these devices can be turned off by the base or gate drive. Because of the absence of commutation circuits, the GTO, MOSFET, and

transistor inverters have lower weight, volume, and cost, and higher efficiency compared to thyristor inverters.

Control of AC Output Voltage by the use of Multiple Inverters

The fundamental component can be varied by adding the output voltages of two six-step inverters with their control signals out of phase. If the phase difference between the control signals is φ, then the phase difference between the inverter output voltages will also be φ. By controlling φ from 0 to 180°, the fundamental component can be changed from the maximum to zero. The resultant voltage has the same harmonics as the individual inverters but their proportion increases rapidly with the decrease in the fundamental component. Hence, the speed range is limited to 25 to 30 percent of base speed. Since the inverters can be supplied by a common diode bridge, the input power factor is high and the drive transient response is fast.

Example 8.1

A 460 V, 60 Hz, 6 pole, 1180 rpm, Y-connected squirrel-cage induction motor has the following parameters per phase referred to the stator: $R_s = 0.19\ \Omega$, $R_r' = 0.07\ \Omega$, $X_s = 0.75\ \Omega$, $X_r' = 0.67\ \Omega$, and $X_m = 20\ \Omega$

The motor is fed by a 6-step inverter, which in turn is fed by a 6-pulse fully-controlled rectifier.

1. If the vectifier is fed by an ac source of 460 V and 60 Hz, what should the rectifier firing angle be to get the rated fundamental voltage across the motor?
2. Calculate the percent increase in copper loss of the machine at 60 Hz, compared to the value when fed by a sinusoidal supply. Neglect skin effect.
3. If the machine is operated at a constant flux,
   (a) Calculate the inverter frequency at 600 rpm and rated torque.
   (b) Calculate the inverter frequency at 500 rpm and half the rated torque. Also calculate the motor current.

Neglect the derating due to harmonics and use the equivalent circuit of figure 6.1d.

Solution:

1. From equation (8.1), the fundamental rms line voltage of a 6-step inverter is given by

$$V_L = \frac{\sqrt{6}}{\pi} V_d \tag{E8.1}$$

Also from equation (3.78), for a six-pulse rectifier,

$$V_d = \frac{3}{\pi} V_m \cos \alpha \tag{E8.2}$$

where V_m = peak of ac source line voltage.
From equations (E8.1) and (E8.2)

$$V_L = \frac{3\sqrt{6}}{\pi^2} V_m \cos \alpha$$

or

$$\cos \alpha = \frac{V_L}{V_m} \cdot \frac{\pi^2}{3\sqrt{6}} \tag{E8.3}$$

Here $V_L = 460$ V, $V_m = 460\sqrt{2}$ V.
Hence $\cos \alpha = \pi^2/6\sqrt{3} = 0.95$ or $\alpha = 18.25°$.

2. From equation (6.1), the synchronous speed in rpm,

$$N_S = \frac{120f}{p} = \frac{120 \times 60}{6} = 1200 \text{ rpm}$$

$$\text{Rated slip} = \frac{1200 - 1180}{1200} = \frac{1}{60} = 0.0167$$

From figure 6.1d

$$I_r' = \frac{V_L/\sqrt{3}}{\sqrt{(R_s + R_r'/s)^2 + (X_s + X_r')^2}}$$

Hence, the rated rotor current is

$$I_{rR}' = \frac{460/\sqrt{3}}{\sqrt{\left(0.19 + \frac{0.07}{0.0167}\right)^2 + (0.75 + 0.67)^2}} = 57.66 \text{ A}$$

Now $\omega_{ms} = \frac{1200}{60} \times 2\pi = 40\pi = 125.66$ rad/sec.

Since

$$T = \frac{3}{\omega_{ms}} I_r'^2 R_r'/s$$

the rated torque is

$$T_r = \frac{3}{125.66}(57.66)^2 \frac{0.07}{0.0167} = 332.7 \text{ N-m}$$

From equation (6.108), the kth harmonic current (which is the same for the stator and rotor) at a per unit frequency a

$$I_{rk}' = \frac{V_k}{k(X_s + X_r')a} \tag{E8.4}$$

where V_k is the harmonic rms phase voltage at a per unit frequency a.
From equation (8.2)

$$V_k = \frac{V_1}{k} \tag{E8.5}$$

where V_1 is the fundamental phase voltage at a per unit frequency a.
From (E8.4) and (E8.5)

$$I_{rk}' = \frac{V_1}{ak^2(X_s + X_r')} \tag{E8.6}$$

Hence rms harmonic current is

$$I_h = \frac{V_1}{a(X_s + X_r')}\left(\sum_{k=5,7,11,13}^{\infty} \frac{1}{k^4}\right)^{1/2}$$

Neglecting harmonics higher than 13 gives

$$I_h = \frac{V_1}{a(X_s + X_r')}\left(\frac{1}{5^4} + \frac{1}{7^4} + \frac{1}{11^4} + \frac{1}{13^4}\right)^{1/2}$$

$$= \frac{0.046V_1}{a(X_s + X_r')} \qquad \text{(E8.7)}$$

Substitution of the known values gives

$$I_h = \frac{0.046 \times 265.6}{(0.75 + 0.67)} = 8.6 \text{ A}$$

Hence, the rms rotor current

$$I_{rr}' = \sqrt{(57.66)^2 + (8.6)^2} = 58.3 \text{ A}$$

Copper loss with sinusoidal source

$$= 3(57.66)^2(0.19 + 0.07) = 2593 \text{ W}$$

Copper loss with inverter supply

$$= 3(58.3)^2(0.19 + 0.07) = 2651 \text{ W}$$

Percent increase in copper loss

$$= \frac{2651 - 2593}{2593} = 2.2 \text{ percent}$$

3. **(a)** It has been shown in section 6.4.2 that for a given torque the motor operates at a fixed slip speed for all frequencies when the flux is maintained constant. The slip speed in rpm at the rated torque

$$N_{s\ell} = 1200 - 1180 = 20 \text{ rpm}$$

Hence, synchronous speed at 600 rpm

$$N_s = N + N_{s\ell} = 600 + 20 = 620 \text{ rpm}$$

The inverter frequency = $(620/1200) \times 60 = 31$ Hz

(b) The back emf at the rated operation

$$E_{rated} = I_r'[(R_r'/s)^2 + X_r'^2]^{1/2}$$

$$= 57.66\left[\left(\frac{0.07}{0.0167}\right)^2 + (0.67)^2\right]^{1/2} = 244.76 \text{ V}$$

Torque at a constant flux is given by equation (6.44):

$$T = \frac{3}{\omega_{ms}}\left[\frac{E_{rated}^2 R_r'/(as)}{R_r'^2/(as)^2 + X_r'^2}\right] \qquad (6.44)$$

Note that ω_{ms} and X_r' in this equation are for rated frequency. Substituting the known values gives

$$\frac{332.7}{2} = \frac{3}{125.66}\left[\frac{(244.76)^2 \times 0.07/(as)}{\left(\frac{0.07}{sa}\right)^2 + (0.67)^2}\right]$$

or

$$\left[\left(\frac{0.07}{as}\right)^2 + (0.67)^2\right] = \frac{0.6}{as}$$

or

$$\left(\frac{1}{as}\right)^2 - 122.44\frac{1}{(as)} + 91.61 = 0$$

which gives

$$\frac{1}{as} = 121.7 \tag{E8.8}$$

From equation (6.43)

$$a = \frac{\omega_m}{(1-s)\omega_{ms}} = \frac{500}{1200}\frac{1}{(1-s)} = \frac{0.417}{(1-s)}$$

Substituting from equation (E8.8) for a gives

$$\frac{1}{121.7s} = \frac{0.417}{(1-s)}$$

which gives $s = 0.0193$
From equation (E8.8), $a = 0.425$
Thus, frequency $= 0.425 \times 60 = 25.5$ Hz
Substituting the known values in equation (6.42) gives

$$I_r' = \frac{244.76}{[(0.07 \times 121.7)^2 + (0.67)^2]^{1/2}} = 28.64 \text{ A}$$

Machine fundamental phase voltage

$$V_1 = I_r'\left[\left(R_s + \frac{R_r'}{s}\right)^2 + a^2(X_s + X_r')^2\right]^{1/2}$$

$$= 28.64\left[\left(0.19 + \frac{0.07}{0.0193}\right)^2 + (0.425 \times 1.42)^2\right]^{1/2}$$

$$= 110.68 \text{ V}$$

Taking V_1 as a reference vector

$$\bar{I}_r' = 28.64\underline{/-9^\circ} \text{ A}$$

$$\bar{I}_m = \frac{V_1}{aX_m}\underline{/-90^\circ} = \frac{110.68}{0.425 \times 20}\underline{/-90^\circ} = 13.02\underline{/-90^\circ} \text{ A}$$

$$\bar{I}_s = \bar{I}_r' + \bar{I}_m = 28.64\underline{/-9^\circ} + 13.02\underline{/-90^\circ}$$

or

$$I_s = 33.27 \text{ A}$$

From equation (E8.7),

$$I_h = \frac{0.046 V_1}{a(X_s + X_r')} = \frac{0.046 \times 110.68}{0.425 \times 1.42} = 8.44 \text{ A}$$

The rms input current $= (I_s^2 + I_h^2)^{1/2}$

$$= [(33.27)^2 + (8.44)^2]^{1/2} = 34.32 \text{ A}$$

Example 8.2

If the drive of example 8.1 is operated at a constant (V/f) ratio, calculate the inverter frequency and the stator current at half the rated torque and 500 rpm. Neglect derating due to harmonics and use the equivalent circuit of figure 6.1d.

Solution: The torque for a constant (V/f) ratio is given by equation (6.54):

$$T = \frac{3}{\omega_{ms}} \left[\frac{V_1^2 R_r'/(as)}{\left(\frac{R_s}{a} + \frac{R_r'}{as}\right)^2 + (X_s + X_r')^2} \right] \tag{6.54}$$

Note that V_1, ω_{ms}, and $(X_s + X_r')$ in this equation are for the rated machine frequency. From example 8.1, $\omega_{ms} = 125.66$ rad/sec., $V_1 = 265.6$ V

$$T = 0.5 \times 332.7 = 166.35 \text{ N-m}$$

Subsituting the known values in equation (6.54) gives

$$166.35 = \frac{3}{125.66} \cdot \frac{(265.6)^2 \times 0.07/(as)}{\left(\frac{0.19}{a} + \frac{0.07}{as}\right)^2 + (1.42)^2}$$

or

$$\left(\frac{0.19}{a} + \frac{0.07}{as}\right)^2 + 2 = \frac{0.71}{as} \tag{E8.9}$$

Now $\omega_m = 500 \times 2\pi/60 = 52.36$ rad/sec.
From equation (6.43)

$$a = \frac{\omega_m}{(1 - s)\omega_{ms}}$$

or

$$a = \frac{52.36}{(1 - s)125.66} = \frac{0.417}{(1 - s)} \tag{E8.10}$$

Equations (E8.9) and (E8.10) can be solved by iteration. For an assumed value of s, a is obtained from equation (E8.10). These values of a and s are substituted in equation (E8.9) to check whether it is satisfied. If not, the calculations are repeated for another value of s. The iteration gives the following values:

$$s = 0.0181 \qquad a = 0.4247$$

Hence, the inverter frequency $= 0.4247 \times 60 = 25.5$ Hz

At 25.5 Hz, $V_1 = a \times 265.6 = 112.76$ V

$$\bar{I}_r' = \frac{V_1}{\sqrt{(R_s + R_r'/s)^2 + a^2(X_s + X_r')^2}} \angle -\tan^{-1} \frac{a(X_s + X_r')}{(R_s + R_r'/s)}$$

$$= \frac{112.76}{\sqrt{\left(0.19 + \frac{0.07}{0.0181}\right)^2 + (0.4247 \times 1.42)^2}} \angle -\tan^{-1} \frac{0.4247 \times 1.42}{0.19 + 0.07/0.0181}$$

$$= 27.49 \angle -8.4° \text{ A}$$

$$\bar{I}_m = \frac{V_1}{aX_m} \angle -90° = \frac{112.76}{0.4247 \times 20} \angle -90° = 13.27 \angle -90° \text{ A}$$

$$\bar{I}_s = \bar{I}_r' + \bar{I}_m = 27.49 \angle -8.4° + 13.27 \angle -90° \text{ A}$$

Hence, $I_s = 32.2$ A

From equation (E8.7) of example 8.1,

$$I_h = \frac{0.046 V_1}{a(X_s + X_r')} = \frac{0.046 \times 112.76}{0.4247 \times 1.42} = 8.6 \text{ A}$$

The rms input current $= (I_s^2 + I_h^2)^{1/2}$

$$= (32.2^2 + 8.6^2)^{1/2} = 33.3 \text{ A}.$$

Example 8.3

If the drive of example 8.1 is driving a load requiring rated power for all speeds greater than base speed, calculate the torque, frequency, stator current, efficiency, fundamental power factor, and motor power factor for a speed of 1300 rpm. Use the equivalent circuit of figure 6.1d and neglect friction, windage, core loss, skin effect, and derating due to harmonics.

Solution: From example 8.1,

$$\text{Rated power} = 332.7 \times \frac{(1180 \times 2\pi)}{60} = 41.11 \text{ kW}$$

$$\omega_m = \frac{1300 \times 2\pi}{60} = 136 \text{ rad/sec}.$$

$$\text{Torque} = \frac{41.11 \times 10^3}{136} = 302 \text{ N-m}$$

From equation (6.56) for $a > 1$

$$T = \frac{3}{\omega_{ms}} \left[\frac{V_{rated}^2 R_r'/(as)}{\left(R_s + \frac{R_r'}{s}\right)^2 + a^2(X_s + X_r')^2} \right]$$

Substituting known values gives

$$302 = \frac{3}{125.66} \cdot \frac{(265.6)^2 \times 0.07/(as)}{\left(0.19 + \frac{0.07}{s}\right)^2 + a^2(1.42)}$$

which gives

$$\left(0.19 + \frac{0.07}{s}\right)^2 + a^2(1.42)^2 = \frac{0.39}{as} \tag{E8.11}$$

From equation (6.43),

$$a = \frac{\omega_m}{\omega_{ms}} \cdot \frac{1}{(1-s)} = \frac{1300}{1200} \frac{1}{(1-s)} \tag{E8.12}$$

Solution of equations (E8.11) and (E8.12) by iteration gives

$$s = 0.0172 \quad \text{and} \quad a = 1.102$$

Hence, frequency = $60 \times 1.102 = 66.1$ Hz
Now

$$I_r' = \frac{V_1}{\sqrt{(R_s + R_r'/s)^2 + a^2(X_s + X_r')^2}}$$

$$= \frac{265.6}{\sqrt{\left(0.19 + \frac{0.07}{0.0172}\right)^2 + (1.102 \times 1.42)^2}} = 58.53 \text{ A}$$

As expected this is equal to the rated rotor current. Phase angle of $\bar{I}_r'$ with respect to the reference vector V_1

$$= \tan^{-1} \frac{a(X_s + X_r')}{R_s + R_r'/s} = \tan^{-1}\left(\frac{1.102 \times 1.42}{0.19 + 0.07/0.0172}\right) = 20.2°$$

$$\bar{I}_r' = 58.53 \angle 20.2° \text{ A}$$

$$\bar{I}_m = \frac{V_1}{jaX_m} = \frac{265.6}{j1.102 \times 20} = 12.05 \angle -90° \text{ A}$$

$$\bar{I}_s = \bar{I}_r' + \bar{I}_m = 58.53 \angle -20.2° + 12.05 \angle -90° = 63.7 \angle -30.4°$$

From equation (E8.7) of example 8.1

$$I_h = \frac{0.046 V_1}{a(X_s + X_r')} = \frac{0.046 \times 265.6}{1.102 \times 1.42} = 7.8 \text{ A}$$

Input current $= I = (I_s^2 + I_h^2)^{1/2}$

$$= (63.7^2 + 7.8^2)^{1/2} = 64.2 \text{ A}$$

Power developed $= T\omega_m = 302 \times 136 = 41.07$ kW
According to the equivalent circuit of figure 6.1d,

copper loss $= 3(I_r'^2 + I_h^2)(R_s + R_r')$

$$= 3(58.53^2 + 7.8^2)(0.19 + 0.07) = 2.72 \text{ kW}$$

Power input $= 41.07 + 2.72 = 43.79$ kW

$$\text{Efficiency} = \frac{41.07}{43.79} = 94 \text{ percent}$$

Fundamental power factor = cos 30.4° = 0.86
From equations (8.3) and (8.4), rms phase voltage

$$V = \frac{\pi}{3} V_1 \qquad \text{(E8.13)}$$

$$= 1.047 \times 265.6 = 278.1 \text{ V}$$

$$\text{Apparent power} = 3VI = 3 \times 278.1 \times 64.2 = 53.56 \text{ kVA}$$

$$\text{Power factor} = \frac{\text{real power}}{\text{apparent power}} = \frac{43.79}{53.56} = 0.82.$$

8.1.4 PWM Inverters

The drawbacks of 6-step inverter drives, described in the previous section, are eliminated in pulse-width modulated (PWM) inverter drives, shown in figure 8.3. The PWM inverters also have provision for the control of the output voltage; hence they can be supplied from a fixed dc voltage. When the mains supply is dc, the scheme of figure 8.3a is used. When the supply is ac, the inverter is supplied from a diode bridge as shown in figure 8.3b, giving unity fundamental power factor. Because of a low harmonic content in the output voltage of a diode bridge and in the input current of a PWM inverter, the filter components are small; consequently drive response is fast.

Because of a low harmonic content in the inverter output voltage, the drive has smooth low-speed operation, free from torque pulsations and cogging, and with a lower derating of the motor and higher efficiency. Because of a constant dc bus voltage, a number of PWM inverters with their associated motors can be supplied from a common diode bridge and the commutation problem associated with a 6-step thyristor inverter, as explained in the previous section, is eliminated. These advantages, however, are obtained at the expense of a complex control and a higher switching loss due to a higher frequency operation of the switches.

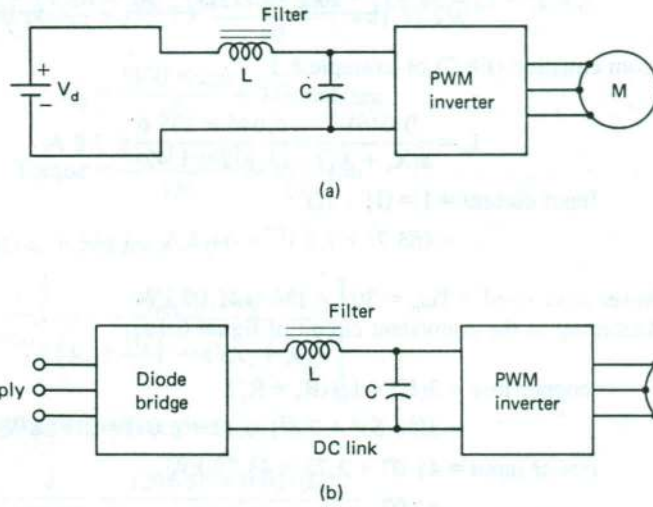


Figure 8.3 Pulse width modulated inverter drives.

There are a number of schemes of PWM. Prominent among these are sinusoidal PWM,[1] PWM with uniform sampling,[2-4] "optimal" PWM techniques based on the minimization of certain performance criteria[5-11] — for example, selective harmonic elimination,[5-7] optimization of efficiency,[8-10] minimization of torque pulsations,[11] and so on. While the sinusoidal pulse-width modulation and the modulation with uniform sampling can be implemented using analog techniques, the optimal pulse-width modulation techniques require the use of a microprocessor, with the exception of the scheme described in reference 7.

All these schemes use the power circuit of figure 8.1a.

Sinusoidal Pulse-Width Modulation

Three-phase reference voltages v_a, v_b, and v_c of a variable amplitude A are compared in three separate comparators with a common isosceles triangular carrier wave v_T of a fixed amplitude A_m, as shown in figure 8.4a. The outputs of comparators 1, 2, and 3 form the control signals for the three legs of the inverter formed by switch pairs (S_1, S_4), (S_3, S_6), and (S_5, S_2), respectively (fig. 8.1a). Let us consider the operation of the pair (S_1, S_4), which controls the voltage of the machine phase A with respect to the imaginary middle point of the dc source, O. This is explained in figure 8.4b where the reference wave v_a and the carrier wave v_T are drawn on a common time axis for a positive half-cycle of v_a. Switch S_1 receives the control signal when $v_a > v_T$, and switch S_4 receives it when $v_a < v_T$. The resultant waveform of v_{AO} is shown in the figure. The waveforms of figure 8.4b are drawn for the case when a cycle of the reference wave consists of 12 cycles of the triangular wave. One can similarly draw voltages v_{BO} and v_{CO} by considering the operation of switch pairs (S_3, S_6) and (S_5, S_2), respectively. The modulation is called sinusoidal PWM because the pulse width is a sinusoidal function of its angular position in the cycle. The modulation is also known as triangulation or PWM with natural sampling. The line voltage v_{AB} is obtained by subtracting v_{BO} from v_{AO}. Similarly the line voltages v_{BC} and v_{CA} are obtained. Production of the line voltage waveform v_{AB} when each cycle of the reference wave has 6 cycles of triangular wave is shown in figure 8.5.

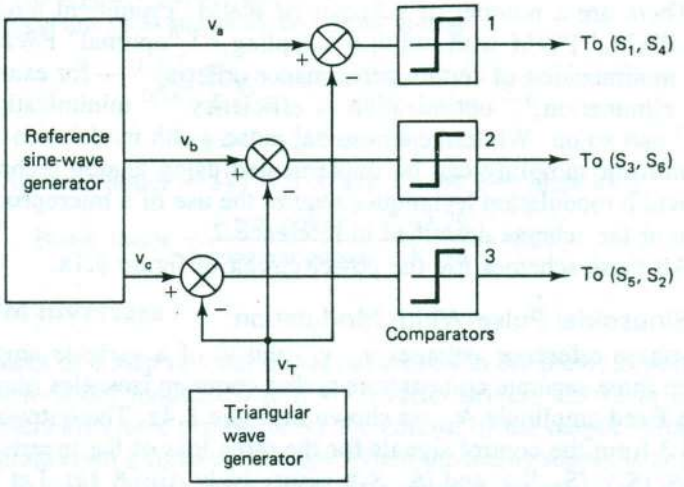
The frequency of the fundamental component of the motor terminal voltage is the same as that of the reference sinusoidal voltages. Hence, the frequency of the motor voltage can be changed by changing the frequency of the reference voltages. The ratio of the amplitude of the reference wave to that of the carrier wave, m, is called the modulation index, Thus,

$$m = \frac{A}{A_m} \tag{8.5}$$

The fundamental (rms) component in the waveform v_{AO} is given by

$$V_1 = \frac{mV_d}{2\sqrt{2}} \tag{8.6}$$

Thus, the fundamental voltage increases linearly with m until $m = 1$ (that is, when the amplitude of the reference wave becomes equal to that of the carrier wave). For $m > 1$, the number of pulses in v_{AO} becomes less and the modulation ceases to be sinusoidal PWM.



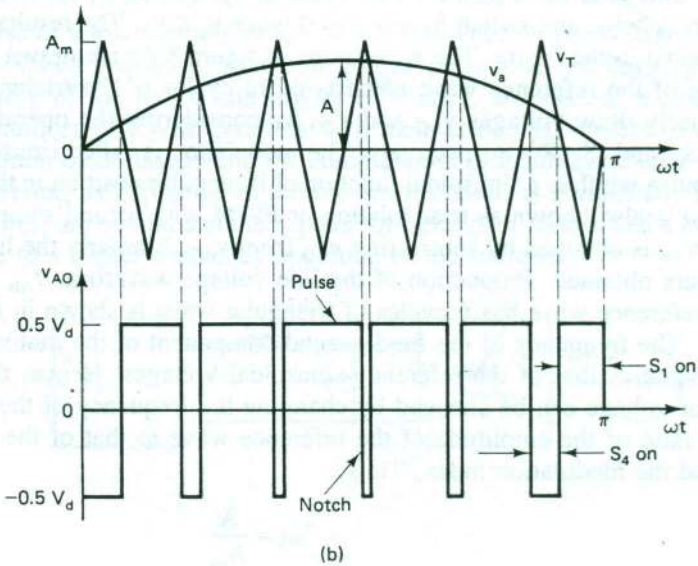


Figure 8.4 Principle of sinusoidal pulse width modulation: (a) generation of control signals, (b) modulated waveform for p = 6.

The waveform v_{AO} contains harmonics which are odd multiples of the carrier frequency f_c (that is, f_c, $3f_c$, $5f_c$, and so on). The harmonics which are even multiples of the carrier frequency are zero. The waveform also contains sidebands centered around multiples of f_c and given by

$$f_h = Kf_c \pm kf \tag{8.7}$$

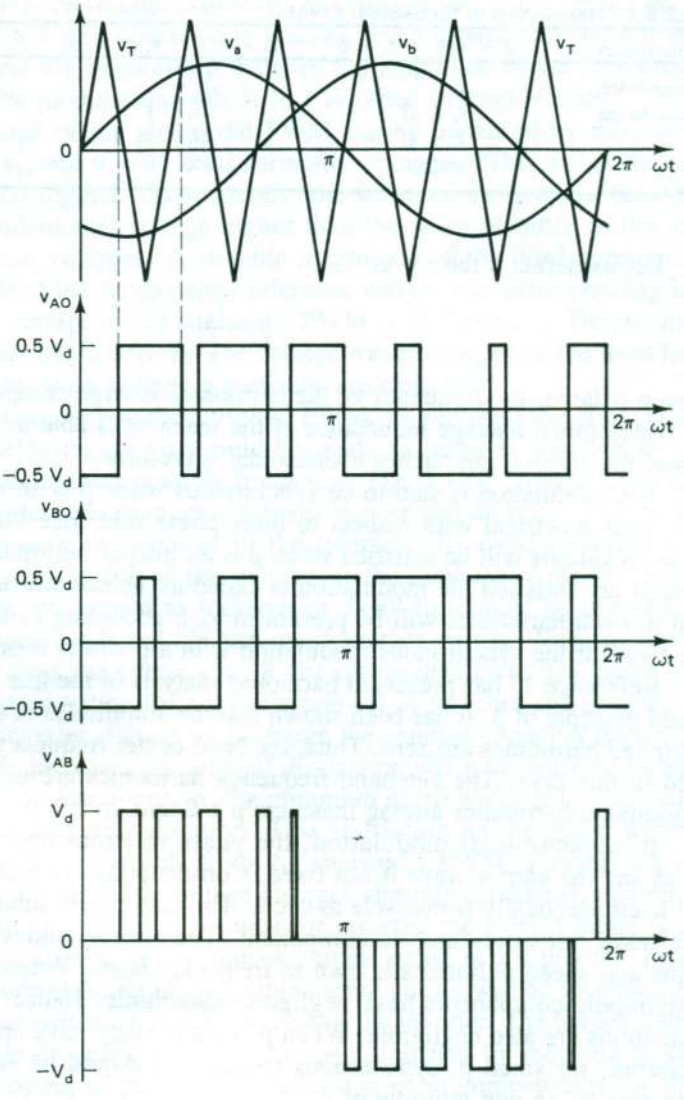


Figure 8.5 Sinusoidal pulse width modulation.

where f_h and f are the frequencies of the sidebands and the reference signal, respectively, in Hz. K and k are integers and K + k is always an odd number. The harmonics and sidebands are listed in table 8.1. The bands which are centered on the carrier frequency and its odd multiples comprise upper and lower sideband components of equal amplitudes and displaced by even multiples of the reference frequency. The bands centered around the even multiples have upper and lower sidebands displaced by odd multiples of the reference frequency. The magnitudes of the band frequency harmonics decrease rapidly with the increasing distance from the band center. Furthermore, the width of a band increases with the increase in the modulation index.

TABLE 8.1 Harmonics in Sinusoidal PWM

Harmonics	K = 1, k = 2, 4, 6	K = 2, k = 1, 3, 5, . . .	K = 3, k = 2, 4, 6, . . .
harmonics	f_c	—	$3f_c$
side bands	$f_c \pm 2f$	$2f_c \pm f$	$3f_c \pm 2f$
	$f_c \pm 4f$	$2f_c \pm 3f$	$3f_c \pm 4f$
	etc.	etc.	etc.

Let us define a ratio p as

$$p = \frac{f_c}{f} \tag{8.8}$$

When p is large, the frequency of the harmonics is large compared to the fundamental. The nominal leakage inductance of the machine is able to filter out the harmonics and the current approaches a sinusoidal waveform.

The modulation is said to be synchronous when p is an integer and the carrier wave is symmetrical with respect to three-phase reference voltages v_a, v_b, and v_c. These conditions will be satisfied when p is an integer multiple of 3. When this condition is not satisfied the modulation is called asynchronous or free-running. For an even p, even harmonics will be present in v_{AO}, according to table 8.1. This will be true even for the synchronous modulation with a p which is an even multiple of 3.

Reference 17 has presented harmonic analysis of the line voltage for p which is an odd multiple of 3. It has been shown that the amplitudes of the pth and its integer-multiplied harmonics are zero. Thus, the band center frequency harmonics are eliminated in this case. The sideband frequency harmonics are given by table 8.1. The predominant harmonics among these are $p \pm 2$ and $2p \pm 1$.

In asynchronous modulation, the phase relationship between the reference waves and the carrier wave is not fixed. Consequently the pulse pattern does not repeat itself identically from cycle to cycle. This introduces subharmonics of the reference wave frequency and dc component. The subharmonics cause low-frequency torque and speed pulsations known as frequency beats. When p is large, the dc and subharmonic components have negligible magnitude. Hence, the torque and speed fluctuations are also negligible. When p is small, they have appreciable magnitudes. Therefore, for small p, synchronous modulation should be employed and p should preferably be an odd multiple of 3.

As mentioned earlier, the boundary of the sinusoidal modulation is reached when the modulation index m = 1. For m = 1, the amplitude of the fundamental in the waveform v_{AO} (fig. 8.5) is $(V_d/2)$. An increase of m beyond 1 increases the fundamental component with a relationship which is no longer linear. The harmonics which are odd multiples of the fundamental frequency (or reference frequency) are also introduced. The number of pulses in the waveform also reduces. When m is made sufficiently large v_{AO} becomes a square wave and the line voltage waveform becomes a 6-step waveform, and the inverter operates like the 6-step inverter described earlier. The amplitude of the fundamental component of v_{AO} reaches the maximum value of $(2V_d/\pi)$. Note that the amplitude of the fundamental on the boundary of the sinusoidal PWM is only 78.5% of the maximum value. To make full

use of the inverter voltage capability, the modulation index should be increased beyond 1 until a 6-step operation is reached, even though this introduces lower-order harmonics and the relationship between the amplitude of the fundamental and m is nonlinear. The modulation with $m > 1$ is called overmodulation.

The range of the sinusoidal PWM can be increased by mixing the reference voltages v_a, v_b, and v_c with third harmonic voltages.[12] This makes the resultant reference wave flat-topped. Consequently, the width of the notch at the center becomes zero at a fundamental voltage higher than the value obtained in the absence of the third harmonic voltages. A suitable magnitude of the third harmonic is one-sixth the amplitude of the fundamental reference waves. The corresponding increase in the range of the voltage of the sinusoidal PWM is 15.5 percent. The mixing of the third harmonic does not distort the line voltage waveforms, since the third harmonic components in the phase voltage waveforms are cancelled.

When operating with sinusoidal PWM, to minimize the effect of harmonics on the motor performance, p should be made as large as possible by operating the switches at the highest possible frequency. The device capabilities, and the modulation and inverter operation requirements impose certain restrictions on the value of p and the frequency of operation of the switches. To avoid a short-circuit due to the simultaneous conduction of the switches in the same leg of the inverter (fig. 8.1a), a fixed time delay, known as lock-out or dwell time must be provided between the turn-off of one switch and the turn-on of another switch. When m approaches 1, the notch width near the center (fig. 8.4b) approaches zero. The notch should have a minimum time duration for a reliable turn-off of switches and snubber relaxation. In a transistor inverter, failure to discharge the snubber capacitor during the on period of the switch, increases the turn-off dissipation and may cause its failure by the second breakdown. For a large p, the minimum notch width in radians will be large for a given minimum time duration. This restriction on the notch width reduces the maximum voltage available from the sinusoidal PWM. Further, when the notches are dropped, the machine voltage changes abruptly producing a current surge and a consequent torque surge. In a transistor inverter, the current rating is increased due to a low ratio of peak to continuous current rating of a transistor. In a thyristor inverter, the current rating of the commutation circuit will be increased, consequently increasing the switching losses and cost of the inverter. The abrupt change in voltage caused by the dropping of pulses can be eliminated by reducing m simultaneously with the dropping of pulses. The reduction of m compensates for the increase in voltage caused by the dropping of pulses. Alternatively, one can use methods described in reference 13. However, these additional provisions make the inverter control more complex.

With an increase in the frequency of operation of switches, the machine losses decrease but the inverter switching losses increase. Thus beyond a certain frequency, the drive efficiency falls. The increase in switching losses also derates the switching devices and the associated components in the snubber and commutation circuits.

According to the foregoing discussion, the maximum value of p will have to be restricted, depending on the switching capability of the device used to realize the switches. A large value of p is realized using a fast device with low switching losses. For these reasons, the power transistor is widely used in low-power and GTO

may be preferred in medium-power applications. The MOSFET is the best from the above considerations (very low switching loss, and so on); however, because of its low power capability and large on state drop, it is yet to find application in drives.

The modes of induction motor operation with sinusoidal PWM inverter control will now be described with the help of figure 8.6. In this figure, the frequency of the carrier wave has been plotted against the ratio f/f_b, where f_b is the base frequency. The base frequency is the inverter frequency on the boundary between the constant torque and constant power region of a variable frequency drive. At low motor speeds for which the fundamental frequency is low, the asynchronous or free-running modulation is employed. The carrier frequency f_c is maintained constant at the highest possible value. As the modulation index m and the reference frequency f are increased to increase the motor speed, p reduces but remains sufficiently large to maintain a nearly sinusoidal current through the motor. A smooth operation, free from torque pulsations, cogging, and frequency beats is obtained. At a suitable value of reference frequency, the operation is shifted to synchronous modulation. For this, the carrier frequency f_c is reduced to get the frequency ratio $p = p_1$. Now f_c is changed with f at a constant ratio p_1. Before f_c becomes large enough to violate the minimum notch time restriction, it is reduced to obtain $p = p_2$ where $p_2 < p_1$. With the frequency ratio at p_2, the operation is brought close to the boundary of the sinusoidal PWM. Now with the frequency ratio set at p_3, at the base frequency the modulation ceases and 6-step operation of the inverter is reached. The motor operation changes from constant torque to constant power operation. For higher frequencies, the inverter functions as a 6-step inverter.

Beyond the boundary of sinusoidal PWM, because of the high harmonic content, the machine losses increase substantially causing its derating. Torque pulsations are also present. Although they are filtered out by the motor inertia, giving ripple-free speed, they adversely affect the life of the motor.

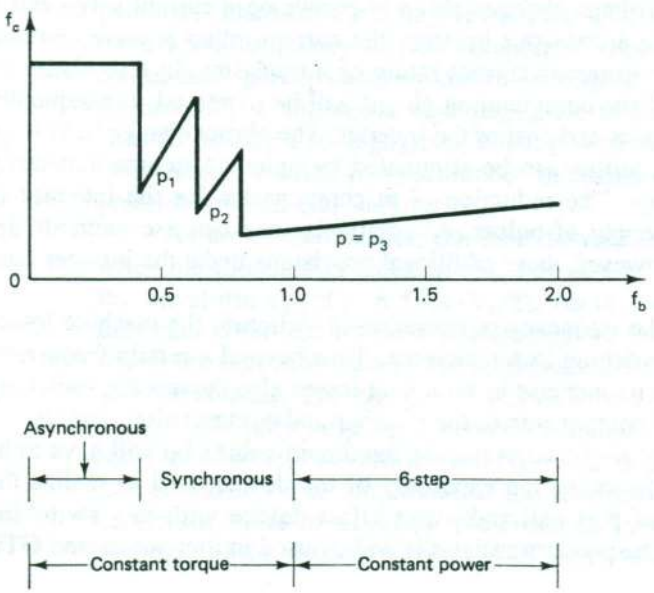


Figure 8.6 Variation of the carrier frequency f_c with (f/f_b) ratio.

Pulse-Width Modulation with Uniform Sampling

The PWM modulation with uniform sampling is shown in figure 8.7. Like sinusoidal PWM, it also uses a set of three-phase reference voltages v_a, v_b, and v_c and a common triangular carrier wave with a period T_c. The figure shows the production of the modulated waveform v_{AO} of phase A with respect to the imaginary neutral of the dc supply (fig. 8.1a). The reference wave v_a is sampled by a sample-and-hold circuit at a regular interval T_c. This produces the waveform v_R (fig. 8.7a). The waveform v_R is compared with the triangular carrier wave (fig. 8.7b). The inverter switch S_1 is

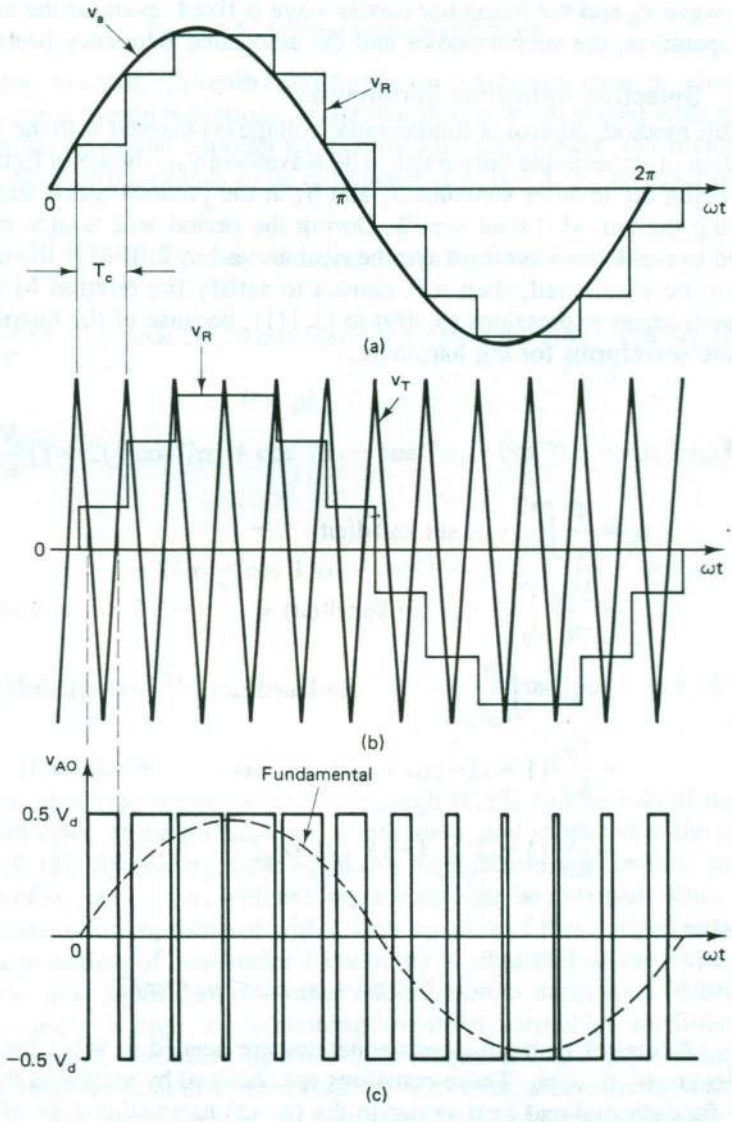


Figure 8.7 Pulse width modulation with uniform sampling.

given the control signal when $v_R > v_T$ and switch S_4 is given the control signal when $v_R < v_T$. As a result the modulated waveform v_{AO} is produced (fig. 8.7c). The fundamental component of v_{AO} is also shown by the dotted line. It has the same frequency as that of the reference wave, and its amplitude depends on the amplitude of the reference wave. Hence the magnitude and frequency of the motor voltage can be controlled by controlling the frequency and amplitude of the reference wave. Since the pulse widths are proportional to the amplitude of the reference wave at uniformly spaced sampling times, the modulation is said to have uniform or regular sampling.

Compared to the sinusoidal PWM, the modulation with uniform sampling has lower low-frequency harmonics. Since the phase relationship between the modulating wave v_R and the triangular carrier wave is fixed, even for the asynchronous mode of operation, the subharmonics and the associated frequency beats are not present.

Selective Harmonic Elimination

In this method, control of fundamental voltage is obtained with the simultaneous elimination of undesirable harmonics. The waveform v_{AO} shown in figure 8.8 is produced by firing the inverter switches S_1 and S_4 at the predetermined angles $\alpha_1, \alpha_2, \ldots, \alpha_n$ during the period $0 \le \omega t \le \pi/2$. During the period $\pi/2 \le \omega t \le \pi$, the switches are fired to make the waveform symmetrical around $\pi/2$. If M is the number of harmonics to be eliminated, then n is chosen to satisfy the relation $M = n - 1$. From the Fourier series expressions (3.109) to (3.111), because of the quarter wave symmetry of the waveforms for kth harmonic,

$$b_k = 0 \tag{8.9}$$

and

$$a_k = \frac{4}{\pi}\int_0^{\pi/2} v_{AO} \sin k\omega t \, d(\omega t) \tag{8.10}$$

$$= \frac{2V_d}{\pi}\left[\int_0^{\alpha_1} (+1) \sin k\omega t \, d(\omega t) + \int_{\alpha_1}^{\alpha_2} (-1) \sin k\omega t \, d(\omega t) \right.$$

$$\left. \ldots + \int_{\alpha_{n-1}}^{\alpha_n} (-1)^{n-1} \sin k\omega t \, d(\omega t) + \int_{\alpha_n}^{\pi/2} (+1) \sin k\omega t \, d(\omega t)\right]$$

$$= \frac{2V_d}{k\pi}[1 + 2(-\cos k\alpha_1 + \cos k\alpha_2 - \ldots + \cos k\alpha_n)]$$

$$= \frac{2V_d}{k\pi}\left[1 + 2\sum_{p=1}^{n} (-1)^p \cos k\alpha_p\right] \tag{8.11}$$

Further,

$$0 < \alpha_1 < \alpha_2 \ldots < \alpha_n < \pi/2 \tag{8.12}$$

A total of n simultaneous equations are needed to solve for the values of variables $\alpha_1, \alpha_2, \ldots, \alpha_n$. These equations are obtained by assigning the required value to the fundamental and zero values to the $(n - 1)$ harmonics to be eliminated. The procedure is explained by the following example.

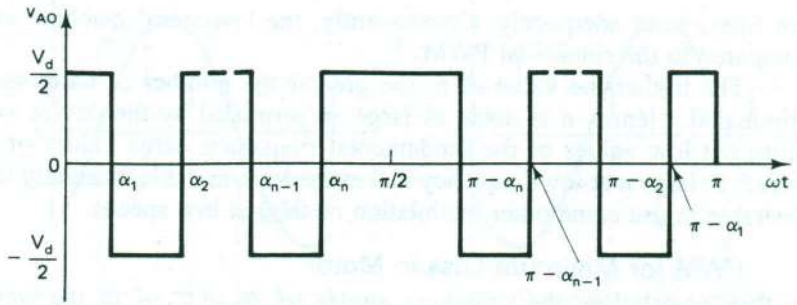


Figure 8.8 Selective harmonic elimination.

The fifth, seventh, eleventh, and thirteenth harmonics must be eliminated to remove low-speed torque pulsations. If the machine is Y-connected with an isolated neutral, then the third and tripplen harmonics will not produce any motor current. Since the number of harmonics to be eliminated is 4, n = 5. From equation (8.11),

$$a_1 = \frac{2V_d}{\pi}[1 + 2(-\cos\alpha_1 + \cos\alpha_2 - \cos\alpha_3 + \cos\alpha_4 - \cos\alpha_5)] \tag{8.13}$$

$$a_5 = \frac{2V_d}{5\pi}[1 + 2(-\cos 5\alpha_1 + \cos 5\alpha_2 - \cos 5\alpha_3 + \cos 5\alpha_4 - \cos 5\alpha_5)] = 0 \tag{8.14}$$

$$a_7 = \frac{2V_d}{7\pi}[1 + 2(-\cos 7\alpha_1 + \cos 7\alpha_2 - \cos 7\alpha_3 + \cos 7\alpha_4 - \cos 7\alpha_5)] = 0 \tag{8.15}$$

$$a_{11} = \frac{2V_d}{11\pi}[1 + 2(-\cos 11\alpha_1 + \cos 11\alpha_2 - \cos 11\alpha_3 + \cos 11\alpha_4 - \cos 11\alpha_5)] = 0 \tag{8.16}$$

$$a_{13} = \frac{2V_d}{13\pi}[1 + 2(-\cos 13\alpha_1 + \cos 13\alpha_2 - \cos 13\alpha_3 + \cos 13\alpha_4 - \cos 13\alpha_5)] = 0 \tag{8.17}$$

The nonlinear algebraic equations (8.13) through (8.17) can be solved numerically for a specified value of the fundamental amplitude a_1 and subjected to the constraints of equation (8.12). By solving these equations for different fundamental amplitudes, the variation of $\alpha_1, \alpha_2, \ldots, \alpha_5$ with the fundamental can be obtained. Since the variation is nonlinear, a microprocessor with a look-up table of these angles may be used.

The elimination of low-order harmonics is obtained at the expense of an increase in the next significant low-order and high-order harmonics. When the fundamental frequency is high, the remaining low-order harmonics are filtered out by the machine leakage reactance, because their frequencies are large enough for the machine reactance to dominate its resistance. However, when the fundamental frequency is low, the frequencies of the remaining low-order harmonics are not large enough for the resistance to be negligible compared to the reactance. Hence they are

not filtered out adequately. Consequently, the low-speed machine losses are more compared to the sinusoidal PWM.[15]

The higher the value of n, the greater the number of harmonics that can be eliminated. Hence, n is made as large as permitted by the device switching capability. At low values of the fundamental frequency, large values of n will be preferred. A large n at low frequency makes the look-up table unusually large, making it desirable to use some other modulation method at low speeds.

PWM for Minimum Loss in Motor

In this modulation, the switching angles $\alpha_1, \alpha_2, \ldots, \alpha_n$ of the waveform of figure 8.8 are selected to obtain a specified value of the fundamental amplitude with a minimum harmonic copper loss. Such an approach allows improvement in efficiency compared to the selective harmonic elimination method and the sinusoidal PWM.

From equations (6.108) and (6.117), the harmonic loss is given by

$$P_h = \sum_{k=3,5,\ldots}^{\infty} \frac{3(R_{sk} + R'_{rk})}{(X_s + X'_r)^2} \left(\frac{V_k}{k}\right)^2 \tag{8.18}$$

where R_{sk} and R'_{rk} are respectively the stator and stator referred rotor resistances for the kth harmonic. The resistance values are harmonic dependent because of skin effect. $(X_s + X'_r)$ is the leakage reactance of the motor at the fundamental frequency. V_k is the rms value of the kth harmonic, thus

$$V_k = \frac{a_k}{\sqrt{2}} \tag{8.19}$$

where a_k is given by equation (8.11)

If skin effect is neglected

$$P_h = K \sum_{k=3,5,\ldots}^{\infty} \left(\frac{V_k}{k}\right)^2 \tag{8.20}$$

where K is a constant.

Equation (8.11) gives the amplitudes of the fundamental and harmonics. Substitution from equations (8.11) and (8.19) into equation (8.20) gives the harmonic loss as a function of switching angles $\alpha_1, \alpha_2, \ldots, \alpha_n$. Using a suitable optimization technique, such as steepest descent, the switching angles can be obtained to minimize P_h for a specific value of fundamental. By repeating these calculations for several values of the fundamental, the variation of switching angles with the fundamental can be obtained. Since the variation is nonlinear, a microprocessor with a look-up table of these angles may be used.

8.1.5 Braking and Multiquadrant Control

The phase-voltage and phase-current waveforms of a Y-connected motor fed by a six-step inverter of figure 8.1a, assuming ideal filtering of the current waveform is shown in figure 8.9a. The devices under conduction during a cycle of the voltage wave are also shown in the figure. The power input to the motor is given by

$$P_{in} = 3V_1 I_s \cos \phi \tag{8.21}$$

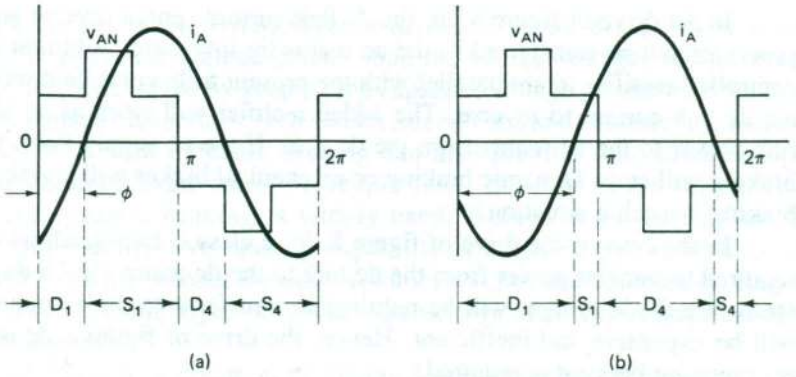


Figure 8.9 Induction motor phase voltage and phase current waveforms: (a) motoring, (b) generating.

where

V_1 = rms value of the fundamental component of the phase voltage, V

I_s = rms fundamental phase current, A

and ϕ = phase angle between V_1 and I_s

A ϕ less than 90° gives motoring operation. A reduction in frequency makes the synchronous speed less than the motor speed. The relative speed between the rotating field and the rotor reverses. Hence, the rotor current and the corresponding component of the stator current reverse, making ϕ more than 90°. Consequently, the power flow reverses and the machine operates as a generator, producing braking torque. The phase voltage and phase current waveforms, and the devices under conduction for generator operation are shown in figure 8.9b. Both switches and diodes take part in the motoring and generating operations. The direction of the instantaneous current on the dc side of the inverter is positive when the switches are on and it is negative when the diodes are on. The average value of this current changes from positive to negative when the operation changes from motoring to generating. Since the polarity of the dc link voltage remains the same, the change of the average current direction reverses the flow of power.

Regenerative braking is obtained when the power flowing from the inverter to the dc link is usefully employed, either by absorbing it in the dc link itself or transferring it to the ac mains as the case may be. Dynamic braking is obtained when the power flowing into the dc link is wasted in a resistance.

Let us return to the drives of figures 8.2 and 8.3 and consider how we can make them operate under regenerative and dynamic brakings.

From the drive of figure 8.2a, a drive with regenerative braking capability is obtained when a class C two-quadrant chopper is employed. The class C two-quadrant chopper permits the dc link current to reverse and thus allows the transfer of the power supplied by the inverter to the dc mains. However, regenerative braking occurs only when the source is able to use this energy, either by storing it when it is a battery or by delivering it to the parallel loads.

In the drive of figure 8.2b, the dc link current cannot reverse and the generated power cannot be transferred to the ac mains by inversion. Addition of another fully controlled rectifier in antiparallel with the present fully controlled rectifier will allow the dc link current to reverse. The added rectifier will work as an inverter transferring power to the ac mains from the dc link. If the ac supply fails, the regenerative braking will stop. Dynamic braking or mechanical brakes will be required to provide braking in such a situation.

In the case of the drive of figure 8.2c, a class C two-quadrant chopper will be required to transfer power from the dc link to the dc mains, and a dual converter, instead of a diode bridge, will be required to transfer it to the ac mains. Such a drive will be expensive and inefficient. Hence, the drive of figure 8.2c is not used when regenerative braking is required.

Regenerative braking with a PWM inverter occurs in the same way as with a 6-step inverter. No additional circuitry is needed for regenerative braking of the drive of figure 8.3a. However, regenerative braking will occur only when the source can use the power delivered by the inverter. In the case of the drive of figure 8.3b, regenerative braking can occur when the power delivered by the inverter to the dc link can be supplied to parallel loads. This can happen when a number of inverter drives are fed from a common diode bridge. When the power cannot be used by parallel loads, it can be transferred to the ac mains by inversion. For this, the diode bridge in figure 8.3b should be replaced by a dual converter. During motoring, the converter is operated with a zero firing angle to maintain the fundamental power factor at unity. During regenerative braking, the firing angle is kept at the highest permissible value, again to maintain a good power factor. Instead of a dual converter, one can also use a fully controlled rectifier in antiparallel with the diode bridge (fig. 8.3). During motoring the rectifier is deactivated, and during braking it operates with the diode bridge in the simultaneous control mode (circulating-current operation). Here also braking will fail if the ac supply fails.

Four-quadrant operation can be obtained by any drive with regenerative braking capability. A reduction of the inverter frequency, to make synchronous speed less than motor speed, transfers the operation from quadrant I to II. The inverter frequency and voltage are reduced to brake the machine to zero speed. Now the phase sequence of the inverter output voltage is reversed by interchanging the control signals between the switches of any two legs of the inverter — for example, between the pairs (S_1, S_4) and (S_3, S_6). This transfers the operation to quadrant III. The inverter frequency and voltage are increased to get the required speed in the reverse direction.

For dynamic braking, a braking resistor with a switch in series is connected across the filter capacitor in the drives of figures 8.2 and 8.3. The generated power charges the filter capacitor and its voltage rises. When the filter capacitor voltage reaches a prescribed maximum value, the switch is turned on, connecting the braking resistor across the filter capacitor. The energy generated and the energy supplied by the filter capacitor are dissipated in the braking resistor. The capacitor voltage falls. When the voltage reaches a prescribed minimum value, the switch is turned off. Again the capacitor voltage starts rising and the cycle repeats. The energy should be prevented from flowing from the source to the braking resistor and the filter capacitor. In the drive schemes of figure 8.2, this is achieved by deactivating the

chopper/rectifier. In the PWM inverter drives of figure 8.3a, this is achieved by keeping the capacitor voltage greater than the source voltage. In the drive of figure 8.3b, this is achieved by keeping the capacitor voltage greater than the peak of the ac source voltage. Alternatively, a switch can be used to disconnect the mains supply.

The drive with dynamic braking can also provide four-quadrant operation. It can be obtained even when the supply fails. Because of the simplicity and the reliability, dynamic braking is widely used. Except in the case of the drive of figure 8.3a, regenerative braking capability is acquired by the addition of extra equipment, which increases the cost, volume, weight, and complexity. Regenerative braking should be employed only when the savings in energy is large enough to justify these expenses.

In dc traction, a common dc supply feeds a number of trains with their own inverter-motor drives. When a train is regenerating, there may not be other trains to consume all this power. Then the remaining power may be dissipated by dynamic braking. When regenerating, the power which is not consumed is stored in the filter capacitor and its voltage rises. When it reaches a prescribed maximum value, the braking resistor is switched in. Now the generated power and a part of the energy stored in the capacitor are used to supply the train loads and the braking resistor. When the capacitor voltage falls below a prescribed minimum value, the braking resistor is switched off and the cycle repeats. Thus, only the amount of energy which cannot be used is dissipated in the braking resistor. The capacitor voltage at which the braking resistor is turned off must be set higher than the dc suppply voltage to prevent the flow of energy from the dc supply to the braking resistor. It may be recalled from section 4.5.2, that the combination of regenerative and dynamic brakings is called composite braking.

Example 8.4

For the braking operation of the drive of example 8.1 at a constant (V/f) ratio, calculate

1. The speed for a braking torque of 350 N-m and the inverter frequency of 40 Hz.
2. The inverter frequency, stator current, regenerated power, efficiency, and power factor for a motor speed of 1000 rpm and a braking torque of 400 N-m.

Neglect friction, windage, core loss, and skin effect.

Solution: From example 8.1 at 60 Hz, $V_l = 265.6$ V, $\omega_{ms} = 125.66$ rad/sec.

1. At 40 Hz, $a = 40/60 = 2/3$

$$V_l = a \times 265.6 = \frac{2}{3} \times 265.6 = 177 \text{ V}$$

Torque for a constant (V/f) ratio is given by equation (6.54)

$$T = \frac{3}{\omega_{ms}} \left[\frac{V_l^2 R_r'/(as)}{\left(\frac{R_s}{a} + \frac{R_r'}{as}\right)^2 + (X_s + X_r')^2} \right], \qquad a < 1 \tag{6.54}$$

Note that V_1, ω_{ms}, and $(X_s' + X_r')$ in this equation are for the rated frequency. Substituting the known values

$$-350 = \frac{3}{125.66}\left[\frac{(265.6)^2 \times 0.07/(2s/3)}{\left(\frac{0.19 \times 3}{2} + \frac{0.07 \times 3}{2s}\right)^2 + (1.42)^2}\right]$$

or

$$\left(0.285 + \frac{0.105}{s}\right)^2 + 2 = -\frac{0.505}{s} \tag{E8.14}$$

This gives

$$x^2 + 51.22x + 188.74 = 0 \tag{E8.15}$$

where $x = 1/s$
From equation (E8.15) $x = -47.22$ or -4
Hence $s = -0.02$ or -0.25
Since the higher slip is for the operating point beyond the breakdown torque, the feasible value of s is -0.02.
Now, synchronous speed $N_s = (40/60) \times 1200 = 800$ rpm
Slip speed $N_{s\ell} = -0.02 \times 800 = -16$ rpm
Hence, the motor speed $= N_s - N_{s\ell} = 800 - (-16) = 816$ rpm

2. Substituting the known values in equation (6.54) (just given) gives

$$-400 = \frac{3}{125.66}\left[\frac{(265.6)^2 \times 0.07/(as)}{\left(\frac{0.19}{a} + \frac{.07}{as}\right)^2 + (1.42)^2}\right]$$

or

$$\left(\frac{0.19}{a} + \frac{0.07}{as}\right)^2 + 2 = -\frac{0.295}{as} \tag{E8.16}$$

Also from equation (6.43)

$$a = \frac{\omega_m}{(1 - s)\omega_{ms}} = \frac{1000}{1200}\frac{1}{(1 - s)} = \frac{0.83}{(1 - s)} \tag{E8.17}$$

Solution of equations (E8.16) and (E8.17) by iteration gives

$$s = -0.0205 \qquad a = 0.813$$

Hence, frequency $= 0.813 \times 60 = 48.8$ Hz and

$$V_1 = 0.813 \times 265.6 = 215.9 \text{ V}$$

Now

$$I_r' = \frac{V_1}{\sqrt{\left(R_s + \frac{R_r'}{s}\right)^2 + a^2(X_s + X_r')^2}}$$

$$= \frac{215.9}{\sqrt{\left(0.19 + \frac{.07}{-0.0205}\right)^2 + (0.813 \times 1.42)^2}} = 63 \text{ A}$$

Phase angle of $\overline{I_r'}$ with respect to the phase voltage

$$= \tan^{-1}\frac{a(X_s + X_r')}{R_s + R_r'/s} = \tan^{-1}\frac{0.813 \times 1.42}{0.19 - \dfrac{0.07}{0.0205}} = 160°$$

Hence, $\overline{I_r'} = 63\underline{/-160°}$ A

$$I_m = \frac{215.9}{aX_m} = \frac{215.9}{0.813 \times 20} = 13.28 \text{ A}$$

$$\bar{I}_s = 63\underline{/-160°} + 13.28\underline{/-90°}$$

$$= -59.2 - j21.55 - j13.28$$

$$= -59.2 - j34.8 = 68.67\underline{/-149.5°} \text{ A}$$

Note that the phase difference between the fundamental stator voltage and current is more than 90°. Hence, power flows from the machine to the inverter.
From equation (E8.7) of example 8.1,

$$I_h = \frac{0.046V_1}{a(X_s + X_r')} = \frac{0.046 \times 215.9}{0.813 \times 1.42} = 8.6 \text{ A}$$

Hence the input current $I = (I_s^2 + I_h^2)^{1/2} = (68.67^2 + 8.6^2)^{1/2} = 69.2$A.
According to the equivalent circuit of figure 6.1d,

$$\text{copper loss} = 3(I_r'^2 + I_h^2)(R_s + R_r')$$

$$= 3(63^2 + 8.6^2)(0.19 + 0.07) = 3.15 \text{ kW}$$

$$\omega_m = \frac{1000 \times 2\pi}{60} = 104.72 \text{ rad/sec.}$$

Power developed $= -400 \times 104.72 = -41.89$ kW
Power output $= -41.89 + 3.15 = -38.74$ kW
Efficiency $= (-38.74/-41.89) \times 100 = 92$ percent
From equations (8.3) and (8.4)
RMS input phase voltage $V = 1.047V_1 = 1.047 \times 215.9 = 226.0$ V
Apparent power $= 3VI = 3 \times 226 \times 69.2 = 46.92$ kVA
Power factor = Real power/Apparent power $= 38.74/46.92 = 0.83$.

8.1.6 Voltage Source Inverter Variable Frequency Drives

Except in the case of the minimum loss control, the basic control strategy is to operate the machine with nearly constant flux up to the base speed and a fixed terminal voltage above the base speed. The variation of the motor terminal voltage required to implement this strategy is shown in figure 6.14. The operation at a constant maximum stator current gives constant torque operation up to the base speed and constant power operation from the base speed to the critical speed ω_{mc} (fig. 6.13). During the constant torque operation, the slip speed is held constant. It increases linearly with frequency during the constant power operation and reaches the breakdown value at the critical speed ω_{mc}. Operation beyond the critical speed is carried out at the breakdown slip speed with reduced stator current and power.

As explained in section 1.3.2, the inverter current rating is chosen to match the transient current rating of the motor. For fast transient response, the motor current is allowed to exceed the rated current. Maximum motor current is chosen anywhere between the rated current and the current at breakdown torque. The inverter rating is chosen equal to the maximum motor current. In drives where fast transient response is not necessary, the inverter current rating is made equal to the rated motor current. This allows considerable savings in the cost of the inverter and the associated controlled rectifier. As in the case of dc drives, a current control loop is always provided to restrict the transient motor current within the inverter current rating. When the transient current allowed is more than the rated motor current, the current control loop cannot protect the motor against sustained torque overloads. Then some type of inverse time overcurrent protection is provided, in addition to the current control loop. The current control may consist of either the current-limit control or the inner-current control (section 3.9).

Current control can be implemented indirectly by controlling slip speed. At a constant flux, the motor slip speed for a given stator current is constant (fig. 6.14). Hence, in the constant torque region, the maximum value of the current can be limited indirectly by limiting the slip speed below a specified maximum value. In the constant power region, the slip speed for a given motor current varies linearly with frequency. Hence, the maximum current can be limited by limiting slip speed below a limit which varies linearly with frequency. If operation is required above the critical speed ω_{mc}, the drive can be operated at a constant slip speed which is slightly less than the breakdown slip speed. The current will then be less than the permissible value.

To operate the machine with a good power factor, high torque per ampere, and high efficiency, the drive must operate at a slip speed less than its breakdown value, both during motoring and braking operations. This also prevents runaway speed of the drive when regenerating against an active load torque, as explained in section 6.3.1.

In variable frequency inverter drives, braking is mandatory. For negative slips, the induction motor generates, charging the filter capacitor. In the absence of braking, the capacitor voltage attains unusually large values. This damages the semiconductor devices and explodes the filter capacitor, posing a serious hazard to the operator and the equipment around.

A number of schemes are available for voltage source inverter drives. They differ in the type of inverter (six-step or PWM), type of braking (regenerative or dynamic), and current control (direct current control or slip speed control). Here a few schemes employing the PWM inverter are described. They can be easily extended to six-step inverter drives.

An open-loop variable frequency drive with current-limit control and dynamic braking is shown in figure 8.10. A speed command ω_m^* sets the inverter frequency f. From the inverter frequency f, the inverter voltage V_1 is set according to the relation $V_1 = Kf + V_0$ to operate the machine at nearly constant flux up to the base speed. At the base speed, the motor terminal voltage saturates. Consequently the machine operates at a constant terminal voltage above the base speed. The offset voltage V_0 is chosen to produce the nominal flux at zero speed and the constant K is

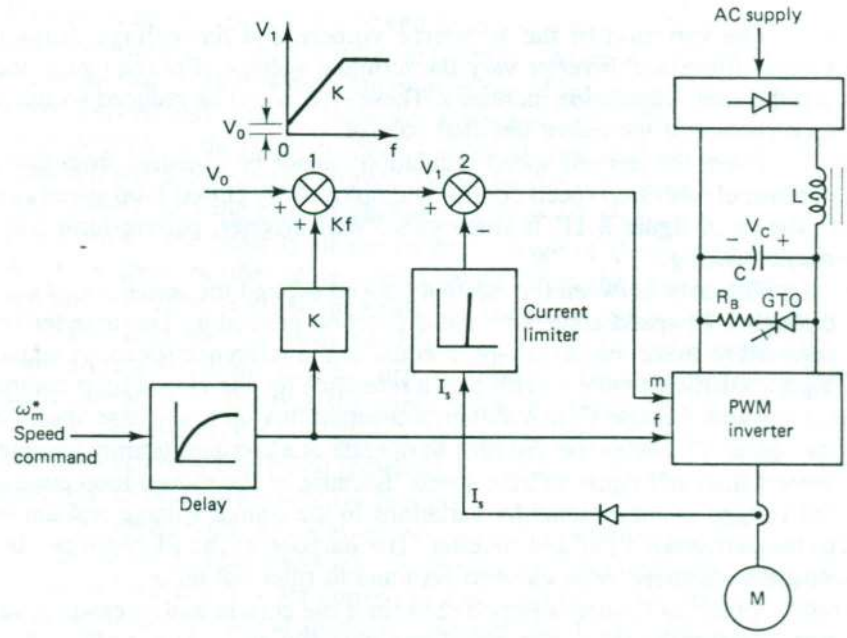


Figure 8.10 Open-loop variable frequency PWM inverter drive with dynamic braking.

chosen to get the rated (or the maximum available if less than the rated) voltage at the base speed.

Current-limit control is provided to prevent the motor current from exceeding a safe value. The stator current I_s is sensed by a three-phase current transformer and a three-phase diode bridge. As long as I_s is less than the permissible value, the current limiter output remains zero and m is set according to signal V_1. Whenever I_s crosses the permissible value, the current limiter output reduces m, which reduces I_s. Thus, the drive operates around a maximum value of I_s, until the speed reaches a value for which I_s is lower than the permissible value.

In the absence of the delay circuit, a step change in speed command will cause the motor slip to exceed the breakdown value. The motor current will tend to exceed the safe value, but will be prevented from doing so by the current-limit control. The motor terminal voltage will decrease, reducing the motor torque. This may lead to unstable operation and the motor may stall. Similarly a step decrease in speed command may shift the braking operation beyond the breakdown slip, leading to runaway speeds. Therefore, the speed command is applied through a delay circuit. The inverter frequency now changes slowly, allowing the motor speed to track the changes in frequency. Consequently, the slip does not exceed the breakdown value. The delay circuit, however, slows down the transient response.

A reduction in speed command during the steady-state operation causes the drive to go into the dynamic braking mode. A GTO is employed to connect and disconnect the braking resistor as the voltage of the filter capacitor reaches the prescribed maximum and minimum values. When the drive speed reaches the command value, the operation changes from braking to motoring.

The variation of the ac source voltage and the voltage drops in the diode bridge, filter, and inverter vary the terminal voltage. Consequently, the flux varies and the speed regulation increases. These effects can be reduced by using the closed-loop control of the motor terminal voltage.

When the desired speed regulation cannot be obtained from open-loop speed control, closed-loop speed control is employed. A closed-loop speed control scheme is shown in figure 8.11. It employs a PWM inverter, current-limit control, and dynamic braking.

The error between the reference speed ω_m^* and the actual speed ω_m is processed through a PI speed controller and a function generator. The inverter frequency f is adjusted to make the actual speed equal to the reference speed. A signal $V_1^* = Kf + V_0$ is constructed from f. This sets a reference for the closed-loop control of the motor terminal voltage V_1 to maintain a constant flux up to the base speed. Saturation in the signal V_1^* causes the machine to operate at a constant terminal voltage for speeds greater than and equal to base speed. Because of the closed-loop control, the terminal voltage is not affected by variations in the source voltage and the voltage drops in the converter, filter and inverter. The purpose of the PI controller is to make the steady-state speed error close to zero and to filter out noise.

A current limiter is provided to limit the current and to ensure that the inverter frequency tracks the motor speed and thus slip speed is not allowed to exceed the breakdown value. A step increase in speed command activates the current limiter.

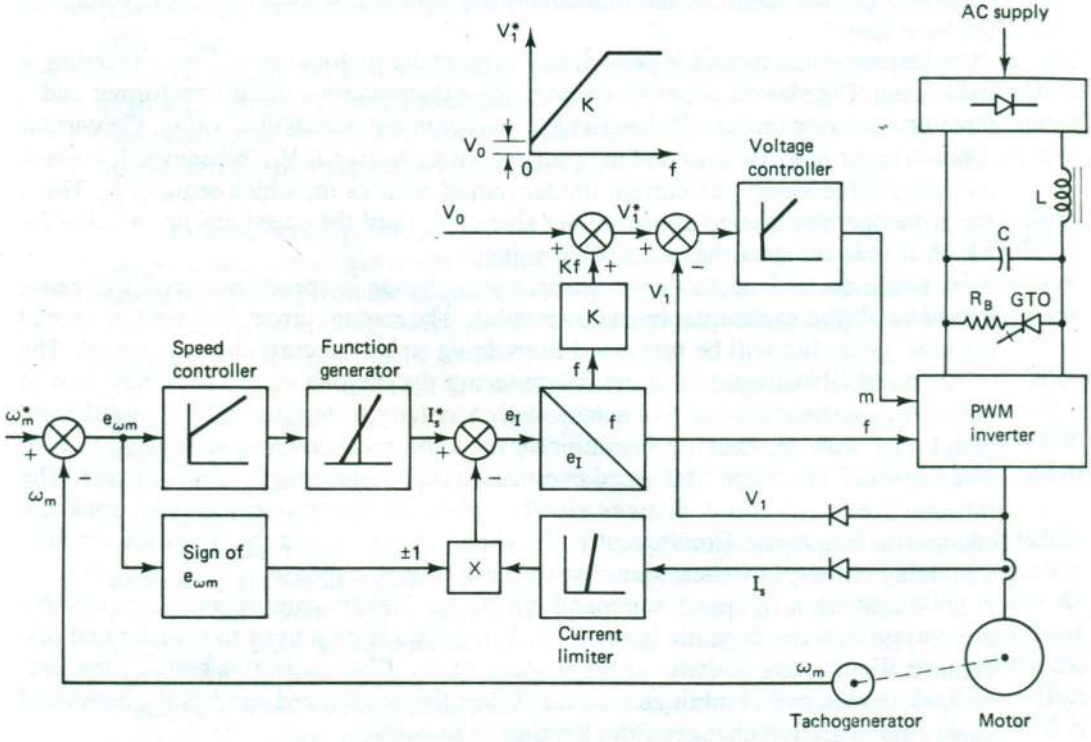


Figure 8.11 Closed-loop variable frequency PWM inverter drive with dynamic braking.

Consequently, frequency is allowed to increase slowly. The machine accelerates at the maximum allowable current and torque. When the speed reaches close to the command value, the current limiter goes out of action and the machine settles at the desired speed and current where the motor torque equals the load torque. Similarly, in response to a decrease in speed command, the machine decelerates at the maximum allowable torque and current due to the current limiter. Now the current limiter must slow down the rate of decrease in inverter frequency. Hence, for negative speed errors, the polarity of the output voltage of the current limiter is reversed. When the speed reaches close to the desired value, the current limiter goes out of action and operation shifts to motoring and the drive settles at the desired speed.

The drive has fast response because the speed error is corrected at the maximum allowable torque due to current limiting. The current limiting also ensures stable operation by limiting slip speed below the breakdown value.

A closed-loop control scheme, where the slip speed $\omega_{s\ell}$ is controlled directly and thus current is controlled indirectly, is shown in figure 8.12. A dual converter is employed to obtain regenerative braking.

The speed error is processed through a PI controller and a slip speed regulator. The slip speed regulator sets the slip speed command $\omega^*_{s\ell}$, whose maximum value is limited to limit the inverter current to a permissible value. The synchronous speed, obtained by adding actual speed ω_m and slip speed $\omega^*_{s\ell}$, determines the inverter frequency. The reference signal for the closed-loop control of the machine terminal voltage V_1^* is generated from frequency f using a function generator. It ensures nearly a

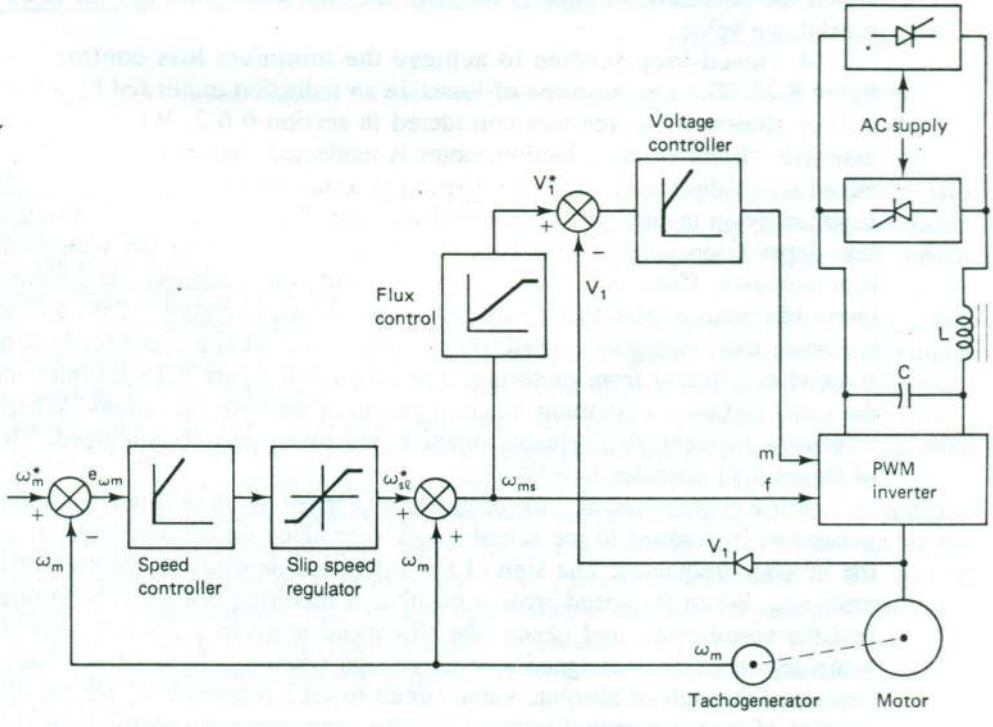


Figure 8.12 Closed-loop slip speed-controlled PWM inverter drive with regenerative braking.

constant flux operation up to the base speed and the operation at a constant terminal voltage above the base speed.

A step increase in speed command ω_m^* produces a positive speed error. The slip speed command $\omega_{s\ell}^*$ is set at the maximum value. The drive accelerates at the maximum permissible inverter current, producing the maximum available torque, until the speed error is reduced to a small value. The drive finally settles at a slip speed for which the motor torque balances the load torque.

A step decrease in speed command produces a negative speed error. The slip speed command is set at the maximum negative value. The drive decelerates under regenerative braking, at the maximum permissible current and the maximum available braking torque, until the speed error is reduced to a small value. Now the operation shifts to motoring and the drive settles at the slip speed for which the motor torque equals the load torque.

The drive has fast response because the speed error is corrected at the maximum available torque. Direct control of slip speed assures stable operation under all operating conditions. The dynamic performance of this drive is somewhat superior to the drive of figure 8.11. It also avoids an expensive current sensing circuit.

For operation beyond the base speed, as explained at the beginning of the present section, the slip speed limit of the slip speed regulator must be increased linearly with the frequency until the breakdown value is reached. This is achieved by adding to the slip speed regulator output an additional slip speed signal, proportional to frequency and of appropriate sign. For frequencies higher than the frequency for which the breakdown torque is reached, the slip speed limit is kept fixed near the breakdown value.

A closed-loop scheme to achieve the minimum loss control is shown in figure 8.13. The minimization of losses in an induction motor fed by a variable frequency sinusoidal source was considered in section 6.6.2. When saturation of the magnetic circuit of an induction motor is neglected, the motor efficiency is maximized when slip speed varies with speed in a manner shown in figure 6.25b. When supplied by an inverter, the harmonic loss must also be considered. As the harmonic loss depends on voltage, like core losses, the proportion of the voltage dependent loss increases. Consequently the optimum operation is obtained at a lower voltage. One can obtain a relationship similar to that shown in figure 6.25b, accounting for harmonic loss. At a given speed, the optimum value of slip speed for braking will be somewhat different from motoring. The scheme of figure 8.13 assumes them to be the same and uses a common function generator for both operations. When it is significantly different, two separate function generators may be employed. The scheme of figure 8.13 operates as follows.

For a given speed ω_m, the optimum slip speed $\omega_{s\ell}$ is obtained from the function generator. It is added to the actual speed to produce synchronous speed, which sets the inverter frequency. The sign of the slip speed depends on the sign of the speed error $e_{\omega m}$. When the speed error is positive, a motoring operation is required to correct the speed error, and hence, the slip speed is given a positive sign. Using the same argument, it is assigned a negative sign when $e_{\omega m}$ is negative. Speed error is processed through an absolute value circuit to set a reference V_1^* for the closed-loop control of motor terminal voltage V_1. An inner-current control loop is provided

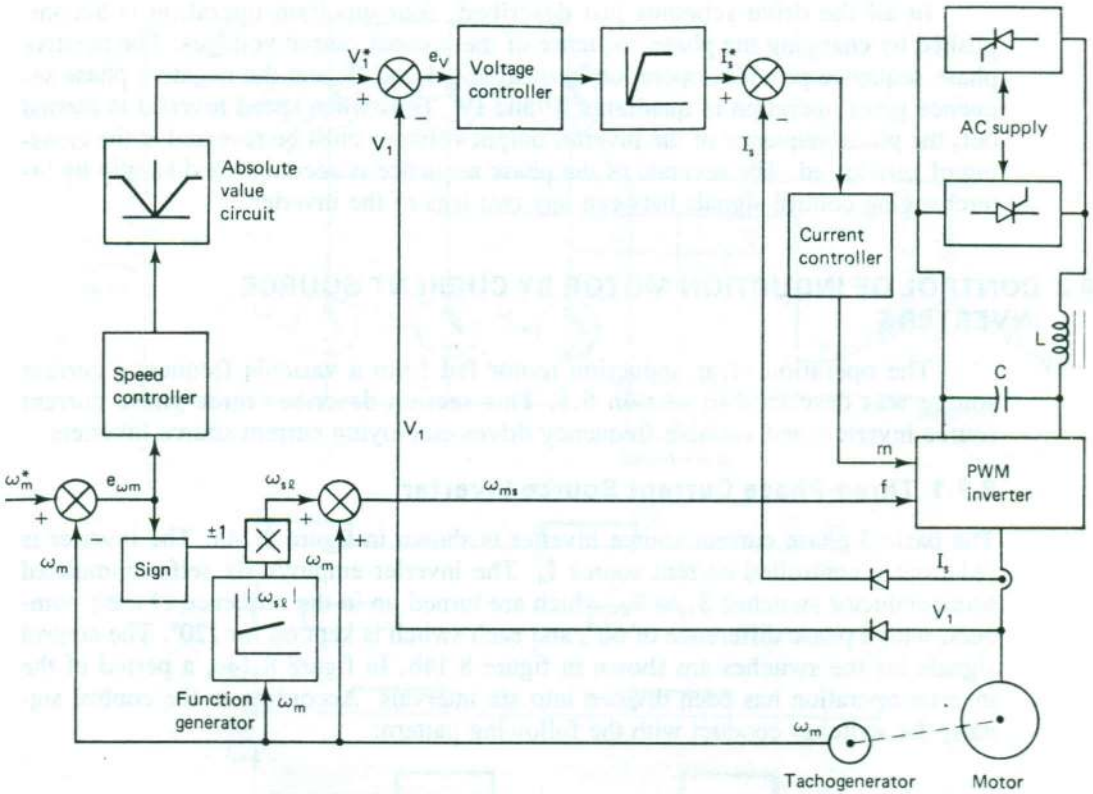


Figure 8.13 Variable frequency PWM inverter drive with regeneration and minimum loss control.

within the voltage loop. Thus, while the frequency loop ensures an optimum slip speed for each motor speed, the voltage loop adjusts the voltage to make the motor torque equal to the load torque at the desired speed. The speed, voltage, and current controllers employ PI controllers.

A step increase in the speed command produces a large speed error $e_{\omega m}$ and consequently a large voltage error e_v. The reference current I_s^* is set at the maximum value and the drive accelerates at the maximum allowable current. As the speed changes, the slip speed is adjusted automatically at the optimum value. When $e_{\omega m}$ approaches zero, the current reference I_s^* reduces and the motor current is adjusted to produce a torque equal to the load torque at the desired steady-state speed.

A step decrease in speed command makes synchronous speed less than motor speed and regenerative braking takes place. The machine decelerates under the current control at the maximum permissible current. When close to the desired speed, the motor current reduces and the operation shifts to motoring. Finally, the drive settles at the desired speed. Because of the slow response of flux to a change in motor voltage, the transient response is slow when initially operating at a reduced flux.

In all the drive schemes just described, four-quadrant operation is accomplished by changing the phase sequence of the inverter output voltages. The positive phase sequence provides operation in quadrants I and II, and the negative phase sequence gives operation in quadrants III and IV. Thus when speed reversal is carried out, the phase sequence of the inverter output voltages must be reversed at the crossing of zero speed. The reversal of the phase sequence is accomplished simply by interchanging control signals between any two legs of the inverter.

8.2 CONTROL OF INDUCTION MOTOR BY CURRENT SOURCE INVERTERS

The operation of an induction motor fed from a variable frequency current source was described in section 6.5. This section describes three-phase current source inverters and variable frequency drives employing current source inverters.

8.2.1 Three-Phase Current Source Inverter

The basic 3-phase current source inverter is shown in figure 8.14a. The inverter is fed from a controlled current source I_d. The inverter employs six self-commutated semiconductor switches S_1 to S_6, which are turned on in the sequence of their numbers, with a phase difference of 60°, and each switch is kept on for 120°. The control signals for the switches are shown in figure 8.14b. In figure 8.14c, a period of the inverter operation has been divided into six intervals. According to the control signals, the switches conduct with the following pattern:

Interval	Switches Conducting	Interval	Switches Conducting
I	S_1, S_6	IV	S_4, S_3
II	S_2, S_1	V	S_5, S_4
III	S_3, S_2	VI	S_6, S_5

Thus, two switches conduct at a time—one from the upper group (S_1, S_3, and S_5) and one from the lower group (S_2, S_4, and S_6). The waveform of the line current i_A is shown in figure 8.14c. It is a six-step square wave. The waveforms for the line currents i_B and i_C will be identical with a phase difference of 120° and 240°, respectively.

At $\omega t = 0$, the current source forces i_A to jump instantaneously from 0 to I_d. In the absence of the capacitor bank C, the machine phase current will also jump instantaneously. Due to a large rate of change of the phase current, the machine leakage inductance will produce a sharp positive spike in the phase voltage v_{AN}. Similarly, when switch S_1 is turned off at $\omega t = 2\pi/3$, i_A is suddenly forced to become zero. Consequently, the leakage inductance will produce a sharp negative spike in v_{AN}. These voltage spikes, which are produced each time the phase current changes, cause a manifold increase in the voltage rating of the switches. The capacitor bank, by providing alternative paths for the currents to flow, reduces the rate of

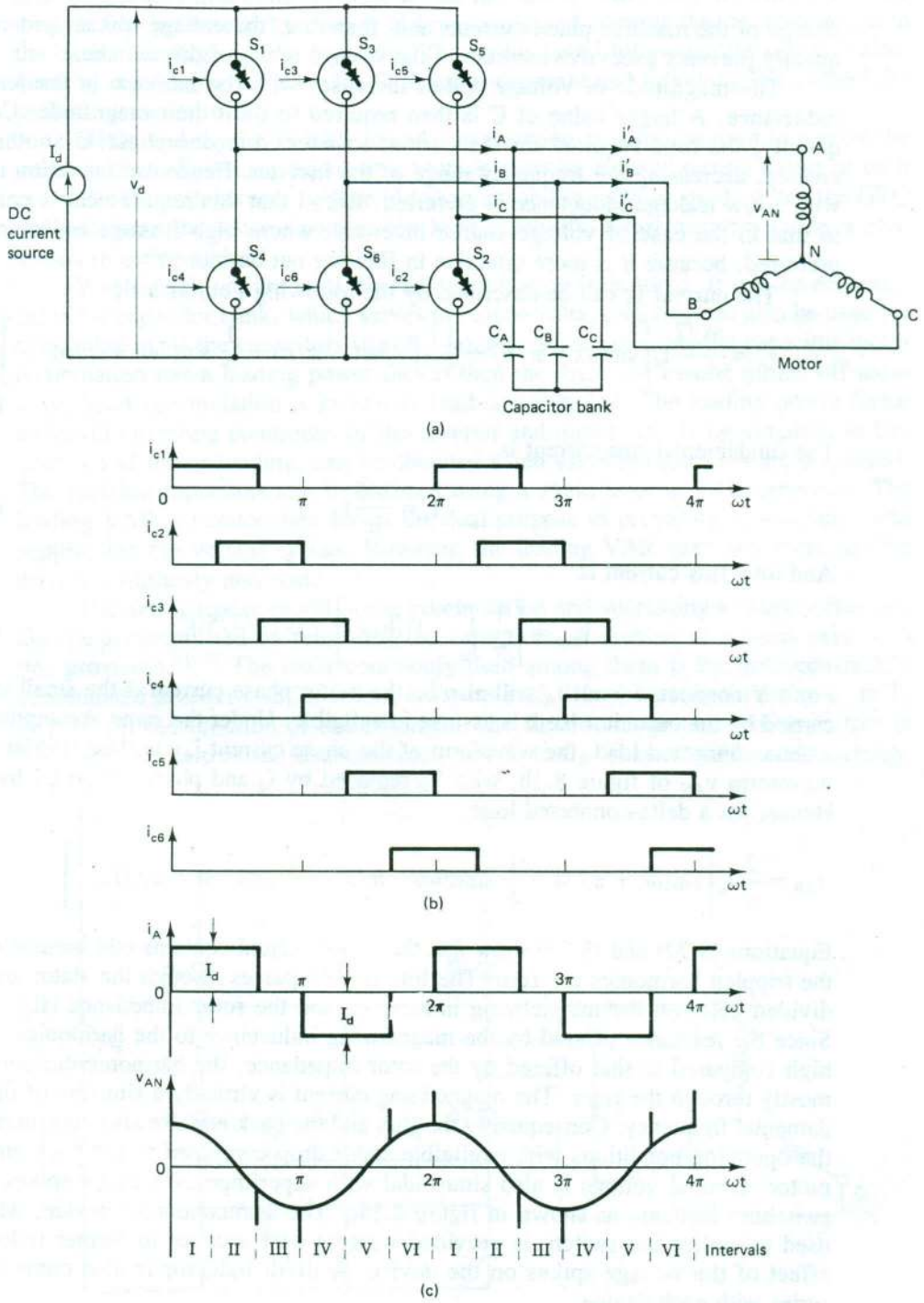


Figure 8.14 Three-phase current source inverter.

change of the machine phase currents and, therefore, the voltage spikes, and consequently prevents excessive increase in the voltage rating of the switches.

The magnitude of voltage spikes increases with the increase in the leakage inductance. A larger value of C is then required to limit their magnitude. Consequently, the time required for the current transfer from one phase to another increases, decreasing the frequency range of the inverter. Hence, an induction motor with a low leakage inductance is preferred. Recall that this requirement is contrary to that in the case of voltage source inverters, where high leakage inductance is preferred, because it is more effective in filtering out harmonics.

The current i_A can be described by the following Fourier series:

$$i_A = \frac{2\sqrt{3}}{\pi} I_d \left[\sin(\omega t + \pi/6) + \frac{1}{5} \sin(5\omega t - \pi/6) + \frac{1}{7} \sin(7\omega t + \pi/6) \ldots \right] \tag{8.22}$$

The fundamental rms current is

$$I_s = \frac{\sqrt{6}}{\pi} I_d \tag{8.23}$$

And total rms current is

$$I_{rms} = \left[\frac{1}{\pi} \int_0^{2\pi/3} I_d^2 \, d(\omega t) \right]^{1/2} = (\sqrt{2/3}) I_d \tag{8.24}$$

For a Y-connected load, i_A will also be the motor phase current if the small current carried by the capacitor bank is assumed negligible. Under the same assumption, for a delta-connected load, the waveform of the phase current i_{AB} will be similar to the waveform v_{AN} of figure 8.1b, with V_d replaced by I_d and phase advanced by $\pi/3$. Hence, for a delta-connected load,

$$i_{AB} = \frac{2}{\pi} I_d \left[\sin(\omega t + \pi/3) - \frac{1}{5} \sin(5\omega t - \pi/3) + \frac{1}{7} \sin(7\omega t + \pi/3) \ldots \right] \tag{8.25}$$

Equations (8.22) and (8.25) show that the motor current contains odd harmonics and the tripplen harmonics are zero. The line current passes through the stator and gets divided between the magnetizing inductance and the rotor impedance (fig. 6.1b). Since the reactance offered by the magnetizing inductance to the harmonics is very high compared to that offered by the rotor impedance, the harmonic currents flow mostly through the rotor. The magnetizing current is virtually a sinusoid of the fundamental frequency. Consequently the flux and the back emf are also sinusoidal. For the operating conditions with negligible stator drop compared to the back emf, the motor terminal voltage is also sinusoidal with superimposed voltage spikes at the switching instants, as shown in figure 8.14c. The semiconductor device, which is used to realize the switch, is provided with its own snubber to further reduce the effect of the voltage spikes on the device. A dI/dt inductor is also connected in series with each device.

Because of the sinusoidal flux, only the fundamental component of the current contributes to the developed torque and power. The harmonics increase the copper

and core losses and consequently derate the motor. They also produce pulsating torques which cause jerky motion at low speeds and cogging during reversal, as in the case of voltage source inverters. The motor speed-torque curves can be calculated by considering only the fundamental component and following the method described in section 6.5.

GTOs and thyristors with forced commutation circuits are used to realize the switches S_1 to S_6. Since the reverse voltage must be blocked during a part of each cycle, GTOs with reverse voltage blocking capability must be used. When the GTO does not have the reverse voltage blocking capability, a fast recovery diode is connected in series with it.

When thyristors are used, forced commutation is required. It will be economical if the capacitor bank, which serves to reduce voltage spikes, can also be used for commutation. If the capacitors (fig. 8.14a) are chosen such that the capacitor-motor combination has a leading power factor, then the thyristors can be turned off naturally. Such commutation is known as load commutation. The leading power factor under all operating conditions of the inverter and motor, involving variation in frequency and motor loading, can be obtained when variable capacitors are employed. The variable capacitors can be realized using a static leading VAR generator. The leading VAR generator then serves the dual purpose of providing commutation and suppressing the voltage spikes. However, the leading VAR generator increases the drive's complexity and cost.

The dual purpose of providing commutation and supressing voltage spikes can also be accomplished by relocating the capacitors. A number of circuits exist with this provision.[21–23] The most commonly used among them is the autosequentially commutated inverter (ASCI) shown in figure 8.15. It employs six thyristors T_1 to T_6 to perform the function of the switches. The forced commutation of thyristors is done with the help of six identical capacitors C_1 to C_6. These capacitors also arrange

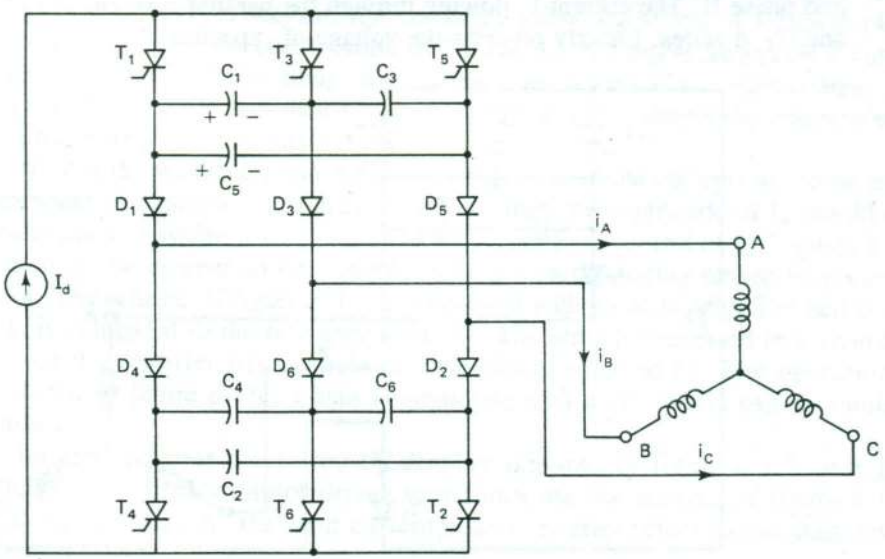


Figure 8.15 Auto-sequentially commutated current source inverter.

gradual transfer of current between the motor phases at the instants of switching, and thus also perform the function of suppression of switching spikes. The diodes help in retaining charge on the capacitors with polarities appropriate for the commutation of thyristors. Each thyristor is also provided with its own dI/dt inductor and the snubber circuit. These are not shown in the figure. The thyristor gate signals will be the same as the control signals i_{C1} to i_{C6} of figure 8.14b. Thus, the thyristors will be fired in the sequence of their numbers with a phase difference of 60°, and two thyristors will be in conduction at a time.

In the steady state, the sequence of events will be identical at each switching. Hence, it will be adequate to examine the inverter operation for one switching. Let thyristors T_1 and T_2 be conducting initially. Then, the source current I_d will be flowing through the path consisting of thyristor T_1, diode D_1, phase A, phase C, diode D_2, and thyristor T_2. The next thyristor to be turned on is T_3. When the commutation and transfer of current are completed, thyristors T_2 and T_3 will be conducting and the source current I_d will be flowing through the path consisting of thyristor T_3, diode D_3, phase B, phase C, diode D_2, and thyristor T_2. Thus the switching operation involves the turn-off of thyristor T_1 and the transfer of the source current I_d from phase A to phase B. We will now examine how this transfer of current is achieved in the circuit under consideration.

As just stated, initially devices T_1, D_1, T_2, and D_2 are in conduction. Let capacitors C_1 and C_5 be charged with the polarities shown in the figure during the previous commutation. When thyristor T_3 is turned on, full voltage of capacitor C_1 is applied to reverse bias thyristor T_1, which is then commutated. The circuit that is effective during the transfer of current is shown in figure 8.16. The current now flows through thyristor T_3, a parallel circuit formed by C_1, and C_3 and C_5 in series, diode D_1, phase A, phase C, diode D_2, and thyristor T_2. Diode D_3 does not conduct due to a reverse bias applied by C_1 through the path consisting of diode D_1, phase A, and phase B. The current I_d flowing through the parallel combination of C_1, and C_3 and C_5 in series, linearly reverses the voltage of capacitor C_1 until it is sufficient to

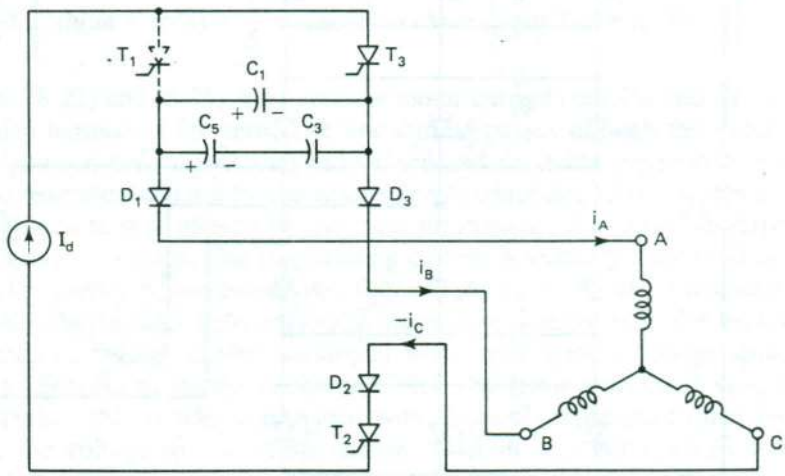


Figure 8.16 Equivalent circuit of inverter of Fig. 8.15 during commutation.

forward bias diode D_3. When diode D_3 conducts, the current in phase B begins to flow. As capacitor C_1 charges, the current in phase A reduces and the current in phase B increases, at the rate determined by the value of the capacitors, and thus a gradual transfer of current from phase A to phase B takes place. During the current transfer, capacitor C_1 is charged with the right plate positive, and capacitor C_3 (fig. 8.15) is charged with the left plate positive. Thus, C_3 is charged with a polarity which is appropriate for the commutation of T_3 when T_5 is turned on for the next comutation in the upper thyristor group.

In the commutation process of T_1 just described, when T_1 ceases to conduct, capacitor C_1 applies reverse bias across T_1 for a duration which depends on the value of C_1 and I_d. By selecting an appropriate value of C_1, the reverse bias across C_1, for all values of I_d, can be maintained for a sufficient time to turn it off.

To suppress the switching spikes adequately to keep the device voltage ratings reasonable, large capacitors are required. The value of capacitors is such that, with careful design, the inexpensive converter grade thyristors can be turned off. Hence, converter grade thyristors are used in the current source inverter. This is different from the voltage source inverters where expensive inverter grade thyristors are employed. Because of large capacitors, the maximum frequency of operation of a current source inverter is rather low.

The ASC inverter finds wide applications in medium and large power current source inverter drives.

8.2.2 Current Sources

Let us examine the scheme shown in figure 8.17a. The inverter is fed from a dc source through a large inductance L_d. Because of the large value of L_d, the current I_d can be assumed to be ripple-free dc. When the inverter is controlled as described in the previous section, the six-step current waveform of figure 8.14b is obtained. Therefore, the combination of L_d and the inverter is known as a current source inverter. Strictly speaking, this scheme does not act as a current source. Any change in the machine impedance, with a change in slip, changes the magnitudes of I_d and machine phase currents.

If both the waveform and the magnitudes of machine currents are to be made independent of changes in machine operation, then the magnitude of I_d should also be maintained constant. This is achieved by closed-loop control of I_d. Figures 8.17b and c show the current source inverter schemes incorporating closed-loop current control. The scheme of figure 8.17b is employed with an ac supply. The actual current I_d is compared to the reference value I_d^*. The error is processed in a controller to adjust the rectifier firing angle α, to make I_d equal to I_d^*. The operation of the scheme of figure 8.17c, which is employed with a dc supply, can be similarly explained.

Induction motor drives mostly employ schemes of figure 8.17b or 8.17c. Single-phase induction motor drives sometimes use the scheme of figure 8.17a. Hence, in this chapter, the term current source inverter refers to the schemes of figures 8.17b and c only.

Usually a 3-phase fully controlled rectifier (fig. 8.17b) is employed. A single-phase controlled rectifier may be used for low-power applications. An exception is

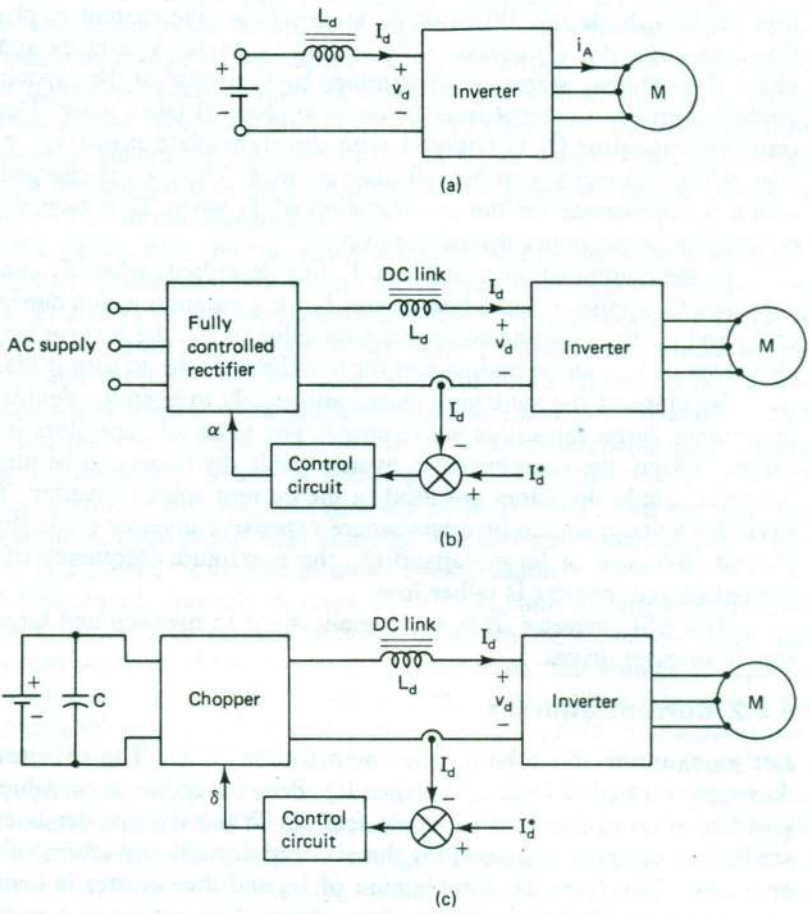


Figure 8.17 Current sources.

made for traction applications where, owing to the availability of the 1-phase supply only, the single-phase rectifier is employed at substantially high power levels.

The major drawback of the rectifier current source is the poor power factor at low dc link voltages. This can be overcome by using a rectifier with controlled flywheeling[27] or pulse-width modulation.[28]

8.2.3 Braking

The phase voltage and phase current waveforms of a Y-connected induction motor fed by a current source inverter are shown in figure 8.14c. As explained in section 8.1.5, the phase current lags behind the phase voltage by an angle ϕ, which is less than 90° for motoring and greater than 90° for generation. In a current source inverter, the phase of the current waveform remains fixed with respect to the switching instants. Hence the machine phase voltage is produced to lead the current waveform by an angle ϕ which is less than 90° for motoring and more than 90° for generation. The machine phase current waveform and the line voltage waveforms

are shown in figure 8.18 for motoring and generation. Since v_{AN} leads i_A by an angle ϕ, line voltage v_{AB} leads i_A by an angle of $(\phi + 60°)$ and voltages v_{BC} and v_{CA} lag behind v_{AB} by 120° and 240°, respectively. From the line voltages one can obtain the inverter input voltage v_d. For example, during the interval $0 \le \omega t \le \pi/3$, switches S_1 and S_6 conduct, hence $v_d = v_{AB}$; during the interval $\pi/3 \le \omega t \le 2\pi/3$ switches S_1 and S_2 conduct, hence $v_d = v_{AC}$ (fig. 8.14), and so on. Figure 8.18 shows that the average value of v_d (V_d) is positive for motoring and negative for generation. This is true for both the drive schemes of figure 8.17. Since the direction of the dc link current is fixed, the flow of dc link power reverses as operation of the machine is shifted from motoring to generation.

In the drive of figure 8.17b, the closed-loop control of the rectifier current automatically adjusts the rectifier firing angle and keeps the dc link current constant. When the operation shifts from motoring to generation, and hence V_d becomes negative, the rectifier firing angle automatically changes to make its output voltage negative. Consequently, the rectifier works as an inverter transferring power from the

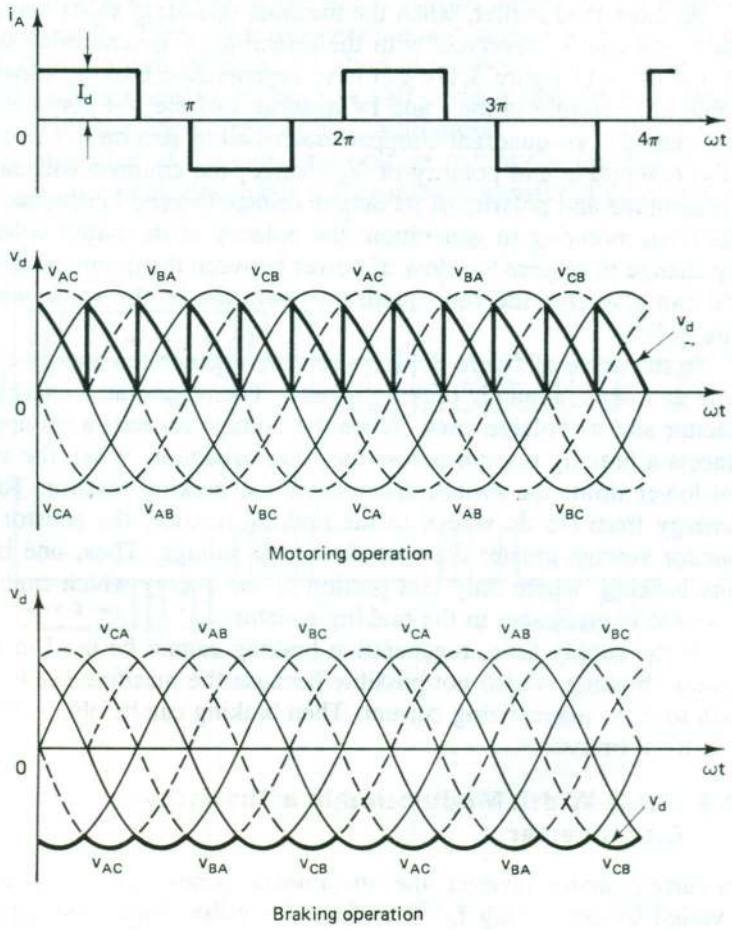


Figure 8.18 Braking operation of CSI-fed induction motor.

dc link to the ac supply, and regenerative braking is achieved. Thus, regenerative braking of a current source inverter drive is very simple. One has simply to reduce the inverter frequency to make the synchronous speed less than the rotor speed. The inverter and the rectifier automatically adjust their operations to arrange the transfer of the generated power to the ac supply.

Regenerative braking capability is inherent in the drive of figure 8.17b and no additional equipment is required to achieve it. This should be compared with the voltage source inverter drive where the regenerative braking capability is achieved by the addition of a fully controlled rectifier in antiparallel with the existing controlled rectifier, which adds to the drive's cost and complexity.

The drive of figure 8.17b can also provide four-quadrant operation. With a given phase sequence, the drive operates in quadrants I and II. When the motor is at standstill, the phase sequence is reversed simply by interchanging the control signals between any two legs of the inverter, causing the motor to run in the reverse direction and to provide operation in quadrants III and IV.

As explained earlier, when the machine operation shifts from motoring to generation, voltage V_d reverses, with the direction of I_d remaining unchanged. Therefore, the drive of figure 8.17c can have regenerative braking capability if a chopper capable of operating in the I and IV quadrants of the V-I plane is employed. Recall that a class D two-quadrant chopper, described in section 4.7.2, has this capability. As the magnitude and polarity of V_d change, the chopper will automatically adjust the magnitude and polarity of its output voltage to keep I_d constant. As the operation shifts from motoring to generation, the polarity of its output voltage will automatically change to reverse the flow of power between the dc mains and the dc link. This drive can also provide four-quadrant operation in the same way as the drive of figure 8.17b.

In the drive of figure 8.17c, when the regenerated energy cannot be absorbed by the dc mains, dynamic braking is used. The regenerated energy charges the filter capacitor and its voltage rises. When the voltage exceeds a set upper limit, a switch connects a braking resistor across the filter capacitor. When the voltage falls below a set lower limit, the switch disconnects the braking resistor. To prevent the flow of energy from the dc source to the braking resistor, the resistor is turned off at a capacitor voltage greater than the dc source voltage. Thus, one is able to get composite braking, where only that portion of the energy which cannot be accepted by the source is dissipated in the braking resistor.

If the supply fails, regenerative braking cannot be used in both the schemes. Dynamic braking is also not possible because the machine needs some source from which to draw magnetizing current. Then braking can be obtained only by the use of mechanical brakes.

8.2.4 Pulse-Width Modulation in a Thyristor C.S. Inverter

In a current source inverter, the fundamental component of the machine current can be varied by controlling I_d. Therefore, the pulse-width modulation (PWM) is only required to improve the current waveforms.

As explained in section 8.2.1, for the thyristor inverter of figure 8.15, switching spikes are suppressed with the help of capacitors by delaying the transfer of current between the phases. The delay in the transfer of current restricts the frequency of operation of thyristors and inverter. The higher the value of capacitors, the greater the reduction in switching spikes and the greater the delay in the transfer of current. The value of capacitors required for the adequate suppression of spikes is rather high; consequently the frequency of operation of thyristors is rather low. The frequency restriction is imposed by the circuit configuration and not by the switching capability of thyristors. Hence, replacing the converter grade thyristors by the faster thyristors, such as inverter grade thyristors, is of little value.

Because of the low operating frequency of thyristors, the PWM is possible only at low inverter frequencies. Hence, it has been used only at low motor speeds — less than 10 percent of the base speed — mainly to eliminate torque pulsations.

The implementation of the PWM in ASCI of figure 8.15 is shown in figure 8.19. In section 8.2.1, a typical switching operation was considered involving the turn-on of T_3 and the commutation of T_1, resulting in the transfer of a positive current from line A to line B, while line C carried current in the negative direction through thyristor T_2. The switching operation also caused the voltage of capacitor C_1 to reverse. It also signified the termination of the positive half-cycle of line current i_A and the commencement of the positive half-cycle of line current i_B while the negative half-cycle of the line current i_C was in progress.

An examination of the capacitor voltage polarity at the end of the foregoing switching operation indicates that it will enable the transfer of a positive current back

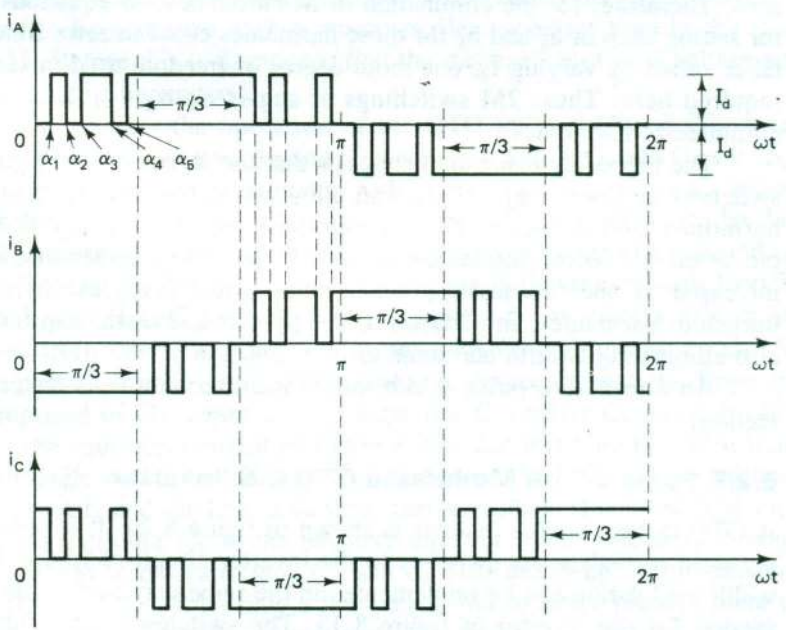


Figure 8.19 Current waveforms of a PWM current source inverter.

to line A if thyristor T_1 is gated again. It is, therefore, possible to transfer a positive current back and forth between lines A and B by alternately gating T_1 and T_3 while the other line C is carrying a negative current through T_2. Thus, the trailing side of the positive half-cycle of i_A and the leading side of the positive half-cycle of i_B can be pulse-width modulated in such a way that the pulse-off period in one line is equal to the pulse-on period in the other. By repeating this process in all the upper thyristors, T_1, T_3, and T_5, both ends of the positive half-cycles of the three line currents can be pulse-width modulated. Similarly, both ends of the negative half-cycles of the three line currents can be pulse-width modulated by repeating this process for the lower group of thyristors, T_2, T_4, and T_6.

If the three phases are modulated symmetrically, then the pulse widths in the leading side of a phase will be equal to the notch widths in the trailing side. Hence in general the modulation will have half-wave symmetry. When two phases are being modulated, the third phase cannot be modulated. Thus the wide middle pulse cannot be less than $\pi/3$ as shown in the figure.

The switching angles $\alpha_1, \alpha_2, \ldots$ can be chosen to implement the selective harmonic elimination (SHE) method, described in section 8.1.4. In the SHE method of section 8.1.4, the waveform, had quarter-wave symmetry. Hence the Fourier coefficient b_k was zero. Therefore, for the elimination of M harmonics, M equations were needed to set the Fourier coefficient a_k for all these harmonics equal to zero. One degree of freedom was used to vary the fundamental. Thus, $(M + 1)$ switchings at angles $\alpha_1, \alpha_2, \ldots, \alpha_{M+1}$ were required to implement the SHE method. In the present case, the waveforms have half-wave symmetry. Hence, both a_k and b_k will not be zero. Therefore, for the elimination of M harmonics, M equations will be required for setting each of a_k and b_k for those harmonics equal to zero. Since the fundamental is varied by varying I_d, one more degree of freedom used in section 8.1.4 is not required here. Thus, 2M switchings at angles $\alpha_1, \alpha_2, \ldots, \alpha_{2M}$ will be needed to eliminate M harmonics.

The preceding discussion suggests that the waveforms of figure 8.19 with five switching angles $\alpha_1, \alpha_2, \ldots, \alpha_5$ can allow the elimination of the fifth and seventh harmonics and consequently can eliminate the dominant sixth harmonic torque pulsation. At lower fundamental frequencies, the number of switchings can be increased. If one chooses to produce nine switching angles, then the eleventh and thirteenth harmonics, in addition to the fifth and seventh, can be eliminated, thus also eliminating twelfth harmonic torque pulsation.

An alternative pulse-width modulation technique is described in the next section.

8.2.5 Pulse-Width Modulated GTO C.S. Inverter

A GTO current source inverter is shown in figure 8.20. It is obtained when GTOs are used to realize the switches in the inverter circuit of figure 8.14a. The pulse-width modulation can be implemented in the same way as described in the previous section for the inverter of figure 8.15. The switching angles may be selected to achieve the selective harmonic elimination method. Alternatively, the method shown in figure 8.21a may be employed. The figure shows the modulation of line current i_A in its positive half-cycle. A carrier wave v_C is compared with a modulating reference

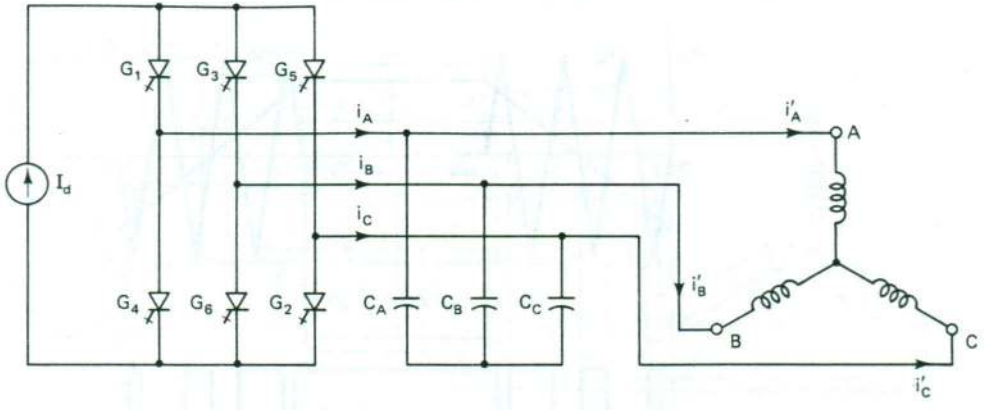
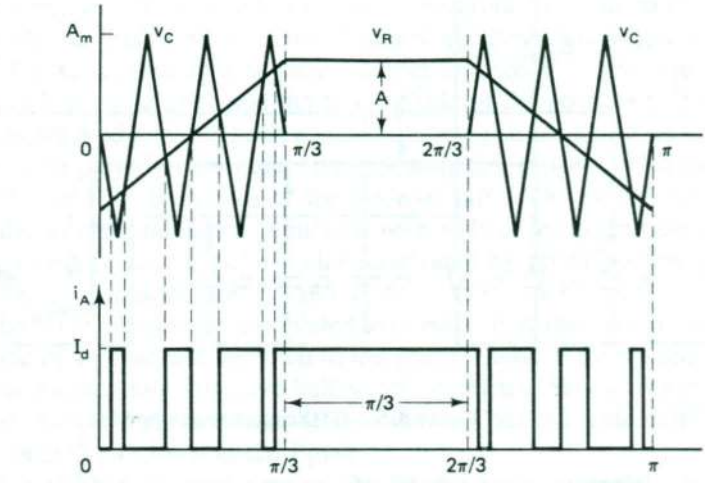


Figure 8.20 GTO current source inverter.

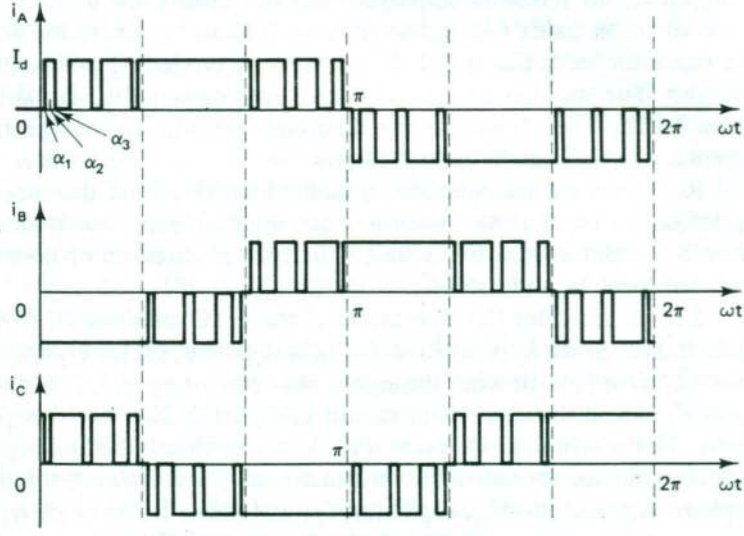
wave v_R. When $v_R > v_C$, a pulse of current is produced. In this case, the waveform has quarter-wave symmetry. Since the fundamental component can be varied by changing I_d, the PWM is employed only to improve the current waveform. Hence, the modulation index (A/A_m) is fixed at 0.82 to minimize the harmonic content.[32] The capacitor bank filters out the harmonics, producing a sinusoidal current in the machine. The modulated waveforms of line currents i_A, i_B, and i_C are shown in figure 8.21b. The sinusoidal machine currents will lag behind the respective line currents, due to a small leading current drawn by the capacitors.

Reference 32 has described a method which allows the suppression of switching spikes to be achieved with a much smaller capacitor than in the inverter of figure 8.15. Before this is discussed, first the mechanism of how voltage spikes are produced must be understood.

Let us consider the operation of the GTO inverter (fig. 8.20) at $\omega t = 0$ in figure 8.21b. Since i_C is positive and i_B is negative, GTOs G_5 and G_6 are on, and the source current I_d is flowing through a path consisting of G_5, phase C, phase B, and G_6 as shown in the equivalent circuit of figure 8.22a. Let this path be designated loop 1. The machine phase current i'_A, which is sinusoidal and lags behind i_A, is still negative. Hence, another relatively small current is flowing through loop 2 formed by phase A, capacitor C_A, capacitor C_B, and phase B. At angle α_1 (fig. 8.21b), I_d is transferred from line C to line A, by turning off G_5 and turning on G_1. Since the machine phase current i'_A is still negative, the source current, I_d, flows through a loop comprised of G_1, capacitor C_A, capacitor C_C, phase C, phase B, and G_6, as shown in the equivalent circuit of figure 8.22b. Let this loop be called loop 3. The current through loop 2 continues to flow as before. Now C_A is charged both by the source current I_d and the loop 2 current, and its voltage shoots up. The loop 2 current charges capacitor C_B in the negative direction. The voltage v_{AB}, which is the sum of voltages across capacitors C_A and C_B also shoots up. The modulated waveform of figure 8.21b shows only a few pulses, but in actual practice there would be many more pulses. As the source current is transferred back and forth between loop 1 and loop 3, by alternate conductions of G_5 and G_1, the voltages across capacitors C_A and C_B build up. Consequently, voltage v_{AB} rises to a very high value. The



(a) Principle of modulation



(b) Modulated current waveform

Figure 8.21 PWM current source inverter.

build-up of voltage v_{AB} continues as long as the machine phase current i'_A remains negative. After i'_A reverses, the loop 2 current flows to discharge capacitors C_A and C_B, while the loop 3 current continues to charge them. At a certain value of i'_A, the build-up of the capacitor voltage is checked and v_{AB} decreases after attaining a maximum value. This is the maximum spike in the voltage v_{AB} and it occurs soon after the reversal of the machine current i'_A. Since the circuit works symmetrically, identical spikes will be produced at the reversal of currents i'_B and i'_C.

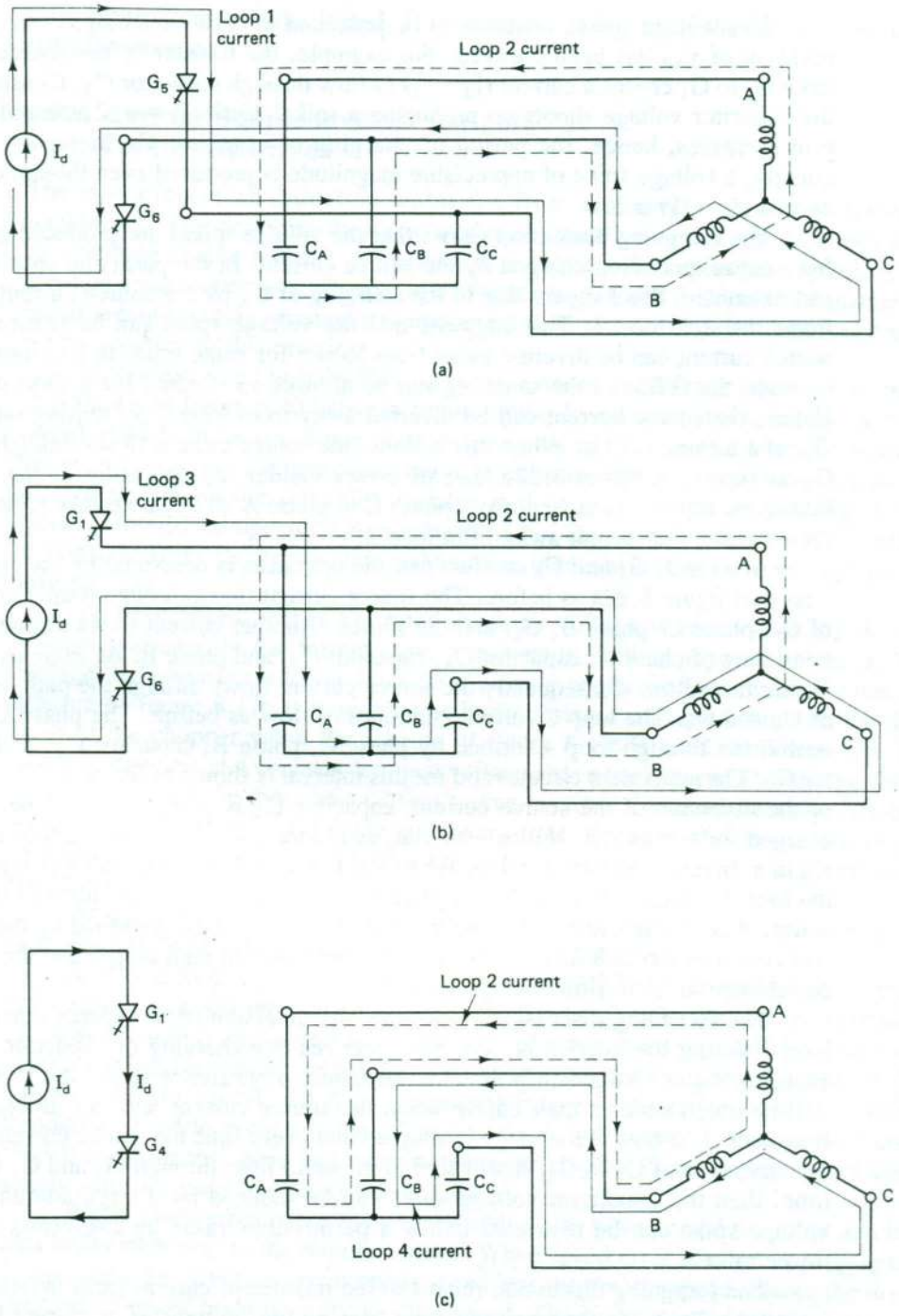


Figure 8.22 Equivalent circuits of the inverter of Fig. 8.20.

The voltage spikes continue to be produced at each switching, even after the build-up of v_{AB} has been checked. For example, the transfer of the source current from G_5 to G_1 causes a current $(I_d - i'_A)$ to flow through capacitor C_A. Consequently, the capacitor voltage shoots up producing a spike. Until $\omega t = \pi/2$, the pulse duration increases; hence, the period of charging of capacitor C_A increases. Consequently, a voltage spike of appreciable magnitude is produced even though charging current $(I_d - i'_A)$ is low.

The foregoing discussion shows that the voltage spikes are produced primarily from capacitors being charged by the source current. In the particular case we have just examined, this happens due to the charging of C_A by the source current when it flows through loop 3. This suggests that the voltage spike can be reduced if the source current can be diverted away from loop 3 for some time. In a current source inverter, the GTOs in the same leg can be allowed to conduct for a short duration. Hence, the source current can be diverted away from loop 3 by turning on G_1 and G_4, and turning off G_6. When this is done, the source current flows through G_1 and G_4 as shown in figure 8.22c. Let us now consider the operation of the inverter during the transfer of current from phase C to phase A when an arrangement is made for diverting the current away from loop 3.

At $\omega t = 0$, G_5 and G_6 conduct and the operation is described by the equivalent circuit of figure 8.22a as before. The source current flows through loop 1 consisting of G_5, phase C, phase B, G_6, and the source. Another current flows through loop 2 consisting of phase A, capacitor C_A, capacitor C_B, and phase B. At angle α_1, G_1 and G_4 are turned on. Consequently the source current flows through the path consisting of G_1 and G_4. The loop 2 current continues to flow as before. The phase C current now flows through loop 4 formed by phase C, phase B, capacitor C_B, and capacitor C_C. The equivalent circuit valid for this interval is shown in figure 8.22c. Because of the diversion of the source current, capacitor C_A is prevented from being overcharged for some time. Notice also that, as before, loop 2 current charges capacitor C_B in a direction to increase v_{AB}. However, the loop 4 current, which is higher than the loop 2 current, charges C_B to reduce voltage v_{AB}. At α_2, G_4 is turned off and G_6 is turned on. G_1 is already on. The inverter operation is now governed by the equivalent circuit of figure 8.22b. The source current flows through loop 3 and the phase A current continues to flow through loop 2.

The foregoing discussion shows that the diversion of the source current from loop 3 during the interval $\alpha_1 < \omega t < \alpha_2$ reduces overcharging of capacitor C_A, and capacitor C_B is charged in a direction to reduce v_{AB}. Hence at α_3, the voltage v_{AB} will be much smaller than before when the source current was not diverted away from loop 3. If now the strategy is adopted that every time the source current is to be transferred from G_5 to G_1, it will first be made to flow through G_1 and G_4 for some time, then the maximum voltage spike can be reduced by a large amount, or the voltage spike can be restricted below a permissible value by capacitors of much lower value.

The foregoing discussion relates to the transfer of current from inverter line C to line A. The same approach much be adopted for the transfer of current from any one inverter line to another. Obviously, the control becomes complex. The motor operates with sinusoidal voltage and current, and hence the problems of derating and

torque pulsations are eliminated. However, the diversion of the source current derates the inverter.

8.2.6 Current Source Inverter Variable Frequency Drives

The operation of an induction motor fed from a current source is described in section 6.5. The drive has the same modes of operation as the induction motor fed by a voltage source inverter. It operates at a constant flux up to base speed, giving constant torque operation. Operation above the base speed is done at a constant terminal voltage, giving constant power operation. However, the minimum loss control is an exception to this practice.

Operation at a constant flux up to base speed is realized when the relationship shown in figure 6.20 between stator current I_s and rotor slip speed $\omega_{s\ell}$ is maintained at all frequencies. Since, I_d is proportional to I_s, a similar relationship exists between I_d and $\omega_{s\ell}$. When operating at a constant flux, the operating points are located mostly on the statically unstable part of speed-torque curves (fig. 6.19). Hence, closed-loop operation is mandatory. Since the motor is constrained to operate at constant flux, its steady-state behavior is identical to that of a voltage source inverter case.

At base speed, either the rated machine voltage is reached or the dc link voltage saturates. In the former case, a maximum limit is imposed on the dc link voltage; by imposing a lower limit on the firing angle α of the rectifier when the scheme of figure 8.17b is employed or by imposing a maximum limit on the duty ratio of the chopper when the scheme of figure 8.17c is used. In either case, the machine operates at a constant terminal voltage above base speed. Though the waveforms continue to be those of a current source inverter, the motor behavior is somewhat similar to that of a voltage source inverter. Because of a low value of the maximum operating frequency of a current source inverter, the maximum drive speed is rather low.

At a constant value of I_s, various parameters change with frequency in the same way as shown in figure 6.14 for a voltage source.

A closed-loop CSI drive is shown in figure 8.23.[23] The speed error signal, after being processed through the speed controller, is given to the slip speed regulator, which controls slip speed $\omega_{s\ell}$. The sum of rotor speed ω_m and slip speed $\omega_{s\ell}$ gives synchronous speed, which determines the inverter frequency. Based on the value of the slip speed, the flux controller produces a reference signal I_d^*, which through a closed-loop current control adjusts the dc link current I_d to maintain a constant flux. Both the speed and current controller employ PI controllers to get good steady-state accuracy and to attenuate noise.

When speed error is positive, slip speed is also positive and the drive accelerates under motoring to the required speed. When speed error is negative, slip speed is also negative, giving a synchronous speed less than motor speed. Hence the drive decelerates under braking to the desired speed. The limit imposed on the output of the slip regulator limits I_d at the inverter (and converter) rating. Thus, the transient operation of the drive, below base speed, both in motoring and braking, is carried

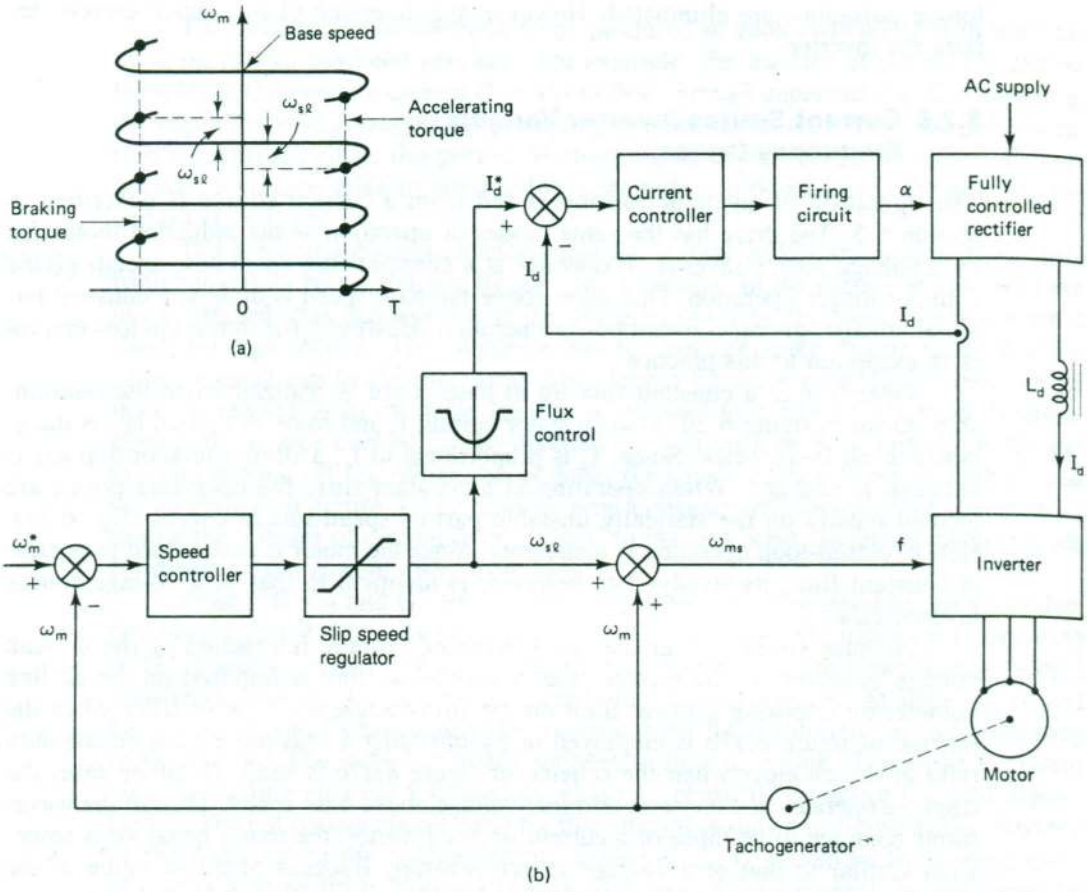


Figure 8.23 CSI variable frequency drive with slip speed control.

out at the rated inverter current. Since the flux is constant, the drive operates at the maximum available torque, as shown by dotted lines in figure 8.23a. Consequently, fast transient response is obtained.

Beyond base speed, the machine terminal voltage saturates as stated earlier. Consequently, flux control becomes ineffective and the machine operates in a constant power mode. To operate the drive up to the rated inverter current, the slip speed limit of the slip speed regulator must increase linearly with frequency. This is achieved by adding to the slip speed regulator output an additional slip speed signal proportional to frequency and of appropriate sign.

An alternative scheme is shown in figure 8.24.[55] The speed error is processed through a speed controller and an absolute value circuit to get the current reference I_d^*. Through a closed-loop current control, the dc link current I_d is made to track I_d^*. Based on the value of I_d, the flux control block produces a slip speed $\omega_{s\ell}$ to maintain a constant flux. The addition of actual speed ω_m and slip speed $\omega_{s\ell}$ gives synchronous speed ω_{ms}, which determines inverter frequency. When the speed error is positive, the slip speed is assigned a positive sign for the motoring operation.

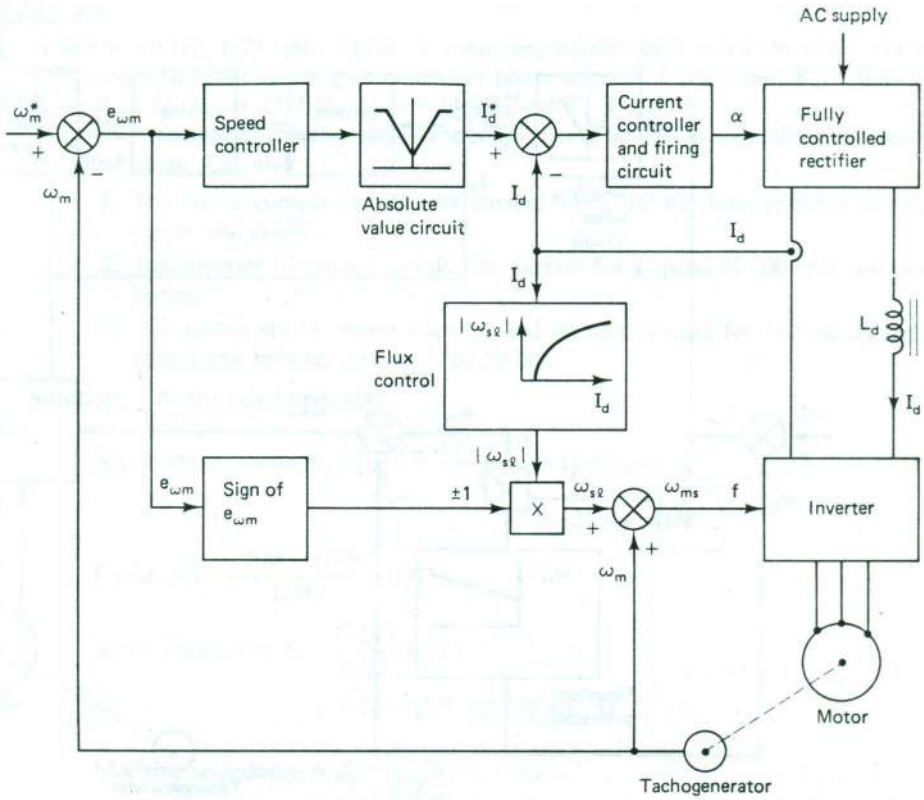


Figure 8.24 CSI variable frequency drive with current control.

When the speed error is negative, the slip speed is assigned a negative sign for the braking operation.

Saturation in the absolute value circuit ensures that speed error is corrected at the rated inverter current. Since the flux is constant, the drive operates at the maximum torque, as shown by dotted lines in figure 8.23a. Hence, fast transient response is achieved. In a thyristor inverter, a minimum value of I_d must be retained to achieve thyristor commutation. This is ensured by the absolute value circuit which maintains a minimum value of I_d^*.

Both the drives just described have a similar dynamic response.[34] Reference 35 gives a method of designing controllers for these drives.

A drive with minimum loss control is shown in figure 8.25. The relationship between optimum slip speed and motor speed is the same as for the case of a voltage source inverter. This relationship is stored in the function generator. It is assumed that for a given speed, the optimum slip has the same magnitude for motoring and regenerative braking operations. When these magnitudes are sufficiently different, two separate function generators may be used—one for motoring and another for braking. The function generator determines the magnitude of optimum slip speed $\omega_{s\ell}$ based on motor speed ω_m. When speed error is positive, a motoring operation is required. When it is negative, a braking operation is required. Hence, the optimum

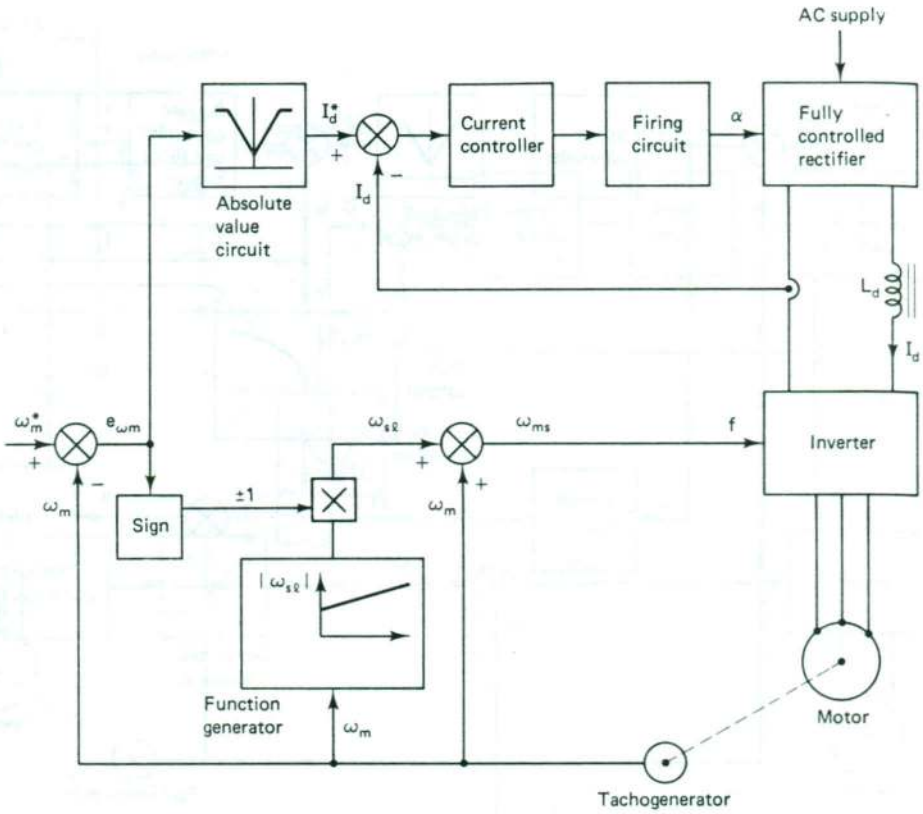


Figure 8.25 CSI variable frequency drive with minimum loss control.

slip speed is assigned the same sign as that of the speed error $e_{\omega m}$. The sum of ω_m and $\omega_{s\ell}$ determines the inverter frequency. An inner-current loop within the speed loop is provided to serve three purposes: to provide a current source for the inverter, to prevent current from exceeding the safe value, and to vary the drive torque for a given speed. The absolute value circuit has the same functions as described for the drive of figure 8.24.

When the speed command ω_m^* is increased, the reference current I_d^* is set at the highest value. The drive accelerates at the maximum allowable current. The slip speed is always maintained at the optimum positive value. When near the desired speed, I_d^* decreases. The drive finally settles to the desired speed with I_d^* adjusted to cause the motor torque to equal the load torque.

When the speed command ω_m^* is reduced, again the reference current I_d^* is set at the highest value. Since the speed error $e_{\omega m}$ has a negative sign, the optimum slip speed also has a negative sign. Hence, the motor decelerates under regenerative braking at the maximum permissible current. When close to the desired speed, I_d^* decreases, reducing motor torque. Speed continues to fall until $e_{\omega m}$ changes sign and the operation shifts to motoring. The drive finally settles at the desired speed.

Example 8.5

A 460 V, 60 Hz, 1176 rpm, 6 pole, Y-connected squirrel-cage induction motor has the following equivalent circuit parameters per phase referred to the stator: $R_s = 0.29\ \Omega$, $X_s = 0.21\ \Omega$, $X_m = 13.3\ \Omega$, $R_r' = 0.145\ \Omega$, and $X_r' = 0.5\ \Omega$
The motor is supplied from a current source inverter. The flux is maintained constant at the rated value. Calculate

1. The stator current and dc link current when the machine operates at rated torque and 60 Hz.
2. The inverter frequency and dc link current for a speed of 600 rpm and rated torque.
3. The motor speed, stator current, and dc link current for half of the rated torque and inverter frequency of 30 Hz.

Solution: At the rated operation,

$$\text{Synchronous speed } N_s = \frac{120f}{p} = \frac{120 \times 60}{6} = 1200 \text{ rpm}$$

$$= 125.66 \text{ rad/sec.}$$

$$\text{Rated slip} = \frac{1200 - 1176}{1200} = 0.02$$

$$\text{Rotor impedance } Z_r' = \frac{0.145}{0.02} + j0.5$$

$$= 7.25 + j0.5 = 7.27\underline{/3.9^\circ}\ \Omega$$

$$\text{Machine impedance} = Z_s + \frac{Z_r' Z_m}{Z_r' Z_m}$$

$$= 0.29 + j0.21 + \frac{(7.25 + j0.5)(j13.3)}{7.25 + j13.8}$$

$$= 5.57 + j3.46 = 6.557\underline{/31.84^\circ}\ \Omega$$

$$\bar{I}_s = \frac{460\sqrt{3}}{6.557\underline{/31.84^\circ}} = 40.5\underline{/-31.84^\circ}\ \text{A}$$

$$\bar{I}_r' = \frac{Z_m}{Z_m + Z_r'} \bar{I}_s = \frac{13.3\underline{/90^\circ}}{15.59\underline{/62.28^\circ}} \times 40.5\underline{/-31.84^\circ} = 34.55\underline{/-4^\circ}\ \text{A}.$$

$$\bar{I}_m = \frac{Z_r'}{Z_m + Z_r'} \bar{I}_s = \frac{7.27\underline{/3.9^\circ}}{15.59\underline{/62.28^\circ}} \times 40.5\underline{/-31.84^\circ} = 18.89\underline{/-90^\circ}\ \text{A}.$$

$$\text{Torque} = \frac{3}{\omega_{ms}} I_r'^2 (R_r'/s)$$

$$= \frac{3}{125.66}(34.55)^2 \left(\frac{0.145}{0.02}\right) = 206.6 \text{ N-m}$$

1. From equation (8.23)

$$I_d = \frac{\pi}{\sqrt{6}} I_s = \frac{\pi}{\sqrt{6}} \times 40.5 = 51.94 \text{ A}$$

From equation (8.24) rms stator current

$$I_{rms} = (\sqrt{2/3})I_d = 0.816 \times 51.94 = 42.4 \text{ A}.$$

2. It was explained in section 6.5.2 that when the motor is controlled at a constant flux for a given torque, the slip speed has a constant value.

The slip speed at the rated torque and frequency is

$$N_{s\ell} = sN_s = 0.02 \times 1200 = 24 \text{ rpm}$$

Hence, at the motor speed of 600 rpm, synchronous speed, $N_s = 600 + 24 = 624$ rpm
Hence the inverter frequency $= (624/1200) \times 60 = 31.2$ Hz.

As explained in section 6.5, when the motor is controlled at a constant flux, for a given torque, the stator current remains constant at all speeds. Since the stator current is constant, the dc link current also remains constant at 51.94 A.

3. Since the flux is constant, for a given torque, the slip speed is constant at all frequencies. Thus, the slip speed for 30 Hz operation at half the rated torque can be found from 60 Hz operation. For 60 Hz operation,

$$E_{rated} = I_m X_m = 18.89 \times 13.3 = 251.2 \text{ V}$$

Now,

$$T = \frac{3}{\omega_{ms}}\left[\frac{E_{rated}^2 R_r'/s}{(R_r'/s)^2 + X_r'^2}\right]$$

Substituting for the known values,

$$\frac{206.6}{2} = \frac{3}{125.66}\left[\frac{(251.2)^2(0.145/s)}{\left(\frac{0.145}{s}\right)^2 + (0.5)^2}\right]$$

or

$$\left(\frac{0.145}{s}\right)^2 + (0.5)^2 = \frac{2.1146}{s}$$

which gives $s = 0.01$
Hence, slip speed $N_{s\ell} = sN_s = 0.01 \times 1200 = 12$ rpm
Now consider the operation at 30 Hz

$$a = 30/60 = 0.5$$

Synchronous speed $N_s = 0.5 \times 1200 = 600$ rpm
Hence, motor speed $= 600 - 12 = 588$ rpm

$$s = \frac{N_{s\ell}}{N_s} = \frac{12}{600} = 0.02$$

$$I_r' = \frac{aE_{rated}}{\sqrt{\left(\frac{R_r'}{s}\right)^2 + (aX_r')^2}} = \frac{0.5 \times 251.2}{\sqrt{\left(\frac{0.145}{0.02}\right)^2 + (0.5 \times 0.5)^2}} = 17.32 \text{ A}$$

From equation (6.72)

$$I_r'^2 = \frac{I_s^2 - I_m^2}{1 + \frac{2X_r'}{X_m}}$$

or

$$17.32^2 = \frac{I_s^2 - 18.89^2}{1 + \frac{2 \times 0.5}{13.3}}$$

which gives $I_s = 26.1$ A
From equation (8.23)

$$\text{DC link current } I_d = \frac{\pi I_s}{\sqrt{6}} = \frac{\pi \times 26.1}{\sqrt{6}} = 33.47 \text{ A}$$

From equation (8.24)

$$\text{The rms stator current } I_{rms} = (\sqrt{2/3}) \times 33.47 = 27.33 \text{ A}$$

8.2.7 Comparison of Current Source and Voltage Source Inverter Drives

The important relative merits and disadvantages of CSI and VSI drives are as follows:

1. In a CSI drive, the simultaneous conduction of two switches in the same leg of the inverter, due to misfiring or commutation failure, does not lead to a shoot through fault because the resultant current, which rises slowly due to the large value of L_d, can be regulated by the control of the rectifier firing angle. Protection for a short-circuit across the load takes place in the same way. In the case of a VSI, such faults can be cleared only by high-speed fuse links. For these reasons, a CSI is more reliable and rugged than a VSI.
2. The regenerative braking capability is inherent in a CSI drive powered by an ac source. In a VSI drive, an additional fully controlled rectifier is required to obtain regenerative braking capability. If the ac supply fails, regenerative braking will not be possible in both the drives. In such an eventuality, a VSI drive can use dynamic braking but not a CSI drive.

   When the drive is powered by a dc source, only the PWM VSI drive has inherent regenerative braking capabilities. The six-step VSI and CSI drives require the replacement of a single-quadrant chopper by a two-quadrant chopper to achieve regenerative braking capability.
3. Because of the large value of L_d, the dynamic response of a CSI drive is slower compared to a PWM VSI drive. Because of the large filter capacitor, the dynamic response of a six-step VSI drive is nearly as slow as that of a CSI drive.
4. In the case of VSI drives, the use of PWM allows efficient and smooth operation, free from torque pulsations and cogging. Because of the low frequency of

operation of a thyristor CSI, PWM is possible only at low speeds. This eliminates torque pulsations at low speeds but not at high speeds. Though the torque pulsations present at high speeds do not cause any speed fluctuations, they do reduce the life of a motor. When GTOs are used, PWM may be employed up to the base speed in CSI also, but then the control becomes complex.

5. When the source is dc, a PWM VSI drive will be much cheaper compared to a CSI drive of the same rating. Because of the large commutation capacitors and large dc link inductor, which is oversized to prevent saturation, the volume and weight of a CSI drive is much larger compared to a PWM VSI drive.
6. The CSI is not suitable for multi-motor drives. Hence, each motor is fed by its own inverter and rectifier. A single-diode bridge or controlled rectifier can be used to feed a number of VSI inverter-motor systems. Alternatively, a single VSI can feed a number of motors.
7. The frequency range of a CSI is lower than VSI. Consequently, the CSI drive has a lower speed range.

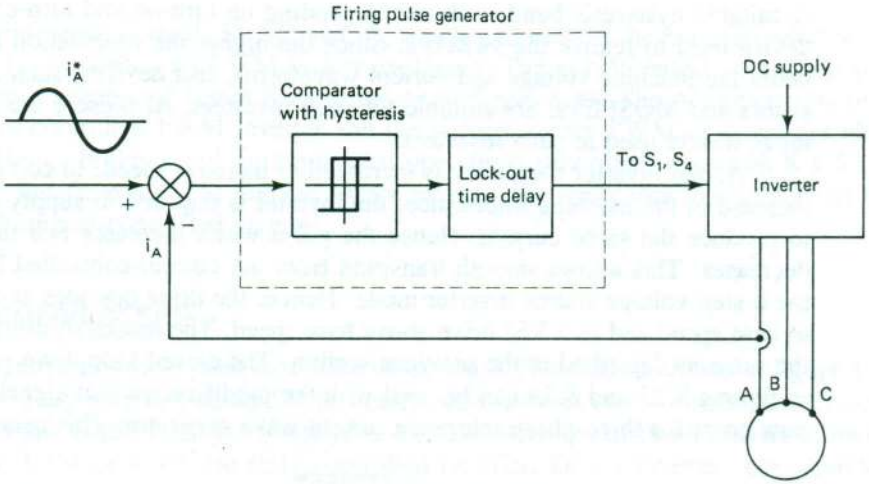
8.3 CURRENT-CONTROLLED PWM INVERTERS

In the inverters discussed so far, a VSI was fed by a dc voltage source, and a CSI was fed by a dc current source. In this section, a PWM CSI fed by a dc voltage source is described. Such an inverter is called a current-controlled PWM inverter.

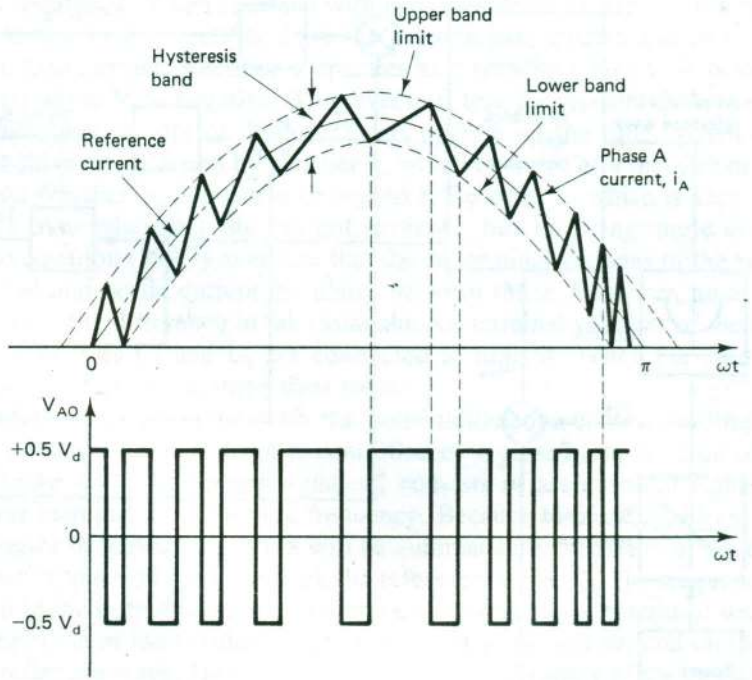
The VSI power circuit shown in figure 8.1a is employed. The dc supply for the inverter is obtained by using the scheme of figure 8.3. The three-phase reference currents i_A^*, i_B^*, and i_C^* are compared with the respective machine phase currents i_A, i_B, and i_C in three separate comparators. The comparator arrangement for phase A is shown in figure 8.26a. The generation of a modulated waveform is shown in figure 8.26b for the positive half-cycle of i_A^*.

Consider the interval when switch S_1 is on. The voltage of machine phase A with respect to the imaginary midpoint of the source v_{AO}, is positive. Hence, the phase current i_A increases. When i_A reaches the upper band limit, the comparator output changes from 1 to 0. This information is used to turn off S_1 and turn on S_4 after the lock-out time. Consequently v_{AO} becomes negative and i_A decreases. When i_A reaches the lower band limit, the comparator output changes from 0 to 1. Switch S_4 is turned off and switch S_1 is turned on after the lock-out time. Thus by alternate conductions of S_1 and S_4, the machine current i_A is made to track the reference current i_A^* within the hysteresis band. The frequency and magnitude of the fundamental component in i_A are the same as in i_A^*. Hence, the magnitude and frequency of the fundamental in machine currents i_A, i_B, and i_C can be controlled by changing the magnitude and frequency of the reference currents i_A^*, i_B^*, and i_C^*. Since the magnitude of the machine currents (which depends on the reference currents) is independent of the load impedance and changes in the source voltage, the inverter essentially operates as a current source inverter.

The peak-to-peak ripple in motor current is restricted to the hysteresis band. Consequently, the harmonic content and the maximum instantaneous current have low values. The low harmonic content reduces motor heating and derating, and torque pulsations. The low value of the maximum instantaneous current is advanta-



(a)



(b)

Figure 8.26 Current controlled pulse width modulation.

geous in transistor inverters, because the ratio of maximum instantaneous to continuous current rating is low for a transistor. It is also beneficial in thyristor and GTO inverters because it reduces the commutation burden. A decrease in hysteresis band reduces the peak-to-peak current ripple, but increases the frequency of modulation.

A suitable hysteresis band is chosen depending on turn-on and turn-off times of the device used to realize the switches. Since the higher the modulation frequency, the better the machine voltage and current waveforms, fast devices, such as power transistors and MOSFETs, are suitable for such inverters. At present the power transistor is widely used in such inverters.

As the inverter frequency is increased to increase speed, to compensate for the increase in the machine impedance, the inverter is required to supply higher voltage to produce the same current. Hence the pulse width increases and the notch width decreases. This allows smooth transition from the current-controlled PWM mode to the 6-step voltage source inverter mode. Hence, the drive operates as a CSI drive up to base speed and as a VSI drive above base speed. The modes of drive operation are the same as described in the previous section. The closed-loop drive schemes shown in figures 8.23 and 8.24 can be used with the modification that signals f and I_d^* will now control a three-phase reference current wave generator. This generator will pro-

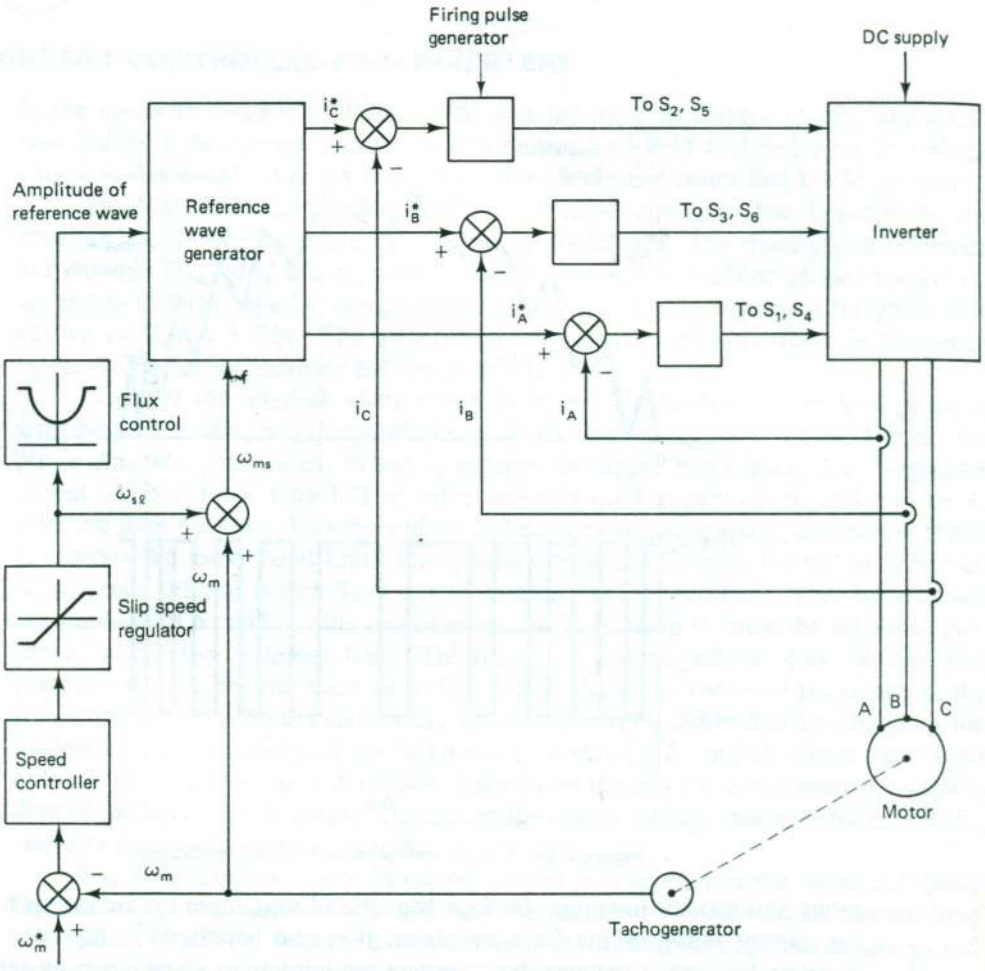


Figure 8.27 Closed-loop variable frequency drive using current-controlled PWM inverter.

duce reference currents i_A^*, i_B^*, and i_C^*, which will control the machine winding currents i_A, i_B, and i_C by PWM as just explained. The system based on the slip speed regulator scheme of figure 8.23 is shown in figure 8.27. The power circuits of the current-controlled PWM inverter and the voltage source PWM inverters are identical. Hence braking and multiquadrant operations described in section 8.1.5 for a motor fed from a PWM voltage source inverter are applicable to a motor fed by a PWM current-controlled inverter.

8.4 CYCLOCONVERTERS

Dual converters have been described in section 3.10.1 and are shown in figure 3.32d and e. They may be operated with simultaneous control or nonsimultaneous control. Rectifiers 1 and 2 may consist of a 6-pulse fully controlled rectifier shown in figure 3.16a or a 3-pulse fully controlled rectifier. Dual converters are capable of providing operation in all four quadrants of the V-I plane, with a variable voltage. The positive load current is supplied by rectifier 1 and the negative load current is supplied by rectifier 2. When operated with nonsimultaneous control, only one rectifier conducts at a time — rectifier 1 for the positive load current and rectifier 2 for the negative load current. Rectifier 1 operates as a rectifier when V_a is positive and as an inverter when V_a is negative. The reverse is true for rectifier 2. When operating with simultaneous control, both rectifiers operate all the time. When the load current is positive, it is carried by rectifier 1, which operates as a rectifier or inverter depending on whether V_a is positive or negative. Rectifier 2, which is kept in readiness to take over whenever the current reverses, has its firing angle controlled according to equation (3.117) to ensure that the dc terminal voltages of the two rectifiers are equal and no dc current circulates between them. However, an ac current circulates due to the difference in the instantaneous terminal voltages of the two rectifiers, and inductors L_1 and L_2 are connected to limit it. When the load current reverses, the two rectifiers reverse their roles.

Consider a dual converter with the nonsimultaneous control feeding an R-L load. The converter output voltage is controlled by a closed-loop voltage control as shown in figure 8.28. Reference signal v_A^* consists of a sinusoidal signal of frequency lower than that of the source frequency. Because of the closed-loop control, the firing angles of various thyristors will be automatically altered to allow the average converter output voltage V_a to track the reference wave v_A^*. The frequency of the fundamental in the output wave will be the same as that of the reference wave. Further, the amplitude of the fundamental in the output wave will depend on the amplitude of the reference wave. Hence, the amplitude and frequency of the output voltage can be controlled by varying the amplitude and frequency of the reference signal. Thus, a dual converter operated in this way works as a single-phase cycloconverter. It allows a variable frequency and variable magnitude voltage source to be obtained from a source of fixed frequency and fixed voltage. A three-phase voltage source cycloconverter is obtained by using three dual converters with their reference waves phase shifted by 120° with respect to each other. To avoid interaction between the dual converters, each dual converter is fed by a separate three-phase winding of a three-phase transformer as shown in figure 8.29. When the dual converters are oper-

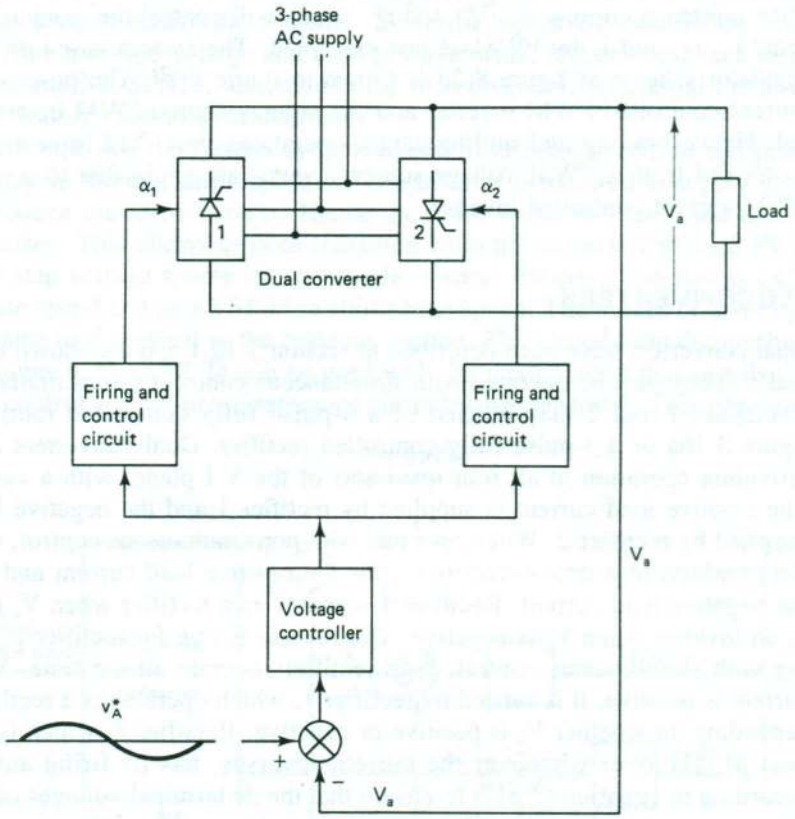


Figure 8.28 Realisation of a 1-phase cycloconverter by closed-loop voltage control of a dual converter.

ated with simultaneous control, the transformer can be avoided and instead interphase inductors may be used as shown in figure 8.30.

The foregoing approach can be used to realize a cycloconverter from off the shelf dual converters. It is also helpful in understanding the principle of a cycloconverter. In actual practice, a cycloconverter is realized without closed-loop voltage control, as explained next.

The inverse cosine firing for a dual converter is described in section 5.2.2. The generation of firing pulses is shown in figure 5.10. The firing pulses for thyristors T_1 to T_6 of rectifier 1 are produced at the intersection of control voltage v_c with signals v_{r1} to v_{r6}, respectively. When v_c is changed from its maximum positive value to 0, the firing angle changes from 0 to 90° and the output voltage changes from its maximum positive value to 0. When v_c is changed from 0 to its negative maximum, the firing angle changes from 90° to (180° − δ) and the output voltage changes from 0 to its maximum negative value. The firing pulses for rectifier 2 are produced at the intersection of v_c with signals v'_{r1} to v'_{r6}. The variations of the firing angle and the output voltage with v_c are opposite to that for rectifier 1.

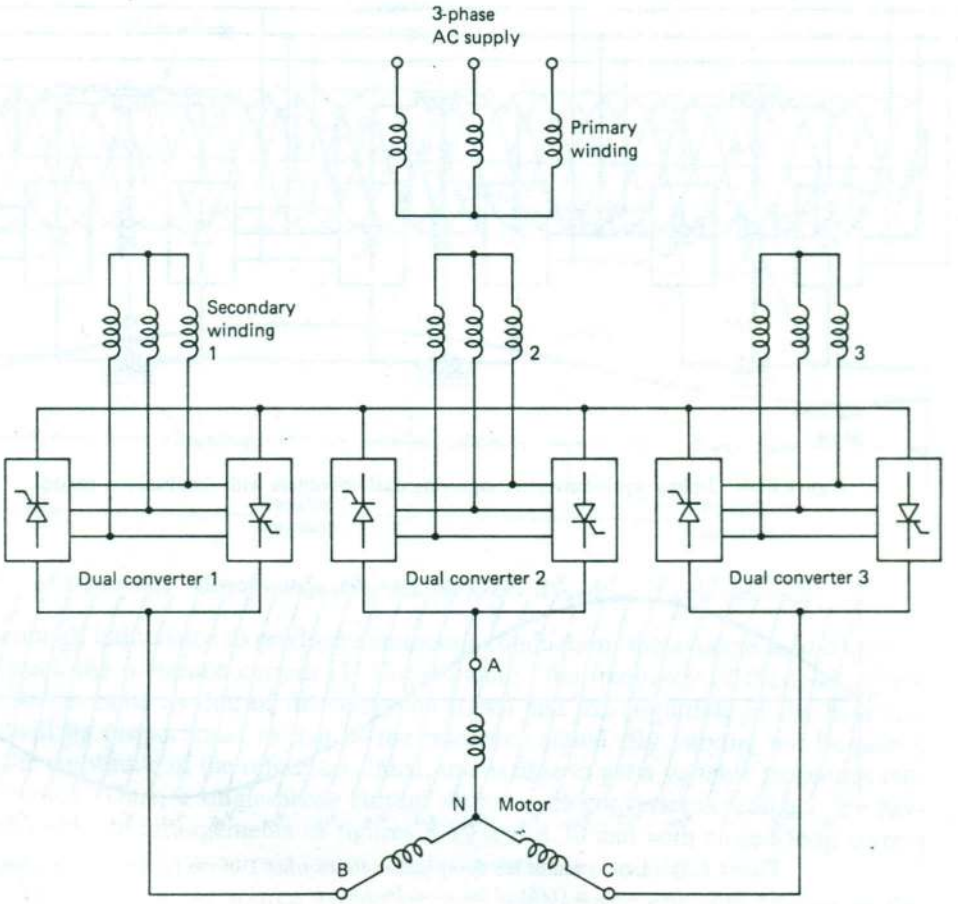


Figure 8.29 3-phase cycloconverter employing dual converters with non-simultaneous control.

In a cycloconverter, the dc control signal is replaced by a sinusoidal signal v_c as shown in figure 8.31. Inverse cosine reference signals v_{r1} to v_{r6} for rectifier 1 are also shown in the figure. As the control signal increases from its zero value, the firing angle of rectifier 1 progressively decreases and hence the output voltage increases, reaching a maximum value at the peak of the control signal. After the peak, the firing angle increases, reaching a value of 90° when $v_c = 0$. Hence, the output voltage also decreases. When the control signal reverses, the firing angle increases from 90°, reaching the maximum value at the negative peak of the control signal. After the peak, the firing angle decreases reaching 90° when $v_c = 0$. Consequently the output voltage also varies in the same way as v_c. The same control signal with reference voltage v'_{r1} to v'_{r6} (not shown) produces somewhat identical variation in the output voltage of rectifier 2. When operating with simultaneous control, both the rectifiers work together. When operating in nonsimultaneous control, rectifier 1 works when the current is positive and rectifier 2 works when the current is nega-

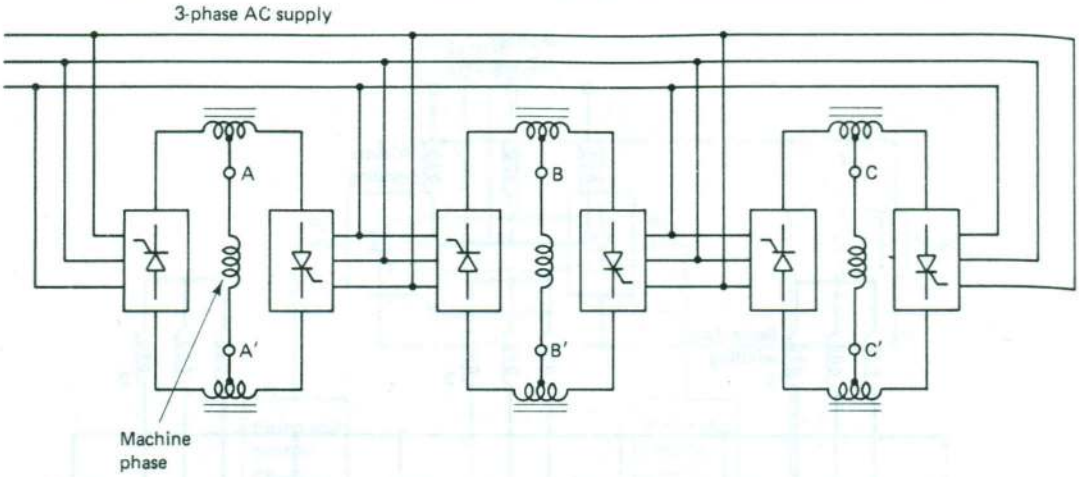


Figure 8.30 3-phase cycloconverter employing dual converters with simultaneous control.

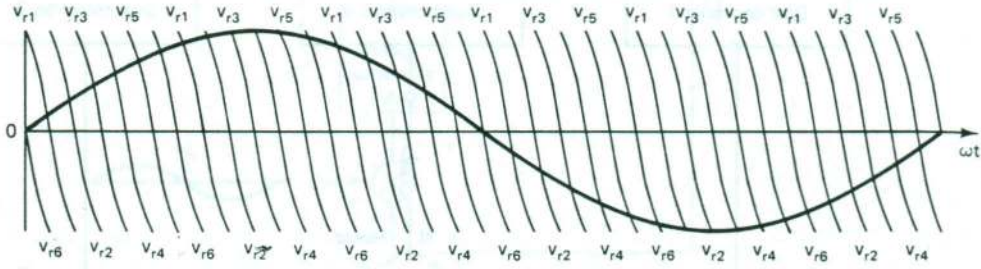


Figure 8.31 Generation of the firing pulses for rectifier 1 of the dual converter when the output frequency is (1/6) of the input frequency.

tive. The load voltage waveform and the load current waveform, assuming ideal filtering, are shown in figure 8.32 for nonsimultaneous control. The rectifier under operation and the mode of operation are also indicated in the figure.

A 3-phase cycloconverter is obtained when the three dual converters in figures 8.29 and 8.30 are controlled by sinusoidal control voltages which form a three-phase set.

The frequency of the output voltage has to be less than that of the source. With an increase in the load frequency, the harmonic content in the load voltage and source current increases. Depending on how much harmonic content can be tolerated by a load, a limit is imposed on the maximum frequency. In general, the output frequency is limited to below 40 percent of the source frequency—that is, below 24 Hz when the supply frequency is 60 Hz and below 20 Hz when it is 50 Hz.

Now consider the operation of the dual converter of figure 8.28 with a closed-loop current control, instead of closed-loop voltage control. Let the reference current i_A^* be sinusoidal and of frequency lower than the source frequency. When the load

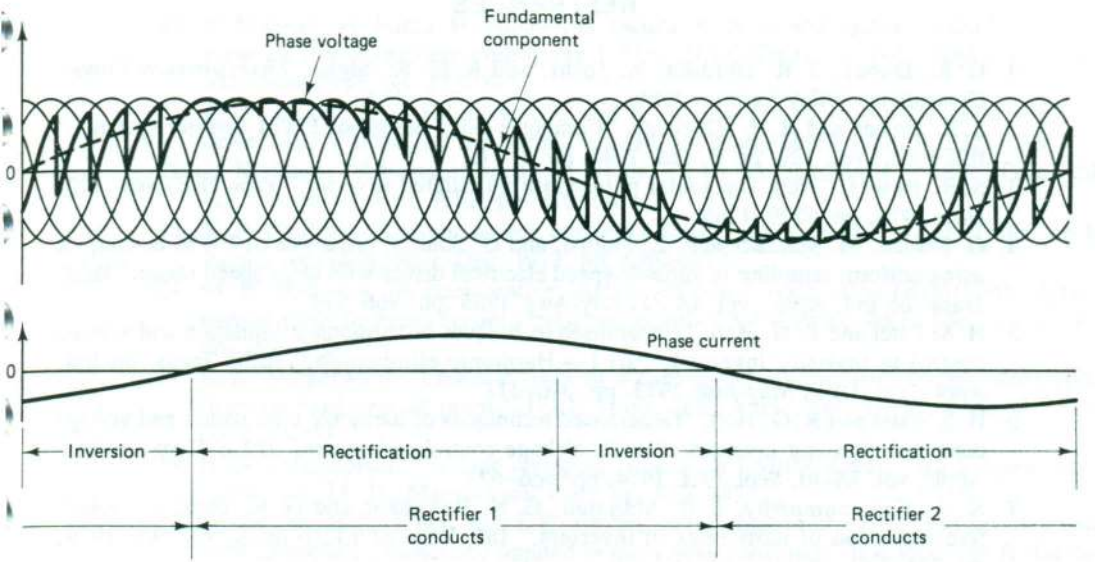


Figure 8.32 Output voltage waveform and modes of operation of a cycloconverter.

has enough inductance to produce continuous conduction, the average load current I_a will track the reference current i_A^*. Consequently, the frequency of the load current will be the same as that of the reference signal and the amplitude of the load current will be proportional to that of the reference signal. By varying the frequency and the amplitude of the reference signal, one is able to get a variable frequency current source. Thus, a single-phase current source cycloconverter is realized. By having three-phase arrangements of figures 8.29 and 8.30 and with closed-loop current control, a three-phase current source cycloconverter is obtained.

With the voltage source cycloconverter, the variable frequency drives of figures 8.10 to 8.13 can be used and with the current source cycloconverter, the schemes of figures 8.23 to 8.25 and 8.27 can be employed. In both the cases, the power can flow in either direction. Hence, regenerative braking is inherent. Full four-quadrant operation of the drive is obtained by reversing of the phase sequence. Because of the low frequency range, the speed range of a cycloconverter drive is low.

A 3-phase cycloconverter will require 36 thyristors. The number of thyristors can be reduced to 18 when a cycloconverter employs 3-pulse fully controlled rectifiers instead of 6-pulse fully controlled rectifiers. This, however, increases the harmonic content. Because of the need for a large number of thyristors, a cycloconverter becomes economical only for large size drives, where, because of the large current rating, a number of thyristors will have to be connected in parallel if an inverter is used.

For the foregoing reasons, cycloconverter controlled induction motor drives are used in high power drives requiring low speed range, such as rolling mills and mine winders. Like controlled rectifiers, cycloconverters also suffer from a low power factor at low-output voltages.

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PROBLEMS

8.1 A 460 V, 60 Hz, 4 pole, 1760 rpm, Y-connected squirrel-cage induction motor has the following parameters per phase referred to the stator: $R_s = 0.14\ \Omega$, $X_s = 0.4\ \Omega$, $R_r' = 0.08\ \Omega$, $X_r' = 0.8\ \Omega$, $X_m = 15\ \Omega$

The motor is fed by a 6-step inverter, which in turn is fed by a 6-pulse fully controlled rectifier with an ac supply of 460 V, 60 Hz.

The motor is controlled by variable frequency control at a constant flux.

1. Calculate the inverter frequency, motor input current, and rectifier firing angle for 1200 rpm and the rated torque.
2. Calculate the motor speed and input current, and rectifier firing angle for an inverter frequency of 30 Hz and the rated torque.
3. If the minimum inverter frequency is restricted to 6 Hz, calculate the starting torque and motor current as a ratio of their values when the motor is started at the rated voltage and frequency.

Use the equivalent circuit of figure 6.1d and neglect derating due to harmonics.

8.2 The drive of problem 8.1 is now controlled at a constant (V/f) ratio, instead of at a constant flux. Calculate

1. The inverter frequency, rectifier firing angle, motor input current, efficiency and power factor, and rectifier power factor for a speed of 1200 rpm and the rated torque.
2. The motor speed, input current, efficiency and power factor, and rectifier power factor for an inverter frequency of 40 Hz and 80 percent of the rated torque.
3. If the minimum inverter frequency is restricted to 6 Hz, calculate the breakdown torque at 6 Hz as the ratio of the breakdown torque at 60 Hz. Also calculate the starting torque as a percentage of the starting torque obtained with constant flux control.

Neglect friction, windage, core loss, skin effect, and motor derating due to harmonics. Assume converter output current to be perfect dc. Use the equivalent circuit of figure 6.1d.

8.3 A 440 V, 50 Hz, 6 pole, 960 rpm, Y-connected squirrel-cage induction motor has the following parameters for the equivalent circuit of figure 6.1d: $R_s = 0.6\ \Omega$, $X_s = 1{\cdot}\Omega$, $X_m = 50\ \Omega$, $R_r' = 0.3\ \Omega$, $X_r' = 2\ \Omega$

The machine is controlled by a 6-step inverter at a constant flux

1. Calculate and plot slip speed against frequency at rated and half of rated torque.
2. Also calculate and plot slip frequency against torque for the frequencies 60 Hz and 30 Hz.

Neglect friction, windage, and core loss.

8.4 The drive of problem 8.3 is controlled at a constant (V/f) ratio. Calculate the motor speed at half of rated torque and inverter frequency of 40 Hz. Also calculate the rectifier firing angle, input ac current and power factor, and motor input current, efficiency, and power factor at this operating point.

Neglect friction, windage, core loss, skin effect, motor derating due to harmonics, and losses in inverter, filter, and rectifier. Assume the rectifier output current to be ripple free.

8.5 A 440 V, 50 Hz, 1470 rpm, 4 pole, Y-connected squirrel-cage induction motor has the following parameters per phase referred to the stator: $R_s = 0.2\ \Omega$, $X_s = 0.5\ \Omega$, $X_m = 17.5\ \Omega$, $R_r' = 0.11\ \Omega$, $X_r' = 0.8\ \Omega$

The motor is controlled by a 6-step inverter. If it is driving a load requiring rated power for all speeds greater than base speed, calculate for the motor

1. The torque, frequency, stator current, efficiency, and power factor for a speed of 1600 rpm.
2. The speed, torque, stator current, efficiency, and power factor for a frequency of 60 Hz.
3. What is the highest speed at which the motor can drive this load?

Use the equivalent circuit of figure 6.1d. Neglect friction, windage, core loss, skin effect, and derating due to harmonics.

8.6 The drive of problem 8.5 drives a load whose torque varies inversely as speed squared. At the rated speed, the load torque is equal to 1.2 times the rated torque. Calculate

1. The torque and frequency for a speed of 2000 rpm.
2. The speed and torque for a frequency of 80 Hz.

Use the same assumptions as described in problem 8.5.

8.7 A 460 V, 60 Hz, 1180 rpm, 6 pole, Y-connected, squirrel-cage induction motor has the following parameters per phase referred to the stator: $R_s = 0.1\ \Omega$, $X_s = 0.3\ \Omega$, $X_m = 10\ \Omega$, $R_r' = 0.06\ \Omega$, $X_r' = 0.6\ \Omega$

The motor is controlled by a 6-step inverter at a constant flux up to base speed and at rated terminal voltage above base speed. For a constant rotor current of the rated value, plot the developed torque, developed power and slip speed against the frequency ratio "a". Use the equivalent circuit of figure 6.1d.

8.8 For the braking operation of the drive of problem 8.7, calculate

1. The speed for a braking torque of 600 N-m and an inverter frequency of 30 Hz.
2. The frequency for a speed of 900 rpm and a braking torque of 750 N-m.
3. The speed for a braking torque of 300 N-m and an inverter frequency of 75 Hz.
4. The frequency for a speed of 1800 rpm and a braking torque of 400 N-m.

Use the equivalent circuit of figure 6.1d and neglect friction, windage, and core loss.

8.9 A 460 V, 60 Hz, 1185 rpm, 6 pole, Y-connected induction motor has the following values for the parameters of the equivalent circuit of figure 6.1d: $R_s = 0.05\ \Omega$, $X_s = 0.3\ \Omega$, $X_m = 8\ \Omega$, $R_r' = 0.04\ \Omega$, $X_r' = 0.4\ \Omega$

The motor is controlled by a 6-step inverter at a constant (V/f) ratio up to the base speed and at the rated motor voltage above base speed.

1. Calculate the inverter frequency, stator current, regenerated power, and motor power factor and efficiency for a speed of 900 rpm and a braking torque of 1000 N-m.
2. Calculate the speed, stator current, regenerated power, power factor, and efficiency for a frequency of 30 Hz and a braking torque of 1000 N-m.

Neglect friction, windage, core loss, and skin effect.

8.10 The inverter-motor combination of problem 8.9 is fed by a 3-phase 6-pulse dual converter with an ac input voltage of 460 V, 60 Hz. Calculate the inverter frequency, power fed to ac supply and input ac current of the rectifier, when the drive is regenerating at a speed of 1300 rpm and a braking torque of 600 N-m. Neglect losses in the rectifier, filter, and inverter, core loss, friction, windage, and skin effect. Assume ideal filtering — that is, a ripple-less rectifier output current and inverter input voltage.

8.11 In problem 8.10, dynamic braking is used instead of regenerative braking. Calculate the value of the braking resistance if the switch which connects and disconects the braking resistor has a duty ratio of 0.5.

8.12 A 400 V, 50 Hz, 4 pole, 1485 rpm, Y-connected squirrel-cage motor has the following parameters for the equivalent circuit of figure 6.1b: $R_s = 0.03\ \Omega$, $X_s = 0.32\ \Omega$, $X_m = 7\ \Omega$, $R_r' = 0.024\ \Omega$, $X_r' = 0.48\ \Omega$

The motor is fed from a current source inverter, which in turn is fed from a 3-phase fully controlled 6-pulse rectifier with an ac supply of 440 V, 50 Hz. The dc link inductance has a resistance of 0.001 ohm. The flux is maintained constant at the rated value. Calculate

1. The stator current, dc link current, and rectifier firing angle when the motor operates at the rated torque and 50 Hz.
2. The inverter frequency, dc link current, and rectifier power factor for a speed of 1200 rpm and 80 percent of the rated torque.
3. The motor speed, stator current, dc link current, and rectifier firing angle for 90 percent of the rated torque and inverter frequency of 25 Hz.

8.13 The drive of problem 8.12 is controlled at the rated motor terminal voltage for a frequency higher than the rated. For an inverter frequency of 75 Hz and a motor torque of one-fourth the rated value, calculate motor speed, stator current, dc link current, rectifier firing angle, and drive efficiency. Neglect friction, windage, core loss, skin effect, derating due to harmonics, and losses in the inverter and rectifier.

8.14 The motor of problem 8.12 is fed from a current source inverter, which in turn is fed from a class D chopper with a dc line voltage of 600 V. The dc link inductor has a resistance of 0.001 Ω. The motor is operated at a constant flux of the rated value. Calculate the duty ratio of the chopper for the rated motor torque and a speed of 1000 rpm.

8.15 A 460 V, 60 Hz, 4 pole, 1782 rpm, Y-connected squirrel-cage motor has the following parameters for the equivalent circuit of figure 6.1b: $R_s = 0.02\ \Omega$, $X_s = 0.22\ \Omega$, $X_m = 6.5\ \Omega$, $R_r' = 0.016\ \Omega$, $X_r' = 0.32\ \Omega$

The motor is fed from a current source inverter, which in turn is fed from a 3-phase fully controlled 6-pulse rectifier with an ac supply voltage of 480 V and 60 Hz. The resistance of the dc link inductor is 0.0008 Ω. The motor flux is maintained constant at the rated value. For an inverter frequency of 40 Hz, calculate motor speed, dc link current, and rectifier firing angle for the developed braking torque of half the rated value.

8.16 The inverter motor system of problem 8.15 is now fed from a dc source of 750 V through a class D chopper. The dc link inductor now has a resistance of 0.001 Ω. The motor flux is maintained constant at the rated value. Calculate the inverter frequency and the duty ratio of the chopper for the developed braking torque of 80 percent of the rated value and a speed of 1000 rpm. Also calculate regenerated power neglecting core loss and skin effect.

9

Slip Power Controlled Wound-Rotor Induction Motor Drives

The methods of induction motor control described in chapters 7 and 8, control the motor from the stator. Hence they are applicable to both squirrel-cage and wound-rotor motors. However, the squirrel-cage motor is always preferred because of the advantages described in chapter 8.

The present chapter considers the methods which control the motor from the rotor. They are applicable to wound-rotor motors only. Compared to a squirrel-cage motor, the wound-rotor motor has a number of disadvantages, such as higher cost, weight, volume, and inertia, and frequent maintenance due to the presence of brushes and slip rings. However, the control of a wound-rotor motor from the rotor allows cheaper drives to be obtained for a few specific applications.

The portion of the air-gap power which is not converted into mechanical power is called slip power. Slip power control methods regulate the amount of slip power. Hence for a given air-gap power, the power converted into mechanical power is altered. Consequently, the speed for a given torque is changed. The methods involving rotor resistance control and voltage injection in the rotor, described in sections 6.4.3 and 6.4.4, come under this category. This chapter considers the implementation of these methods using power semiconductor converters. The following methods are considered:

1. Static rotor resistance control.
2. Static Scherbius drive.
3. Static Kramer drive.

9.1 STATIC ROTOR RESISTANCE CONTROL

The rotor resistance control of a wound-rotor induction motor is described in section 6.4.3. It is inefficient because the speed reduction is obtained by wasting slip power in external resistors. It has, however, advantages of low cost, a good power factor, and a high torque-to-current ratio for a wide range of speed, including starting and braking. The variable frequency control is the only other method of induction motor speed control which gives a high torque-to-current ratio. However, it is very expensive. Hence, rotor resistance control finds application in drives requiring low cost and a high torque-to-current ratio, such as low-power excavators, crane hoists, and so on.

Instead of mechanically varying the resistance, the rotor circuit resistance can be varied statically by using the principle of a chopper. This gives stepless and smooth variation of resistance and consequently of motor speed. As shown in figure 9.1, the slip frequency ac rotor voltages are converted into dc by a 3-phase diode bridge and applied across an external resistance R. The self-commutated semiconductor switch S, connected in parallel with R, is operated periodically with a period T and remains on for an interval t_{on} in each period. The effective value of resistance R changes from R to 0 as t_{on} changes from 0 to T. The filter inductor L_d is provided to minimize the ripple in current I_d. A high ripple in I_d produces high harmonic content in the rotor, increasing copper losses and causing derating of the motor. The filter inductor also helps in eliminating discontinuous conduction at light loads. As in the case of a dc motor, discontinuous conduction makes the speed regulation poor. The main contributor to the ripple is the diode bridge and not the semiconductor switch, because it operates at a sufficiently high frequency.

The diode bridge output voltage V_d changes from its maximum value at standstill to nearly 5 percent of the maximum value at near rated motor speed. If switch S is realized using a thyristor, reliable commutation can only be obtained either by using a bulky commutation capacitor or an auxiliary source for charging the commutation capacitor. Hence, a thyristor is not suitable for this application. Because induction motors are usually designed with a stator-to-rotor turns ratio greater than 1, the volt-

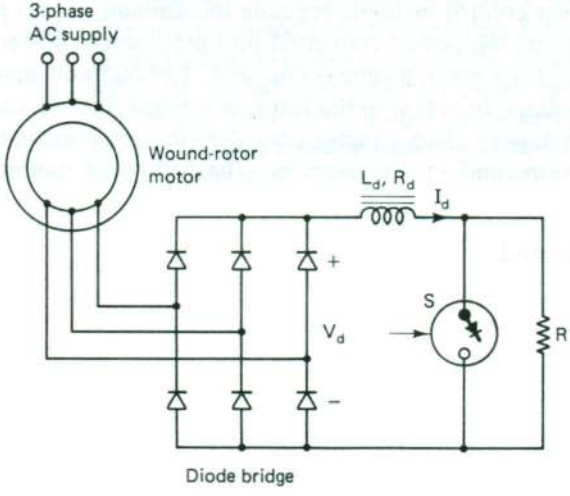

Figure 9.1 Static rotor resistance control of wound-rotor induction motor.

age V_d is small. Hence, a transistor is suitable for low-power drives. A GTO may be employed for ratings beyond the capability of transistors. The self-commutation capability of these devices ensures reliable commutation at all operating points and makes the semiconductor switch compact.

An alternative static rotor resistance control circuit is obtained by using either a 6-pulse or 3-pulse controlled rectifier instead of the diode bridge and semiconductor switch S. The power consumed by R is then controlled by controlling the rectifier firing angle. As the firing angle is increased from 0 to the maximum, the effective rotor resistance increases from R to a maximum value, controlling the speed.

9.1.1 Analysis and Performance

The following assumptions are made:

1. Commutation overlap in the diode bridge due to the motor leakage inductances is ignored. The effect of the commutation overlap is to increase the phase lag in the rotor-induced voltage and the fundamental rotor current.
2. The filter inductor current I_d is assumed to be ripple-free dc.
3. Under assumptions 1 and 2, the rotor phase current will have a six-step waveform shown in figure 9.2. The waveform of the corresponding input phase voltage of the diode bridge is also shown. The fundamental rotor current is in phase with the phase voltage. The rotor phase current waveform is similar to the phase current waveform of a current source inverter (fig. 8.14). It can be described by the Fourier series of equation (8.22). As far as harmonics are concerned, the motor can be considered fed from the rotor by a current source. The harmonics in rotor current cause only a small harmonic current to flow in the stator.[8] Therefore, the machine-induced emf and hence flux can be assumed sinusoidal. When the flux is sinusoidal, the torque is produced only by the fundamental. The harmonics produce only pulsating torques.
4. The losses in the diode bridge and the semiconductor switch are neglected.

Equivalent Circuit

The duty ratio of the switch δ is defined in the same way as for a chopper. Thus,

$$\delta = \frac{t_{on}}{T} \tag{9.1}$$

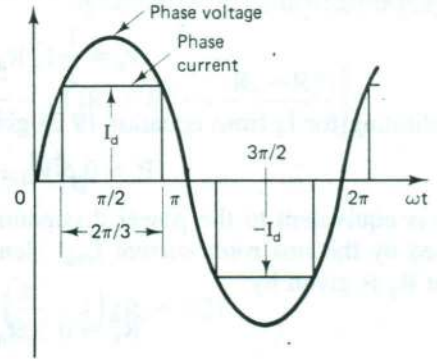

Figure 9.2 Rotor phase current and phase voltage waveforms.

When the ripple in current I_d is neglected, the energy absorbed by resistance R during a period of operation of the switch (T) is given by

$$E_R = I_d^2 R(T - t_{on})$$

The average power absorbed by resistance R during a period T is

$$P_r = \frac{1}{T}[I_d^2 R(T - t_{on})]$$

Substituting from equation (9.1) gives

$$P_r = I_d^2 R(1 - \delta)$$

Hence, the effective value of resistance R

$$R^* = (1 - \delta)R \tag{9.2}$$

From figure 9.2, the rms value of the rotor phase current is

$$I_{rms} = \left[\frac{1}{\pi}\int_0^{2\pi/3} I_d^2 \, d(\omega t)\right]^{1/2} = \frac{\sqrt{2}}{\sqrt{3}} I_d \tag{9.3}$$

From the Fourier series expressions (3.109) to (3.111), because of the quarter-wave symmetry of the rotor phase current i_r, $b_1 = 0$ and

$$a_1 = \frac{4}{\pi}\int_{\pi/6}^{\pi/2} I_d \sin \omega t \, d(\omega t) = \frac{2\sqrt{3}}{\pi} I_d$$

Hence the fundamental rotor current is

$$I_r = a_1/\sqrt{2} = \frac{\sqrt{6}}{\pi} I_d \tag{9.4}$$

From equations (9.3) and (9.4),

$$I_r = \frac{3}{\pi} I_{rms} \tag{9.5}$$

The total resistance across the diode bridge

$$R_e = R_d + R^* = R_d + (1 - \delta)R$$

The per-phase power consumed by resistance R_e

$$P_e = \frac{1}{3} I_d^2 [R_d + (1 - \delta)R]$$

Substituting for I_d from equation (9.3) gives

$$P_e = 0.5[R_d + (1 - \delta)R] I_{rms}^2$$

This is equivalent to the power dissipation in a resistance of $0.5[R_d + (1 - \delta)R]\ \Omega$ caused by the rms rotor current I_{rms}. Hence, the effective per-phase value of resistance R_e is given by

$$R_e^* = 0.5[R_d + (1 - \delta)R] \tag{9.6}$$

An equivalent circuit of a machine is valid for a given frequency. In this analysis a fundamental frequency equivalent circuit of the drive is derived, which, while retaining the relationship between fundamental voltages and currents, also allows calculation of developed torque and total copper loss, including the copper loss caused by harmonic currents.

In a fundamental equivalent circuit, the power transferred across the air-gap (P_g) is given by

$$P_g = 3EI_r' \cos \Theta_r \tag{9.7}$$

where Θ_r is the phase angle between phasors $\bar{E}$ and $\bar{I}_r'$.

In the drive under consideration, the total power consumed in the rotor circuit (P_g') is

$$P_g' = 3I_{rms}^2(R_r + R_e^*) + P_m \tag{9.8}$$

Substituting from equation (9.5) gives

$$P_g' = \frac{\pi^2}{3} I_r^2(R_r + R_e^*) + P_m \tag{9.9}$$

The fundamental equivalent circuit of the drive must satisfy the condition, $P_g = P_g'$. Hence, from equations (9.7) and (9.9)

$$EI_r' \cos \Theta_r = \frac{\pi^2}{9} I_r^2(R_r + R_e^*) + \frac{P_m}{3} \tag{9.10}$$

In the drive under consideration, the slip power due to the fundamental rotor current

$$sP_{g1} = 3I_r^2(R_r + R_e^*) \tag{9.11}$$

where P_{g1} is the fundamental air-gap power in the drive. The mechanical power developed by the fundamental rotor current is given by

$$P_m = (1 - s)P_{g1}$$

Substituting from equation (9.11) gives

$$P_m = 3I_r^2(R_r + R_e^*)\frac{(1 - s)}{s} \tag{9.12}$$

Substituting from equation (9.12) into equation (9.10) and rearranging the terms gives

$$\begin{aligned} EI_r' \cos \Theta_r &= \left[\left(\frac{\pi^2}{9} - 1\right)(R_r + R_e^*) + \frac{(R_r + R_e^*)}{s}\right] I_r^2 \\ &= \left(R_h + \frac{R_f}{s}\right) I_r^2 \end{aligned} \tag{9.13}$$

where

$$R_h = \left(\frac{\pi^2}{9} - 1\right)(R_r + R_e^*) \tag{9.14}$$

$$R_f = (R_r + R_e^*) \tag{9.15}$$

Referring all parameters on the right side of equation (9.13) also to the stator side gives

$$EI_r' \cos \Theta_r = \left(R_h' + \frac{R_f'}{s}\right) I_r'^2 \tag{9.16}$$

where R_h' and R_f' are respectively the values of R_h and R_f referred to the stator side. Thus,

$$R_h' = a_{T1}^2 R_h \quad \text{and} \quad R_f' = a_{T1}^2 R_f \tag{9.17}$$

where a_{T1} is the stator to rotor turns ratio.

The per-phase fundamental equivalent circuit of the drive referred to the stator, as obtained from equation (9.16), is shown in figure 9.3a. Resistance (R_f'/s) accounts for the developed mechanical power and the fundamental rotor copper loss. Resistance R_h' accounts for the rotor harmonic copper loss. The equivalent circuit of figure 9.3a can be simplified to that of figure 9.3b.

Performance

From the equivalent circuit of figure 9.3b,

$$\bar{I}_r' = \frac{\bar{V}}{(R_s + R_h' + R_f'/s) + j(X_s + X_r')} \tag{9.18}$$

$$T = \frac{3}{\omega_{ms}} I_r'^2 (R_f'/s) \text{ N-m} \tag{9.19}$$

Substituting from equation (9.18) gives

$$T = \frac{3}{\omega_{ms}} \left[\frac{V^2 (R_f'/s)}{(R_s + R_h' + R_f'/s)^2 + (X_s + X_r')^2} \right] \tag{9.20}$$

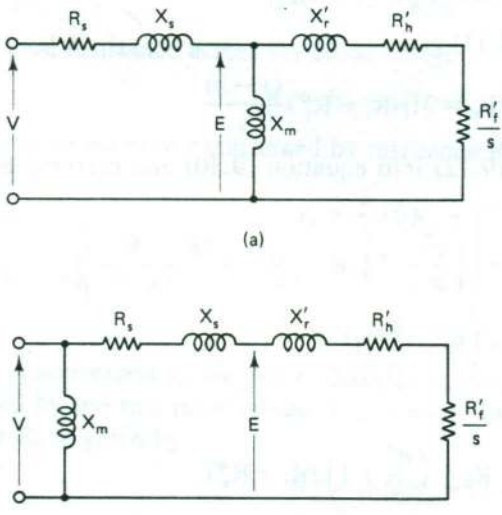

Figure 9.3 Equivalent circuits of wound-rotor motor with static rotor resistance control.

where R_h' and R_f' are given by equations (9.14), (9.15), and (9.17).

For given values of δ and s, the rotor current and torque can be calculated from equations (9.18) and (9.20).

In this analysis, the energy loss in switch S and the diodes have been ignored. This loss is negligible compared to the total rotor loss for low values of δ. For δ close to unity, this loss forms a significant portion of the total rotor loss. Thus, appreciable error may be caused in the calculation of speed-torque curves for values of δ close to unity.

The nature of speed-torque curves for different values of δ is shown in figure 9.4. For $\delta = 1$, R is fully bypassed by the semiconductor switch S. However, due to the additional losses in the switch, resistance R_d, resistance R_h', and diodes, the speed-torque curve for $\delta = 1$ lies below the natural speed-torque curve. For a given torque, speed reduces with δ. The control region consists of the area enclosed in ABCD. Any operating point in this region can be obtained by controlling δ. The operation is not possible in the area AD0. The control region is increased and the area AD0 is decreased when R is increased.

Since only the fundamental rotor current is assumed to contribute to the torque, the same value of the fundamental rotor current is necessary to produce a given torque, whether the current is sinusoidal or nonsinusoidal. For the rated thermal loading, the rms current is fixed, irrespective of whether it is sinusoidal or nonsinusoidal, when the increase in machine resistance due to skin effect is neglected. Therefore, the maximum fundamental current rating of the machine will decrease by a factor (I_r/I_{rms}). The motor power rating will also decrease by this factor. When used in this drive, the motor power rating will reduce to (I_r/I_{rms}) times its rating. Thus,

$$\text{motor derating} = \left(\frac{I_r}{I_{rms}}\right) = \frac{3}{\pi} = 0.95$$

If the ripple in the I_d, commutation overlap in the diode bridge, skin effect, and the reduction in full-load speed due to losses in diode bridge, inverter, transformer, and semiconductor switch are considered, the derating of the motor will be much higher.

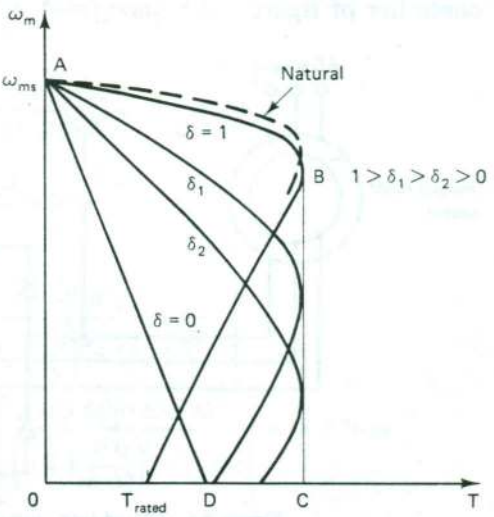

Figure 9.4 Speed-torque curves for static rotor resistance control.

The harmonic torques, both steady and pulsating, have small values, and they can be ignored in a well-designed motor.

Compared to conventional rotor resistance control methods employing contactors, slip regulators, and so on, the static rotor resistance control has the disadvantage of requiring motor derating. But it has many more advantages, such as smooth and stepless control, fast response, less maintenance, longer life, compact size, assured balance between rotor phase currents, simple closed-loop control, and so on.

Because of the fast and simple closed-loop operation, the drive can provide fast transient response during starting and braking. Fast acceleration during starting is obtained by operating the motor at breakdown torque for all speeds. It was shown in section 6.1.1 that the rotor current has a fixed value at the breakdown torque for all speeds. The closed-loop current control scheme of figure 9.5, with reference current set for the breakdown current, is employed to get fast acceleration. Similarly, fast deceleration under plugging is obtained by operating the motor again at breakdown torque. A phase sequence reversal arrangement in the stator will be required to switch over from motoring to plugging. This will also allow motor reversal. The closed-loop control of figure 9.5 can be used for fast deceleration in plugging and also during the entire reversing operation, provided R is chosen large enough to get the breakdown torque at the highest speed in plugging. In the case of dc dynamic braking also, for a given dc current through the stator, the rotor current at breakdown torque has a constant value for all speeds, as explained in section 6.3.3. Therefore, the closed-loop scheme of figure 9.5 can also be used to get fast dc dynamic braking.

In some crane applications, to get good speed regulation and smooth operation, the drive is operated with closed-loop speed control with inner-current control (similar to the scheme of fig. 5.1b or fig. 7.5). This allows speed control with good speed regulation as long as the operating point is located in the control region ABCD of figure 9.4. For lower torques, the operation will take place on line AD and closed-loop control will become ineffective. To extend the constant speed operation to region AD0 and to get four-quadrant control with plugging, the three-phase voltage controller of figure 7.3 is incorporated in the stator circuit. In the region ABCD

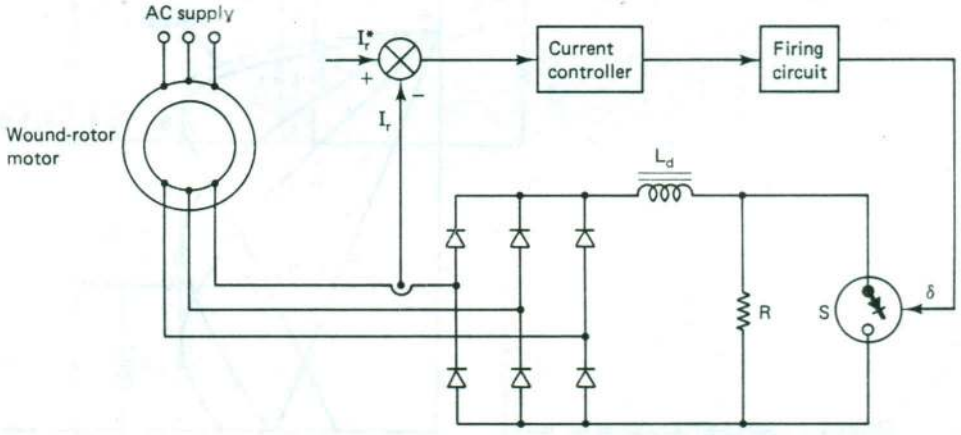

Figure 9.5 Closed-loop current control for starting and braking.

(fig. 9.4), the stator voltage is maintained constant at the rated value and the closed-loop speed control is obtained by controlling δ. In the region AD0, δ becomes 0 and closed-loop speed control is obtained by varying the firing angle α of the voltage controller. The transient operation for a change in speed command is carried out at the rated terminal voltage by the rotor resistance control. This ensures a high torque-to-current ratio, and, hence, fast transient operation. The speed reversal is carried out as follows.

When speed command is reversed, the firing pulses are withdrawn to force the current through the conducting thyristors to zero. After the thyristors cease to conduct, a delay of 5 to 10 ms is provided for the thyristors to regain their forward voltage blocking capability. Now the firing pulses are released to another set of thyristors to cause reversal of the phase sequence. The firing angle is set to provide rated stator voltage and δ is controlled to regulate the current. The motor is braked and then accelerated in the reverse direction at the maximum allowable current and torque by the rotor resistance control. The drive is finally brought to the desired speed by first adjusting δ. However, if the operating point lies in the region AD0 of figure 9.4, δ would reach its lowest limit (that is, 0, on line AD). Now the stator voltage will be reduced by increasing α to get the required speed in the region AD0.

Example 9.1

A 3-phase, 460 V, 60 Hz, 1164 rpm, Y-connected, wound-rotor induction motor has the following parameters: $R_s = 0.4\ \Omega$, $R_r' = 0.6\ \Omega$, $X_s = X_r = 1.8\ \Omega$, $X_m = 40\ \Omega$, stator to rotor turns ratio is 2.5. The motor speed is controlled by static rotor resistance control. Filter resistance is $0.02\ \Omega$ and the external resistance is chosen such that at $\delta = 0$, the breakdown torque is obtained at standstill.

1. Calculate the value of the external resistance.
2. Calculate δ for a speed of 960 rpm at 1.5 times the rated torque.
3. Calculate the speed for $\delta = 0.6$ and 1.5 times the rated torque.

Neglect friction and windage.

Solution:

$$N_s = \frac{120f}{p} = \frac{120 \times 60}{6} = 1200 \text{ rpm}$$

$$V = 460/\sqrt{3} = 265.6 \text{ V}$$

$$\omega_{ms} = \frac{1200 \times 2\pi}{60} = 125.66 \text{ rad/sec.}$$

$$\text{Full-load slip} = \frac{1200 - 1164}{1200} = 0.03$$

Without rotor resistance control,

$$T = \frac{3}{\omega_{ms}}\left[\frac{V^2(R_r'/s)}{\left(R_s + \frac{R_r'}{s}\right)^2 + (X_s + X_r')^2}\right]$$

$$\text{Full-load torque} = \frac{3}{125.66}\left[\frac{(265.6)^2(0.6/0.03)}{\left(0.4 + \frac{0.6}{0.03}\right)^2 + (3.6)^2}\right] = 78.5 \text{ N-m}$$

1. From figure 9.3b, at the breakdown torque

$$\frac{R_f'}{s} = [(R_s + R_h')^2 + (X_s + X_r')^2]^{1/2}$$

When the breakdown torque occurs at standstill,

$$(R_f')^2 = (R_s + R_h')^2 + (X_s + X_r')^2$$

or

$$(R_f')^2 = R_h'^2 + 2R_h'R_s + R_s^2 + (X_s + X_r')^2 \tag{E9.1}$$

From equations (9.14) and (9.15),

$$R_h' = \left(\frac{\pi^2}{9} - 1\right)R_f' = 0.0966R_f' \tag{E9.2}$$

Substituting from (E9.2) and known values in (E9.1) gives

$$R_f'^2 - 0.078R_f' - 13.25 = 0$$

which gives $R_f' = 3.68\ \Omega$; hence, $R_h' = 0.355\ \Omega$
From equation (9.15)

$$R_e'^* = R_f' - R_r' = 3.68 - 0.6 = 3.08\ \Omega$$

$$R_e^* = 3.08/a_{T1}^2 = 0.49\ \Omega$$

From equation (9.6) for $\delta = 0$,

$$R = 2R_e^* - R_d = 2 \times 0.49 - 0.02 = 0.96\ \Omega$$

2. With rotor resistance control, from equation (9.20),

$$T = \frac{3}{\omega_{ms}}\left[\frac{V^2(R_f'/s)}{(R_s + R_h' + R_f'/s)^2 + (X_s + X_r')^2}\right] \tag{9.20}$$

From (E9.2), $R_h' = 0.0966R_f'$

$$s = \frac{1200 - 960}{1200} = 0.2$$

Rated torque as just calculated = 78.5 N-m. Substituting known values in equation (9.20) gives

$$1.5 \times 78.5 = \frac{3}{125.66}\left[\frac{(265.6)^2(R_f'/0.2)}{(0.4 + 0.0966R_f' + R_f'/0.2)^2 + (3.6)^2}\right]$$

or $R_f'^2 - 2.595R_f + 0.5 = 0$ or $R_f' = 2.39\ \Omega$ or $0.2\ \Omega$

The latter value is not feasible because it is less than R_r'
Hence, $R_f' = 2.39\ \Omega$
From equation (9.15),

$$R_e^* = (R_f' - R_r')/a_{T1}^2 = \frac{(2.39 - 0.6)}{(2.5)^2} = 0.29\ \Omega$$

From equation (9.6),

$$(1-\delta)=\frac{2R_e^*-R_d}{R}=\frac{2\times 0.29-0.02}{0.96}=0.58 \quad \text{or} \quad \delta=0.42$$

3. From equations (9.6),

$$R_e^*=0.5[0.02+(1-0.6)\times 0.96]=0.202\ \Omega$$
$$R_f'=R_r'+a_{T1}^2R_e^*=0.6+6.25\times 0.202=1.86\ \Omega$$
$$R_h'=0.0966\times 1.86=0.18\ \Omega$$

Substituting the known values in equation (9.20) gives

$$1.5\times 78.5=\frac{3}{125.66}\left[\frac{(265.6)^2(1.86/s)}{\left(0.4+0.18+\frac{1.86}{s}\right)^2+(3.6)^2}\right]$$

which gives

$$\left(\frac{1}{s}\right)^2-7.06\left(\frac{1}{s}\right)+3.84=0$$

$$\text{giving } \frac{1}{s}=6.47 \quad \text{or} \quad 0.595 \quad \text{or} \quad s=0.15 \quad \text{or} \quad 1.68$$

Discarding the latter value which is not feasible,

$$N=N_s(1-s)=1200(1-0.15)=1020 \text{ rpm}$$

9.2 STATIC SCHERBIUS DRIVES

Instead of wasting the slip power in the rotor circuit resistance, it can be fed back to the ac mains using the approach suggested by Scherbius.[7] A static scheme based on this approach is shown in figure 9.6. It is known as the static Scherbius drive. It is also known as the slip power recovery scheme or subsynchronous converter cascade because it is capable of providing speed control only in the subsynchronous speed range.

A diode bridge converts a portion of slip power into dc which in turn is converted into line frequency ac by a 3-phase line commutated inverter and fed back to the ac mains through a transformer. The filter inductor L_d is provided to eliminate discontinuous conduction and to minimize the ripple in the dc link current I_d to keep the harmonic copper losses, and the consequent derating of the motor, low.

Drive operation and analysis is considered subjected to the assumptions 1 to 4 described in section 9.1.1 for the static rotor resistance control. In addition to these four assumptions, the following assumptions are also made:

5. The transformer is assumed ideal — that is, having no leakage, no loss, and the capability to exactly transform a six-step current wave from the inverter to the ac mains.

6. From the foregoing assumption it automatically follows that the commutation overlap in the inverter will be neglected. In any case it is much smaller com-

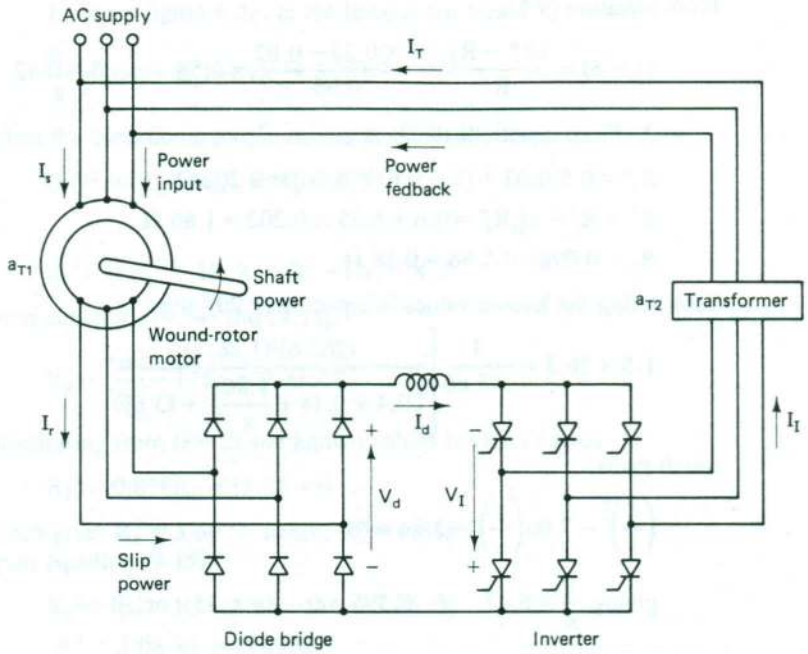

Figure 9.6 Static Scherbius drive.

pared to the commutation overlap in the diode bridge which has also been neglected (assumption 1).

Initially, approximate relations are derived to get an idea of the control range of the drive and to simplify the phasor diagram which is considered in the next section.

If the stator and rotor leakage impedance drops are also neglected, in addition to assumptions 1 to 6, the output voltage of the diode bridge is obtained from equation (3.78) by substituting $\alpha = 0$. Thus,

$$V_d = \frac{3\sqrt{6}}{\pi} \frac{sV}{a_{T1}} \tag{9.21}$$

where V is the stator phase voltage and a_{T1} is the stator to rotor turns ratio.

Again from equation (3.78), the dc output voltage of the line commutated inverter is given by

$$V_I = \frac{3\sqrt{6}}{\pi} \frac{V}{a_{T2}} \cos \alpha \tag{9.22}$$

where a_{T2} is the transformer line side-to-inverter ac side turns ratio and α is the inverter firing angle.

If the resistance of inductor L_d is ignored, then

$$V_d + V_I = 0$$

Substituting from equations (9.21) and (9.22) and rearranging terms gives

$$s = -\frac{a_{T1}}{a_{T2}} \cos \alpha = -a_T \cos \alpha \tag{9.23}$$

where $a_T = a_{T1}/a_{T2}$.

For inverter operation $\alpha \geq 90°$; therefore s is always positive. By varying α from 90° to 180°, s can be varied from 0 to a_T. If a_{T2} is chosen equal to a_{T1}, then the slip will vary from 0 to 1 and the motor speed from synchronous to standstill. Thus the motor speed can be controlled in the subsynchronous region simply by controlling the inverter firing angle.

From assumptions 1 and 2, the rotor current will have a six-step waveform as shown in figure 9.2. Its fundamental component will be in phase with the slip frequency input voltage of the diode bridge, because a diode bridge always operates with $\alpha = 0$. According to equations (9.21) and (9.22), for a given inverter firing angle, V_I, V_d, and therefore the input voltages of the diode bridge, have fixed values. As the inverter firing angle is changed, the diode bridge input voltages are made to change to maintain a balance between V_d and V_I, consequently changing the motor speed. Thus, this method in fact is the speed control of the induction motor by injecting a slip frequency voltage in the rotor circuit, as described in section 6.4.4. Consequently, the power flow diagram of figure 6.17 and equations (6.59) to (6.68) are also applicable to this drive. However, it can only provide the subsynchronous motoring and supersynchronous braking operations. For the subsynchronous braking operation to be obtained, the slip power must flow from the dc link to the rotor, as shown in table 6.2 and explained in section 6.4.4. This is not possible due to the diode bridge, which can allow power flow only from the rotor to the dc link.

Approximate speed-torque relations can be derived as follows. If the rotor copper loss is neglected, then from figure 6.17, the fundamental rotor slip power sP_g is approximately equal to the dc link power. Thus,

$$sP_g = V_d I_d \tag{9.24}$$

or

$$P_g = \frac{V_d I_d}{s}$$

Now

$$T = \frac{P_g}{\omega_{ms}} = \frac{V_d I_d}{s\omega_{ms}}$$

Substituting from equation (9.21) yields

$$T = \frac{3\sqrt{6}}{\pi} \frac{V I_d}{a_{T1}\omega_{ms}} \tag{9.25}$$

which suggests that the torque is approximately proportional to the dc link current I_d. Since the fundamental rotor current is proportional to I_d [equation (9.4)], the torque is directly proportional to the fundamental rotor current.

9.2.1 Power Factor Considerations

The approximate phasor diagram of the drive for operation at the rated torque is shown in figure 9.7. All the phasors are referred to the stator or line side. The addition of the magnetizing current $\bar{I}_m$ and the stator-referred fundamental rotor current $\bar{I}_r'$ gives the stator current $\bar{I}_s$, which lags behind the stator phase voltage $\bar{V}$ by an angle ϕ_s. As just explained, for a given torque, the fundamental rotor current is constant. Hence, for the rated torque operation, $\bar{I}_s$ has a constant magnitude and phase angle. The addition of $\bar{I}_s$ and the transformer line side fundamental current $\bar{I}_T$, gives the fundamental drive input current $\bar{I}_L$. If the dc link current is assumed ripple free, the fundamental inverter ac current $\bar{I}_I$ will lag behind its ac phase voltage by an angle α. Hence, $\bar{I}_T$ also lags behind $\bar{V}$ by an angle α. The phasor diagram has been drawn for two values of "a_T" (that is, $a_T = 1.04$ and $a_T = 0.52$, where $a_T = a_{T1}/a_{T2}$).

First consider the operation for $a_T = 1.04$. For reliable commutation of thyristors, the inverter firing angle is kept less than 180°. It is assumed that reliable commutation is assured if the maximum value of α is restricted to 165°. Now if speed control is required from synchronous speed to stand-still, then from equation (9.23),

$$1 = -a_T \cos 165° \quad \text{or} \quad a_T = 1.04$$

This shows that "a_T" has been chosen such that zero motor speed is obtained when $\alpha = 165°$. Thus for $s = 1$, phasor $\bar{I}_T$ lags behind $\bar{V}$ by 165°. If the machine losses are neglected, then zero speed is obtained when the power fed back equals the motor power input, which gives

$$I_T \cos 165° = I_s \cos \phi_s$$

This condition is satisfied when OA forms the phasor $\bar{I}_T$. For a ripple-free I_d, the rotor and inverter ac currents have identical amplitudes and waveforms. Therefore, I_I

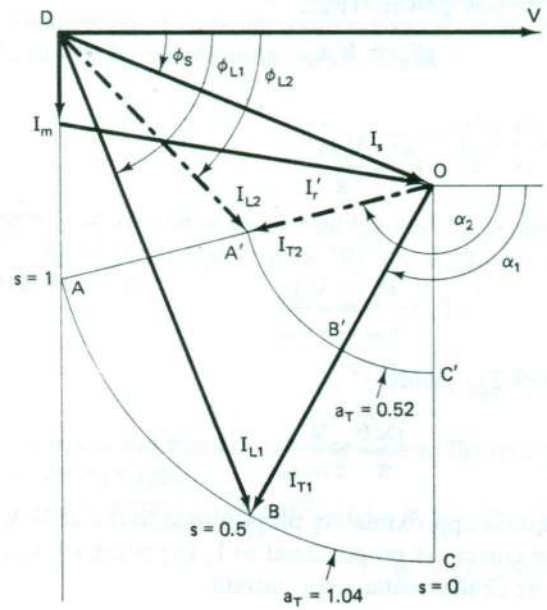

Figure 9.7 Phasor diagram of static Scherbius drive at rated torque.

must be equal to I_r and hence $I_T = a_T I_r'$. Since for a given torque I_r' is constant, I_T must also be constant. This suggests that as the slip is controlled from 1 to 0, the tip of the phasor $\bar{I}_T$ moves along the locus ABC, which is an arc of the circle with the center at O and the radius equal to OA. At point B, the slip is 0.5 because the power returned to the line is 50 percent of the motor power input. Similarly, the slip is ideally zero at point C because no power is returned to the line. The line current $\bar{I}_L$ is obtained by adding phasors $\bar{I}_s$ and $\bar{I}_T$. The tip of phasor $\bar{I}_L$ will also move along the arc ABC as slip is changed from 1 to 0, with the other end fixed at D. For $s = 0.5$, phasor $\bar{I}_T$ is shown as $\bar{I}_{T1}$. The sum of phasors $\bar{I}_S$ and $\bar{I}_{T1}$ gives line current $\bar{I}_{L1}$ which lags behind the line phase voltage $\bar{V}$ by an angle ϕ_{L1}. Note that the fundamental drive power factor cos ϕ_{L1} is much smaller compared to the motor power factor cos ϕ_s. Similarly, if line current phasors are examined for $s = 0$ (DC) and $s = 1$ (DA), one observes that the drive power factor, which is already very low at $s = 0$, decreases further with an increase in slip and becomes zero at $s = 1$. The drive power factor is poor at $s = 0$, because, while the real power drawn by the drive from the line remains the same as that taken by the motor, the inverter draws an additional reactive power $3VI_T$ from the line. The drive power factor decreases with an increase in slip, because, while the real power drawn from the line is the difference between the real powers drawn by the motor and inverter, the reactive power is the sum of the reactive powers drawn by the motor and the inverter. The poor power factor is a major drawback of this drive.

Next consider an application where speed control is required from synchronous to half of synchronous speed. From equation (9.23), the value of "a_T" which gives the lowest speed ($s = 0.5$ here) at the highest permissible value of firing angle (that is, 165°) is given by

$$0.5 = -a_T \cos 165° \quad \text{or} \quad a_T = 0.52.$$

Any value of "a_T" greater than 0.52 will also allow the speed control from synchronous to half-synchronous speed. For example, with $a_T = 1.04$, the speed can be varied from synchronous to half-synchronous speed when α is controlled from 90° to 118.7°. Let us examine the power factor for these two alternatives (that is, $a_T = 0.52$ and $a_T = 1.04$).

Figure 9.7 also shows the phasor diagram for $a_T = 0.52$ and the rated torque operation. Since the torque is the same, the stator current is still represented by the phasor $\bar{I}_s$. As just explained, $I_T = a_T I_r'$; hence for $a_T = 0.52$, the magnitude of the $\bar{I}_T$ phasor will be half of that for $a_T = 1.04$. Thus, as the slip is changed from 0 5 to 0, the tip of $\bar{I}_T$ phasor will move along the locus A′B′C′, which is an arc of a circle with the center at O and the radius equal to $(a_T I_r')$. The line current $\bar{I}_L$ which is the sum of $\bar{I}_s$ and $\bar{I}_T$ will have its tip moving along the locus A′B′C′ when s is changed from 0.5 to 0. For $s = 0.5$, the line current for $a_T = 0.52$ is shown as I_{L2}, and the line current for $a_T = 1.04$ is shown as $\bar{I}_{L1}$. The comparison of the phases of these line current phasors shows that the drive will have a higher power factor for $a_T = 0.52$. Comparison of phase angles for any value of s between 0 and 0.5 will show that the power factor is higher for $a_T = 0.52$. A comparison of the power factor for $a_T = 0.52$ with that for any other value of $a_T > 0.52$ will show that for any slip between 0 and 0.5, the power factor will have the highest value when $a_T = 0.52$. It may be noted

that the choice of $a_T = 0.52$ was done to achieve the drive operation at the maximum permissible firing angle at the lowest speed.

From the preceding discussion, the following important conclusions can be drawn:

1. The drive power factor is maximized when "a_T" is chosen to obtain the drive operation at the maximum permissible firing angle at the lowest speed.
2. The narrower the speed range, the greater the power factor.

The optimum value of "a_T" which satisfies condition 1 can be obtained by the choice of transformer turns ratio. Let us consider a numerical example to fully emphasize the need for a transformer to improve the power factor of the drive. Consider a motor with $a_{T1} = 4$, V = 230 V, and $\bar{I}_s = 40\underline{/-30°}$. The desired control range is from synchronous to 80 percent of synchronous speed. For simplification it is assumed that the highest value of firing angle is 180°.

From equation (9.23), the optimum value of $a_T = 0.2$. Hence,

$$a_{T2} = a_{T1}/a_T = 4/0.2 = 20.$$

Now $I_T = a_T I_r'$
Assuming $I_r' \simeq I_s$

$$I_T = 0.2 \times 40 = 8 \text{ A}$$

Hence, $$\bar{I}_T = 8\underline{/-180°} \text{ A}$$

$$\bar{I}_L = \bar{I}_s + \bar{I}_T = 40\underline{/-30°} + 8\underline{/-180°} = 33.3\underline{/-36.9°} \text{ A}$$

Power factor = cos 36.9° = 0.8 .

Let us examine the power factor when the transformer is not employed and the inverter is directly connected to ac mains. This amounts to setting $a_{T2} = 1$, which gives $a_T = 4$. From equation (9.23) at s = 0.2 and $a_T = 4$, $\alpha = 92.9°$.

$$\text{Now, } I_T = a_T I_r' = a_T I_s = 4 \times 40 = 160 \text{ A}$$

$$\bar{I}_T = 160\underline{/-92.9°} \text{ A}$$

$$\bar{I}_L = 40\underline{/-30°} + 160\underline{/-92.9°} = 181.75\underline{/-81.6°} \text{ A}$$

Power factor = cos 81.6° = 0.146

Note that when the transformer is not used, the power factor at the lowest speed drops to 0.146 from 0.8 and the line current increases from 33.3 A to 181.75 A.

Because a_{T1} is usually greater than 1, a transformer is necessary to achieve the optimum condition 1 just stated. If a transformer is not used, the power factor will deteriorate. The narrower the speed range, the greater the deterioration in the power factor. In other words, when the speed range is narrow, the use of a transformer substantially improves the power factor over its value without a transformer.

The drive power factor can be further improved if the lagging reactive power drawn by the converter from the line can be reduced. A simple alternative is to operate the inverter with controlled flywheeling as described in section 3.5.2.[10–12] In a

3-phase inverter, the controlled flywheeling can be used for the firing angles between 90° and 120°. Thus, the power factor is improved for the speed range covered by this range of firing angle. For $\alpha \geq 120°$, controlled flywheeling cannot be used; hence the drive power factor remains unaltered. A comparison of the power factor with and without controlled flywheeling is shown in figure 9.8. With controlled flywheeling, when the slip is changed from 1 to 0, the tip of the line current phasor $\bar{I}_L$ moves along the path ABO instead of ABC in the absence of the controlled flywheeling.

Greater improvement in the power factor is obtained by operating the inverter with pulse-width modulation. For this, thyristors in the inverter are replaced by self-commutated switches (section 3.8). With pulse-width modulation, the inverter can be operated with zero reactive power. In this case the tip of the line current vector will move along the locus OEF, and thus the drive power factor will be substantially improved. When pulse-width modulation is employed, the inverter can also be operated with a leading reactive power, thus causing further improvement in the drive power factor.

9.2.2 Rating and Applications

Let us assume that the inverter operates at the maximum firing angle at the lowest speed. This maximizes the drive power factor, as explained in the previous section. To simplify the calculations, the largest firing angle is taken to be 180° and the slip at the lowest speed is denoted by s_{max}. Then from equation (9.23),

$$a_T = s_{max} \tag{9.26}$$

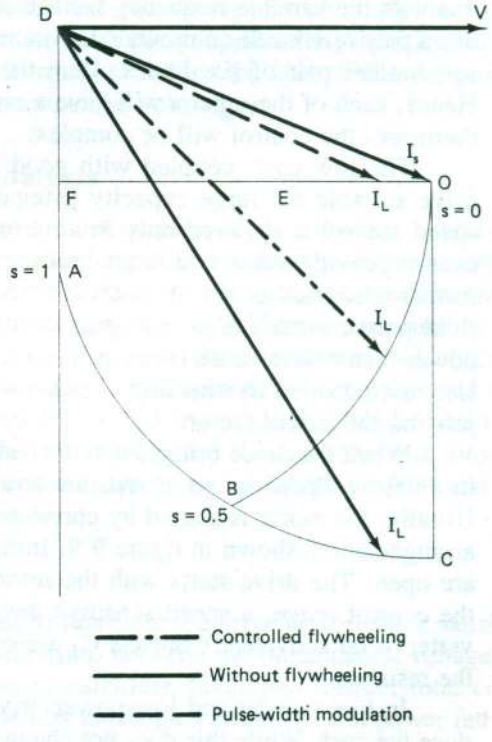

Figure 9.8 Comparison of power factors.

For a given torque, I_r is fixed. Since $I_I = I_r$, the ac current carried by the rotor, diode bridge, inverter, and transformer secondary is the same. The ac voltage across the diode bridge is maximum for the slip s_{max}. It is given by $(s_{max}V/a_{T1})$. The ac side inverter voltage is

$$\frac{V}{a_{T2}} = \frac{V}{a_{T1}} \cdot \frac{a_{T1}}{a_{T2}} = \frac{a_T V}{a_{T1}}$$

Since $a_T = s_{max}$, from equation (9.26), the ac voltage rating of the diode bridge, inverter, and transformer secondary is the same. As the current rating has to be the same, the kVA rating of these three is equal. Since the voltage rating of the diode bridge is s_{max} times that of the rotor, the kVA rating of each of these is s_{max} times that of the motor.

The foregoing conclusion about the kVA rating can also be derived from a different approach. At the rated motor torque, the air-gap power P_g is constant. From figure 6.17, when the rotor copper loss is neglected, the maximum power carried by the diode bridge, inverter, and transformer will be $s_{max} P_g$ (that is, s_{max} times the motor rating).

When the speed control is required only in a narrow range, the kVA rating of the diode bridge, inverter, and transformer will be small. Therefore the capital cost of the drive will be low. For example, when speed control is required from synchronous to 80 percent of synchronous speed, as in some fan and pump drives, the kVA rating of each of these will be 20 percent of the motor kVA rating. Compare this with the variable frequency voltage source PWM inverter drive. An inverter will use a pair of six self-commutated switches and six diodes, and the diode bridge will use another pair of six diodes. Here the rating is independent of the speed range. Hence, each of these pairs will have a rating at least equal to that of the motor. Furthermore, the control will be complex.

The low cost, coupled with good efficiency and simple control, makes this drive suitable for large capacity [Megawatts range] fan and pump drives, where speed control is required only in a narrow range. The drive has two drawbacks. It cannot provide subsynchronous braking, and the power factor is rather poor. The former drawback is not of much consequence in fan and pump drives, because the fluid pressure itself is able to provide adequate braking torque. However, the poor power factor does cause concern, particularly when the drive capacity is very large. Use of controlled flywheeling or pulse-width modulation can be of great help in improving the power factor.

When the diode bridge, inverter, and transformer are chosen to provide control in a narrow speed range, a separate arrangement is required for starting the motor. Usually, the motor is started by connecting external resistors in the rotor. A starting arrangement is shown in figure 9.9. Initially, contacts C_2 are closed and contacts C_1 are open. The drive starts with the rotor resistance control. At the lowest speed in the control range, a speed sensitive device causes contacts C_1 to close and the inverter to be activated. Contacts C_2 are opened after a suitable time delay to remove the resistors.

In low-voltage and low-power drives, the transformer is not employed, to reduce the cost. While this does not change the inverter current rating, the voltage rat-

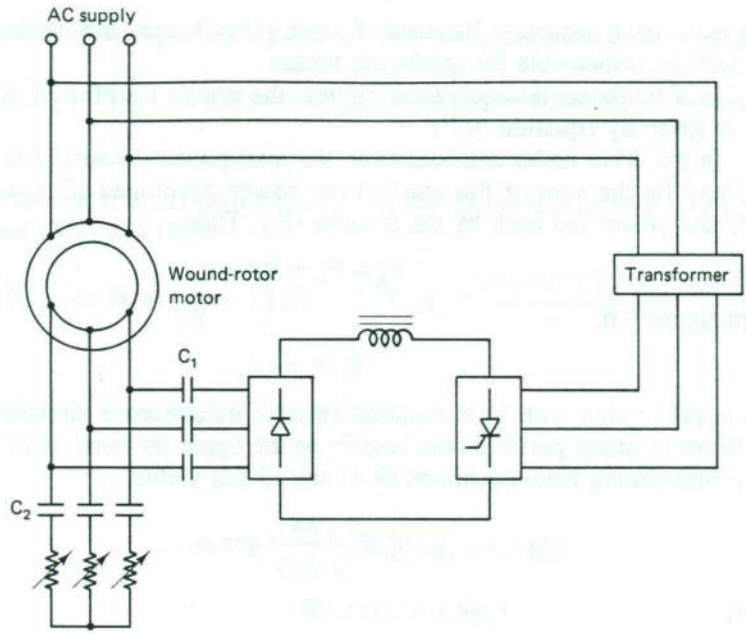

Figure 9.9 Starting of Scherbius drive.

ing is increased to that of the line, and the power factor is greatly reduced. The diode bridge may also be chosen to withstand rotor voltage for unity slip, to dispense with the external starting resistors. For very low power applications (less than 10 kW), a 1-phase inverter is sometimes used, again to reduce the cost. Since the power is now fed to only two lines, the line currents are unbalanced.

9.2.3 Equivalent Circuit and Analysis

Equation (9.25) was derived with the limited purpose of showing that I_d and I_r' remain approximately constant for a constant torque. It is too imprecise to be used for the calculation of the motor speed-torque curves. In this section, a fundamental frequency equivalent circuit and a method of calculating the drive performance more accurately are described. In this analysis, the same six assumptions, which are described in section 9.2, are made. Since the dc link current has been assumed ripple free for a given I_d, the rotor current waveform and amplitude will be identical to that of the static rotor resistance controlled drive of figure 9.1, and the rms rotor current I_{rms} and the fundamental rotor current I_r are given by equations (9.3) and (9.4) and their interrelationship by equation (9.5).

Equivalent Circuit

For the fundamental equivalent circuit to represent the drive performance satisfactorily, it should not only retain the relationship between the fundamental voltages and currents, but it should also allow us to calculate developed torque, total copper loss — including the copper loss caused by harmonic currents — and power fed back

with reasonable accuracy. Because of assumption 3, only the fundamental rotor current will be responsible for producing torque.

In a fundamental equivalent circuit, the power transferred across the air-gap (P_g) is given by equation (9.7).

In the drive under consideration, the total power consumed in the rotor circuit (P_g') will be the sum of the mechanical power developed (P_m), rotor copper loss (P_{cr}), and power fed back by the inverter (P_r). Thus,

$$P_g' = P_m + P_{cr} + P_r \tag{9.27}$$

From figure 9.6,

$$P_r = -V_I I_d$$

The negative sign with V_I is required because its reference direction is chosen with the lower terminal positive with respect to the upper terminal as in any rectifier circuit. Substituting from equations (9.4) and (9.22) yields

$$P_r = -\frac{3VI_r}{a_{T2}} \cos \alpha \tag{9.28}$$

Now,

$$P_{cr} = 3I_{rms}^2 R_r + I_d^2 R_d$$

Substituting from equation (9.3) yields

$$P_{cr} = 3I_{rms}^2 (R_r + 0.5R_d) \tag{9.29}$$

which suggests that the effective rotor circuit resistance per phase is $(R_r + 0.5R_d)$. Substituting from equation (9.5) gives

$$P_{cr} = \frac{\pi^2}{3} I_r^2 (R_r + 0.5R_d) \tag{9.30}$$

The torque and mechanical power are produced only by the fundamental rotor current. The slip power with the fundamental rotor current is

$$sP_{g1} = 3I_r^2 (R_r + 0.5R_d) + P_r \tag{9.31}$$

where P_{g1} is the fundamental air-gap power in the drive.

The mechanical power developed by the fundamental rotor current is given by

$$P_m = (1 - s)P_{g1}$$

Substituting from equation (9.31)

$$P_m = [3I_r^2 (R_r + 0.5R_d) + P_r]\left(\frac{1 - s}{s}\right) \tag{9.32}$$

Substituting from equation (9.28) yields

$$P_m = 3\left[I_r^2 (R_r + 0.5R_d) - \frac{VI_r}{a_{T2}} \cos \alpha\right]\left(\frac{1 - s}{s}\right) \tag{9.33}$$

Substituting from equations (9.28), (9.30), and (9.33) in equation (9.27)

$$P'_g = 3\left(\frac{\pi^2}{9} - 1\right)(R_r + 0.5R_d)I_r^2 + \frac{3}{s}\left[(R_r + 0.5R_d)I_r^2 - \frac{VI_r}{a_{T2}}\cos\alpha\right] \tag{9.34}$$

The fundamental equivalent circuit must satisfy the condition $P_g = P'_g$. Hence, from equations (9.7) and (9.34)

$$EI'_r \cos\Theta_r = \left(\frac{\pi^2}{9} - 1\right)(R_r + 0.5R_d)I_r^2 + \frac{(R_r + 0.5R_d)}{s}I_r^2 - \frac{VI_r}{sa_{T2}}\cos\alpha \tag{9.35}$$

$$= R_h I_r^2 + \frac{R_f}{s}I_r^2 + \frac{V_r}{s}I_r \tag{9.36}$$

where

$$R_h = \left(\frac{\pi^2}{9} - 1\right)(R_r + 0.5R_d) \tag{9.37}$$

$$R_f = (R_r + 0.5R_d) \tag{9.38}$$

and

$$V_r = -\frac{V\cos\alpha}{a_{T2}} \tag{9.39}$$

Also referring all parameters on the right side of equation (9.36) to the stator side yields

$$EI'_r \cos\Theta_r = \left(R'_h + \frac{R'_f}{s}\right)I_r'^2 + \left(\frac{V'_r}{s}\right)I'_r \tag{9.40}$$

where

$$R'_h = a_{T1}^2 R_h \quad \text{and} \quad R'_f = a_{T1}^2 R_f \tag{9.41}$$

and

$$V'_r = a_{T1}V_r = -a_T V\cos\alpha \tag{9.42}$$

The per-phase fundamental equivalent circuit of the drive referred to the stator, as obtained from equation (9.40), is shown in figure 9.10. Resistance (R'_f/s) accounts for the developed mechanical power and the fundamental rotor copper loss.

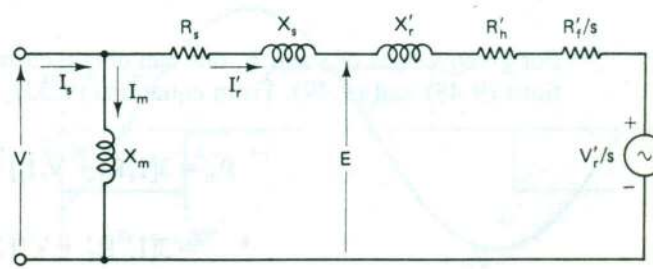

Figure 9.10 Equivalent circuit of wound-rotor motor with static Scherbius control.

Resistance R'_h accounts for the rotor harmonic copper loss. Counter emf (V'_r/s) accounts for the power fed back to the ac mains. It is always in phase with I'_r.

In deriving the equivalent circuit of figure 9.10, the energy loss in the diode bridge and the inverter has been ignored. This loss is negligible except for speeds close to synchronous speed. For accurate prediction of performance for speeds close to synchronous speed, this loss can be accounted for by noting that two diodes and two thyristors will conduct simultaneously causing nearly a constant voltage drop equal to the sum of voltages across the four devices. Let this drop be denoted by V_D. This can be accounted for by modifying V'_r as given next.

$$V'_r = -a_T V \cos \alpha + \frac{\pi a_{T1}}{3\sqrt{6}} V_D \tag{9.43}$$

Speed-Torque Curves

From the equivalent circuit of figure 9.10,

$$\left[\left(R_s + R'_h + \frac{R'_f}{s}\right) I'_r + \frac{V'_r}{s}\right]^2 + [(X_s + X'_r) I'_r]^2 = V^2$$

or

$$[RI'_r + V_{eq}]^2 + [XI'_r]^2 = V^2 \tag{9.44}$$

where

$$R = R_s + R'_h + R'_f/s \tag{9.45}$$

$$V_{eq} = V'_r/s \tag{9.46}$$

$$X = X_s + X'_r \tag{9.47}$$

From equation (9.44),

$$(R^2 + X^2) I'^2_r + 2V_{eq} R I'_r - (V^2 - V^2_{eq}) = 0$$

Hence,

$$I'_r = \frac{-V_{eq} R + \sqrt{V^2_{eq} R^2 + (R^2 + X^2)(V^2 - V^2_{eq})}}{(R^2 + X^2)} \tag{9.48}$$

Also from figure 9.10 [or equation (9.44)]

$$\cos \Theta_r = \frac{RI'_r + V_{eq}}{V} \tag{9.49}$$

For given values of s and α, one can obtain current I'_r and its phase angle from equations (9.48) and (9.49). From equations (9.33), (9.38), and (9.39),

$$\begin{aligned} P_m &= 3[I_r^2 R_f + V_r I_r] \frac{(1-s)}{s} \\ &= 3[I'^2_r R'_f + V'_r I'_r] \frac{(1-s)}{s} \end{aligned} \tag{9.50}$$

Now,

$$T = \frac{P_m}{(1-s)\omega_{ms}} = \frac{3}{s\omega_{ms}}[I_r'^2 R_f' + V_r' I_r'] \tag{9.51}$$

Power Factor

When the supply voltage is assumed sinusoidal,

$$\text{Drive power factor (P.F.)} = \frac{\text{real power input}}{\text{apparent power input}} = \frac{VI_L \cos \phi_L}{VI_{L(rms)}}$$

where I_L = fundamental line current
ϕ_L = phase difference between $\bar{V}$ and $\bar{I}_L$
and $I_{L(rms)}$ = rms line current
Now,

$$\text{P.F.} = \frac{I_L}{I_{L(rms)}} \cos \phi_L = \mu \cos \phi_L \tag{9.52}$$

where μ is the distortion factor and $\cos \phi_L$ is the fundamental power factor.

Equation (9.52) suggests that the P.F. can be calculated if the fundamental line current I_L and its phase angle ϕ_L, and the rms line current $I_{L(rms)}$ are known. These are evaluated as follows.

For a ripple-less I_d, the rotor and inverter ac currents have identical amplitudes and waveforms, but the inverter current lags behind the inverter voltage by an angle α. If the rotor phase current waveform shown in figure 9.2 is shifted by an angle α, the inverter ac current waveform is obtained. The transformer line side current i_T has the same phase, but the amplitude is I_d/a_{T2}. The phase A current waveform of the transformer is shown in figure 9.11. The supply phase A voltage has been taken as the reference phasor. Thus,

$$v_{AN} = \sqrt{2}\, V \sin \omega t \tag{9.53}$$

The stator phase A current i_{sA} has also been shown in the figure. It lags behind v_{AN} by an angle ϕ_s. Thus,

$$i_{sA} = \sqrt{2}\, I_s \sin(\omega t - \phi_s) \tag{9.54}$$

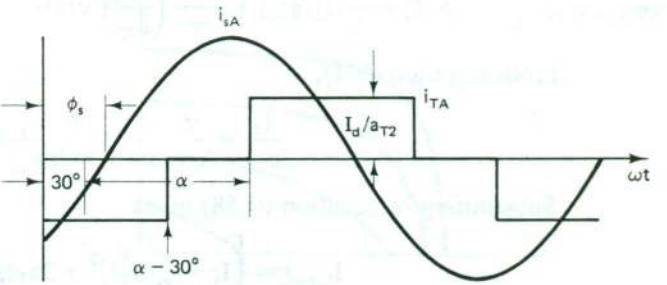

Figure 9.11 Stator current and feedback current waveforms.

Now,

$$\bar{I}_T = \frac{I_r \angle -\alpha}{a_{T2}} = a_T I_r' \angle -\alpha \tag{9.55}$$

and

$$\bar{I}_L = \bar{I}_T + \bar{I}_s$$

or

$$I_L \angle -\phi_L = a_T I_r' \angle -\alpha + I_s \angle -\phi_s \tag{9.56}$$

From figure 9.11, the instantaneous line current i_L is given by the following equation:

$$\begin{aligned} i_L &= \sqrt{2}\, I_s \sin(\omega t - \phi_s) - I_d/a_{T2}, \qquad \phi_s \leq \omega t \leq \left(\alpha - \frac{\pi}{6}\right) \\ &= \sqrt{2}\, I_s \sin(\omega t - \phi_s), \qquad \left(\alpha - \frac{\pi}{6}\right) \leq \omega t \leq (\alpha + \pi/6) \\ &= \sqrt{2}\, I_s \sin(\omega t - \phi_s) + I_d/a_{T2}, \qquad \left(\alpha + \frac{\pi}{6}\right) \leq \omega t \leq (\pi + \phi_s) \end{aligned} \tag{9.57}$$

Now,

$$\begin{aligned} I_{L(rms)}^2 &= \frac{1}{\pi}\left[\int_{\phi_s}^{\alpha-\pi/6} (\sqrt{2}\, I_s \sin(\omega t - \phi_s) - I_d/a_{T2})^2\, d\omega t \right. \\ &\quad + \int_{\alpha-\pi/6}^{\alpha+\pi/6} (\sqrt{2}\, I_s \sin(\omega t - \phi_s))^2\, d\omega t \\ &\quad \left. + \int_{\alpha+\pi/6}^{\pi+\phi_s} (\sqrt{2}\, I_s \sin(\omega t - \phi_s) + I_d/a_{T2})^2\, d\omega t \right] \\ &= \frac{1}{\pi}\left[\int_{\phi_s}^{\pi+\phi_s} 2I_s^2 \sin^2(\omega t - \phi_s)\, d(\omega t) + \int_{\phi_s}^{\alpha-\pi/6} \frac{I_d^2}{a_{T2}^2}\, d(\omega t) + \int_{\alpha+\pi/6}^{\pi+\phi_s} \frac{I_d^2}{a_{T2}^2}\, d(\omega t) \right. \\ &\quad - \int_{\phi_s}^{\alpha-\pi/6} \frac{2\sqrt{2}\, I_s I_d}{a_{T2}} \sin(\omega t - \phi_s)\, d(\omega t) \\ &\quad \left. + \int_{\alpha+\pi/6}^{\pi+\phi_s} \frac{2\sqrt{2}\, I_s I_d}{a_{T2}} \sin(\omega t - \phi_s)\, d(\omega t) \right] \\ &= I_s^2 + \frac{2}{3}(I_d^2/a_{T2}^2) + \frac{2\sqrt{6}}{\pi}\left(\frac{I_d I_s}{a_{T2}}\right) \cos(\alpha - \phi_s) \end{aligned} \tag{9.58}$$

From equation (9.4),

$$\frac{I_d}{a_{T2}} = \frac{\pi}{\sqrt{6}} \frac{I_r}{a_{T2}} = \frac{\pi}{\sqrt{6}} \frac{a_{T1}}{a_{T2}} I_r' = \frac{\pi}{\sqrt{6}} a_T I_r' \tag{9.59}$$

Substituting in equation (9.58) gives

$$I_{L(rms)} = \left[I_s^2 + \frac{\pi^2}{9} a_T^2 I_r'^2 + 2 a_T I_s I_r' \cos(\alpha - \phi_s) \right]^{1/2} \tag{9.60}$$

Substitution from equations (9.56) and (9.60) into equation (9.52) gives the power factor.

9.2.4 Performance

The nature of speed-torque curves of a static Scherbius drive with α as a parameter is shown in figure 9.12. For speeds close to synchronous speed, regulation is good. It becomes poor at low speeds, but the drive is rarely used in this range. Because of the additional losses in the diode bridge, inverter, filter, and resistance R_h', the rated drive speed for $\alpha = 90°$ is lower than the rated motor speed. The harmonic content in the rotor current is the same as for the static rotor resistance control. Hence, their effect on the motor is similar to that described for the static rotor resistance control in section 9.1.1. The important points to be noted are a derating of the motor by more than 5 percent and that the harmonic torques, both steady and pulsating, are negligible.

When the drive has been designed to operate in a limited speed range, the fall of motor speed below the control range, due to a disturbance, may cause the dc link voltage to exceed the peak of the inverter input voltage. This will cause commutation failure in the inverter and consequently the short-circuiting of the dc link and ac source by the inverter's thyristors. Such a situation can be avoided by installing a speed sensitive device which will change the operation to rotor resistance control whenever the speed falls below the control range. The same device can be used to reconnect the diode bridge (fig. 9.9) when the speed reaches the control range. This speed sensitive device can also be used for switching over from resistance control to static Scherbius control during starting.

If a reversible drive is required, a contactor for phase sequence reversal is incorporated in the stator circuit. Speed reversal is accomplished by switching over to rotor resistance control (fig. 9.9).

Some pump and fan drives require closed-loop speed control. Closed-loop control is obtained by using the inner current control scheme of figure 9.13. This is similar to the single-quadrant closed loop speed control of a rectifier-fed dc motor, shown in figure 5.1b. The functions of various blocks in the speed and current loops are the same as described in section 5.1 for the dc motor drive of figure 5.1b. Since the static Scherbius drive does not have subsynchronous braking capability, it is capable of providing only forward motoring operation. Any decrease in speed command causes the inverter counter emf to increase. Consequently, I_d falls to zero,

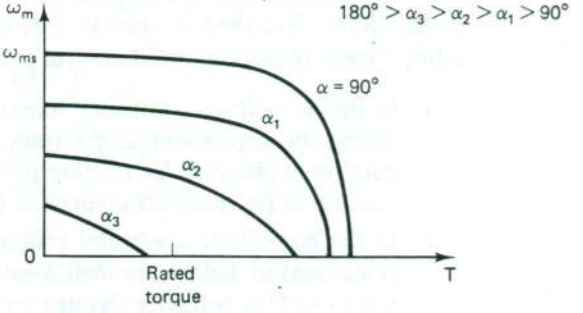

Figure 9.12 Nature of speed-torque curves of static Scherbius drive.

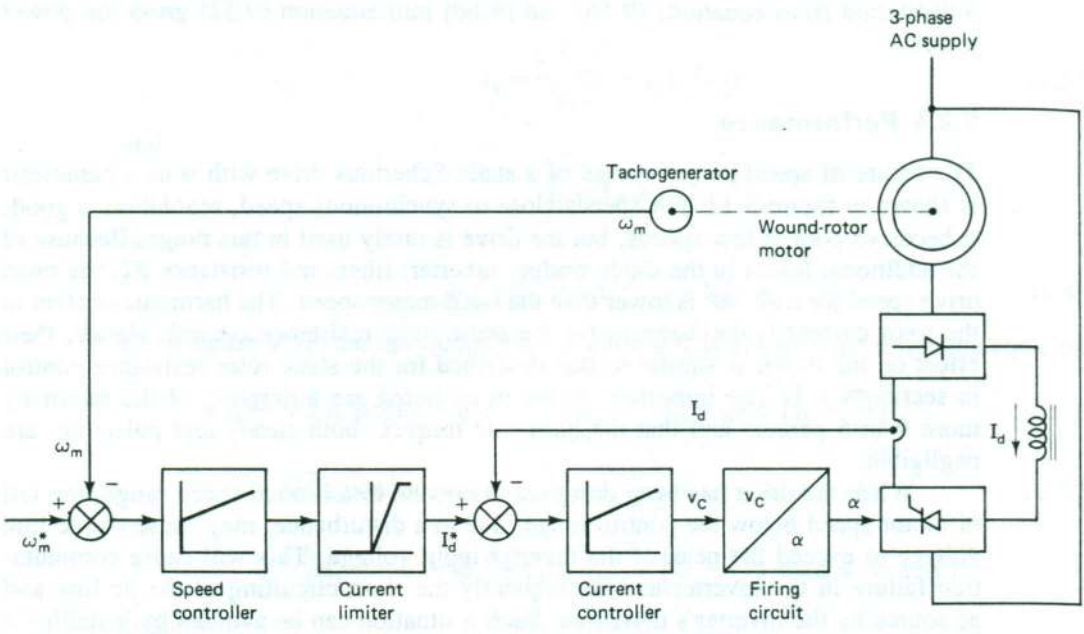

Figure 9.13 Closed-loop speed control of static Scherbius drive.

causing the motor torque to go to zero. The drive, therefore, decelerates only due to the load torque. In fan and pump drives, the load torque is large enough to provide adequate deceleration. When the speed reaches close to the desired speed, the dc link current flows and its value is adjusted to make the motor torque equal to the load torque at the set speed. Any increase in the speed command sets the current reference for the maximum dc link current. The drive accelerates at the maximum allowable current and torque. When near the set speed, the current limiter desaturates and the current value is adjusted to produce a motor torque equal to the load torque at the set speed.

When the inverter is operated with the conventional fully controlled operation, the dc link current is sensed by sensing the inverter ac side current, as is done in dc drives. When operated with controlled flywheeling or pulse-width modulation, the link current is no longer proportional to inverter ac currents; consequently, it must be sensed directly from the dc link.

It is worthwhile to compare the static Scherbius drive with the ac voltage controller drive described in section 7.3, because both are commonly used in fan and pump drives requiring speed control over a small range.

1. In the ac voltage controller scheme, the speed reduction is obtained by dissipating the slip power in the rotor. In the static Scherbius drive, the speed reduction is obtained by feeding the slip power back to the line. Hence the latter scheme is far more efficient than the former.
2. In the ac voltage controller scheme, the motor current increases as the speed is decreased below the full-load value. The maximum value is reached at $s = 1/3$. This behavior derates the motor. To keep the derating low, use of an

inefficient high slip motor, with a full-load slip between 0.1 and 0.2, becomes mandatory. In a static Scherbius drive, the current decreases with speed; hence, it does not suffer with this type of derating. Therefore, a highly efficient low-slip wound-rotor motor can be used. This increases full-load speed, power, and efficiency.

3. The static Scherbius drive power factor is low at and near full-load speed.
4. The ac voltage controller drive employs a squirrel-cage motor; hence, it does not suffer from the limitations of a wound-rotor motor described at the beginning of this chapter.
5. Initial cost of a static Scherbius drive is higher, but the running cost is lower.

Example 9.2

The motor of example 9.1 is now controlled by a static Scherbius drive. The drive has been designed to provide speed control up to 50 percent of the synchronous speed. The maximum value of the firing angle is 170° and $R_d = 0.02\ \Omega$.

1. Calculate "a_T".
2. Calculate the torque and power factor for $\alpha = 120°$ and 720 rpm.
3. Calculate α for the rated torque and 720 rpm.

Solution:

1. If the no-load speed is chosen to be 50 percent of the synchronous speed, then for all loads, speed control up to half-synchronous speed is assured. Thus from equation (9.23),

$$0.5 = -a_T \cos 170°$$

or

$$a_T = 0.508$$

2. $s = (1200 - 720)/1200 = 0.4$

From equations (9.37) to (9.42),

$$R_f' = a_{T1}^2(R_r + 0.5R_d) = R_r' + 0.5R_d a_{T1}^2$$
$$= 0.6 + 0.5 \times 0.02 \times 6.25 = 0.6625\ \Omega$$
$$R_h' = \left(\frac{\pi^2}{9} - 1\right)R_f' = \left(\frac{\pi^2}{9} - 1\right) \times 0.6625 = 0.064\ \Omega$$
$$V_r' = -a_T V \cos\alpha = -0.508 \times 265.6 \cos 120° = 67.46\ \text{V}$$

From equations (9.45) to (9.47),

$$V_{eq} = V_r'/s = 67.46/0.4 = 168.7\ \text{V}$$
$$X = 3.6\ \Omega,$$
$$R = 0.4 + 0.064 + 0.6625/0.4 = 2.12\ \Omega$$
$$V_{eq}R = 168.7 \times 2.12 = 357.7\ \text{V}$$
$$R^2 + X^2 = 2.12^2 + 3.6^2 = 17.45$$
$$V^2 - V_{eq}^2 = 265.6^2 - 168.7^2 = 42{,}084$$

Substituting the known values in equation (9.48) gives

$$I_r' = \frac{-357.7 + \sqrt{357.7^2 + 17.45 \times 42{,}084}}{17.45} = 32.72 \text{ A}$$

From equation (9.51),

$$T = \frac{3}{0.4 \times 125.66}[32.72^2 \times 0.66\,25 + 67.46 \times 32.72]$$
$$= 174 \text{ N-m}$$

Substitution of the known values in equation (9.49) gives

$$\cos \Theta_r = \frac{2.12 \times 32.72 + 168.7}{265.6} = 0.9$$

or

$$\Theta_r = 25.8°$$

$$I_m = \frac{265.6}{40} = 6.64 \text{ A}$$

$$\bar{I}_s = \bar{I}_r' + \bar{I}_m = 32.72\underline{/-25.8} + 6.64\underline{/-90°}$$
$$= 36.1\underline{/-35.4°} \text{ A}$$

$$\bar{I}_T = a_T I_r'\underline{/-\alpha} = 0.508 \times 32.72\underline{/-120°} = 16.62\underline{/-120°} \text{ A}$$

$$\bar{I}_L = \bar{I}_s + \bar{I}_T = 36.1\underline{/-35.4} + 16.62\underline{/-120°} = 41.13\underline{/-59.1°} \text{ A}$$

Substituting the known values in equation (9.60) gives

$$I_{L(rms)} = \left[36.1^2 + \frac{\pi^2}{9}(0.508)^2 \times (32.72)^2 + 2 \times 0.508 \times 36.1 \times 32.72 \cos(84.6°)\right]^{1/2} = 41.46 \text{ A}$$

From equation (9.52),

$$\text{P.F.} = \frac{41.13}{41.46} \cos 59.1° = 0.51$$

3. From example 9.1, rated torque = 78.5 N-m.
From equation (9.42),

$$V_r' = -a_T V \cos \alpha = -0.508 \times 265.6 \cos \alpha$$
$$= -134.9 \cos \alpha \qquad \text{(E9.3)}$$

Substitution of the known values in equation (9.51) gives

$$78.5 = \frac{3}{0.4 \times 125.66}[I_r'^2 \times 0.6625 + V_r' I_r']$$

or

$$0.6625 I_r'^2 + V_r' I_r' = 1315 .$$

Substituting from equation (E9.3) and rearranging the terms gives

$$I_r'^2 - 203.6I_r' \cos \alpha = 1985 \tag{E9.4}$$

From part 2, $R = 2.12\ \Omega$, $X = 3.6\ \Omega$, $R^2 + X^2 = 17.45$
Now $V_{eq} = V_r'/s = -(134.9 \cos \alpha)/0.4 = -337.25 \cos \alpha$
and $2V_{eq}R = -2 \times 2.12 \times 337.25 \cos \alpha = -1429.9 \cos \alpha$

From equation (9.48),

$$(R^2 + X^2)I_r'^2 + 2V_{eq}RI_r' - (V^2 - V_{eq}^2) = 0$$

Substituting known values,

$$17.45I_r'^2 - 1429.9I_r' \cos \alpha - [265.6^2 - (337.25)^2 \cos^2 \alpha] = 0$$

or

$$I_r'^2 - 81.94I_r' \cos \alpha - [4043 - 6518 \cos^2 \alpha] = 0 \tag{E9.5}$$

Subtracting (E9.5) from (E9.4) yields

$$-121.7I_r' \cos \alpha - 6518 \cos^2 \alpha + 2058 = 0 \tag{E9.6}$$

Eqns. (E9.4) and (E9.6) are two simultaneous nonlinear equations in I_r' and α. Their solution by iteration gives $I_r' = 12.89$ A and $\alpha = 134°$.

9.2.5 Supersynchronous Speed Control

It was explained in section 9.2 that the drive of figure 9.6 can provide subsynchronous motoring and supersynchronous regenerative braking operations only, because the power flow can take place only from the rotor to the ac mains. If the changes are made such that the power flow between the rotor and the ac mains becomes bidirectional, then the motoring and braking operations can be obtained both for subsynchronous and supersynchronous speeds. If the rating is chosen to provide operation at the maximum slip s_{max} then with the provision for supersynchronous speeds, the slip can be controlled from $-s_{max}$ to $+s_{max}$, consequently doubling the speed range for the same rating. To realize these operations, the voltage injected into the rotor must track the frequency and phase sequence of the rotor-induced voltages, as explained in section 6.4.4. Thus, the phase sequence of the injected voltage must be reversed for operation at supersynchronous speeds. Next we discuss three possible changes to be made in the drive of figure 9.6 to also get speed control at supersynchronous speeds.

The first change which allows the power flow in either direction is to replace the diode bridge by a 6-pulse fully controlled rectifier. The power flow reverses when the rectifier works as an inverter and the existing inverter is made to work as a rectifier. The increase in speed range is obtained at the expense of an increase in cost and complexity. The cost of a thyristor bridge is substantially higher than that of a diode bridge. A slip frequency gating circuit is needed for the new converter. Near the synchronous speed, the magnitude of the rotor-induced voltages becomes insufficient to provide line commutation of thyristors of the new rectifier, consequently the drive fails to develop any torque near synchronous speed.

The second change which can allow the power flow in either direction is to replace the diode bridge by a current source inverter.[15] The inverter frequency and phase sequence are made to track the frequency and the phase sequence of the rotor induced voltages. As explained in section 8.2.3, a current source inverter and fully controlled rectifier combination is capable of allowing power flow in either direction. Because of the forced commutation, no thyristor commutation problem occurs near synchronous speed. At synchronous speed, the current source inverter supplies the rotor with the dc current necessary to produce torque. The cost and complexity of this drive is substantially higher than the drive of figure 9.6.

The third possible change is to replace the diode bridge and inverter of figure 9.6 by a cycloconverter, which may be operated either as a voltage source or current source.[16,17] The cycloconverter frequency and phase sequence are made to track those of the rotor-induced voltages. Since the thyristors of the cycloconverter depend for their line commutation on the transformer output voltages which are constant and independent of the drive speed, no commutation problem is encountered near synchronous speed. At synchronous speed, the cycloconverter supplies the dc current required to produce torque. When the drive is to be controlled in a limited speed range, the cycloconverter frequency can be restricted to well below one-third of the source frequency, ensuring nearly a sinuosidal rotor current and thus improving efficiency and reducing motor derating.

Because of the reactive power drawn by the cycloconverter, the drive has a low power factor. The explanation for this behavior is the same as that given for the drive of figure 9.6 in section 9.2.1. Sometimes the transformer is eliminated and the cycloconverter is directly connected to the mains to reduce cost, weight, and volume of the drive. But this makes the power factor very poor and increases the cost of the cycloconverter due to an increase in the voltage rating of its thyristors.

Since the cycloconverter employs a large number of thyristors, the drive is suitable only for very large capacity drives.

In all three schemes, when the drive is designed for a limited speed range, resistance control is used for starting. The drive does not have the capability of speed reversal. If speed reversal is required, a phase sequence reversing contactor is connected in the stator and rotor resistance control is used during reversal. The first scheme is not used in practice. The last two are employed in very large capacity pump or fan drives. But they have limited applications because the increase in cost and complexity outweigh the benefits available from the increase in speed range. The subsynchronous static Scherbius drive is generally preferred because of its low cost and simplicity.

9.3 MODIFIED KRAMER DRIVES

Kramer suggested that the slip power taken from the rotor for speed control can be usefully employed by converting it to mechanical power in an auxiliary motor mounted on the induction motor shaft.[7] The mechanical power produced by the auxiliary motor supplements the main motor power, thus allowing the same power to be

delivered to the load at different speeds. In the earlier static Kramer drive, the slip frequency rotor voltages were converted into dc by a diode bridge. The rectified dc voltage was fed to the armature of a dc motor mounted on the induction motor shaft. If the armature resistance drop is neglected, then the dc machine induced voltage is equal to the dc output voltage of the diode bridge, which in turn depends on the induction motor slip. By controlling the field, the dc motor induced voltage, and consequently the induction motor speed, can be controlled. This static Kramer drive is not employed anymore because of problems associated with dc motors, particularly at high power levels. Instead, the drive of figure 9.14 is employed.[18] In this drive, the dc machine is replaced by a commutatorless dc motor, which consists of a synchronous motor fed by a load commutated inverter. The operation of the commutatorless motor is described in chapter 11. The speed is controlled by varying the commutation lead angle. The speed can also be controlled by the field current control. However, it is not preferred because of the following problems associated with it. To drive the system at synchronous speed, the field current must be reduced to zero to reduce the inverter dc terminal voltage to zero. Now the induced voltage will not be sufficient to obtain load commutation. Two other problems associated with field current control are the slow response of the field circuit and increased armature reaction at low field currents. The field current control can, however, be used to control flux and the synchronous motor terminal voltage within the ratings of inverter thyristors.

The drive has a better power factor and lower harmonic content in the line current compared to the static Scherbius drive. In the static Scherbius drives, reactive power and harmonics are associated with the power fed back to the line. In the static Kramer drive, since the power is not fed back to the line, problems associated with the feedback of power are also eliminated.

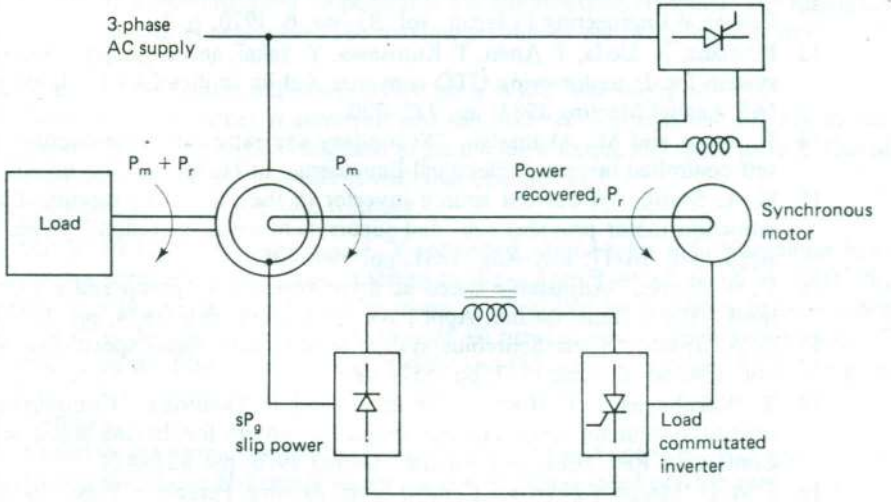

Figure 9.14 Commutatorless Kramer drive.

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PROBLEMS

9.1 A 460 V, 60 Hz, 6 pole, 1170 rpm, Y-connected wound-rotor induction motor has the following parameters per phase referred to the stator: $R_s = 0.1\ \Omega$, $X_s = 0.3\ \Omega$, $X_m = 10\ \Omega$, $R_r' = 0.06\ \Omega$, $X_r' = 0.6\ \Omega$. Stator to rotor turns ratio is 2. The motor speed is controlled by the static rotor resistance control. Filter resistance is $0.01\ \Omega$. The value of external resistance is chosen such that for $\delta = 0$, the breakdown torque is obtained at $s = 2$.

1. Calculate the value of the external resistance.
2. Calculate the torque for $\delta = 0.5$ and $s = 0.5$.

Neglect friction, windage, and core loss.

9.2 For the static rotor resistance controlled drive of problem 9.1, calculate and plot the boundaries of the control region (region ABCD of figure 9.4).

9.3 A 460 V, 60 Hz, 4 pole, 1760 rpm, Y-connected wound-rotor induction motor has the following parameters per phase referred to the stator: $R_s = 0.14\ \Omega$, $X_s = 0.4\ \Omega$, $R_r' = 0.1\ \Omega$, $X_r' = 0.7\ \Omega$, $X_m = 18\ \Omega$. The motor is controlled by the static rotor resistance control. $R_d = 0.01\ \Omega$. The value of the external resistance is chosen such that at standstill and $\delta = 0$, the motor torque is 50 percent of the rated torque. $\alpha_{T1} = 2$.

1. Calculate the value of the external resistance.
2. Calculate δ for a speed of 600 rpm and the rated torque.
3. Calculate the speed for $\delta = 0.5$ and the rated torque.

Neglect core loss, friction, and windage.

9.4 For the drive of problem 9.1

1. Obtain δ for a speed of 1020 rpm and the rated torque.
2. Calculate the speed for $\delta = 0.8$ and 120 percent of the rated torque.

Neglect friction, windage, and core loss.

9.5 The motor of problem 9.3 is controlled by a static Scherbius scheme. The drive is designed for a speed range of 30 percent below the synchronous speed. The resistance of the filter inductor is negligible. The maximum value of the firing angle is 165°.

1. Calculate a_T.
2. Calculate the torque and power factor for $\alpha = 105°$ and 960 rpm.
3. Now the transformer is removed and the inverter is connected directly to the ac mains. The firing angle is adjusted to get the same torque and speed as in 2. Calculate the power factor and compare it with that obtained in 2.

Neglect core loss, friction, and windage.

9.6 A 440 V, 50 Hz, 1470 rpm, 4 pole, Y-connected wound-rotor induction motor has the following parameters per phase referred to the stator: $R_s = 0.2\ \Omega$, $X_s = 0.5\Omega$, $X_m = 20\ \Omega$, $R_r' = 0.1\ \Omega$, $X_r' = 0.8\ \Omega$. The motor is controlled by a static Scherbius scheme. The drive is designed for a speed range of 50 percent below synchronous speed. The resistance of the filter inductor is negligible and the maximum value of the firing angle is 170°.

1. Obtain a_T.
2. Calculate the torque and power factor for $\alpha = 120°$ and speed of 1025 rpm.
3. Obtain the torque and power factor for $\alpha = 100°$ and speed of 1125 rpm.

Neglect friction, windage, and core loss.

9.7 The motor of problem 9.6 is controlled by a static Scherbius scheme. The drive is designed for a speed range from synchronous speed to standstill, assuming the maximum firing angle to be 180°. The filter inductor has a negligible resistance. Calculate and plot the speed-torque, speed-rotor current, and speed-fundamental power factor curves for $\alpha = 90°$, 120°, and 150°. Neglect friction, windage, and core loss.

9.8 A 460 V, 60 Hz, 1185 rpm, 6 pole, Y-connected wound-rotor induction motor has the following parameters per phase referred to the stator: $R_s = 0.05\ \Omega$, $X_s = 0.3\ \Omega$, $X_m = 8\ \Omega$, $R_r' = 0.04\ \Omega$, $X_r' = 0.4\ \Omega$. The motor is controlled by a static Scherbius scheme. The drive is designed for a speed range of 30 percent below synchronous speed, taking the maximum firing angle to be 170°. The resistance of the filter inductor can be neglected.

1. Calculate a_T.
2. Calculate the inverter firing angle α for the rated motor torque and 960 rpm.
3. What will be the motor speed at the rated torque and $\alpha = 120°$?

Neglect friction, windage, and core loss.